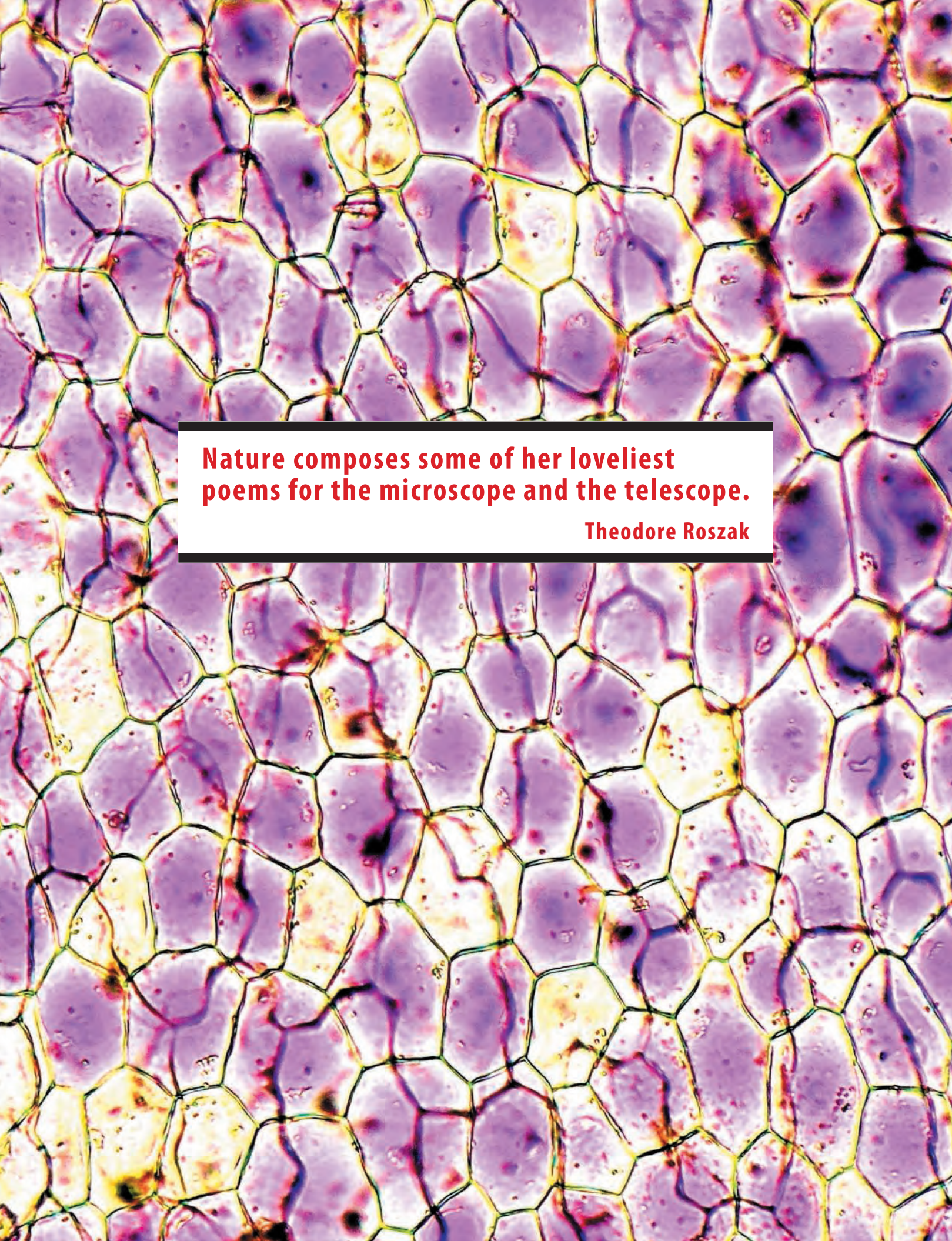


Nature composes some of her loveliest poems for the microscope and the telescope.

Theodore Roszak

1 Science is . . .

Although science is a body of knowledge, it is also a way of solving problems and finding the answers to questions. Scientific knowledge is always growing and changing because scientists design and perform investigations. Observing, measuring, constructing tables, drawing graphs and forming conclusions are just some of the skills used in conducting scientific investigations.

OVERARCHING IDEAS

- Patterns, order and organisation
- Scale and measurement
- Matter and energy
- Stability and change

GENERAL CAPABILITIES

Literacy
Numeracy
ICT competence
Critical and creative thinking
Personal and social competence

SCIENCE AS A HUMAN ENDEAVOUR

Nature and development of science
Use and influence of science

SCIENCE INQUIRY SKILLS

Questioning and predicting
Planning and conducting
Processing and analysing data and information
Evaluating
Communicating

THINK ABOUT SCIENCE

- Why are graphs useful to scientists?
- How do you start your own scientific investigation?
- What do the letters CSIRO stand for?

A scientist studying paper chromatographs

YOUR QUEST

Researching the CSIRO

INVESTIGATE

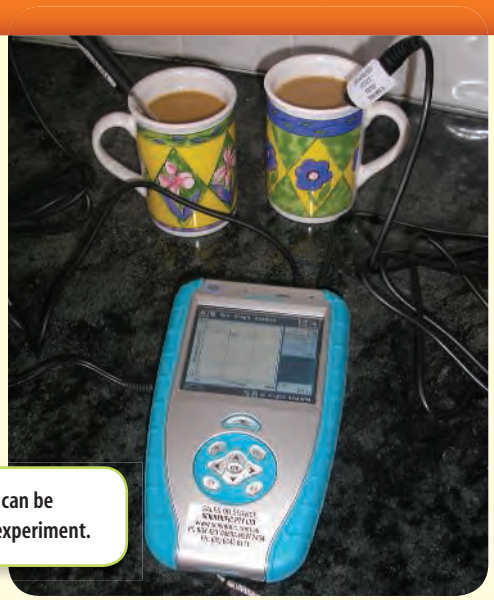
1 What is the CSIRO?

2 The CSIRO's website describes some of the research done by CSIRO scientists. Read the information provided for one area of research that the CSIRO is involved with and summarise this research in point form.

3 Form groups of three. Each student should explain to the other two students the area of research they have just read about. Try doing this without referring to your notes.

eBookplus Weblink

CSIRO
Use the **CSIRO** weblink in your eBookPLUS to answer the questions on the left.



A data logger can be used for this experiment.

INQUIRY: INVESTIGATION 1.1

Milk now or later?

You have just finished making yourself a cup of coffee when the phone rings. So that your coffee is as warm as possible, should you add the milk to your coffee now or after you have finished talking on the phone? Does your answer depend on the length of the phone call?

KEY INQUIRY SKILLS:

- planning and conducting
- processing and analysing data and information

Equipment:

kettle *2 identical cups*
instant coffee *milk*
2 thermometers or a data logger with 2 temperature probes
2 measuring cylinders

- Your teacher will assign a particular 'phone call' time to each group of students.
- Heat some water in a kettle and use it to make two cups of instant coffee. Use the same type of cup and the same amount of hot water and coffee powder.
- Place a thermometer or temperature probe in each cup of coffee. If you are using a data logger, set it to collect results for at least 10 minutes.
- Add 40 mL of milk to one of the cups.
- If you are using thermometers, record the temperature of the coffee in both cups every 30 seconds.
- After your phone call time has passed, add 40 mL milk to the second cup.
- Continue measuring the temperature in both cups every 30 seconds until 10 minutes has passed since you added the milk to the first cup.
- If you used thermometers, record your results in a table.

Time (minutes)	Temperature (°C)	
	Milk added at time 0	Milk added after 'phone call'
0.0		
0.5		
1.0		
1.5		

- Plot line graphs of your results on the same set of axes. Put time on the horizontal axis and temperature on the vertical axis.
- If you used a data logger, a graph is plotted automatically. If necessary, adjust the settings so that the graph shows the temperatures measured by both probes on the same set of axes. Put the graph into the results section of your experiment report.

DISCUSS AND EXPLAIN

1 Does hot coffee cool faster than warm coffee? How can you tell from your graph?

2 Did the two lines on the graph cross at any stage? What does this indicate?

3 Write a conclusion based on your results.

4 Does the length of the 'phone call' affect the results? Compare your graphs with those of other groups.

5 Why was it important to put exactly the same amount of water in both cups and to use the same type of cup?

6 What are the advantages and disadvantages of using a data logger for this experiment?

7 Explain how this experiment could be improved.

Safety first

Conducting scientific investigations in a laboratory can be exciting, but accidents can happen if they are not carried out carefully. There are certain rules that must be followed for your own safety and the safety of others.

Doing experiments in science can be exciting, but accidents can happen if investigations are not carried out carefully. There are certain rules that must be followed for your own safety and the safety of others.

ALWAYS . . .

- follow the teacher's instructions
- wear safety glasses and a laboratory coat or apron, and tie back long hair when mixing or heating substances
- point test tubes away from your eyes and away from your fellow students
- push chairs in and keep walkways clear
- inform your teacher if you break equipment, spill chemicals or cut or burn yourself
- wait until hot equipment has cooled before putting it away
- clean your workspace — don't leave any equipment on the bench
- dispose of waste as instructed by your teacher
- wash your hands thoroughly after handling any substances in the laboratory.

NEVER . . .

- enter the laboratory without your teacher's permission
- run or push in the laboratory
- eat or drink in the laboratory
- smell or taste chemicals unless your teacher says it's ok. When you do need to smell substances, fan the odour to your nose with your hand
- leave an experiment unattended
- conduct your own experiments without the teacher's approval
- put solid materials down the sink
- pour hazardous chemicals down the sink (check with your teacher)
- put hot objects or broken glass in the bin.

Handy hints

- Use a **filter funnel** when pouring from a bottle or container without a lip.
- Never put wooden test-tube holders near a flame.
- Always turn the tap on before putting a **beaker**, **test tube** or **measuring cylinder** under the stream of water.
- Remember that most objects get very hot when exposed to heat or a naked flame.
- Do not use tongs to lift or move beakers.

Working with dangerous chemicals

Your teacher will tell you how to handle the chemicals in each experiment. At times, you may come across warning labels on the substances you are using.

Always wear gloves and **safety glasses** when using chemicals with this symbol. Corrosive substances can cause severe damage to skin and eyes. Acid is an example of a **corrosive** substance.

These substances are easily set on fire so keep them away from flames. Methylated spirits is **flammable**.

Chemicals with this label can cause death or serious injury if swallowed or breathed in. They are also dangerous when touched without gloves because they can be absorbed by the skin. Mercury is a **toxic** substance.

Heating substances

Many experiments that you will conduct in the laboratory require heating. In school laboratories, heating is usually done with a

Bunsen burner. A Bunsen burner provides heat when a mixture of air and gas is lit.

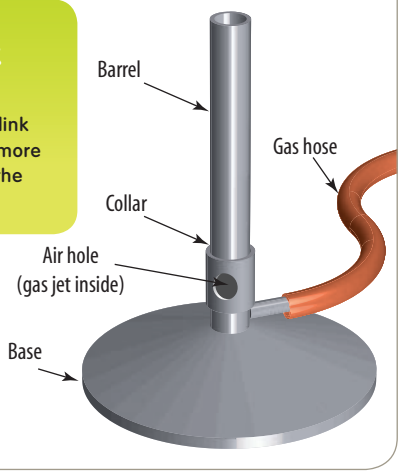
Bunsen burners heat objects or liquids with a naked flame. Always tie hair back, and wear safety glasses and a laboratory coat or apron when using a Bunsen burner.

A GUIDE TO USING THE BUNSEN BURNER

1. Place the Bunsen burner on a heatproof mat.
2. Check that the gas tap is in the 'off' position.
3. Connect the rubber hose to the gas tap.
4. Close the air hole of the Bunsen burner collar.
5. Light a match and hold it a few centimetres above the barrel.
6. Turn on the gas tap and a yellow flame will appear.
7. Adjust the flame by moving the collar until the air hole is open and a blue flame appears.
8. Remember to close the collar to return the flame to yellow when the Bunsen burner is not in use.

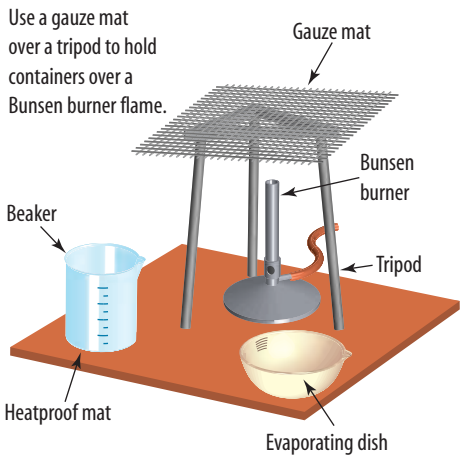
eBookplus Weblink

Robert Bunsen
Use the **Robert Bunsen** weblink in your eBookPLUS to learn more about the man after whom the Bunsen burner was named.



HEATING CONTAINERS

Beakers and evaporating dishes can be placed straight onto a gauze mat for heating. Never look directly into a container while it is being heated. Wait until the equipment has cooled properly before handling it.



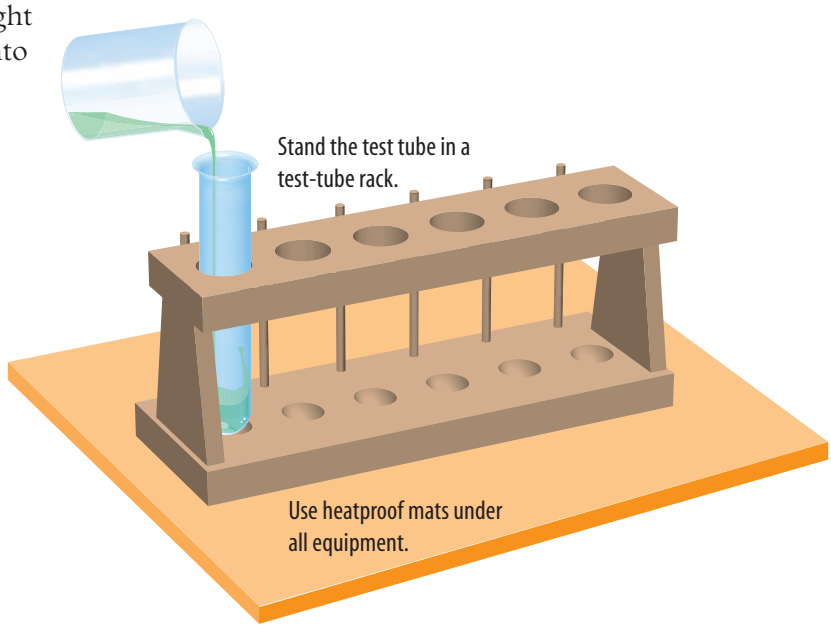
HEATING A TEST TUBE

Tripods and gauze mats are not used when heating test tubes. Hold the test tube with the test-tube holder. Keep the base of the test tube above the flame. Make sure that the test tube points away from you and other students.

Glassware

POURING A LIQUID INTO A TEST TUBE

Pour liquids carefully into the test tube from a beaker or measuring cylinder. Use a filter funnel when pouring from bottles or containers without a lip.

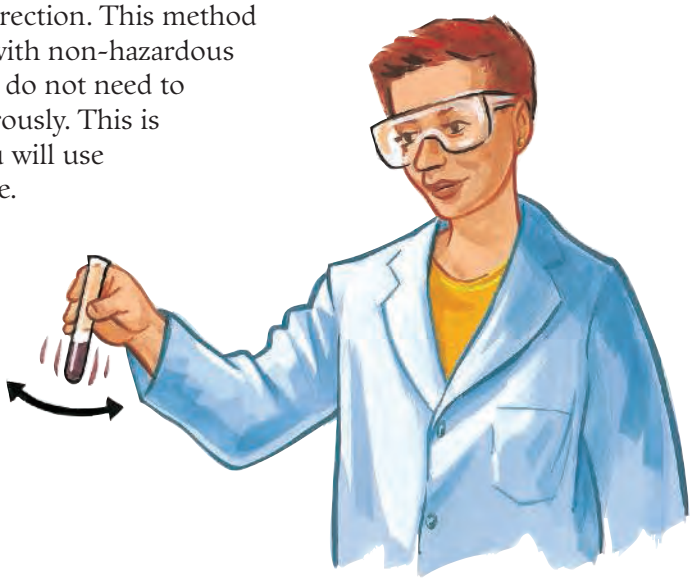


SHAKING A TEST TUBE

There are two ways to shake substances in a test tube.

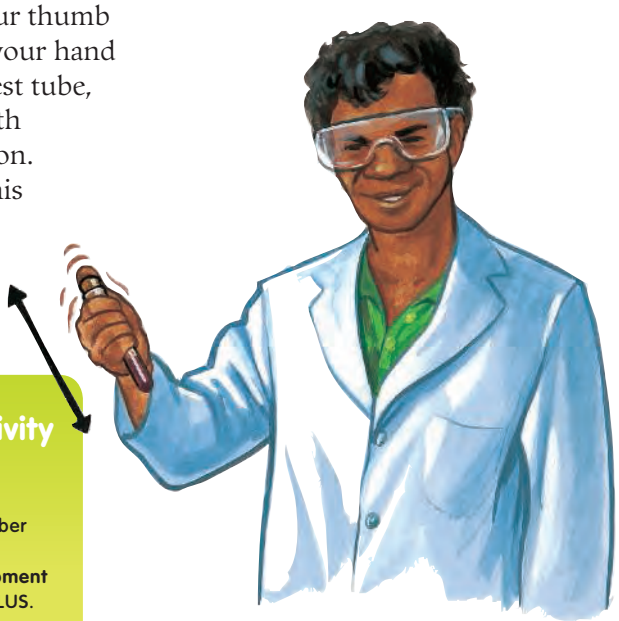
Method 1

Hold the top of the test tube and gently move the base of the test tube in a sideways direction. This method is good to use with non-hazardous substances that do not need to be shaken vigorously. This is the method you will use most of the time.



Method 2

Use a stopper when a substance needs to be mixed by shaking vigorously. Place an appropriately sized stopper into the mouth of the test tube. With your thumb over the stopper and your hand securely around the test tube, shake the test tube with an up and down motion. Shake a test tube in this way only if instructed to do so by the teacher.



eBookplus **Interactivity**

Using equipment
Identify the equipment you will need to perform a number of laboratory processes by completing the **Using equipment** interactivity in your eBookPLUS. **int-0200**

Using electricity safely

Electrical equipment in the science laboratory should be used with great care, just as it should be in the home or workplace. Never:

- place heavy electrical appliances near the edge of a bench or table
- allow water near electrical cords, plugs or power points
- place objects other than the correct electrical plug into a power point
- use appliances with damaged cords or exposed wires.

UNDERSTANDING AND INQUIRING

UNDERSTAND

1 Explain, with the aid of a diagram, how to safely heat a liquid in a test tube using a Bunsen burner. Ensure all relevant safety rules are included in your explanation.

2 How should a substance in a test tube be shaken if you are not instructed to shake it vigorously with a stopper in the mouth of the test tube?

3 Methylated spirits is a flammable liquid. What does this mean?

THINK

4 List the dangers of each of the following examples of unsafe behaviour in the science laboratory.
(a) Not wearing safety glasses while heating a liquid in a beaker
(b) Using an electronic balance to measure the mass of a substance before cleaning up some spilled water on the bench next to it

5 Why should you always wear gloves when working with:
(a) corrosive substances?
(b) toxic substances?

6 Long hair should be tied back when heating or mixing substances in the laboratory. Why is this so?

7 Explain why a test tube should be standing in a test-tube rack when you are pouring a liquid into it.

CREATE

8 Draw a flowchart to illustrate the correct method for lighting a Bunsen burner. Use pictures or cartoons as well as words.

9 Which one safety rule do you feel is the most important when you are mixing two liquids and heating them? Create a poster to illustrate the rule.

eBookplus

10 Identify the equipment you will need to perform a number of laboratory processes by completing the **Using equipment** interactivity in your eBookPLUS. **int-0200**

work sheets
1.1 Safety in the laboratory
1.2 Safety rules

Planning your own investigation

Discuss your ideas with others.

Formulating your question

Before you define your question in detail, you need to find a topic that interests you. Selecting your topic is the first and one of the most crucial steps in conducting your research project.

Finding a topic

1. Start by searching for a general area of interest. List your hobbies and other interests.
2. Do you have a friend or relative who might be able to help you in a scientific investigation? Write down the topic areas in which you could get help.
3. Discuss the possible research topics you have already written down with a group of fellow students. Listen carefully to their ideas. They might help you to decide on your own topic. Write down your ideas.
4. Have a look through the list of ideas below left. Even if none of the suggested topics appeals to you, they may help you to think of other ideas. For example, 'The strength of sticky tape' could lead you to consider topics such as the strength of glass, wood, paper, plastics or some other material. 'Brainstorm' possible topics with your friends and make your own list of suggested investigations.
5. Search in a library or at home for books or newspaper articles about the topic areas that you have already written down. You might also find magazines or journals that include articles about these topic areas. Conduct an internet search.

SOME IDEAS FOR TOPICS

- How do fertilisers affect the growth of plants?
- Can plants grow without soil?
- What makes algae grow in an aquarium?
- What is the best shape for a boomerang?
- What type of wood gives off the most heat while burning?
- What makes iron rust?
- Which paint weathers best?
- Which battery lasts longest?
- The strength of sticky tape
- Which type of glue is best?
- How much weight can a plastic bag hold?
- Which food wrap keeps food fresher?
- The effectiveness of pre-wash stain removers
- Which fabrics burn faster?
- Comparing fabrics for warmth
- How can the growth of mould on fruit be slowed down?
- Which concrete mixture is strongest?
- Comparing the strength of fishing lines
- Does the thickness of a rubber band affect how far it stretches?
- What type of paper aeroplane flies furthest?
- What is the best recipe for soap bubble mixture?
- Do tall people jump higher and further than short people?
- What type of fabric keeps you warmest in winter?

The aim of the game

Your investigation should have a clear and realistic aim. It should be very specific. The aim of an investigation is its purpose, or the reason for doing it. Some examples of aims are:

- to find out how the weight and shape of paper aeroplanes affects how far they will fly
- to compare the effect of different fertilisers on the growth of pea plants
- to find out whether different coloured lights affect the growth of algae in an aquarium
- to find out how the exposure of iron to salty water affects how quickly it rusts.

'To find out if the weight of paper planes makes them fly better' is not a suitable aim because 'fly better' has not been defined. 'Fly better' could mean fly further, fly in a straighter line or stay in the air longer. A better aim would be 'To find out how the weight of paper planes affects their flight distance and time in the air.'

When you have decided what your aim is, make sure that it is realistic. You should be able to answer 'yes' to each of the questions on the following page.

WHAT DOES IT MEAN?

The word *experiment* comes from the Latin word ***experimentum***, meaning 'trial' or 'test'.

- Is my question simple and clear enough?
- Will I be able to get the background information that I need?
- Is the equipment I need for my experiments available or can it be made?
- Is the question a safe one to investigate?

If you answer no to any of these questions you need to 'rethink' your aim.

Forming a hypothesis

A **hypothesis** is a sensible guess about the outcome of an experiment. It should relate to your aim and should be able to be tested with an experimental investigation. The results of your investigation will either support (agree with) or not support (disagree with) the hypothesis. It is not possible to prove conclusively that a hypothesis is correct.

When scientists make a hypothesis, they usually carry out a number of experiments to test that hypothesis. Sometimes, a number of teams of scientists test the same hypothesis with slightly different experiments. Even if the results of each experiment agree with the hypothesis, the scientists could never say that the hypothesis is proven to be correct. They would say that each experiment has provided further evidence to support the hypothesis.

Your hypothesis should be based on what you know about the topic or what you have already observed. For example, if you are trying to design the best parachute for a toy, you should read about parachutes before writing your hypothesis. You might also recall that when you are walking in the rain a cotton T-shirt soaks up a lot of water and becomes heavy, whereas a nylon jacket does not soak up water. As a result, your hypothesis might be: 'Closely woven nylon is a better fabric to use for a parachute than loosely woven cotton.' A statement that cannot be tested with a scientific experiment is not a suitable hypothesis.

The table below shows how problems and observations can lead to hypotheses.

Problem	Observation	Hypothesis
The television remote control doesn't work.	If I press the 'on' button on the remote control, the television won't come on.	The batteries in the remote control are flat.
My hair is sometimes dry and frizzy.	My hair is driest soon after washing it with Mum's shampoo.	Mum's shampoo dries out my hair.
No parrots come to our bird feeder.	There is bread in the bird feeder, and magpies and miner birds feed there.	Parrots prefer wheat seeds.

Will nylon be better? Your own experience might help you form a hypothesis.

UNDERSTANDING AND INQUIRING

UNDERSTAND

1 List four questions you should ask about your aim before it is final.
2 Define the term 'hypothesis'.
3 How can a hypothesis be tested?

THINK

4 Why is 'to find out which glue is best' not a suitable aim? Write a more suitable aim for an investigation about glue.
5 Is each of the following statements a suitable hypothesis? If not, justify your answer.
   (a) White chocolate tastes better than dark chocolate.
   (b) Washing powder X removes tomato sauce stains faster than washing powder Y.
   (c) Plants grow faster under red light than under green light.
   (d) Sagittarians are nicer people than Leos.
   (e) Playing video games increases the muscle strength in your thumbs.
   (f) Playing video games affects the development of social skills.
   (g) Science teachers are more interesting people than English teachers.
   (h) Science teachers perform better in IQ tests than English teachers.
6 Consider the table on the left. Describe how you could test each of the three hypotheses.

work sheet 1.3 Observations and inferences

Record keeping and research

Scientists do experiments to test hypotheses, which are based on observations as well as the previous discoveries of other scientists.

Before designing their experiments scientists do background research, which usually includes reading reports written by other scientists. Scientists also need to keep records of all their observations and any changes they make to the design of their experiments. When you conduct your own research investigation, you will probably be asked to do this by keeping a logbook.

What is a logbook?

A logbook is a document in which you keep a record of all the work you do towards an investigation. Each entry should be dated like a diary. In your logbook, you might include the following items.

- A timeline or other evidence of planning your use of time
- Notes about conversations you have had with teachers, friends, parents or experts on your project and how these conversations affected your project. Make sure you record each person's details so you can acknowledge their contribution in your report.
- Notes from library research you have done for your project. Include all the details you will need for your bibliography.
- A plan or rough outline of the method you will use for your experiment(s)
- Notes about any problems you encountered during your project and how you dealt with these
- Information on any changes you made to your original plan
- Results of all your experiments (these may be presented roughly at this stage)
- A plan or storyboard for your presentation if you are required to present your research to your class

A logbook can be written by hand on paper, or with a word-processing program on a computer, or it can even be in the form of a website. A blog is a website that has dated entries so it can be used as a logbook. It has the added advantage that you can invite other people, such as your friends, parents and teachers, to look at your work and post comments. You should check with your teacher on the format required for your logbook.

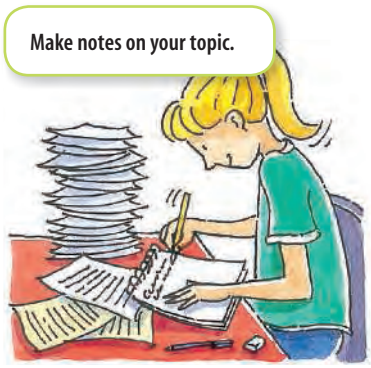
Part of a blog site used by a researcher to share the results of her investigations into acid–base indicators

Researching your topic

Before you start your own experiments, you should find out more about your topic.

As well as increasing your general knowledge of the topic, you need to find out whether your problem has been investigated by others. Information already available about your topic might help you to design your experiments. It might also help you in explaining your results.

Make notes on your topic as you find information. You may be able to include some relevant background information in your report.



Make notes on your topic.

USING THE LIBRARY

The best place to start is the school library. There are several different types of information sources in the library. They usually include those listed on the following page.

Nonfiction books

Use the subject index catalogue to find out where to find books with information about your topic. Your library catalogue is most likely to be stored in a computer database. You might need to ask the librarian to help you use the catalogue at first. It is a good idea to browse through the contents list of science textbooks. Your topic may appear.

Reference books

These include encyclopaedias, atlases and yearbooks. The index of a good encyclopaedia is a great place to start looking for information.

Journals and magazines

There are quite a few scientific journals that are suitable for use by school students. They provide up-to-date information. Your library may have an index for journals, such as 'Guidelines', which you can use to find articles on your topic. You may, however, need to browse. Some journals to look for are: *New Scientist*, *Ecos*, *Australasian Science*, *Habitat*, *Popular Science*, *Choice* and *Helix*.

Information file

Many school libraries keep information files of newspaper articles on topics of interest or even collections of articles on CD-ROM. Ask your school librarian if you don't know how to use these resources.

Audiovisual resources

The library may have slides, videos and audio tapes that can be used or borrowed. These resources can be located using the subject index catalogue.

BEYOND THE LIBRARY

Information on your topic may be available from the following sources.

Your science teacher

This may seem obvious, but many people don't even think to ask. Your science teacher may also be able to direct you to other sources of information.

Government departments and agencies

Federal, state and local government departments and agencies may be able to provide you with information or advice on your topic. Try searching through the government listing at the front of the white pages of the phone book. Addresses to write to are usually listed. A polite letter to the appropriate department or agency is the best way to ask for help.

The internet

The internet provides a wealth of information on almost every topic imaginable. Use a search engine such as Google or Yahoo! The success of your search will depend on a thoughtful choice of keywords. Take the time to find out how to take full advantage of the search engine before you use it.

Industry

Information on some topics can be obtained from certain industries. For example, if you were testing glues for strength, or batteries to find which ones

Look for information beyond the library.

last longest, the manufacturers might have useful information. Use the yellow or white pages of the phone book to find addresses. A polite letter is often the best way to ask for help.

Relatives or friends

Perhaps you or a relative know somebody who works in your area of interest. Let your friends and relatives know about your intended research.

In your logbook complete a checklist like the one below to see if you have thoroughly searched sources of information.

School library:

- nonfiction books ☐
- reference books ☐
- journals and magazines ☐
- information file ☐
- audiovisual resources ☐

Beyond the library:

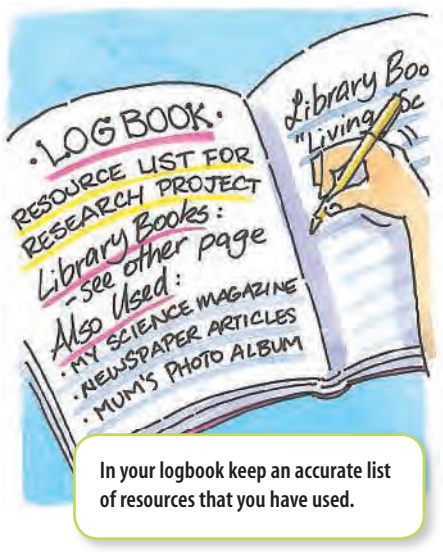
- your science teacher ☐
- government departments and agencies ☐
- the internet ☐
- industry ☐
- relatives or friends ☐
- other sources ☐

HOW TO USE INFORMATION

Make notes on information that is *relevant* to your research topic. Think about what you really need to know. You need information that will help you to:

- plan your experiments
- understand your results later on
- show in your report how your research relates to everyday life or why your research is important.

You will need to keep an accurate list in your logbook of the steps you have taken and the resources that you have used.



In your logbook keep an accurate list of resources that you have used.

UNDERSTANDING AND INQUIRING

UNDERSTAND

1 Why is a logbook a bit like a diary?

2 Define the term 'blog'.

3 List the resources that you could use to research your investigation topic:
(a) in your school library
(b) outside the school library.

THINK

4 Imagine you are a scientist. Assess the advantages and disadvantages of maintaining a blog rather than keeping a logbook in your office.

5 You can find information about science topics in science textbooks and on the internet.
(a) Explain why you would not find the results of scientific research that was done last month in a science textbook.
(b) Outline some advantages and disadvantages of using the internet as a source of information.

6 Your school and local library probably have an online catalogue.
(a) Explain what an online catalogue is.
(b) Distinguish between online catalogues and search engines such as Google.

7 Use the **Wikipedia** weblink in your eBookPLUS to answer the following questions about Wikipedia, an online encyclopedia.

eBookplus

(a) When did Wikipedia start?
(b) Who founded Wikipedia?
(c) How is Wikipedia different from other online encyclopedias such as World Book Online and Encarta Online?
(d) Some of the information in Wikipedia is not reliable. Explain what 'reliable' means and why that might be the case for Wikipedia.
(e) How could you assess whether a particular piece of information you found on Wikipedia was reliable?
(f) Outline some reasons why people may wish to put inaccurate information on Wikipedia.
(g) Discuss whether certain people should be banned from contributing to Wikipedia. Give reasons for your answer.
(h) Which Wikipedia entries are most likely to be highly reliable: those that have been edited many times by a large number of people, or those that have been written by only one person and never edited? Explain your answer.

8 Use the **Wiki** weblink in your eBookPLUS to have a go at writing a class wiki on a topic of your choice.

Scale and measurement

In order to answer a question scientifically, a controlled investigation needs to be performed. The investigation must also be reliable.

In the simple investigation of a swinging pendulum below, variables are controlled. However, to be reliable as well, measurements in the investigation need to be accurate, repeated and averaged.

INQUIRY: INVESTIGATION 1.2

The period of a pendulum

KEY INQUIRY SKILLS:

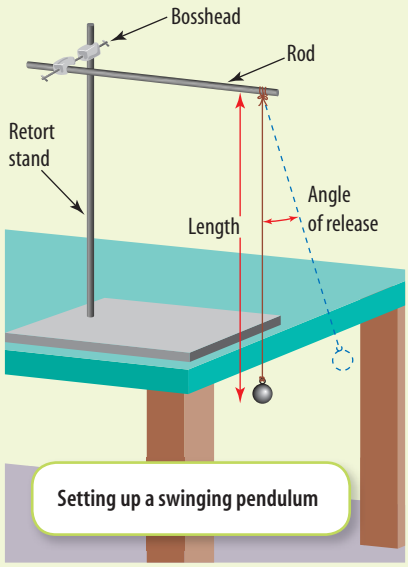
- planning and conducting
- processing and analysing data and information

Equipment:

length of string (at least 80 cm long)
set of slotted masses
retort stand with bosshead
pair of scissors and a one-metre ruler
stopwatch or clock with a second hand

Part 1: The effect of mass

- Set up your pendulum so it can swing freely. Start with the largest possible length and the smallest weight.
- Copy the table below right into your workbook and record the mass and the length of the pendulum in it. The length should be measured from the top of the pendulum to the bottom of the swinging mass, as shown in the diagram.
- Pull the mass aside so that the angle of release is about 20°. Take note of the height from which the mass is released so that this angle of release is used throughout the experiment.
- Release the pendulum. Measure the time taken for 10 complete swings of the pendulum. Repeat your measurement at least twice so that you can find the average time for 10 swings. Use this average to calculate the time taken for one complete swing (the period). Record all the measurements in your table.



Setting up a swinging pendulum

- Repeat this procedure for three larger masses, completing the table as you go.

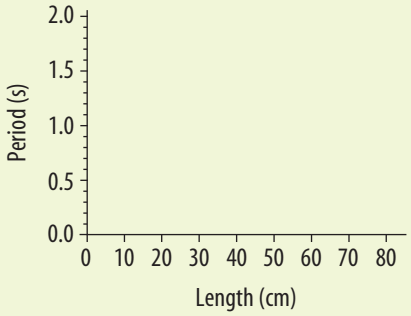
Part 2: The effect of length

- Construct a table like the one on page 13 to identify all of the variables that need to be considered for an investigation of the effect of length on the period of a pendulum.

The effect of mass on the period of a pendulum

Length of pendulum = _____ cm
Angle of release = 20°

- Construct a second table in which to record your measurements. Remember that this time you'll be testing four different lengths without changing the mass. Use the same procedure as you did in part 1 for measuring the period.
- Draw a line graph to show how the period of the pendulum is affected by its length.



DISCUSS AND EXPLAIN

1. How does the mass of the pendulum affect its period?
2. How does the length of the pendulum affect its period?
3. The period of most standard clock pendulums is one second. Use your graph to predict the length of a standard clock pendulum.
4. Explain why it is a good idea to measure the time for 10 swings rather than just one.

Mass (grams)	Time taken for 10 complete swings (seconds)				Period (seconds)
	Trial 1	Trial 2	Trial 3	Average	

In the swing of things

When was the last time you were on a swing? A playground swing is simply a large **pendulum**. Pendulums are used mainly as measuring instruments. Their most well-known use is in clocks, such as grandfather clocks.

A playground swing is simply a large pendulum.

A pendulum is a suspended object that is free to swing to and fro. Each complete swing is called an **oscillation**. The time taken for one complete oscillation of a pendulum is called its **period**.

Variables

There are several factors that could affect the period of a pendulum. They include:

- the length of the pendulum
- the total mass that is swinging
- the height from which the pendulum is released.

These factors are called **variables**. The variable that you are measuring (in this case the period of the pendulum) is called the **dependent variable**. The variable that you are investigating is called the **independent variable**. In Investigation 1.2 on the previous page, you investigated two independent variables, the mass of the pendulum and the length of the pendulum.

Fair testing

Scientific investigations must be **fair tests**. In a fair test, only one variable is changed at a time — the independent variable. In the first part of Investigation 1.2, the independent variable is the mass of the pendulum. All variables other than the dependent variable must be controlled; that is, they must be kept the same. If they were not, you couldn't tell which variable was affecting the period of the pendulum. You might find it helpful when designing your own investigations to use a table like the one below to identify all the variables.

Experiment: How does mass affect the period of a pendulum?

Independent variable	• The mass of the pendulum
Dependent variable	• The period of the pendulum
Controlled variables	• The length of the pendulum • The angle of release • The method of release

UNDERSTANDING AND INQUIRING

UNDERSTAND

1 What is a variable?

2 Explain the difference between a dependent variable and an independent variable.

3 Why is it important to control variables in a scientific investigation?

THINK

4 In Investigation 1.2 you conducted three trials for each measurement and calculated an average. List two or more reasons for the repetition.

5 A metronome is an 'upside-down' pendulum. To make the period of the metronome longer, should you move the sliding mass up or down?

A metronome's period is changed by moving the sliding mass up or down.

6 What are (i) the independent variable and (ii) the dependent variable in:
(a) part 1 of Investigation 1.2?
(b) part 2 of Investigation 1.2?

INVESTIGATE

7 Predict whether the angle of release affects the period of a pendulum and write down your hypothesis. Perform an investigation to test your hypothesis and write a brief report. In your conclusion, state clearly whether your hypothesis was supported by your results.

eBook*plus*

8 Identify the temperature on a number of different thermometers by completing the **Reading scales** interactivity in your eBookPLUS. **int-0201**

work sheet 1.4 Fair testing

The main game

Now that you've selected your topic, defined your aim, formed a hypothesis, done some research and understood the importance of variables, it's almost time to start gathering data.

Getting approval

Almost all scientists need the approval of their employer before they commence an investigation. As a student, you should not commence an investigation until your plan has been approved by your teacher.

1. Title
Decide on the likely title — you may decide to change it before your work is completed.

2. The aim or problem
Briefly state what you intend to investigate or the question you intend to answer.

Aim: To study the behaviour of slaters
Problem: What makes algae grow in an aquarium?

3. Hypothesis
Make an educated guess about the answer to your problem or what you expect to find out. It is important to be creative and objective and to use logical reasoning when devising a hypothesis and testing it.

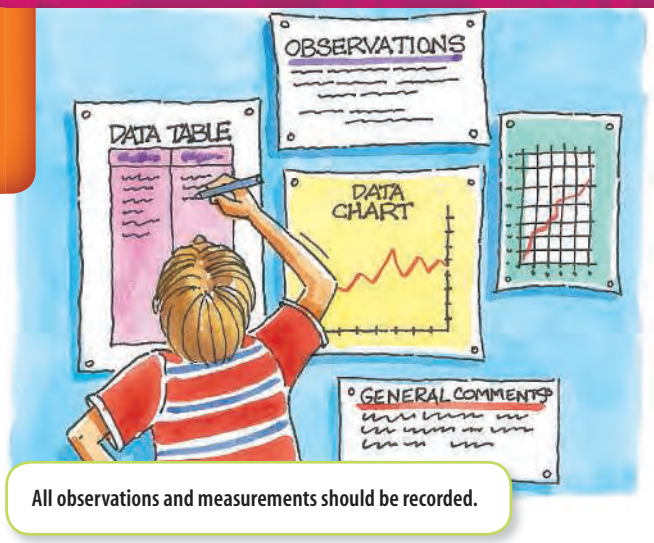
4. Outline of experiment
Explain how you intend to test your hypothesis, and briefly outline the experiments that you intend to conduct.

5. Equipment
List any equipment that you will need for your experiments.

6. Resources
List the sources of information that you have used or intend to use. This list should include library resources, organisations and people.

Performing your experiments

Once your plan has been approved by your teacher, you may begin your experiments. Detail how you conducted your experiments in your logbook. All observations and measurements should be recorded.



All observations and measurements should be recorded.

Use tables where possible to record your data. Use graphs to display your data.

Some information about using tables, graphs and data loggers is provided on pages 16–23.

Where appropriate, measurements should be repeated and an average value determined. All measurements — not just the averages — should be recorded in your logbook.

Photographs should be taken if appropriate.

You might need to change your experiments if you get results you don't expect. If things go wrong, record what happened. Knowing what went wrong allows you to improve your experiment and technique. Any major changes should be checked with your teacher.

Writing your report

You can begin writing your report as soon as you have planned your investigation, but it cannot be completed until your observations are complete. Your report should be typed or neatly written on A4 paper. It should begin with a 'table of contents', and the pages should be numbered. Your report should include the following headings (unless they are inappropriate for your investigation).

ABSTRACT

Briefly describe your experiments and your main conclusions. Even though this appears at the beginning of your report it is best not to write it until after you have completed the rest of your report.

INTRODUCTION

Present all relevant background information. Include a statement of the problem that you are investigating, saying why it is relevant or important. You could also explain why you became interested in the topic.

AIM

State the purpose of your investigation — that is, what you are trying to find out.

HYPOTHESIS

Using the knowledge you already have about your topic, make a guess about what you will find out by doing your investigation.

MATERIALS AND METHOD

Describe in detail how you carried out your experiments. Begin with a list of the equipment used and include photographs of your equipment if appropriate. The description of the method must be detailed enough to allow somebody else to repeat your experiments. It should also convince the reader that your investigation is well controlled. Labelled diagrams can be used to make your description clear. Using a step-by-step outline makes your method easier to follow.

RESULTS

Observations and measurements (data) are presented in this section. Wherever possible, present data as a table so that it is easy to read. Graphs can be used to help you and the reader interpret data. Each table and graph should have a title. Ensure that you use the most appropriate type of graph for your data (see pages 16–21).

DISCUSSION

Discuss your results here. Begin with a statement of what your results indicate about the answer to your question. Explain how your results might be useful. Any weaknesses in your design or difficulties in measuring could be outlined here. Explain how you could have improved your experiments. What further experiments are suggested by your results?

CONCLUSION

This is a brief statement of what you found out and may link with the final paragraph of your 'Discussion'. It is a good idea to read your 'Aim' again before you write your conclusion. Your conclusion should also state whether your hypothesis was supported. Don't be disappointed if it is not supported. Some scientists deliberately set out to reject hypotheses!

BIBLIOGRAPHY

Make a list of books and other printed or audiovisual material to which you have referred. The list should include enough information to allow the source of information to be easily found by the reader. Arrange the sources in alphabetical order.

For each printed resource, list the following information in the order shown:

- author(s) (if known)
- title of book or article
- publisher or name of journal/ magazine (if not in title)
- place of publication (if given)
- date of publication
- chapter or pages used.

For example:

Breidahl, H., Australia's Southern Shores, Lothian, Melbourne, 1997, Chapter 2.

For websites, list the following:

- name of the website
- date the site was updated
- URL address
- date accessed.

For example:

'Millipede Mayhem', last updated 14 March 2008, http://www.csiro.au/csiro/channel/pchgb.html, accessed 30 November 2011.

ACKNOWLEDGEMENTS

List the people and organisations who gave you help or advice. You should state how each person or organisation assisted you.

UNDERSTANDING AND INQUIRING

UNDERSTAND

1 In which section of your investigation report should you write each of the following?
   (a) A list of the books and other resources you used to find information for your project
   (b) A table showing all the measurements you recorded
   (c) A diagram of the equipment you used
   (d) The purpose of the experiment
   (e) A brief summary of your investigation and findings
   (f) A statement that relates the results back to the aim and outlines what your results show

THINK

2 When scientists write up their investigations for publication in a scientific journal, the abstract is one of the most important parts of the report. Explain why the abstract is usually read by many more people than the full report.

3 Explain why it is important for scientists to publish their investigations in scientific journals and to read the reports written by other scientists.

INVESTIGATE

4 There have been instances where scientists have faked their results or committed other types of scientific misconduct.
   (a) Use the words 'scientific misconduct' in a search engine to find examples of such instances.
   (b) Why do you think that some scientists might be tempted to fake or fabricate their results?
   (c) Explain why cases of scientific misconduct are damaging to all scientists.
   (d) What do you think might happen to scientists who are found to have faked their results?

work sheet 1.5 Scientific reports

Presenting your data

Observations and measurements obtained from an investigation are called **data**. Having collected the data, it is important to present it clearly so that another person reading or studying it can understand it. Tables and graphs are a great way to organise data.

Using tables

When data is organised in a table it is easier to read and trends are more easily identified. An example of a simple table is shown below; it includes all the features you need to remember when constructing a table.

Always include a title for your table.

Temperature of the Earth at different depths

Include the measurement units in the headings.

Depth (km)	Temperature (°C)
0	15
1	44
2	73
3	102
4	130
5	158
6	187
7	215
8	242

The column headings show clearly what has been measured.

Use a ruler to draw lines for rows, columns and borders.

Enter the data in the body of the table. Do not include units in this part of the table.

You may need to construct more complex tables, like the one below, to present your student research project results.

Do large paper aeroplanes fly further than small paper aeroplanes?

Labels

Units

Width of paper (cm)		21	15	9
Length of paper (cm)		14	10	6
Distance (m)	Trial 1	4.5	6.2	3.2
	Trial 2	4.9	5.9	3.6
	Trial 3	4.6	5.8	3.5
	Average			

Using graphs

Organising data as a graph is a widely recognised way of making a clear presentation. It makes the information easier to read, interpret, show trends and make conclusions.

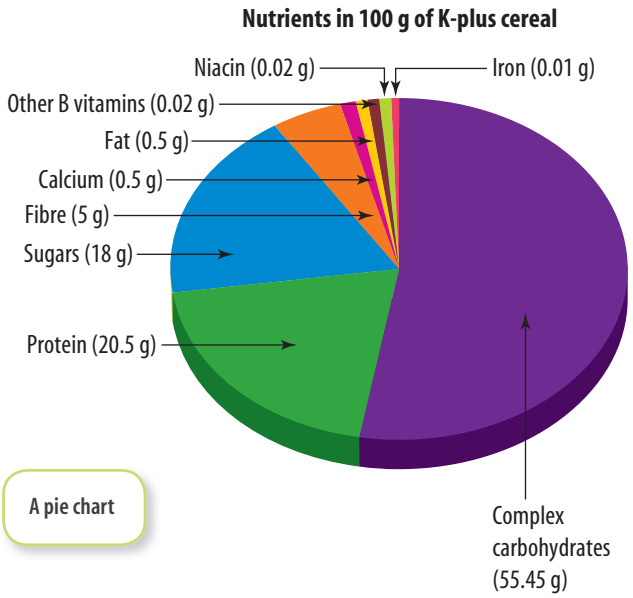
A graph, especially a line graph, can also be used to find values other than those used in the investigation. This can be done by interpolation or extrapolation (see the section later in this unit).

Types of graphs

Five different types of graphs commonly used in scientific reports are pie charts, column graphs and bar graphs, divided bar graphs, histograms and line graphs.

PIE CHARTS (OR SECTOR GRAPHS)

A pie chart (also known as a sector graph) is a circle divided into sections that represent parts of the whole. This type of graph can be used when the data can be added as parts of a whole. The example below shows the food types, vitamins and minerals that make up the nutrients in a breakfast cereal.

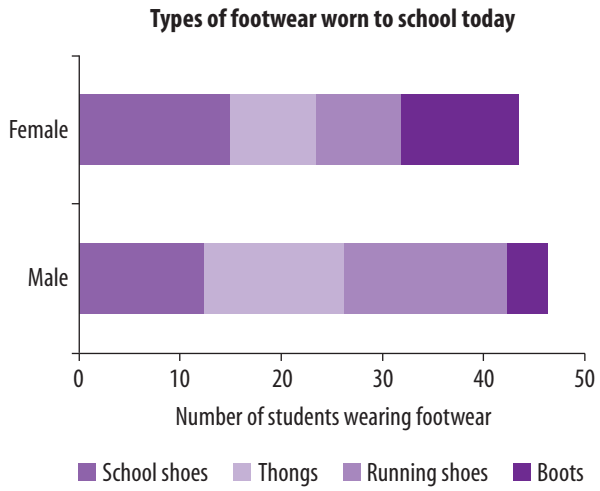


A pie chart

DIVIDED BAR GRAPHS

Divided bar graphs are also used to represent parts of a whole. However, the data is represented as a

long rectangle, rather than a circle, divided into sections. The example below shows the type of footwear worn to school today by male and female students.

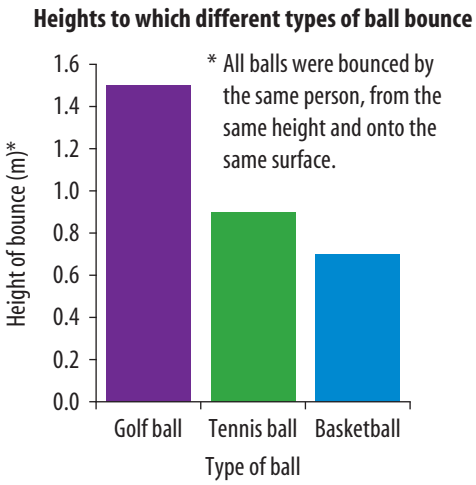


A divided bar graph

COLUMN GRAPHS AND BAR GRAPHS

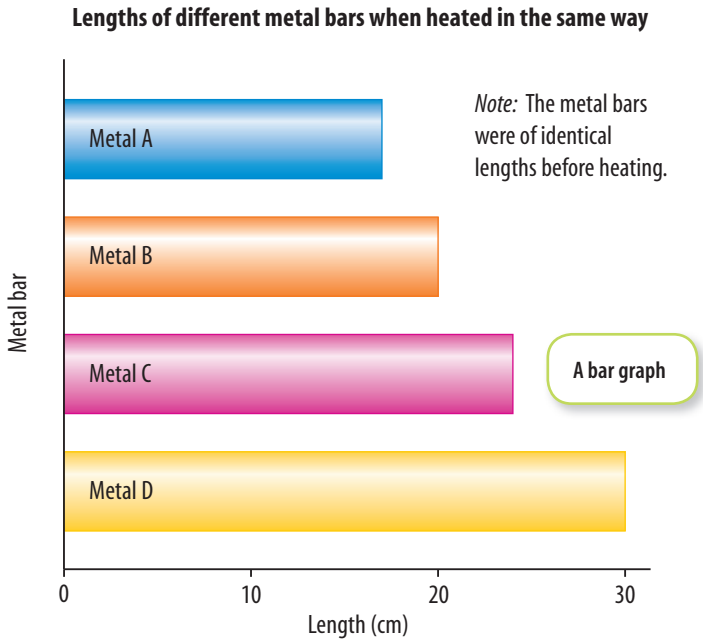
A column graph (sometimes called a bar graph) has two axes and uses rectangles (columns or bars) to represent each piece of data. The height or length of the rectangles represents the values in the data. The width of the rectangles is kept constant. This type of graph can be used when the data cannot be connected and is therefore not continuous.

The example below shows data on the average height to which different balls bounced during an experiment. Each column represents a different type of ball.



A column graph

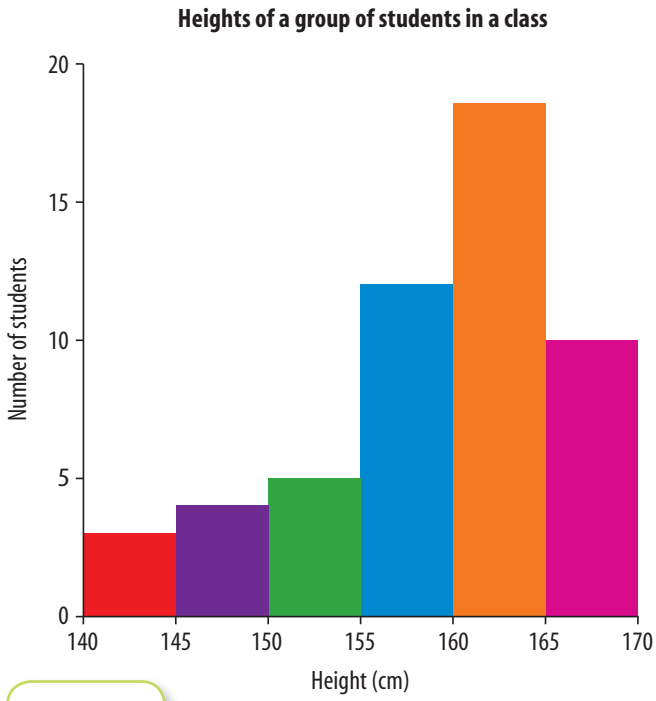
The example below shows the lengths of different metal bars when heated. Each bar represents a different metal.



A bar graph

HISTOGRAMS

Histograms are similar to column graphs except that the columns touch each other because the data is continuous. They are often used to present the results of surveys. In the histogram below, each column represents the number of students that reached a particular height.



A histogram

LINE GRAPHS

A line graph has two axes — a horizontal axis and a vertical axis. The horizontal axis is known as the *x*-axis, and the vertical axis is known as the *y*-axis. The line graph is formed by joining a series of points or drawing a line of best fit through the points. Each point represents a set of data for two variables, such as height and time. Two or more lines may be drawn on the same graph. Line graphs are used to show continuous data — that is, data in which the values follow on from each other. The features of line graphs are shown below.

3. Setting up and labelling the axes

Graphs represent a relationship between two variables. When choosing which variable to put on each axis, remember that there is usually an independent variable (which the investigator chooses) and a dependent variable. For example, if students wish to find out how far a runner could run in 15 seconds, they may choose to measure the distance covered every five seconds. The time of each measurement has been chosen by the students and is the independent variable. The distance that is measured is therefore the dependent variable. Usually the independent variable is plotted on the horizontal x-axis and the dependent variable on the vertical y-axis.

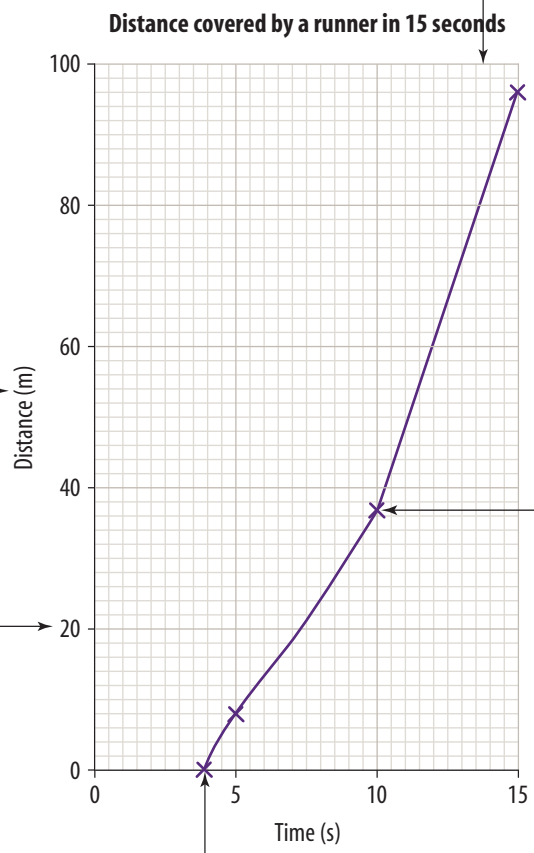
After deciding on the variable for each axis, you must clearly label the axes with the variable and the units in which the variables are measured. The unit is written in brackets after the name of the variable.

2. Title

Tell the reader what the graph is about! The title should describe the results of the investigation or the relationship between variables.

1. Grid

Graphs should always be drawn on grid paper to ensure that the values are accurately placed. Drawing freehand on lined or plain paper is not accurate enough for most graphs.



Data table

Distance (m)	Time (s)
0	4
8	5
37	10
96	15

5. Putting in the values

A point is made for each pair of values (the meeting point of two imaginary lines from each axis). The points should be clearly visible. Include a point for (0, 0) only if you have the data for this point.

4. Setting up the scales

Each axis should be marked into units that cover the entire range of the measurement. For example, if the distance ranges from 0 m to 96 m, then 0 m and 100 m could be the lowest and highest values on the vertical scale. The distance between the top and bottom values is then broken up into equal divisions and marked. The horizontal axis must also have its own range of values and uniform scale (which does not have to be the same scale as the vertical axis).

The most important points about the scales are:

- they must show the entire range of measurements
- they must be uniform: that is, show equal divisions for equal increases in value.

6. Drawing the line

A line is then drawn through the points.

A line that follows the general direction of the points is called a 'line of best fit' because it best fits the data. It should be on or as close to as many points as possible. Some points follow the shape of a curve, rather than a straight line. A curved line that touches all the points can then be used.

The type of data you are graphing may lead you to expect either a straight line or a curve. For example, you might expect the increase in temperature of water being boiled to be presented as a straight line because the temperature increases at a steady rate. A graph of the growth rate of a red panda (see page 27) would be curved and irregular because pandas have growth spurts. Inspection of the data will help you to decide whether your line should be a straight line or a smooth, curved line.

Interpolation

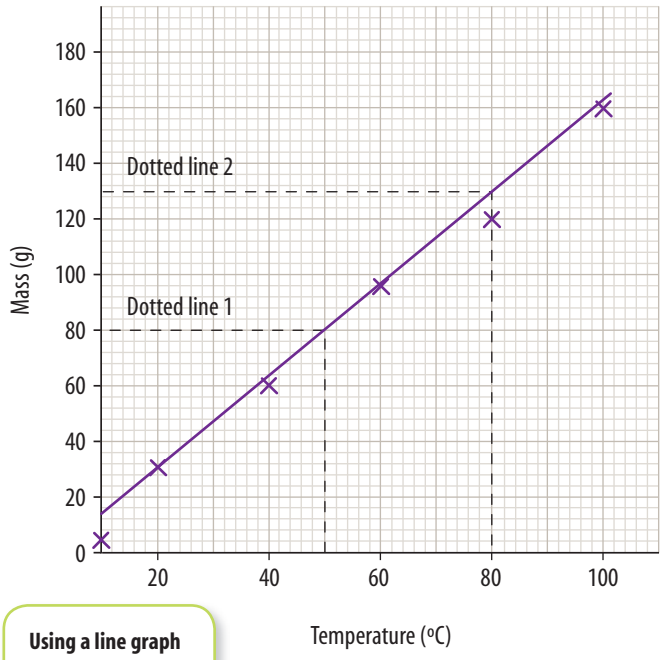
Line graphs can be used to estimate measurements that were not actually made in an investigation. The table below shows the results of an experiment in which a student measured how many spoons of sugar dissolved in a cup of tea at various temperatures.

Amount of sugar that dissolves in one cup of tea at different temperatures

Temperature (°C)	Mass of sugar dissolved (g)
0	4
20	30
40	60
60	98
80	120
100	160

The student did not measure how much sugar dissolved at 50°C, but we can work this out by interpolation. First we need to plot the data collected in the experiment. Then we read off the graph the amount of sugar that would dissolve at 50°C (shown by dotted line 1). The same procedure can be used to work out the water temperature that would be needed to dissolve 130 g of sugar in one cup of tea. This is shown in the graph below by dotted line 2.

Effect of temperature on the amount of sugar dissolved in tea



Using a line graph for interpolation

Extrapolation

In many cases it is also possible to assume that the two variables will hold the same relationship beyond the values that have been plotted. This is called extrapolation. Consider the table below, which shows the results obtained when different masses were attached to a spring and the increase in length of the spring was measured.

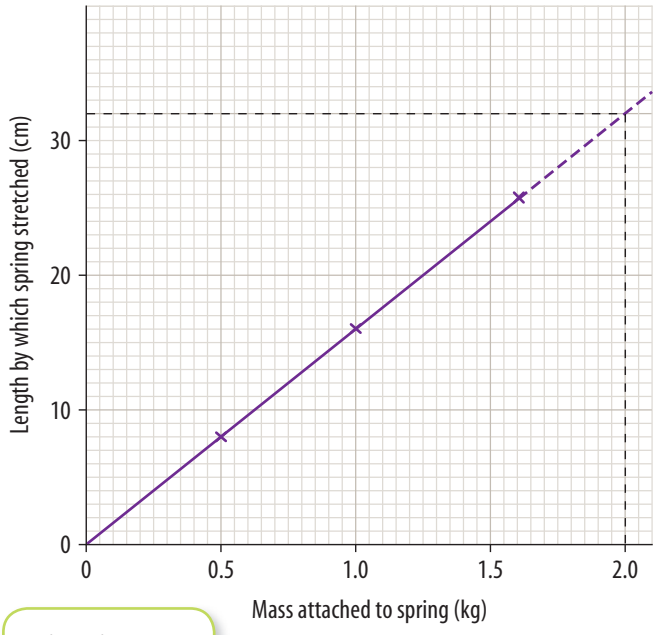
Amount that a spring stretched when various masses were attached to it

Mass attached to the spring (kg)	Length by which spring stretched (cm)
0.0	0
0.5	8
1.0	16
1.6	26
?	32

If you want to predict the mass needed to stretch the spring by 32 centimetres, you need to plot the data on a graph and extrapolate the value.

The data in the table above has been plotted on the graph below. Values have been plotted up to a mass of 1.6 kg and an increase in length of 26 centimetres. The line on the graph has been projected onwards (as the dotted lines show). This extrapolation shows that a mass of 2 kg will stretch the spring 32 centimetres.

Effect of mass on spring stretch



Using a line graph for extrapolation

INQUIRY: INVESTIGATION 1.3

Drawing a line graph

KEY INQUIRY SKILLS:

- processing and analysing data and information

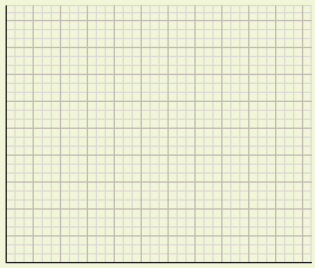
A student conducted an experiment to see how temperature affected the amount of sugar that would dissolve in a cup of tea. Each cup contained the same volume of tea, and the sugar was stirred in at an equal rate for each cup. The following results were obtained.

Graph the data in the table at right using the steps and diagrams below.

Amount of sugar dissolved in one cup of tea

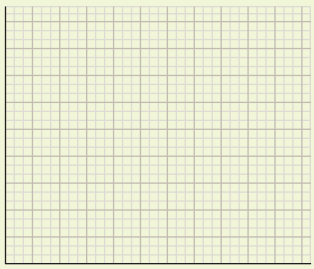
Temperature (°C)	Mass of sugar dissolved (g)
0	4
20	30
40	60
60	98
80	120
100	160

1 Set up the grid.



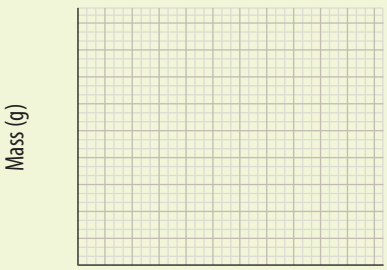
2 Give the graph a title.

Effect of temperature on the amount of sugar dissolved in tea



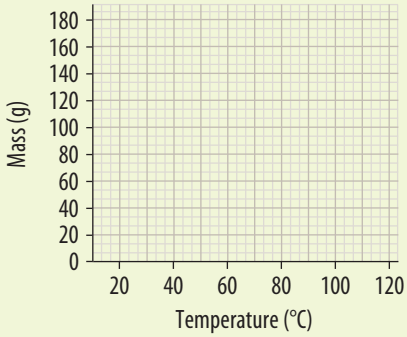
3 Set up the axes and label them.

Effect of temperature on the amount of sugar dissolved in tea



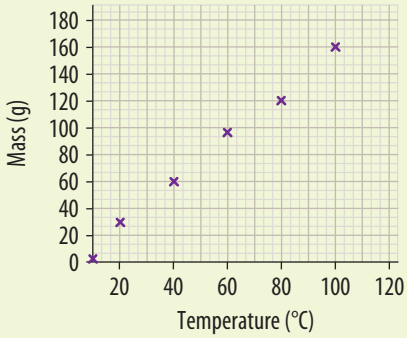
4 Place the scales on the axes.

Effect of temperature on the amount of sugar dissolved in tea



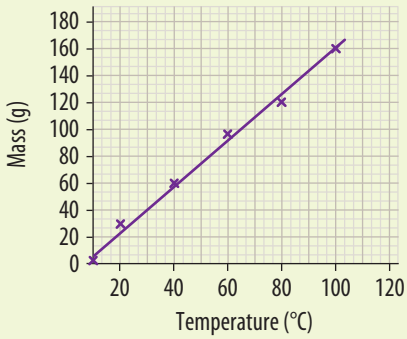
5 Plot each pair of values as a point marked with an x. Make sure each point is clearly visible. Don't forget to plot (0, 4) because you have the data for this point.

Effect of temperature on the amount of sugar dissolved in tea



6 Draw a line of best fit; that is, a line drawn in between the points so that some points are on the line, some are below it and some are above.

Effect of temperature on the amount of sugar dissolved in tea



UNDERSTANDING AND INQUIRING

ANALYSE AND EVALUATE

1 The following table shows the uses of plastics in Australia.

Uses of plastics in Australia

Use	Percentage (%)
Agriculture	4.0
Building	24.0
Electrical/electronic	8.0
Furniture and bedding	8.0
Housewares	4.0
Marine, toys and leisure	2.0
Packaging and materials handling	31.0
Transport	5.0
Others	14.0

(a) Select a suitable graph type and prepare a graph from this table.
(b) Choose two uses of plastic from your graph. For each use, state a particular item that is made of plastic.
(c) There has been recent controversy about the waste products that humans create.
  (i) List some uses of plastics that would contribute to waste products.
  (ii) Suggest some action people can take to reduce the amount of plastic waste products.

2 The data in the following table relates the speed of a car to its stopping distance (the distance the car travels after the brakes are applied).

Relationship between the speed of a car and its stopping distance

Speed of car (m/s)	Stopping distance (m)
0	0
10	12
20	36
30	72
40	120

(a) Graph the data.
(b) Make a conclusion about the information in the graph.
(c) How could this information be applied to your everyday life?

3 The boiling point of water changes with air pressure. For example, water may not boil at 100°C at the top of Mount Everest, where the air pressure is less than the pressure at sea level. The following data shows the boiling point of water at various air pressure values.

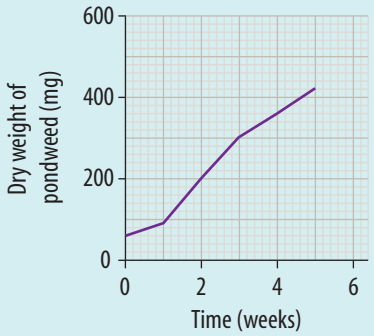
Boiling point of water at different air pressures

Air pressure in kilopascals (kPa)	Boiling point of water (°C)
0	0
1	20
7	40
21	60
45	80
101	100
200	120

(a) Graph the data.
(b) Describe the shape of your graph.
(c) What is the pressure of the atmosphere at sea level?
(d) Would it take a longer or shorter time to boil water at the top of Mount Everest, compared with the time it would take at sea level? Explain your answer.

4 The following graph shows the increase in mass of a growing pondweed.

Increase in mass of pondweed with time



(a) What was the mass of the plant after 3 weeks of growth?
(b) How long did it take for the plant to grow to 250 grams?
(c) Predict the mass of the plant after 6 weeks of growth.
(d) Can you be sure that your extrapolation for part (c) is accurate? Suggest reasons why it may not be accurate.
(e) Would the interpolations from parts (a) and (b) be more accurate than your extrapolation? Discuss your ideas in class.

work sheets
1.6 Scientific drawing skills
1.7 Data analysis

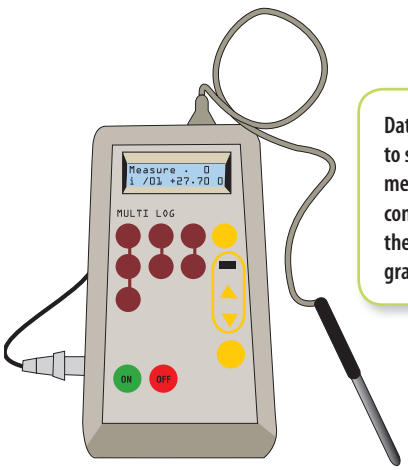
Using data loggers

What is a data logger?

A data logger is a type of scientific recording instrument. It collects and stores measurements. These are called data because they are numbers. A data logger has to be attached to a measuring instrument called a **sensor**. The sensor does the measuring and sends the measurements to the data logger.

The real advantage of working with a data logger is that it can store thousands of individual measurements. The measurements can be taken in quick succession or over a long period of time, and the data logger can be programmed to do this automatically. This is why scientists often use data loggers in their work.

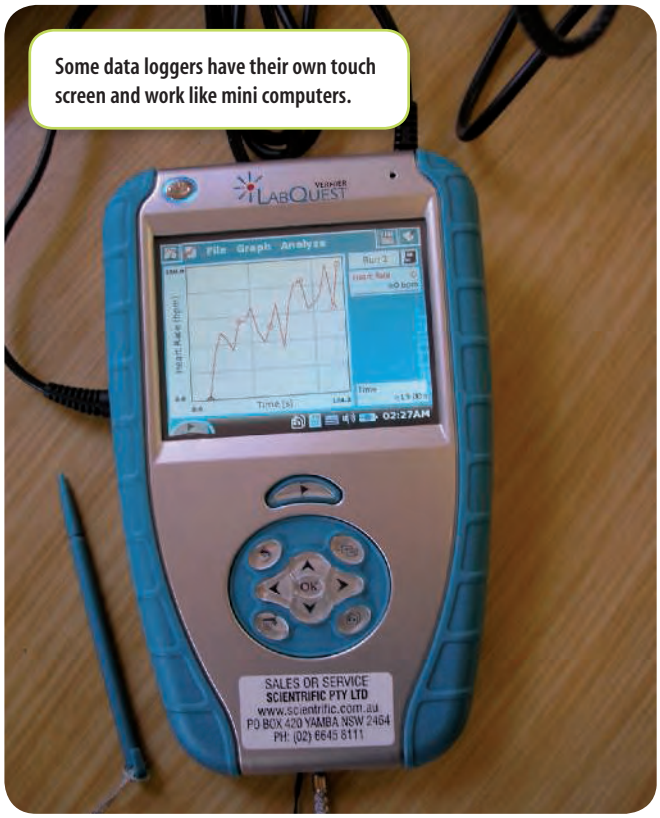
Of course, to be at all useful, the stored measurements must be easily accessed. That is why the data logger is also attached to either a computer or an instrument called a graphics calculator. The computer or calculator takes the data and, using special software that comes with the data logger, it shows the data as a table, a graph or both.



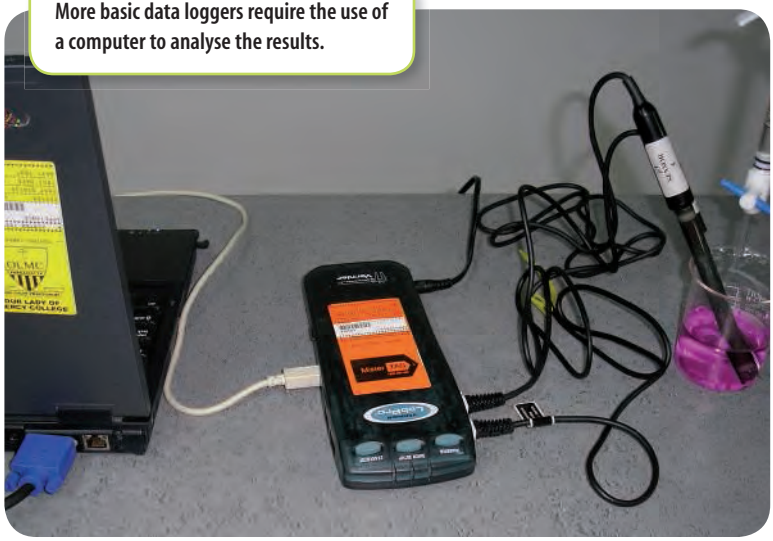
Data loggers are attached to sensors that take the measurements, and to computers that can show the data in tables and graphs.

Data loggers in temperature measurement

In Investigation 1.1 on page 3, the measuring instrument you used was a thermometer. You looked at the thermometer every thirty seconds and observed the temperature, which you wrote down in a table. You then made a line graph of temperature against time. If you had used a data logger with a temperature sensor instead of the thermometer, it could have taken the temperature every second and sent it to a computer that would have automatically tabulated the temperature data and graphed it as well.



Some data loggers have their own touch screen and work like mini computers.



More basic data loggers require the use of a computer to analyse the results.

Other uses for data loggers

Data loggers can be used for just about any experiment where measurements are taken. All that is needed is the appropriate sensor to be plugged in. It is even possible to plug in several sensors to take different measurements at the same time.

Some of the many different sensors that are available include:

- temperature sensors capable of measuring several hundred degrees
- light intensity sensors
- soundwave sensors (microphones)
- motion sensors
- magnetic field sensors
- acceleration sensors
- force sensors
- electric current and voltage sensors
- humidity sensors
- blood pressure sensors
- heart rate sensors.

One type of sensor that isn't necessary is a time sensor (stopwatch) because the data logger has its own inbuilt clock that is very accurate. In fact, one of the most useful things about data loggers is their ability to collect measurements at very small and precise time intervals, even as many as a thousand measurements in one second!

UNDERSTANDING AND INQUIRING

REMEMBER

1 Match each of the words listed below with its meaning:

Word	Meaning
(a) Sensor	(i) You may need to download the data from the data logger to one of these.
(b) Data logger	(ii) Piece of information
(c) Computer	(iii) These are plugged into the data logger and they take the measurements.
(d) Data logger software	(iv) Allows you to input data into the data logger or computer by touching it with your finger or a stylus
(e) Touch screen	(v) Software that allows you to process the data collected by the data logger
(f) Data	(vi) A device that can collect and store data from sensors connected to it

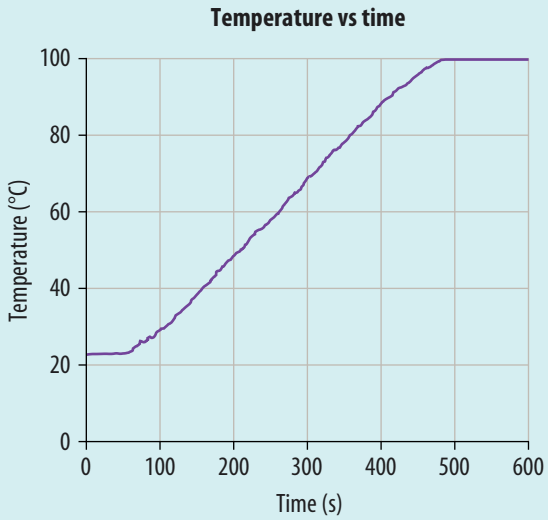
2 Sensors are the devices that take the measurements that the data logger collects. Outline scientific investigations that could use data collected by sensors that measure:
(a) electric current
(b) heart rate
(c) motion
(d) sound waves
(e) light intensity.

ANALYSE AND EVALUATE

3 The graph at right shows data collected by a data logger for an experiment in which water is heated to boiling point in a beaker. A temperature sensor was used to take the measurements.

If you were at this computer, you could scroll through every temperature measurement in the table. The computer has graphed all this data. Now let's see how much you've learned about interpreting line graphs.

(a) How long did the whole experiment go for?
(b) About when did the heating of the water begin?
(c) What was the temperature of the water when heating began?
(d) What was the temperature of the water when heating finished?
(e) About when did the water begin to boil?
(f) Between 100 and 400 seconds, at what rate (in degrees per second) did the water temperature rise during heating?
(g) The water continued to be heated even when its temperature reached boiling point, yet its temperature did not rise beyond 100°C. What has happened to all the energy that was being put into the water if it isn't causing the water temperature to rise?
(*Hint:* Think about what happens to water while it is boiling.)



Greats from the past

eBookplus eLesson
Career spotlight: scientist
Watch this video lesson to learn what it takes to be a scientist.
eles-0766

Who is the greatest scientist of all time? Is it Curie, Einstein, Newton or Pasteur? Or is it one of the people who saved millions of lives by discovering X-rays, penicillin or vaccination?

The slow starter

Albert Einstein (1879–1955) is most well known for his theory of relativity (there are actually two theories of relativity) and the equation $E = mc^2$, which describes how mass can be converted into energy.

Albert Einstein was certainly a slow starter. Although he was fascinated by mathematics, Einstein performed badly at school and left at the age of 15. He returned later and trained as a teacher in Switzerland. Einstein often failed to attend lectures and passed university exams by studying the notes of his classmates.

Einstein's first job was as a junior clerk in a patent office. His work was not demanding and he spent a lot of time doing 'thought' experiments.

Einstein's first wife, Mileva, was a mathematician. He discussed many of his new ideas with her.

At the age of 26, Einstein began to publish his ideas. These ideas altered our view of the nature of the universe by changing existing laws and discovering new ones.

Einstein explained the photoelectric effect, in which light energy is transformed into electrical energy, and was awarded the Nobel Prize in Physics in 1921 for this explanation.

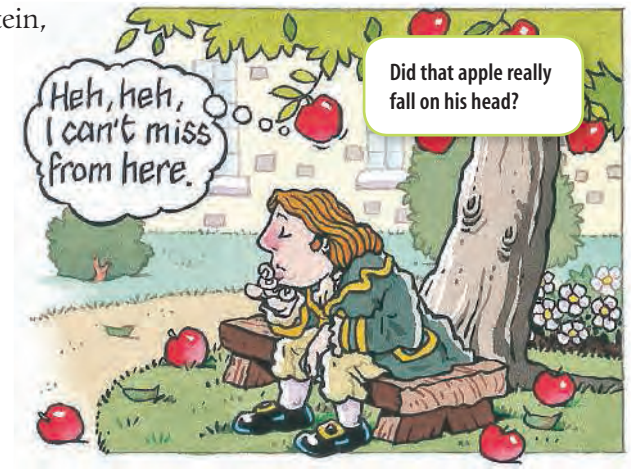
Einstein's theories of relativity were so different from earlier theories about the universe that they were not believed or understood by most scientists. His theory of special relativity explains the behaviour of objects that travel at speeds close to the speed of light. His theory of general relativity explains the effect of gravity on light and predicts that time 'slows down' in the presence of large gravitational forces. These theories provide useful clues about the development and future of the universe.

Einstein's theories suggested that mass could be converted into energy. This idea led to the development of the atomic bomb and nuclear power. Einstein, who was Jewish, fled Germany in 1933 to live and work in the United States. Einstein was an active opponent of nuclear weapons and was involved in the peace movement long before atomic bombs destroyed Hiroshima and Nagasaki at the end of World War II.

Did that apple really fall on his head?

Sir Isaac Newton (1642–1727) is probably most well known for his laws of gravitation, which explain the motion of the planets around the sun. According to some historians, his ideas about gravity arose after an apple fell on his head. We'll probably never know if this is true.

Isaac Newton was sent to Cambridge University at the age of 18. When the university closed down in 1665 as a result of the Great Plague, young Isaac went home for two years. There he developed his laws of gravitation and his three laws of motion. During his life, he also made discoveries about the behaviour of light and invented a whole new branch of mathematics, called calculus. Much of the scientific knowledge that has been acquired since the seventeenth century is built upon Newton's discoveries during that amazing two-year period.



A family affair

Marie Curie (1867–1934) became the first scientist to win two Nobel Prizes when she was awarded the Nobel Prize in Chemistry in 1911 for her discovery of two new elements, polonium and radium. Radium was used in the treatment of cancer until cheaper and safer radioactive materials were developed. Marie Curie's first Nobel Prize, for the study of radioactivity, was shared with her husband, Pierre, and fellow scientist Antoine-Henri Becquerel in 1903.

As a child, Marie Sklodowska (her birth name) wanted to study science. However, girls were forbidden to attend university in her native country of Poland. She worked as a private tutor for three years so that she could earn enough money to study at the University of Paris. It was there that she met her future husband, Pierre. They were very poor and spent most of their money on laboratory equipment, leaving them with very little money for food. In fact, they often couldn't afford to eat. After Pierre was knocked down and killed by a speeding wagon, Marie continued her research in radioactivity, pioneering the development of radioactive materials for use in medicine and industry. She became the first female teacher at the University of Paris and worked hard to raise money for scientific research.

Marie Curie in her laboratory

The germ of an idea

Louis Pasteur (1822–1895) proved that infectious diseases were caused by microbes. His ideas became known as 'germ theory'. He also developed several vaccines that made people immune to diseases such as rabies and smallpox. In doing this he has been responsible for saving the lives of millions of people and countless animals.

Pasteur removed the body fluids of rabbits infected with rabies and made a vaccine for the disease from their dried-out bodies.

Pasteur began his scientific career in physics and chemistry, but became interested in microbes when he was using light to investigate the differences between chemicals in living and non-living things.

Pasteur's next challenge was to rescue the French wine industry. Wine (and beer) became sour very quickly and this was beginning to have an impact on the French economy, which relied heavily on the export of wine. Pasteur showed that the souring was caused by acids produced by the action of bacteria in the wine. Pasteur invented a process that rapidly heated some of the ingredients of the wine. The rapid heating killed most of the offending microbes without altering the flavour of the wine. The process, known as pasteurisation, was later adapted to slow down the souring of milk.

UNDERSTANDING AND INQUIRING

THINK

1 Make a quick list of your 'Top 3' scientists of all time. For each one, answer the following questions.
   (a) What impact does their work have on your life?
   (b) Did they just happen to be in the 'right place at the right time'?
   (c) Did they work under adverse conditions?
   (d) Did their work save lives?
   (e) Did their work have any destructive influence?
   (f) What other special qualities make them great?

2 Is it fair to select the single 'greatest' scientist of all time? Explain your answer.

3 Louis Pasteur conducted many of his experiments on animals. Many of them would now be considered cruel. However, the experiments saved many human lives.
   (a) Present the arguments for and against the use of animals in such experiments.
   (b) Were the animal experiments justified? Write a brief statement supporting your opinion.
   (c) Use the **Louis Pasteur** weblink in your eBookPLUS to learn more about this French scientist.

eBookplus

IMAGINE

4 Imagine that you are one of the three scientists whom you have chosen as the greatest scientists of all time. Write a short speech (3–5 minutes) about your life and work, and deliver it to your class. Illustrate your speech with models, diagrams or photographs.

INVESTIGATE

5 Write a biography similar to the four presented on these pages about one of the following scientists:
Michael Faraday (1791–1867)
Charles Darwin (1809–1982)
Lise Meitner (1878–1968)
Barbara McClintock (1902–1992)
Peter Doherty (1940–)
Stephen Hawking (1942–)

STUDY CHECKLIST

THE LABORATORY

- identify and safely use a range of equipment used to perform scientific investigations
- use specialised equipment to make accurate observations and measurements
- use digital technology such as dataloggers to make and record measurements

PLANNING AND CONDUCTING INVESTIGATIONS

- work individually and with others to identify a problem to investigate
- use information from investigations and scientific knowledge to make predictions and form hypotheses
- undertake research using a variety of sources
- develop a logical procedure for undertaking a controlled experiment
- recognise the need to control variables and distinguish between dependent and independent variables
- use repetition of measurement to increase the reliability of data

PROCESSING AND ANALYSING DATA AND INFORMATION

- accurately record observations and measurements
- organise data clearly using tables and spreadsheets
- construct an appropriate type of graph to present your data
- use tables, spreadsheets and graphs to identify trends and patterns, and assist in the formation of conclusions
- identify data that support or discount a hypothesis
- form conclusions based on experimental results

EVALUATING AND COMMUNICATING

- reflect on your methods and make suggestions for improvements to your investigations
- use information from investigations and scientific knowledge to evaluate claims
- discuss ideas and investigations with others
- use a scientific report with scientific language, clear diagrams, tables and graphs where necessary to describe your investigations and their findings

SCIENCE AS A HUMAN ENDEAVOUR

- identify the contributions of individual scientists, including Australians, to scientific knowledge
- describe some scientific discoveries that have had a major impact on our understanding of the world

INDIVIDUAL PATHWAYS

eBookplus

Activity 1.1
Investigating
doc-2861

Activity 1.2
Analysing investigations
doc-2862

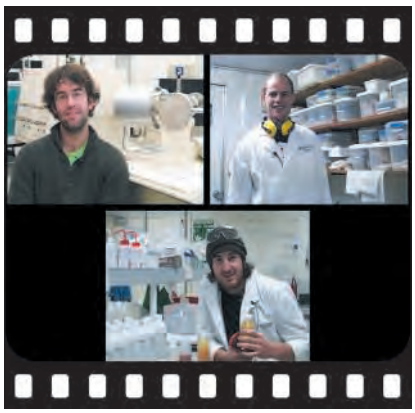
Activity 1.3
Designing investigations
doc-2863

ICT

eBookplus Summary

eLESSON

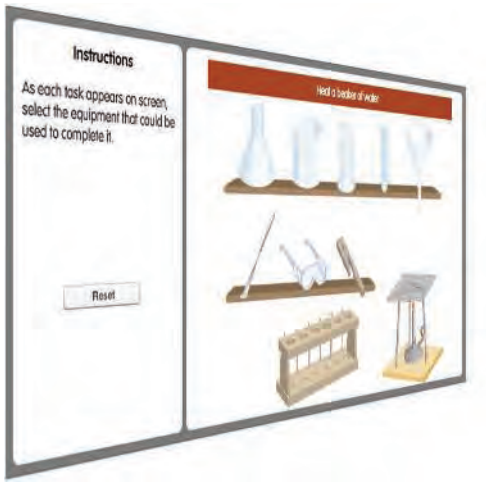
Career spotlight: scientist
In this video lesson you will meet a water scientist, a palaeontologist and a geologist, and learn what it takes to be a scientist. With insight into work in the lab and in the field, you will get some useful advice to help you decide whether this could be an attractive career for you.



Searchlight ID: eles-0766

INTERACTIVITIES

Using equipment
In this interactivity, you are given a number of scientific processes and you must indicate which equipment from a selection of items commonly found within a laboratory you would use to complete the processes. Instant feedback is provided.



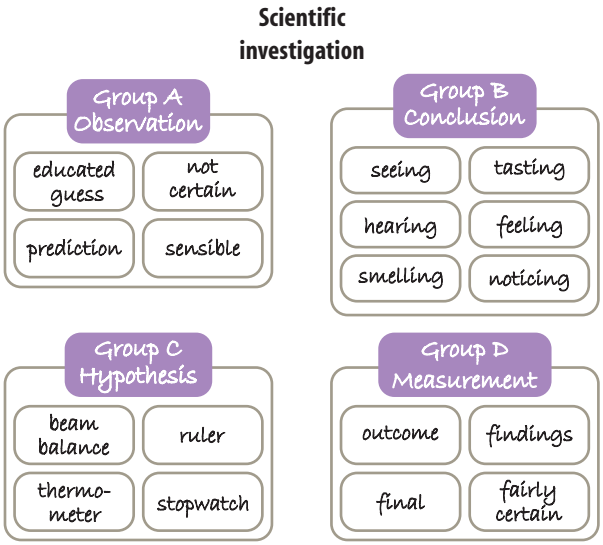
Searchlight ID: int-0200

Reading scales
This interactivity challenges your knowledge of scales by testing your skill in identifying temperatures on a number of different thermometers. Instant feedback is provided.

Searchlight ID: int-0201

LOOKING BACK

1 The affinity diagram below organises some of the ideas used by scientists into four groups. Each category name is a single word and represents an important part of scientific investigations. However, the category names have been jumbled up. What are the correct categories for groups A, B, C and D?



2 Bahir was sick of being bitten by mosquitoes. He counted several bites each evening when he sat outside to have dinner. He had heard that a burning citronella candle was a good way to keep mosquitoes away. Design an experiment to test Bahir's idea. List the independent and dependent variables and the controlled variables needed to make this a fair test. Suggest a control for your experiment.

3 Singalia and Sallyana are two red panda cubs born at Sydney's Taronga Zoo. The table above right shows their masses during their first 22 weeks. The photograph below shows one of the cubs being weighed.

(a) Graph both sets of data onto a grid. Use different symbols for the points for each panda and label each line with the panda's name. You may have to extend the vertical axis to fit in the scale for the pandas' masses (or else convert the masses to kilograms and plot in kilograms).

(b) Describe the growth of each of the panda cubs. How do they compare with each other?

(c) How long did it take the cubs to double their mass measured in week 1?

(d) Did the pandas grow at the same rate during the 22 weeks?

(e) Which were the fastest and slowest growth periods for each panda?

(f) What age was each of the cubs when it reached 1 kg?

(g) At what age would you predict each cub to reach 1.5 kg? Explain how you made your prediction. What assumption did you make to answer the question?

Red panda cubs' masses (grams)

Week	Singalia	Sallyana
1	213	219
2	285	290
3	330	349
4	365	377
5	403	408
6	465	452
7	536	514
8	564	576
9	594	610
10	650	637
11	703	680
12	714	740
13	814	796
14	872	812
15	956	806
16	1111	786
17	1043	890
18	1130	1000
19	1163	1083
20	1182	1162
21	1225	1218
22	1335	1270

4 Four students each measured the temperature in the same classroom using a thermometer. Their results are shown in the table below.

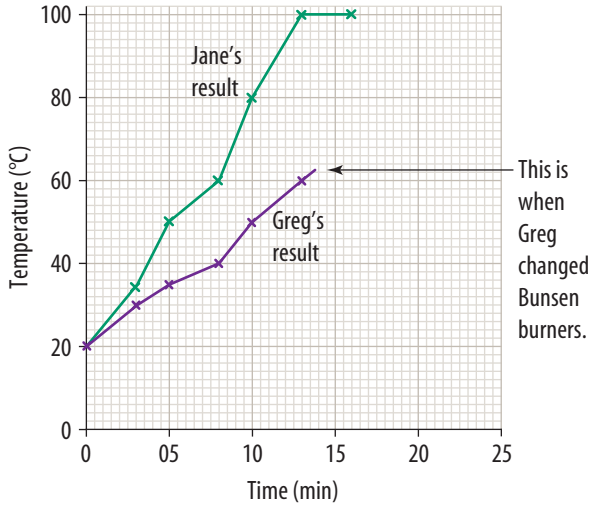
Temperature as measured by each of four students in the same classroom

Student	Temperature (°C)
1	23.5
2	24.0
3	25.0
4	22.0

(a) Construct a bar graph of these results.

(b) Propose some possible reasons for the differences between measurements.

5 Jane and Greg decided to test how quickly water would boil when using either the yellow flame or blue flame of the Bunsen burner. They set up identical experiments, except that Jane used a blue flame and Greg used a yellow flame. Their results are graphed below.



(a) How long did it take for Jane's water to boil?
(b) What was the temperature of Greg's water when Jane's water boiled?
(c) In your own words, explain how you worked out the answers for these two questions.
(d) Jane removed her beaker and Greg quickly placed his beaker over Jane's Bunsen burner. Assuming that the temperature of Greg's beaker did not drop while swapping Bunsen burners, predict the time at which his water would boil. Using your own words, explain how you predicted this.

6 The following table shows the winning times for the men's 400 m freestyle swimming event. The data is from various Olympic Games from 1896 to 2008.

Year	Name, country	Time (min:s)
1896	Paul Neumann, Austria	8:12.60
1908	Henry Taylor, Great Britain	5:36.80
1920	Norman Ross, USA	5:26.80
1932	Buster Crabbe, USA	4:48.40
1948	Bill Smith, USA	4:41.00
1960	Murray Rose, Australia	4:18.30
1972	Bradford Cooper, Australia	4:00.27
1984	George DiCarlo, USA	3:51.23
1996	Danyon Loader, New Zealand	3:47.97
2000	Ian Thorpe, Australia	3:40.59
2004	Ian Thorpe, Australia	3:43.10
2008	Taehwan Park, Korea	3:41.86

(a) Is data available for each Olympics every 4 years?
(b) Construct a line graph of the times for the men's 400 m freestyle over these years. Take into account your answer to part (a).
(c) Use your graph to estimate the winning time for this event in the 1956 Melbourne Olympic Games.
(d) Discuss how the winning times have changed over the 112-year period.
(e) Suggest some reasons for the change in winning times.
(f) Discuss how you believe the times for the men's 400 m freestyle might change over the next 40 years.

7 Create a storyboard that tells the story of the main events in the life of one of these famous scientists.
(a) Albert Einstein
(b) Sir Isaac Newton
(c) Marie Curie
(d) Louis Pasteur

8 Below is part of a report on an experiment about dissolving sugar.

Date: 29 February

Dissolving sugar

Aim:

To find out how much sugar will dissolve in hot water compared with cold water

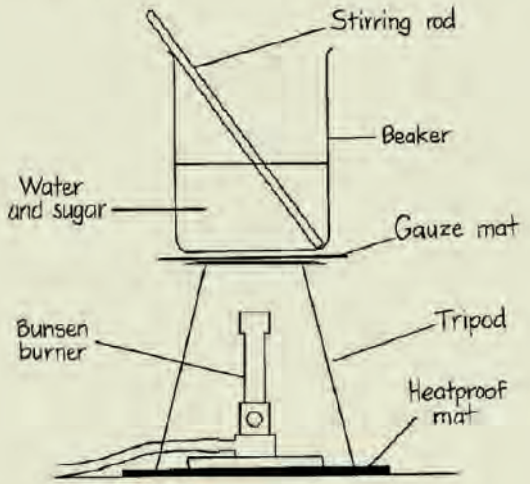
Materials:

Beaker, heatproof mat, Bunsen burner, tripod, gauze mat, matches, spatula, stirring rod, sugar, water

Method:

1. A spatula was used to add sugar to 100 mL of cold water in a beaker. The sugar was stirred and more added until no more would dissolve. The amount of sugar dissolved was recorded.

2. The mixture of sugar and water was heated with a Bunsen burner for 4 minutes and the extra amount of sugar that could be dissolved was recorded.



Results:

Amount of sugar dissolved in cold water = 2 spatulas
Extra amount of sugar dissolved in hot water = 4 spatulas
Total amount of sugar dissolved in hot water = 6 spatulas

(a) Write a 'Discussion' section for this report.
(b) Write a conclusion for this report.
(c) How could this investigation be improved?

work sheets
1.8 Investigating: Puzzle
1.9 Investigating: Summary

projectsplus

An inspiration for the future

SEARCHLIGHT ID: PRO-0071

Scenario

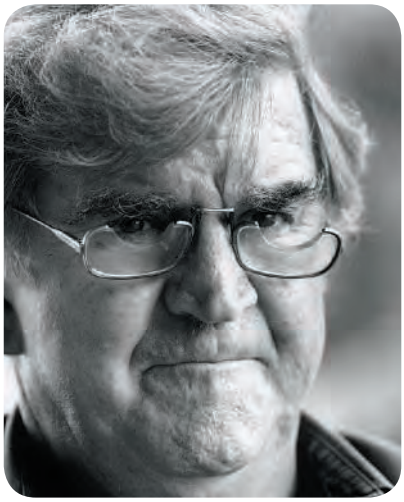
The Florey Medal was established in 1998 by the Australian Institute of Policy and Science in honour of the Australian Nobel Prize-winning scientist Sir Howard Florey, who developed penicillin. It is awarded biennially to an Australian biomedical researcher for significant achievements in biomedical science and human health advancement.

In a similar spirit, the Australian Academy of Science hopes next year to establish an award for outstanding science students. The AAS wishes to name the medal after an Australian scientist who provides the greatest inspiration for young people considering a future career in science. After months of consultation, they have narrowed the choices down to the following:

- David Unaipon (1872–1967): Inventor

- Peter Doherty (1940–): Veterinarian and immunologist

- Fred Hollows (1929–1993): Ophthalmologist

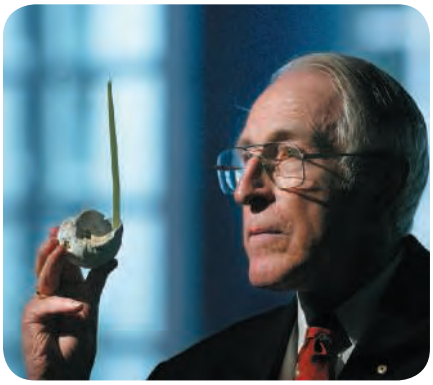


- Andrew Thomas (1951–): Astronaut

- Fiona Wood (1958–): Plastic surgeon and burns specialist

- Ian Frazer (1953–): Immunologist

- Graeme Clark (1935–): Otolaryngeal surgeon and engineer



Your task

You will create a podcast of between eight and ten minutes' duration that is in the format of an interviewer discussing with a number of different people which of these scientists would be the best choice to name the AAS medal after. The interviewees (played by group members) should be people who would be likely to have an interest or stake in the award. Examples could include a member of the AAS medal panel, the Minister for Innovation, Industry, Science and Research, the head of a university science or science education department, a high school science teacher, or even a high school science student. Each interviewee should have their own preference as to which scientist should be selected and at least four scientists should be discussed during the interview.

Process

- Open the ProjectsPLUS application for this chapter in your eBookPLUS. Watch the introductory video lesson, click the 'Start Project' button and then set up your project group. You can complete this project individually or invite other members of your class to form a group. Save your settings and the project will be launched.
- Navigate to your Research Forum. Here you will find topic headings that will be the starting points for your research. In this case, you will need to start by researching the life and work of each of the scientists on the shortlist. You may also add your own research topics.
- Start your research. Make notes about different aspects of the life and work of each of the scientists and what characteristics make them an inspiration for future scientists. Enter your findings as articles under your topic headings in the Research Forum. You should each find at least three sources (other than the textbook and at least one offline, such as a book or encyclopaedia) to help you discover extra information. You can view and comment on other group members' articles and rate the information they have entered. When your research is complete, print your Research Report to hand in to your teacher.

SUGGESTED SOFTWARE

- ProjectsPLUS
- GarageBand (Mac users) or Audacity (PC users)
- Word or other word processing software
- Internet access

Your ProjectsPLUS application is available in this chapter's Student Resources tab inside your eBookPLUS. Visit www.jacplus.com.au to locate your digital resources.

- Visit your Media Centre. Listen to the sample podcast *Maths in crisis* from the Audio section to get a feel for how your interview/discussion should be formatted. The transcript of this interview can be downloaded from the Template section.
- Use the transcript as a template to help you write a script for your podcast. All members of your group must play a role, either as the interviewer or as one of the interviewees. Note that it is assumed that all of the interviewees are in the same studio, so there can be discussion and comments between interviewees as well as between interviewer and interviewee.
- Download the article *Create your own podcast* from the Documents section of the Media Centre and use it as a guide to turn your scripted interview into a podcast.

MEDIA CENTRE

Your Media Centre contains:

- a sample podcast
- a 'Create your own podcast' document
- weblinks to research sites on these Australian scientists
- an assessment rubric.

2 Language of learning

What do you know about 'big picture' learning? Your brain may have trouble taking on a whole concept or big idea at once … so you can 'nibble away' at a concept and build a model up over time. To do this, it often helps to be able to map out your learning journey so that you know where you are going.

OVERARCHING IDEA

- Patterns, order and organisation

GENERAL CAPABILITIES

Literacy

Critical and creative thinking

Personal and social competence

KEY INQUIRY SKILLS

Processing and analysing data and information

Communicating

THINK ABOUT LEARNING

- Which coloured hat should you 'wear' to think creatively?
- What's the difference between a 'fat' and a 'skinny' question?
- What can you do to make a good first impression?
- How do we communicate our feelings without using words?
- Who gave dinosaurs their names and why?
- What do emotions have to do with memory?

What does your learning journey look like?

YOUR QUEST

The three-floor thinking model

On what floor is your thinking?

First-floor thinkers gather information. It is on this floor that the groundwork is laid.

Second-floor thinkers process the information. On this floor, thinkers decide which information is relevant and which is not, and then try to make some sense of it. This may involve brainstorming and playing with the ideas, looking for patterns or analysing data.

Third-floor thinkers apply information. They understand what needs to be done and complete it. It is on this floor that tasks are prioritised and further ideas are synthesised or evaluated. This may also be the floor for assembling the parts and adding the creative finishing touches.

Are you on your first, second or third thinking floor?

INVESTIGATE, DESIGN AND CREATE

1 Select a project topic from one of the following and use the three-floor thinking model to gather, process and apply your information. Include information on the chemical and physical properties of your selected topic. Invent, design and construct a model of a device or method that would help to:
   (a) identify a range of common rock types
   (b) sort household wastes
   (c) recycle household wastes
   (d) test the effectiveness of detergents
   (e) test the effectiveness of toothpaste.

2 Use the three-floor thinking model to gather, process and apply information about an example of how science informs laws and guidelines about health or our environment. Present your findings as an advertisement that incorporates multimedia or animation to effectively communicate the relevant scientific understanding behind the law or guideline. You may select one of the following examples or identify your own example. Some examples of laws and guidelines influenced by our scientific knowledge include:
   - quarantine laws
   - food handling laws
   - bushfire safety guidelines
   - laws about the wearing of seatbelts
   - chemical storage guidelines
   - fire restriction day laws.

Problem solving with thinking hats

What's the problem? What are the facts? What other information do we need to find out? Does this new information help us to suggest possible solutions to the problem? Of these suggestions, which is the best idea and why?

1. What is the problem?

If there is a problem, this often suggests that something could be improved or made better. Your black thinking hat is a good hat to put on for thinking about this. (For more information on thinking hats, refer to *Science Quest 7*, chapter 2.)

Once you have pinpointed your problem, you can start thinking about what end results or outcomes you would like. It is sometimes also helpful to consider the hurdles that may impede your success and how high you will need to jump to overcome them!

It is often useful to write a problem statement to help you focus on the issue. For example, you may state 'who', 'does' (action verb) and 'what' for your problem. However you express it, it's important that you take ownership of the problem and phrase it in your own words.

2. What are the facts?

This is the time to put on your white thinking hat and move into your first-floor thinking room. It is here that you lay the groundwork. If you are accurate, persistent and thorough in this stage then your progress through the other phases is likely to be much smoother.

Start with what you already know about the problem. Ask yourself questions about who, where, when, why, how and what has been tried so far. You can document this initial information as a KND chart by listing what you 'know', 'need to know' and 'need to do' in a table or columns with these title headings.

3. After the facts

Once you are clear about what the problem is, you need to look at it from different perspectives. Remember that there is usually more than one side to a problem. When you are finding your information, consider different points of view so that you can better understand the issue. It is also useful to talk to others to find out how they feel about it.

During this phase, the information that you find may require you to modify or rephrase your problem statement.

4. What are some possible solutions?

Now it is time to move up to your second floor of thinking and put on your blue and yellow thinking hats.

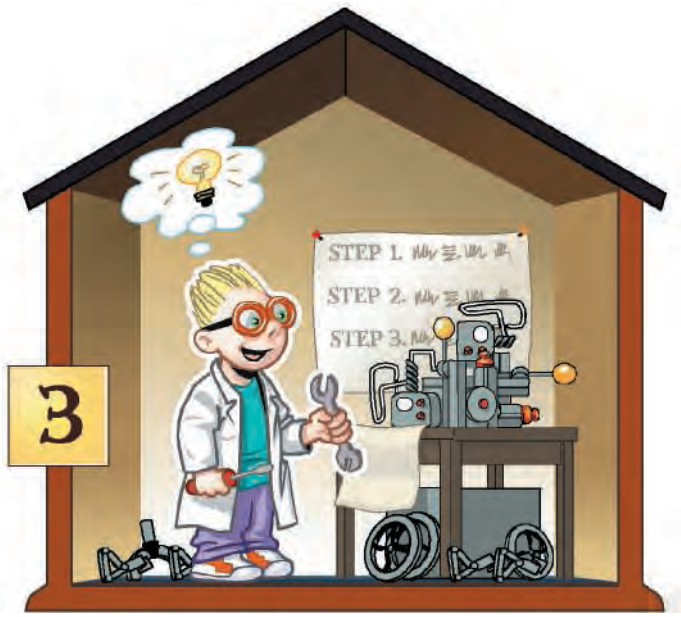
With your blue hat on, you can think about what you now know and try to make sense of it. You can use target maps to determine what is relevant to possible solutions and what is not.

Then you can put on your yellow hat. This is when you and your team can become a real 'think tank'. You may find thinking keys and visual thinking tools very useful in this creative part of problem solving. Some of the really useful keys are the 'What if' and the 'Forced relationship' keys. (See *Science Quest 7*, section 2.3 for more information on thinking keys.) Most brainstorming webs help you to develop your creative thinking. Brainstorming can be well recorded on mind maps, cluster maps or single bubbles.

5. Which is the best idea?

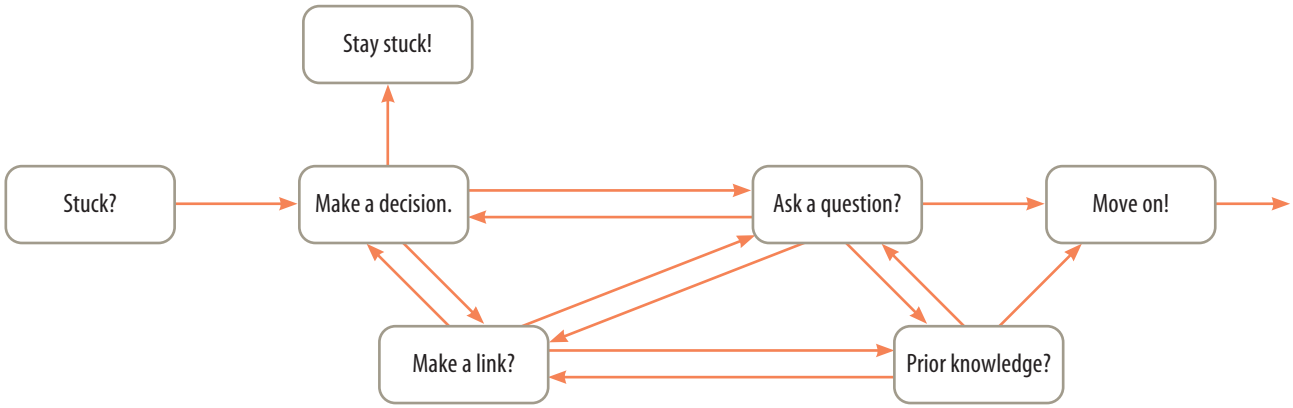
Your critical thinking becomes very important in selecting the best possible solution to your problem. Many of the **habits of mind** may help you to use intellectual behaviours that support the development of critical thinking. This is when task-specific graphic organisers such as priority grids can help you to structure your thinking, organise your ideas and stay focused.

Selection criteria can also be useful to help you to select the most appropriate alternative or idea. These may include consideration of its usefulness in achieving something, effects that it may have or how feasible it is.



6. Your action plan

You are up on the third floor now. This is where your green thinking hat can be put to good use. The time has come to divide and prioritise tasks and check your timelines. You can use thinking process maps to help both creative and critical thinking. You may use them to decide what needs to be done and then to order or sequence this within a time frame. Visual maps that may help you with this include timelines and flowcharts.



When you are working on trying to solve a problem, it is easy to get 'stuck'. The important thing is not to give up and to develop your own strategies to help you 'move on'.

UNDERSTANDING AND INQUIRING

REMEMBER

1 If there is a problem, what does this often suggest?

2 Suggest a way that a problem statement can be constructed.

3 Which thinking hat and thinking floor are used for fact finding?

4 Suggest how target maps can help you in the problem-solving process.

5 How can yellow thinking hats and visual tools help you to solve problems?

6 Which visual tools may be useful in helping you develop your action plan?

THINK, DISCUSS AND INVESTIGATE

7 (a) Appropriate use of language can also help you to clarify problems. Some personal problem statements may begin as shown below. For each of these, complete the sentence to describe a personal problem that may be experienced by someone your age.
- 'I feel angry when . . .'
- 'It worries me that . . .'
- 'I'd love to be able to . . .'
- 'I hate it when . . .'
- 'It stresses me that . . .'

(b) Select one of these problems and go through the problem-solving stages 2–6 shown on pages 33–4.

8 (a) In your team or class, brainstorm examples of problems that would fit under each of the problem categories below.
(i) What is unjust in Australia?
(ii) What takes too long?
(iii) What costs too much?
(iv) What is disorganised?

(b) Select one of these problems to research and put together a suggested action plan solution.

9 (a) Some believe that teaching people to think critically is the best training for informed and intelligent democratic citizens. Do you agree with this idea? Why? Show your responses in a SWOT analysis.

(b) Are asking questions, probing assumptions and seeking reasons valued in all cultures? Give examples to support your response.

10 (a) Brainstorm a list of at least 10 things that need improving at your school.

(b) For each item, suggest a reason why it needs to be improved.

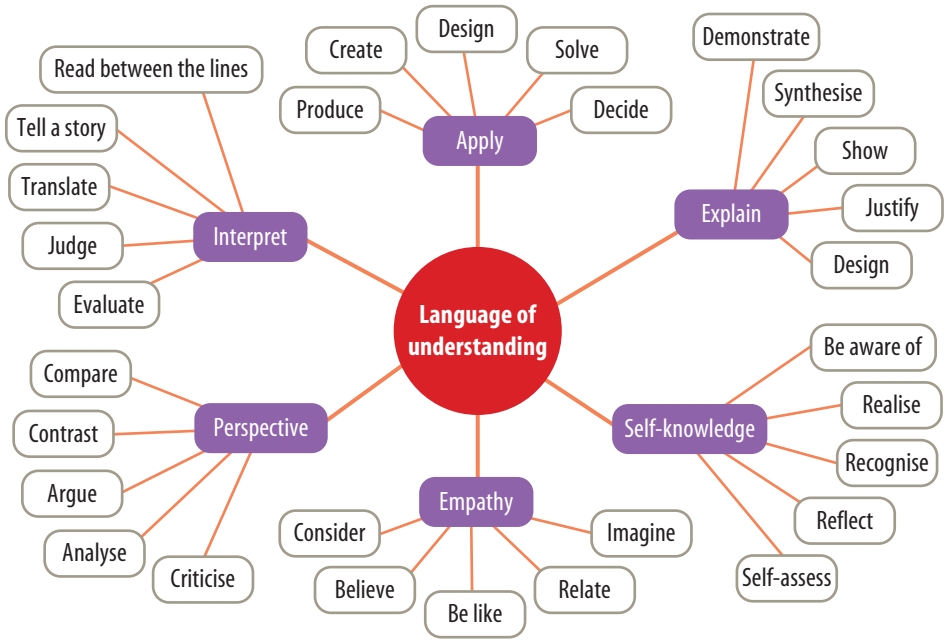
(c) Select one of these things to investigate and creatively problem solve.

11 Select one of the following categories of problems and then focus on one problem that is relevant to it. Use your creative problem-solving skills to come up with an action plan.
(a) Social problems (examples include drug abuse, bullying, racism)
(b) Environmental problems (examples include pollution, endangered species, bushfires, droughts, floods)
(c) Global problems (examples include terrorism, disease/health, national disasters)

work sheet 2.1 Thinking keys

The language of understanding

To understand something, you need to know more than 'what' and 'when'; you need to be able to explain the 'why' and 'how'. Evidence of understanding comes when you are able to apply or transfer the new knowledge to a different situation and are able to explain it.



Are you wearing your words?

The words that you use to communicate reveal a lot about you — your beliefs and assumptions, and your feelings about yourself and others. When speaking to others, try to use language that is inclusive (that doesn't make people feel left out and uncomfortable), avoid putting yourself or others down, refrain from labelling others and use proactive rather than reactive language. For example, instead of saying 'There's nothing I can do about it', say 'Let's think of the alternatives'.

Understanding language

When you are trying to make sense of words associated with understanding, it is helpful to divide them up into categories:

- **apply** — How and where can I use this knowledge?
- **explain** — Why is it so? How does it work?
- **self-knowledge** — What are my weaknesses? How do I best learn?
- **interpret** — What does this mean?
- **perspective** — Is it reasonable? Whose point of view is this?
- **empathy** — What are others aware of that I am missing?

An awareness and working knowledge of these types of thinking can increase the depth of your understanding.

Is your learning bag packed?

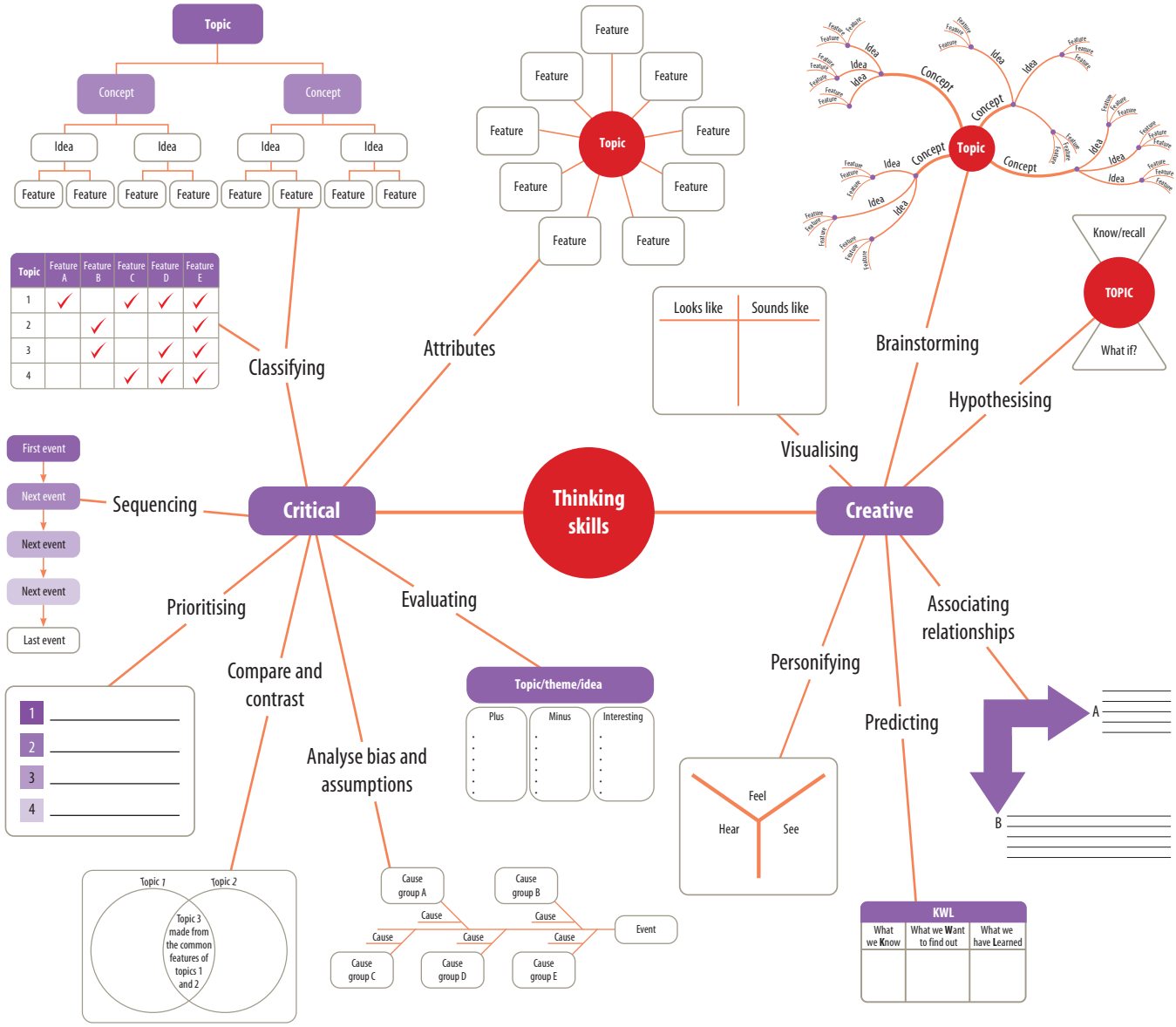
One of the goals of learning is to make information portable, so that it can be used in new situations and locations. Transfer of learning occurs when learning is taken into new contexts and applied in innovative ways.

At first glance, transfer of learning may seem to be difficult and out of reach. You may need to use your critical and creative thinking to make the task more manageable. Visual thinking tools can provide you with opportunities to practise these types of thinking.

'Fat' and 'skinny' questions

If you don't ask the right question, you won't get the right answer! 'Skinny' questions usually require only a brief, simple response. For example, 'Name your best friend' or 'Have you had science class yet?'.

'Who influences you and how?' is an example of a 'fat' question. These questions take time to think through. This is because they require deeper understanding. They need a lot of discussion, and explanations that include relevant examples.



UNDERSTANDING AND INQUIRING

THINK AND DISCUSS

1 Suggest ways that you can use language positively when you are communicating with others.

2 For each of the questions below, provide a response and suggest which type of understanding it belongs to.
   (a) Why is it illegal to smoke cigarettes in restaurants and other public areas?
   (b) What are the different points of view about stem cell research?
   (c) What would it be like to be a genetic engineer?

3 When you 'think about your thinking' you are being metacognitive. You can develop this type of learning by writing your thoughts and reflections in your learning journal. In your team, discuss specific strategies to help incorporate metacognition in your learning.

4 Australia is often described as being a multicultural country because it contains people from many different cultural groups.
   (a) In teams, identify examples of two different cultural groups in Australia.
   (b) Use a KWL tool to record what you already know, what you need to find out and what you have learned after researching about the different perspectives each of these cultural groups may have on science.
   (c) Suggest three 'fat' questions that would help you to deepen your understanding of the different cultural perspectives.
   (d) Share your findings with others in the class and create a class collage of the different cultural perspectives of science.

At first glance

Your understanding can be influenced by trust. When you meet someone for the first time, you may be lucky enough to have **rapport**; however, trust needs time to develop. Your verbal and non-verbal communication can influence not only how others see you, but also how you see yourself.

Rapid rapport

You meet someone for the first time and want to make a good impression. In fact, you *really* want to make a good first impression! Good rapport may be initiated by non-verbal signals such as eye contact, mirroring behaviour, posture or breathing. In fact, non-verbal cues are thought to make up about 65 per cent of your first impression. Only 35 per cent of your impact is due to verbal cues such as words, pace, inflection, pitch and volume.

If you want to develop rapport, then you can pay attention to another person's body language and subtly mirror it. For example, you could mirror their gestures, postures, muscle tension and facial expressions. You can also show active listening skills in both verbal and non-verbal ways.

When you meet someone for the first time, what sorts of things make a good impression on you? What sorts of things make a bad impression?

Listening skills

THE BIG BAD Ws

Are you a good listener? Do you really listen to what the other person is saying, or do you:

- want to interrupt?
- want to have your own conversation within (rehearsing what you want to say)?
- wait for an opportunity to make public only your own wishes and wants?

THREE Ps OF LISTENING SKILLS

When you are listening, do you:

- pause and not talk over the other person?
- probe what they are really saying? Are there generalisations, omissions or distortions in their comments? Do you need to ask questions to gain clarity about what they really mean?
- paraphrase what they have said to show that you have been listening and to develop rapport?

INQUIRY: INVESTIGATION 2.1

Another perspective

KEY INQUIRY SKILLS:

- evaluating
- communicating

Scientists often observe animals in the natural wild habitat. Some animals react aggressively to the presence of humans, although the scientists make painstaking efforts to be unobserved. Often the presence of humans changes the animals' normal behaviour.

DISCUSS AND EXPLAIN

1 Find out more about scientific observations of animals in the wild and summarise your findings into a PMI chart.

2 (a) State and justify your opinion about scientists observing animals in the wild.
(b) Suggest comments that a scientist involved may have regarding your opinion.

3 Use your imagination to write a story from the perspective of the animal being observed.

INQUIRY: INVESTIGATION 2.2

Active listening

KEY INQUIRY SKILLS:

- communicating
- processing and analysing data and information

There are many ways to help you develop your active listening skills. Some people use the PACTS method:

Paraphrase or play back what has been said in your own words.
Affirm or appreciate what has been said.
Clarify or check on specific details of what has been said.
Test options and prompt talk that tempts new ideas.
Seek out and be sensitive to the talker's feelings.

Practise the PACTS method:

1 Research a scientific discovery, summarising your findings into a cluster map or mind map.
2 In a team of three, decide who will be persons A, B and C.
3 Person C watches and records a PMI on what happens between person A and person B.
   - Person A talks about 'their scientific discovery' for two minutes, while person B listens.
   - Person B then paraphrases what person A has said, asks questions to clarify, and provides affirming comments to show what they particularly appreciated.
   - Person C shares their PMI chart with persons A and B and the team discusses it.

4 Now, swap roles, with person A watching and recording a PMI chart on what happens between person B and person C. Person B talks about 'their scientific discovery' for two minutes, while person C listens. Repeat the steps from the previous exercise.
5 Individually reflect on what you have learned, and how you could use this to become a more effective listener.

What are you saying?

Whether you are talking informally or giving a formal presentation, it is good to consider how you communicate your thoughts, emotions and attitudes. For instance, when you are having a conversation, do you map what you or the other person is saying in your mind? This can help to clarify and give structure to your shared dialogue.

If you were to look at a road map, you would find that locations are indicated by different fonts (for example, larger or coloured letters) and areas are indicated by colours or shading. In a conversation, your non-verbal signs indicate parts that have greater influence or meaning (for example, when you increase your volume or alter your tone). There are often hidden meanings (assumptions and beliefs) below the surface of language. Can you think of any examples?

UNDERSTANDING AND INQUIRING

THINK, DISCUSS AND CREATE

1 (a) Think about an occasion when you met someone for the first time. List the sort of things you noticed about them in the first few minutes.
   (b) Compare your list with others in your team. Use a Venn diagram to show any similarities and differences.
   (c) With your team, discuss the similarities and suggest reasons for your shared first impressions. Use a cluster map to summarise the key points of your discussion.
   (d) With your team, discuss whether the features of a good first impression are always the same or if they change in different situations.
   (e) Use a diagram, sketch or caricature of yourself to describe the first impressions that you give to others.
   (f) On the basis of this activity, are you likely to change anything you do or say prior to meeting someone you want to impress? Jot these ideas down in your personal learning journal and think about how you can go about incorporating them into situations when meeting someone for the first time.

2 (a) Score yourself between 0 and 4 on the following questions (where 0 is *never* and 4 is *always*).
   (i) I fidget when someone is talking to me.
   (ii) I finish other people's sentences.
   (iii) I talk over other people.
   (iv) My body language suggests that I'm not interested in what people are saying.
   (v) I interrupt when someone else is talking.
   (vi) I don't look at the speaker when they are talking to me.
   (vii) I think about what I will say next, rather than listening to the person talking.
   (viii) While listening, I sneak a look at my watch.
   (b) Comment on what you have learned from this quiz and what you could do to be a more effective listener.

Coded communication

Have you ever been misunderstood? It's not always because of what you say, but sometimes the way that you say it!

Turning on your transmission

Interpersonal communication is the transmission of information between two or more people. The sender encodes the message and the receiver decodes the message. Verbal communication involves speaking or writing the words whereas non-verbal communication relies on other methods.

VERBAL COMMUNICATION

Verbal communication includes not only *what* we say, but also *how* we say it. We often modify our verbal messages to match our perceptions of the intended receiver. You can also learn a lot about the attitudes of others not only by what they say or write, but also by interpreting how they communicate it.

NON-VERBAL COMMUNICATION

Non-verbal communication can be divided into three general categories: kinesics, personal space and paralinguistics. **Kinesics** involves the use of bodily movements or actions to convey a specific meaning or idea and is often referred to as body language. Personal space is the 'invisible' physical area surrounding your body that you regard as your personal territory. **Paralinguistics** involves *how* or the *way* that something is said.

INQUIRY: INVESTIGATION 2.3

Do you get the message?

KEY INQUIRY SKILL:

- communication

(a) Look at the body posture gestures and facial expressions on these two pages and select the attitude or emotion that best matches them from the following list.

curiosity	shyness
empathy	determination
enthusiasm	confidence
puzzlement	happiness
welcome	excitement
suspicion	scared
confused	anger
upset	rejection

(b) In your team:

(i) Write the terms in (a) onto different cards.
(ii) Shuffle the cards and place the pack face down.
(iii) Take turns to pick up a card and mime the emotion for other members of the team to guess.
(iv) Keep scores for each team member to see who gets the most correct.
(v) Brainstorm other examples of emotions or attitudes and write these on additional cards.
(vi) Repeat steps (ii) to (iv) with your new set of cards.
(vii) Were some emotions or attitudes easier to guess than others? Suggest reasons for this.
(viii) Suggest how this activity may be used to increase the effectiveness of your communication with others.

UNDERSTANDING AND INQUIRING

REMEMBER

1 Describe what is meant by 'interpersonal communication'.
2 What is the difference between verbal and non-verbal communication?
3 Is verbal communication related only to what we say? Explain.
4 Use a cluster map to show details of the three general categories of non-verbal communication.

THINK AND DISCUSS

5 (a) In your team, discuss examples of appropriate and inappropriate methods of verbal and non-verbal communication. Summarise your discussion into a visual thinking map such as a mind map or cluster map.
(b) Share a summary of your discussion with other teams.
(c) Back in your teams, reflect on similarities and differences of the discussions.
(d) Suggest what your class can do with the information gained and communicated during this activity.

INVESTIGATE, DISCUSS AND PRESENT

6 Identify three advertisements that include 'scientific claims' to support a particular product or practice.
(a) State the scientific claim.
(b) Investigate other resources to see whether there is unbiased evidence to support the claim.
(c) Describe any verbal or non-verbal communication that is used to support the scientific claim.
(d) Design a 'fair test' that could be used to collect evidence that may support or disprove the scientific claim used in the advertisement.
(e) Construct a PMI chart about the advertisement based on how effectively it provides data to support the scientific claim made about it.
(f) Based on your findings, in which ways could you improve:
(i) the advertisement so that it is more scientifically accurate?
(ii) the product so that it more effectively meets the scientific claim?

7 Find out more about kinesics, personal space and paralinguistics. Discuss with your team some rules or guidelines that could be used to make your communications with each other more effective. Present your findings and discussion summary as a communication guideline or rule book, cards, brochure or poster.

Telling tales

To be an effective learner, you need to develop both 'I' and 'we' learning. This means that you need to consider activities and strategies that tap into both your **intrapersonal** and **interpersonal intelligences**.

Intrapersonal and interpersonal intelligences deal with people — yourself and others. Some may suggest that these two intelligences are intimately related because you cannot fully know others until you know yourself!

Morals — learning how to live together

The idea of morals and ways of behaving may have originated from not wanting others to experience something that we would not like to experience ourselves. By identifying with others and treating them with the respect and care that we would want ourselves, we can become valued members of our communities. This can also lead to a happier life for both yourself and those around you.

One way in which morals have been passed down over many generations has been by telling stories. Many fairytales and fables have such messages woven into them.

Telling tales

Hans Christian Andersen (1805–1875) was one of the first known creators of fairytales and wrote more than a hundred stories. A number of his stories dealt with the main character feeling uncomfortable with themselves or wishing to be somewhere or something else. You may know two of his stories: *The Little Mermaid* and *The Ugly Duckling*.

The Ugly Duckling **is a story about a very ugly baby 'duck'. It is another story about viewing the world from different perspectives. The other ducks teased her and made fun of her because they thought that she was so ugly. As she got older, however, instead of growing into an adult duck, she became a beautiful swan. It was a pity that she didn't have the classification knowledge that you will have by the end of Year 8. She would have been saved a lot of pain and suffering — but then so would she have if the ducks had shown some empathy!**

Aesop's fables are further examples of stories with morals that suggest ways or behaviours to help people live and get on together. *The Crow and the Pitcher* is one example. Other examples of some fables and their morals are shown in the table below.

Examples of Aesop's fables and their morals

Fable	Moral of the story
The Boy and the Nettles	Whatever you do, do with all your might.
The Crow and the Pitcher	Necessity is the mother of invention.
	Little by little does the trick.
The Dancing Monkeys	Not everything you see is what it appears to be.
The Eagle and the Fox	Treat others as you would like to be treated.
The Four Oxen and the Lion	United we stand, divided we fall.
The Fox and the Goat	Look before you leap.
The Mule	Every truth has two sides.
The Wolf and the Kid	It is easy to be brave from a safe distance.

UNDERSTANDING AND INQUIRING

INVESTIGATE AND THINK

1 Aboriginal and Torres Strait Islander people have traditional stories that are passed from one generation to the next. Find examples of traditional stories:
   (a) used as a basis to understand their ecosystem
   (b) used to describe their special relationship with the land, native flora and fauna
   (c) about the formation of Australian landform features (for example, the Twelve Apostles, Cradle Mountain, Pinnacles, Flinders Ranges or Warrumbungles).

2 Use the 'thinking floor model' to collect information, organise it and then construct your own story on one of the following.
   - Why Australian flora and fauna are different from those of many other countries
   - Why kangaroos have a pouch and hop
   - The impact of cane toads on our native animals
   - Why southern Australian koalas have different sized ears from northern Australian koalas
   - How cutting down trees can affect the survival of other Australian organisms
   - How to encourage children (and adults) to appreciate, respect and take care of one of the following:
     — the diversity of living things on our planet
     — their environment
     — their bodies.

3 Identify a product that you would like to sell and develop an advertising campaign for it that includes claims from scientific perspectives and uses story telling.

Tell a tale

Did your parents read to you when you were little? Reading or being read to stimulates your imagination and takes you on an emotional journey. Stories can help you to learn about other people and how they communicate, express themselves, think and feel, and ways that they can support each other.

Stories can also help you to discover more about yourself. You may explore your strengths and vulnerabilities and your likes and dislikes. Stories can even help you to develop your imagination and creativity, and to think up your own ideas and be flexible in your thinking. This type of open-ended thinking can open up lots of new possibilities, rather than focusing on a single right answer. As a storyteller, you can provide a framework for your reader's imagination to flourish.

The Crow and the Pitcher **is a fable about a crow dying of thirst. The crow found a pitcher and excitedly flew down to drink from it. Sadly, when he got there, it contained only a very small amount of water at the bottom that he could not reach. He tried all sorts of ways to get to the water, but his efforts were in vain. Then he had a brainwave; he collected some stones and dropped them one by one into the pitcher. Finally, the water level rose high enough for him to drink it. His life had been saved.**

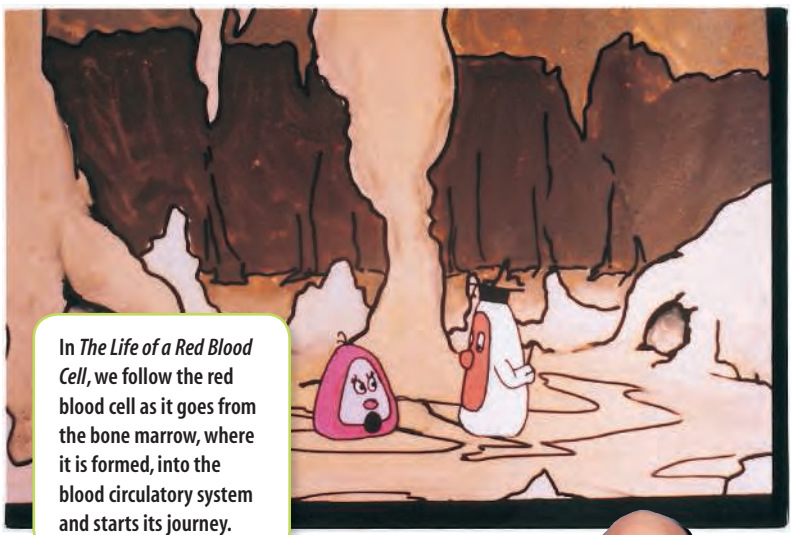
Cartoon quest

Telling the 'story' . . .

You may be wondering how telling fairytales and fables fits into the seriousness of learning science and being an effective scientist. No matter how clever you are or how much you can remember or understand, it is very important for you to be able to communicate your ideas to others effectively. Wouldn't it be great if you could have lots of fun at the same time?

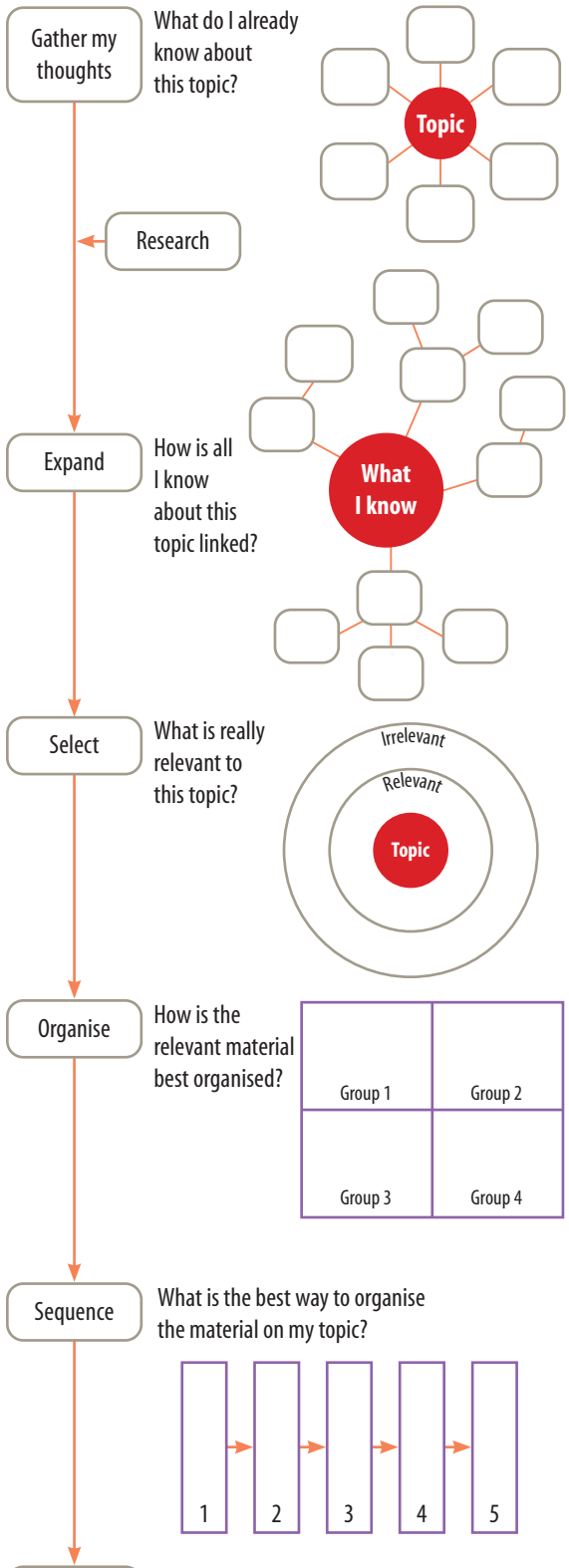
One man who has found a way to do this is Bruno Annetta, a filmmaker, graphic designer, actor and science teacher. Through his animated cartoons, Bruno can communicate scientific ideas effectively *and* share his excitement about the wonders of science. His cartoon *Foodchain* is shown on the next page and an image from his animation *The Life of a Red Blood Cell* is shown below.

'I've always found it easier to remember information by doing something with it that was personal to me. The basic idea for the film *The Life of a Red Blood Cell* came out of me trying to remember the circulatory system when I first encountered it in Year 9 (Form 3). I created the animation in my head and later turned that idea into a film.'



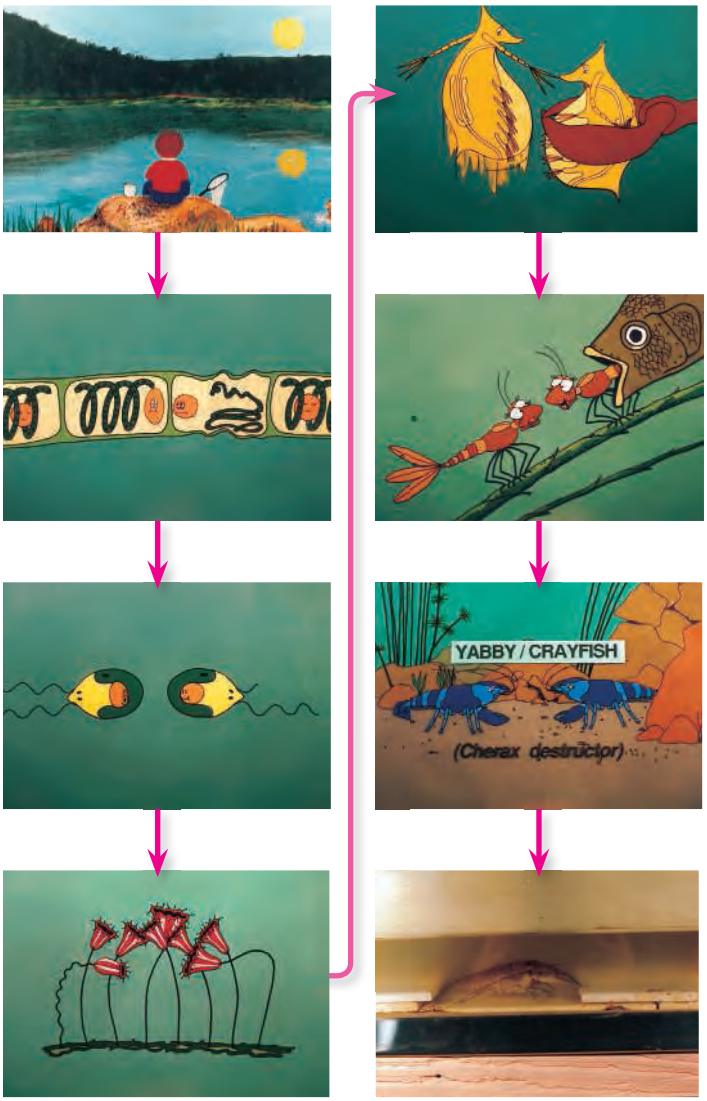
In *The Life of a Red Blood Cell*, we follow the red blood cell as it goes from the bone marrow, where it is formed, into the blood circulatory system and starts its journey.

'I believe that, through the use of animation, humour, music and story telling, learning can be fun and easy. I have used science in the making of animated films. An understanding of light, lenses and different coloured filters is crucial in exposing film correctly. I believe that through combining the sciences with the arts, one has a richer and more fulfilling life.'



You may find visual thinking tools useful in planning your story.

SCENES FROM *FOODCHAIN*



This is a musical and wacky animated look at a food chain from a freshwater ecosystem. It starts with a zoom in to a small section of pond in front of a child fishing. A series of cartoon character organisms are encountered and information is presented in the form of images, song and comical dialogue.

The zoom takes us to a microscopic view of *Spirogyra crassa* (commonly used filamentous algae). The 'crass' nuclei refer to chloroplasts, sugar production and aerobic respiration. They also allude to the fact that they are a 'duck's dinner'.

We are then introduced to *Chlamydomonas* (a mobile single-celled plant), which is the first organism of the food chain being investigated. Through their dialogue we learn about *Euglena* and are introduced to the terms 'autotroph' and 'heterotroph'.

Vorticella 'suck in' the *Chlamydomonas* and lead us to the second level of the food chain — the first-order consumer. Eight *Vorticella* clinging to a twig sing an opera tune — 'O sole mio'. They each represent a different note from an octave. Their scientific name is *Vorticella octava*.

The names and personalities have been carefully chosen to aid memory of each organism and their scientific name. *Daphnia magna* (water flea) are portrayed as 'toffy', upper middle-class women.

Damselfly nymphs feed on *Daphnia* and are portrayed as street smart, young girls. Incidental characters such as the Australian pygmy perch are introduced to allow for discussion to flow into food webs.

Yabbies are presented as 'true blue' Aussies and scavengers (scientific name — *Cherax destructor*). The story ends with one of the yabbies being caught by the child who we saw sitting on the bank at the start of the video. This finishes the food chain with 'man' as the last organism.

Videomicroscopy of the real organisms is provided at the end of the program, along with their scientific names.

UNDERSTANDING AND INQUIRING

REMEMBER

1 How can 'telling stories' help you in your learning of science and being a scientist?

2 Describe how Bruno Annetta shares his passion for science and learning.

3 In which part of the body is a red blood cell formed?

4 Use information in the *Foodchain* cartoon to answer the following questions:
   (a) Give a scientific name for filamentous algae.
   (b) What is special about *Euglena*?
   (c) Name an organism that eats *Chlamydomonas*.
   (d) How did Bruno use music to help you remember the species name *Vorticella octava*?
   (e) What is the common name for *Daphnia magna*?

INVESTIGATE, THINK, DESIGN AND CREATE

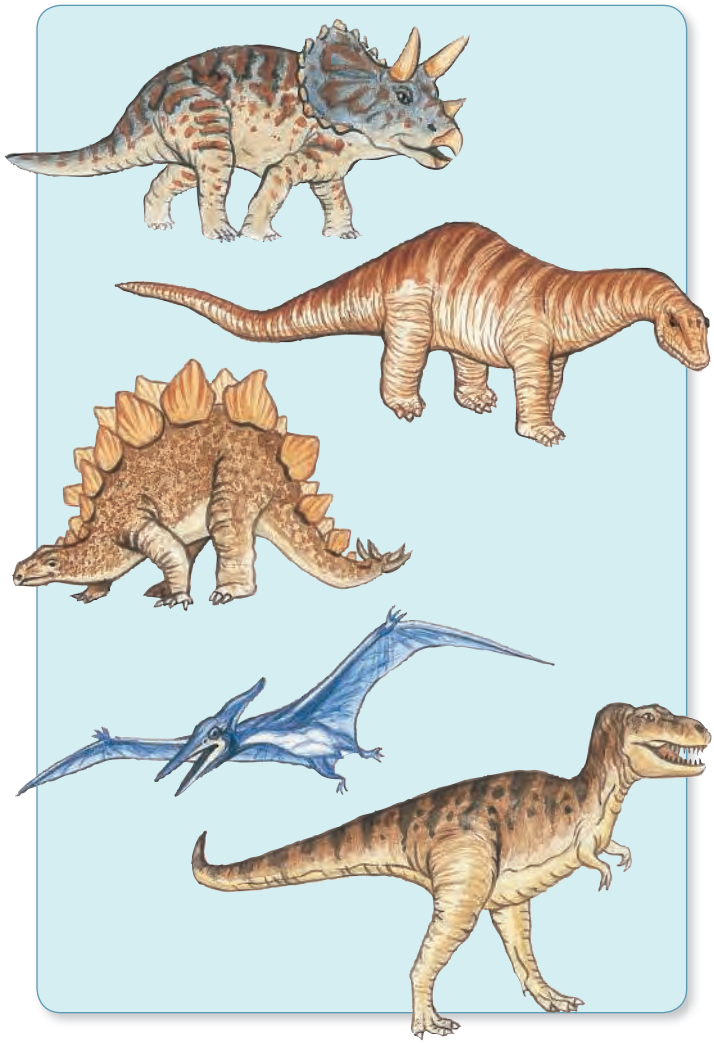
5 (a) Select one of the following topics for a story: skin cells, neurons, chloroplasts, nucleus, amoeba, fungi, muscle tissue, connective tissue, white blood cells, swine flu, solar-powered cars, oxygen, electronic optics, mitosis.
   (b) Brainstorm and list the key ideas or facts that could be included in your story.
   (c) Construct a storyboard or flowchart to show how your story will be sequenced.
   (d) Add more details (picture, songs, words) to your story.
   (e) Present your story to the class as a cartoon, PowerPoint presentation, Flash animation or storybook, or in some other creative way.

Unlocking meaning

Scientists use a language that may appear alien, but once you find some of the keys and can unlock the meaning of words, it starts to make sense.

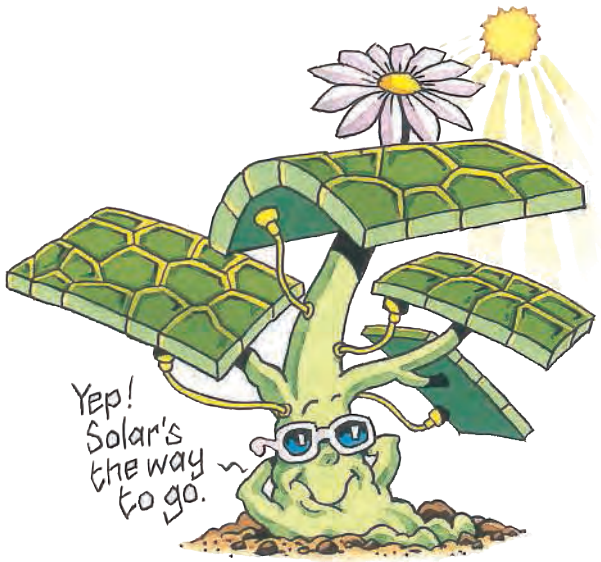
Name me a dino . . .

Tyrannosaurus, Pterosaurus, Stegosaurus . . . Did you ever wonder why the dinosaurs had such big names? Do you know what they mean? The term 'dinosaur' was actually decided on by the British anatomist and paleoontologist, Sir Richard Owen in 1842. *Dino* means 'terrifying' and *saur* means 'lizard'. Some dinosaurs were named for their unusual head or body features, others for their teeth or feet, or after a person or place.



Unlocking patterns

Many scientific terms, like others in our language, begin with a particular **prefix** and end with specific **suffix**. These can give you hints as to what the words mean. For example, the terms 'chlorophyll' and 'chloroplast' both begin with *chloro*, which comes from the Greek word *chloros*, meaning 'green'. Chlorophyll is the green pigment found in the chloroplasts of plant cells. This green pigment captures light energy so that plants can made their own food using the process of photosynthesis (*photo* = light and *synthesis* = to make). The presence of chlorophyll in the chloroplasts is the reason that they (and plants) appear to be green.



Leucoplasts (*leuco* = 'white') and chromoplasts (*chromo* = 'colour'), like chloroplasts, are plastids found in plant cells. Leucoplasts are not coloured as they do not contain coloured pigments. Chromoplasts are coloured and contain pigments other than chlorophyll. They are responsible for pigment synthesis and storage and are found in the coloured parts of plants, such as fruit and petals, giving them their characteristic colours. These pigments can be extracted and used as plant dyes.

Chemicals such as those in foods that you eat also have clues in their names that help you to work out what they are made of. You may have heard of glucose, sucrose and starch. Gluc*ose* and sucr*ose* are both sugars. Glucose is a *mono*saccharide

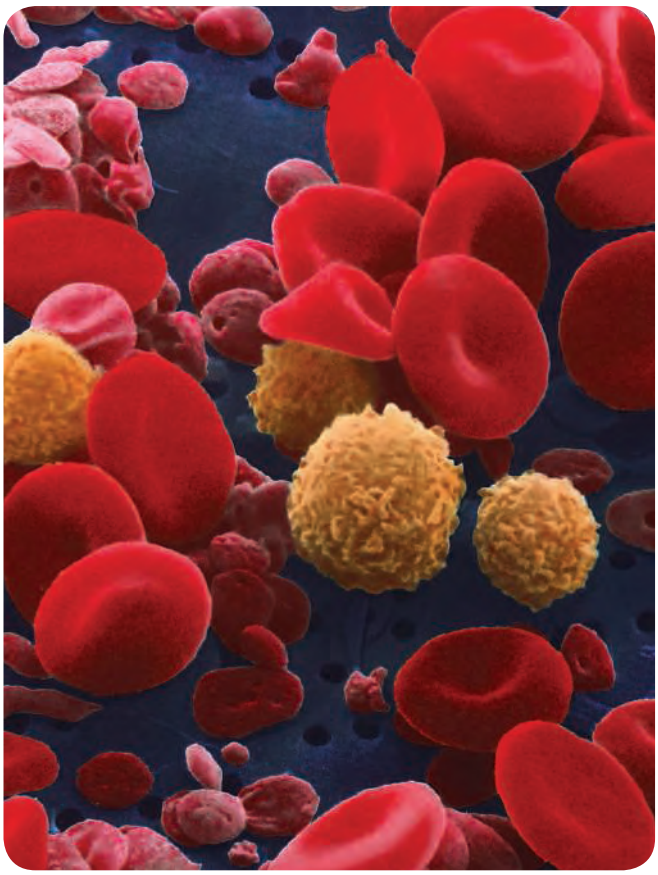
(*mono* = 'one' and *saccharide* = 'sweet'). Sucrose is a *di*saccharide and made up of two monosaccharides. Starch is a *poly*saccharide and is made up of many monosaccharides.

There are other prefixes that provide you with clues about size and number. A microscope (*micro* = 'small' + *'scope'* = view) and megafauna (*mega* = 'large' + *fauna* = 'animal') are examples of terms that indicate size in their names. Can you deduce which numbers are indicated in the following words: *uni*cellular, *bi*nary, *di*chotomous, *tri*pod, *quad*rant, *deci*mal, *centi*grade, *milli*pede?

'CELL SPEAK'

Later, when you study different types of blood cells, you will come across terms containing the suffix or prefix *cyte*. This is a variation of *cyto*, which means 'cell'. Examples of terms that you may come across include:

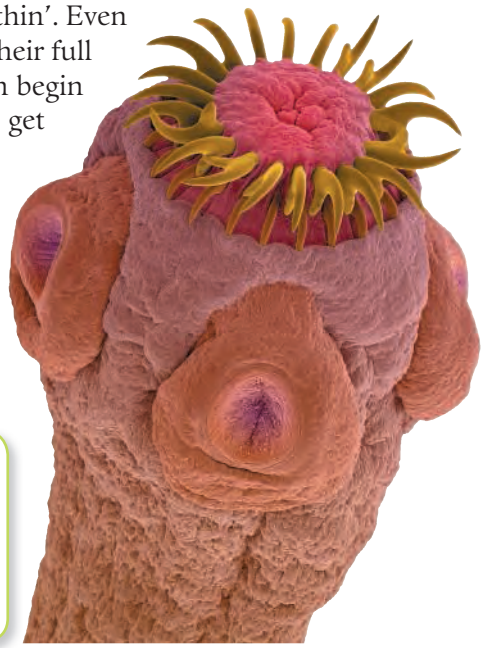
monocytes	cytosol	phagocytosis
phagocytes	cytoplasm	endocytosis
leucocytes	cytology	exocytosis
lymphocytes	cytoskeleton	
erythrocytes	cytotoxic	



Erythrocytes (red blood cells) and leucocytes (white blood cells)

INSIDE OR WITHIN

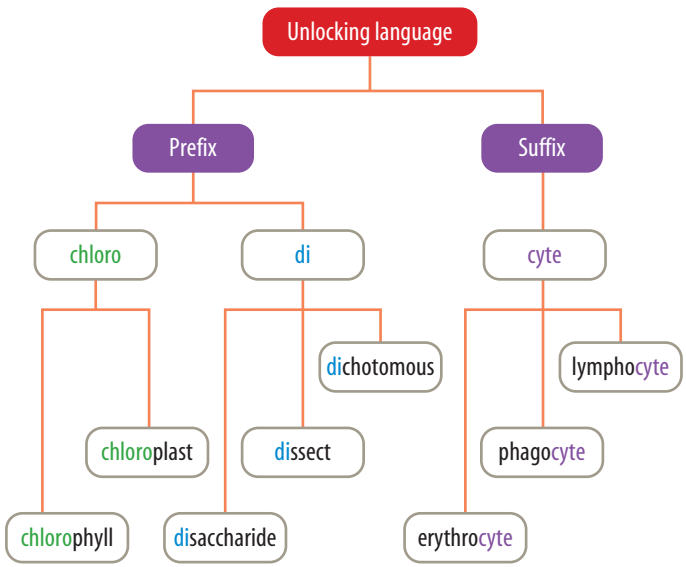
In science, you will learn about endoskeletons, endocytosis, endoplasmic reticulum, the endocrine system and endoparasites. The prefix *endo* in these words tells you that they all have something to do with 'inside' or 'within'. Even without knowing their full definitions, you can begin to see patterns and get an idea as to what they may refer to.



Tapeworms are endoparasites and live inside their host. You can see the hooks and suckers that it uses to attach itself to its host.

NUMBERS OR WORDS?

In science you also need to know the difference between two different ways of describing your data. One of these is *quali*tative and the other is *quanti*tative. Qualitative data describes your observations in words (describing the 'qualities' of the data), whereas quantitative data uses numbers (or 'quantities').



The prefixes and suffixes of scientific terms often give you hints as to what they mean.

UNDERSTANDING AND INQUIRING

REMEMBER

1 Suggest what type of animal may have *saur* in its name.

2 Outline the difference between the terms 'prefix' and 'suffix'.

3 Identify the shared meaning between the terms:
(a) chlorophyll and chloroplast
(b) monocytes, leucocytes, erythrocytes.

4 (a) Where are you likely to find leucoplasts, chromoplasts and chloroplasts?
(b) Describe how they differ.

5 Outline the difference between qualitative and quantitative data.

6 Suggest the suffix that sugars may share in their names.

7 Distinguish between monosaccharides, disaccharides and polysaccharides.

8 If you came across two words and one began with *micro* and the other with *mega*, predict what the difference between them would be.

THINK, INVESTIGATE AND DISCUSS

9 (a) Carefully examine the dinosaur names below and discuss with your partner any clues that may help you predict their meaning.
Triceratops, *Spinosaurus*, *Ceratosaurus*, *Heterodontosaurus*, *Pentaceratops*, *Microdontosaurus*, *Microceratops*
(b) Find out the meaning behind the dinosaur names above.
(c) Based on your findings, comment on any patterns and suggest groups you could divide them up into.

10 Find out about the life and scientific contributions of Sir Richard Owen in 1842. Report your findings in a timeline.

11 Research examples of Australian megafauna. Report on clues within their names that help describe what they may have looked like.

12 Find out what palaeontologists do, investigate their distinct ways of working and representing their specialised knowledge, and give an example of a contribution an Australian palaeontologist has made to our understanding of ancient life in Australia.

13 Find out prefixes for one, two, three, four, ten and hundred that have originated from Latin or Greek words. State an example of a scientific term that is used for each.

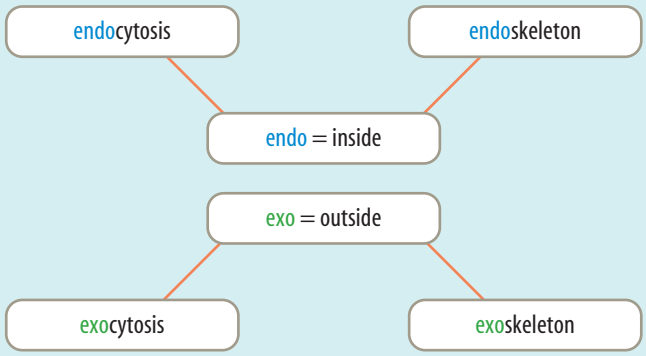
14 Find out the meaning of and similarities and differences between the following:
(a) microscopes, telescopes, periscopes
(b) millimetre, centimetre, nanometre, kilometre
(c) binary fission, dichotomous key, binocular
(d) *Tyrannosaurus*, *Pterosaurus*, *Stegosaurus*
(e) anatomist, scientist, palaeontologist
(f) cardiac, renal, pulmonary
(g) dehydrated, deoxygenated, denatured

15 Throughout history, coloured pigments from plants and animals have been used by humans. Find out about two plant and two animal examples. Identify the scientific names of the pigments and what they mean.

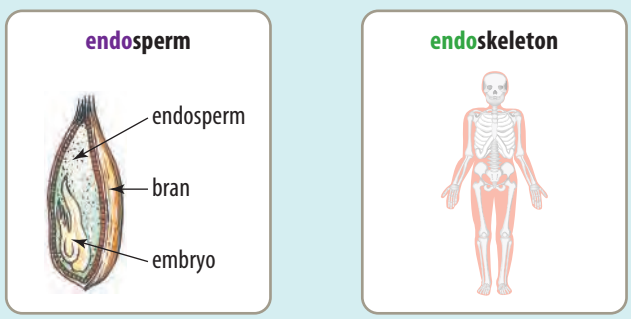
16 Chlorine is an element. Suggest what colour it may be.

17 Find out the history behind the names of the elements in the chemistry periodic table. Present your findings as a picture book.

18 Find out the definition and two key points for each of the 'endo' and 'exo' words shown in the diagram below.



19 (a) Find at least five examples of scientific terms that begin with the following prefixes: endo, bio, anti, chloro, thermo, bi, hetero.
(b) Create your own set of scientific terminology cards, using a particular colour for each prefix and adding a diagram or image for each that provides a hint of its meaning. The illustration below shows what your cards may look like.
(c) Design a game that uses the cards to teach students about scientific terminology. Include an instruction brochure or rule book with your game.
(d) Play your game and those of others.
(e) Construct a PMI chart for each game that you play.



An example of what your cards may look like

Total recall?

eBookplus eLesson

Teen brain
Take a fascinating look inside the teen brain.
eles-0224

Find out how you remember, to help you to remember how to remember better!

What is memory?

If a friend gave you her phone number, how long could you remember it without writing it down? While learning is about gaining new knowledge, memory is about retaining and then retrieving that learned information.

That is, for us to remember something, we have to be able to **record** the experience and **store** it in an appropriate part of the brain. If we are unable to **retrieve** or pull out that information, we have forgotten it.

Building memories

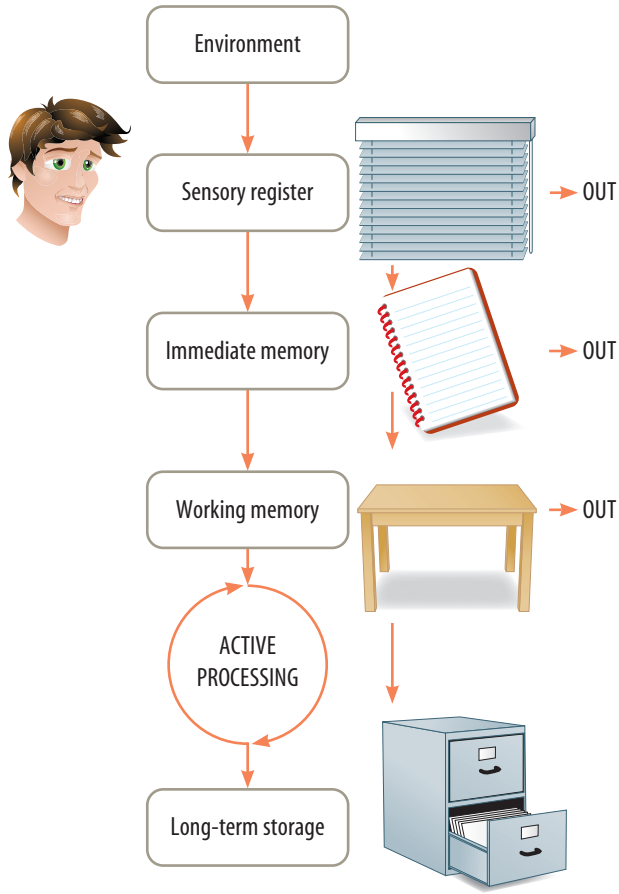
MODELLING MEMORY?

Scientists construct models to communicate ideas. Models can be concrete (for example, a plastic model of the brain) or symbolic (for example, a map or diagram). Models provide the opportunity for learners to bring previous knowledge into their working memory. This helps learners to attach meaning to, and make sense of, their new learning.

A model can be used to represent various stages in how information is processed by your brain. Shown at right is a simplified version of an information processing model. While this model has many limitations, it provides a framework that can be used to help you attach previous knowledge to new learning about the stages of memory.

FILTERING

What was that? You use your senses (for example, sight and hearing) to detect various stimuli in your environment. Incoming information is filtered through a system called the **sensory register** (shown in the model as venetian blinds). This system filters incoming information on the basis of its importance to you. Your sensory register involves your **thalamus** (a part of the limbic system of your brain) and a portion of your brain stem called the **reticular activation system** (RAS). The more important the information is to your survival, the higher the chance that it will get through for further processing in your brain.



The information processing model provides a simplified explanation of how your brain deals with external stimuli (information) from your environment.

REMEMBER ME?

Even if information has made it through your sensory register, it doesn't mean that you will remember it. You will remember information only if you have stored it in **long-term storage** (shown in the model as a filing cabinet). It is the job of your **hippocampus** to encode it and send it to one or more of the long-term storage areas in your brain. This encoding takes time and is usually done during deep sleep.

Memories are not stored as a whole or in one place. When you retrieve and reconstruct memories, storage areas distributed throughout your brain are activated. While long-term storage can be thought

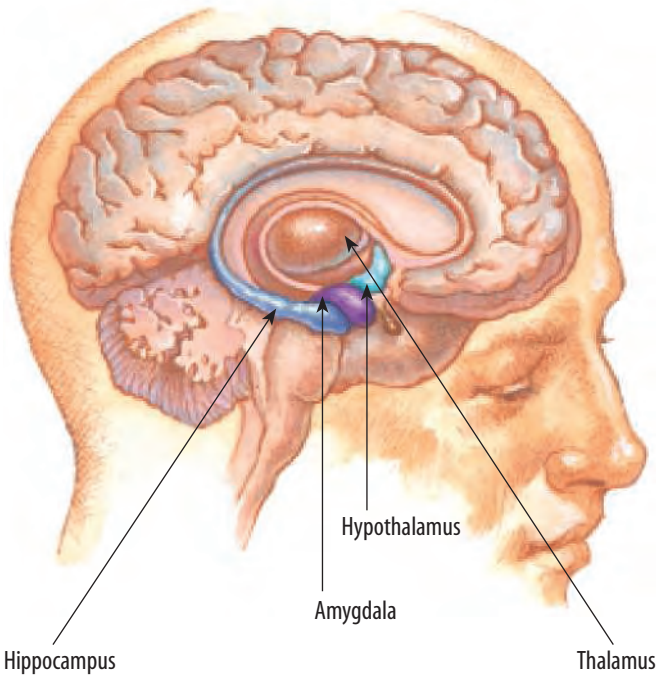
of as *where* your memories are stored in your brain, your long-term memory relates to the dynamic process of *sorting* and *retrieving* the information.

HOW ABOUT THAT!

Memory is not stored in just one place in your brain. It is currently thought that memories exist as patterns of connections at the synapses between the brain's neurons. To store a particular memory, nerve signals travel along a specific pathway through certain synapses. Each time this memory is remembered, nerve signals are reactivated to again travel along this pathway.

JUST VISITING?

Before any information is stored in long-term storage, it needs to pass through your temporary short-term memory. Examples of short-term memory include immediate memory (shown as a notepad in the model) and working memory (shown as a table in the model).



'Notepad' memory

Information that has made it past your thalamus moves to your immediate memory where a decision is made about what to do with it. Your past experience helps to determine its importance. An example of the length of time information will stay in this type of memory is when you temporarily remember a phone number and ring it. After this time the information may be lost or, if considered important enough, moved to your working memory.

'Working table' memory

It is within your working memory that information generally captures your focus and demands attention. There is a limited capacity (amount of information dealt with) and time limit for this type of memory. Research suggests that this capacity changes with age. Between the ages of 5 and 14 years there is a range of about 3–5 'chunks' that can be dealt with at one time; after this age it increases to about 5–7 chunks. This limited capacity is one of the reasons why you need to memorise songs, poems or other information in stages. By memorising a few lines at a time and repeating them (or rehearsing them) you are able to increase the number of items in your working memory. This is an example of chunking.

Studies have suggested that the time limit in working memory is about 10–20 minutes. This is often the amount of time you can spend on one activity. This time, however, can be influenced by interest and motivation. Both of these can have emotional elements and also involve a special part of your brain called the **amygdala**.

Remember to learn

Your past experiences influence new learning. What you already know acts as a filter to help you focus on things that have meaning and ignore those that don't. Your self-concept (how you see yourself in the world) is also shaped by your past experiences. It is your self-concept that often determines how much attention you will give to new information.

You can transfer things from your short-term memory into your long-term memory by rehearsing information (practising) and applying meaning to it. The two key questions asked in the decision of whether to move information into long term memory are:

- does it make sense?
- does it have meaning?

I don't understand! This is the type of comment made when a learner is having trouble making sense of new learning. Determining whether new

information 'makes sense' is related to whether the new information fits in with what you already know.

Why do I have to know this? Whether the new information 'has meaning' relates to whether it is relevant to you and whether you consider that the purpose of remembering it is worthwhile. You can improve the chance that you remember something by making connections between the new learning and your previous knowledge.

If both sense and meaning are present, there is a very high chance that the information will be sent to long-term storage.

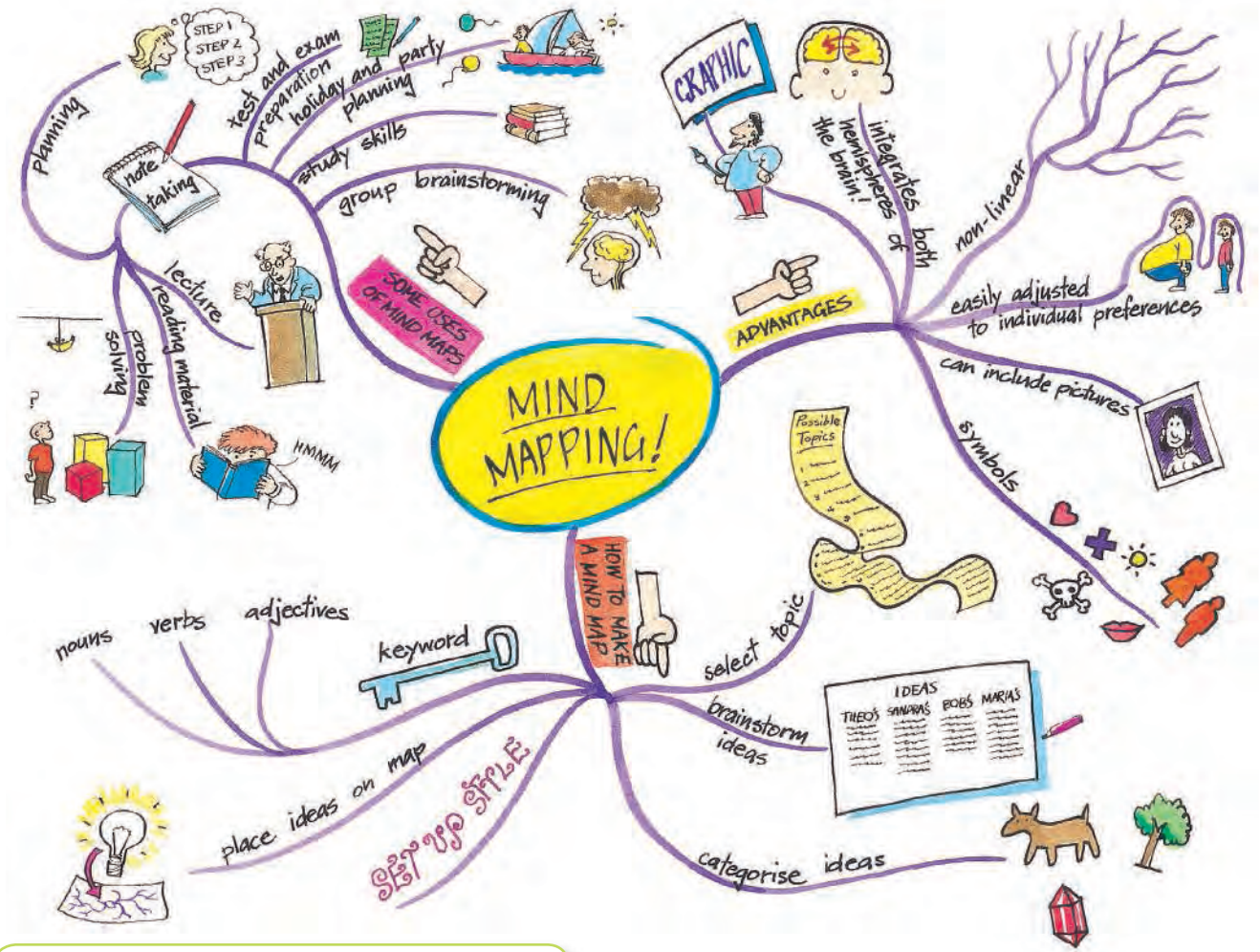
Is meaning present?		Is sense present?	
		No	**Yes**
Yes		Moderate to high	Very high
No		Very low	Moderate to high

Memory aids

Have you used a **mnemonic** in your learning today? A mnemonic is something that helps you to remember something else. It may take the form of a word, poem, story or image. Here are some examples of mnemonics:

- *Tell a tale:* Make up a story using the words or information that you need to learn.
- *Link-it:* Link the words or pieces of information together with images.
- *First letter:* Acronyms use the **f**irst letters of the words to make a new word. The example below shows how this can be used to help you remember some of the different memory systems for different types of learning:

Spatial memory
Procedural memory
Episodic memory
Working memory
Semantic memory



Mind mapping can help you find sense and meaning in your learning.

Mind mapping

Mind maps help create memory (learning) in a way similar to that of the brain by presenting information in a visual and connected form. Mind maps contain information in a predigested form that the memory can most easily assimilate and access.

WHY USE THEM?

Mind maps appeal to the right side of the brain, which processes colours, relationships, pictures and symbols. Using mind maps can increase your understanding of information and boost your recall of it dramatically.

UNDERSTANDING AND INQUIRING

REMEMBER

1 Give two reasons scientists use models.
2 What is the key role of the sensory register in the information processing model?
3 Identify the part of the brain involved in transferring information from short-term memory to long-term memory.
4 Outline the difference between long-term storage and long-term memory.
5 State the approximate capacity and time limits of working memory.
6 List the five different memory systems described in this section.
7 What is a mnemonic? Give an example.
8 State the two questions 'asked' to determine whether information moves into long-term memory.
9 Suggest the advantage of traumatic or emotionally charged events being remembered more deeply.

INVESTIGATE, THINK AND DISCUSS

10 Outline some research on chemicals that can affect memory.
11 Get a pencil and paper and then concentrate on the number below for 7 seconds. After 7 seconds, look away and write the number down. Did you get it right? Compare with others in your team.

5167340

Now repeat the procedure again with the number below. Did you get it right? Compare with others in your team. Did you get the same results for each number? Discuss your results with your team. Suggest a reason for any differences between results.

3847918362

12 Although we all use the same senses to collect information from our environment, they do not contribute equally to our learning. Learners develop preferences for certain senses over others. This is where terms such as 'visual', 'auditory' and 'kinaesthetic' learners originate.
(a) Research each of these and develop a set of questions that can be used to determine which preferences you and others in your class have.
(b) Discuss the impact that these differences can have on your learning.
(c) Suggest how you can use this knowledge to be a more effective learner.
13 Find out the possible effects of the following chemicals on learning: adrenaline, phenylalanine, norepinephrine, calpain and choline.
14 Find out more about memory drugs. Construct a PMI chart to summarise and share your findings.
15 Write a newspaper article, cartoon or web page on ways to improve your memory.
16 Find out more about research on memory and chemicals that may be used to enhance or erase it. Organise a class debate on one of the following statements.
(a) Drugs that have an effect on memory should be illegal.
(b) Everyone should have access to drugs that erase memories.
(c) Research on drugs that alter memories should be stopped.

INVESTIGATE, CREATE AND PRESENT

17 Find out more about the information processing model; then, in teams of 8, discuss how you could act it out. Include the following roles: sensory register, immediate memory, working memory, long-term storage (two people), incoming information (three people).
18 Research the structure and function of the thalamus, amygdala or hippocampus and construct a model to communicate your findings to others.

eBookplus

19 Use the **Memory games** and **Brain games** weblinks in your eBookPLUS to test your memory and learn more about the brain and nervous system.
20 Learn how your brain controls all your thoughts, actions and feelings by completing the **Brain control** interactivity in your eBookPLUS. **int-0010**

Venn diagrams

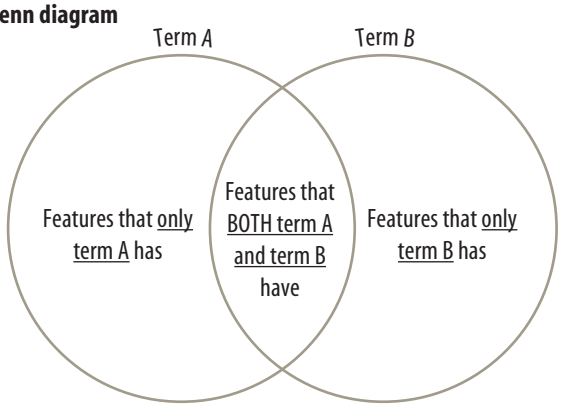
1. Draw two circles that overlap in the middle.
2. Write one of the words that you are comparing above each circle.
3. Write in the things that they have in common in the overlapped middle section.
4. Write the differences between them in each of the non-overlapped sections.

how to ...?

why use?

To show similarities and differences between two separate topics, terms or ideas

Venn diagram

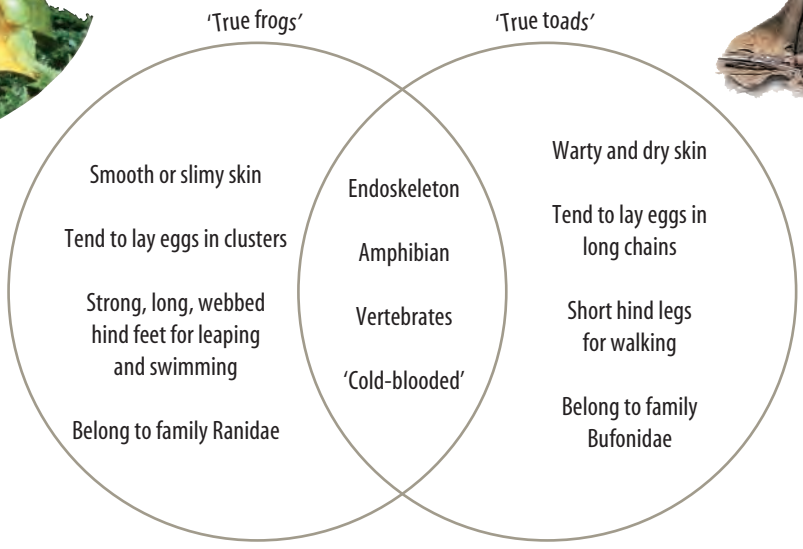


question

Compare these two topics.

What are the similarities and differences between term A and term B?

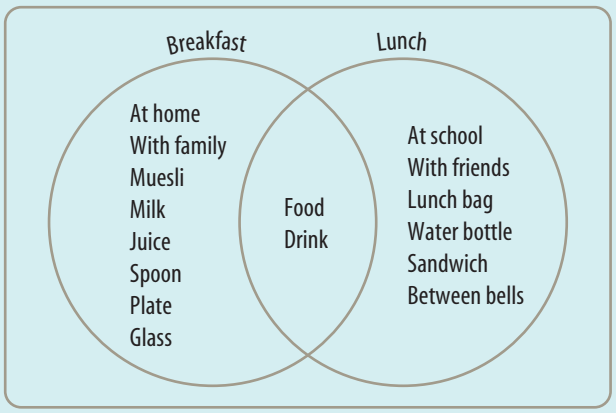
example



UNDERSTANDING AND INQUIRING

THINK AND CREATE

1 Using the figure below, construct your own Venn diagram about your breakfast and lunch during school days.



2 Construct Venn diagrams on the following topics:
(a) 'school day activities' and 'weekend activities'
(b) 'books' and 'movies'
(c) 'fruits' and 'vegetables'.

3 Research the meanings of the words below and then construct a Venn diagram to show similarities and differences between them.
(a) 'chlorophyll' and 'chloroplast'
(b) 'leucocyte' and erythrocyte'
(c) 'prefix' and 'suffix'
(d) 'endoparasite' and 'ectoparasite'
(e) 'cilia' and 'flagella'
(f) 'plant cells' and 'animal cells'
(g) 'vertebrates' and 'invertebrates'

4 (a) Carefully observe the information in the boxes above right, then construct Venn diagrams to compare the different types of dinosaurs.
(i) *Microceratops* and *Triceratops*
(ii) *Triceratops* and *Pentaceratops*
(iii) *Microdontosaurus* and *Heterodontosaurus*
(iv) *Microdontosaurus* and *Microceratops*

(b) Suggest the meanings of the prefixes *micro*, *tri*, *penta*, *micro* and *hetero*.
(c) What do you think the prefix *donto* might refer to? Justify your response.

Microceratops
- Name means 'small-horned face'
- Only 76 cm long
- Lived about 83–65 million years ago

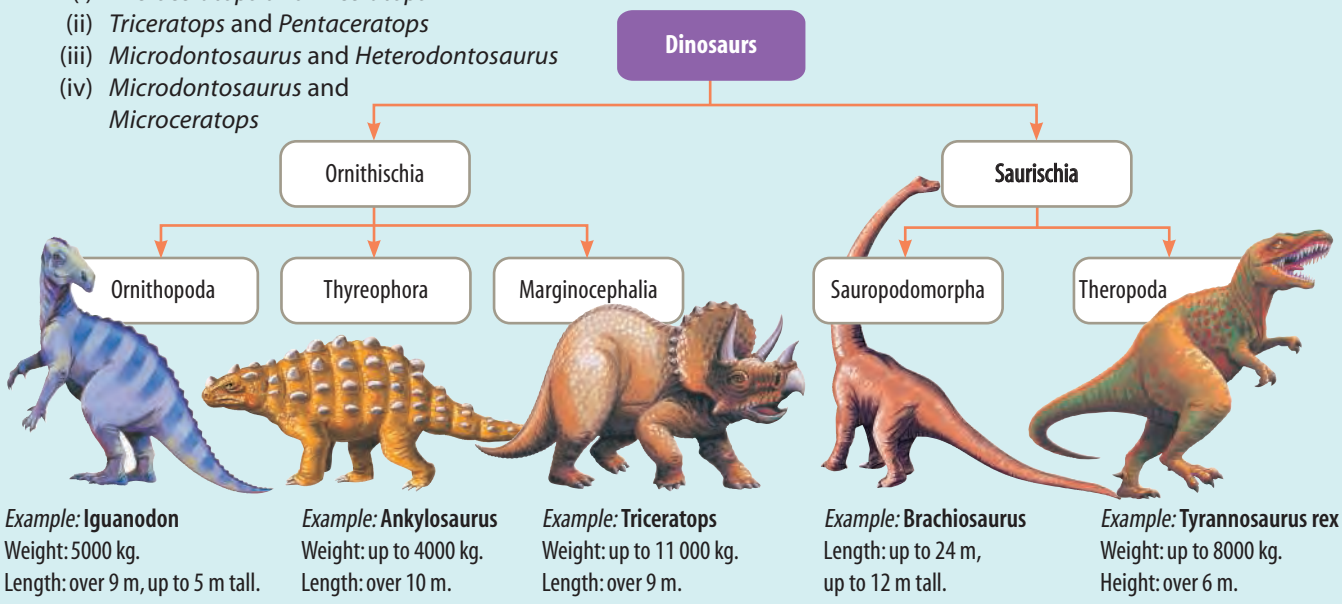
Heterodontosaurus
- Name means 'different-toothed lizard'
- Had three types of teeth
- 2.2 m long
- Lived about 208–200 million years ago

Pentaceratops
- Name means 'five-horned face'
- Had three horns on its head
- 8 m long
- Lived about 75–65 million years ago

Triceratops
- Name means 'three-horned face'
- 'Frilled' dinosaur
- Had three horns on its head
- 8 m long
- Lived about 75–65 million years ago

Microdontosaurus
- Name means 'tiny-toothed lizard'
- 8 m long
- Lived about 75–65 million years ago

5 Use the information in the diagram below to construct the following Venn diagrams.
(a) Dinosaurs under 10 metres long and dinosaurs over 10 metres long
(b) Dinosaurs with horns and spikes and dinosaurs without horns and spikes
(c) Dinosaurs that weighed less than 8000 kilograms and dinosaurs that weighed more than 8000 kilograms



Example: **Iguanodon**
Weight: 5000 kg.
Length: over 9 m, up to 5 m tall.

Example: **Ankylosaurus**
Weight: up to 4000 kg.
Length: over 10 m.

Example: **Triceratops**
Weight: up to 11 000 kg.
Length: over 9 m.

Example: **Brachiosaurus**
Length: up to 24 m, up to 12 m tall.

Example: **Tyrannosaurus rex**
Weight: up to 8000 kg.
Height: over 6 m.

STUDY CHECKLIST

LANGUAGE OF LEARNING

- describe a strategy that helps a problem or issue to be considered from different perspectives
- identify terms associated with each of these categories: apply, explain, self-knowledge, empathy, perspective, interpret
- using a diagram example, identify a visual thinking tool associated with each of these types of critical thinking: compare and contrast, evaluating, prioritising, sequencing, classifying
- using a diagram example, identify a visual thinking tool associated with each of these types of creative thinking: personifying, brainstorming, visualising, hypothesising
- outline the key difference between 'fat' and 'skinny' questions, providing an example of each
- suggest questions that would fit into each of the following categories: apply, explain, self-knowledge, interpret, perspective, empathy
- suggest skills important for effective listening
- provide an example that supports the statement: *Verbal communication includes not only 'what' we say, but also 'how' we say it*
- list examples of activities that involve intrapersonal intelligences and interpersonal intelligences
- describe how prefixes and suffixes can help unlock patterns in scientific terminology
- outline the involvement of the amygdala and hippocampus in building memories
- define the term 'mnemonic' and give an example of one that you have used
- list five different memory systems and suggest a way of remembering them
- construct a Venn diagram to compare and contrast: short-term memory and long-term memory, red thinking hats and white thinking hats, 'explain' and 'empathy', 'fat' questions and 'skinny' questions, leucocytes and erythrocytes

INDIVIDUAL PATHWAYS

eBookplus

Activity 2.1
Learning
doc-6045

Activity 2.2
Investigating learning
doc-6046

Activity 2.3
Developing learning skills
doc-6047

ICT

eBookplus **Summary**

eLESSONS

Teen brain

Take a fascinating look inside the teen brain and discover why teenagers seem to excel at impulsive behaviour that defies adult understanding. The latest science says their brains are physically unable to control some of these impulses until they are at least 25 years of age. A worksheet is included to enhance your understanding of the key concepts from your course.

Searchlight ID: eles-0224

INTERACTIVITY

Brain control

Learn how your brain controls all your thoughts, actions and feelings.

Searchlight ID: int-0010

LOOKING BACK

1 Use your three floors of thinking to help you create your own magazine cover of the key ideas that you have learned from this chapter. Remember to:
(a) gather your information — first-floor thinking
(b) process your information — second-floor thinking
(c) apply your information — third-floor thinking.
(d) **Reflecting on your learning**
Metacognition is the act of 'thinking about your own thinking'. This is when you notice and scrutinise what you are thinking. It also involves thinking about how you are thinking.
- Ask yourself questions like:
- What am I thinking about?
- What am I thinking about what I am thinking about?
- What caused me to think about this?
- What have I learned?
- How do I know that I'm right?
- How might I think about this differently?

2 (a) Use the rating guide below to assess your presentations throughout this chapter.

When I presented information to the class or my team ...

1. I spoke in a clear and effective voice.

2. I was well organised.

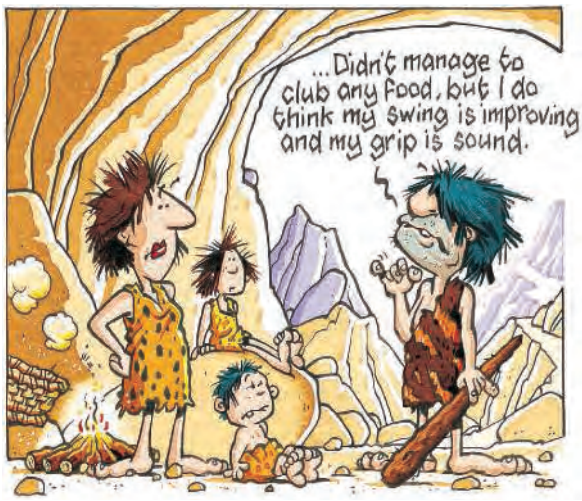
3. I was confident on my topic.

4. I presented information in an interesting and entertaining way.

Adapted with permission from Kagan Publishing from Dr. Spencer Kagan's book: Cooperative Learning © 1994, 1 (800) 933-2887, www.KaganOnline.com

(b) Describe what you liked about how others in your class presented information.
(c) From your new experiences in presenting and communicating ideas (and observing others), describe things that you could do to be a more effective communicator.

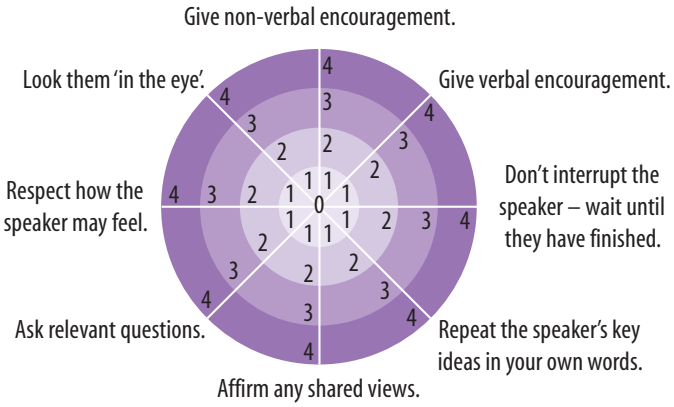
3 Carefully study the cartoon below.
(a) Paraphrase what the caveman is saying.
(b) Predict what his wife will say in return.
(c) Suggest what the caveman may say to his wife's comment.
(d) Describe a time when you were in a similar situation.
(e) Create your own cartoon to describe a situation in which you effectively learned something.



4 (a) Suggest how people remembered and communicated ideas before pen and paper.
(b) What evidence is there that humans have tried to pass knowledge from generation to generation?
(c) Which type of memory system would have been most useful to cave people? Justify your selection with examples.
(d) Suggest examples of inventions that would have been useful to cave people.
(e) Design and construct an 'invention' for cave people. Describe how and when it would be used.

1. Generation of ideas			
1	**2**	**3**	**4**
Few ideas generated	Some of group pushed ideas	Most of group gave creative ideas	All of group gave creative ideas
2. Time management			
1	**2**	**3**	**4**
Lot of time wasted	Often got distracted	Progressed well once on track	Stayed on target; no time wasted
3. Decision making			
1	**2**	**3**	**4**
Group conflicts not resolved	One person took over	Group compromised on important points	All group supportive and encouraged agreement
4. Overall			
1	**2**	**3**	**4**
Goal not achieved	Goal nearly achieved	Goal achieved but no more	Goal exceeded

5 Sit with a partner and position yourself 'Eye-to-Eye and Knee-to-Knee'. This will help you to maintain eye contact and will promote active listening. While positioned this way, take turns to tell each other about the sort of things that you think are important to make a good first impression. Use the active listening wheel below to score how effectively you think you actively listened and then score your partner for their active listening. Discuss what you found easy or difficult, and what you would try next time to enhance your listening skills.



6 Use the rating guide above to assess your individual and team activities throughout this chapter.
(a) How productive was I?
(b) How productive was my team?

7 Use the guides below to reflect about yourself.

How do you most want to be?

Honest	1	2	3	4	5
Caring	1	2	3	4	5
Creative	1	2	3	4	5
Logical	1	2	3	4	5
Independent	1	2	3	4	5
Cooperative	1	2	3	4	5
Playful	1	2	3	4	5
Adventurous	1	2	3	4	5

What are you and what do you prefer ...?

I am a:

Doer ⟷ Thinker

Night person ⟷ Morning person

Listener ⟷ Talker

Follower ⟷ Leader

I prefer:

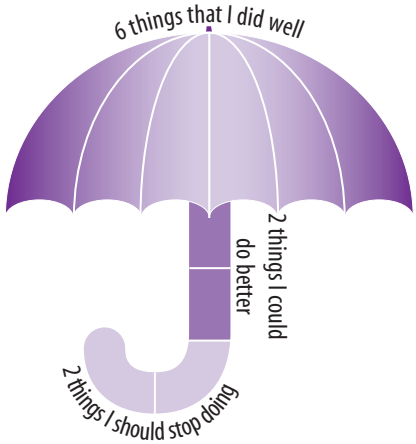
Quiet ⟷ Noise

Dark ⟷ Light

To work alone ⟷ To work in teams

8 Use diagrams and thinking tools to describe yourself in the past, present and future. Explain how and why you arrived there.

9 Use the reflection umbrella to list six things that you do well, two things you could do better and two things that you should stop doing.



10 Self-analysis reflection

I tried to have an open mind and think flexibly.	0 1 2 3 4 5
I understood what I was to do.	0 1 2 3 4 5
I asked when I was not sure what was expected of me.	0 1 2 3 4 5
I persisted, even when the going was tough.	0 1 2 3 4 5
I used my time effectively.	0 1 2 3 4 5
I was well organised.	0 1 2 3 4 5
I contributed to team discussions.	0 1 2 3 4 5
I encouraged others.	0 1 2 3 4 5
I actively listened to others.	0 1 2 3 4 5

The most important thing that I learned during this chapter was ...

work sheets
2.2 Personal learning
2.3 Summing up
2.4 Looking back

3 Cells

Cells are the basic units of all living things. The first cell appeared on Earth about 3.5 billion years ago. Today, there are many different kinds of cells. The differences in the cells of organisms are sometimes used to classify them into groups. Although cells may vary in their size, shape, contents and organisation, they all perform functions that are involved in keeping the organism to which they belong alive.

OVERARCHING IDEAS

- Patterns, order and organisation
- Form and function
- Stability and change
- Systems

SCIENCE UNDERSTANDING

Cells are the basic units of living things and have specialised structures and functions.

Elaborations

Examining a variety of cells using a light microscope, by digital technology or by viewing a simulation

Distinguishing plant cells from animal or fungal cells

Identifying the organs and overall function of a system of a multicellular organism in supporting the life processes

Identifying structures within cells and describing their function

Recognising that cells reproduce via cell division

Describing mitosis as cell division for growth and repair

THINK ABOUT CELLS

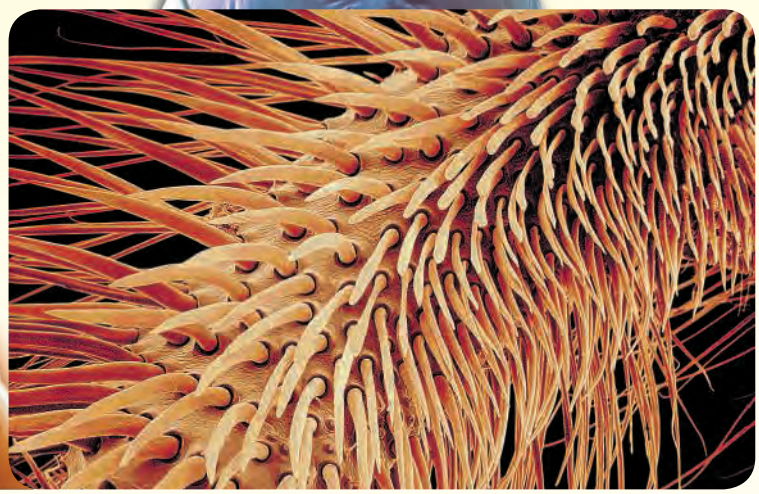
- What's all the fuss about stem cells?
- How can you make small things look bigger?
- Which are bigger, viruses or bacteria?
- What does Schwann have to do with cells?
- Why are beaches tested for the presence of *E. coli*?
- How does a cell become a clone?
- Why don't all cells look the same?

YOUR QUEST

Who am I?

Microscopes are responsible for opening a whole new world to us. They have allowed us to see beyond our own vision. The more developed these microscopes become, the more detail and wonder we are able to observe —but often, rather than answering our questions, they provide us with many more.

The three photos below and the one on the previous page show parts of different animals. They were taken with a scanning electron microscope, which allows us to see more detail of the surface of specimens.



OBSERVE, THINK AND SHARE

1 Look carefully at the photos of each animal part and think about:
   (a) what they could be
   (b) what they may do
   (c) who they may belong to.

2 Talk through your suggestions with your partner, adding all of the details that you have both observed onto a sheet of paper.

3 Two of these photos show parts of one type of animal, and the other two are of different animals. Does that information change the way that you look at the details? Which animal do you think two of the parts belong to? Brainstorm to decide which two animals the other parts could belong to.

4 Suggest other sorts of information that may be helpful in determining who these parts belong to and what they are used for.

A whole new world

A whole new world was discovered just over 400 years ago when an English inventor and scientist used magnifying lenses to observe the basic units of which all living things are made. This led to a new way of thinking about living things that required a new scientific language, new classifications and new inventions to find out more about this new world.

An early microscope used by Robert Hooke

The discovery of cells

In the seventeenth century, Robert Hooke looked at thin slices of cork under a **microscope** (= very small + view) that he had made himself from lenses. He observed small box-like shapes inside the cork. He called the little boxes that he saw **cells**. Microscopes opened up a whole new world that had never been seen before.

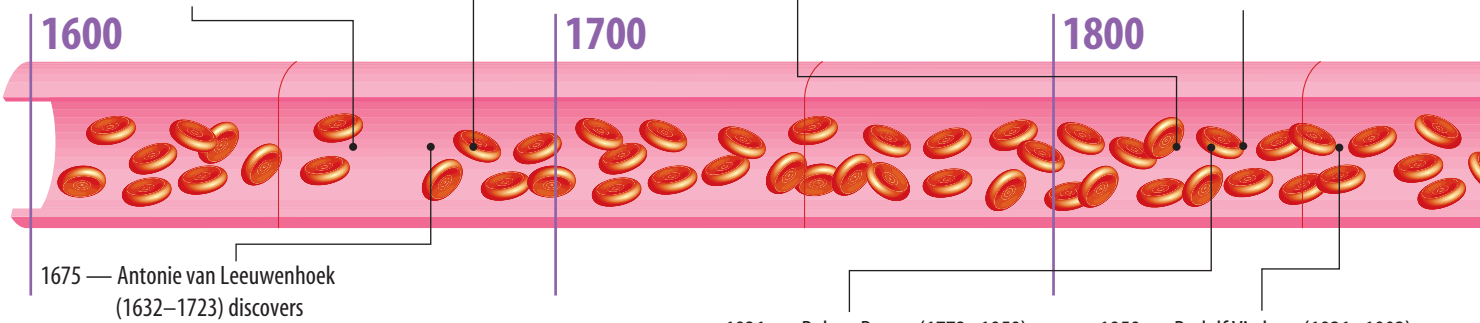
Using microscopes to carefully observe different living things showed that they also were made of these tiny basic units. As the **magnification** provided by microscopes increased, it could be seen that, although the basic structure of cells was similar, there were quite a few differences. Different groups of organisms often contained different types of cells. It was also discovered that different types of cells could be found within an individual organism.

HOW ABOUT THAT!

There are many different types of scientists who study cells. Examples include bacteriologists, cell biologists, clinical microbiologists, cytologists, electron microscopists, genetic scientists, medical microbiologists and virologists.

WHAT DOES IT MEAN?

The word *microscope* comes from the Greek words ***micrós***, meaning 'small', and ***skopein***, meaning 'to view'.



1665 — Robert Hooke (1635–1703) uses the term 'cell' to describe the tiny box-like units in a thin slice of cork.

1675 — Antonie van Leeuwenhoek (1632–1723) discovers unicellular microscopic organisms in stagnant water, which he calls 'animalcules'. (We now call these bacteria.)

1683 — Leeuwenhoek discovers bacteria in saliva.

1824 — René Dutrochet (1776–1847) states that all plants and animals are made up of cells.

1831 — Robert Brown (1773–1858) reports on his observation of the nucleus in both plant and animal cells.

1839 — Theodor Schwann (1810–1882) suggests that all animals are also made of one or more cells and that the cell is the basic unit of structure for all living things.

1858 — Rudolf Virchow (1821–1902) suggests that all cells arise from cells that already exist.

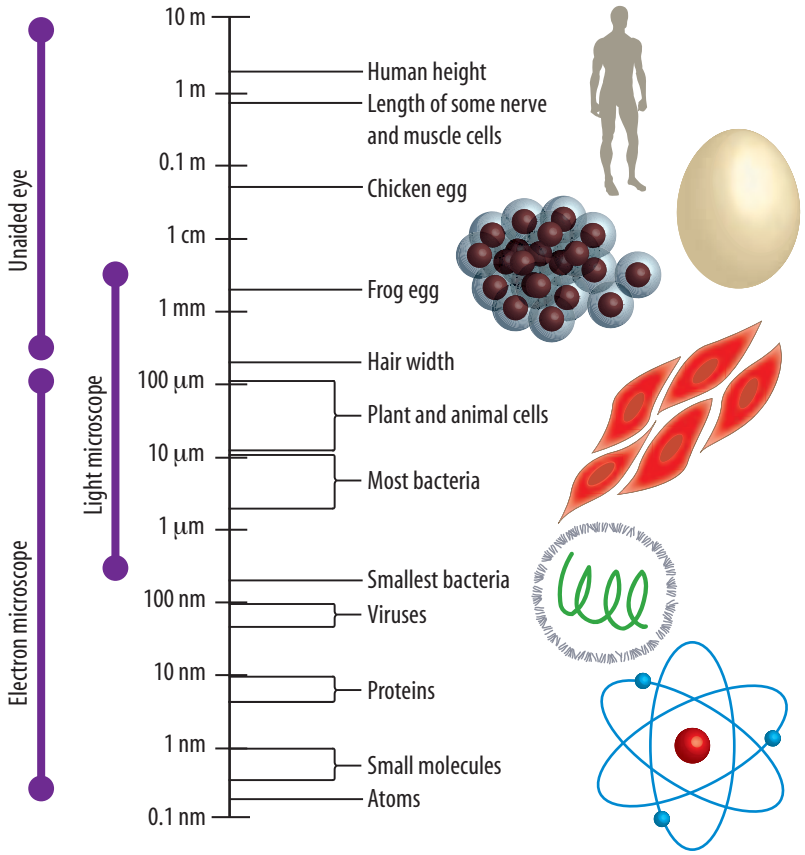
Timeline showing the development of microscope and cell theory

Little, littler, littlest . . .

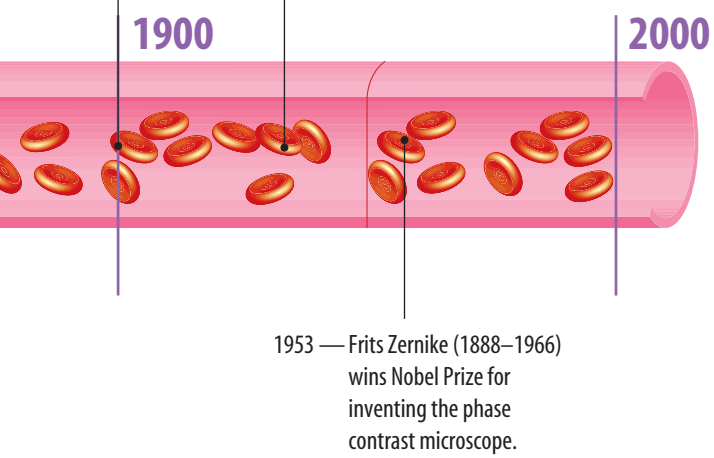
With the development of instruments such as microscopes, scientists needed to find words to describe some of the tiniest lengths and time scales in nature. They wanted some simple names to describe, for example, a billionth of a billionth of a metre.

In the microscopic world, there is often a need to describe things in much smaller terms than the units of measurement that you already know, such as metre, centimetre and millimetre. In describing cells, other units of measurement, such as micrometre (μm, also called micron) and nanometre (nm), are often used.

In 1964, experts in weights and measures from 35 nations met at a palace near Paris and decided to officially adopt a new prefix of 'atto', which means 1/1 000 000 000 000 000 000 (or 10^{-18}) of something. At the time it seemed impossible to measure something so small — but it doesn't seem so difficult now. Scientists are developing technology that can measure attoscale quantities. There are a number of very exciting 'attoworld' investigations and discoveries underway. With new technologies yet to be developed, how much more will we be able to 'see' and what will we find?



Unit	Symbol	No. units in 1 m
Millimetres	mm	1000
Micrometres	μm	1 000 000
Nanometres	nm	1 000 000 000



PRESENT DAY — Development of:

- transmission electron microscopes, which show the internal structures of cells
- scanning electron microscopes, which show images of the surface features (often involve coating the specimen with a very thin layer of metal atoms)
- superfast electron microscopy, which enables scientists to capture the movement of atoms (visit the **Electron strobe** weblink in your eBookPLUS).

Scope 'to go'

There are an increasing number of portable microscopes being invented that can open our eyes to the microbial world no matter where we are. Some of these are connected to camera-phones and can be used to diagnose a variety of diseases. The images can then be sent by internet to specialists for additional investigation. This may become an important field tool for the diagnosis of malaria, tuberculosis and other dangerous diseases.

eBookplus eLesson

Inside cells
Learn about cells and organelles in this animated video lesson.
eles-0054

Portable microscope for spotting and tracking disease

Looking like a grotesque eyeball, this hand-held microscope magnifies your specimens to two hundred times their normal size.

UNDERSTANDING AND INQUIRING

REMEMBER

1 State a feature that all living things have in common.
2 Suggest why Hooke called the objects that he observed under the microscope 'cells'.
3 Do all cells look the same?
4 Explain the importance of the microscope to biology.
5 List five types of scientists that study cells.
6 Suggest a function for portable microscopes.

THINK AND REASON

7 Use the timeline on pages 60–1 to answer the following questions.
   (a) In which year did Hooke use the term 'cells' to describe his observ ations of cork slices?
   (b) What did Virchow suggest in 1858?
   (c) In which substance did Leeuwenhoek discover bacteria?
   (d) When did Ruska build the first electron microscope?
   (e) State the differences between cell observations made with a scanning electron microscope and those with a transmission electron microscope.

INVESTIGATE AND CREATE

8 If you were living 300 years ago, how might you react to being told that you were made up of cells? What might cause these reactions? Construct a story, play or cartoon that answers these questions.
9 Find out how a light microscope works and design and construct your own model.
10 Create your own picture book or PowerPoint presentation that would teach primary students about the discovery of cells.

INVESTIGATE

11 Robert Hooke has been described as having 'a mechanic's mind and an artist's heart'. He wove these traits into his work as a scientist and inventor. Research and report on one of the following:
   - his work with Robert Boyle
   - his debate with Isaac Newton on the nature of light and gravity
   - Hooke's Law
   - his invention of the iris diaphragm in cameras
   - Hooke and micrography.

12 Research and report on one of the following:
   - Ernst Ruska's Nobel Prize for Physics
   - reasons why Frits Zernike has been described as 'a pioneer in forensic science'
   - examples of recent microscope inventions
   - examples of Australian research using electron microscopes.

13 Research one of the following types of scientists; imagine you are such a scientist and create a journal that describes a discovery you have made: bacteriologist, cell biologist, clinical microbiologist, cytologist, electron microscopist, genetic scientist, medical microbiologist, virologist.
14 In 2009, expatriate Australian Professor Elizabeth Blackburn was jointly awarded the Nobel Prize for Medicine for her discovery of telomeres and telomerase. Find out more about her discovery and its relevance to cells and microscopes. Report your findings in a PowerPoint presentation.

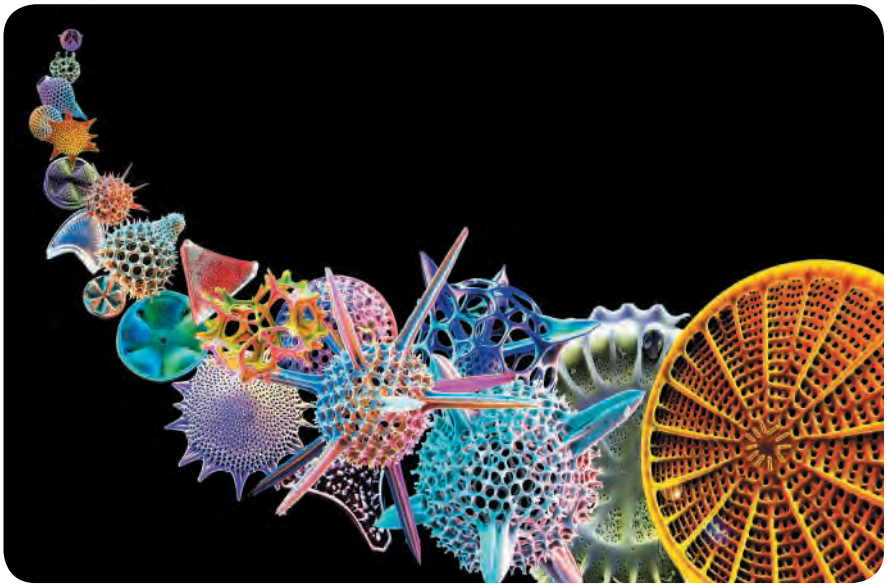
eBook*plus*

15 Visit the **Robert Hooke** weblink in your eBookPLUS and investigate why he used the term 'cells' for the little box-shaped structures he observed in cork. What did people think living things consisted of before Hooke's discovery of cells? Write a story about your findings.

work sheet 3.1 History of the light microscope

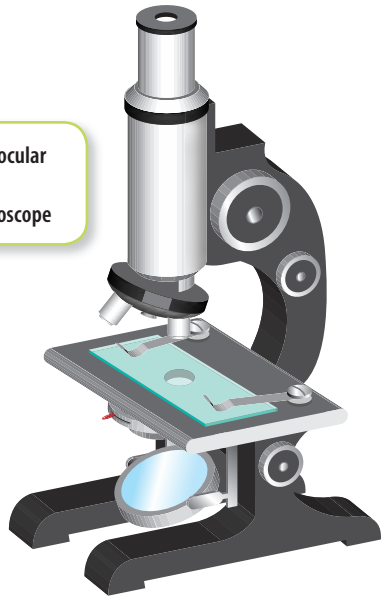
Focusing on a small world

Just because you can't see something doesn't mean that it's not there. Microscopes can be used to make small objects appear bigger, so that we can see what was previously invisible to us.



This coloured 'electron micrograph art' shows examples of diatoms, which belong to a group of photosynthetic, single-celled algae.

You may have light microscopes at your school. These may be either **monocular** (using one eye) or **binocular** (using two eyes). It is important that the specimen you observe using these is very thin, so that the light can pass through it. One type of binocular microscope, however, a **stereo** microscope, allows you to see the detail of much larger specimens. Stereo microscopes can be used to observe various objects including living organisms or parts of them.



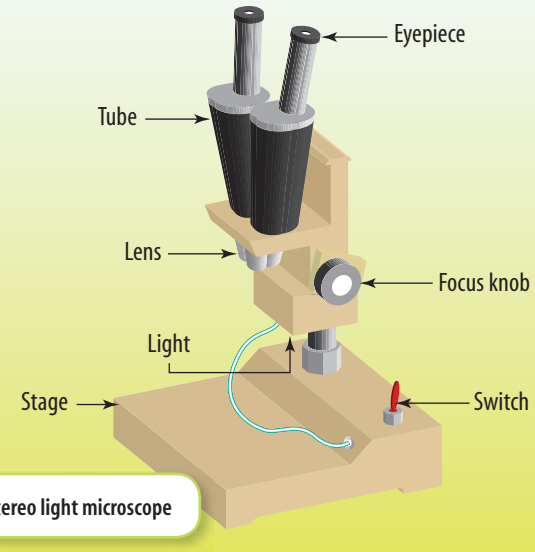
Monocular light microscope

Types of microscopes

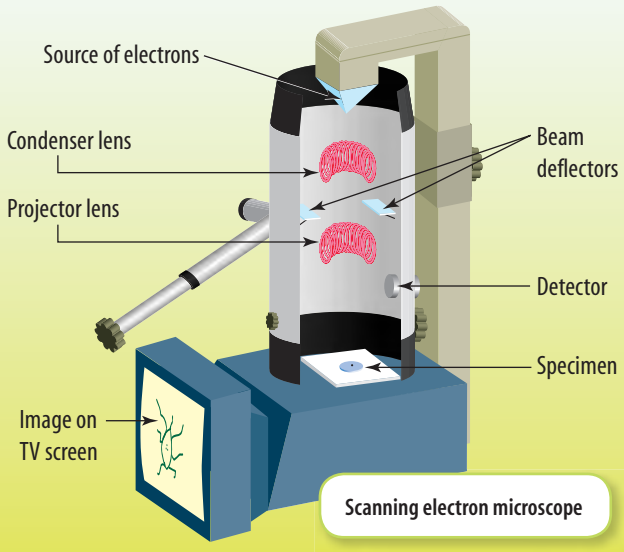
The two main types of microscopes are light microscopes and electron microscopes. **Light microscopes** use light rays whereas **electron microscopes** use small particles called electrons.

Some comparisons between light microscopes and electron microscopes

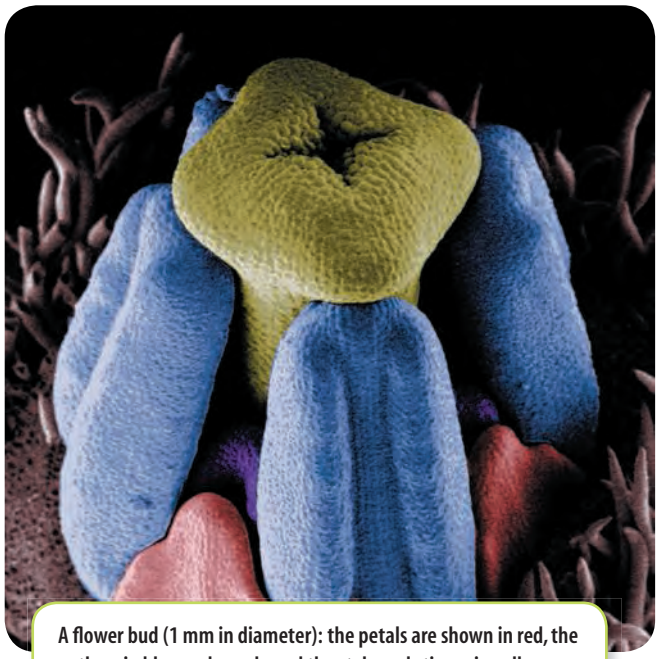
Type of microscope	Magnification (how many times bigger)	Resolution (how much detail can be seen)	Advantage(s)	Disadvantage(s)	Examples of detail that can be seen
Light microscope	Up to ×1500	Up to about 500 times better than the human eye	Samples prepared quickly; coloured stains can be used; living cells can be viewed	Limited visible detail	Shapes of cells; some structures inside cells, e.g. nucleus and chloroplasts
Electron microscope	Up to ×1 000 000	Up to about 5 million times better than the human eye	High magnification and resolution	Only dead sections can be viewed	All parts of cells; viruses



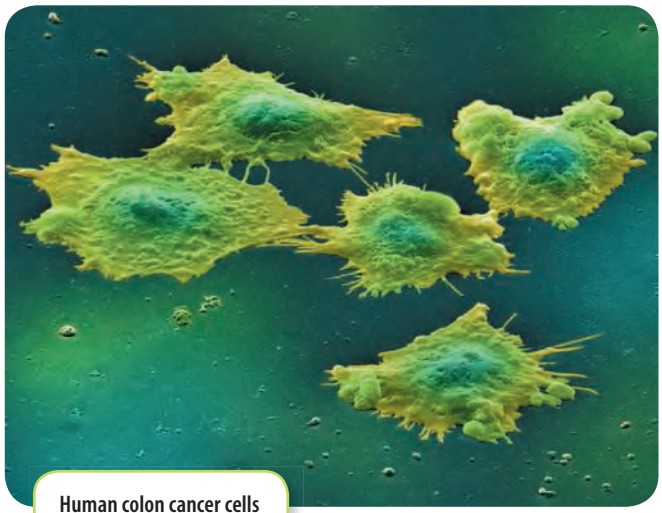
Stereo light microscope



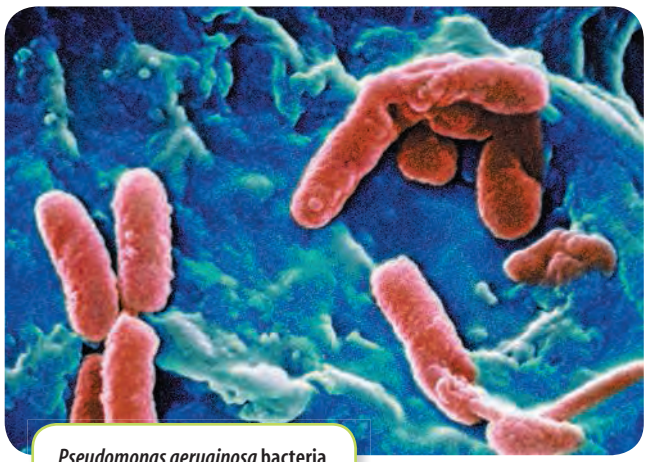
Scanning electron microscope



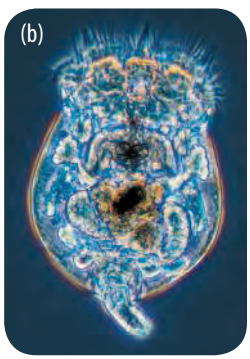
A flower bud (1 mm in diameter): the petals are shown in red, the anthers in blue and purple and the style and stigma in yellow.



Human colon cancer cells



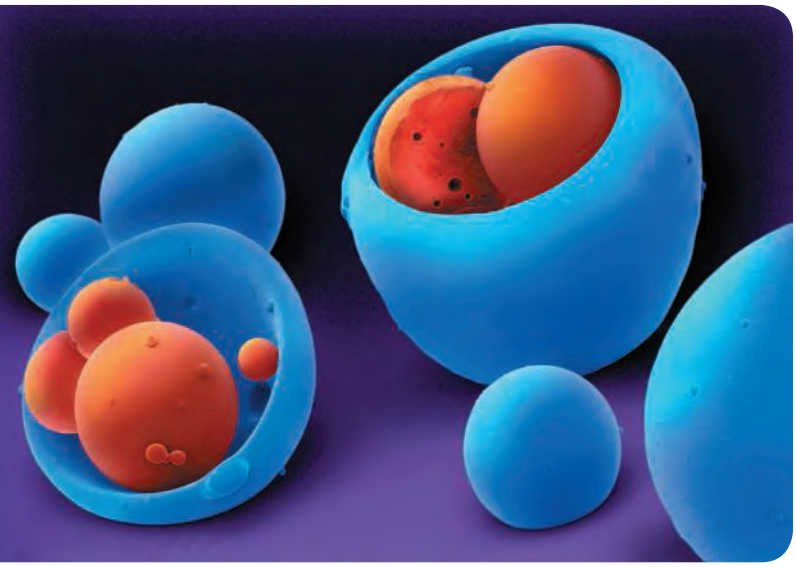
Pseudomonas aeruginosa bacteria



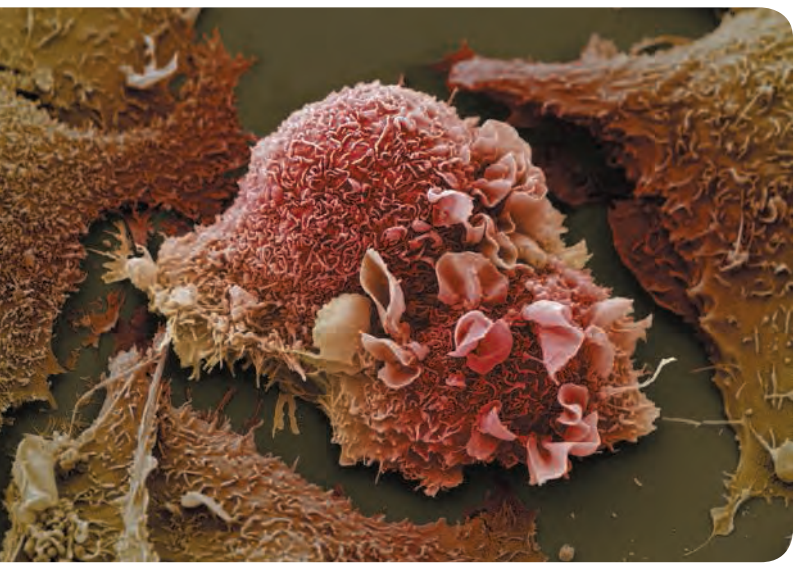
What's in your water? These images show zooplankton viewed through a scanning electron microscope.
(a) *Chaetognath*
(b) *Daphnia*
(c) A rotifer

Award-winning images

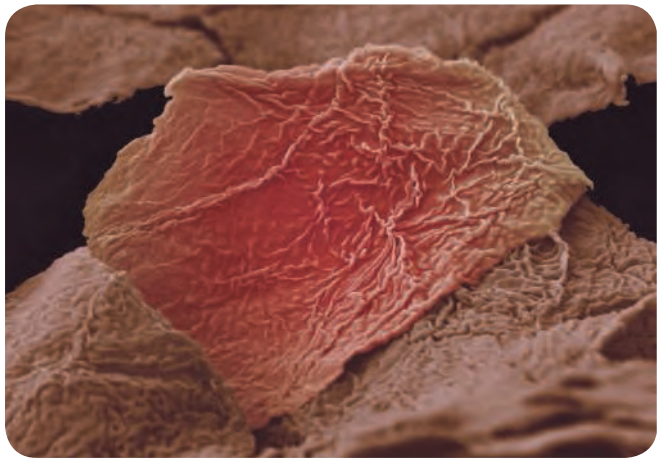
Microscopes are used not just to observe images of organisms, but also in many other areas of science. Some microscope images win awards recognising not just expertise but also creativity. For example, the Wellcome Trust, a charity that funds health research, presents awards for images that creatively explore the fields of medicine, health care and biology. The figures on this page show examples of some of the 2009 winners.



Electron micrograph of copolymers that can be used in drug delivery. The blue outer particle is the copolymer that encapsulates the inner orange particle loaded with the drug prednisolone, which is used to treat inflammatory bowel disease. (*Image:* Annie Cavanagh and Dave McCarthy)



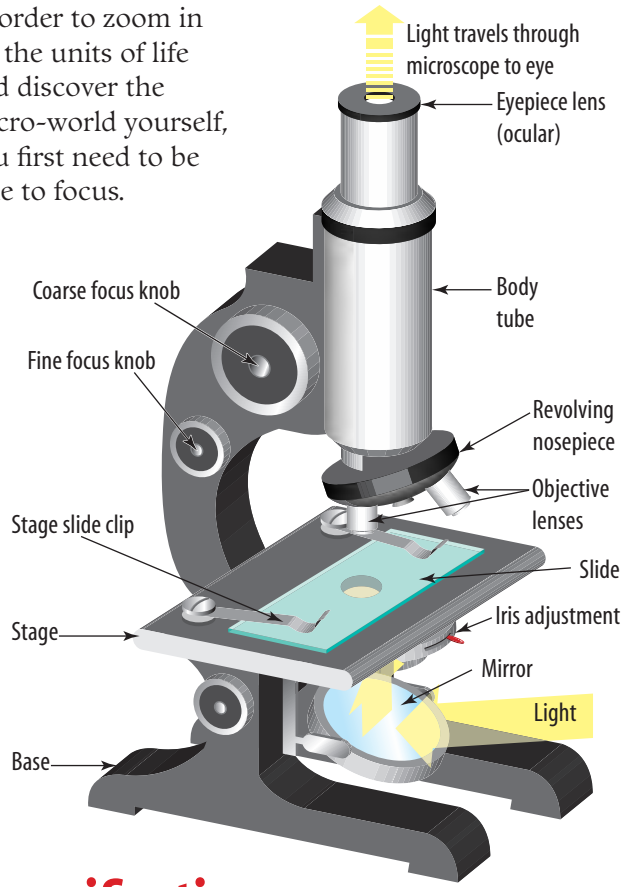
Electron micrograph of a single cell grown from a culture of lung cancer cells (*Image:* Anne Weston)



Electron micrograph of burn damaged skin cells; the burn was caused by the photographer, Anne Weston, spilling boiling soup on her hand. Anne Weston considers that this kind of curiosity is important in image-making as 'you never know what you are going to find'. (*Image:* Anne Weston)

Your turn as microbiologist

In order to zoom in on the units of life and discover the micro-world yourself, you first need to be able to focus.

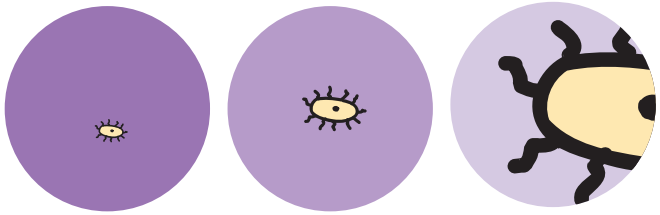


Magnification

The two lenses that determine the magnification of your microscope are the eyepiece lens and the objective lens. Each lens has a number on it that signifies its magnification. Multiplying the eyepiece

number by the objective lens number will give you the magnification of the microscope. For example:

- eyepiece: ×10
- objective: ×40
- magnification = ×400.



Field of view 4 mm (4000 μm) Magnification x40

Field of view 1.6 mm (1600 μm) Magnification x100

Field of view 0.4 mm (400 μm) Magnification x400

As the field of view gets smaller, the magnification gets larger.

INQUIRY: INVESTIGATION 3.1

Getting into focus with an 'e'

KEY INQUIRY SKILL:

- processing and analysing data and information

Equipment:

1 cm square piece of newsprint containing the letter 'e'

monocular light microscope

microscope slide *clear sticky tape*

- Carefully stick the 1 cm square of newsprint onto a clean microscope slide using sticky tape.
- Using the microscope directions, get the paper into focus using the coarse focus knob and the lowest power objective lens (smallest magnification).
- Carefully move the slide until you have a letter 'e' in focus.
- Change to a higher level of magnification by rotating to a higher power objective lens.

DISCUSS AND EXPLAIN

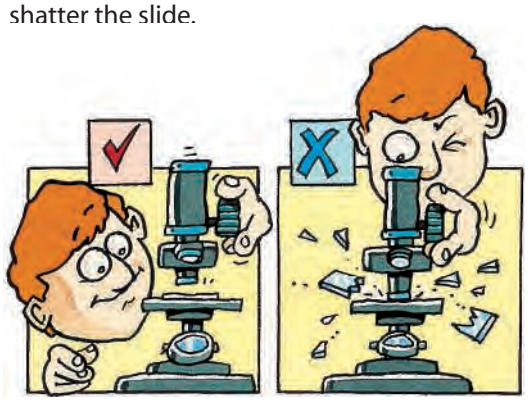
1. In which direction did the paper under the microscope move when you moved the slide (a) towards you or (b) to the left?
2. What does the letter 'e' look like under the microscope? Draw a pencil sketch of what you see.
3. Record the magnification that you are using, and estimate how much of the viewed area is covered by the letter 'e' at this magnification.
4. Suggest what the letters 'P' and 'R' would look like under the microscope. Sketch your predictions, and then view examples of these under the microscope. Were your predictions correct?
5. Summarise all your results in a table, using descriptive diagrams.

Important points to remember when using a microscope

1. When lifting the microscope, put one hand on the body of the microscope and one hand under its base.
2. The microscope should be used on a flat surface and not too close to the edge.
3. Take care that the light intensity is not too high, or it might damage your eye.
4. When you have finished using the microscope, return the shortest objective lens into position.
5. Remove the slide, and ensure that the stage is clean.
6. Make sure that, when your microscope is not in use, it is always clean and carefully put away.

Using a microscope

1. Adjust your mirror so the appropriate amount of light passes through the hole in the stage.
2. Place the glass microscope slide (with a single hair specimen on top) onto the stage.
3. While watching from the side, use the coarse focus knob to lower the objective lens until it is just above the slide. Moving it down too far may shatter the slide.



How to focus your microscope — and how not to!

4. While looking through the eyepiece lens, carefully turn the coarse focus knob until the specimen is seen clearly.
5. Carefully use the fine focus knob so that you can see the details of your specimen as clearly as possible.
6. Sketch what you see.
7. Suggest by how many times your specimen has been magnified.

UNDERSTANDING AND INQUIRING

REMEMBER

1 Construct Venn diagrams or double bubble maps to show the similarities and differences between:
   (a) a monocular microscope and a stereo microscope
   (b) a light microscope and an electron microscope.

2 Other than just to observe images of organisms, how else can microscopes be used?

3 Suggest why it is important not to have the light intensity setting too high on a light microscope.

4 Explain the importance of watching from the side of the microscope while using the coarse focus knob.

5 As the field of view of your microscope gets smaller, what happens to the magnification?

THINK AND REASON

6 Use the 'field of view' diagrams on page 66 to answer the following questions. (1000 µm = 1 mm)
   (a) Estimate the length of the specimen shown in the diagram at ×40, ×100 and ×400 magnification.
   (b) Describe the differences in your observations of the three different magnifications.

THINK

7 When you are looking down the microscope, what happens when you move the microscope slide (a) to the left, (b) to the right, (c) towards you or (d) away from you?

8 If you are using an eyepiece with a magnification of ×10 and an objective lens of ×10, how many times will the specimen viewed under the microscope be magnified?

9 If a specimen is 1 mm in length, how big will it appear if it is magnified ×100?

10 If a specimen takes up the entire field of view at ×100, how much of it will be seen at ×400?

11 (a) Sketch a line diagram of your microscope and label as many of its parts as you can, using the diagram on page 65 to help you.
   (b) How does your microscope differ from the one shown on page 65?
   (c) Suggest the advantages and disadvantages of the differences.

12 Copy and complete the table below.

Ocular lens (eyepiece)	Objective lens	Magnification
×5	×5	×25
×5	×10	
×10		×100
	×40	×400

CREATE

13 (a) Investigate developments in our understanding of cells.
   (b) Suggest how this knowledge has influenced areas in health or medicine.

14 Design and make a poster which shows either how a microscope should be used or what happens when you use it the wrong way.

15 Make a model of a microscope.

16 Design a microscope of the future. Prepare an instruction booklet and develop exciting promotional material.

INVESTIGATE

17 (a) Draw up a table that summarises the functions of the different parts of the microscope.
   (b) Construct a crossword that could be used to help students learn this information.
   (c) Share copies of your crossword with others in the class to try to find the solutions, and attempt to solve the crosswords of others in your class.
   (d) In a team, act out the parts and functions of a microscope.

18 Observe 10 different specimens (for example, hair, fingernail, pencil, insect, plant) under a stereo microscope. Sketch or describe what you see. Comment on the similarities and differences observed, and on any interesting findings.

19 Find out how specimens are prepared for examination under an electron microscope. Construct a PMI chart on your findings.

20 Prepare a report on the activities of scientists involved in these studies of cells: (a) cytology, (b) biochemistry, (c) microbiology, (d) histology.

eBookplus

21 Visit the **Wellcome Trust award winners** weblink in your eBookPLUS to find out more about the award winners and listen to Anne Weston talk about her work as an electron microscopist.

22 Test your knowledge of the functions of different parts of a microscope by completing the **Microscope parts** interactivity in your eBookPLUS. **int-0205**

23 Use the **Electron microscope** weblink in your eBookPLUS to view some electron micrographs. Analyse how the images produced by an electron microscope are different from those produced by a light microscope.

work sheet 3.2 In focus

3.3 OVERARCHING IDEA

Form and function: Cell make-up

Same building blocks, but different structures?

Similar, but different

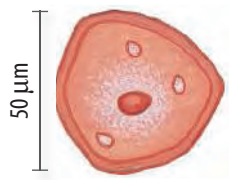
Cells are the building blocks that make up all living things. Organisms may be made up of one cell (**unicellular**) or many cells (**multicellular**). These cells contain small structures called organelles that have particular jobs within the cell and function together to keep the organism alive.

Cells can be divided on the basis of the presence and absence of particular organelles and other structural differences. Organisms can be classified by the different types of cells they are made up of.

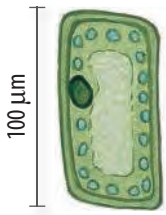
How big is small?

The size of cells may vary between organisms and within a multicellular organism. Most cells are too small to be seen without a microscope. Cells need to be very small because they have to be able to quickly take in substances they need and remove wastes and other substances. The bigger a cell is, the longer this process would take.

Very small units of measurement are used to describe the size of cells. The most commonly used unit is the **micrometre** (µm). One micrometre equals one millionth (1/1 000 000) of a metre or one thousandth (1/1000) of a millimetre. Check out your ruler to get an idea of how small this is! Most cells are in the range of 1 µm (bacteria) to 100 µm (plant cells).



Human cheek cell



Plant cell

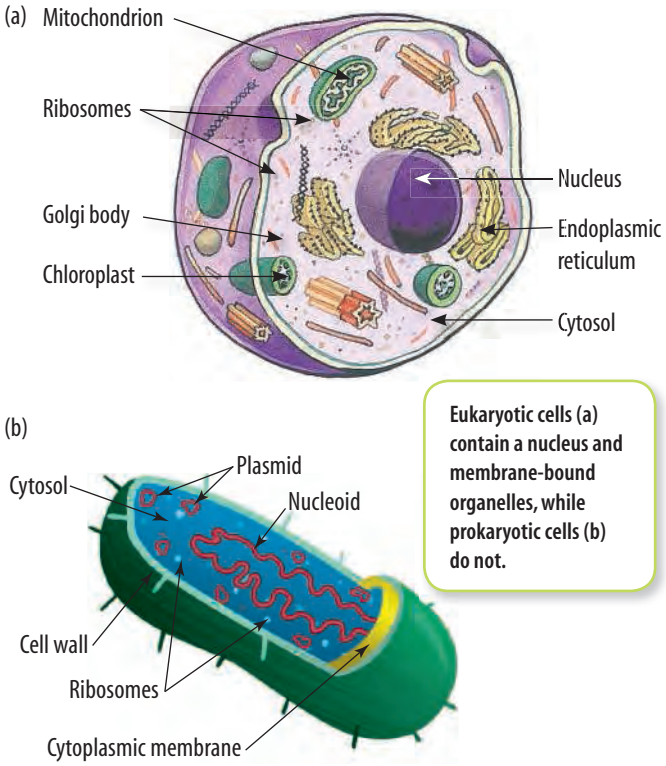
Advances in technology are creating an increased need for the use of the **nanometre** (nm) as a unit. One nanometre (nm) equals 1 billionth (1/1 000 000 000) of a metre. Investigating the organelles within cells and the molecules they are reacting with requires this level of measurement.

Nanotechnology is a rapidly developing field that includes studying and investigating cells at this 'nanolevel'. While it requires lots of creative, exciting and futuristic 'what if' thinking, it also requires an understanding of the basics of information and ideas currently known.

Have it or not?

Prokaryotes such as bacteria were the first type of organism to appear on Earth. The key difference between the prokaryotes and all other kingdoms is that members of this group do not contain a nucleus or other membrane-bound organelles. The word prokaryote comes from the Greek terms *pro*, meaning 'before', and *karyon*, meaning 'nut, kernel or fruit stone', referring to the cell nucleus.

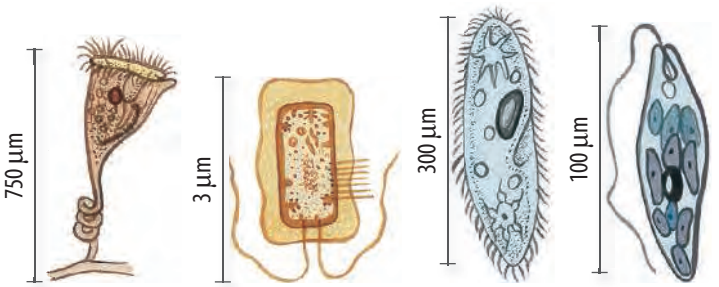
Eukaryotic organisms made up of eukaryotic cells appeared on Earth billions of years later. As *eu* is the Greek term meaning 'good', **eukaryote** can be translated as 'true nucleus'. Members of the four kingdoms Animalia, Plantae, Fungi and Protoctista are eukaryotes and are made up of cells containing a nucleus and other membrane-bound organelles.



Eukaryotic cells (a) contain a nucleus and membrane-bound organelles, while prokaryotic cells (b) do not.

What do we share?

What most cells have in common is that they are made up of a **cell membrane** containing a fluid called **cytosol** and small structures called **ribosomes**. The collective term used to describe the cytosol and all the organelles suspended within it is **cytoplasm**. The hundreds of chemical reactions essential for life that occur within the cytoplasm are referred to as the cell's **metabolism**. The ribosomes are where proteins such as enzymes are made, which regulate the many chemical reactions important to life. The cell membrane regulates the movement of substances into and out of the cell. This enables the delivery of nutrients and substances essential for these reactions, and the removal of wastes.

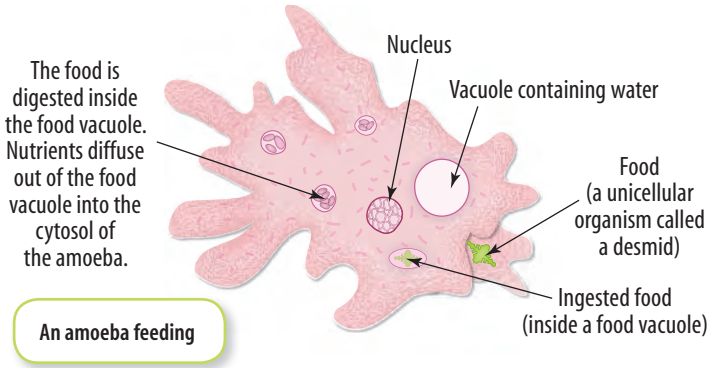


Vorticella | Bacterium | A *Paramecium* is a unicellular animal. | *Euglena*

All on my own

Unicellular organisms such as bacteria, *Amoeba*, *Euglena* and *Paramecium* need to carry out all the required processes themselves. They even reproduce themselves by dividing into two. This process is called **binary fission**.

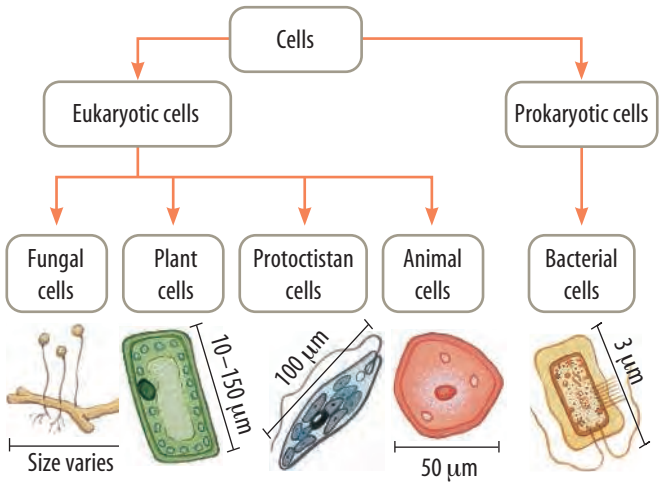
To live long enough to reproduce, unicellular organisms need to be able to function on their own. They need to be able to obtain their nutrients and remove their wastes. The solution to this requirement has resulted in the wonderful diversity of unicellular organisms that are alive on our planet today or have lived in our planet's history.



An amoeba feeding

Five kingdoms?

Living things can be divided into five kingdoms — **Animalia** (animals), **Plantae** (plants), **Fungi** (for example, mushrooms), **Protoctista** (also called Protista) and **Prokaryotae** (also called Monera). While this system provides an opportunity to classify organisms into these groups, information from currently developing technologies means that it will not be long until a new extended classification system evolves.



A key characteristic used to classify organisms into kingdoms is the structure of their cells.

WHAT DOES IT MEAN?

The prefix *uni-* comes from the Latin term meaning 'one'. The prefix *multi-* comes from the Latin term meaning 'many'.

Specialist workers

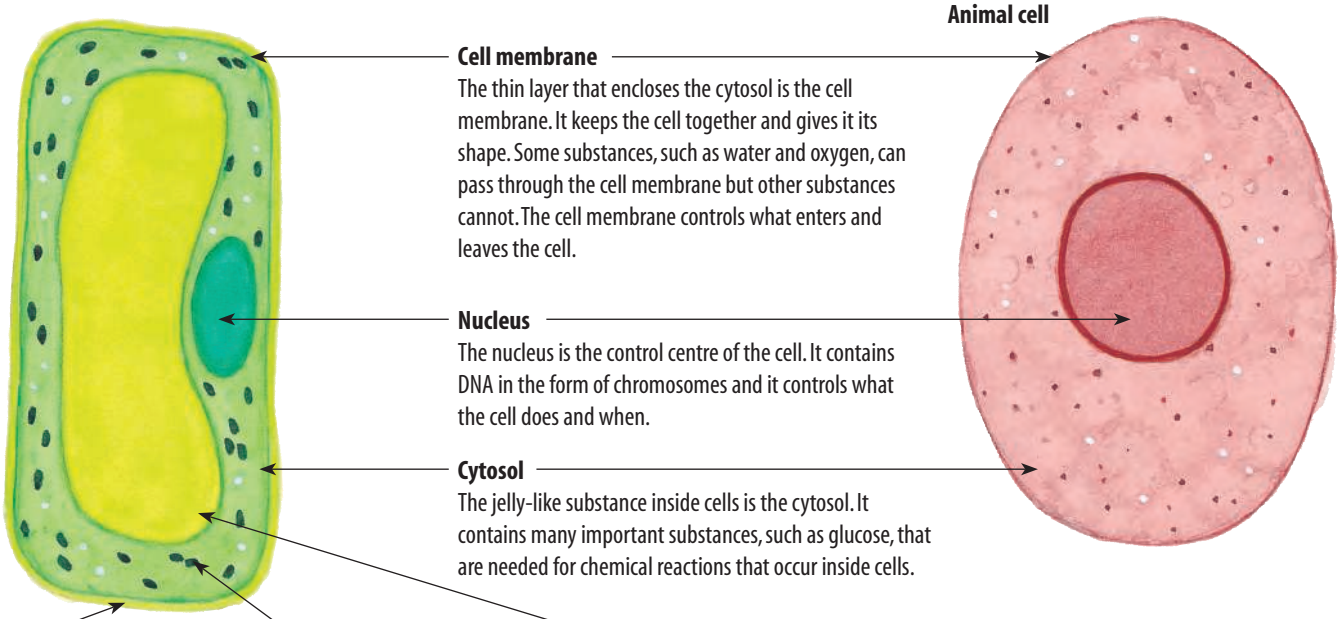
Multicellular organisms are made up of many different types of cells that have different jobs to do. Each of these cells has a particular structure so that it is able to do the job it is specialised for. This may include the presence and number of particular organelles or additional external structures to assist with movement (such as flagella or cilia).

Microfactories

Mitochondria and **chloroplasts** are examples of membrane-bound organelles found in eukaryotic cells. While all eukaryotic cells contain mitochondria, because they are all involved in **cellular respiration**, only those involved in **photosynthesis** (such as those in plant leaves) contain chloroplasts. Chloroplasts contain the green pigment **chlorophyll**. This pigment is used to trap light energy so that it can be converted into chemical energy and used by the cells.

Plant cell

Animal cell



Cell membrane
The thin layer that encloses the cytosol is the cell membrane. It keeps the cell together and gives it its shape. Some substances, such as water and oxygen, can pass through the cell membrane but other substances cannot. The cell membrane controls what enters and leaves the cell.

Nucleus
The nucleus is the control centre of the cell. It contains DNA in the form of chromosomes and it controls what the cell does and when.

Cytosol
The jelly-like substance inside cells is the cytosol. It contains many important substances, such as glucose, that are needed for chemical reactions that occur inside cells.

Cell wall
The tough covering around plant cells is the cell wall. It gives plant cells strength and holds them in shape. Plant cell walls are made of a substance called **cellulose**. Water and dissolved substances can pass through the cell wall. Animal cells do not have a cell wall.

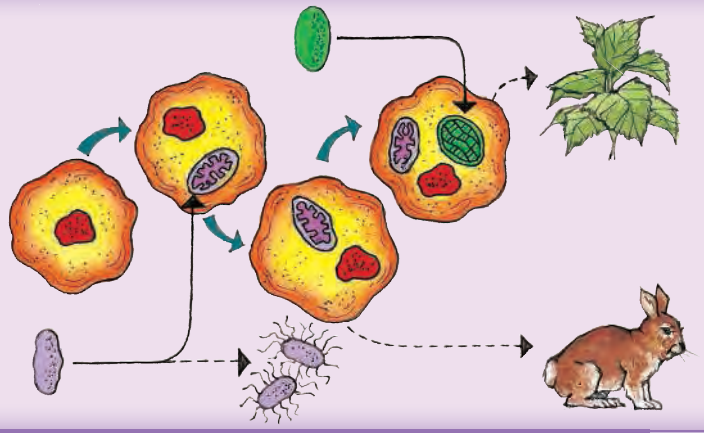
Chloroplasts
Chloroplasts are the oval-shaped organelles found only in plant cells. Chloroplasts contain a green substance called chlorophyll. Chloroplasts use energy from the sun to make food. Not all plant cells contain chloroplasts. They are found only in leaf and stem cells.

Vacuole
The vacuole is an organelle used to store water and dissolved substances. Vacuoles can look empty, like an air bubble. Plant cells usually have one large vacuole. The mixture inside a plant's vacuoles is called **cell sap**. The red, blue and violet colours that you often see in plant leaves and flowers are due to the substances stored in vacuoles. Most animal cells don't have vacuoles.

HOW ABOUT THAT!

There is a theory called the endosymbiotic theory that suggests that mitochondria and chloroplasts were once prokaryotic organisms. This theory suggests that, at some time in the past, these organisms were engulfed by another cell and over time they evolved to depend on each other.

The origin of the eukaryotic cell? Some scientists are also suggesting that our nucleus may have come from a giant viral ancestor.



Some differences in the basic cell design in the five kingdoms

Characteristic	Kingdom				
	Animalia (animals: e.g. lizards, fish, spiders, earthworms, sponges)	**Fungi** (e.g. yeasts, moulds, mushrooms, toadstools)	**Plantae** (plants: e.g. ferns, mosses, conifers, flowering plants)	**Protoctista** (e.g. algae, protozoans)	**Prokaryotae** (bacteria and cyanobacteria)
Number of cells	Multicellular	Usually multicellular but some unicellular	Most multicellular	Unicellular or multicellular	Unicellular
Nucleus	Present	Present	Present	Present	Absent
Cell wall	Absent	Present	Present	Present in some	Present
Large vacuole	Absent	Absent	Present	Present in some	Absent
Chloroplasts	Absent	Absent	Present in leaf and stem cells	Present in some	Absent (but chlorophyll may be present in some)

UNDERSTANDING AND INQUIRING

REMEMBER

1 What do all living things have in common?
2 Why is the nucleus important to the cell?
3 State the names of the five kingdoms and use the table above to determine which kingdoms contain organisms that are eukaryotes.
4 What is the purpose of the cell membrane?
5 Identify where enzymes are made in a cell and state why are they important.
6 Construct a triple Venn diagram (three overlapping circles) to compare plants, animals and fungi.
7 Suggest why organisms are divided into five kingdoms rather than just the plant and animal kingdoms.

THINK AND REASON

Use the table above to answer the following questions.

8 In which kingdom(s) do the cells of an organism:
(a) not have a cell wall, large vacuole or chloroplasts?
(b) have a cell wall, large vacuole and chloroplasts?
(c) have a cell wall, but no large vacuole or chloroplasts?
(d) have a cell wall and a nucleus without a membrane around it?
9 List two examples of each of the five kingdoms.

Use the diagrams of cells on page 69 to complete activities 10 and 11.

10 Construct a table with the following headings: 'Name of organism' or 'Type of cell', and 'Cell size (µm)'.
11 Show the sizes of the cells on a graph, with the horizontal axis representing the type of cell and the vertical axis representing the size of cell. Sketch an outline of each cell as accurately as you can in the correct position on the graph. Which cell is the biggest and which is the smallest?
12 Fungi were once classified as plants. On the basis of their cells, suggest why they are no longer classified in this group.

CREATE

13 Make a labelled model of a cell from one of the kingdoms. Use materials available at home, such as drink bottles, egg cartons, cottonwool, wool, cotton or dry foods.

INVESTIGATE AND CREATE

14 What does the endosymbiotic theory suggest? Formulate questions to ask about it. Research and report on your questions.
15 Research and report on:
(a) examples of prokaryotic cells and interesting survival strategies
(b) mitochondrial DNA and haplogroups.
16 Research two of the organelles or cells listed below. Create a play and construct puppet models for your characters. Present your play to the class.
- Nucleus
- Mitochondrion
- Chloroplast
- Prokaryotic cell
- Protoctistan cell
- Animal cell
- Plant cell

17 Investigate the different types of cells and create your own picture book about them using the following steps.
(a) Construct a matrix table (see page 340) to show the differences between the cells of the different kingdoms.
(b) Construct a storyboard for a picture book about them.
(c) Create the picture book.

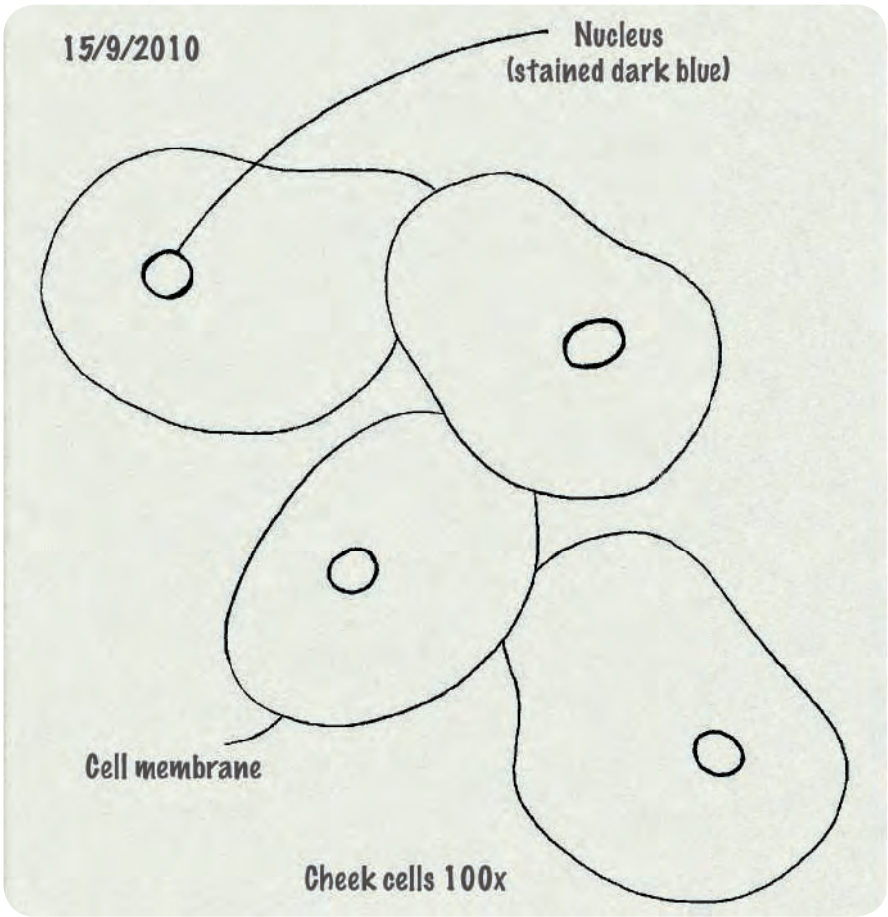
Zooming in on life

Now that you are in focus, let's zoom in on life.

Sketching what you see under the microscope

SOME POINTS TO REMEMBER

1 Use a sharp pencil.
2 Draw only the lines that you see (no shading or colouring).
3 Your diagrams should take up about a third to half a page each.
4 Record the magnification next to each diagram.
5 State the name of the specimen and the date of observation.
6 A written description is also often of considerable value.
7 When you are viewing many cells at one time, it is often useful to select and draw only two or three representative cells for each observation.



An example of a sketch of a microscope specimen

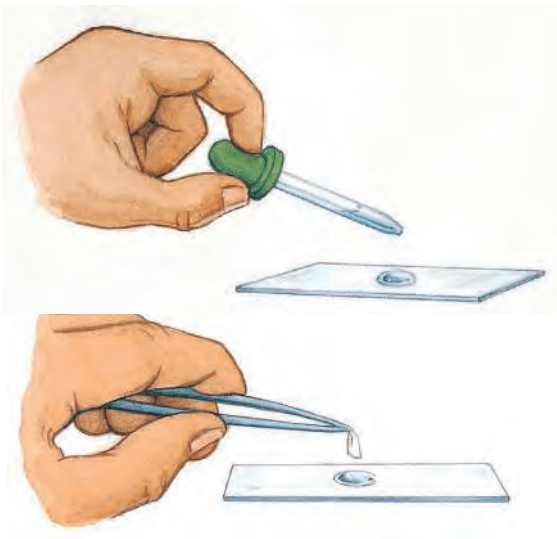
Preparing a specimen

Light microscopes function by allowing light to pass through the specimen to reach your eye. If the specimen is too thick, the object cannot be seen as clearly or may not be seen at all.

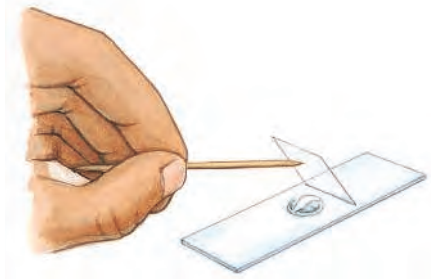
Careful peeling, scraping, slicing or squashing techniques can be used to obtain thin specimens of the object to be studied.

PREPARING A WET MOUNT SLIDE

1 Place a drop of pond water on a clean glass slide. Place the specimen on top.

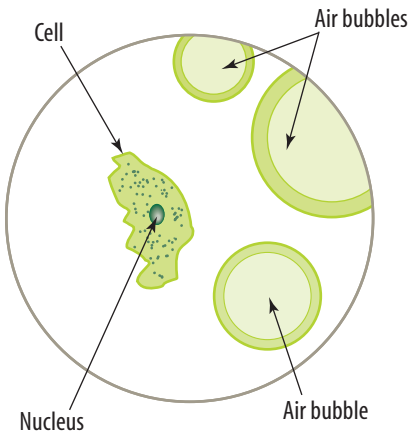


2 Gently place a coverslip (a very thin piece of glass that holds the specimen in place) on the pond water by putting one edge down first. Use a toothpick to lower the coverslip to avoid air bubbles.



3 Use a tissue or blotting paper to soak up any water that has escaped.

4 Incorrect placing of the coverslip can result in air bubbles.



STAINING A SPECIMEN

Many objects are colourless when viewed down the microscope, so specimens are often stained to make them easier to see. Methylene blue, iodine and eosin are some examples of stains often used.

Each stain reacts with different chemicals in the specimen. For example, iodine stains starch a blue-black colour.

Take care when using these stains, because they can stain you as well!

1 Add a few drops of iodine to a very small amount of squashed banana on a microscope slide.
2 Gently place a coverslip on top as shown in the diagram on page 72.

INQUIRY: INVESTIGATION 3.2

All in one cell!

KEY INQUIRY SKILLS:

- processing and analysing data and information
- planning and conducting

Equipment:

light microscope

living specimens: Paramecium, Euglena, Amoeba

- Use a light microscope to prepare a wet mount of the specimens.
- Draw sketches of what you see, identifying any cell components. Remember to include the magnification used.
- Construct a matrix to show similarities and differences between the specimens.

DISCUSS AND EXPLAIN

Which kingdom do you think each specimen may belong to? Why?

INQUIRY: INVESTIGATION 3.3

Kitchen and wardrobe detective

KEY INQUIRY SKILLS:

- questioning and predicting
- planning and conducting

Equipment:

light microscope and microscope slides

spatula

selection of white powders (e.g. flour, salt, sugar, baking soda)

ground, instant and freeze-dried coffee

different brands or types of spices and leaf tea

fibres (e.g. cotton, linen, silk, wool, polyester, nylon, various combinations of these)

hairs (of members of your class or family, or of animals)

- Examine a variety of the substances and materials listed above under the microscope.
- Try to determine the contents of an 'unknown mixture' that contains three or four of the substances and materials previously examined.

DISCUSS AND EXPLAIN

1 Draw sketches of what you see, and record comments or descriptions next to each.
2 Estimate the size of the specimens that you see.
3 Summarise your results in a table.
4 Describe how your specimens differed.

INQUIRY: INVESTIGATION 3.4

Pond water

KEY INQUIRY SKILL:

- processing and analysing data and information

Equipment:

light microscope
microscope slides
coverslips
pond water
pipette

- Prepare a wet mount of the pond water on a microscope slide.
- Examine the pond water under the microscope.
- Draw sketches and describe what you see.

INQUIRY: INVESTIGATION 3.5

Peel or squash and stain

KEY INQUIRY SKILL:

- processing and analysing data and information

Equipment:

light microscope
microscope slides
coverslips
pipette
onion, banana
water, methylene blue, iodine

- Peel some onion skin and carefully place on a slide with a drop of methylene blue.
- Cover carefully with a coverslip.
- Prepare a banana squash on a slide and add a drop of iodine.

- Cover with a coverslip.

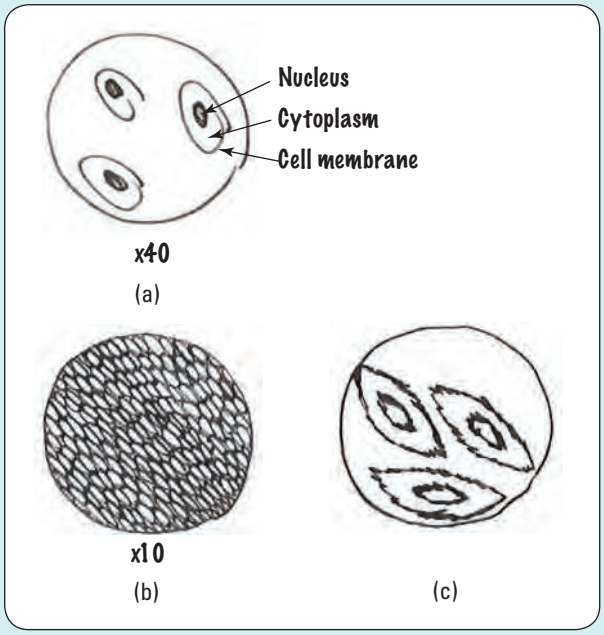
DISCUSS AND EXPLAIN

1 Record what you see under the microscope for each slide.
2 Describe the similarities and differences between your observations of the banana and the onion cells.

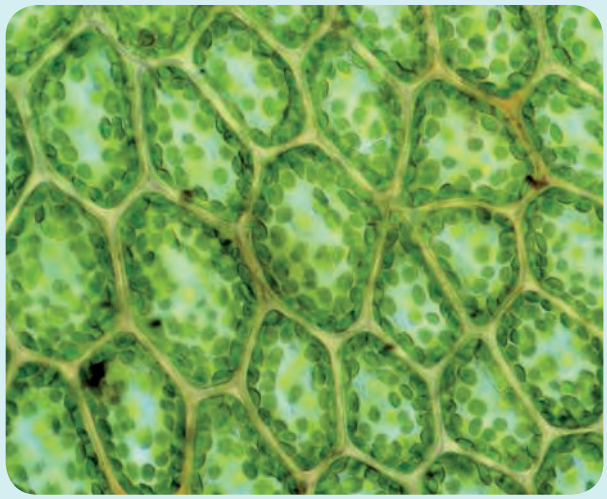
UNDERSTANDING AND INQUIRING

THINK, DISCUSS AND INVESTIGATE

1 Carefully observe the student sketches shown below. For each diagram, list what is wrong with it and suggest how it could be improved.



2 Carefully observe the figure of plant cells below and construct a sketch of one of the cells.



CREATE

3 Design a poster that shows others how to prepare a variety of specimens to be viewed under a microscope.

work sheet 3.3 Preparing a stained wet mount

3.5 SCIENCE UNDERSTANDING

Focus on animal cells

What do the cells inside your body look like? Why aren't they all the same size and shape?

In all shapes and sizes

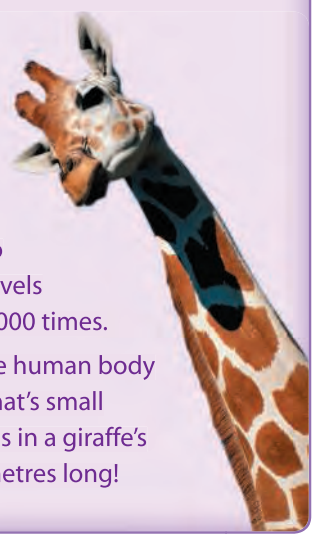
Cells within an organism may differ in their shape and size. This difference may be due to the particular jobs or functions that the cells carry out within the organism. The human body is made up of more than 20 different types of cells, with each type suited to a particular function.

Nerve cells develop long, thin fibres that quickly carry messages from one cell to another. Cells lining the trachea have hair-like cilia that move fluid and dust particles out of the lungs. Muscle cells contain fibres that contract and relax, and the human sperm cell has a tail or flagellum that helps it swim to the egg cell.

HOW ABOUT THAT!

Did you know these facts about human cells?

- Hair and nails are made of dead cells, and because they are not fed by blood or nerves you can cut them without it hurting.
- A human baby grows from one cell to 2000 million cells in just nine months.
- Red blood cells live for one to four months and each cell travels around your body up to 172 000 times.
- Some of the nerve cells in the human body can be one metre long. But that's small compared with the nerve cells in a giraffe's neck. They are two to three metres long!



INQUIRY: INVESTIGATION 3.6

Animal cells — what's the difference?

KEY INQUIRY SKILL:

- processing and analysing data and information

Equipment:

light microscope

prepared animal slides:

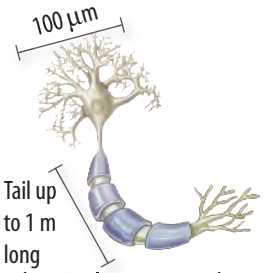
blood cells, muscle cells, cheek cells, nerve cells

- Construct a table like the one below, making it large enough for all of your results.
- Use a microscope to observe the prepared slides, recording your observations in the table as you make them.
- Prepare a summary table that describes the similarities and differences observed between the different cells examined.

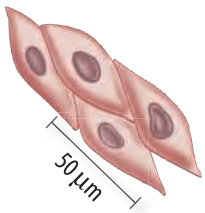
DISCUSS AND EXPLAIN

1. Which features did the animal cells have in common?
2. In what ways did the animal cells differ from each other?
3. Why are there some features that all cells possess?
4. Find out the functions of the different types of cells examined.
5. Suggest how the shape or size of the cells may assist the cell in doing its job.
6. Suggest reasons for some of the differences observed between the cells.

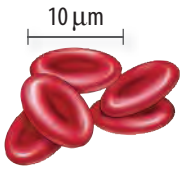
Source of specimen	Type of specimen	Sketch of specimen	Description of specimen
Animal	Cheek cells	[Allow as much space as you can; draw only 2 or 3 cells, in pencil, and include magnification and estimated size.]	[Describe in words what the specimen looked like.]



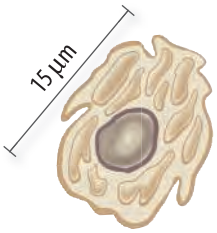
Nerve cells
Nerve cells are very long and have a star shape at one end. The long shape of nerve cells helps them detect and send electrical messages through the body at the speed of a Formula 1 racing car. There are nerve cells all over your body. They allow you to detect touch, smell, taste, sound, light and pain.



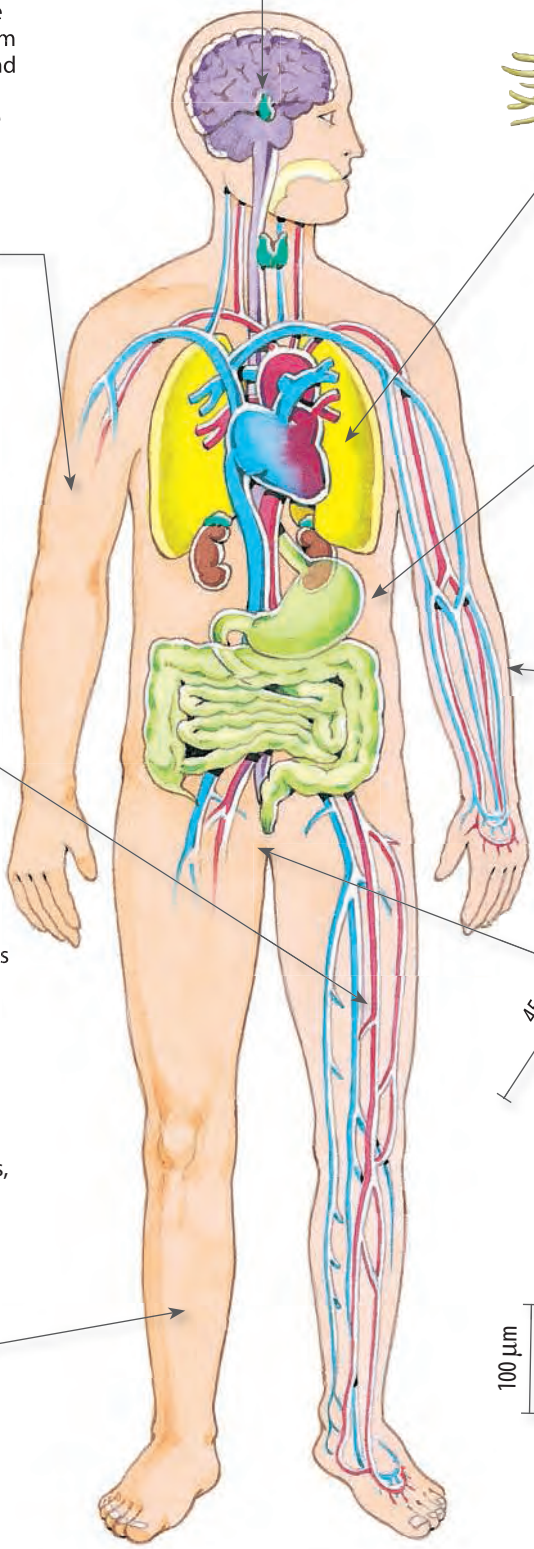
Muscle cells
Muscle cells are long and elastic. Long thin cells can slide further over each other to allow you to move. There are different types of muscle cells. The walls of your blood vessels and parts of your digestive system have 'smooth muscle' cells. The muscles that are joined to your bones are called 'skeletal muscles'. Skeletal muscles work in pairs — one muscle contracts (shortens) and pulls the bone in one direction while the other muscle relaxes.

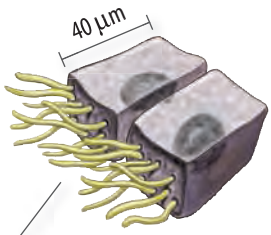


Red blood cells
Red blood cells carry oxygen around the body. Their small size allows them to move easily through blood vessels. The nucleus in a red blood cell dies soon after the cell is made. Without a nucleus, red blood cells live for only a few weeks. The body keeps making new blood cells to replace those that have died. Red blood cells are made in bone marrow at the rate of 17 million cells per minute! This is why most people can donate some of their blood to the Red Cross without harm. White blood cells, which are larger than red blood cells, are also made in the bone marrow. Their job is to rid the body of disease-causing organisms and foreign material.



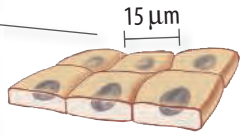
Bone cells
Minerals such as calcium surround your bone cells. The minerals help make bone cells hard and strong. Bone cells need to be hard so that they can keep you upright.





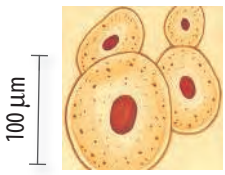
Lung epithelial cells
The cells that line your nose, windpipe and lungs are a type of lining cell. They have hair-like tips called cilia. These cells help protect you by stopping dust and fluid from getting down your windpipe. The cilia can also move these substances away from your lungs. You remove some of these unwanted substances whenever you sneeze, cough or blow your nose.

Adipose tissue cells
Some cells store fat. Fat stores a lot of energy for cells to use later. Round shapes are good for holding a lot of material in a small space. Fat cells are mostly found underneath your skin, especially in the chest, waist and buttocks.



Skin cells
Special cells line the outside surfaces of your body. These are the cells that form your skin. These cells have a flattened shape so they can better cover and protect your body.

Sperm cells
Sperm cells have long tails that help them swim towards egg cells. Only males have sperm cells.



Egg cells
Egg cells are some of the largest cells in a human body. Their large round shape helps them store plenty of food. Only females have egg cells. When a sperm cell moves into an egg cell, the egg cell is fertilised.

UNDERSTANDING AND INQUIRING

REMEMBER

1 Match the types of cells with their functions.

Type of cell	Function
Muscle	Outside covering for protection
Skin	contracts and relaxes
Red blood	Carries messages from cell to cell
Nerve	Carries oxygen from lungs to other parts

2 Which features do most cells have in common?

3 Describe some ways in which cells may differ.

4 Suggest why the cells in a multicellular organism are not all the same. Give examples in your answer.

THINK AND REASON

5 (a) Use the illustrations on the opposite page to find the size of the following different types of animal cells.
- (i) Adipose tissue cell
- (ii) Epithelium cell
- (iii) Cheek cell
- (iv) Nerve cell
- (v) Human blood cell
- (vi) Smooth muscle cell
- (vii) Sperm cell

(b) Summarise your data in a table.
(c) Determine the average size of an animal cell.
(d) Use a bar graph to plot the sizes of the different types of cells.
(e) Comment on the differences between the cells.

THINK

6 Copy and complete the table below.

Cell parts and functions in animal cells

Part	Function
Cytoplasm	
Nucleus	
Cell membrane	

INVESTIGATE, IMAGINE AND CREATE

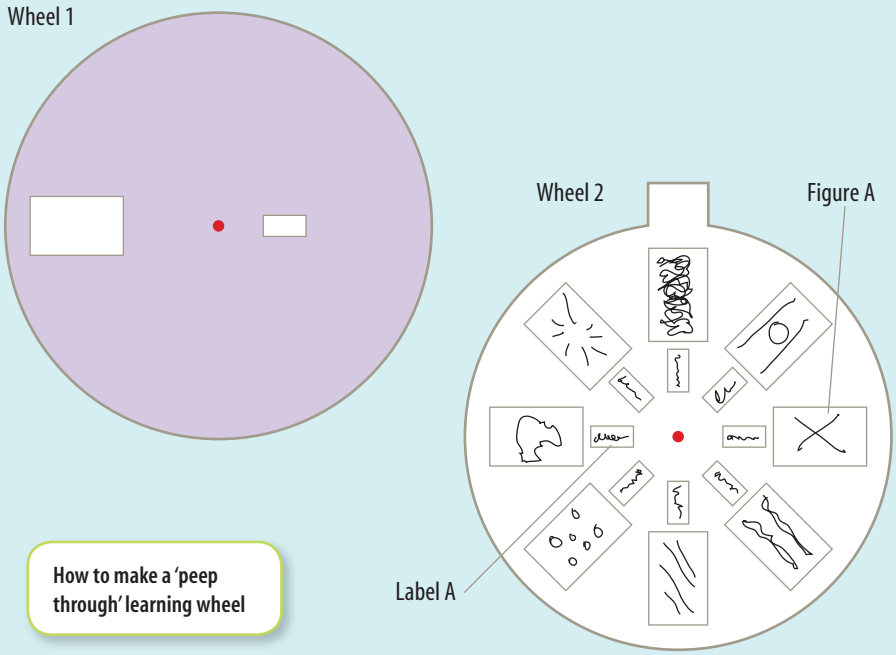
7 Find out more about a particular type of cell and use this information to write a play, poem or story about a day in the life of this type of cell.

8 Construct a model of a nerve cell using food as your construction material.

eBookplus

9 Match each cell with its purpose in the body by completing the **Cell jobs** interactivity in your eBookPLUS. **int-0206**

10 Using your own research and the information on page 76, construct a 'peep through' learning wheel that shows the structure and function of the different types of animal cells. Instructions for making a 'peep through' learning wheel are given below.



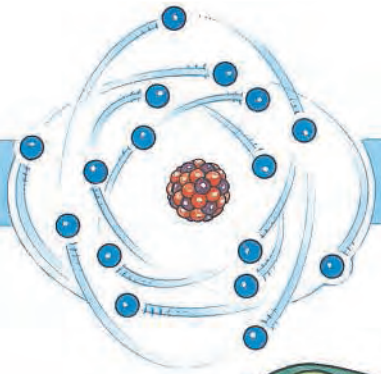
How to make a 'peep through' learning wheel

a. On an A4 piece of white paper or card draw two circles, one with a 'tab' (wheel 2) and one without (wheel 1).
b. Cut out the two rectangular box areas as shown on wheel 1.
c. Draw in the large and the small rectangles as shown on wheel 2.
d. Write the animal cell types in the small boxes on wheel 2. Sketch matching diagrams of examples of these cell types in the corresponding large box opposite.
e. Attach the two wheels, with wheel 1 on top, using a paper fastener.
f. Rotate your wheel to view examples of types of animal cells.

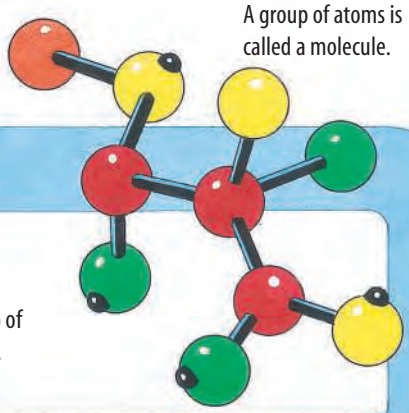
Animal cells — made to order?

Like all matter in the universe, you are made up of atoms. Collections of atoms make up **molecules**, molecules make up organelles, which make up cells, which make up tissues, which make up organs, which make up systems, which make up you.

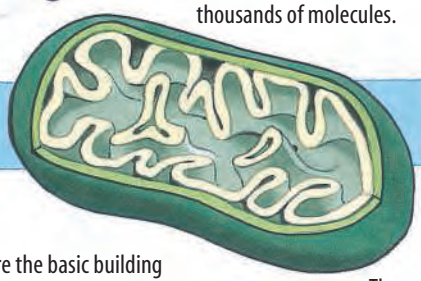
All matter in the universe is made up of atoms.



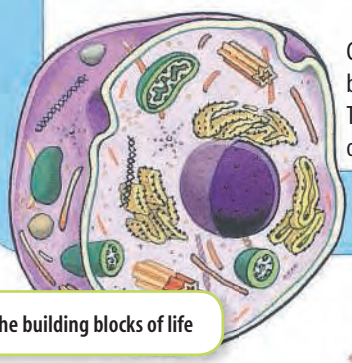
A group of atoms is called a molecule.



An organelle is made up of thousands of molecules.



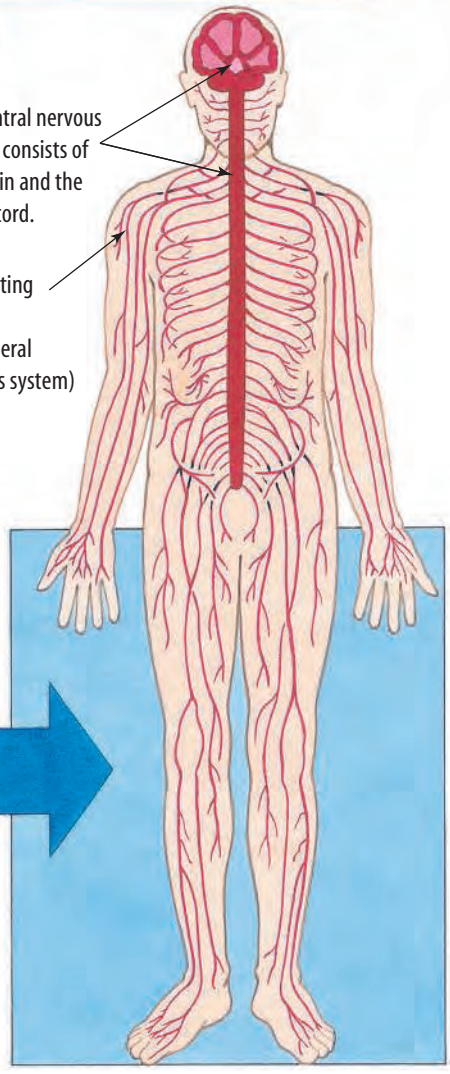
Cells are the basic building blocks of all living things. They contain different types of organelles.

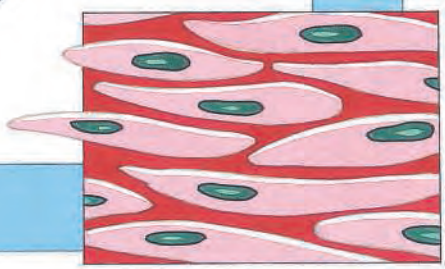


The building blocks of life

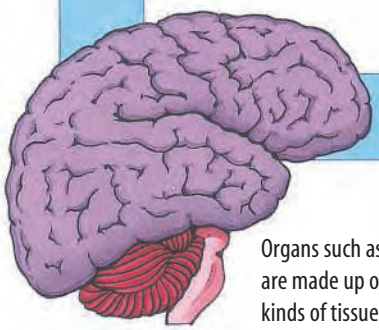
The central nervous system consists of the brain and the spinal cord.

Connecting nerves (peripheral nervous system)





Groups of cells that do a specialised job are called tissues. The smooth muscle in your body is a tissue.



Organs such as the human brain are made up of different kinds of tissue.

Several organs working together make up a system, such as the central nervous system and peripheral nervous system.

Organelles

Thousands or millions of molecules make up organelles. Each organelle has a particular job to do. Mitochondria, for example, are organelles in which the chemical energy in glucose is transformed into energy that our cells can use.

Cells

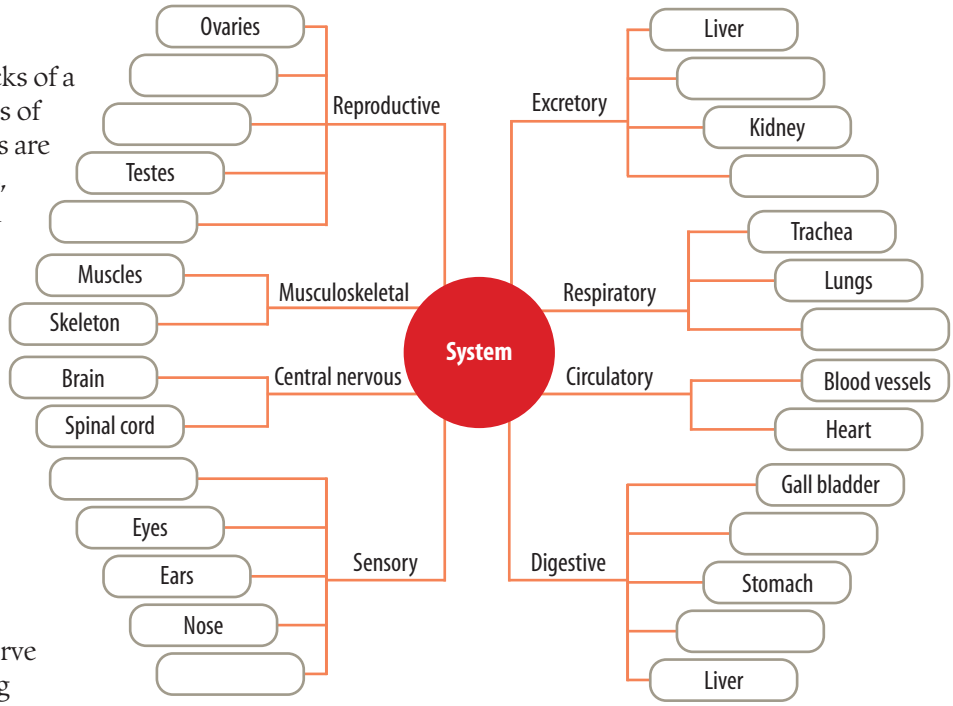
Just as bricks are the basic building blocks of a house, cells are the basic building blocks of all living things. Multicellular organisms are made up of many different types of cells, each with different jobs to do. Although these cells may have similar basic structures, they may differ in size, in shape and in the number and types of organelles that they contain.

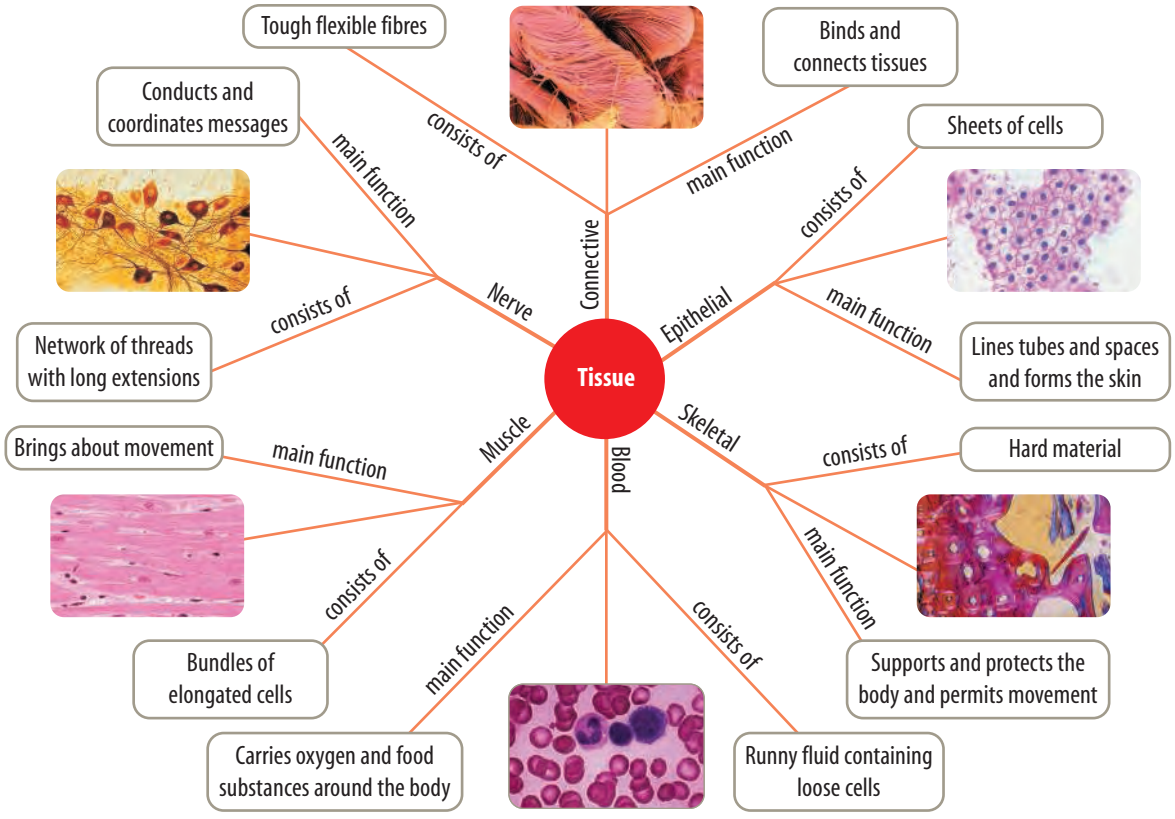
Tissues

Groups of similar cells that carry out a specialised job are called **tissues**. Muscle tissue contains cells with many mitochondria so that the energy requirements of the tissue can be met. Nerve tissue consists of a network of nerve cells with extensions to assist in carrying messages throughout your body.

Organs

Organs are made up of one or more different kinds of tissue and carry out one (or sometimes more) main function or job. Our brain, ears, eyes and skin are examples of organs. Several organs working together make up a **system**.



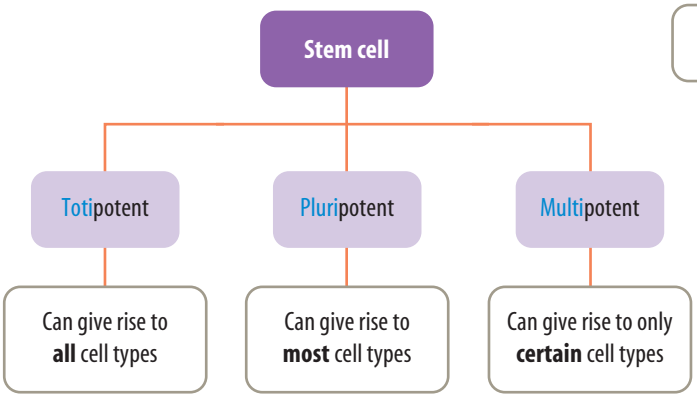


Stem cells

Have you heard or read about the issues regarding stem cells? What's the stem of the trouble? What are stem cells and why are people arguing about them?

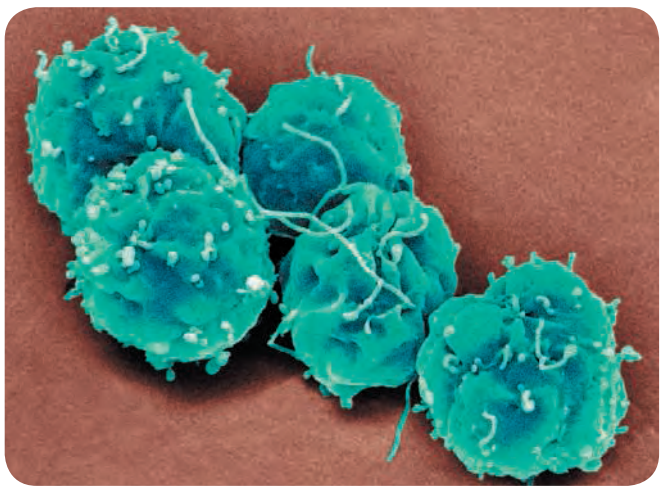
WHAT ARE STEM CELLS?

Stem cells are unspecialised cells that can reproduce themselves indefinitely. They have the ability to differentiate into many different and specialised cell types. Stem cells in a fertilised egg or zygote are **totipotent** — they have the ability to differentiate into *any* type of cell. The source of the stem cell determines the number of different types of cells that it can differentiate into.



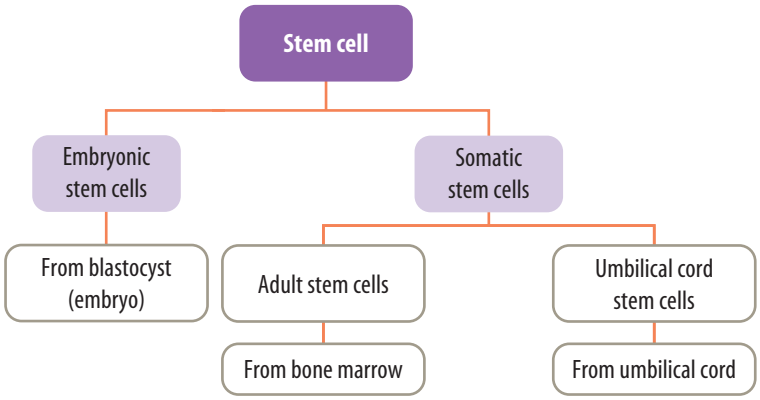
Stem cells can be divided into categories on the basis of their ability to produce different cell types.

The ability to differentiate into specific cell types makes stem cells invaluable in the treatment and possible cure of a variety of diseases. For example, they may be used to replace faulty, diseased or dead cells. The versatility of stem cells is what makes them very important.

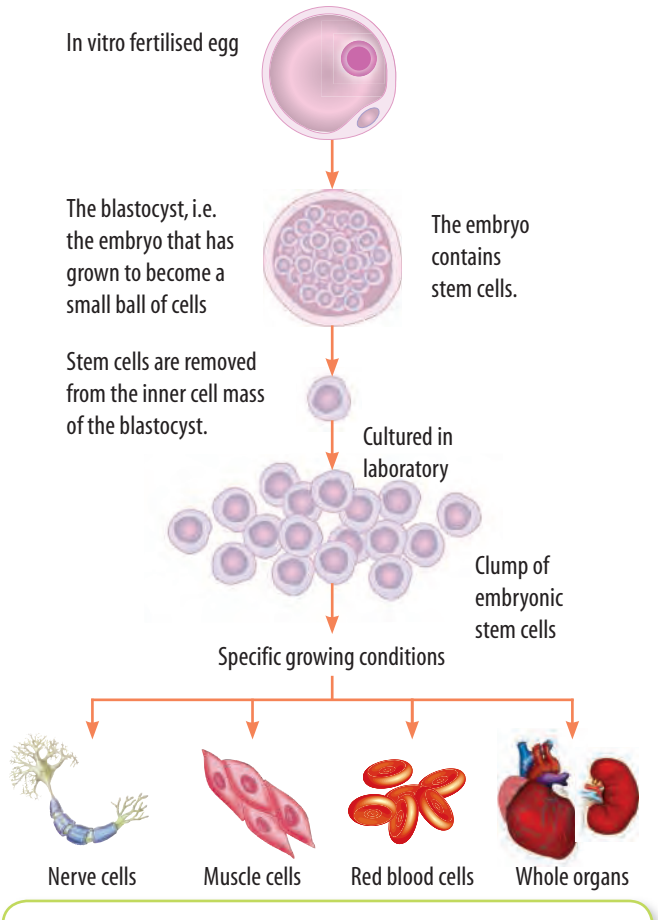


WHAT ARE THE SOURCES OF STEM CELLS?

Embryonic stem cells can be obtained from the inner cell mass of a blastocyst. The blastocyst is the term used to describe the mass of cells formed at an early stage (5–7 days) of an embryo's development. Embryonic stem cells are **pluripotent** and can give rise to most cell types; for example, blood cells, skin cells, nerve cells and liver cells.



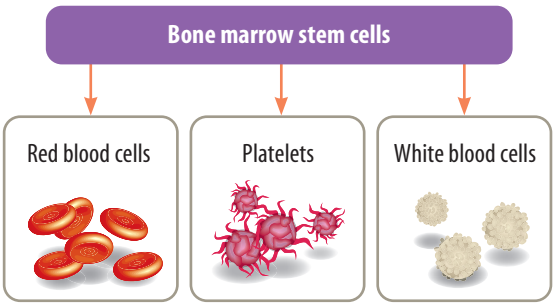
Stem cells can be described in terms of their source.

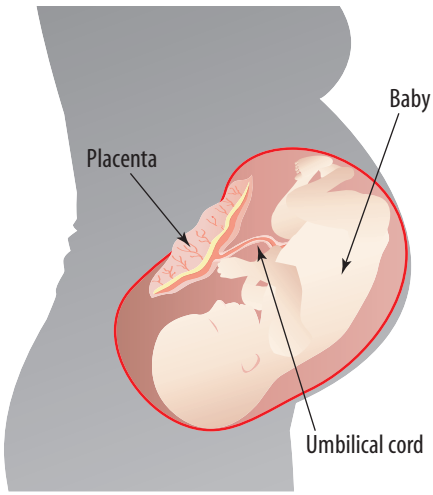


Embryonic stem cells are removed from the blastocyst and cultured to produce different cell types.

Somatic stem cells can be obtained from bone marrow and umbilical cord blood. Stem cells obtained from the bone marrow are often referred to as **adult stem cells**. These cells are **multipotent** and can develop into many kinds of blood cells.

The umbilical cord is the cord that connects the unborn baby to the placenta. This is how the baby gets nutrients and oxygen while it is still inside the mother's body. This cord contains stem cells that can develop into only a few types of cells, such as blood cells and cells useful in fighting disease. **Umbilical cord stem cells** can be taken from this cord when the baby is born.





STEM CELLS — MADE TO ORDER?

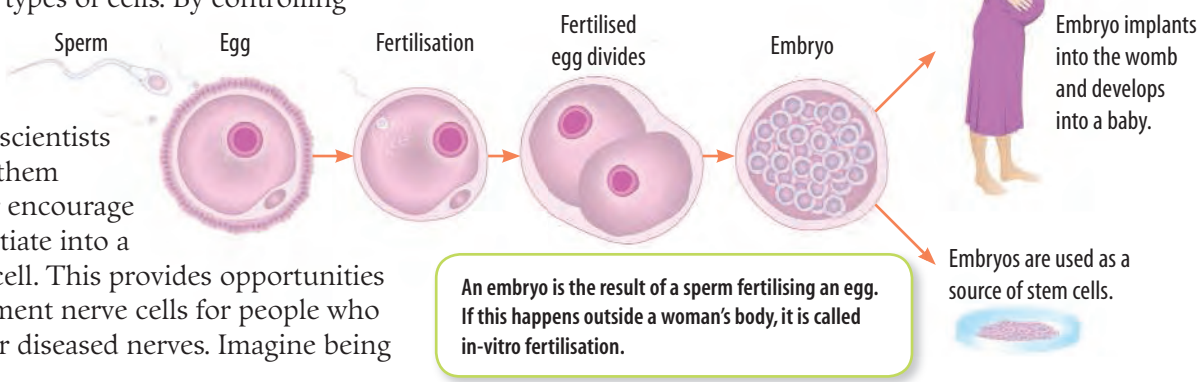
While the information in your genetic instructions tells your cells which types of cells they should become, scientists have also been able to modify the 'future' of some types of cells. By controlling the conditions in which embryonic stem cells are grown, scientists can either keep them unspecialised or encourage them to differentiate into a specific type of cell. This provides opportunities to grow replacement nerve cells for people who have damaged or diseased nerves. Imagine being able to cure paralysis or spinal cord injury. In the future, stem cells may also be used to treat and cure Alzheimer's disease, motor neurone disease, Parkinson's disease, diabetes and arthritis.

HOW ABOUT THAT!

Professor Alan Trounson is an Australian scientist who has spent a great part of his working life perfecting the technique for creating embryos outside the human body. He was part of the team that produced the first test-tube baby in Australia in 1980. He has also done a lot of work on embryonic stem cells. In 2000, his team showed that it was possible to produce nerve cells from embryonic stem cells.

He was recently appointed as the president of a Californian institute that specialises in stem cell research. It is the best-funded facility of its kind in the world, so Trounson will have the best facilities at his disposal to move stem cell research forwards.

Alan Trounson, an Australian scientist who is one of the world's top stem cell research scientists



An embryo is the result of a sperm fertilising an egg. If this happens outside a woman's body, it is called in-vitro fertilisation.

SO WHAT'S THE PROBLEM?

It is the source of embryonic stem cells that raises so many issues. Embryonic stem cells can be taken from spare human embryos that are left over from fertility treatments or from embryos that have been cloned in the laboratory. Some argue that this artificial creation of an embryo solely for the purpose of obtaining stem cells is unethical. There has also been concern about the fate of the embryo. In the process of obtaining stem cells, the embryo is destroyed.

Some parents have decided to have another child for the sole purpose of being able to provide stem cells for a child who is ill or has a disease. In this case, the blood from the umbilical cord or placenta is used as the source. Some suggest that this is not the 'right' reason to have a child and that they should not be considered to be a 'factory' for spare parts for their siblings.

Open minds are crucial in the debate on stem cell research

Politicians who decide on issues involving complex science need to be thoroughly briefed . . .

Sydney Morning Herald, 24 March 2006

We must move ahead on stem cells

To maintain its world-leading reputation, Australia needs to change the law to allow the therapeutic cloning of cells.

The Age, 30 September 2005

Discovery offers hope on cancer

Melbourne researchers have taken a giant step towards understanding the genesis of breast cancer by inducing female mice to grow new mammary glands from stem cells.

The Age, 5 January 2006

RADICAL BID TO GROW NEW BONE

STEM CELL THERAPY A WORLD FIRST

Michelle Pountney
Health reporter, 6 April 2006

A MELBOURNE man is the first person in the world whose own stem cells are being used to try to mend a broken leg.

The cutting-edge stem-cell technology has helped Jamie Stevens, 21, back on his feet.

A motorcycle crash nine months ago left him with a severely broken left thigh bone. Part of the femur stuck through his leg, and other parts of the bone were missing.

The bone failed to heal and Mr Stevens's leg was held together by a large titanium plate.

Royal Melbourne Hospital orthopedics director Richard de Steiger decided Mr Stevens was the ideal first patient for a revolutionary stem-cell trial at the hospital.

About seven weeks ago, Mr de Steiger harvested bone marrow from Mr Stevens' pelvis. The adult stem cells were then separated from the other cells. A sub-group of stem cells called mesenchymal precursor cells — those that can transform into tissues including bone cartilage and heart — were isolated and grown.

Last week about 30 million of these cells were implanted into the 5 cm × 3 cm hole in Mr Stevens' thigh bone. The cells are expected to regenerate new bone and grow through the calcium phosphate.

Yesterday, just four days after surgery, Mr Stevens went home.

'It's good to be part of something that is on the brink,' he said. 'I wouldn't say I understand it. It's all pretty cool.'

It will be three to four months before the result of the operation is known.

'This is radical and the first procedure in the world to use a patient's own stem cells and make them turn into bone-forming cells that are the patient's own cells, to stimulate healing of a fracture,' Mr de Steiger said.

The cells are harvested, cultured and expanded using Australian biotechnology company Mesoblast's specialist adult stem-cell technology.

Mr de Steiger hopes that eventually the technique will be refined to a simple injection, avoiding further surgery.

Using a patient's own cells avoids the potential problem of the body's rejection of foreign cells.

The only other alternative to repair Mr Stevens' leg would have been a painful bone graft, taking a chunk of bone from his hip and plugging it into the hole in his thigh.

'The conventional way used over many years involves a large incision at the pelvis and taking out quite a large amount of bone in Jamie's case,' Mr de Steiger said.

'In this situation there is the risk of a separate incision, reported continuing pain from that incision, and separate infection risk at that site.'

Mr de Steiger said orthopedic specialists at the Royal Melbourne hospital treated about 200 fractures of the long leg bones each year.

About 19 per cent become 'non-union' fractures that fail to heal; 10 of these patients will be recruited to be part of the year-long trial.

Mr Stevens said he was no more nervous about being the first patient in the world to have the procedure than he would have been having a graft.

'I think the benefit outweighs the old procedure, and being able to avoid having a big chunk of bone taken out of the hip … the recovery period of it was a lot quicker.'

After living in limbo for nine months, Mr Stevens said he was looking forward to resuming the life he enjoyed before his accident.

If there are too many arguments about using someone else's stem cells, why not grow your own to mend, replace and renew?

UNDERSTANDING AND INQUIRING

REMEMBER

1 What do organisms have in common with other matter in the universe?
2 Describe the relationship between atoms, molecules and organelles.
3 In what ways can cells be different from each other? Why do cells differ?
4 What are the differences between cells, tissues, organs and systems?
5 What are stem cells?
6 Distinguish between the terms 'totipotent', 'pluripotent' and 'multipotent'.
7 Outline the importance of stem cells.
8 List sources of stem cells.
9 Outline issues regarding stem cell research.
10 Describe a scientific contribution made by the Australian scientist Professor Alan Trounson.

THINK AND REASON

Use the mind maps on page 79 to answer the following questions.

11 What kind of tissue:
   (a) has the function of conducting and coordinating messages around the body?
   (b) binds and connects tissues?
   (c) supports the body?
   (d) carries oxygen around the body?
12 Which body system has the function of:
   (a) detecting stimuli?
   (b) supporting and moving the body?
   (c) taking in oxygen and getting rid of carbon dioxide?
   (d) conducting messages from one part of the body to another?

INVESTIGATE, CREATE AND DESIGN

13 Find out about the different tissues and systems that exist in plants. Present your information, using diagrams and lots of colour, on a poster or as a PowerPoint presentation. Be as creative as you can.
14 Select one of the systems in the mind map on page 79 and find out more about what it does and how it works.
15 Design and construct a model of one of the following systems: respiratory, excretory, reproductive, digestive.

INVESTIGATE, SHARE AND DISCUSS

16 Investigate some of the following questions.
   (a) Which inherited genetic diseases are potentially treatable with stem cells?
   (b) How many different kinds of adult stem cells exist and in which tissues can they be found?
   (c) Why have the adult stem cells remained undifferentiated?
   (d) What are the factors that stimulate adult stem cells to move to sites of injury or damage?

eBookplus

17 View an animation about stem cells by going to the website for this book and clicking on the **Genetic Science Learning Center** weblink in your eBookPLUS. Use these ideas to construct your own story, cartoon, PowerPoint presentation or animation on stem cells.
18 Click on the **Stem Cell** weblink in your eBookPLUS to investigate the views of the major world religions on stem cell research.
19 In your team, discuss the following questions to suggest a variety of perspectives.
   (a) Is it morally acceptable to produce and/or use living human embryos to obtain stem cells?
   (b) Each stem cell line comes from a single embryo. A single cell line allows hundreds of researchers to work on stem cells. Suggest and discuss the advantages and disadvantages of this.
   (c) If the use of human multipotent stem cells provides the ability to heal humans without having to kill another, how can this technology be bad?
   (d) Parents of a child with a genetic disease plan a sibling whose cells can be used to help the diseased child. Is it wrong for them to have another child for this reason?
20 Find out how stem cell research is regulated in Australia and in one other country. What are the similarities and differences of the regulations? Discuss the implications of this with your team-mates.
21 Investigate aspects of stem cell research and put together an argument for or against the research and its applications. Find a class member with the opposing view and present your key points to each other. Ask questions to probe any statements that you do not understand or would like to clarify. Construct a PMI to summarise your discussion.
22 Investigate and report on research at the Australian Stem Cell Centre.
23 (a) Use a bubble map or mind map to summarise the key points in the article *Stem cell therapy a world first* on page 82.
   (b) As a team, discuss the article and construct a PMI.
   (c) Formulate questions that may help you to develop an informed opinion.
   (d) Research your questions.
   (e) State your opinion on the use of stem cell therapies like the one in the article. Give reasons for your opinion.
   (f) Do you have the same opinion about other types of stem cell therapies? Explain.

Focus on plant cells

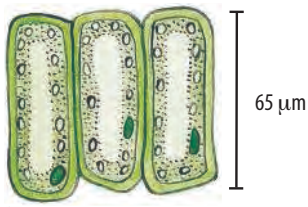
If all living things are made up of cells, why aren't they all the same?

Have or have not

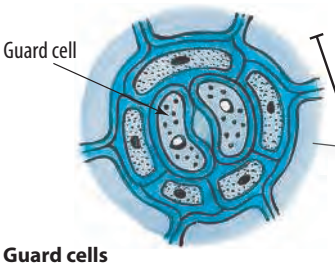
Like animal cells, plant cells have cytoplasm, a membrane and a nucleus. Unlike animal cells, plant cells have a cellulose cell wall and a large central vacuole filled with cell sap. Often plant cells also contain chloroplasts, which enable them to make their own food.

On the surfaces of leaves, there are pairs of special cells called guard cells, which surround tiny pores called stomata. The guard cells can change shape, opening or closing the stomata. Special cells on the roots extend into microscopic hairs that penetrate between soil particles. The hairs provide a big surface area through which water may be absorbed from the soil.

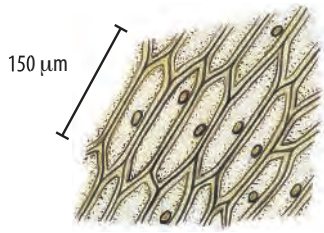
Leaf cells (palisade cells)
The main function of leaf palisade cells is to photosynthesise, so they are packed with chloroplasts and are usually green.



Leaf cell



Guard cells
Guard cells are kidney-shaped cells found on the surface of leaves. They can change shape to either open or close the small hole between them. The small holes, called stomata (or stomates), allow substances such as carbon dioxide to enter the leaf. They also let water out of the leaf. Most plants open their stomata at night; they close their stomata during the day (when it is hotter) to conserve water.

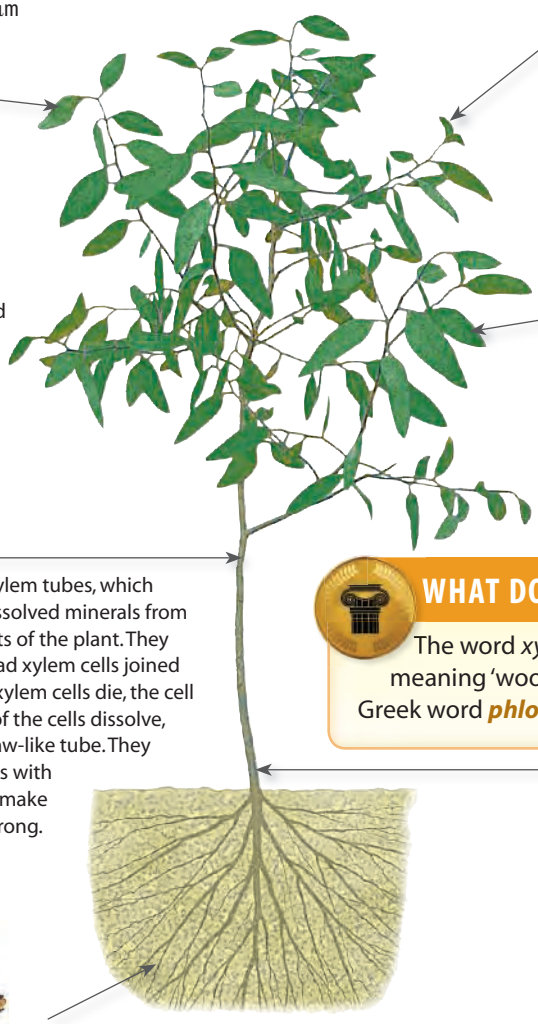


Epidermal cells

Epidermal cells
Epidermal cells are found on the outside of the plant. They form an outer skin for the plant and protect the cells underneath. This explains why they need a flat shape and why they interlock like tiles. Epidermal cells do not usually photosynthesise so they lack chloroplasts. Light needs to pass through them, and they are usually transparent. The cells in the diagram above are onion epidermal cells.

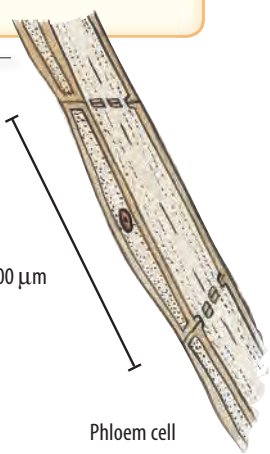
Xylem cells

Xylem cells
Xylem cells form xylem tubes, which carry water and dissolved minerals from the roots to all parts of the plant. They are made up of dead xylem cells joined end to end. When xylem cells die, the cell walls at each end of the cells dissolve, forming a long straw-like tube. They have thick cell walls with lots of cellulose to make the xylem tubes strong.



WHAT DOES IT MEAN?

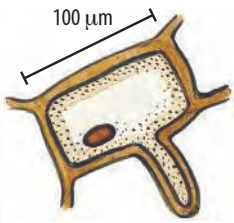
The word *xylem* comes from the Greek word ***xulan***, meaning 'wood'. The word *phloem* comes from the Greek word ***phloos***, meaning 'bark'.



Phloem cell

Phloem cells
Like xylem cells, phloem cells form tubes. The tubes formed by phloem cells carry the food made in the leaves to all parts of the plant. Phloem cells do not need to die to do this job. The ends of phloem cells have holes and look like sieves.

Some of the types of cells found in plants



Root hair cell

Root hair cells
Root hair cells absorb water and dissolved minerals from the soil. They have small hairs, called root hairs, on their surface. This increases the surface area of the root cells so that they can soak up water more quickly.

INQUIRY: INVESTIGATION 3.7

Plant cells in view

KEY INQUIRY SKILL:

- processing and analysing data and information

Equipment:

light microscope

prepared plant slides: leaf epidermal cells, root hair cells, stomata/guard cells

- Construct a table like the one below, making it large enough for all of your results.
- Use a microscope to observe the prepared slides, recording your observations in the table as you make them.
- Prepare a summary table that describes the similarities and differences observed between the different cells examined.

DISCUSS AND EXPLAIN

1 Which features did the plant cells have in common?

2 In what ways did the plant cells differ from each other?

3 Why are there some features that all cells possess?

4 Find out the functions of the different types of cells examined.

5 Suggest how the shape or size of a cell may assist it in doing its job.

6 Suggest reasons for some of the differences observed between the cells.

Source of specimen	Type of specimen	Sketch of specimen	Description of specimen
Plant	Leaf epidermal cells	[Allow as much space as you can; draw only 2–3 cells, in pencil, and include magnification and estimated size.]	[Describe in words what the specimen looked like.]

UNDERSTANDING AND INQUIRING

REMEMBER

1 Match the types of cells with their functions.

Type of cell	Function
Root hair	Changes shape to open and close pores in the leaf
Xylem	Increases surface area for efficient absorption
Guard	Carries water and minerals up the plant

2 Which features do most cells have in common?

3 Describe some ways in which cells may differ.

4 Suggest why the cells in a multicellular organism are not all the same. Give examples in your answer.

THINK AND REASON

5 (a) Use the illustrations on page 84 to find the sizes of these different types of plant cells.
  (i) Leaf cell
  (ii) Onion epidermal cell
  (iii) Phloem cell
  (b) Summarise your data in a table.
  (c) Determine the average size of a plant cell.
  (d) Use a bar graph to plot the cell sizes of the different types of cells.
  (e) Comment on the differences between the cells.

THINK

6 Copy and complete the table below.

Cell structures and functions in plant cells

Structure	Function
Cell wall	
Cytoplasm	
Chloroplast	
Nucleus	
Large vacuole	
Cell membrane	

INVESTIGATE, IMAGINE AND CREATE

7 Find out more about guard cells or leaf epidermal cells. Write a story about what happens to them over 24 hours.

8 Construct a model of a pair of guard cells, using balloons.

9 Using your own research and the information on page 84, construct a 'peep through' learning wheel that shows the structure and function of the different types of plant cells. Instructions for making a 'peep through' learning wheel are given on page 77.

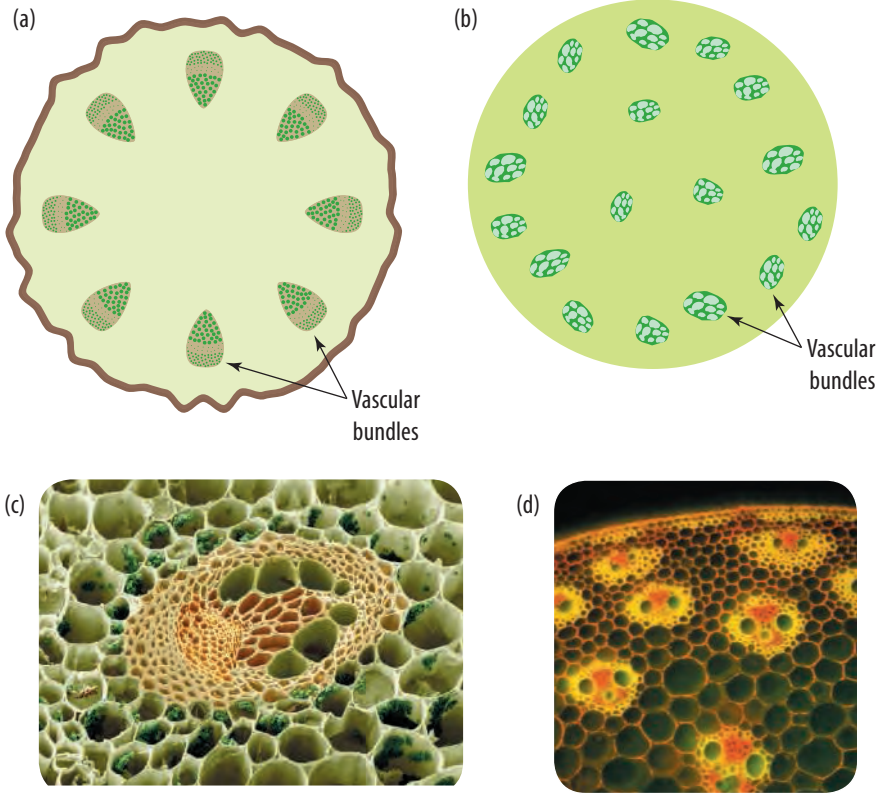
work sheet 3.4 Plant transport highways

Plant cells — holding, carrying and guarding

As in animals, plant cells can work together for a variety of functions to meet their survival needs. Plants have their own transport systems, which consist of many thin tubes made up of different types of cells. Other types of plant cells are involved in water regulation and exchange of important gases, like oxygen and carbon dioxide with their environment.

Sweet transport: phloem

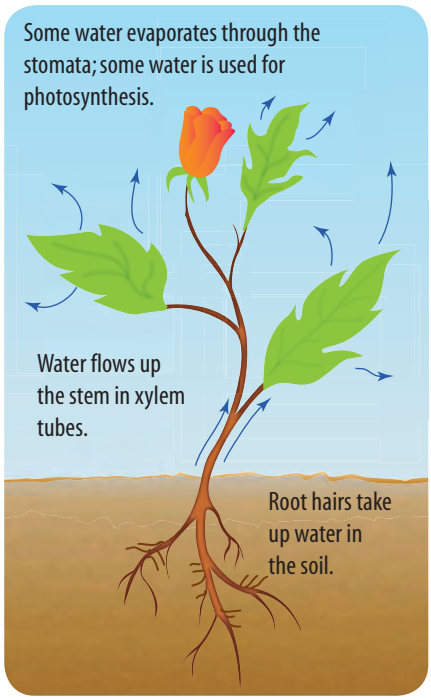
Using the process of photosynthesis, plants make sugar in their leaves. The system of thin-walled tubes that carries this sugar (in the form of glucose or sucrose) from the leaves to other parts of the plant is called **phloem**. Phloem consists of living cells called sieve tubes and companion cells. The transport of the sugar solution up and down the plant is called **translocation**.



Diagrams of typical cross-sections of the stem of (a) a young dicot and (b) a monocot. The photographs show how the cells of (c) a dicot (buttercup) appear when viewed under an electron microscope and (d) a monocot (corn) appear under a light microscope.

Water pipes: xylem

Flowering plants also have tubes with strong, thick walls that carry water and minerals up from the roots through the stem to the leaves. These are called **xylem vessels**. These tubes are formed from the empty remains of dead cells, the walls of which are strengthened with a woody substance called **lignin**. The



The movement of water from roots to leaves is known as the transpiration stream.

INQUIRY: INVESTIGATION 3.8

Stem transport systems

KEY INQUIRY SKILLS:

- processing and analysing data and information
- planning and conducting

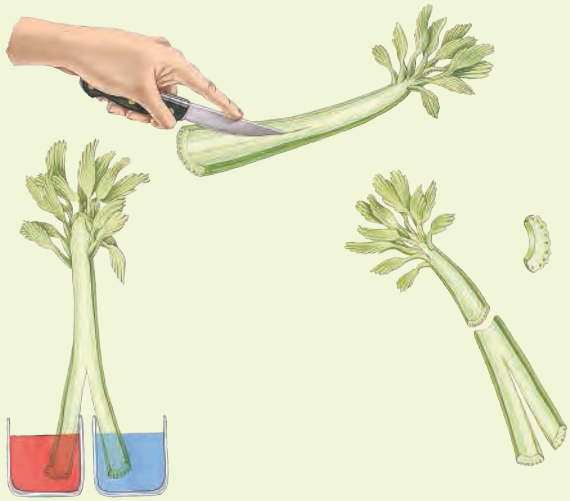
Equipment:

celery stick (stem and leaves)
knife
two 250 mL beakers
water
blue food colouring
red food colouring
hand lens

- Slice the celery along the middle to about halfway up the stem.
- Fill two beakers with 250 mL of water. Colour one blue and the other red with the food colouring.
- Place the celery so that each 'side' of the celery is in a separate beaker.
- Leave for 24 hours and then observe the celery.
- Cut the celery stick across the stem.
- Use the hand lens to look at the inside of the stem.

DISCUSS AND EXPLAIN

1 Look at where the water has travelled in the celery. Draw a diagram to show your observations.
2 Draw a diagram to show what you can see when you cut across the stem.
3 Where is the differently coloured water found in the stem?
4 Where are the different colours found in the leaves?
5 Draw a diagram of the whole celery stick and trace the path of the water through each side to the leaves.
6 How could you turn a white carnation blue? Try it.



xylem is therefore a 'dead' one-way street, rather than a 'living' two-way highway like the other transport tubes you have studied.

Water moves up from the roots of the plant, through its stem and to its leaves, where some may pass out of the plant as water vapour through pores called **stomata**. This movement of water is called the **transpiration stream**.

XYLEM FOR SUPPORT

The phloem and the xylem vessels are located together in groups called **vascular bundles**. The strong, thick walls of the xylem vessels are also important in helping to hold up and support the plant. The trunks of trees are made mostly of xylem. Did you know that the stringiness of celery is due to its xylem tissues?

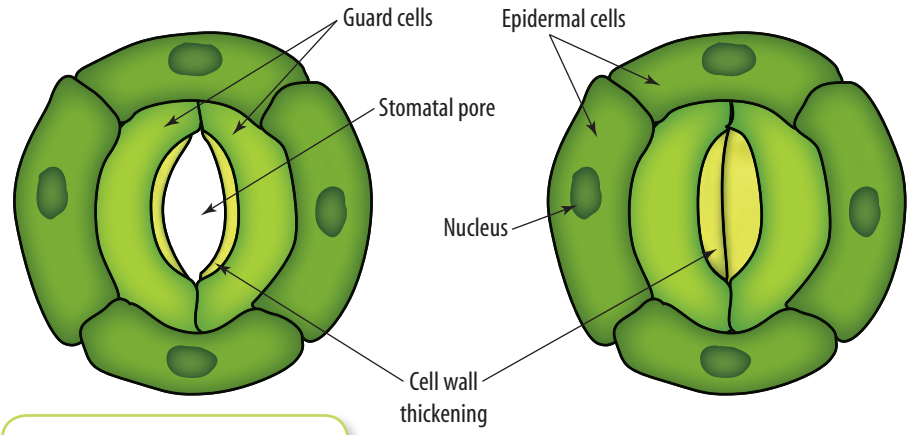
Leaf doorways: stomata

Water transport occurs within the xylem vessels. Some of the water that is transported through the xylem to the leaves is used in photosynthesis. Some water is also lost as water vapour through tiny holes or pores in the leaves. These tiny pores, called stomata (or stomates), are most frequently found on the underside of the leaves. Evaporation of water from the stomata in the leaves helps pull the water up the plant. Loss of water vapour through the stomata is called **transpiration**.

'Stomata art': the arrangement of stomata in a plant

GUARD CELLS IN CONTROL

Oxygen and carbon dioxide gases also move in and out of the plant through the stomata. **Guard cells**, which surround each stoma, enable the hole to open and close, depending on the plant's needs. When the plant has plenty of water, the guard cells fill up with water and stretch lengthways. This opens the pore. If water is in short supply, however, the guard cells lose water and they collapse towards each other. The pore is then closed. This is one way in which the plant can control its water loss.



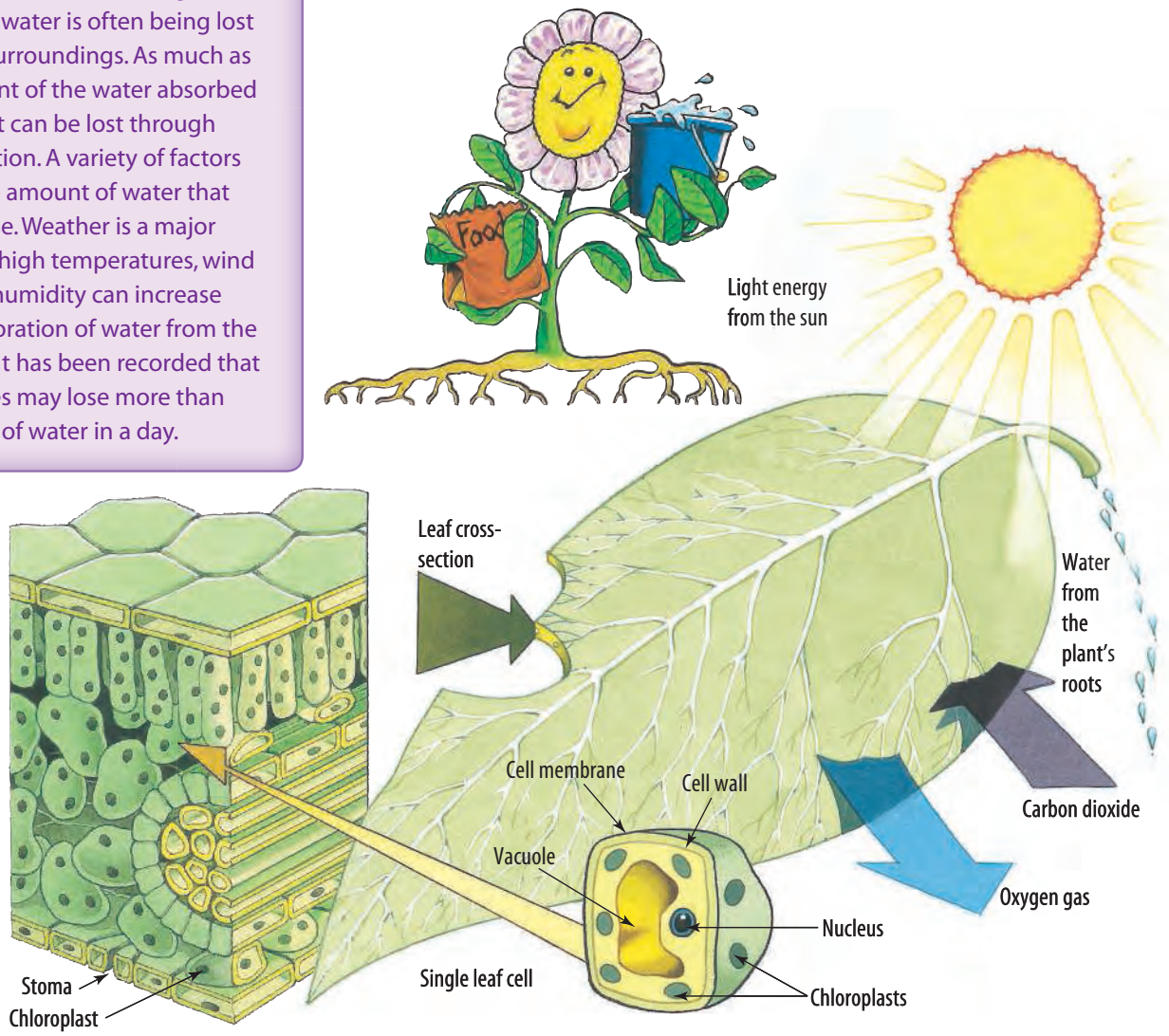
Stomata can close to conserve water.

HOW ABOUT THAT!

Although water makes up about 90–95 per cent of the living tissues of plants, water is often being lost to their surroundings. As much as 98 per cent of the water absorbed by a plant can be lost through transpiration. A variety of factors affect the amount of water that plants lose. Weather is a major factor, as high temperatures, wind and low humidity can increase the evaporation of water from the stomata. It has been recorded that large trees may lose more than 400 litres of water in a day.

DUSTY DOORS

Air pollution can result in particles of dust settling on the leaves of plants. This may limit the amount of light reaching the leaf and so reduce photosynthesis. If these dust particles block up stomata, they can also affect transpiration and gaseous exchange.



INQUIRY: INVESTIGATION 3.9

Observing leaf epidermal cells

KEY INQUIRY SKILLS:

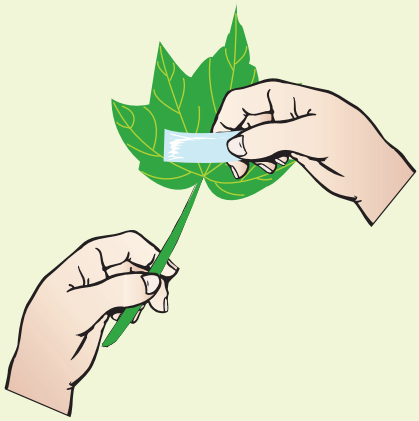
- processing and analysing data and information
- planning and conducting

Equipment:

leaf *clear sticky tape*
microscope slide *microscope*

You can make a 'slide' of leaf epidermal cells with sticky tape.

- Put some sticky tape over a section of the underside of a leaf.
- Press the sticky tape firmly onto the leaf.
- Tear the tape off. Some of the lining cells should come off with the sticky tape.
- Press the tape, sticky side down, onto a microscope slide.
- View the sticky tape under the microscope.
- Try to find a pair of guard cells and one of the stomata.



DISCUSS AND EXPLAIN

1 Is the stoma (the opening) open or closed?
2 Make a drawing of a group of cells, including the guard cells. Include as much detail in your drawing as possible.
3 Label the guard cells and stomata.
4 Title and date your drawing. Write down the magnification used.

HOW ABOUT THAT!

Scientists have used genetic engineering technology to produce plants that glow particular colours when they have mineral deficiencies. This provides farmers with information about which soils need extra minerals added.

INQUIRY: INVESTIGATION 3.10

Looking at chloroplasts under a microscope

KEY INQUIRY SKILLS:

- processing and analysing data and information
- planning and conducting

Equipment:

tweezers
moss, spirogyra or elodea
water
light microscope, slides, coverslips
dilute iodine solution

- Using tweezers, carefully remove a leaf from a moss or elodea plant or take a small piece of spirogyra.
- Place the plant material in a drop of water on a microscope slide and cover it with a coverslip.
- Use a light microscope to observe the leaf.
- Put a drop of dilute iodine solution under the coverslip. (Iodine stains starch a blue-black colour.)
- Using the microscope, examine the leaf again.

DISCUSS AND EXPLAIN

1 Draw what you see before staining.
2 Label any chloroplasts that are present.
3 Describe the colour of the chloroplasts before staining.
4 What gives chloroplasts their colour?
5 Did the iodine stain any part of the leaf a dark colour?
6 If so, what does this suggest?
7 What conclusions can you make about chloroplasts?

Moving in or out?

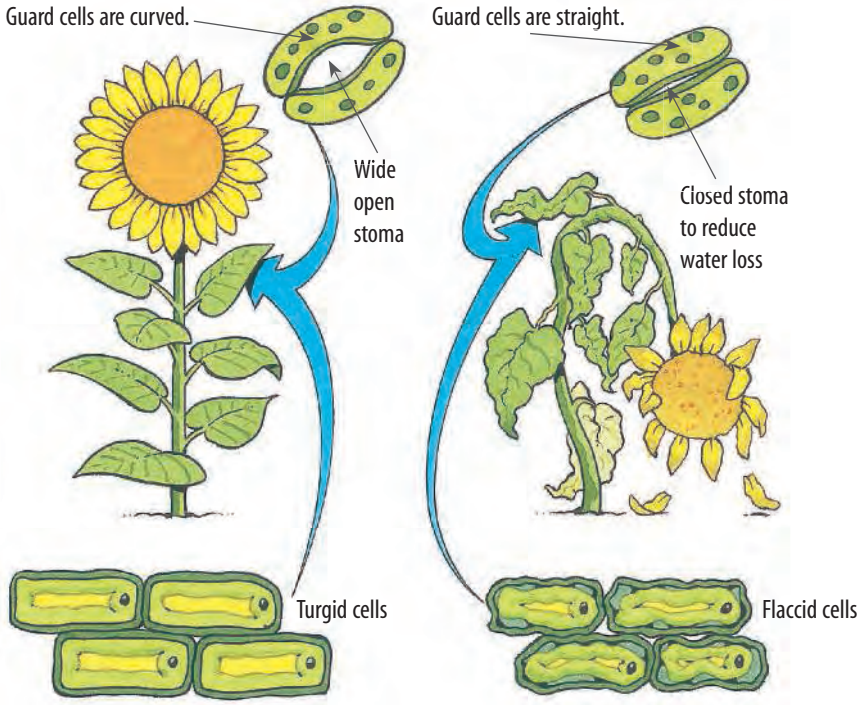
The movement of substances into and out of cells is controlled by the cell membrane. This enables useful substances to be delivered into the cells and waste products to be moved out. Some types of movements require energy and others do not.

Oxygen and carbon dioxide enter the cell by a process called diffusion. Diffusion moves substances from where they are in a high concentration to where they are in a low concentration and so does not require energy. Water moves across the membrane by a special type of diffusion called osmosis. This movement of water into and out of the guard cells is responsible for opening and closing the stomata.

If the cells of a plant do not contain enough water, they become flaccid and the plant wilts.

Flaccid or firm?

If too much water is lost or not enough water is available, the plant may **wilt**. When this occurs, water has moved out of the cell **vacuoles** and the cells have become soft or **flaccid**. The firmness in the petals and leaves is due to their cells being firm or **turgid**.



INQUIRY: INVESTIGATION 3.11

Moving in or out?

Equipment:

two 20 cm lengths of dialysis tubing
starch solution *iodine solution*
scales *2 beakers*

- Soak the dialysis tubing in water so it becomes soft.
- Tie a knot at one end of each piece of dialysis tubing. This will form two small bags.
- Pour water into bag A until it is one-third full. Pour the same amount of starch solution into bag B and add 10 drops of iodine solution.
- Tie a knot at the top of each bag to seal them.
- Weigh both bags.
- Put bag A in a beaker of starch solution. Add enough iodine to the starch solution to produce a dark blue colour.
- Put bag B in a beaker of water.
- Leave the two bags undisturbed for at least two hours (or overnight).
- Weigh the bags again.

DISCUSS AND EXPLAIN

1. Draw up a table to record the weights of the bags before and after being left in the beakers.
2. What happens to iodine when it is added to starch solution?
3. Draw bags A and B in the beakers they were left in. On your diagram, label where blue and yellow colour can be seen.
4. In this experiment, we made a model of a cell. Which part represented the cell membrane?
5. Dialysis tubing allows some substances, but not others, to pass through. Which of the following substances could pass through the dialysis tubing and which could not? What evidence supports this?
   (a) Starch (b) Water (c) Iodine
6. Did the masses of the two bags change? What caused it to change?
7. When water moves in or out of cells by osmosis, it moves in the direction that balances the concentrations of substances inside and outside the cell. Use this information to explain why the masses of the bags changed.

UNDERSTANDING AND INQUIRING

REMEMBER

1 What is the name for the tubes that carry sugar solution around the plant?

2 Describe the difference between:
   (a) phloem and xylem vessels
   (b) sugar and water transport in plants
   (c) the arrangement of vascular bundles in dicots and monocots.

3 In what ways are the vascular bundles important to plants?

4 State two things that may happen to water in a plant.

5 On which part of the plant are stomata usually found? Can you suggest why?

6 What helps 'pull' water up the plant?

7 Describe how the guard cells assist the plant in controlling water loss.

8 Describe the difference between flaccid cells and turgid cells.

9 Copy and complete the table below.

Tissue	What it carries	Direction of movement	Name of cells that form tubes	Are cells that form tubes living?
Xylem				
Phloem				

THINK AND DISCUSS

10 Carefully examine the reaction for photosynthesis as shown below.

$$\text{Carbon dioxide} + \text{water} \xrightarrow[\text{Chlorophyll}]{\text{Light energy}} \text{Glucose} + \text{oxygen} (+ \text{water})$$

   (a) Suggest why water and carbon dioxide are so important to plants.
   (b) Suggest why guard cells are important to plants.
   (c) Predict consequences for a plant if the guard cells close the stomata for long periods of time.

INVESTIGATE

11 Describe the patterns in which the vascular tissue is arranged in the stems of different plants. Obtain your information by:
   (a) examining stained cross-sections
   (b) finding and examining diagrams of the stems of different plants in cross-section.

12 Find out the relationship between 'wood' and xylem tissue.

13 How long do you think it would take for a plant to take up 50 mL of water? What conditions might speed it up? Put forward a hypothesis, and then design an experiment to test your hypothesis.

14 Design an experiment to test the time taken for different volumes of water to be taken up by the plant.

15 Some plants have special features that help them reduce water loss. Some leaves have a thick, waxy layer (cuticle). Others have a hairy surface or sunken stomata. Plants that are able to tolerate extremely dry environments are called xerophytes. Find out some ways in which plants in dry environments, such as deserts, reduce their water loss. Present your information on a poster or as a model.

16 Using sticky tape, remove a layer of cells from the underside of a leaf and then place it on a glass microscope slide. Examine the leaf cells for stomata under a light microscope. Repeat the procedure with as many different types of plants as you can. Summarise your findings in a poster.

17 Design an experiment to measure the amount of water lost through the leaves of a plant.

18 Place a plastic bag over the leaves of plants growing in the school grounds. Seal the bag and record the amount of water collected over 24 hours. What conclusions can you draw from your results?

19 Suggest ways in which plants can trap as much light as possible, presenting them in a mind map.

20 Research and report on one of the following types of science.
- Phycology (study of seaweeds)
- Plant marine biology
- Plant physiology
- Plant pathology
- Ecophysiology

CREATE

21 In a group, write and then act out a play or simulation of the way water moves through a plant.

22 Write a story about a group of water molecules that travels from the soil, through a plant and then into the atmosphere as water vapour.

23 Design experiments to determine the answer to one of the following questions.
   (a) Is carbon dioxide needed for photosynthesis?
   (b) Do plants need chlorophyll for photosynthesis?
   (c) If part of a leaf is covered, will the leaf still photosynthesise?
   (d) If a plant is covered with a plastic bag, will it still photosynthesise?
   (e) How could you determine the best light (colour or intensity) for photosynthesis?

work sheets
3.5 Leafy exchanges
3.6 Photosynthesis

Cell division

All cells arise from pre-existing cells.

Cell division in eukaryotes

Ouch! Did you burn or cut yourself? What about those skin cells you left on the towel when you dried yourself and those hairs you left behind in your brush? Have you replaced these cells? Throughout the life of multicellular organisms, cell division takes place to enable growth, development, repair and replacement of cells. Cell division also plays an important role in reproduction.

NUCLEUS, CHROMOSOMES AND DNA

All eukaryotic cells have a **nucleus**, which contains genetic information with instructions that are necessary to keep the cell (and organism) alive. This information is contained in structures called **chromosomes**, which are made up of a chemical called **deoxyribonucleic acid** (**DNA**).

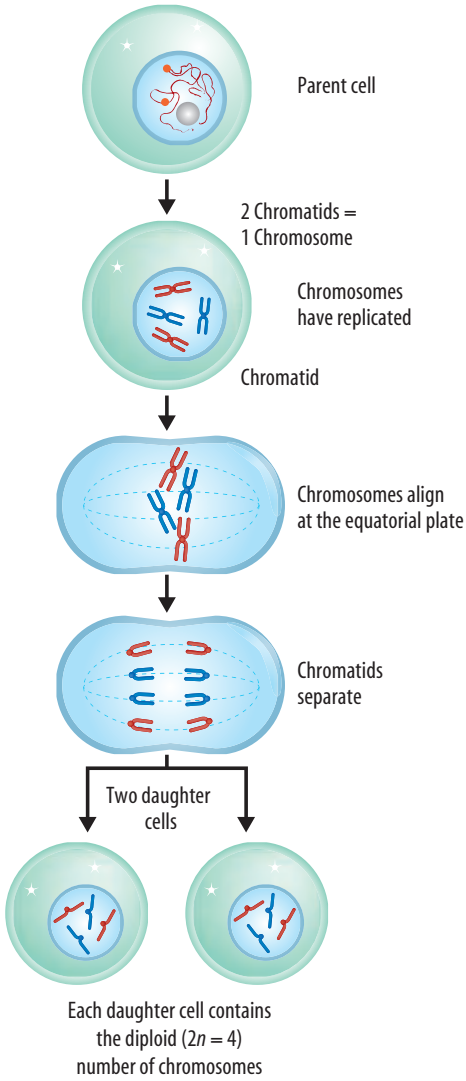
MITOSIS

Mitosis is the name of a process involved in cell division in eukaryotic cells. Some organisms use this type of cell division to asexually reproduce. Multicellular organisms also use mitosis to produce cells for growth, development, repair and replacement.

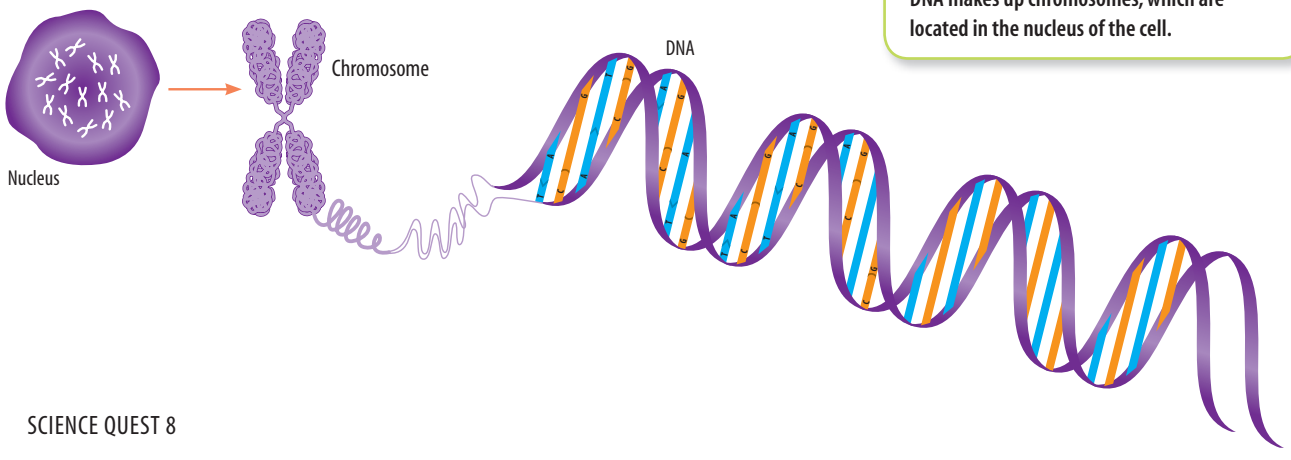
The cells produced by mitosis are genetically identical to each other and to the original cell. They have the same number and types of chromosomes and DNA instructions. As they have identical genetic information, they are described as being **clones**.

CYTOKINESIS

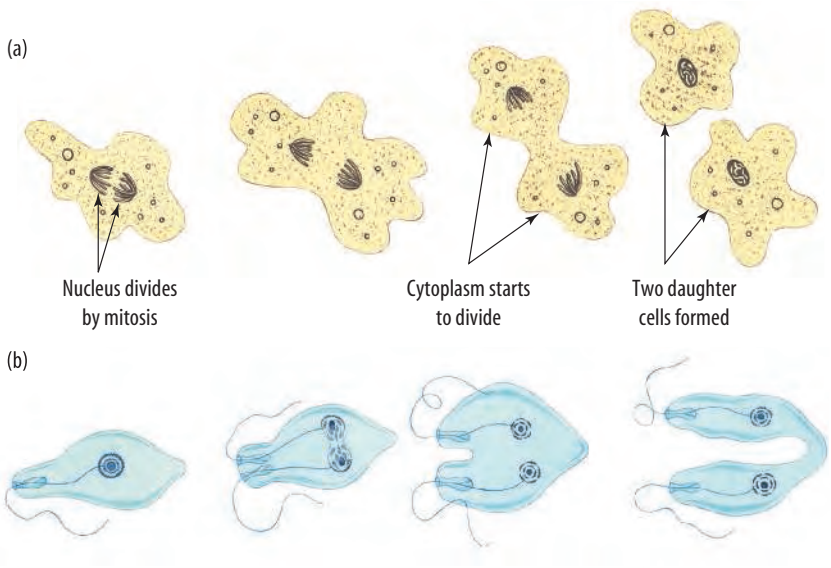
Mitosis is a process that involves division of the nucleus. Once a cell has undergone this process, the cell membrane pinches inwards so that a new membrane is formed, dividing the cell in two. This process of the division of the cytoplasm is called **cytokinesis**.



Mitosis is a type of cell division that produces identical cells.



DNA makes up chromosomes, which are located in the nucleus of the cell.



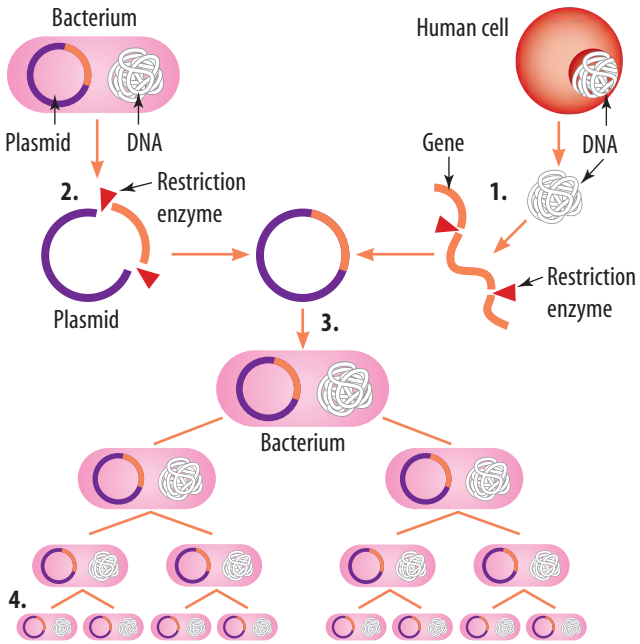
Eukaryotic unicellular organisms like (a) *Amoeba* and (b) *Euglena* divide by binary fission involving mitosis.

CELL DIVISION IN PROKARYOTES

Prokaryotes (such as bacteria) reproduce themselves by dividing into two using a process called binary fission. Although, binary fission also occurs in some eukaryotes, this type is much less complex, as prokaryotes do not have a nucleus. The cells produced are clones; they are identical to each other and to the cell from which they originate.

USING BACTERIA TO MAKE HUMAN PROTEINS

Knowledge of how bacteria reproduce can be used to get them to make human proteins. Scientists can insert genetic instructions from other organisms (including humans) into bacterial cells. When these bacterial cells divide, they produce cells that also contain the inserted foreign DNA and are able to make the protein that it codes for.



Bacteria can produce human insulin if the insulin gene is inserted into the bacterial cells.

This technology can be used in the production of insulin, a protein used in the treatment of a type of diabetes. The rapid rate of bacterial reproduction results in many cells with the human DNA and the production of useful quantities of this useful human protein.

CELL DIVISION AND DISEASE

Diseases can be divided according to whether they are infectious or non-infectious. **Infectious diseases** can be transferred from one organism to another. Tetanus and tuberculosis are examples of infectious diseases in which the cells are damaged by a bacterial infection. **Non-infectious diseases** are not transferred between organisms. Cancer is an example of a non-infectious disease that can be considered as a form of uncontrolled cell division or a disease of mitosis.

Scientists use their knowledge of cell division of disease-causing organisms to control or kill them. **Antibiotics** can be used to kill bacteria inside your body. **Disinfectants** can be used to kill bacteria on surfaces of non-living objects. Disinfectants should not be used on your skin as they can damage your cells. **Antiseptics** can be used on your skin. Antiseptics that kill bacteria are referred to as **bactericidal**, and those that stop bacteria from growing or dividing (but do not kill them) are called **bacteriostatic**.

INQUIRY: INVESTIGATION 3.12

Mitosis: patterns of order

KEY INQUIRY SKILLS:

- planning and conducting
- processing and analysing data and information
- evaluating
- communicating

Equipment:

light microscope

prepared onion root tip cells showing various stages of mitosis

- Use a light microscope to observe the prepared slides.
- Construct labelled diagrams to record your observations.

DISCUSS, EXPLAIN AND INVESTIGATE

1 Carefully observe your diagrams, noting any similarities or differences between your observations for each slide.

2 Discuss your observations with at least two other students.

3 Construct a table that summarises similarities and differences between the slides showing various stages of mitosis.

4 Comment on any patterns that you have observed.

5 Use the mitosis diagram on page 92 to suggest a sequence or order for the mitosis slides observed.

6 Construct a PMI for this investigation that summarises the pluses (strengths), minuses (limitations) and suggested improvements (if you were to do it again).

7 Find out more about where and when mitosis occurs in plants. Create a poster to communicate your findings.

8 The cells that result from mitosis are identical to each other and the original cell. Suggest advantages and disadvantages of this feature.

INQUIRY: INVESTIGATION 3.13

Where are those germs?

KEY INQUIRY SKILL:

- processing and analysing data and information

Equipment:

sterile cotton buds
nutrient agar plates in Petri dishes (3 per group)
sticky tape
sterile Pasteur pipette
marker pen

CAUTION Agar plates should not be opened after incubation.

- Swipe a sterile cotton bud across a surface of your choice (such as canteen counter, computer keyboard, phone mouthpiece or bin lid).
- Swipe the cotton bud across the surface of the agar. Be careful not to push down too hard. The cotton bud should not leave a mark on the agar.
- Use sticky tape to seal the plate around the edge.
- Use a marker pen to write your group's name and where you collected the sample from.
- Use a different cotton bud to swipe a part of your body (such as the inside of your nose, your teeth, inside your ear or your scalp).
- Swipe the cotton bud on the surface of the second agar plate, then seal and label it as before.
- Use the sterile Pasteur pipette to collect about 1 mL of water from a location of your choice (such as a fish tank, puddle, local creek, school swimming pool or drain pipe).
- Pour the sample of water over the surface of the agar and swish it around. Seal and label the agar as before.
- Incubate the three plates upside down at 30 °C for 48 hours. Remove the plates from the incubator and observe the colonies of bacteria through the lid of the Petri dishes (do not open the Petri dishes).

DISCUSS AND EXPLAIN

1 Draw a diagram of each Petri dish showing the location and size of the colonies.

2 Colonies of bacteria tend to be smooth whereas colonies of fungus appear furry and are often larger. Do you have colonies of bacteria or fungi or both on your plates?

3 Look at the other groups' plates.
   (a) Which of the surfaces tested by your class had the most microbes? How can you tell?
   (b) Which body part tested had the most microbes?
   (c) Which of the water samples tested contained the most microbes?

4 Explain why it would be dangerous to unseal the agar plates and lift the lid to look at the colonies of microbes.

5 Find out from your teacher how the plates are disposed of safely at your school.

6 Design an experiment to test whether antibacterial surface spray really does kill bacteria.

UNDERSTANDING AND INQUIRING

REMEMBER

1 Provide three reasons for cell division.

2 Suggest the relationship and sequence of the following: nucleus, DNA, cell, chromosome.

3 State a feature that the daughter cells produced by mitosis share with the parent cell.

4 If a cell is described as being a clone, what does this mean?

5 Identify a type of unicellular organism that:
(a) uses mitosis to reproduce
(b) does not use mitosis to reproduce.

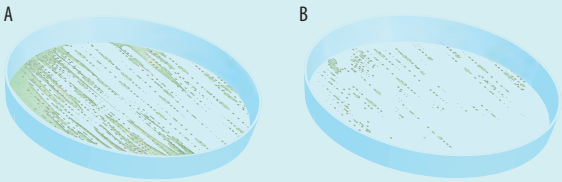
6 Identify the difference between mitosis and cytokinesis.

7 Suggest a way that scientists can apply their knowledge of cell reproduction to benefit humans.

8 Outline the differences between disinfectants, antiseptics and antibiotics.

THINK AND DISCUSS

9 Charlotte wanted to find out if antibacterial soap really works. She prepared two agar plates. She swiped her fingers over the surface of plate A. She then washed her hands with antiseptic soap and wiped her fingers over the surface of plate B. She incubated both plates. Her results are shown below.



(a) Write a conclusion for Charlotte's experiment.
(b) Which plate was the control?
(c) What were the independent and dependent variables in this experiment?
(d) Which variables need to be controlled in this experiment so that it is a fair test?

10 Before mitosis begins, the DNA in the cell is replicated. Suggest why this replication step needs to occur.

11 Suggest the advantages and disadvantages of being a clone.

INVESTIGATE AND CREATE

12 Find some examples of disinfectants and antiseptics. Select one and research how it works, reporting your findings as a 'scientific journal article'.

13 Find out more details about mitosis and then use wool, plasticine or pipe cleaners to create your own model of the process of mitosis.

14 Find out where and when mitosis occurs in plants. Create a poster to communicate your findings.

15 *Entamoeba histolytica* is a unicellular organism that is a cause of diarrhoea among travellers to developing countries.
(a) Find out more about the disease that it causes, its life cycle and what you can do to avoid being infected by it.
(b) Prepare a brochure, poster or PowerPoint presentation that could be used to inform travellers.

16 Investigate cell division in *Amoeba*, *Euglena* or *Paramecium* and create an animation to show how they reproduce.

17 Find out how DNA is replicated in a cell and create an animation or PowerPoint presentation to communicate your findings to others.

18 Investigate the differences between the structure of chromosomes in prokaryotes and eukaryotes, creating a model of each type of chromosome.

19 Investigate the development of the microscope and its impact on our understanding of cell function and division. Present your findings as a short 'documentary' or play.

20 Research examples of genetic engineering in which bacteria have foreign DNA inserted into them so that they produce human proteins. Communicate your findings as a newspaper article.

21 *Clostridium perfringens* is one of the fastest growing bacteria, having an optimum generation time of about 10 minutes.
(a) If you started with one bacteria, plot on a graph how many bacteria there would be each hour over a 24-hour period.
(b) Find out more about the structure and reproduction of this bacterium.
(c) Find out why it is sometimes referred to as a 'flesh-eating' bacterium.
(d) Write a story that includes features of *Clostridium perfringens* as a key part of the storyline.

22 (a) Find out why *Escherichia coli* (*E. coli*) counts at beaches are often stated in newspapers.
(b) How is the concentration of *E. coli* measured?
(c) Find out more about the structure and reproduction of *E. coli*.
(d) Create a model of this organism.

23 (a) Find out the differences between disinfectants and antiseptics, providing examples of each.
(b) Select one of your examples and prepare an advertisement that could be used to market it.

24 Search the internet for animations, songs and games that involve mitosis or binary fission. Use these to guide you in the development of your own creative lesson on this theory.

Skin 'n' stuff

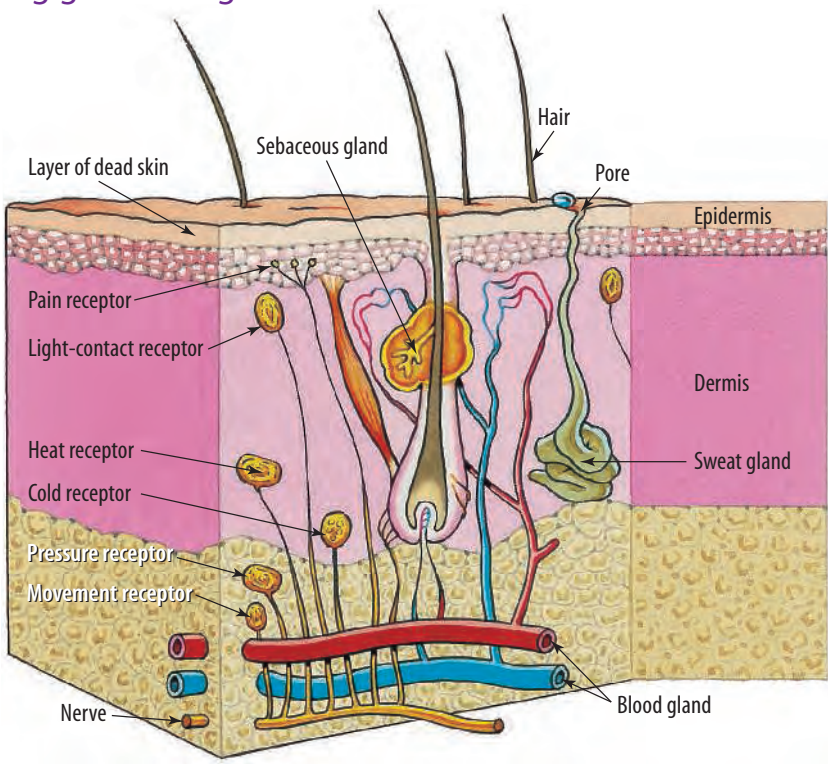
eBookplus eLesson
A cure?
Learn about the revolutionary new Australian trials to find a vaccine for skin cancer. **eles-0070**

Your skin is made up of lots of cells that work together to keep you alive. A type of cell division called mitosis enables you to make skin cells for growth, repair and replacement. But what happens when something goes wrong?

Skin deep

Your skin is the largest organ of your body. As well as holding the insides of your body in, it also:

- protects your body from microbes that could cause disease
- is almost completely waterproof
- protects the inside of your body from chemicals and harmful radiation from the sun
- detects heat, cold, pain, pressure and movement
- helps control your body temperature
- forms vitamin D in sunlight
- releases water and other waste products.



The skin is divided into three layers.

Your skin varies in thickness between about 0.5 millimetres and 5 millimetres. The thickest part is on the soles of your feet. Skin consists of three layers.

The **epidermis** is the top layer. It contains several layers of cells. At the very top is a layer of dead skin cells, which flake off continually. At the bottom of the epidermis, new cells are always being produced. They push upwards on the older cells, moving them towards the surface. Below the epidermis is the **dermis**, which contains **receptors** for the sense of touch. It also contains **sweat glands** and many small blood vessels. Beneath the dermis is a thicker layer of fatty tissue, which acts as an insulator to help keep the body temperature constant. This fat has been stored by the body and can be used when needed to provide extra energy.

When you get hot, it is important that your body cools itself down so the blood remains at its constant temperature of about 37°C. Your sweat glands produce a liquid that is released through the **pores** at the surface of your skin. When the water in your sweat **evaporates**, it takes some of the heat out of your body.

Are you ticklish?

Are you more ticklish on some parts of your skin than others?

Below the surface of your skin there are many receptors that are attached to nerves. The nerves send messages to the brain. There are different receptors for heat, cold, light contact, pain, pressure and movement. They are all receptors to the sense of touch.

The light-contact receptors are nearer to the surface and closer together in some parts of your skin than others. It is those parts that are most sensitive to tickling. Some parts of the skin are also more sensitive to pain, heat, cold, pressure and movement than others. Your sensitivity depends a lot on how close together the receptors are to each other and how deep they are.

HOW ABOUT THAT!

Sweat doesn't actually smell until it is consumed by the bacteria that live on the surface of your skin. The regions around your armpits and external sex organs are warm and moist, providing ideal conditions for bacteria to grow and feed on the sweat. This is why these areas can get smelly.

HOW ABOUT THAT!

Spray-on skin?

Burns can now be treated with skin that is literally sprayed on! Dr Fiona Wood (Australian of the Year in 2005) pioneered the development of 'spray-on skin'. Using the patient's own skin cells, it can be sprayed onto burnt areas and reduces scarring.

Dr Fiona Wood, pioneer of 'spray-on skin'

'Spray-on skin' in action

Sunsense

Skin cancer is the most common form of cancer in Australia. In fact, two out of three Australians are likely to get skin cancer at some time during their lives. The most serious forms of skin cancer are responsible for about 1000 deaths each year in Australia.

WHAT IS CANCER?

As the body's cells die, new cells are made to replace them. In a healthy person, just the right number of new cells are formed using a type of cell division called mitosis. Cancer can be considered as a disease of mitosis. Damage to the DNA in a cell can cause the normal regulatory processes in cell division to be ignored or overridden. This can result in uncontrolled cell division, a condition we call **cancer**.

This uncontrolled cell division can form a mass of cells called a **tumour**. The cells of a tumour are not specialised and cannot do the jobs of the cells that they are replacing. Some tumours still respond to the body's control mechanisms and do not spread to other parts of the body. These tumours are called **benign**. Others have uncontrolled cell growth and do spread, damaging vital organs. These are called **malignant** tumours or cancer. If cancer is detected early, the diseased cells can be removed or destroyed by chemotherapy or radiation. However, once cancer spreads, it is very difficult to control.

WHAT CAUSES SKIN CANCER?

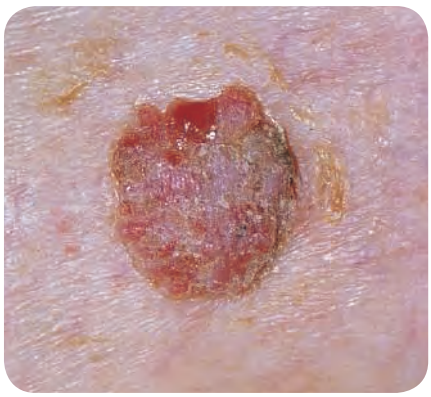
The main cause of skin cancer is exposure to the sun. The ultraviolet radiation reaching Earth from the sun is not visible. Ultraviolet radiation, which is also the cause of sunburn, is at its peak in the middle of the day when the sun is directly overhead. Ultraviolet radiation causes cancer in the cells of the epidermis, the top layer of the skin, because it damages the cells' genetic material, called DNA.

EARLY DETECTION

The key to curing skin cancer is early detection. Even melanomas can be cured in more than 95 per cent of patients if they are detected quickly. If you see a new lump or spot, or a changing freckle or mole, see a doctor promptly.

The three main types of skin cancer include the following.

Squamous cell carcinoma

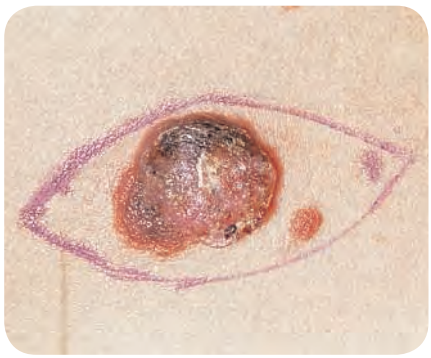


- Less common and more dangerous than basal cell carcinoma
- Appears as a red, scaly sore
- Usually found on the hands, forearms, face and neck, but can spread to other parts of the body
- Mostly affects people over the age of 40 who have been exposed to the sun for many years
- Kills about 200 Australians each year

Basal cell carcinoma

- Most common form of skin cancer and also the least dangerous
- Appears as a red, flaky lump on the skin
- Rarely spreads to other parts of the body but needs to be treated before it grows large or forms a deep sore

Melanoma



- Least common but most dangerous form of skin cancer
- First sign is a change in size, shape or colour of a freckle or mole, or the appearance of a new spot on normal skin
- Can spread quickly to other parts of the body
- Most common in adults aged between 30 and 50 years, usually caused by long periods of exposure to the sun during childhood and adolescence
- Cause of the most deaths from skin cancer — about 800 each year in Australia

Some questions about fun in the sun

Q: *Is a suntan healthy?*

A: No. A suntan is evidence that you have been exposed to the sun for too long. A suntan will not protect you from skin cancer. Fake suntan lotions do not offer protection from skin cancer either.

Q: *Do I need to worry about sunburn or skin cancer when it's cloudy?*

A: Yes. Although clouds block out a lot of the sun's visible light, they do not block out enough ultraviolet radiation to protect your skin completely, especially during summer. The graphs below show that light cloud cover has little effect on the harmful ultraviolet radiation reaching the ground on a summer's day in Sydney. Heavy cloud, however, decreases the amount of ultraviolet radiation reaching the ground by over 90 per cent.

Q: *Do I need to use sunscreen when I wear a hat?*

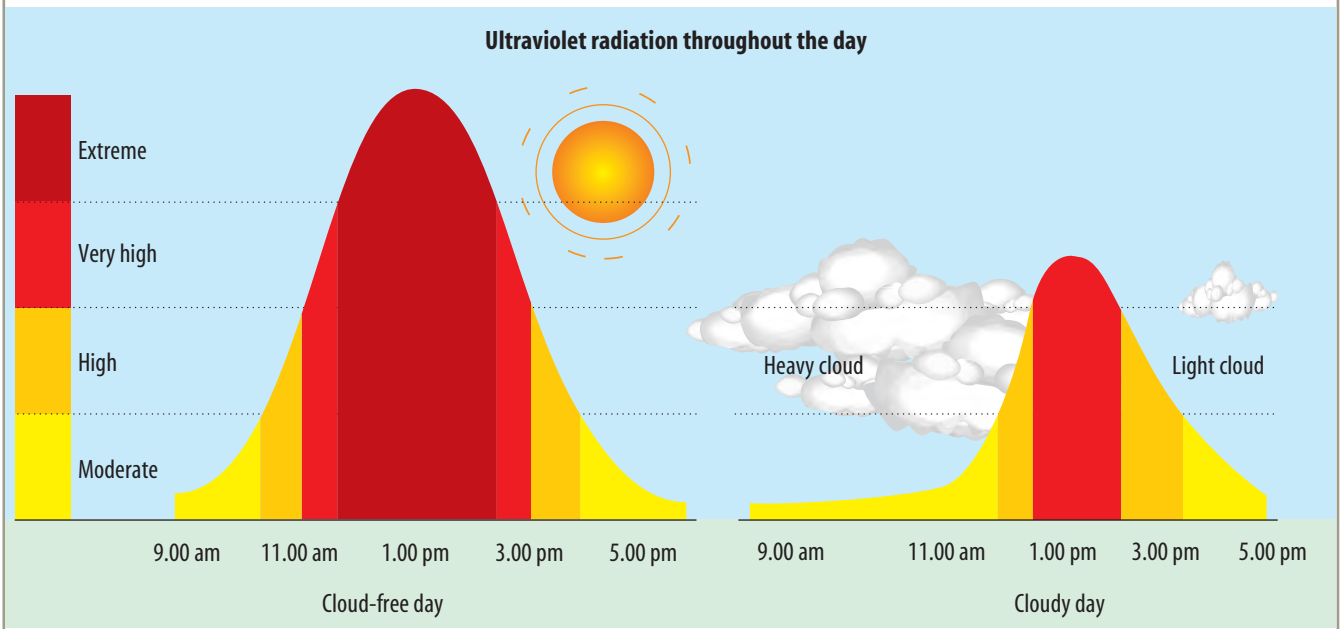
A: Yes. The sun's radiation is reflected from the ground and from water. Snow and sand reflect a lot of radiation, even on cloudy days. In addition, many hats, including baseball hats, do not protect you from direct radiation. Wide-brimmed hats or 'legionnaire' hats provide the best protection because they shade the neck and ears.

Q: *What does SPF 30+ mean?*

A: SPF stands for 'sun protection factor.' It allows you to estimate how long you can stay in the sun before your skin starts to go red. This period can be estimated by multiplying the amount of time that it takes your skin to redden by the SPF factor. For example, if your unprotected skin starts to burn after 10 minutes in the hot sun, proper use of SPF 4 sunscreen would allow you to remain in the sun for $10 \times 4 = 40$ minutes before burning starts. After that 40 minutes, you would burn, even with more sunscreen applied. An SPF water-resistant 30+ sunscreen reapplied every 2 hours would allow you to remain in the sun for at least $10 \times 30 = 300$ minutes before burning starts. SPF 30+ sunscreen blocks out about 97 per cent of the sun's radiation.

Q: *What does 'broad spectrum' mean?*

A: The Cancer Council NSW recommends a broad spectrum SPF 30+ sunscreen. Broad spectrum sunscreens offer protection from the three different types of ultraviolet radiation: UVA, UVB and UVC.



These graphs show how the ultraviolet radiation reaching the ground changes on a typical summer day in Sydney.

HOW ABOUT THAT!

An Australian drug company called Peplin has developed a gel made from a common weed called *Euphorbia peplus*, which has been used successfully to treat some skin cancers. The plant contains a cancer-fighting chemical. Peplin has found a way of extracting this chemical from the plant and making it into a gel that can be applied to skin cancers. When the gel was tested, it cleared most skin lesions after just two days.

UNDERSTANDING AND INQUIRING

REMEMBER

1 Identify the most serious form of skin cancer and how it is caused.

2 Outline the difference between a benign tumour and a malignant tumour.

3 State which part of the sun's radiation is the major cause of skin cancer and sunburn.

4 Identify the most dangerous time of day to be out in the sun.

5 Describe what you should look for on the skin when checking for signs of skin cancer.

6 List five ways that you can help protect yourself from skin cancer.

THINK AND DISCUSS

7 Melanomas occur mostly in adults over the age of 30. Why is it so important that young children and adolescents are aware of the dangers of the sun's radiation?

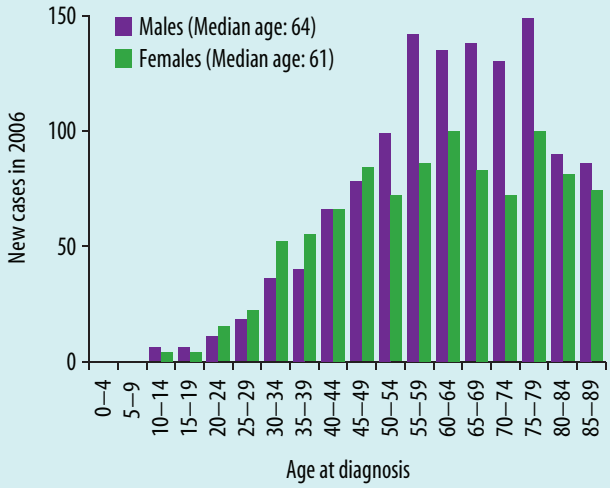
8 Daniel has very pale, sensitive skin that begins to burn after only eight minutes in the summer sun. He goes to the beach and takes a tube of SPF 6 sunscreen with him.
   (a) If he doesn't go swimming, how long would he be able to sit in the sun before getting burnt, assuming that he applies his sunscreen correctly?
   (b) If he used SPF 30+ sunscreen instead, would he be safe sitting in the sun all day? Give a reason for your answer.

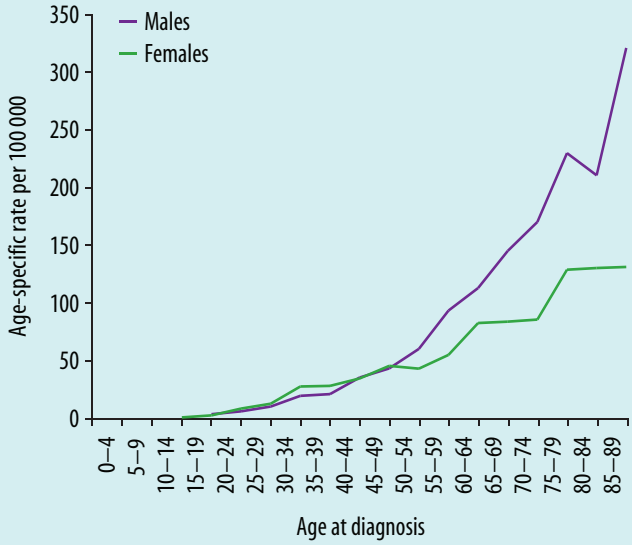
9 Discuss the validity of the following statements.
   (a) Cancer can be considered as a disease of mitosis.
   (b) Melanomas are less dangerous than carcinomas.

10 Tissues are grouped together to form organs to help you function effectively. Describe how the following are important to your survival.
   (a) Muscles
   (b) Bones
   (c) Skin

THINK AND REASON

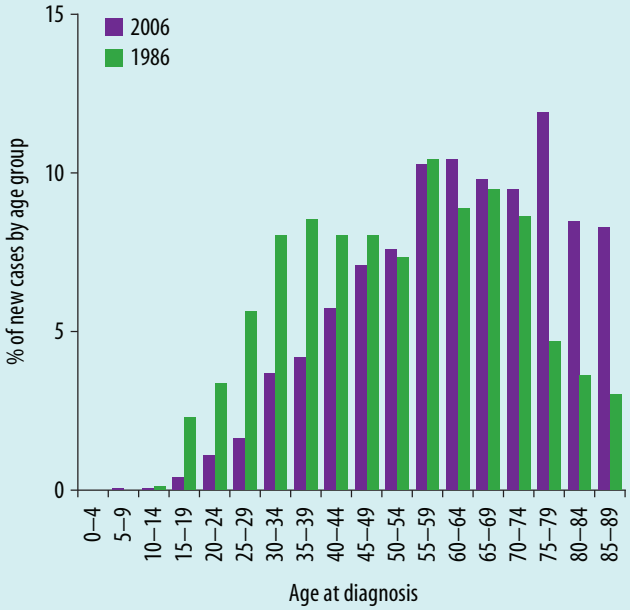
11 Use the 2006 melanoma incidence graphs below to complete the following.
   (a) Describe the pattern of melanoma incidence in Victoria in 2006.
   (b) Suggest reasons for this pattern.





(c) Identify the age group that has the highest number of new cases of melanomas:
  (i) for males
  (ii) for females.
(d) Does the data suggest that there is a higher incidence of melanomas in males or females?
(e) Suggest possible reasons for this data.

12 Use the 1986 and 2006 melanoma incidence graph below to complete the following.
(a) State the age group and year with the highest incidence of melanoma cancer.
(b) State the age group and year with the lowest incidence of melanoma cancer.
(c) Discuss your responses to parts (a) and (b) and suggest possible reasons for your findings.
(d) Describe differences in the patterns of melanoma incidence in Victoria between 1986 and 2006.
(e) Discuss your observations and suggest reasons for this difference.



CREATE AND DESIGN

13 Design and construct a model that shows details of the three layers of skin.

14 Construct a mind map that shows links between key scientific terms that are relevant to skin cancer. Use this map as a framework to create a song or poem about skin cancer that will increase awareness of types, causes, detection and prevention.

15 Design a colourful poster that would encourage people to protect themselves from the sun's harmful radiation.

16 Design and construct a multipurpose hat that shades the head and has at least one other purpose. Give your multipurpose hat a name. Prepare an advertising brochure and instruction manual for your hat.

INVESTIGATE AND DESIGN

17 Design and carry out a survey (consisting of a series of questions) to find out whether people of different age groups protect themselves from the danger of skin cancer by wearing hats, shirts and sunscreens. By sharing your data with other members of your class, you may be able to form a sound conclusion.

18 A number of schools describe themselves as being 'sunsmart'. Find out what the criteria are and who decides whether they are 'sunsmart'.

19 Design your own 'Sunsmart' advertising campaign. Present your advertisement as a video clip with music.

20 Research and report on one of the following science careers: dermatologist, plastic surgeon, skin cancer research scientist.

21 Professor Graham Giles is a scientist at the Cancer Control Research Institute and is involved in Health 2020. Investigate and report on his research and that of other scientists at the institute.

22 Associate Professor Greg Woods is investigating relationships between immune cells and cancer cells. Investigate and report on his research into skin cancer at the Menzies Research Institute.

23 Dr Lisa Ebert at the Ludwig Institute for Cancer Research is investigating responses to a melanoma vaccine. Investigate and report on her research and that of other scientists at the institute.

24 Use the **2020health** weblink in your eBookPLUS to research and report on examples of their investigations.

25 Research and report on the difference between an oncologist and a pathologist.

26 Use the **Molemate** weblink in your eBookPLUS to research and report on the 'Molemate' studies at the University of Western Australia.

27 Obtain a selection of at least five different sunscreens. Identify criteria that could be used to determine their similarities and differences. Construct a matrix table with these criteria and record details for each sunscreen. Determine which sunscreen you would use. Which criteria did you use? Give reasons for your criteria. If you were to invent the 'ideal' sunscreen, what features would it have and why? Design an advertising brochure or advertisement for your 'ideal' sunscreen.

INVESTIGATE

28 With a partner, play a guessing game to see how well you can use your sense of touch alone to identify 10 unknown objects.

You will need a blindfold and 10 objects of about the same size. Sandpaper, plastic, coins and pieces of carpet, polystyrene, nylon and wool would be ideal. See who can identify the most objects correctly.

Tiny size, big trouble

Just because you are tiny, doesn't mean that you can't cause trouble — think about how annoying mosquitoes can be! Sometimes it's the small things in life that make the biggest difference. The presence, absence or extreme levels of particular microbes can reflect the health of their habitat, whether it be an organism or an entire ecosystem.

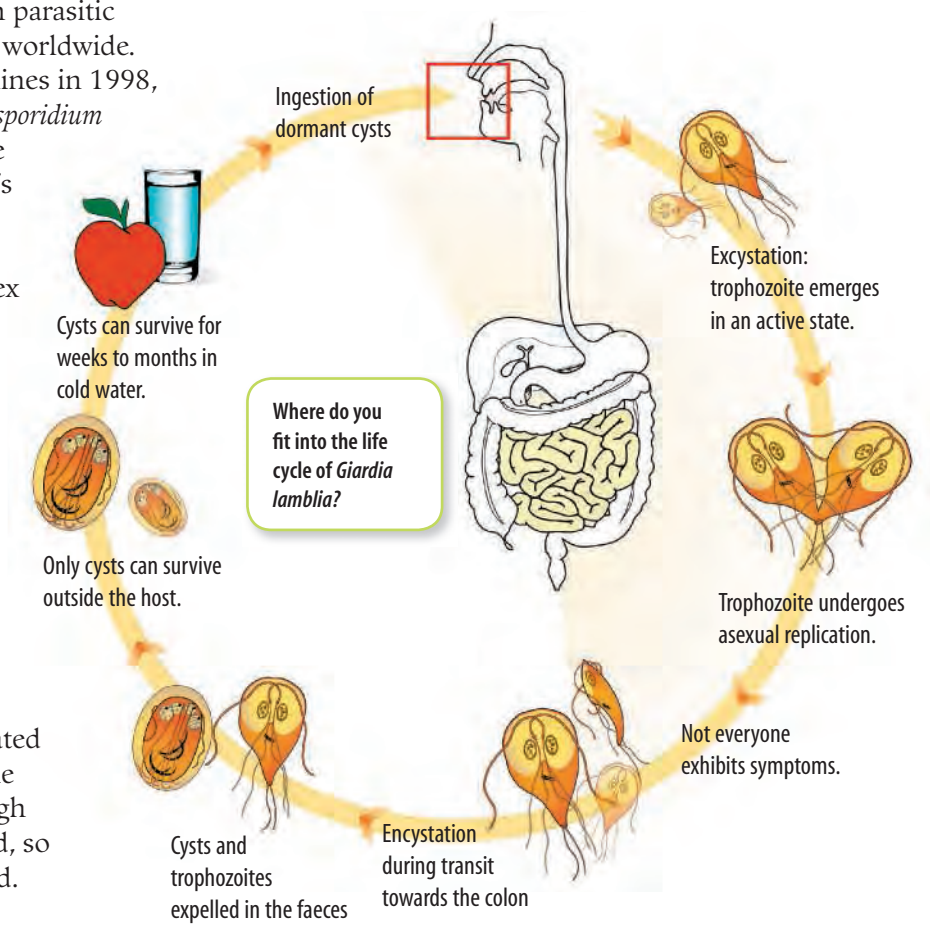
***Giardia lamblia* are pear-shaped and quite large, usually being more than 6 µm in size.**

'Down the hatch'

What's in that gulp of water? You may have swallowed a little more than you thought! As well as water, you may have also 'invited in' viruses, bacteria, protozoans, phytoplankton and zooplankton.

A protoctistan (protozoan) by the name of *Giardia lamblia* has become quite well known for 'hitching a ride' in that gulp of water. These parasites are the cause of one of the most common parasitic gastrointestinal infections in humans worldwide. *Giardia lamblia* made newspaper headlines in 1998, along with another protozoan, *Cryptosporidium parvum*, when high levels of both were reported to be contaminating Sydney's drinking water supply.

Giardia lamblia use their flagella to move around, and they have a complex life cycle. The parasite can survive for a long period outside the body in an inactive form called a cyst. Once swallowed, the cyst is activated by your stomach acid and develops into the disease-causing stage. Using a huge sucker, they then attach themselves to the lining of your intestine, sucking your blood as their food source. After about ten days of infection you could have a million of them living off your blood supply and causing symptoms associated with gastrointestinal complaints. Some of their reproductive cysts pass through your digestive system and are excreted, so that another host can become infected.



Water wise

Living things need water to survive. Some living things also need to live in water to survive. In these ecosystems, there are links between the inhabitants to keep them balanced and healthy. Sometimes, however, these links can be broken or disrupted. This is when problems can occur that may result not only in an unbalanced ecosystem, but also in death.

TAKE A SWIM WITH ME

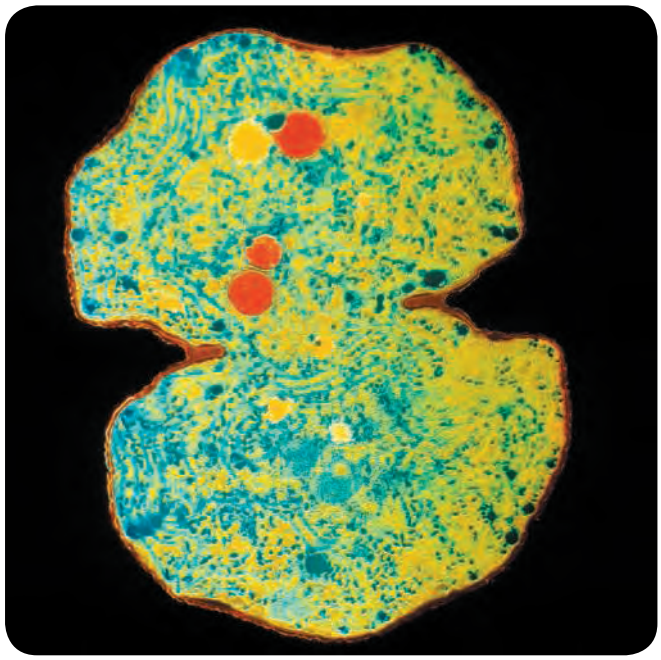
Escherichia coli (*E. coli*) is a type of bacterium found in our intestine. It usually causes us no harm and passes through our digestive system to be excreted. This enables it to be used as an indicator of sewage contamination in water. Contaminated sewage may contain dangerous or even deadly micro-organisms. It is for this reason that *E. coli* levels are tested and reported on at various beaches and swimming locations.

Electron micrograph of *E. coli*

BLOOM'N ALGAE

Algal blooms occur naturally and provide food for many aquatic organisms. Sometimes, however, they can cause harm. Algal blooms can cause large fluctuations in the levels of oxygen and pH (acidity) of the water and block sunlight penetrating through it. The species that cause these blooms may also be toxic to some of the aquatic organisms or even toxic or a skin irritant to humans. The presence of algal blooms can have economic as well as biological consequences.

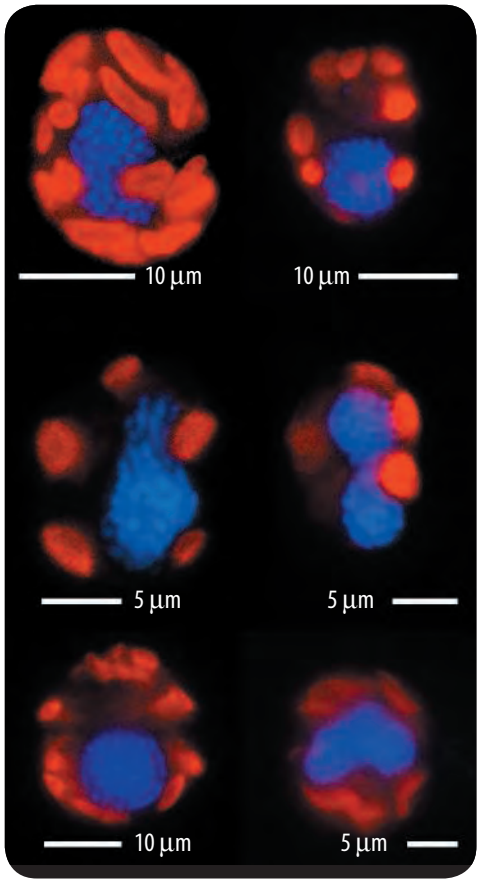
Algal bloom at Warragamba Dam, Sydney, NSW



Microcystis is an example of blue-green algae.

RED TIDE OF DEATH

Scientists at the University of Tasmania have discovered that two new types of algae are killing fish in the Southern Ocean. These algae have been found in abundance and when in full bloom produce a distinctive red-coloured tide. The university's scientists are working with the Australian Antarctic Division to try to establish the number of fish that are suffocating from the algae. This will enable sustainable fishing levels to be maintained.



Fluorescence microscopy can be used to identify the species that a cell belongs to, and can therefore help determine whether the cell may be of a dangerous type. In this image, the position of the chloroplasts shows as red and the nucleus as blue.

WHAT DOES IT MEAN?

The word *pathogen* comes from the Greek terms ***pathos***, meaning 'disease', and ***gen-***, meaning 'birth'.

INQUIRY: INVESTIGATION 3.14

Teeming with tiny . . .

KEY INQUIRY SKILL:

- processing and analysing data and information

Equipment:

light microscope

living protoctistan specimens, such as Paramecium, Euglena, Amoeba

- Place a drop of a solution containing a particular type of protoctistan onto a microscope slide and cover with a coverslip.
- Carefully observe under a light microscope.
- Draw, label and annotate your diagrams.
- Repeat the procedure so that you have observed three different types of protoctistans.
- Repeat the procedure using pond water.

DISCUSS AND EXPLAIN

1 While the protoctistans observed are all unicellular, they have many differences. Use a matrix (see page 340) to summarise these.

2 Discuss an interesting observation with your partner and formulate three questions for further investigation.

3 Discuss your observations of the pond water. Investigate to identify some of the forms of life observed.

HOW ABOUT THAT!

Professor Gustaaf Hallegraeff

Professor Gustaaf Hallegraeff is a scientist at the University of Tasmania. In his research he uses light, scanning and transmission electron microscopes to help him classify tiny marine organisms called phytoplankton. On the basis of his findings, he has produced publications to communicate information about harmful Australian microalgae.

HOW ABOUT THAT!

Snap frozen

Australian Antarctic Division scientists are using a new field emission scanning electron microscope that magnifies specimens by up to 650 000 times. This allows them to observe features two million times smaller than the head of a pin. Snap-freezing technology is used, with the specimen being snap-frozen in super-cooled liquid nitrogen in temperatures of –210°C! The snap-freeze process allows scientists to study more delicate specimens than were previously able to be observed.

UNDERSTANDING AND INQUIRING

REMEMBER

1 (a) Describe the shape and size of *Giardia lamblia*.
(b) Describe how *Giardia lamblia* obtains its food.
(c) In which form can *Giardia lamblia* survive for weeks outside its host?

2 Identify the genus to which each of the following organisms belongs.
(a) *Escherichia coli*
(b) *Giardia lamblia*
(c) *Microcystis* spp.

3 Explain why levels of *E. coli* are measured in water.

4 Are algal blooms always harmful? Explain.

5 Outline problems associated with harmful algal blooms.

6 Identify the type of organism that is responsible for causing red tides and killing fish in the Southern Ocean.

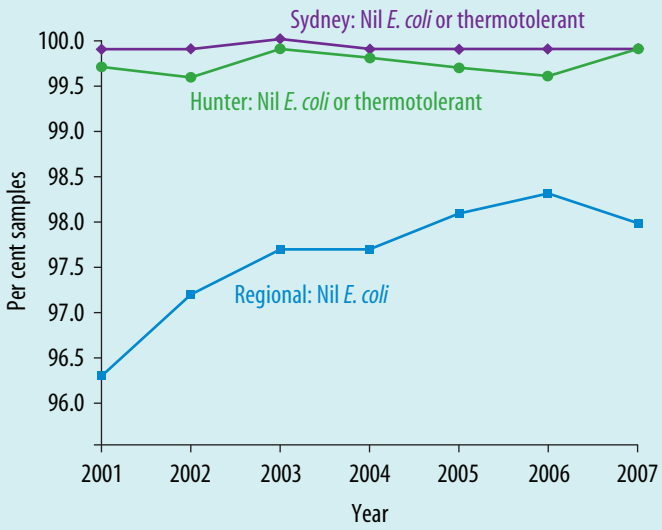
7 Explain why scientists are trying to determine the actual numbers of fish being killed by the effects of red tides.

8 State a function of fluorescence microscopy.

INVESTIGATE, THINK AND DISCUSS

9 Use the graph below to answer the following questions.
(a) Describe the pattern of indicator bacteria in each location between 2001 and 2007.
(b) Identify which location has the highest bacterial counts. Suggest a reason for this observation.
(c) Identify which location has the lowest bacterial counts. Suggest a reason for this observation.
(d) Find examples of *E. coli* counts in your local community waters.
(e) Graph your findings and discuss patterns in your class data.
(f) Suggest reasons for the patterns that you have observed.

Indicator bacteria counts in drinking water, Sydney, Hunter and regional water supplies, NSW 2001 to 2007



10 Pathogens are disease-causing organisms. Research and report on microbial pathogens in Australian waters.

11 Research and report on one of the following.
- *Escherichia coli*
- *Microcystis* spp.
- *Giardia lamblia*
- *Cryptosporidium parvum*

12 Research and report on an example of Australian research on one of the following topics.
- Algal blooms
- Microalgal pigments
- Harmful Australian microalgae
- Micropalaeontological studies
- Coastal eutrophication
- Taxonomy using light and electron microscopy

13 Find out how *E. coli* levels are measured and examples of other micro-organisms they may indicate the presence of.

14 (a) Research and report on the diagnosis, symptoms and treatment of:
(i) cryptosporidiosis
(ii) giardiasis.
(b) Construct a Venn diagram to show how the conditions are similar and how they are different.

15 (a) Identify and research an issue related to micro-organisms in Australian waters.
(b) Construct a PMI chart to summarise your findings.
(c) Discuss your PMI chart with others in your team and add their comments.
(d) Organise a class or team debate on the issue.

16 Do microbes reflect marine health? Investigate this question by researching online and justify your response.

eBook*plus*

17 Use the **Water quality** weblink in your eBookPLUS to check out the latest daily water quality results from water filtration plants in Sydney and obtain current *Giardia* and *Cryptosporidium* data.

Target maps and single bubble maps

1. Draw three concentric circles on a sheet of paper.
2. Write the topic in the centre circle.
3. In the next circle, write words and phrases that are relevant to the topic.
4. In the outer circle, write words and phrases that are not relevant to the topic.

how to ...?

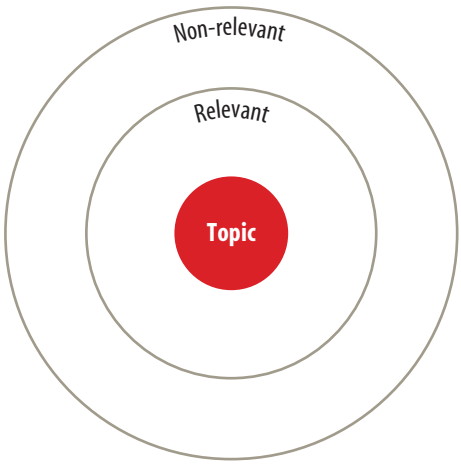
To identify (target) what is part of (relevant to) the topic and what is not

why use?

How can we find out what is relevant?

question

Target map



Circle map

also called

comparison

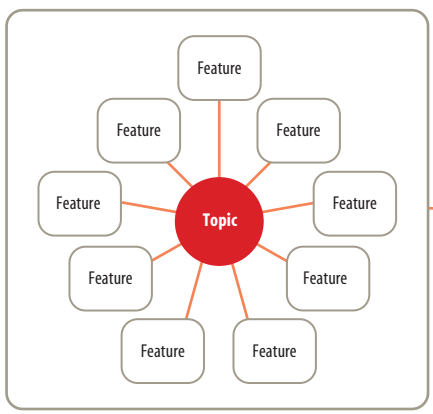
Similarity

Both identify and describe the range of the content.

Difference

Single bubble maps do not identify the non-relevant material.

Single bubble map



example

UNDERSTANDING AND INQUIRING

THINK AND CREATE

1 Use the target map below to answer the following questions.

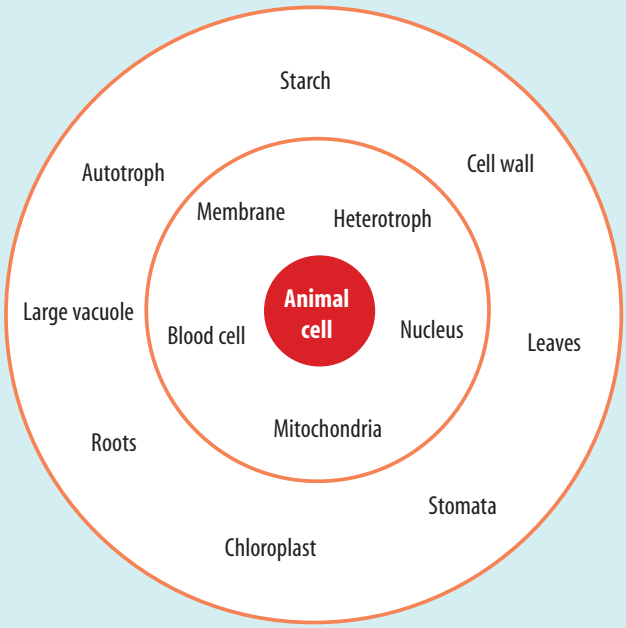
(a) List the content that is relevant to animal cells.

(b) List the content that is not relevant to animal cells.

(c) Using the words in the target map, construct a target map that is relevant to plant cells.

(d) Identify which words are relevant to both plant and animal cells.

(e) Suggest why plant and animal cells both have these features in common.



2 Use the terms in the box below to construct target maps that are relevant to:

(a) plants
(b) animals
(c) fungi
(d) protoctistans
(e) prokaryotes (monera).

multicellular	chloroplast	*Euglena*
prokaryote	bacteria	mushroom
eukaryote	fern	yeast
nucleus	alga	lizard
cell wall	*Paramecium*	sponge
large vacuole	unicellular	moss
xylem	cell membrane	blood cells
possum	stomata	phloem

3 Construct a single bubble map to identify:

(a) types of plant cells

(b) types of animal cells

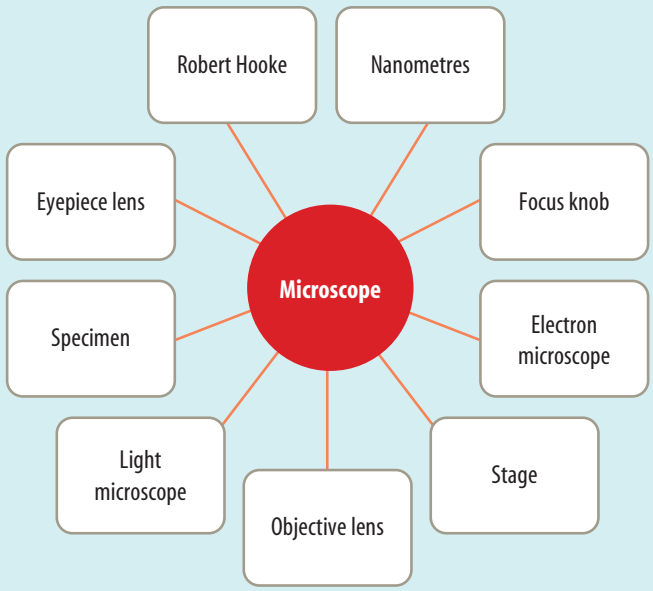
(c) scientists who have contributed to our knowledge of cells

(d) examples of body systems

(e) functions of skin

(f) issues related to stem cells.

4 Use the information in the single bubble map below to construct a target map of the parts of a light microscope.



5 (a) Use the internet to find images of at least five different types of zooplankton.

(b) Carefully observe your zooplankton images, recording key features of each in single bubble maps.

(c) Construct target maps for each of your zooplankton to show how they are different from the other zooplankton.

STUDY CHECKLIST

CELLS

- name and state the function of the parts of a light microscope
- describe how to prepare a specimen for observation under a light microscope
- observe and sketch labelled diagrams of cells and other specimens as viewed under a light microscope
- suggest why stains are used in the preparation of microscope slides
- explain the significance of the invention of the microscope to biology
- outline the contributions of three scientists to our understanding of cells
- state examples of different types of cells and relate their structure to their function
- describe the differences between unicellular and multicellular organisms
- name a type of cell division involved in growth and repair
- explain why not all cells have the same structure
- use differences in cell structure to classify organisms into groups
- explain how cell structure can provide us with evolutionary information

MULTICELLULAR ORGANISMS

- sort the following in order of relative size, from biggest to smallest: tissue, atom, organ, molecule, cell, system, organelle
- name a system in the human body, describe its function and give examples of the types of cells it contains

ECOSYSTEMS

- give examples of how unicellular organisms can have big impacts on ecosystems

SCIENCE AS A HUMAN ENDEAVOUR

- outline developments in the understanding of cells and how this knowledge has affected research areas such as health and medicine
- describe the development of the microscope
- outline the effect microscopes have had on our understanding of cell functions and division
- describe how people use understanding and skills from across the disciplines of science in their occupations
- provide an example of the role of knowledge of cells and cell divisions in the area of disease treatment and control

INDIVIDUAL PATHWAYS

eBookplus

Activity 3.1 Investigating cells **doc-6048**	**Activity 3.2** Analysing cells **doc-6049**	**Activity 3.3** Researching cells **doc-6050**

ICT

eBookplus **Summary**

eLESSONS

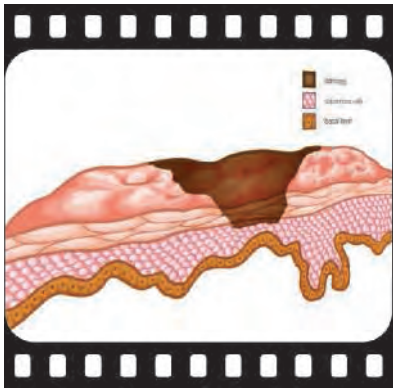
Inside cells

Learn about the building blocks of life called cells and organelles in this animated video lesson, looking closely at the difference between the make-up of animal and plant cells. A worksheet is attached to further your understanding.

Searchlight ID: eles-0054

A cure?

Skin cancer is the most common form of cancer in Australia with around 400 000 people diagnosed each year. Learn about the revolutionary new trials led by Australian scientists to find a vaccine for skin cancer. A worksheet is included to further your understanding.



Searchlight ID: eles-0070

INTERACTIVITIES

Microscope parts

This interactivity focuses on the microscope. You must select the parts of the microscope that best fit a series of descriptions. Instant feedback is provided.

Searchlight ID: int-0205

Cell jobs

This interactivity tests your ability to match a number of different types of cells with their roles in the body. Instant feedback is provided.

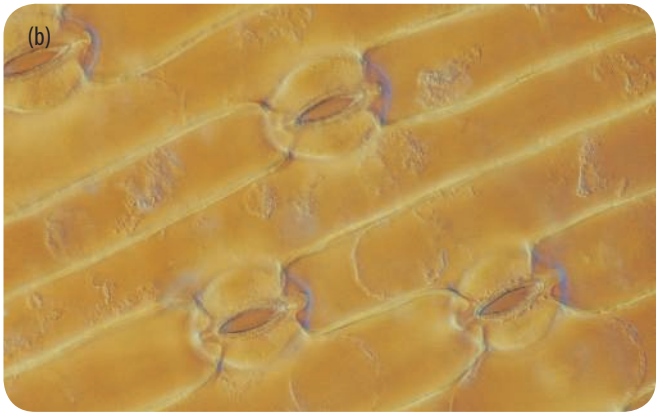
Searchlight ID: int-0206

LOOKING BACK

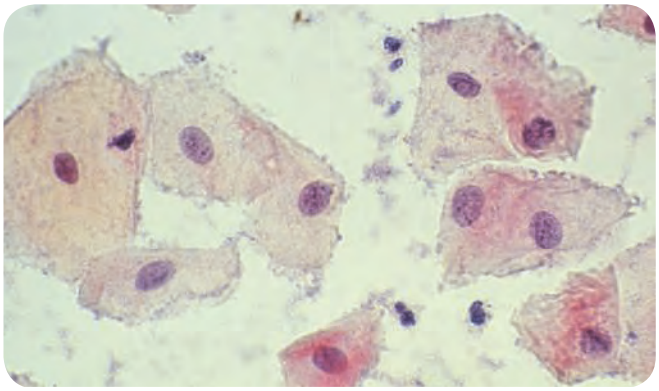
1 (a) Brainstorm as many 'cell'-related words as you can, writing them on a piece of paper.
(b) Pair up with another class member and add any of their words that you missed. Ask your partner what these words mean if you are unsure.
(c) On a new piece of paper, work with your partner to group or link words to make a concept, cluster or mind map.
(d) Compare your map with that of another pair in the class, adding as many more bits and pieces as you can.

2 Which of the following types of microscopes were used to take the photos shown below?
- Scanning electron microscope
- Light microscope

Give reasons for your answers.



3 Make a sketch of these human cheek cells.



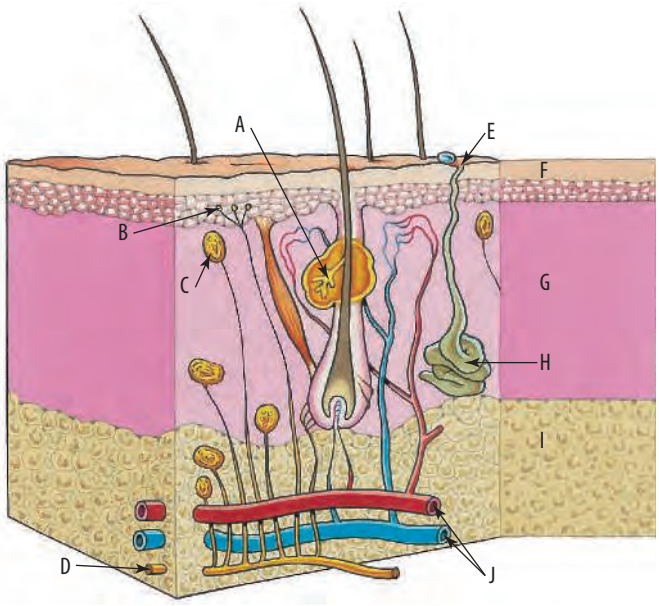
4 (a) Why do you think that cells have been described as 'living factories'?
(b) Think of a typical plant or animal cell. Make a list of all of the different parts and organelles. If the cell was a living factory, what might be the job of each listed part?
(c) Write a play to act out what happens in cells and perform it with others in your class. What sorts of things were easy to show? What sorts of things were hard to show? If you were to rewrite the play, what might you change and why?
(d) Convert the classroom into a giant cell! Take photos and then add information to them on a poster.

5 Construct a Venn diagram to show the similarities and differences between light microscopes and electron microscopes.

6 Explain the significance of the invention of the microscope in terms of how we see the world.

7 Suggest why the invention of microscopes led to the development of new scientific language and classifications.

8 Carefully observe the diagram of the skin below. Match the letters on the diagram with the following labels: epidermis, dermis, fatty layer, light receptor, sweat gland, blood vessels, sebaceous gland, pain receptor, nerve, pore.



9 The body tissues and their main jobs have been jumbled in the table below. Redraw the table and correctly match the type of tissue to its main job.

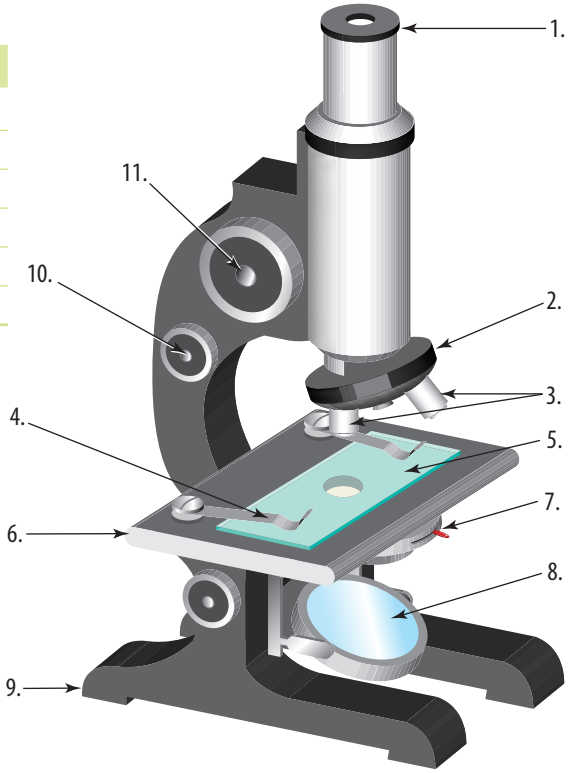
Tissue	Main job
Skeletal	Conducts and coordinates messages
Nerve	Supports and protects
Blood	Enables movement
Muscle	Carries oxygen around the body

10 Copy and complete the table below:

Cell feature	Plant cells	Animal cells	Fungal cells
Cell wall	✔	✘	
Cytoplasm			
Cell membrane			
Chloroplast			
Nucleus			
Large vacuole			

11 Draw and label a typical plant cell and a typical animal cell.

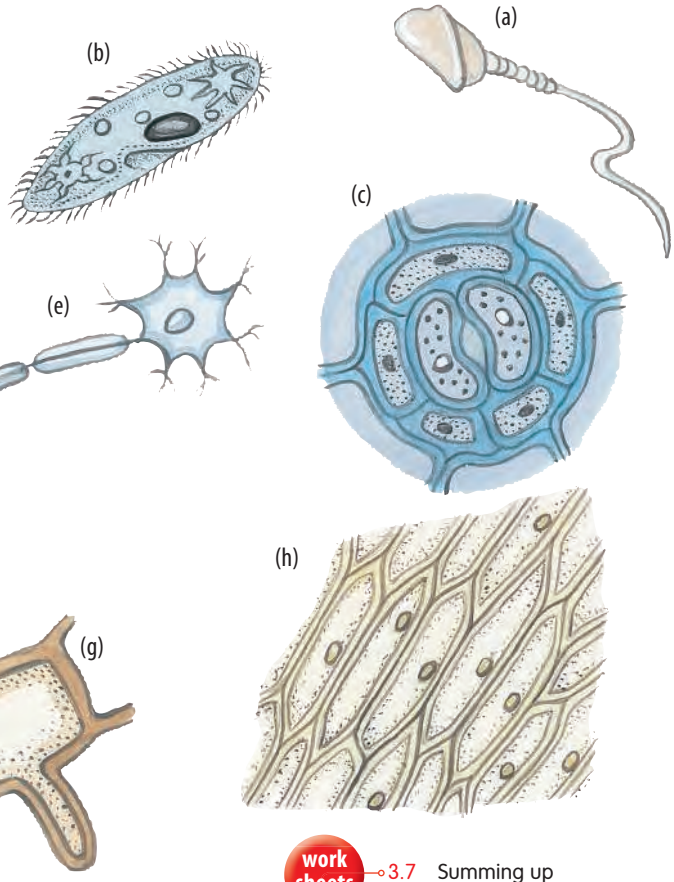
12 What's green and eats porridge? Identify the parts of the microscope on the right and use the code below to find out the answer to this riddle.
Code:
O = revolving nose piece; U = objective lenses;
S = coarse focus knob; K = fine focus knob;
D = microscope slide; L = stage slide clip;
C = base; O = mirror; L = iris adjustment;
I = stage; M = eyepiece lens



1	2	3	4	5	6	7	8	9	10	11

13 Unscramble the words using the clues provided.
(a) Control centre of the cell SEUNCLU
(b) Surrounds the cell ERAMMBNE
(c) Contains cell sap OCVAUEL
(d) Part of the cell between the cell membrane and the nucleus CATOPLMYS
(e) Building blocks of all living things LELSC
(f) Living things ASMOGNIRS

14 (a) Match the following cell names to the diagrams on the right.
Euglena
Paramecium
onion epidermal cells
nerve cell
sperm cell
guard cells
root hair cell
bacterium



(b) To which kingdom does each of these cells belong?

work sheets 3.7 Summing up
3.8 Looking back

4 Systems — living connections

Inside your body is a very busy place. The many complex processes need energy. To convert energy into a form you can use requires transport highways to take nutrients to where they are needed and to carry wastes away. Different parts of your body have different jobs to do — they may work together and rely on each other. It is because of the curiosity, imagination, passion and persistence of humans throughout history that we know today what we do about our bodies and how they function. What sort of questions do you have about what is inside your body and how it works? What exciting new discoveries will scientists make about our bodies in the future?

OVERARCHING IDEAS

- Patterns, order and organisation
- Form and function
- Stability and change
- Systems

SCIENCE UNDERSTANDING

Multicellular organisms contain systems of organs that carry out specialised functions that enable them to survive and reproduce.

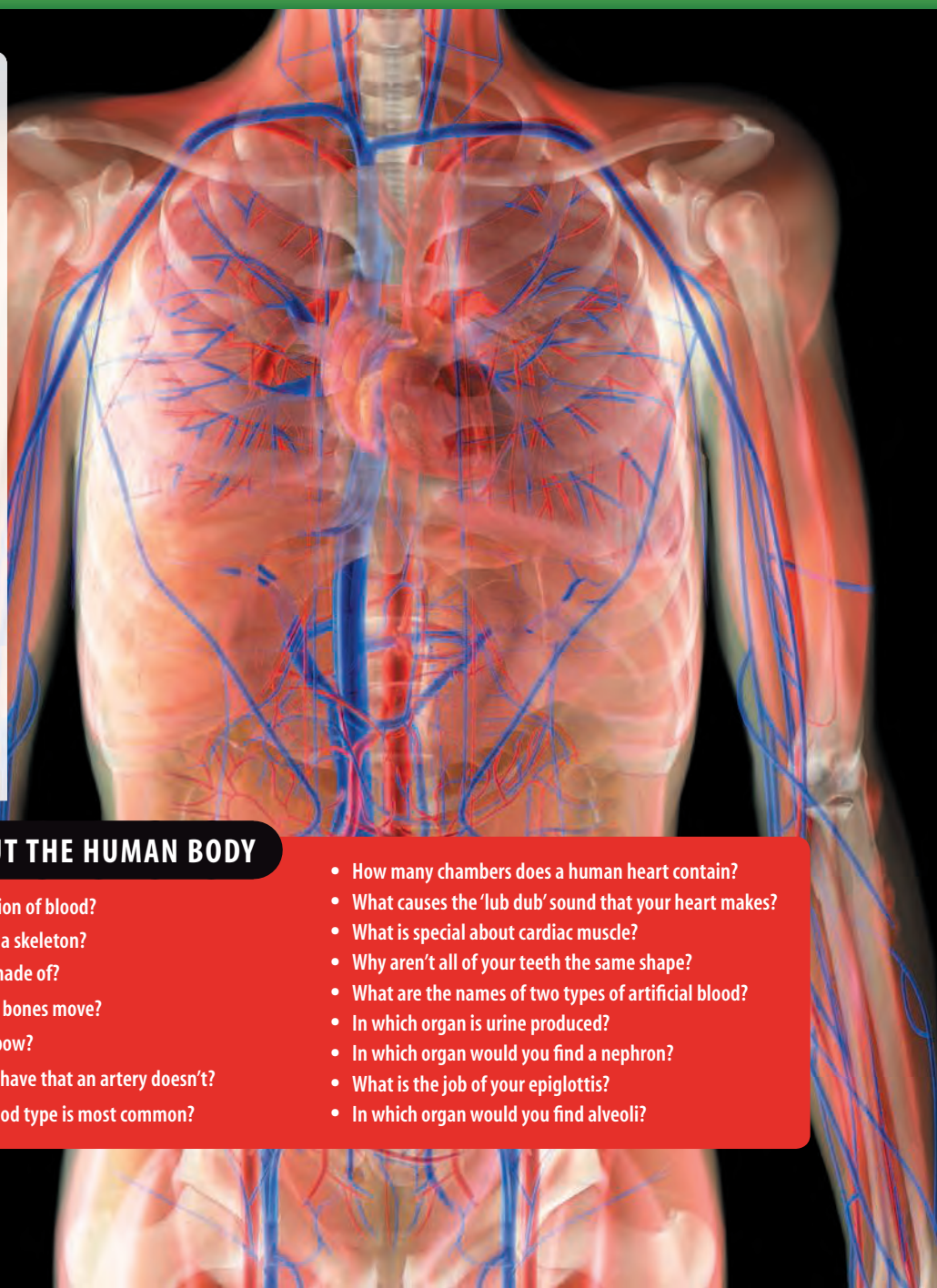
Elaborations

Identifying the organs and overall function of a system of a multicellular organism in supporting the life processes

Describing the structure of each organ in a system and relating its function to the overall function of the system

Examining the specialised cells and tissues involved in structure and function of particular organs

Comparing similar systems in different organisms such as digestive systems in herbivores and carnivores, respiratory systems in fish and mammals



Outside looking in — the human body

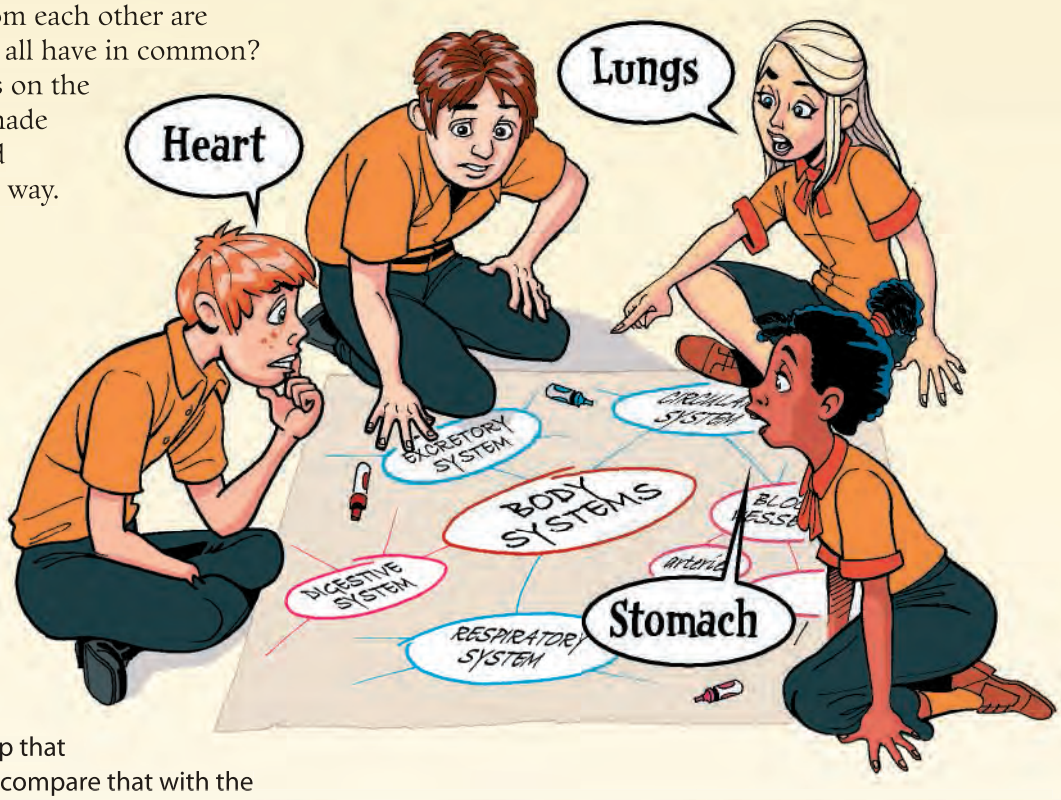
THINK ABOUT THE HUMAN BODY

- What is the function of blood?
- Why do you need a skeleton?
- What are bones made of?
- What makes your bones move?
- What is tennis elbow?
- What does a vein have that an artery doesn't?
- Which human blood type is most common?
- How many chambers does a human heart contain?
- What causes the 'lub dub' sound that your heart makes?
- What is special about cardiac muscle?
- Why aren't all of your teeth the same shape?
- What are the names of two types of artificial blood?
- In which organ is urine produced?
- In which organ would you find a nephron?
- What is the job of your epiglottis?
- In which organ would you find alveoli?

YOUR QUEST

Getting below the surface

Have a look at the other students in your classroom. How different from each other are you? Which features do you all have in common? Perhaps there are differences on the outside, but inside you are made up of all of the same bits and pieces organised in the same way.



THINK AND CREATE

Some of the things that you have in common with your other classmates are your body systems.

1 Use a mind map to summarise all that you know about human body systems.

2 Compare your mind map with those of at least three team members.

3 Create a new team mind map that combines all your ideas and compare that with the mind map of another team. Add any comments that you think help you learn more about human body systems.

THINK AND INVESTIGATE

4 In your team, make a list of ten questions about human body systems. Select four questions and place these on the class noticeboard with those of other teams to make a 'class question gallery'. Arrange these questions into groups or themes.

5 Browse the class question gallery and select one question that interests you most. Research your selected question and report back to the class on your findings in an interesting and creative way.

Driven by curiosity?

Imagine if your curiosity was so intense that you spent hour upon hour painstakingly exploring, sketching and recording layer upon layer of a rotting dead body among other decaying and quartered corpses in the 'dead of night'.

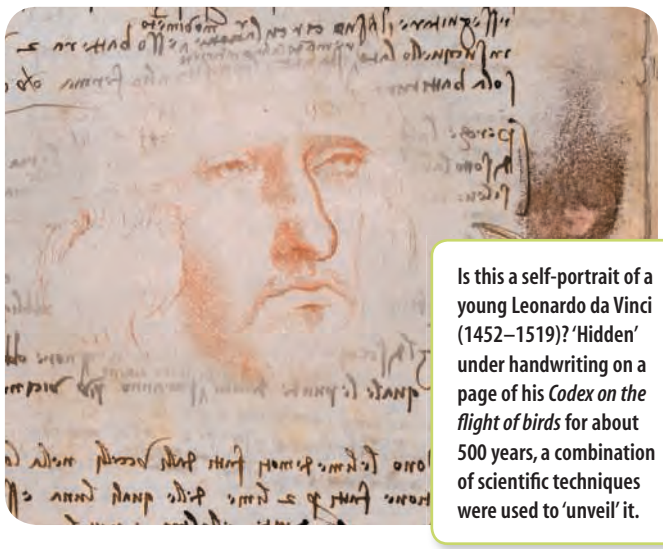
Leonardo da Vinci spent hours amid rotting corpses to draw amazingly detailed observations of body structures.

Intensely curious . . .

> . . . in the medical faculty he learned to dissect the cadavers of criminals under inhuman, disgusting conditions . . . because he wanted [to examine and] to draw the different deflections and reflections of limbs and their dependence upon the nerves and the joints. This is why he paid attention to the forms of even very small organs, capillaries and hidden parts of the skeleton.
>
> Paolo (the first biographer of Leonardo da Vinci), 1520

Leonardo da Vinci was arguably one of the best scientific minds of his time. He was intensely curious and painstaking in his observations. He used close observation, repeated testing and precise illustrations with explanatory notes. Using pen, chalk and brush, his scientific illustrations offered visual answers to mysteries that had escaped others for centuries. His volumes of amazing notes of scientific and technical observations in his handwritten scripts led to the birth of a new systematic and descriptive method of scientific study.

Leonardo da Vinci questioned everything. He may have been the most relentlessly curious man in history. He asked questions such as: Why do birds fly? Why can seashells be found in mountains? What is the origin of the wind and clouds? Why do people die? Where is the human soul found?



Is this a self-portrait of a young Leonardo da Vinci (1452–1519)? 'Hidden' under handwriting on a page of his *Codex on the flight of birds* for about 500 years, a combination of scientific techniques were used to 'unveil' it.

Dissecting, details and drawing

Leonardo's anatomical studies of human muscles and bones began around 1490. His exploration of embryology and cardiology came later, with his astonishingly detailed image of a fetus within the womb (around 1505) providing details for obstetric surgery hundreds of years later. His observations were not just of bodies — later generations have been in awe of his sketches of inventions that were centuries ahead of their time.

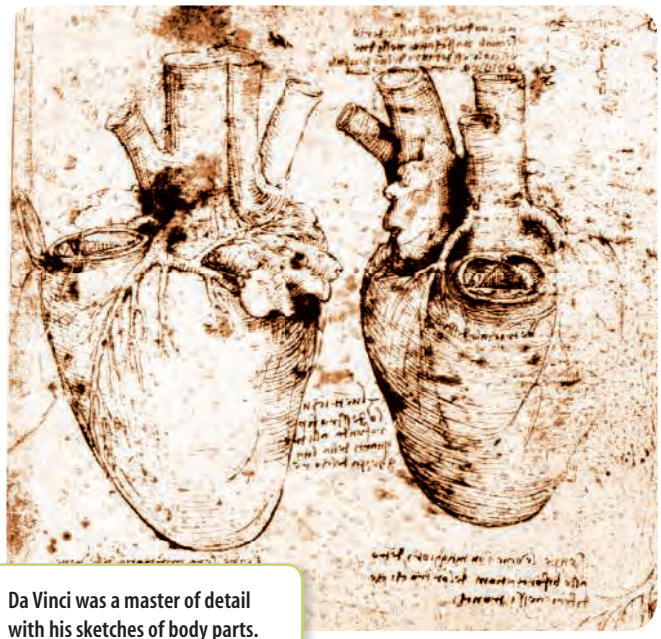
Challenging 'knowledge'

Knowledge of the human body was very different in Leonardo's day from what we accept today. The heart was thought to be made up of two chambers and its function to warm the blood, which was thought to be made in the liver. It was also thought that sperm were produced in the marrow of the spinal column and that the human soul may be located in the spine. Leonardo had questions he wanted to answer. He wanted to find out more. His investigations challenged the accepted knowledge of his day.

Visions and models

Leonardo also emphasised the significance of visual observations and model making — he believed that reality needed to be reconstructed before it could be represented. His models of hands or legs were used to reveal the structural relationships between different layers of arteries, muscles and bones. Leonardo also made a glass model of the heart and used

Leonardo drew this diagram around 1510 — can you see his secretive, reversed form of handwriting?



Da Vinci was a master of detail with his sketches of body parts.

water with different coloured dyes to trace its flow through the heart. His investigations linked anatomy (structure) and physiology (function).

Analogies are sometimes used to help people to connect new learning to previous knowledge. Leonardo used analogies to compare arteries in human bodies to 'underground rivers in the earth' and described the bursting of blood from a vein like 'water rushing out a burst vein of the earth'.

Leonardo's dissections led to changes in the knowledge and understanding about the structure and function of the heart, including that:

- the heart was a muscle
- the heart did not warm the blood
- the heart had four chambers
- left ventricle contractions were connected to the pulse in the wrist.

To locate cavities around the brain and cranium Leonardo used innovative techniques, such as injecting molten wax into them. Although Leonardo did not find the location of the human soul, his studies led him to the discovery that the brain and spine were connected.

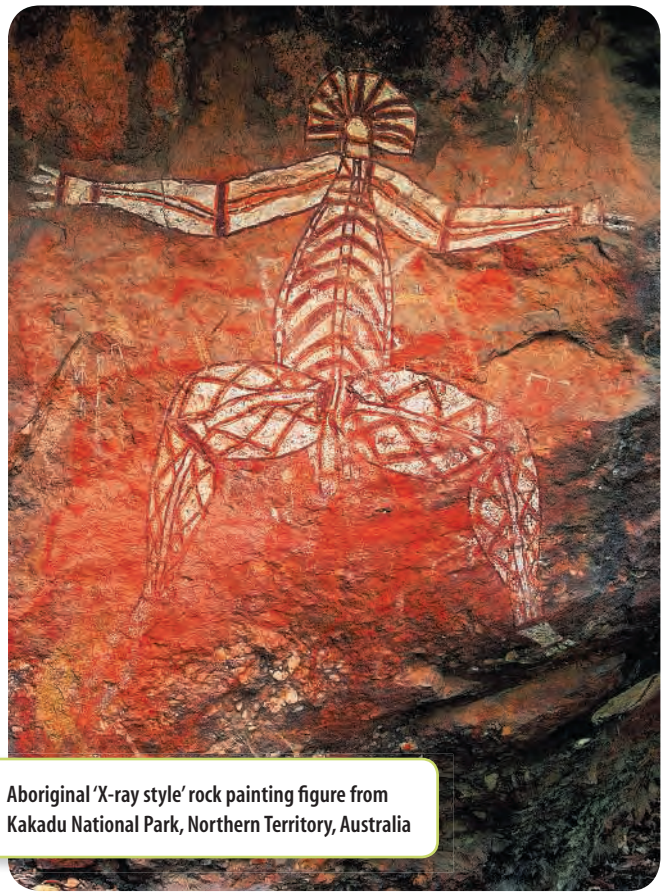
Leonardo's curiosity, determination, creativity and persistence did more than make an amazing contribution to our current scientific knowledge of our bodies. These features also helped mould the way in which scientific frameworks were developed to structure our investigations to explore our questions.

Curiosity throughout time and space

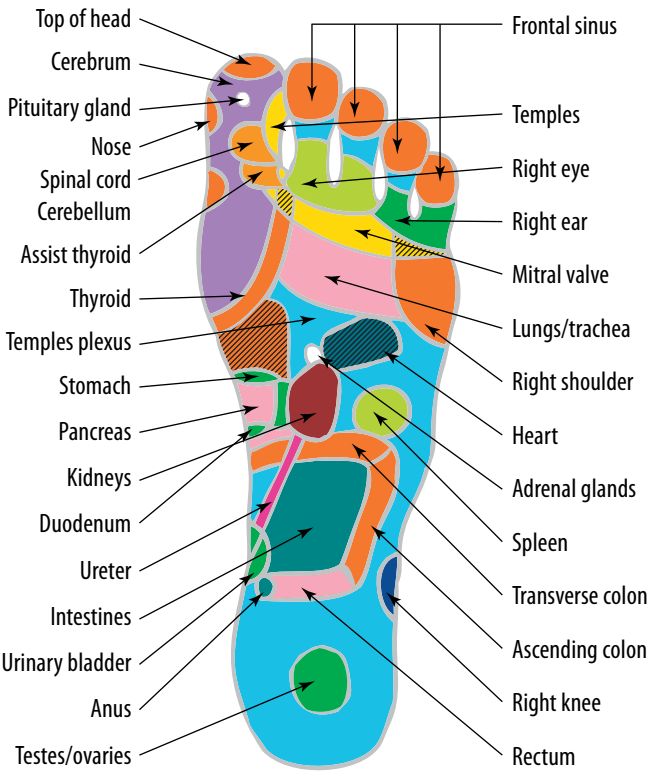
Curiosity is one of the features of humans that have contributed to our survival. Some of this curiosity has been about the structure and function of our own bodies. Evidence of this curiosity is woven throughout history and is often found in art. While Leonardo da Vinci provides one example of curiosity driving a search to find out more, it is not the only example. Nor is human curiosity limited to the place or time in which you live.

Knowledge of the internal biology and physiological process in art appears in rock paintings in caves in Australia that are thousands of years old. Examples of Aboriginal X-ray art also provide evidence that this type of knowledge dates back to more than 6000 years ago.

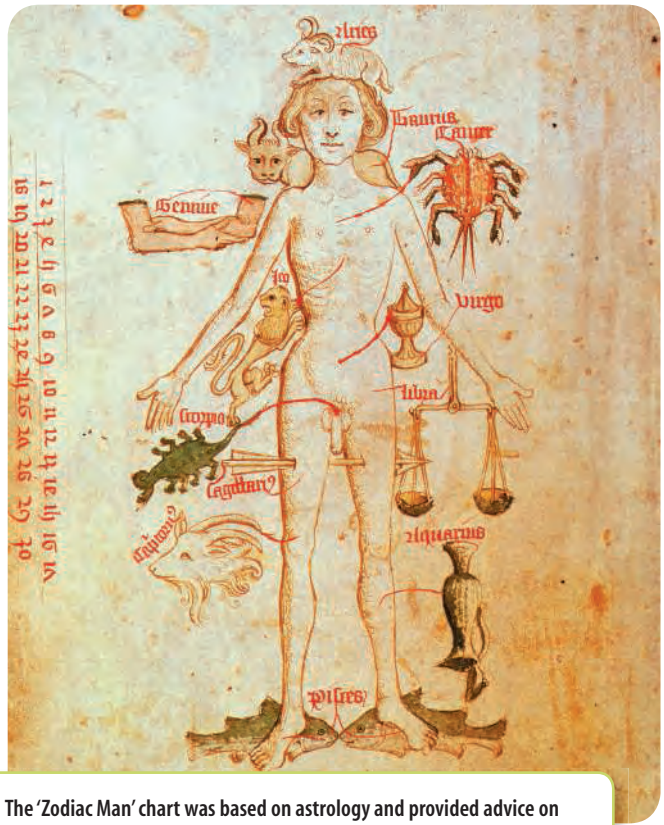
The culture and scientific knowledge of the times often determines the types of treatment given for various diseases of the human body. In medieval times, astrology played a key role in medicine and medical prognosis. It was believed that the 'movement of the heavens' could influence human physiology, with each part of the body being associated with a different astrological sign. An image of the 'Zodiac Man' in the medical texts of the time was used to assist practitioners in their medical treatments.



Aboriginal 'X-ray style' rock painting figure from Kakadu National Park, Northern Territory, Australia



An example of an acupressure reflexology chart



The 'Zodiac Man' chart was based on astrology and provided advice on when, for example, to 'bloodlet' (a medical treatment involving bleeding the patient), regulated by the position of the moon.

Chinese traditional medicine is an ancient medical system that has been practised for over five thousand years and applies understanding of the laws and patterns of nature to the human body. It views health as the changing flow throughout the body of vital energy (*qi*) that, if hindered, can lead to illness. Acupressure is an application of this practice that aims to release blocked energy by stimulating specific points along the body's energy channels.

Scientists are curious

Scientists are also often driven by the thirst to find answers to their questions. With increased technology and knowledge, the answers to these questions often result in even more questions.

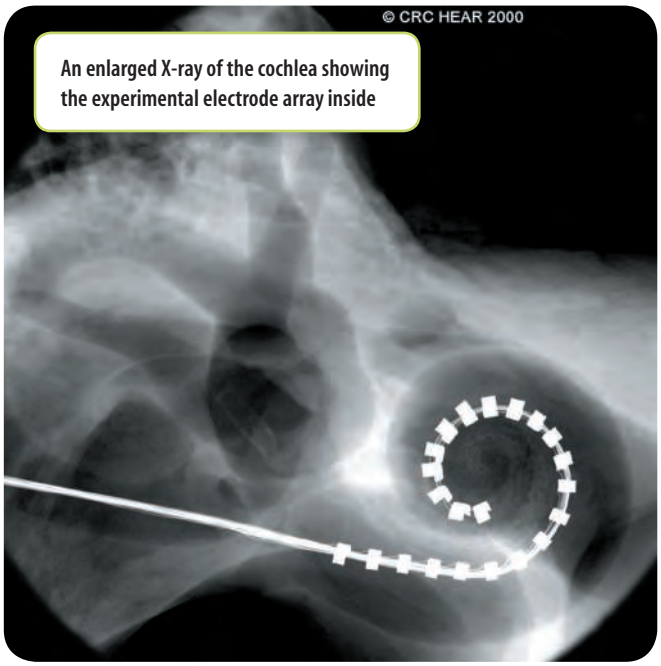
Compared with the situation in Leonardo's day, there are now an amazing number of different types of careers that involve investigations, explorations and applications of science to the human body. Australian scientists are involved in medical research and intervention. They are also involved in the invention and development of medical equipment that assists in the development and understanding of our body systems.

Australian scientists: creative inventors and explorers

Australian scientists have made significant contributions to medical discoveries and inventions. Howard Florey and his team discovered how penicillin could be extracted, purified and produced to be used as an antibiotic to help fight bacterial infections. Barry Marshall and Robin Warren showed that a certain type of bacteria caused stomach ulcers that could be treated with antibiotics. Professor Graeme Clark and his team were involved in the invention of an effective 'bionic ear'. Dr Fiona Wood pioneered a new treatment for burns in her development of spray-on skin that used the patient's own skin cells. Professor Ian Frazer developed the world's first vaccine against cervical cancer.

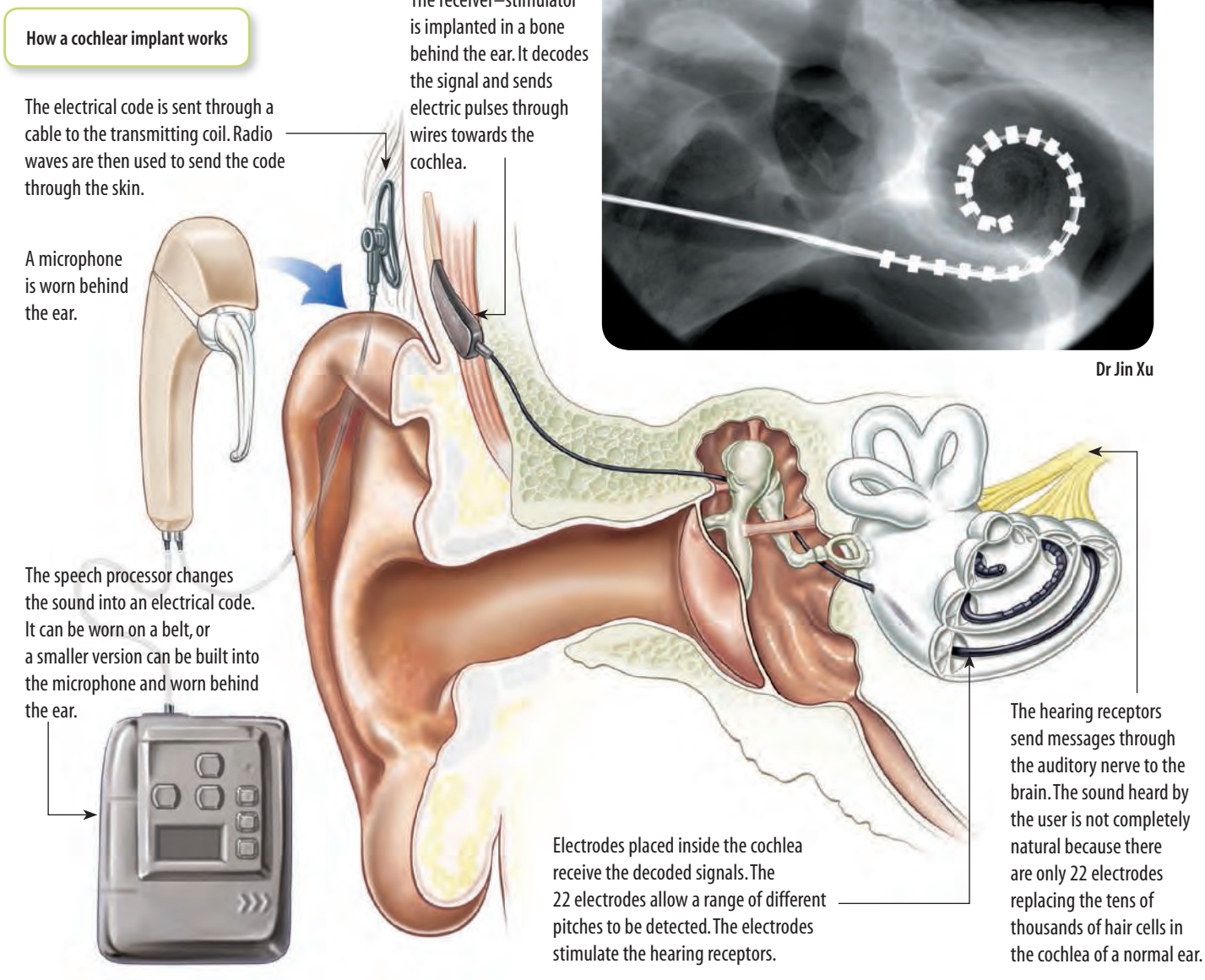
THE BIONIC EAR

The cochlear implant, also known as the bionic ear, has allowed some people with inner-ear problems to hear sound for the first time. When deafness results from serious inner-ear damage, no sounds are heard at all. Normal hearing aids, which make sound louder, do not help in these cases because the cochlea cannot detect the vibrations. However, the cochlear implant can often help by changing sound energy from outside the ear into electrical signals that can be sent to the brain.



An enlarged X-ray of the cochlea showing the experimental electrode array inside

Dr Jin Xu



DAVID UNAIPON

David Unaipon (1872 –1967) has been described as 'Australia's Leonardo'. He was born in South Australia, the fourth of nine children of James Ngunaitponi and his wife Nymbulda. Both of David's parents were Yaraldi speakers from the lower Murray River region.

Interested in Aboriginal mythology, philosophy and science, David was a preacher, author and inventor. He compiled his own versions of Aboriginal legends such as *Hungarrda* (1927), *Kinie Ger — The Native Cat* (1928) and *Native Legends* (1929). David's published poetry and legends pre-dated the work of other Aboriginal writers by over thirty years.

Obsessed with discovering the secret of perpetual motion, David made ten patent applications between 1909 and 1944 for inventions including a modified handpiece for shearing, a centrifugal motor, a multiradial wheel and a mechanical propulsion device.

David Unaipon

UNDERSTANDING AND INQUIRING

INVESTIGATE

1 Research and construct a model of an invention from one of Leonardo da Vinci's sketches.

2 Research examples of Leonardo da Vinci's inventions and make your own variation of one of them, presenting it as a series of annotated sketches.

3 Research an invention sketch by Leonardo da Vinci that is related to something that we use today. Use a tree diagram to show how it may be linked.

4 Find out what the *da Vinci*® Mitral Valve Repair is and suggest why it is named after da Vinci.

5 Find out about the history behind *Treatise on painting* and how it relates to science.

6 Find out more about *Codex on the flight of birds* and summarise your findings in a newspaper article.

7 Research three Australian scientists involved in medical research and intervention and present your findings as curricula vitae.

8 Find an example of how Australian scientists have been involved in the development of medical equipment. Produce a brochure to advertise this equipment to prospective buyers.

9 Suggest how scientific understanding of human body systems can determine how we respond to public health issues such as the 2009 swine flu pandemic.

10 Research traditional Chinese medicine and find out about the knowledge of the structure and function of human body systems. Present your findings in a scroll with wooden sticks at each end.

11 Research two of the following applications of traditional Chinese medicine:
- Qigong
- herbal therapy
- acupressure
- healing foods
- Chinese psychology.

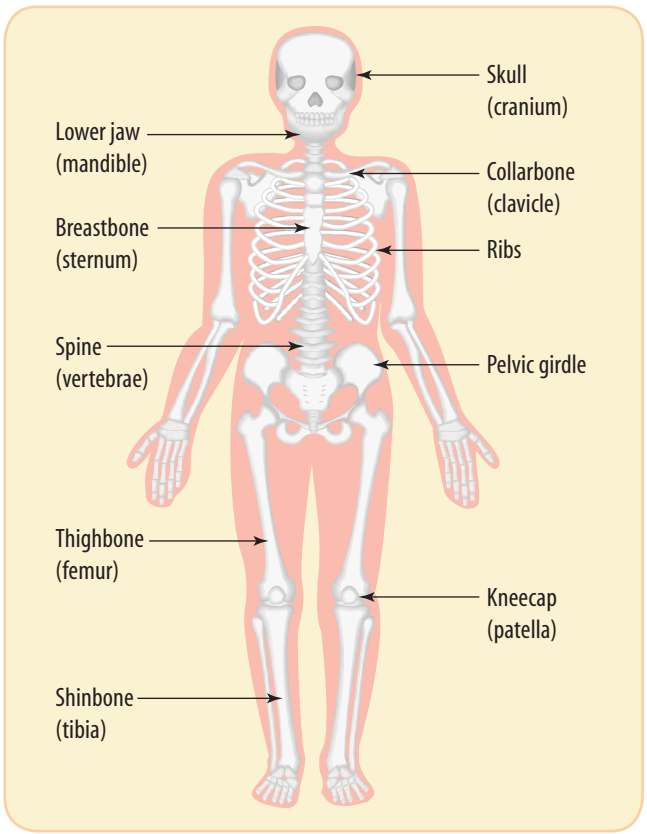
12 Research Aboriginal X-ray art to investigate examples of Aboriginal knowledge of the internal biology and physiological processes of animals.

13 The Warlpiri are one of the largest Aboriginal groups in the Northern Territory. Research and report on their traditional health system and the involvement of ngangkayikirili (or ngangkari or ngangkayi) and Yawulyu ceremonies of the Warlpiri.

14 Research and report on the processes involved in the preparation of mummies in ancient Egypt. Include what happened to specific body organs and why.

Keeping in shape

Without your skeleton you would resemble a jelly-like blob! Your **skeleton** provides support and forms a frame that gives your body shape. It also protects the organs inside your body and with the help of muscles allows you to move.



What's in a bone?

The **bones** that make up your skeleton are alive. They contain living cells and need a blood supply to provide oxygen and other nutrients. If bones were not alive, how would you grow taller? How would a broken arm or leg mend?

The soft **cartilage** that made up your skeleton when you were born is gradually replaced over the first twenty years of your life. Cartilage is very soft and rubbery — it is not as hard or solid as bone. Not all cartilage changes into bone. The ends of your bones remain covered in cartilage, and your trachea (windpipe), nose and ears are made mostly of cartilage.

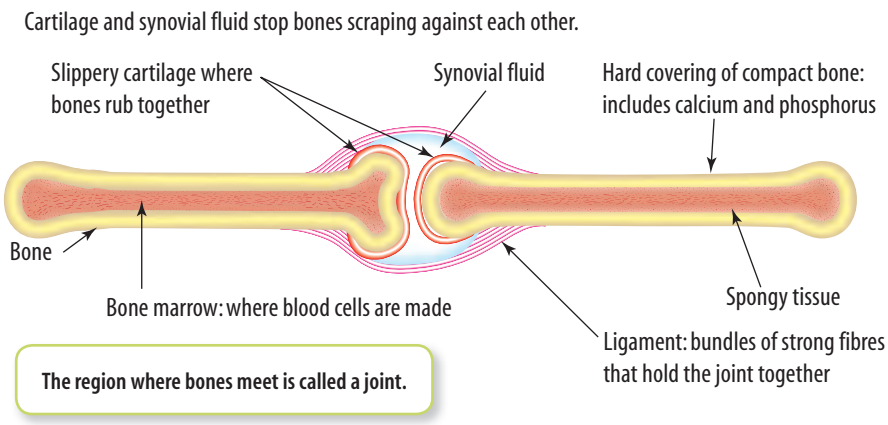
To remain hard, your bones need an adequate supply of two important **minerals**: **calcium** and **phosphorus**. Vitamin D is also important for healthy bone development and strength. Investigation 4.1 on the next page shows what could happen to your bones without a supply of these important minerals.

The hardening of your bones as you get older is called **ossification**. After ossification, the bone is made up of about 70 per cent non-living matter and 30 per cent living matter. As you get even older, your bones may get dry and **brittle**. That is why older people break their bones more easily.

Broken bones

When a bone breaks, the ends of the bone need to be put back into place (set) so that they can grow together. If a bone is shattered into several pieces, it is sometimes possible to use pins or wire to hold the pieces in place while the bone heals. A **greenstick fracture** occurs when the bone cracks but does not break. This type of **fracture** is common in children because their bones are more flexible.

New technologies are being researched and developed to help fix broken bones. Some of these involve special cells called **stem cells**, while others involve the use of special 'glues' that hold the bones together and aid the healing process. Scientists at CSIRO are currently working on a liquid gel called NovoSorb that glues the fractured bone together so



that it is supported while it heals. As this gel degrades naturally, it does not require follow-up surgery to remove pins as is needed with older technologies.

Osteoporosis

Osteoporosis is a loss of bone mass that causes bones to become lighter, more fragile and easily broken. It occurs in middle-aged or elderly men and women. In Australia, about 60 per cent of women and about 30 per cent of men are affected by osteoporosis. It is believed to be caused by lack of calcium in the diet.

In your teenage years, you can help protect yourself from getting osteoporosis later in life by having a healthy diet and exercising. Your diet should include dairy products such as milk, cheese and yoghurt and other foods high in calcium. Such a diet will help ensure that your bone mass is adequate as an adult.

Joints

The bones in your skeleton must be able to move so that your body can move. Bundles of strong fibres called **ligaments** hold your bones together. The region where two or more bones meet is called a joint. Most joints allow your bones to move. The amount and direction of movement allowed depends on the type of joint.

Some joints, such as those that join the plates in your skull, do not move. Such joints are called **immovable joints**. While not allowing movement, these joints provide a thin layer of soft tissue between bones. Their job is to absorb enough energy from a severe knock to prevent the bone from breaking.

INQUIRY: INVESTIGATION 4.1

Rubbery bones

KEY INQUIRY SKILLS:

- planning and conducting
- processing and analysing data and information

Equipment:

2 chicken or turkey bones *2 jars* *vinegar*

- Clean the two chicken or turkey bones and leave them to dry overnight. Place one bone in a jar of vinegar and the other in a jar of water.
- Allow the bones to soak for at least three days. Then remove the bones and observe any changes.

Vinegar is an acid and dissolves minerals such as calcium and phosphorus, removing them from the bone.

- Return the bone to the jar of vinegar for another week, then remove and observe any further changes in the bone. Try to tie the bone in a knot.

DISCUSS AND EXPLAIN

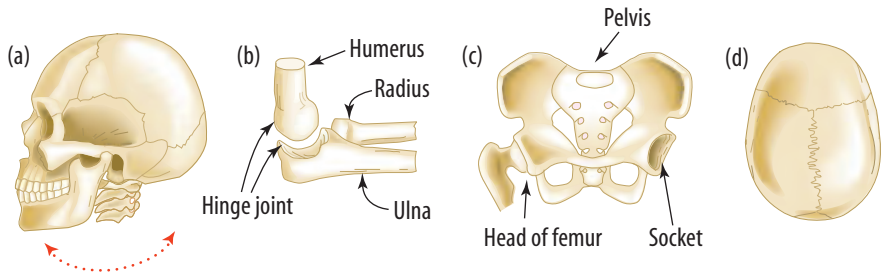
1 What changes occurred in each of the two bones?
2 How did the bone change after more than a week in vinegar?
3 Why was the jar of water used in the first part of this experiment?

Muscles

Tough, elastic fibres called **muscles** are connected to the bones of your skeleton by bundles of tough fibres called **tendons**. Muscles pull on bones by **contracting**, or shortening. Muscles never push.

CONTROL OF MUSCLES

The movement of muscles is controlled by the brain, which sends signals through your nerves.

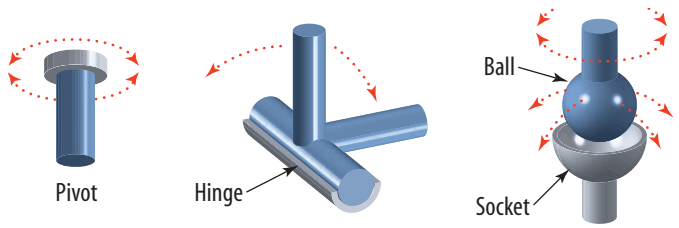


The joint between your skull and spine is a **pivot joint**, allowing a twisting type of movement.

Your knees and elbows are **hinge joints**, like those in a door, allowing movement in only one direction.

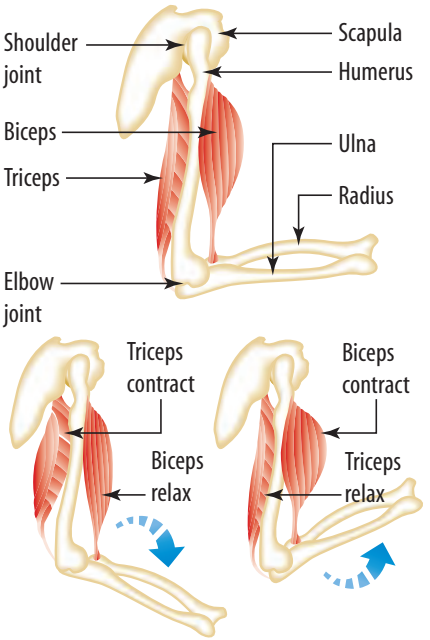
Your hip and shoulder joints are **ball and socket joints**, allowing movement in many directions.

Joints such as those between the plates of the skull are immovable joints.



Different types of joints:
(a) pivot joint
(b) hinge joint
(c) ball and socket joint
(d) immovable joint

Muscles such as those that make your heart pump and those that control your breathing are called **involuntary muscles**. They work without you having to think. The muscles that are connected to bones are called **voluntary muscles** because you have to choose to use them.



When your biceps contract, your arm bends upwards. When your triceps contract, your arm straightens.

OUCH!

- **Sprains** occur when ligaments joining bones at a joint are torn or stretched. Sprains usually happen when you fall onto a joint, such as an elbow or an ankle, and twist it.
- **Arthritis** is a swelling of the joints that makes movement difficult. Osteoarthritis occurs mainly in elderly people and is caused by the wear and tear of the joints. The cartilage gradually breaks down, thus allowing bare bones to grate against each other instead of sliding or turning smoothly. Rheumatoid arthritis is a swelling of the tissue between the joints. The swelling causes the joints to slip out of place, which then causes great pain and deformities.
- **Tennis elbow** is caused when the lining of the elbow joint swells and produces too much synovial fluid. The joint becomes swollen and painful. This occurs when the joint is used a lot and is most common in tennis players.

Torn hamstrings are a common but painful sporting injury.

- **Torn hamstrings** are a common sporting injury. The hamstring muscle joins the pelvis to the bottom of the knee joint, running along the back of the thigh. It controls the bending of the knee and straightening of the hips. A sudden start or turn in sport often stretches the hamstring muscle too far. It tears, causing great pain. Cold and unprepared muscles are more likely to tear. Proper warming up before strenuous sporting activity is one way of reducing the chances of tearing a muscle.

INQUIRY: INVESTIGATION 4.2

Chicken wing dissection

KEY INQUIRY SKILLS:

- planning and conducting
- processing and analysing data and information

Equipment:

chicken wing	*dissection tray or board*
scalpel	*newspaper*
scissors	*disposable gloves*

- Using the scissors and scalpel, gently pull away the skin from the chicken wing. Put the tip of the scalpel blade between the skin and the muscle to separate the skin from the muscle.
- When you have completely removed the skin from one joint, inspect it carefully. Follow each muscle near this joint from one end of the muscle to the other. Try pulling on the muscle. Can you get the bones to move by pulling on the muscle?
- Use scissors to cut through the joint. As you do so, look for tendons and shiny white cartilage.

DISCUSS AND EXPLAIN

1. Sketch one of the joints in the chicken wing. Label the bones, the tendons and the muscles. Show clearly where the muscle inserts (attaches to the bones). Use arrows to show how the bones move when the muscle is shortened.
2. Feel the cartilage with a gloved hand. Does the cartilage feel rough or slippery? Why does it need to be slippery?
3. Is cartilage harder or softer than bone?

INQUIRY: INVESTIGATION 4.3

Inside or out?

KEY INQUIRY SKILLS:

- planning and conducting
- processing and analysing data and information

Equipment:

2 cardboard tubes, each at least 30 cm long

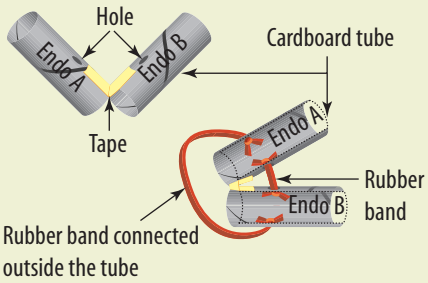
sticky tape

rubber bands

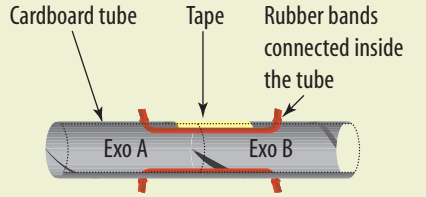
large nail or other pointed object

Being vertebrates, we have internal skeletons called endoskeletons. Many invertebrates (such as insects, spiders and crustaceans) have an external skeleton or exoskeleton. Make these models to investigate differences in how the muscles are joined to the skeletons.

- Cut each cardboard tube into two pieces about 15 cm long.
- Using the nail, make two holes on opposite sides of each tube. These should be about 5 cm from one end of each piece.
- Label two pieces 'Endo A' and 'Endo B' and the other two pieces 'Exo A' and 'Exo B'.
- Tape Endo A and Endo B together on one side, so that they form a hinge at the ends with the small holes.
- Cut two rubber bands and thread the cut ends through the holes from the outside.
- Tie knots so that the rubber bands can't pull back through the holes.
- Tape Exo A and Exo B together in the same way as Endo A and Endo B.
- Cut another two rubber bands and thread the cut ends through the holes so that they run *inside* the tube.
- Make sure that they are stretched very tightly, and then tie knots on the outside of the tubes.



The rubber bands are like the muscles in your arm. They are attached to the bones on either side of your elbow. The arm bends at the joint when the muscle contracts.



The rubber bands are like the muscles in an insect's limb. When a muscle contracts, the joint on which it operates straightens.

DISCUSS AND EXPLAIN

1. When one rubber band contracts, what happens to the one on the opposite side?
2. Draw sketches of each and record your observations when the joint is moved.
3. Describe how the two skeletons are different.

UNDERSTANDING AND INQUIRING

REMEMBER

1. Cover up the diagram of the human skeleton on page 119 and test your memory of the names of some of your important bones by completing the table below.

Scientific name	Common name
Vertebrae	
	Skull
Clavicle	
	Breastbone
Mandible	
	Thighbone
Patella	
	Shinbone

2. Describe the job done by each of the following parts of a joint.
   (a) Ligament
   (b) Cartilage
   (c) Synovial fluid
3. Some joints are referred to as immovable joints. What is the use of having joints that don't move?
4. Write down an example of each of the following types of joint.
   (a) Hinge
   (b) Ball and socket
   (c) Pivot
   (d) Immovable
5. Ligaments and tendons are bundles of tough fibres. What is the major difference between a ligament and a tendon?
6. Describe the action of the biceps and triceps muscles as you bend your elbow to raise your forearm.

THINK AND INVESTIGATE

7 Your **musculoskeletal system** consists of your **skeletal system** (bones and joints) and your **skeletal muscle system** (voluntary or striated muscle). Working together, these two systems protect your internal organs, maintain posture, produce blood cells, store minerals and enable your body to move.

Use information in this section and other resources to relate structural features to the functions of the following parts of these systems.

Part of system	Structural features	Function
Bones		
Cartilage		
Joints		
Skeletal muscles		

8. Find out more about the structure and function of skeletal, smooth and **cardiac muscle** tissue.

9 What is dietary rickets and how is it caused?

10 Look carefully at each of the skeletons below. Three of them are incomplete. Identify the incomplete skeletons and name the missing parts.

11 Apart from warming up just before a game, how do the best basketball and netball players reduce the likelihood of torn muscles and tendons?

12 What would happen if the cartilage in your knee joint wore out?

13 Research and report on one of the following science careers: orthopaedic surgeon, physiologist, physiotherapist, occupational therapist, rheumatologist, fitness trainer.

INVESTIGATE AND DISCUSS

14 (a) In teams of four, use a 'lucky dip' system to match each member of the team with one of the following topics.
- Deviated septum
- Broken nose
- Nose cancer
- Rhinoplasty

(b) Each member should investigate their topic and report their findings to their team.

(c) As a team, construct a cluster map or mind map to summarise the key points of your team findings.

(d As a team, select one of the four topics for further research. Brainstorm questions for your selected topic and research these questions.

(e) Report your team's findings to the class.

CREATE

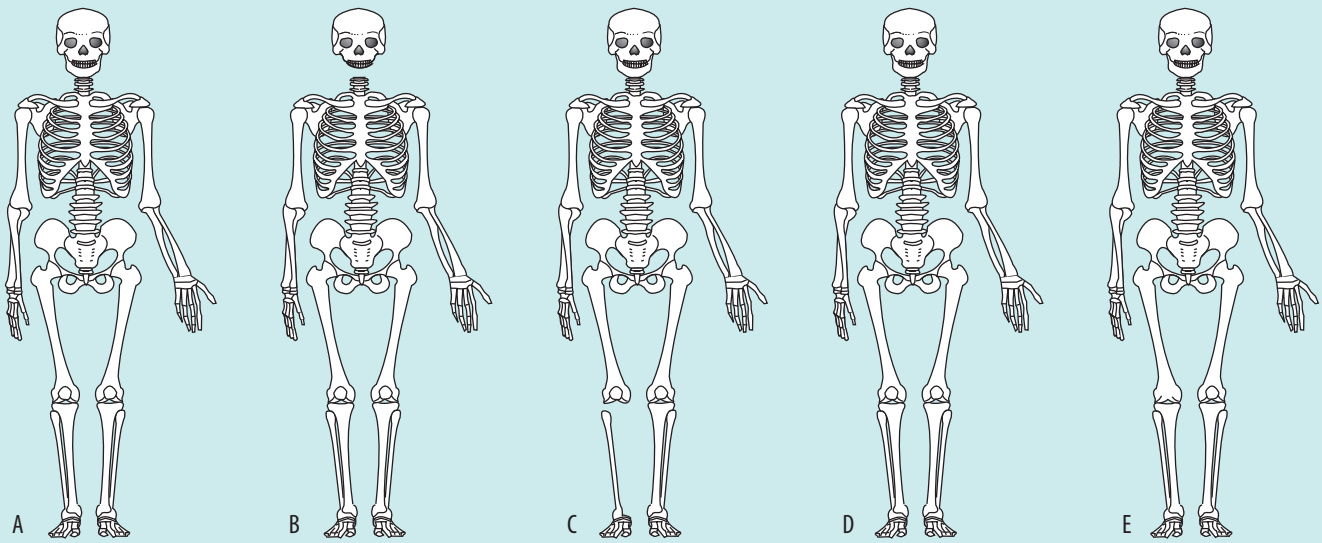
15 Make a skeleton mobile to hang from the ceiling.

(a) Trace the skeleton diagram on page 119 (or a larger one from another book), colour it and cut it into a number of sections.

(b) Paste each section onto cardboard and thread the sections together to make a skeleton mobile.

16 Use a cut-out human skeleton to make a new, imaginary animal by rearranging the bones. Suggest a name and describe the lifestyle of your animal.

17 Make a working model of an arm to show how the biceps and triceps work. Use the illustration on page 121 as a guide. Materials you might use include icy-pole sticks or stiff cardboard (for bones), split pins (for ligaments), string or rubber bands (for muscles), polystyrene foam (for cartilage) and glue. Draw a labelled diagram of your model.



Identify the incomplete skeletons and name the missing parts.

work sheet 4.1 Bones, joints and muscles

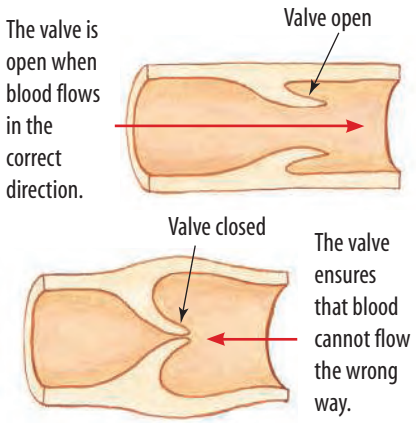
Blood highways

If you cut yourself, you bleed. The red liquid that oozes, trickles or bursts from your body is called **blood**.

Blood by the bucketful

An average-sized human has about five litres of blood; that's about a bucketful. It travels around the body in tubes called **blood vessels**. If these vessels were laid end to end, they would encircle the Earth two and a half times. These tubes enable materials in your body to be transported from one place to another.

Some of these blood vessels are called **arteries**. They have thick, elastic, muscular walls and carry blood under high pressure away from your heart. Some other vessels are called **veins**. They have thinner walls and valves that prevent the blood from flowing backwards. Veins carry blood to the heart.



Veins have valves to ensure that blood flows in only one direction.

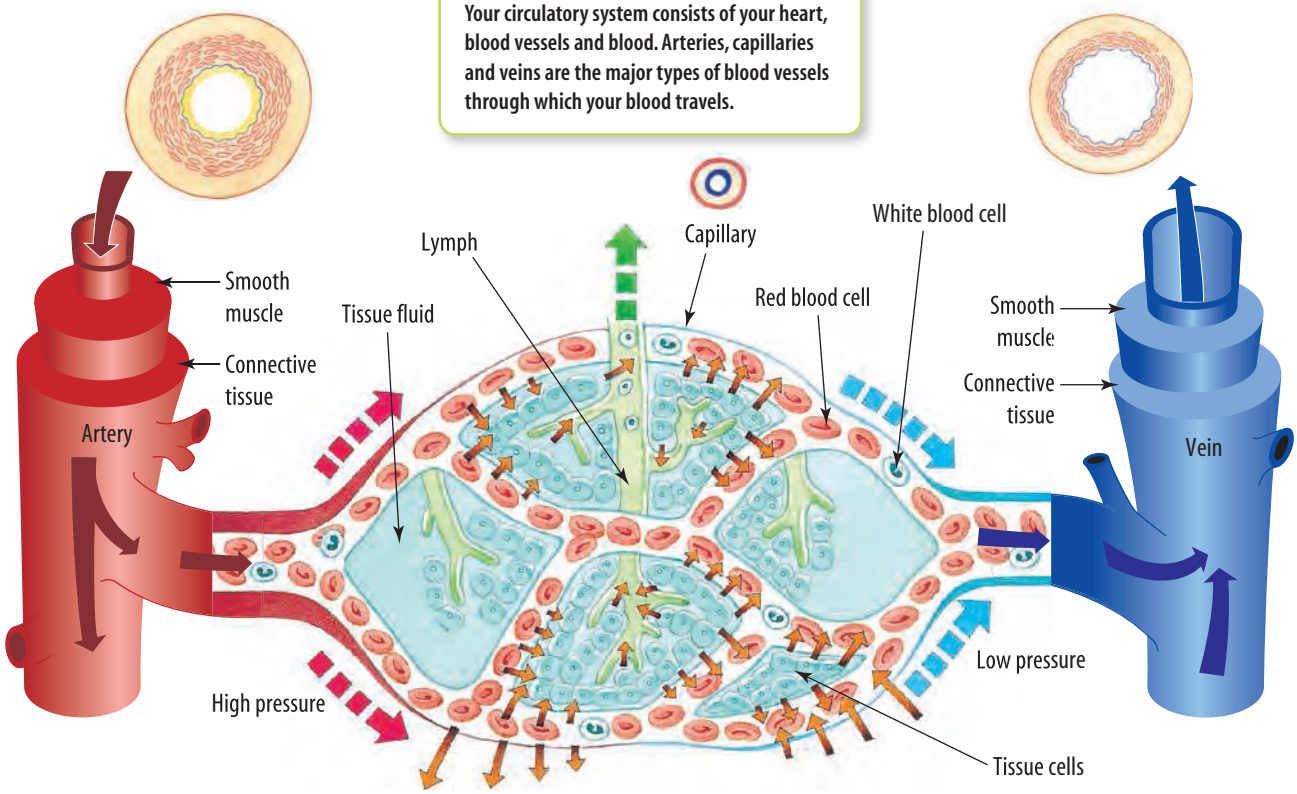
The most numerous and smallest blood vessels are called **capillaries**. Your body contains about 1 000 000 km of capillaries, which penetrate almost every tissue. No cell is very far away from one. Capillaries are very important blood vessels because they carry materials such as oxygen and nutrients to the cells and remove wastes such as carbon dioxide.

What's in blood?

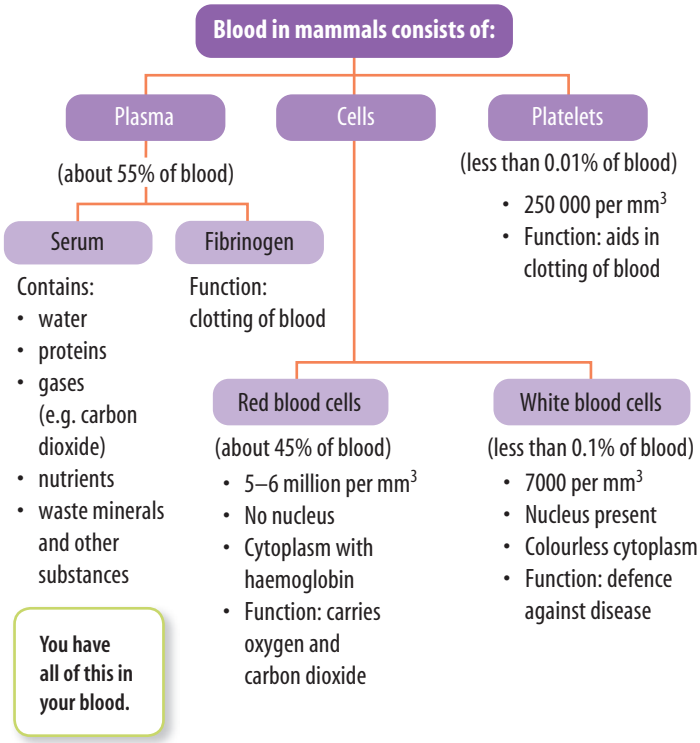
Blood is made up of red blood cells (**erythrocytes**), white blood cells (**leucocytes**), blood platelets and the straw-coloured fluid that they all float in, called **plasma**.

IN A DROP OF BLOOD

Each drop of blood contains about 300 million red blood cells.



Your circulatory system consists of your heart, blood vessels and blood. Arteries, capillaries and veins are the major types of blood vessels through which your blood travels.



These red blood cells travel around the body up to 300 000 times, or for about 120 days. After this they literally wear out and die. Each second you are manufacturing about 1.7 million replacement red blood cells in your bone marrow.

There is a pigment containing iron in the red blood cells, called **haemoglobin**. This is what gives them their reddish colour. The job of haemoglobin is to carry oxygen around your body. When haemoglobin is carrying oxygen it is called **oxyhaemoglobin**. Red blood cells can pack in a lot of haemoglobin because they have no nucleus taking up room, and also because of their special shape.

Someone who has an iron deficiency may become **anaemic**. In this illness, less haemoglobin is carried in the blood. Because less oxygen is available to the cells, the person becomes tired and lethargic.

Blood barriers

Blood also helps us to prevent and fight diseases. White blood cells are the 'soldiers' in the blood. They eat bacteria and other foreign matter and help us to fight infections. Platelets in blood help to heal cuts. They form sticky clots that plug the damaged blood vessel. This seals the cut to prevent germs getting in.

Mix and match?

How much do you know about the red stuff that flows throughout your body? Did you know that your blood might not mix too well with that of your friends? Blood can be grouped into eight types using the ABO system and the Rhesus (Rh) system. Your blood type is inherited from your parents.

These classification systems are based on whether particular chemicals (antigens) are present or absent on your red blood cells. If you are Rh-negative, you do not have the Rhesus factor on your red blood cells; if you do, you are Rh-positive.

INQUIRY: INVESTIGATION 4.4

Viewing blood cells

KEY INQUIRY SKILLS:

- planning and conducting
- processing and analysing data and information

Equipment:

prepared slide of blood smear

microscope

- View the prepared slide under the microscope on high power.
- Find a white blood cell on the slide.

DISCUSS AND EXPLAIN

1. Sketch a few red blood cells and one white blood cell.
2. Estimate how many red blood cells would fit inside a white blood cell.
3. Estimate the number of red blood cells that can fit across the field of view.

HOW ABOUT THAT!

If you bump yourself but haven't cut your skin, a bruise may form. Bruises are caused by burst blood capillaries under your skin. The bruise changes from black to purple to yellow as the blood clears away.

The ABO system divides blood into groups A, B, AB and O. If you need a blood transfusion, it is very important to know your blood type and that of the donor because some blood types cannot be mixed. If the wrong types are mixed, the blood cells may clump together and cause fatal blockages in blood vessels.

HOW ABOUT THAT!

Insect blood looks a little like raw eggwhite, because it contains no pigment. The blood of crabs and crayfish, however, contains the pigment haemocyanin. This pigment has copper in it and is blue when combined with oxygen. This is different from haemoglobin in humans, which is red when combined with oxygen.

Patterns, order and organisation

Multicellular organisms contain organised systems of organs that perform specialised functions that enable them to survive. Your circulatory system, for example, contains organs (such as your heart and blood vessels) that are involved in transporting substances in your blood to where they are needed and wastes are removed.

Multicellular organisms show a pattern of organisation in which there is an order of complexity.

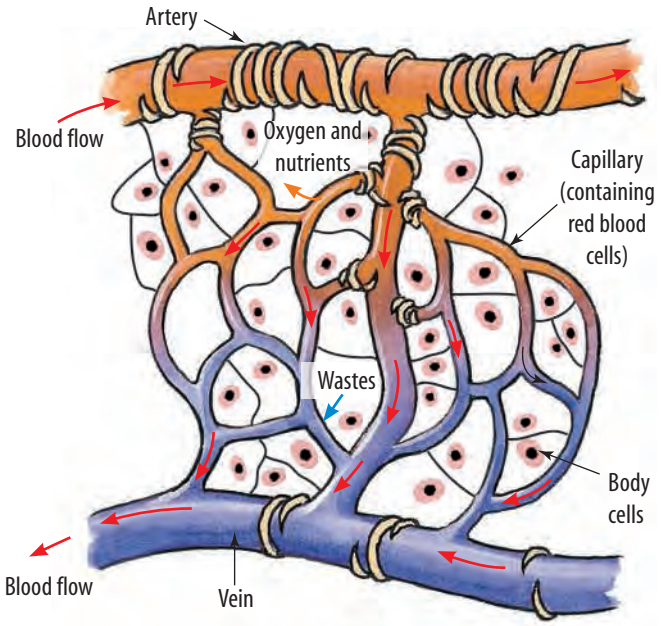
HOW ABOUT THAT!

Could the study of the green stuff in snot help treat heart disease? Research by Australian scientists at the Heart Research Institute in Sydney suggests that this may be the case. Inflammation of the artery wall is a key factor in a type of heart disease called atherosclerosis. This inflammation is caused by an inappropriate defence system response in which activated white blood cells release myeloperoxidase, the enzyme that gives nasal mucus and pus its green colour. The research team is investigating inhibitors to control this enzyme. Their research may result in the discovery of new drugs that could treat not only heart disease but also other inflammatory diseases such as asthma, rheumatoid arthritis and some cancers.

William Harvey published his ideas on how the blood circulated through the body in 1628.

Connected pathways

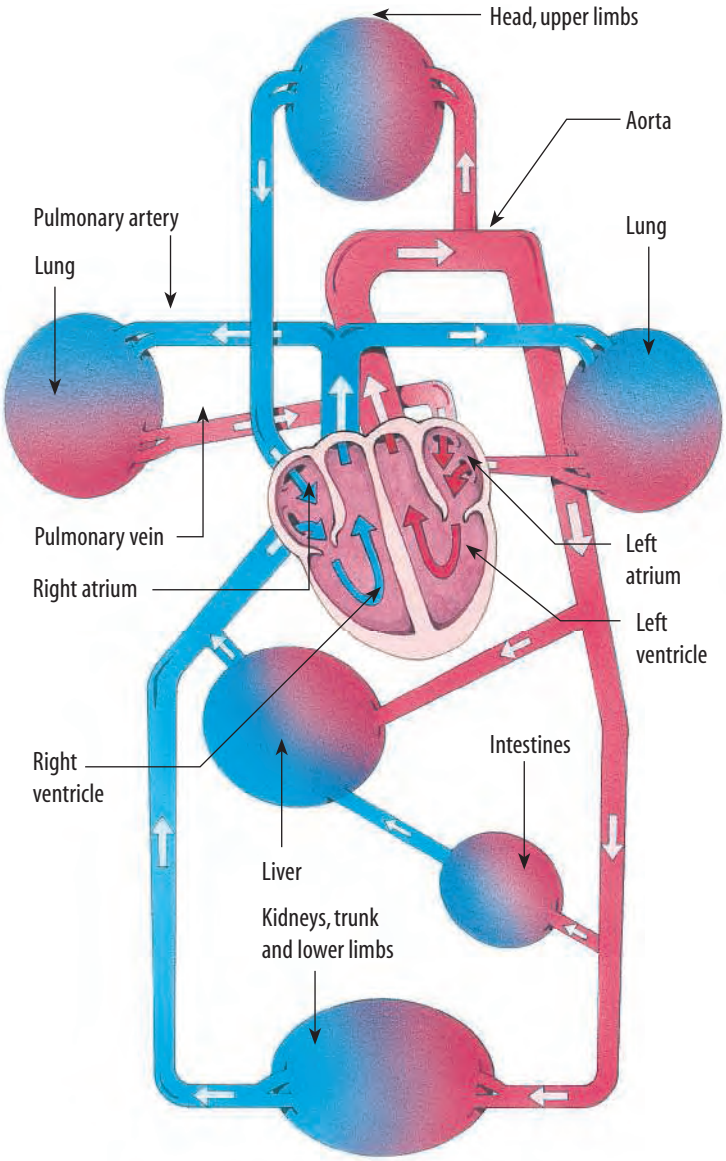
Arteries take blood away from your heart and veins take blood towards your heart, but in between these are capillaries. Capillaries have a very close relationship with your cells. It is through the walls of these tiny blood vessels that oxygen and nutrients move into your cells from your blood, and waste products, such as carbon dioxide, are moved out into your blood.



In the capillaries, oxygen diffuses out of the blood and waste produced by cells diffuses into the bloodstream.

Connected highways

Substances in your blood travel through your blood's busy highway system to areas for use, storage or removal. Blood carries carbon dioxide, a waste product from cellular respiration, to your lungs so that it is removed when you exhale. A variety of metabolic wastes also travel in blood to your kidneys so that they may be excreted through your urinary system.



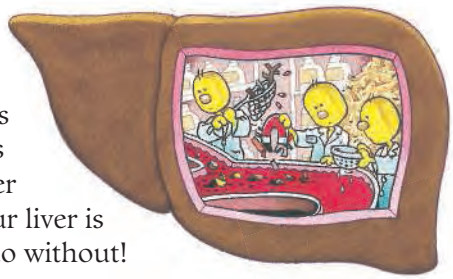
Connected highways — the routes for blood circulation

eBookplus Weblink

Use the **Interviews with Australian scientists** weblink in your eBookPLUS to find out more about the work of vascular biologist Professor Julie Campbell.

Not without my liver I don't!

Over a litre of blood passes through your liver each minute. Your liver is like a chemical factory, with more than 500 different functions. Some of these include sorting, storing and changing digested food. It removes fats and oils from the blood and modifies them before they are sent to the body's fat deposits for storage. It also helps get rid of excess protein, which can form toxic compounds dangerous to the body. The liver converts these protein wastes into urea, which travels in the blood to the kidneys for excretion. It also changes other dangerous or poisonous substances so that they are no longer harmful to the body. Your liver is something you cannot do without!



Gassy blood

Your body is more sensitive to changes in levels of carbon dioxide than of oxygen. If there is too much carbon dioxide in your body, it dissolves in the liquid part of blood and forms an acid. The resulting acidic blood can affect the functioning of your body.

The amount of carbon dioxide in your blood influences your breathing rate. The level of carbon dioxide in the blood is detected by receptors in the walls of some arteries and in the brain. If the level of carbon dioxide in your blood increases, your breathing rate will be increased so that carbon dioxide can be exhaled from your lungs and passed out of your body.

If you were to climb up high on a mountain, you would need time for your body to adjust. Initially you would feel tired and out of breath because you would be restricted by the limited amount of oxygen available to your cells. Your breathing and heart rate would increase, in an effort to get more oxygen around your body. In time, your body would begin to produce more red blood cells and hence more haemoglobin. After this, your breathing and heart rate would return to normal.

HOW ABOUT THAT!

The amount of oxygen carried by haemoglobin varies with altitude. At sea level, about 100 per cent of haemoglobin combines with oxygen. At an altitude of about 13 000 metres above sea level, however, only about 50–60 per cent of the haemoglobin combines with oxygen.

UNDERSTANDING AND INQUIRING

REMEMBER

1 Outline what blood is, and what it does.
2 Name and describe the types of blood vessels in which blood travels around your body.
3 Compare red blood cells, white blood cells and blood platelets.
4 Explain the advantage of not having a nucleus.
5 Explain why haemoglobin is important.
6 Suggest why blood is not the same colour in all animals.
7 Suggest why our blood system is described as a 'transport highway'.
8 Describe the relationship between arteries, capillaries and veins.
9 Outline the route that carbon dioxide travels to get out of your body from your cells.
10 Explain why you can't live without your liver.
11 Is your breathing rate more sensitive to levels of carbon dioxide or of oxygen in the blood? Explain.
12 Explain why mountain climbers sometimes find it difficult to breathe during a climb.

THINK

13 Use information in this section and other resources to relate structural features to the functions of the following parts of the circulatory system.

Part of system	Structural features	Function
Arteries		
Veins		
Capillaries		
Red blood cells		

14 Think of other ways that information about the components of blood could be organised visually. Organise the material in one of these ways.

THINK AND CREATE

15 (a) Copy the 'connected highways' diagram on the previous page into your workbook.
(b) Use a coloured pencil to show the path taken for a red blood cell to travel from the pulmonary vein to the pulmonary artery, if it goes via the intestines.
16 Mark the following sites (a, b, c, d) on your diagram. In which blood vessel(s) would you expect the highest:
(a) blood pressure?
(b) blood glucose levels?
(c) blood carbon dioxide level?
(d) oxygen level?
17 List the following in the order that a red blood cell would reach them after leaving the aorta: pulmonary artery, left ventricle, right atrium, intestine, lung, pulmonary vein, left atrium, liver, right ventricle.
18 Convert your classroom or sports oval into a 'circulatory highway system'. Pretend to be a red blood cell and travel along the route it would take around the body.
19 Divide the class into groups, with each group making a model of a type of blood cell or vessel. Combine all of the models to convert the room into a circulatory system. In teams, write a play to use your circulatory system. Act it out for the class or get the class to act it out for you.

THINK AND DISCUSS

20 (a) Some people have religious grounds for disagreeing with the use of blood transfusions. Imagine a four-year-old child with a life-threatening condition. Her parents will not allow her to have the blood transfusion that she needs. What should the doctors do? Discuss this with your team and report your decision to the class. If there are any differences of opinion, organise a class debate on the issue.
(b) Would your response be different if the child was 18 years old and wanted the blood transfusion, but her parents would not allow it?
21 A day after donating blood, a person finds that they have an infectious disease that can be transmitted by blood. What should they do? Discuss this with your team, giving reasons for your opinions.

INVESTIGATE

22 The higher the altitude, the less oxygen there is in the air. Propose a reason people living at high altitudes usually have more red blood cells than people living at low altitudes.
23 Research one of the following circulation topics and report your findings to the class: blood transfusions, rhesus babies, varicose veins, leukaemia, haemophilia, thrombosis, embolisms, aneurisms.
24 Find out more about how blood circulates in insects and lobsters.
25 Plasma is the liquid part of blood. It can be used to make a number of products to help others in need. Report back on the uses of three of the following: Intragram, Anti D, Albumex 20, Factor VIII, Monofix, Prothrombinex, Thrombotrol VF, normal immunoglobulin, hyper-immunoglobulin.
26 Examine prepared blood slides under a light microscope.
27 Construct a bar graph to show the proportions of the different parts of blood.

28 Find out what happens when people donate their blood at a blood bank. How often can you donate blood, how long does it take and how much blood do they take? Prepare a brochure, storyboard, PowerPoint presentation or cartoon to share your findings.

29 Use the internet to research waste removal and excretion for an animal of your choice. Present your findings in PowerPoint or as a poster.

USING DATA

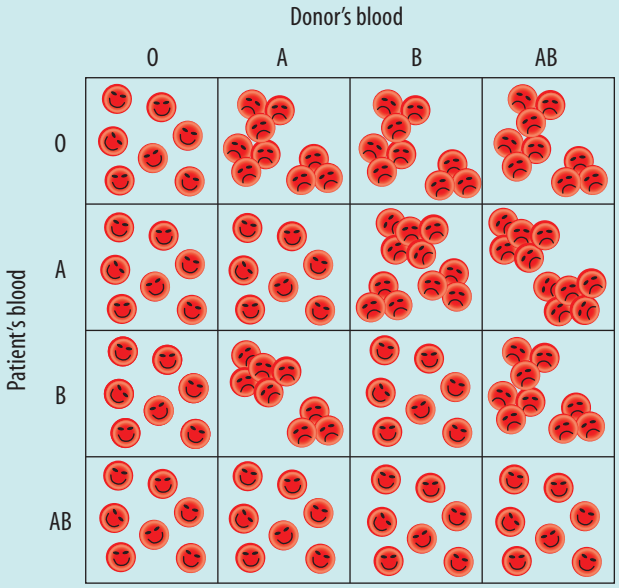
Use the table and illustration below to answer the following questions.

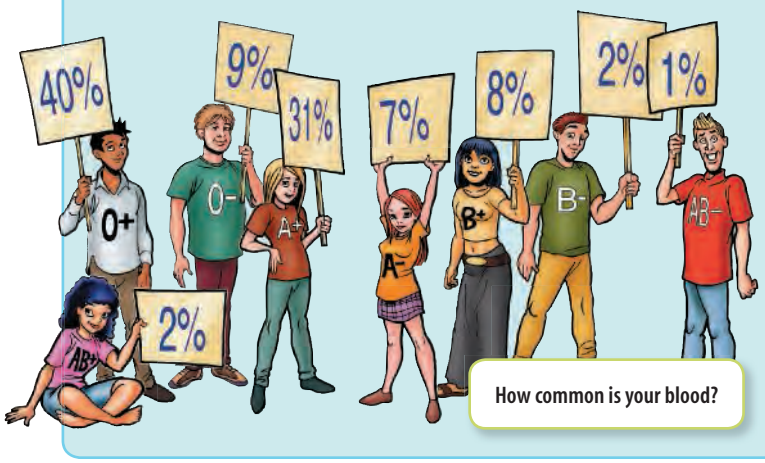
30 Which blood type is the most common? Which is the least common?

31 Which blood group(s), A, B, AB or O, can be accepted by:
(a) all blood groups?
(b) blood group AB?
(c) blood group A?

32 Which blood group, A, B, AB or O, can receive transfusions from all blood types?

33 Convert the information in the table below into a Venn diagram, target map or another visual thinking tool.





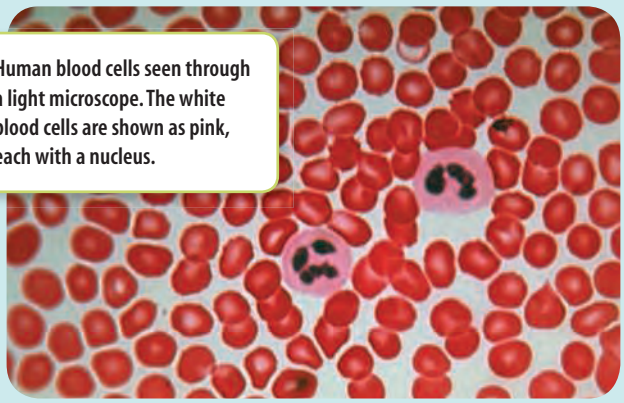
How common is your blood?

INVESTIGATE, THINK AND DISCUSS

34 Imagine that you have a friend who is anaemic. She is constantly tired and very pale.
(a) Using the internet and other resources, find out what you could do to help her improve her health.
(b) Report back to your team, sharing your ideas and any other relevant information. Have your team scribe summarise your ideas in a cluster map or mind map.
(c) As a team, decide on a strategy for helping your anaemic friend.
(d) Share your strategy with other teams as a mind map, flowchart, concept map or another visual tool.
(e) Reflect on what you have learned during this activity. How might it influence your future behaviour or thinking? Could any of the strategies designed by the groups be used to solve any other problems? If so, which ones?

35 With a partner, construct a PMI chart for a law that makes it compulsory for everyone over 16 to donate blood at least once a year.

36 Observe the image of human blood cells.



Human blood cells seen through a light microscope. The white blood cells are shown as pink, each with a nucleus.

(a) Identify which are white blood cells and which are red blood cells.
(b) Describe how you distinguished between the two types of blood cells.
(c) Which are in the greatest abundance? Suggest a reason for this.

37 Dr Mary Kavurma and Dr Seana Gall are Tall Poppy Science Award winners. This award recognises young scientists who excel at research, leadership and communication. Dr Kavurma is a scientist at the University of New South Wales involved in research into atherosclerosis and cardiovascular disease. Dr Gall is based at the Menzies Research Institute, University of Tasmania, and her research field is cardiovascular epidemiology.
(a) Find out more about their research and that of other scientists in this field of science.
(b) Find out more about Australia's Tall Poppy Science Awards and other winners.

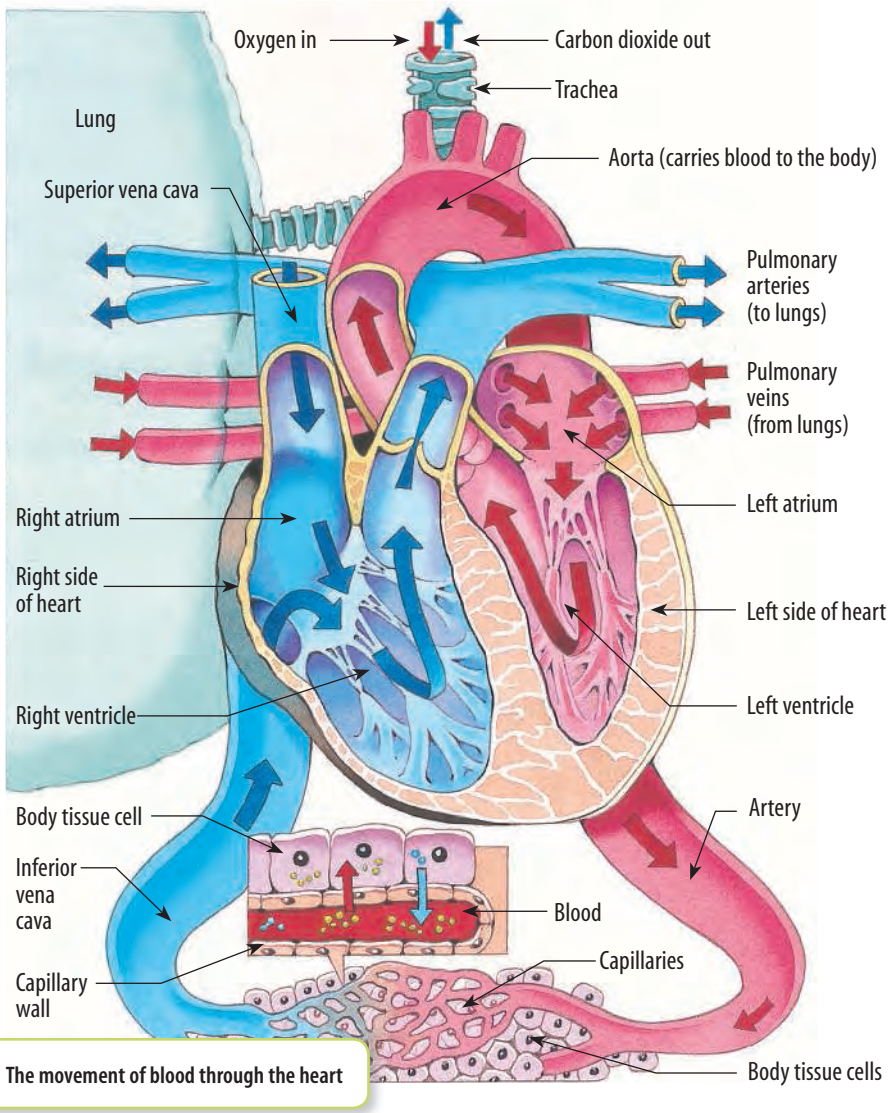
4.4 OVERARCHING IDEA

Form and function: Have a heart

Often linked with emotions, love and courage, the heart has a special meaning for most of us. In a clinical sense, however, it is merely a pump about the size of your clenched fist.

Two pumps in one

To be more precise, the human heart is actually *two* pumps. One side contains oxygenated blood and the other deoxygenated. Your veins bring 'used' **deoxygenated** blood (stripped of oxygen and bluish in colour) back to your heart. All of these veins join up into a larger vein called the **vena cava**. Entering the top right chamber of your heart, blood is pumped into the bottom right chamber. It is then pumped out to your lungs where it picks up oxygen and becoming **oxygenated** and more reddish in colour. It also loses some of the carbon dioxide from it. The oxygenated blood then returns to the left-hand side of your heart to be pumped out through arteries to your body tissues, where it delivers oxygen and nutrients. The deoxygenated blood then returns to the right-hand side of the heart for the cycle to be repeated.



The movement of blood through the heart

Four chambers

The human heart has four chambers. The upper two chambers are called the **left atrium** and **right atrium** (plural = atria), and the lower two chambers are the **left ventricle** and **right ventricle**. The two sides of the heart are different. The walls of the left side are thicker and more muscular because they need to have the power to force the blood from the heart to the rest of the body.

Flap-like structures attached to the heart walls, called **valves**, prevent the blood from flowing backwards and keep it going in one direction. If you listen to your heart beating, you will hear a **'lub dub'** sound. The 'lub' sound is due to the valves between the ventricles and atria shutting. The 'dub' sound is due to the closing of the valves that separate the

eBookplus eLesson

Heart valve

Watch this video eLesson to learn more about replacing heart valves without the substantial trauma of open heart surgery, a procedure set to revolutionise cardiovascular medicine for patients.

eles-0858

heart from the big blood vessels that lead to the lungs and the rest of the body.

Blood pressure

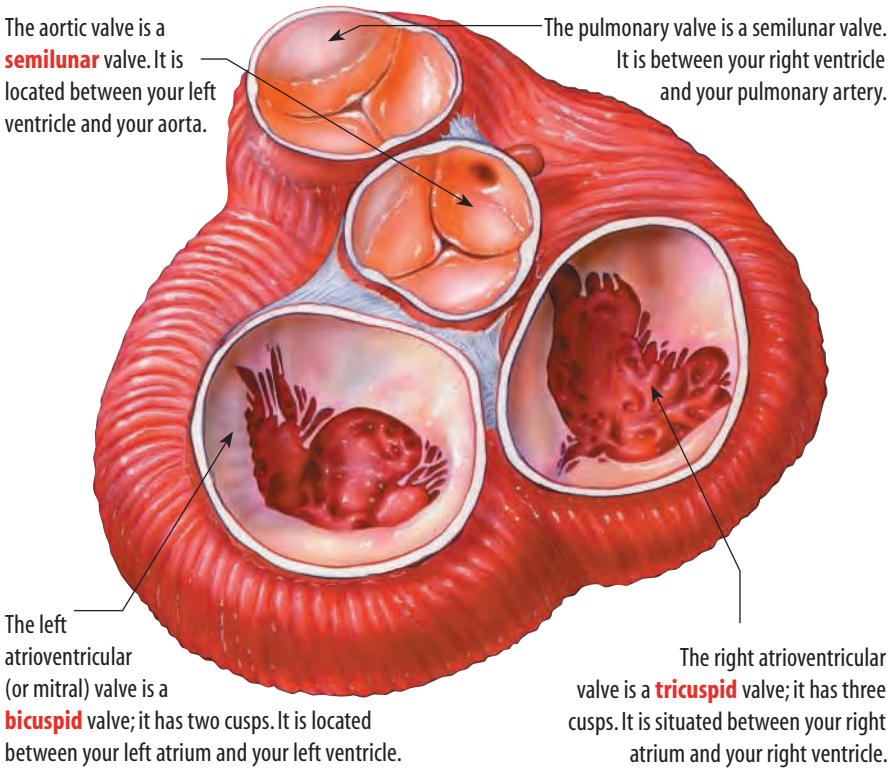
The heart's pumping action and the narrow size of the blood vessels result in a build-up of considerable pressure in the arteries. The force with which blood flows through the arteries is called **blood pressure**. It is affected by different activities and moods. It also goes up and down as the heart beats, being highest when the heart contracts (**systolic pressure**) and lowest when the heart relaxes (**diastolic pressure**). A person's blood pressure is expressed as a fraction. This fraction is the systolic pressure over the diastolic pressure, such as 120/70.

Keeping the pace

During each minute that you are sitting and reading this, about 5–7 litres of blood completes the entire circuit around your body and lungs. In a single day, your heart may have beaten about 100 000 times and pumped about 7000 litres of blood around your body.

A normal human heart beats about 60–100 times a minute, this rate increasing during exercise or stress. With each **heartbeat**, a wave of pressure travels along the main arteries. If you put your finger on your skin just above the artery in your wrist, you can feel this **pulse** wave as a slight throb. Your pulse rate immediately after exercise can be used as a guide to your physical fitness. The fitter you are, the less elevated your heart rate will be after vigorous exercise.

The regular rhythmic beating of the heart is maintained by



INQUIRY: INVESTIGATION 4.5

Dissect a heart

KEY INQUIRY SKILLS:

- planning and conducting
- processing and analysing data and information

Equipment:

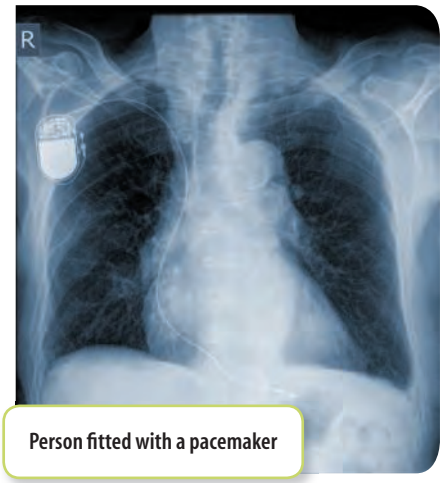
sheep's heart preferably with the blood vessels still attached
dissecting instruments
dissecting board

- Identify the parts of the heart using the illustration on the previous page.
- Try to locate where blood enters and leaves the heart:
  (a) to and from the lungs
  (b) to and from the rest of the body.
- Sketch and label the heart and use arrows to show the direction of blood flow.
- Cut the heart in two so that both halves show the two sides of the heart (similar to the illustration on the previous page).
- In a diagram, record your observations of the thickness of the walls on the left side of the heart compared with the right side.
- Suggest reasons for the differences observed.
- Try to locate the valves in the heart.

DISCUSS AND EXPLAIN

1 Describe the valves and suggest their function.
2 Write a summary paragraph about the structure and function of the heart.

electrical impulses from the heart's **pacemaker**, which is located in the wall of the right atrium. Some people with irregular heartbeats are fitted with artificial, electronic pacemakers to regulate the heart's actions and correct abnormal patterns.



Person fitted with a pacemaker

Try clenching your fist every second for five minutes. Getting a little tired? The heart is made up of special muscle called cardiac muscle, which never tires. Imagine having a 'cramp' or 'stitch' in your heart after running to catch the bus! Owing to its unique electrical properties, heart muscle will continue to beat even if it has been removed from the body. Scientists have shown that even tiny pieces of this muscle cut from the heart will continue to beat when they are placed in a test tube of warm salty solution.

HOW ABOUT THAT!

Not all animals have four-chambered hearts; in fact, some don't have hearts at all! A fish heart has two chambers, while amphibians and reptiles have three-chambered hearts. Can you suggest any advantages or disadvantages of these hearts over a four-chambered mammalian heart?

INQUIRY: INVESTIGATION 4.6

Check your heart

KEY INQUIRY SKILLS:

- planning and conducting
- processing and analysing data and information

Equipment:

stopwatch

blood pressure monitor

- Find your pulse, either on the inside of your wrist or in your neck (see the illustrations). Make sure you use two fingers, not your thumb, to find your pulse.

Two places where your pulse should be easy to find:
(a) radial location (wrist)
(b) carotid location (neck)

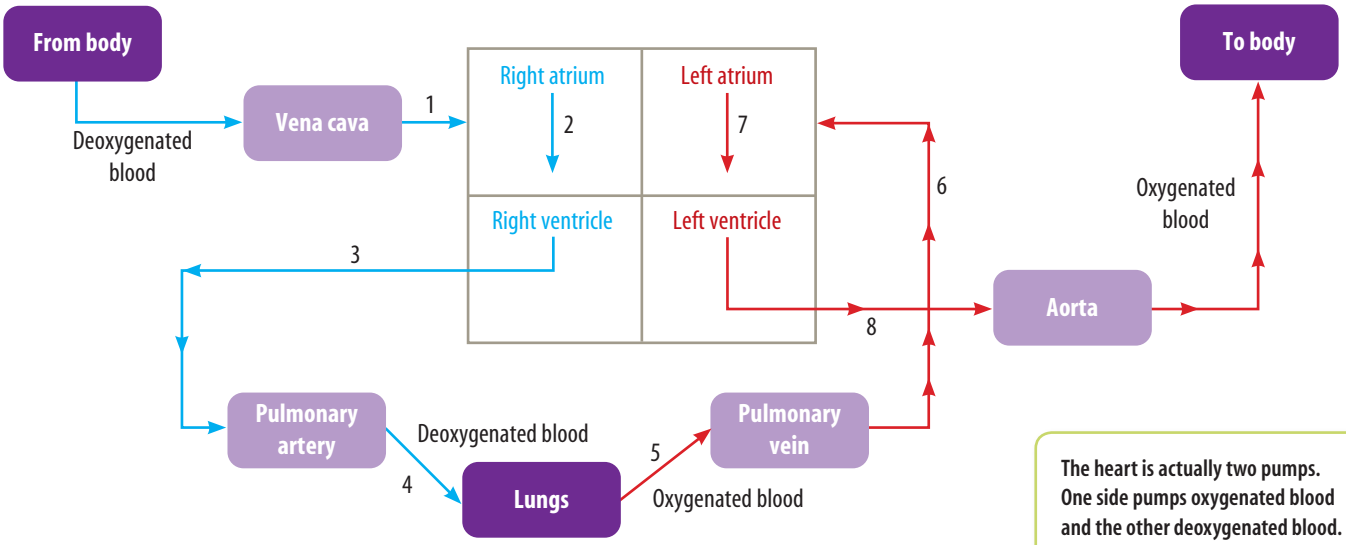
- Measure your heart rate in beats per minute (bpm) by counting the number of times your heart beats in 15 seconds and then multiplying this number by 4.
- Measure your blood pressure using the blood pressure monitor.
- Go for a walk in the playground or around the school oval. Measure your heart rate and blood pressure again.
- Run up and down a flight of stairs. Measure your heart rate and blood pressure again.
- Copy the table below into your workbook and enter your own results.

Test	Heart rate (bpm)	Blood pressure (mm Hg)
Before exercise		
After walking		
After running up stairs		

DISCUSS AND EXPLAIN

1 What effect does exercise have on heart rate and blood pressure?

2 Design and carry out an experiment to test the following hypothesis: 'There is a link between a person's resting heart rate and the number of hours the person spends exercising each week'.



The heart is actually two pumps. One side pumps oxygenated blood and the other deoxygenated blood.

UNDERSTANDING AND INQUIRING

REMEMBER

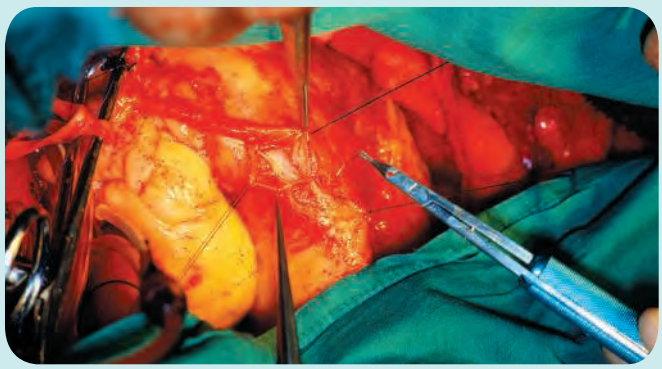
1 What is the difference between:
   (a) the blood in the two sides of the heart?
   (b) the structure of the two sides of the heart?
   (c) systolic and diastolic pressure?
2 Explain why there are valves in the heart.
3 Outline what blood pressure is caused by.
4 (a) How many times does a normal human heart beat each minute?
   (b) Suggest what may cause the heartbeat to increase.
   (c) Explain how the rhythmic beating of the heart is maintained.
5 What is unusual about cardiac muscle?

INVESTIGATE AND CREATE

6 In your team, design and perform an experiment to investigate the effect of different types of activities on your heartbeat.
7 Doctors use a stethoscope to listen to heartbeats. Make and test your own stethoscope using rubber tubing and a plastic funnel.
8 (a) Construct a model of the human heart using clay or bread dough.
   (b) Use information in this section and other resources to relate the structural differences between the atria and ventricles of the heart to their function.
9 Make up a rhyme or poem to describe the flow of blood through the heart. Use as many of the **bold red** words in this section as possible.
10 Construct a 'working' model of a human heart that shows the movement of blood through the various chambers and blood vessels.
11 What does a cardiologist do? Find an example of an Australian cardiologist and write a 'diary entry' for a day at work for them.
12 Find articles in the media that advertise foods or drinks that can reduce heart disease. In a team, research their claims and summarise your findings in a SWOT diagram. As a class, be involved in a debate that includes members from different interest groups or with different perspectives or biases.
13 Can diet, exercise or lifestyle choices change the chances of you having a heart attack? Research this question, summarising your findings in a PMI chart. On the basis of your data, what is your personal answer to this question? Give reasons for your opinion.
14 Test your ability to label the parts of the heart by completing the **Beat It!** interactivity in your eBookPLUS. **int-0210**

eBookplus

15 Use the **Heart surgery** weblink in your eBookPLUS to complete the animation activity and find out more about heart surgery and the anatomy of a human heart.



work sheet 4.2 Blood and blood highways

Transport technology

Heart and blood vessel diseases are the major killers in Australia. They claim twice as many lives as cancer and 20 times more than traffic accidents.

Modern medicine and technology have produced techniques and procedures that attempt to minimise the effects of diseases and disorders of the circulatory system.

Faulty heart and vein valves

The heart, like many other pumps, depends on a series of valves to work properly. These valves open and close to receive and discharge blood to and from the chambers of the heart. They also stop the blood from flowing backwards. If any of the four heart valves becomes faulty, the function of the heart may be impaired.

Veins throughout the body may also contain valves that keep the blood flowing in one direction. Defective valves in leg veins can cause blood to drain backwards, and to pool in the veins closest to the skin surface. These veins can become swollen, twisted and painful and are called **varicose veins**.

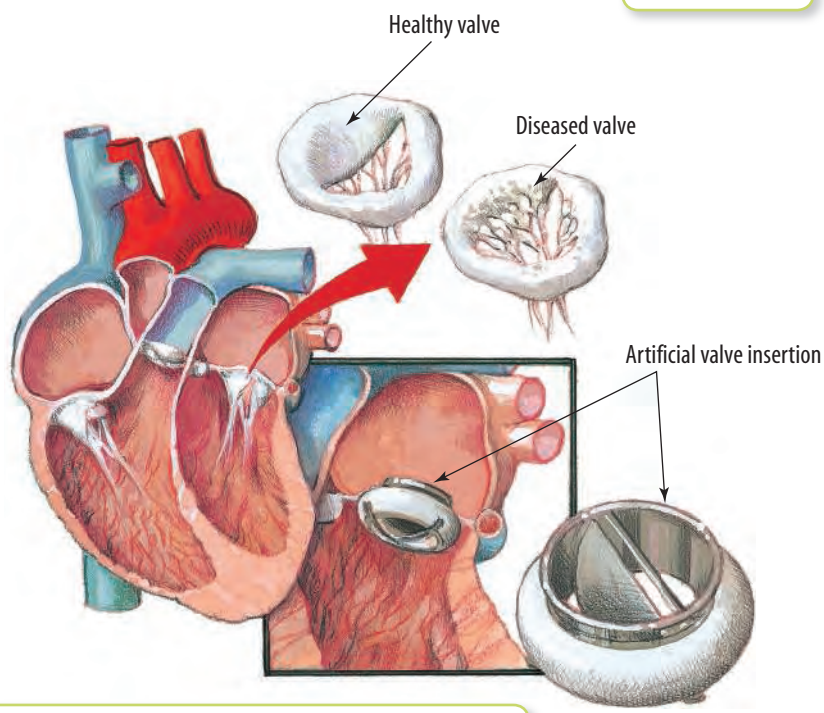
HOW ABOUT THAT!

About 15 per cent of Australians aged between 20 and 65 have hypertension (high blood pressure). This increases their chances of developing heart disease and strokes. To prevent this, people should maintain a healthy body weight, take regular exercise and eat a diet that is low in fat and salt.

'If I only had a heart . . .'

The tin man from *The Wizard of Oz* would have been very happy with the development of an artificial heart. This mechanical device can be made of titanium and plastic. Surgeons also implant a small electronic device in the abdominal wall to monitor and control the pumping speed of the heart. An external battery is strapped around the waist and can supply about 4–5 hours of power. An internal rechargeable battery is also implanted inside the wearer's abdomen. This is so they can be disconnected from the main battery for about 30–40 minutes for activities such as showering.

An artificial heart



A faulty heart valve may be replaced by an artificial valve. Why are the heart valves so important to the functioning of the heart?

A heart — but no pulse?

If only the left ventricle is damaged, and the rest of the heart is in good working order, a back-up pump may be implanted alongside the heart. One model of these devices results in its wearers having a gentle whirr rather than a pulse. This is the sound of the propeller spun by a magnetic field to force a continuous stream of blood into the aorta.

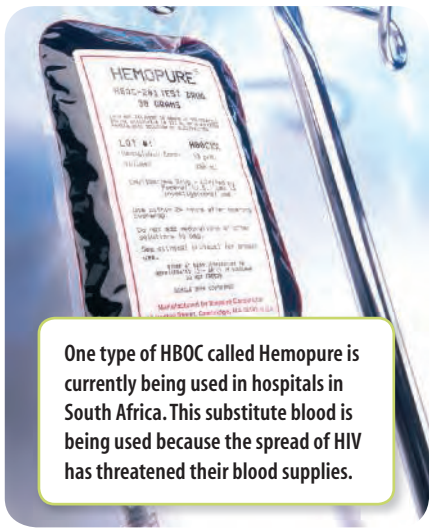
Getting the beat!

An **electrocardiogram (ECG)** shows the electrical activity of a person's heart. ECG patterns are valuable in diagnosing heart disease and abnormalities.

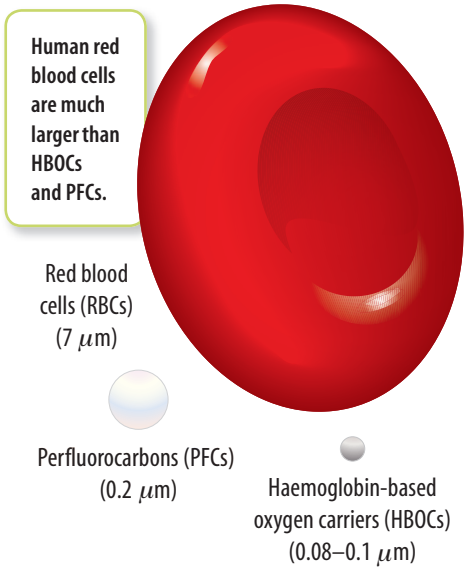
Artificial blood

A current wave of interest in vampire movies and books has brought with it discussion about the merits of artificial blood sources. The interest in artificial blood, however, is not new. There has been interest in the idea of blood transfusions for hundreds of years. William Harvey's description in 1628 of how blood circulated through the body prompted a variety of unsuccessful investigations into the use of alternative fluid substitutes. Shortage of blood supplies during war and disease epidemics has fired up the quest for an artificial blood substitute. Currently the two most promising red blood cell substitutes are haemoglobin-based oxygen carriers (HBOCs) and perfluorocarbon-based oxygen carriers (PFCs).

PFCs are usually white, whereas HBOCs are a very dark red. Although PFCs are entirely synthetic, HBOCs are made from sterilised haemoglobin. The haemoglobin may be from human or cow blood, human placentas or bacteria that have been genetically engineered to produce haemoglobin. As the haemoglobin doesn't have a cell membrane to protect it, various techniques such as cross-linking and polymerisation are used to make it less fragile. Some scientists are even investigating the idea of wrapping it in an artificial membrane.



One type of HBOC called Hemopure is currently being used in hospitals in South Africa. This substitute blood is being used because the spread of HIV has threatened their blood supplies.



Human red blood cells are much larger than HBOCs and PFCs.

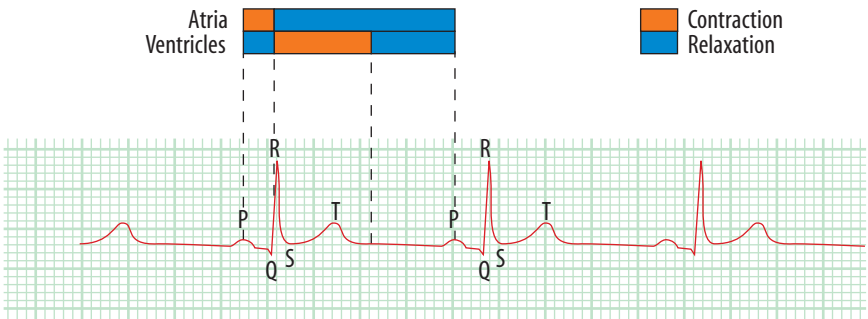
Red blood cells (RBCs) (7 μm)

Perfluorocarbons (PFCs) (0.2 μm)

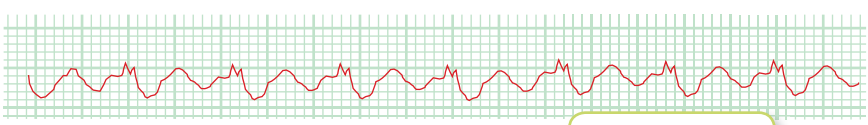
Haemoglobin-based oxygen carriers (HBOCs) (0.08–0.1 μm)

ARTIFICIAL BLOOD VESSELS?

Will the artificial blood vessels of the future be made by bacteria? Molecular biologist Helen Fink, working in Sweden, has suggested this may be the case. The cellulose produced by *Acetobacter xylinum* bacteria is strong enough to cope with blood pressure and function within our bodies, and could be used for artificial blood vessels in heart bypass operations in the future.



(a) Normal electrocardiogram



(b) Abnormal electrocardiogram

Electrocardiograms

In the future, will artificial blood vessels like this one be made by bacteria?

Image courtesy of the University of Gothenburg

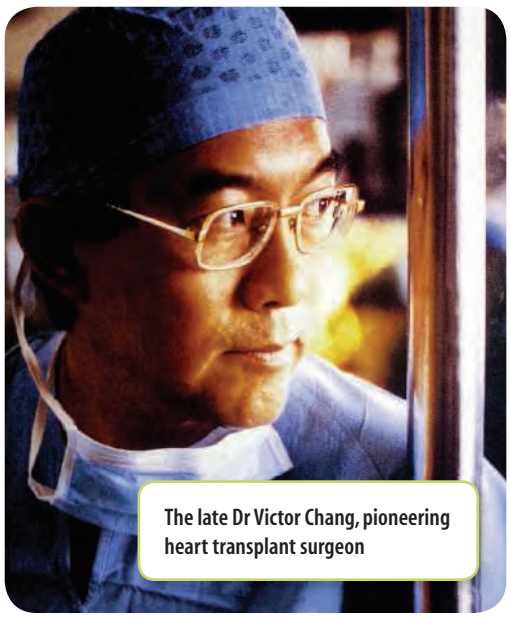
Transplant pioneer

If your heart or lungs were not working properly and you had needed a heart or lung transplant in the 1980s, the doctor to see was Victor Chang.

Victor Chang was an Australian doctor who was awarded a Companion of the Order of Australia for his contribution to medicine. Dr Chang played an important role in establishing the heart transplant unit at St Vincent's Hospital in Sydney. He set up a team of 40 health professionals who were the finest in their field and developed new procedures and techniques that led to an improved rate of success. Of his patients, 92 per cent were still alive one year after their heart or lung transplant operation and 85 per cent were still alive five years later.

The first heart transplant operation that Victor Chang carried out at St Vincent's Hospital was in 1984 on a young girl called Fiona Coote. Fiona is now an adult and, although she has since needed a second heart transplant, she owes her life to Dr Chang.

Dr Victor Chang also developed an artificial heart valve, called the St Vincent heart valve, and was working on developing an artificial heart. Unfortunately his life was tragically cut short in 1991 when he was murdered by gunshot.



The late Dr Victor Chang, pioneering heart transplant surgeon

UNDERSTANDING AND INQUIRING

REMEMBER

1 Identify which group of diseases is the major killer in Australia.
2 What are varicose veins and what causes them?
3 What is an electrocardiogram, and when is it useful?
4 Explain why valves are important to the functioning of the heart.
5 Describe how an ECG is used to detect heart abnormalities.
6 Outline how heart valves are similar to the valves in veins.
7 Construct a matrix table (see page 340) to show the differences between red blood cells, HBOCs and PFCs.

THINK

8 Look at the electrocardiograms on the previous page.
(a) At 'P', are the muscle cells of the atria contracted or relaxed?
(b) After the 'QRS' wave, is the ventricle relaxed or contracted?
(c) How does the normal electrocardiogram differ from the abnormal electrocardiogram?
(d) Suggest what might be wrong with the heart activity shown on the abnormal electrocardiogram.

INVESTIGATE

9 What are artificial hearts made of and how do they work?
10 How can blood loss cause death?
11 Use the internet to research one of the following: strokes, heart murmurs, 'hole in the heart', atherosclerosis, angina, heart attack, arrhythmias, pericarditis, hypertension. Present your findings as a PowerPoint presentation or a poster.
12 Find out more about Dr Victor Chang and report your findings to others in the class.
13 There are a number of issues surrounding the development and use of artificial blood. Find out what these are and then construct a PMI chart as a summary. What is your opinion about artificial blood? Provide reasons for your opinion.
14 Find out and report on 'transport technology' research that Australian scientists are currently involved in.
15 (a) Which organs are most successfully transplanted into humans?
(b) List sources of the organs for transplant and identify associated issues.
(c) Describe how donors and organ recipients are matched.
(d) Organ recipients can require specific treatment after the operation. Outline what this involves and why it is needed.
16 If you required a new heart, would you prefer an artificial one or one from a human or other natural source? Provide reasons for your response.
17 Outline your opinion on being an organ donor yourself.
18 Find out issues related to organ transplants and construct a PMI summary.

eBookplus

19 Use the **Electrocardiogram game** weblink in your eBookPLUS to simulate performing ECGs on patients referred to you by medical doctors.

Sucked in and blown out

Breathe in deeply … now breathe out. You have just introduced some extra oxygen into your body and removed some unwanted carbon dioxide. You do this about 15–20 times per minute without thinking. The muscle movements required for **breathing** are automatic and controlled by the respiratory centre in the brain.

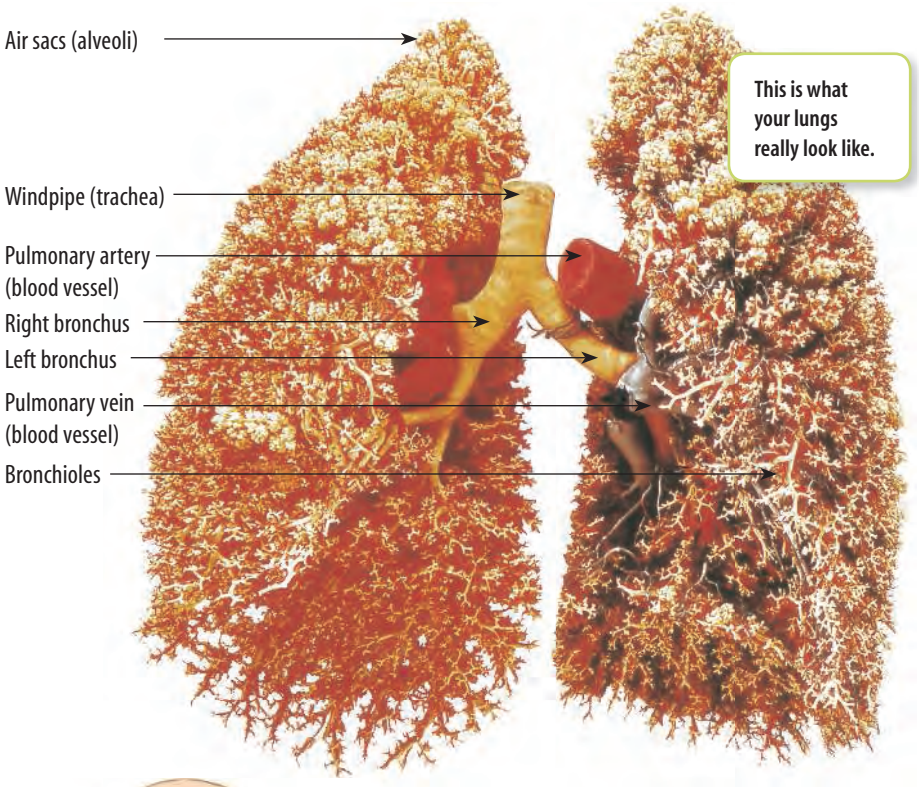
When you breathe in, you take in the mixture of gases called air. Oxygen and carbon dioxide are gases found in the air around you. Oxygen makes up about 21 per cent of the air, while carbon dioxide makes up only about 0.04 per cent. Your body uses some of the oxygen you take in. The table above right shows that the air that you breathe out contains less oxygen and more carbon dioxide than the air you breathe in. The percentages in the table are approximate and vary a little with weather conditions and height above sea level.

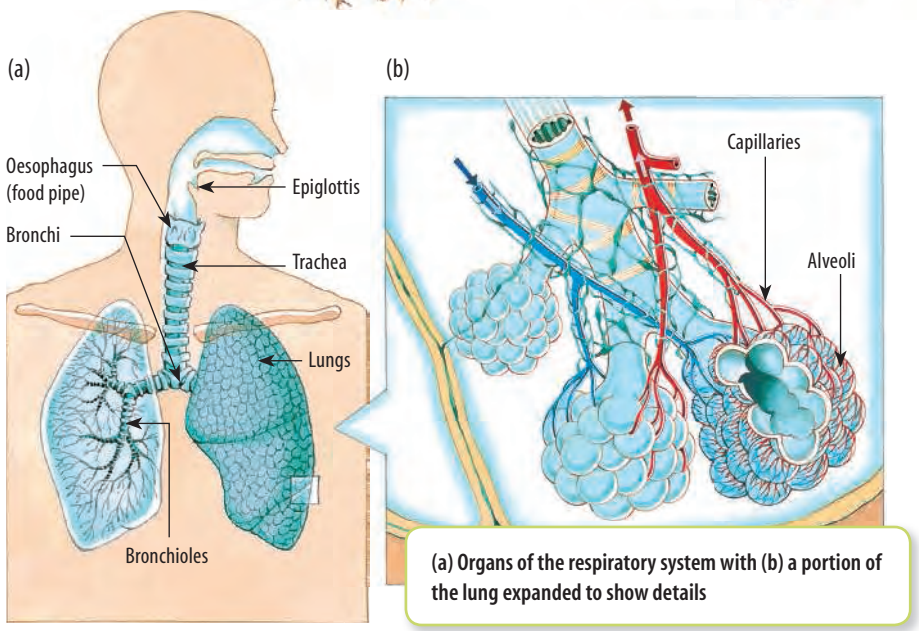
All animals and plants need energy to grow. Animals also need energy both to move and to keep their body systems working. Energy is obtained from food. However, oxygen is also needed to break down the food so that the energy in it can be used by the body. The process of using oxygen to release the energy from food is called cellular respiration. Cellular respiration takes place in the cells of living things.

glucose + oxygen ⟶ carbon dioxide + water + energy

What goes in and what comes out

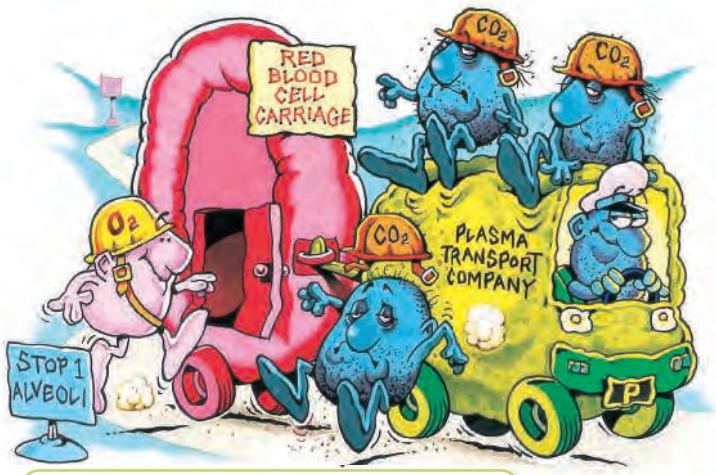
Gas	Oxygen (%)	Carbon dioxide (%)	Water vapour (%)	Nitrogen (%)
Air breathed in	21	0.04	1	78
Air breathed out	15	4	5	76



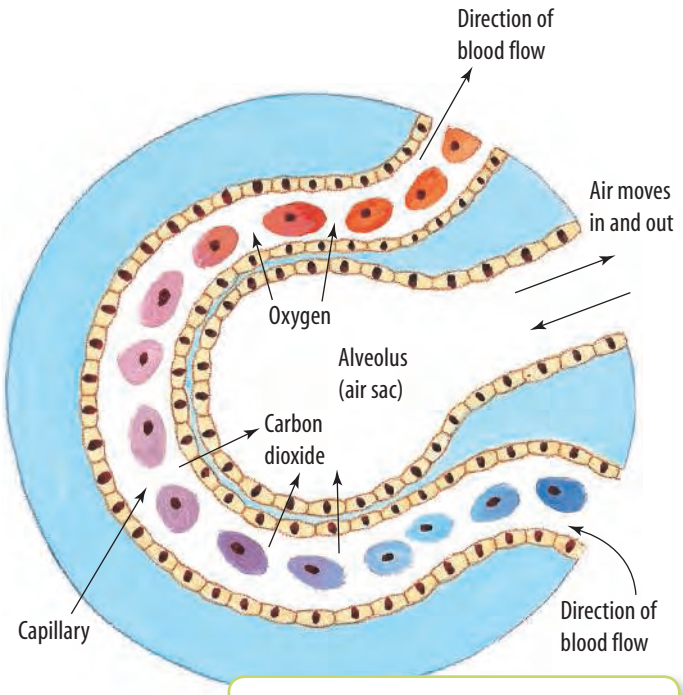


HOW ABOUT THAT!

The air that you breathe enters your body through your nose and mouth. Unless your nose is clogged up by a cold, it is the most important airway. The hairs and sticky mucus in your nose trap dust and dirt and other harmful material such as disease-causing bacteria. Breathing in through your mouth gets the air in faster but without being filtered by the nose. When you play sport, your body uses oxygen more quickly and it is often necessary to breathe in through your mouth, bypassing the filter system in your nose.



Where the action is. **Gas exchange** occurs at the alveoli.



In an alveolus, oxygen diffuses into the blood and carbon dioxide diffuses out of the blood.

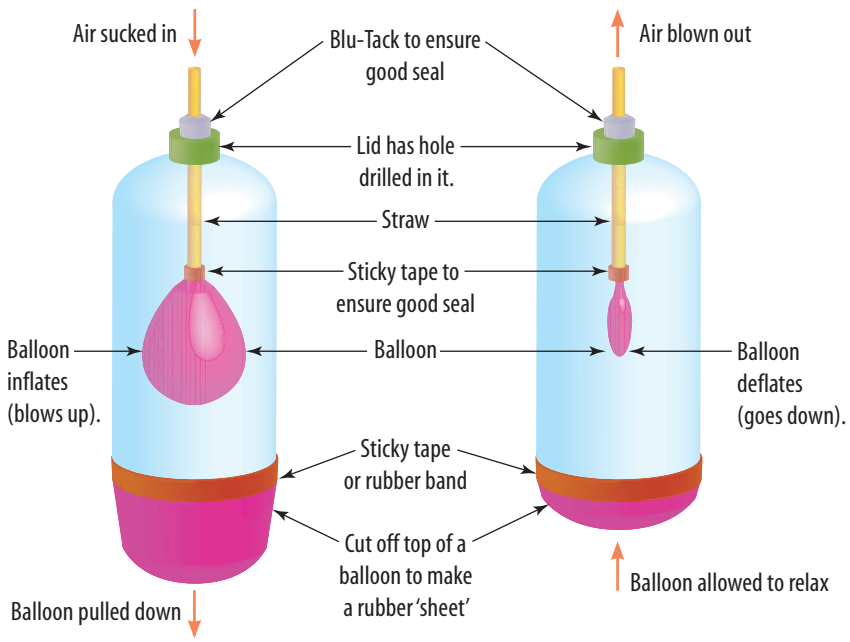
Going down?

After entering your body, the air moves into a narrow tube called the **trachea**, which is more commonly known as the windpipe. At the top of this tube is a flap of tissue called the **epiglottis**. The job of this tissue is to stop food from going down into your lungs. If food does manage to pass it and 'go down the wrong way', a cough soon brings it back up again.

The trachea divides into two narrower tubes called the bronchi. Each of these tubes leads to a lung. Inside the lung, each tube divides into many smaller tubes called **bronchioles**. The bronchioles branch out, getting smaller and smaller until they end at thousands of tiny air sacs called **alveoli**.

Where the action is

It is at the alveoli that the exchange of gases actually takes place. Some of the oxygen in the air that you breathe in moves through the alveoli into narrow capillaries. From there it moves into the red blood cells.



A model lung. When the rubber sheet at the bottom is pulled down, the pressure inside the jar drops and air is sucked into the balloon. The balloon inflates (blows up).

Carbon dioxide takes the opposite journey, moving out of the red blood cells through the capillaries into the alveoli so that it can be breathed out.

The movement of a muscle called the **diaphragm** helps the lungs do their job by sucking in and pushing out air. The images on page 141 show how this happens.

Lung capacity

Although adults can breathe out up to 5 litres of air with each breath, they usually breathe out only about half a litre. The largest volume of air that you can breathe in or out at one time is called your **vital capacity**.

INVESTIGATION 4.7

Measuring your vital capacity

KEY INQUIRY SKILLS:

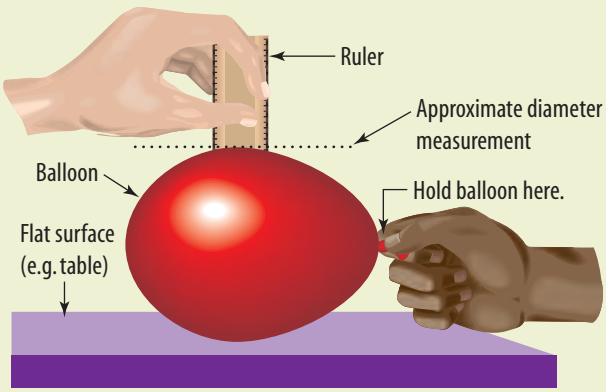
- planning and conducting
- processing and analysing data and information
- evaluating

Equipment:

balloon

ruler

- Blow up a balloon to about 20 cm in diameter two or three times to stretch it. Release the air each time.
- Take the biggest breath you can, then blow out all the air you can into the balloon. Tie up the end of the balloon to hold in your 'blown out' air.
- Use a ruler to measure the diameter of the balloon as shown below.



How to measure the diameter of the balloon

- Use the table at the bottom of this page to determine your approximate vital capacity in litres.
- Release the air from the balloon and repeat your measurement of vital capacity three more times. Average your results to get your best estimate of the maximum 'blow-out' of your lungs.

DISCUSS AND EXPLAIN

1 Why were you asked to stretch the balloon first?

2 Why did you measure your vital capacity four times?

3 (a) Draw up a table with the following headings.

Name	Male or female?	Does this student play a wind instrument?	Lung capacity (L)

(b) Collect results from all the students in your class and complete the table.

(c) Calculate the average lung capacity for all the girls and all the boys. Do girls have a bigger or smaller lung capacity than boys in your class?

(d) Calculate the average lung capacity for all the students in your class who play a wind instrument. Compare that with the average value for the other students in the class. Does playing a wind instrument have an effect on lung capacity?

4 Suggest another way of measuring the amount of air exhaled with each breath.

Determining vital capacity

Balloon diameter (cm)	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Approx. vital capacity (litres)	0.3	0.4	0.5	0.7	0.9	1.2	1.4	1.8	2.1	2.6	3.0	3.6	4.2	4.8

HOW ABOUT THAT!

Phew … garlic breath! Have you ever heard someone say this? Garlic or onion breath comes from further down than your mouth! It has travelled through a number of your body systems. After you have eaten food containing either of these, and it has been digested, it is absorbed through the walls of your intestines and then into your blood. When the smelly onion or garlic blood reaches your lungs through your circulatory system, you breathe out the smelly gas.

INQUIRY: INVESTIGATION 4.8

Hands on pluck

KEY INQUIRY SKILLS:

- planning and conducting
- processing and analysing data and information

Equipment:

sheep's pluck (heart and lungs) with part of the liver and trachea attached

newspaper and tray to place the pluck on

plastic disposable gloves

balloon pump and rubber tubing

- Copy the table below into your workbook.

Organ	Shape (sketch)	Approx. size (cm)	Colour	Texture	Other comments?	System to which the organ belongs
Trachea						
Lung						
Heart						
Liver						

- Carefully observe and record the shape, size, colour and texture of the sheep's trachea, lung, heart and liver. Include notes on how they are connected. Can you see any blood vessels?
- Push a piece of rubber tubing down the trachea to the lungs and use a balloon pump to blow some air into the trachea. **CAUTION: For hygiene reasons, it is not recommended that you use your mouth to blow into the tube inserted in the trachea.**
- Cut through the lung, heart and liver tissue. Make a record of your observations describing how they are similar and how they are different. Discuss possible reasons for the differences with your team members.
- Using a scalpel or scissors, cut off a small piece of heart, lung and liver. Place each piece into a beaker of water and observe what happens. Discuss possible reasons for your observations with your team members.

DISCUSS AND EXPLAIN

1 Could you see any blood vessels? Try to find out their names and what sort of blood they carry.

2 Suggest why there are rings of cartilage around the trachea.

3 Suggest reasons for the differences in texture between the heart and lungs.

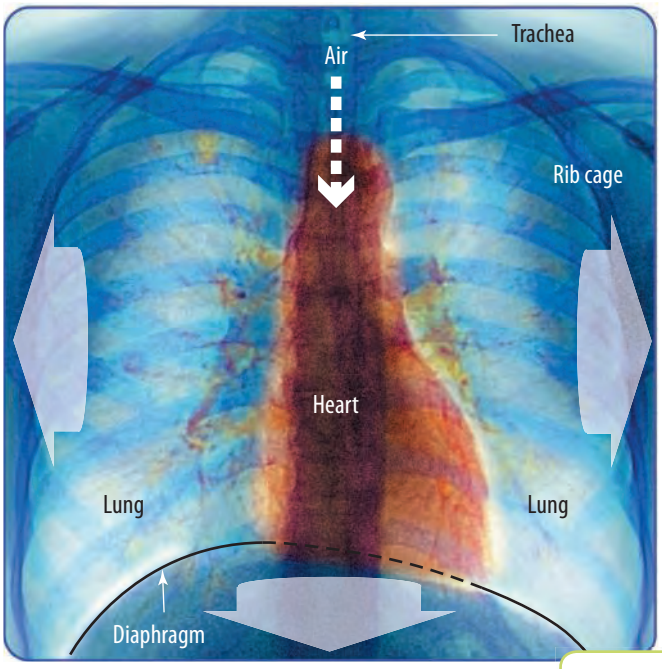
4 Suggest reasons for the differences in the shapes of the organs that you have observed.

5 Comment on something that you learned or found particularly interesting from this investigation. Share your comment with others.

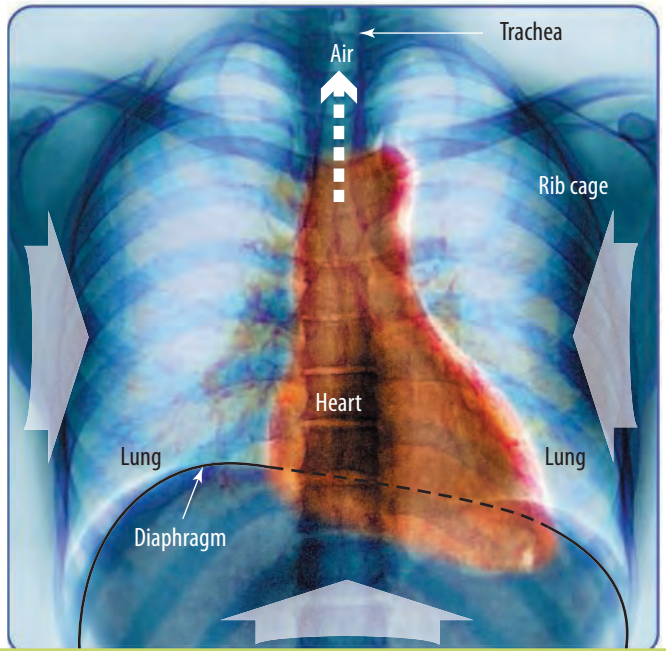
6 Research and report on the following points for each of the organs in this investigation:

- its function and how it carries this out
- the system to which it belongs
- a disease relevant to it.

(a) Breathing in



(b) Breathing out



(a) Breathing in. The diaphragm tightens, allowing the lungs to expand, and the air is sucked in.
(b) Breathing out. The diaphragm relaxes, making the lungs smaller, and the air is pushed out.

UNDERSTANDING AND INQUIRING

REMEMBER

1 Construct a mind map to summarise the key points in this section.

2 Describe the job done by each of the following parts of the respiratory system.
(a) Epiglottis
(b) Diaphragm
(c) Alveoli
(d) Lungs

THINK AND INVESTIGATE

3 Use information in this section and other resources to relate structural features to the functions of the following parts of the respiratory system.

Part of system	Structural features	Function
Trachea		
Alveoli		
Lungs		
Capillaries		

4 Some people describe the structure of the lungs as an upside-down hollowed-out tree. Which parts of the lungs might be the following parts?
(a) Trunk
(b) Branches
(c) Twigs
(d) Leaves

5 Give reasons for each of the following pieces of advice.
(a) It is better to breathe through your nose than your mouth.
(b) You should blow your nose when you have a cold rather than sniff it back.
(c) You should not talk while you are eating or drinking.

6 Find out what a spirometer is.

7 Did you know that mountain climbers often find it difficult to breathe? Some wear oxygen tanks to allow them to climb very high mountains. Research the effects of high altitude on breathing and report your findings.

8 Some singers can hold a musical note for a very long time — investigate what muscles and techniques they use in order to be able to do this.

CREATE

9 Construct a working model of the lungs using the following items.
- Clear plastic soft-drink bottle
- 2 balloons
- 2 rubber bands
- Plasticine or 'Blu-Tack'
- Scissors
- 2 plastic drinking straws

work sheet 4.3 Breathing — constructing a report

Short of breath?

If you do not suffer from **asthma**, it is very likely that you know someone who does. Asthma is a very common condition, which affects about one in seven Australian adolescents.

About one in ten, or 10 per cent, of adults are affected. Young children are the greatest sufferers of asthma with one in four affected. One alarming fact about asthma is that the number of people who suffer from it has doubled in the last 30 years. The reasons for this increase are not clear but you might have some ideas of your own after reading this article.

What is asthma?

Asthma is a narrowing of the air pipes that join the mouth and nose to the lungs. The pipes most affected are the bronchi. They become narrower as:

- the muscle wall of the air pipes contracts
- the lining of the air pipes swells
- too much mucus is produced.

The narrow pipes make breathing difficult and can result in wheezing, coughing and a tight feeling in the chest. The coughing is usually worse at night.

What causes asthma?

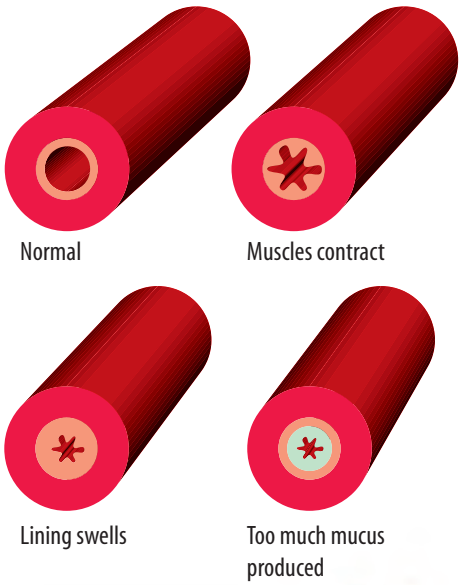
It is not known why some people get asthma and others do not. It seems that it can be inherited, but many people from families without a history of asthma are affected. Asthma is certainly the result of sensitive airways. An asthma attack occurs when those sensitive airways are 'triggered'. If the sufferer has a cold, the airways are already inflamed and are more likely to be triggered.

Triggers

Some of the common triggers of an asthma attack are:

- vigorous exercise
- cold weather
- cigarette smoke
- dust and dust mites
- moulds
- pollen
- air pollution
- some foods and food additives
- some animals.

Not all asthma sufferers are affected by the same triggers. Some people suffer attacks only as a result of exercise. Others might be affected by any one or more of the triggers. It is important that those who get asthma try to find out what triggers the attacks. Many of the triggers can be avoided.



Asthma is a narrowing of the air pipes.

Controlling the triggers

The best way to control asthma is to avoid the triggers. Of course, it is not always possible to identify the triggers. People who are affected by cold weather, pollen or air pollution often move to other places to avoid their triggers.

Pollen from some grasses and trees is very light and becomes airborne on even slightly windy days. The inhaling of pollen can be reduced by avoiding outdoor activities and keeping windows and doors closed on breezy spring days. Moulds live in warm, humid conditions and thrive in bathrooms, kitchens and bedrooms. Their spores are easily breathed in, triggering attacks in some asthma sufferers. Moulds can be reduced by airing the house regularly.

Pollen and moulds are also the main triggers of hay fever. Hay fever clogs up the nose in much the same way as asthma affects the air pipes. It causes sneezing and a runny nose.

Those asthma sufferers whose attacks are triggered by air pollution are warned to remain indoors as much as possible and avoid vigorous activity on smoggy days. If tobacco smoke is a trigger, the cigarette smoke of others needs to be avoided.

Dust mites are a common trigger of asthma attacks. Dust mites are microscopic animals that live in their thousands in warm, moist and dark places like doonas, sheets, pillows, carpets and curtains. Dust mite droppings float in the air and are easily inhaled.

Since you share so much of yourself and where you live with this fellow Australian, you should probably know its name. It is the most common dust mite (a relative of spiders and ticks) in Australia, *Dermatophagoides pteronyssinus*. The good news is that it is half a millimetre long and doesn't bite. The bad news is that there may be thousands of them living in your pillow, each defecating about 20 faecal pellets a day, reproducing (each female laying about 30 eggs in her lifetime), dying and decomposing. The fact that dust mites mate for 24 hours at a time (perhaps because their penis is only about as wide as their sperm) may make this particularly disturbing!

HOW ABOUT THAT!

Dust mites thrive best in bedding and carpets because these contain plenty of dead human skin cells. Humans shed a complete layer of dead skin cells every month. That amounts to about 1 kilogram of skin cells each year. In fact, most of the dust in your house consists of dead skin cells.

Dr Peter Dingle (Associate Professor in Health and the Environment at Murdoch University's School of Environmental Science) suggests that our skin scales are the main food source of these dust mites, so wherever we are, they are. Dr Janet Rimmer (a respiratory physician and Director of the National Asthma Council Australia) also suggests that, of the 45 per cent of Australians who are affected by allergies, about 80 per cent are allergic to dust mites. But not all researchers have bad news about dust mites. Dr Matthew Colloff, a CSIRO researcher, has found them so interesting that he wrote a book (called *Dust mites*) about them.

A house dust mite

Even the cleanest house has dust mites, but their numbers can be reduced by:

- exposing the mattress to the sun, because dust mites are susceptible to drying out
- washing bedding materials and bedclothes in tea-tree or eucalyptus oil or in hot water (more than 55°C)
- removing soft toys that collect dust or hot washing them weekly
- regularly vacuuming curtains and carpets
- airing the bedroom by keeping doors and windows open
- replacing carpets with hard flooring.

Chemicals in foods, and animals, are less common triggers of asthma attacks. However, they are easier to avoid if they have been identified as a trigger.

Asthma medication

Asthma medications can be divided into two main groups: preventers and relievers. Preventers make the lining of the airways less sensitive and therefore less likely to be triggered. Relievers open up the airways once an attack has commenced. Most asthma medications are applied with inhalers or 'puffers', which direct the medication straight

into the air tubes for fast action. Severe attacks of asthma require other drugs and sometimes extra oxygen needs to be supplied.

Up in smoke

About 18 000 Australians die each year as a result of diseases caused by smoking. In fact, smoking is the largest preventable cause of death and disease in Australia.

Just one cigarette

There are clearly many long-term effects of smoking. However, the diagram below shows what happens to you after smoking just one cigarette.

There are some more obvious effects such as bad breath, body odour and watery eyes. After several cigarettes, your teeth and fingers become stained. Your sense of taste is reduced. Even your stomach is affected as acid levels increase.

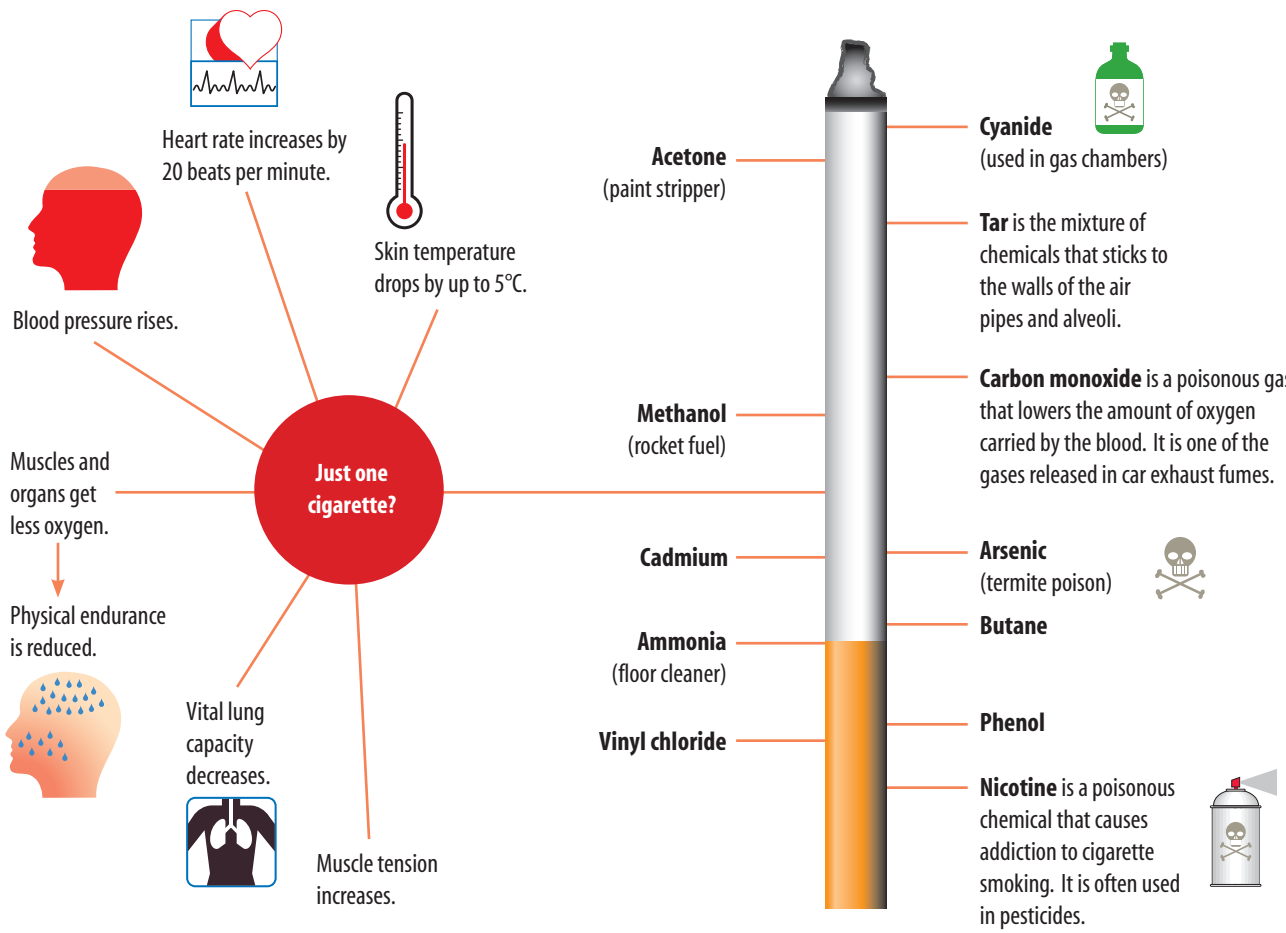
Smoking and your lungs

Lung cancer is the most well-known disease caused by smoking. Chemicals that cause cancer are called **carcinogens**. Cigarette tobacco contains a number of carcinogens. The chemicals in cigarettes also clog up the fine hairs in your air tubes with a mixture of mucus and foreign chemicals.

Cough it up?

Coughing is the body's way of trying to clear the air tubes. However, not all of the clogging can be cleared by coughing. The dirty mixture remains in the air tubes, causing swelling, making them sensitive and slowing down the passage of air. Eventually the sticky mixture sinks down into the lungs, where it blocks some of the pathways to the alveoli, where freshly breathed air should deliver oxygen to the blood.

The diseases caused by this blocking process are called chronic obstructive pulmonary diseases, or COPD. **Emphysema** is the worst of these diseases and results in the eventual destruction of the alveoli.



Professor Robyn E. O'Hehir

BSc, MBBS (Hons I), FRACP, PhD, FRCP, FRCPath

1 What is your current science-related title?

I am a Professor of Medicine, with particular responsibilities for allergy, clinical immunology and respiratory medicine, at Monash University, Melbourne. I am also the Director of the Department of Allergy, Immunology and Respiratory Medicine at the Alfred Hospital in Melbourne.

2 What field of science are you in?

Allergy, cellular immunology and respiratory medicine. I was appointed to the first Chair in Allergy and Clinical Immunology in Australia.

3 Describe some 'science' that you are involved in at the moment.

Millions of people around the world suffer from allergies. I am sure you know several friends who have asthma or hay fever, or you may even have them yourself. Asthma and hay fever are usually triggered by proteins called allergens, from house dust mites or grass pollens. Allergies to peanuts and shellfish are less common but often more serious, because they can trigger life-threatening allergic reactions called anaphylaxis. Allergies are caused by reactions between white blood cells ('T cells') and environmental proteins that are usually harmless. My research group is trying to find ways to damp down the allergic T-cell responses.

Allergen immunotherapy (allergy shots) is the only treatment that can prevent allergic diseases, but currently it can't be used for peanut, even though this is one of the most serious allergens. To develop a safe and effective vaccine against peanut allergies, we are identifying parts of critical peanut proteins that can build up tolerance in allergic patients without risking anaphylaxis.

4 What do you enjoy about being a scientist?

I enjoy the fact that my research not only is laboratory-based, exploring novel methods for switching off allergic responses, but also lets me see patients and train other doctors in how to do research from bench to bedside to the community. I head an active clinical department, still carry out clinics with patients, and am actively engaged in national and international tests of new preventions and treatments for allergies. My combined research and clinical duties allow translation of our research findings into better clinical practice.

5 What triggered your interest in science?

I decided to specialise in allergy and respiratory medicine, focusing on asthma, following my experiences as a young trainee physician at the Alfred Hospital in Melbourne. Asthma was a huge problem in Australia at that time, and many times I resuscitated young adults in the hospital emergency room — and I watched them return, with appropriate medication and careful education, to confident, full lives. Some remain my patients today. The ability to dissect underlying mechanisms of disease and then work towards new therapeutics and practices to benefit patients is a great excitement and honour. The diversity of patients and their needs ensures that every day is quite different.

6 Do you have any other comments that may be of interest to Year 8 Science students?

A career in science combined with medicine may take a bit longer in terms of training, but it gives you a fantastic ability to do interesting work that is intellectually demanding and also involves working with lots of people who need your help. I am very glad that I chose a career in science and medicine.

UNDERSTANDING AND INQUIRING

ANALYSE AND EVALUATE

1 If you were part of an audience of 1400 Year 8 students in a concert hall for a music excursion, how many of them would be likely to be asthma sufferers? (*Hint:* Use the information at the beginning of this section.)

2 The number of people who suffer from asthma has doubled in the last 30 years. Put forward one or more hypotheses suggesting why this might have happened. Write down any evidence that supports each of your hypotheses.

REMEMBER

3 What happens to the air pipes to the lungs during an asthma attack to make breathing difficult?

4 Why is an asthma attack more likely to be triggered in a person with a cold?

5 What is an asthma trigger?

6 What are the two major types of asthma medication and how are they different from each other?

THINK AND DISCUSS

7 Why is it unlikely that you would ever rid a house completely of dust mites? With your partner, brainstorm as many ideas as you can into a cluster map.

8 (a) In your team, brainstorm ideas about the common triggers of asthma that can be controlled. Summarise your discussion in a bubble map.
(b) Construct a table similar to the one below.
(c) Again as a team, add suggestions to your table of ways that the trigger could be controlled.

Trigger	How the trigger can be controlled
Moulds	Air the house regularly.

9 (a) If you suffer from asthma, prepare a talk for the rest of your class explaining:
(i) what it is
(ii) how it affects you
(iii) how you control it or try to prevent attacks.
(b) If you do not suffer from asthma, write a set of at least five questions that you could ask an asthma sufferer in an interview. If possible, conduct the interview and record the answers in writing, on audio cassette or on videotape.

INVESTIGATE AND DISCUSS

10 Propose a series of questions to find out more about each of the areas below, investigate them, and then share your findings with others.
(a) Allergies
(b) Asthma
(c) Anaphylaxis
(d) Allergen immunotherapy
(e) Clinical immunology and respiratory medicine

Peanuts are a common cause of allergies.

11 (a) Find out more about Anaphylaxis Australia Inc.
(b) Outline the topics covered in a first aid course for management of anaphylaxis.
(c) What is the EpiPen® and how is it used?

ANALYSE AND EVALUATE

12 The table below shows how the popularity of smoking has changed over the past 65 years or so.

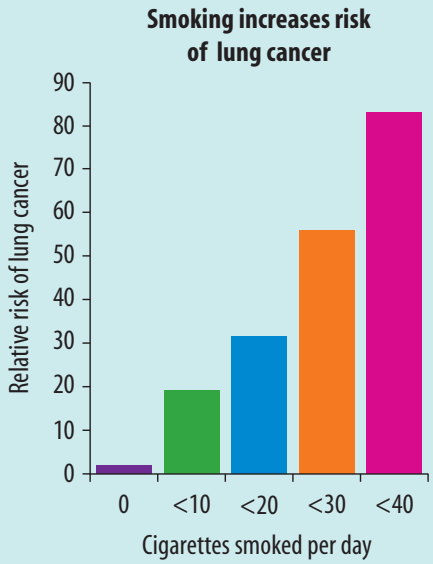
Percentage of adult Australians who smoke

Year	1945	1964	1969	1974	1976	1980	1983	1986	1989	1992	1998	2001	2004
Males (%)	72	58	45	41	40	40	37	33	30	28	29	28	26
Females (%)	26	28	28	29	31	31	30	28	27	24	24	21	20

(a) Draw a line graph of the data in the table. Use 'Year' on the *x*-axis and '% of adult Australians who smoke' on the *y*-axis. Draw lines for males and females in different colours.

(b) Why do you think that the percentage of females who smoke has changed little while the percentage of males who smoke has declined greatly?

(c) Use dotted lines to show your prediction of the trends up to the year 2020. What percentage of males and females do you predict will be smoking in the year 2020?



Graph 1: The risk of getting lung cancer increases with the number of cigarettes smoked daily.

13 Study graph 1 above.

(a) Copy and complete the following statements:

(i) People who smoke 10 cigarettes a day are _____ times more likely to develop lung cancer than non-smokers.

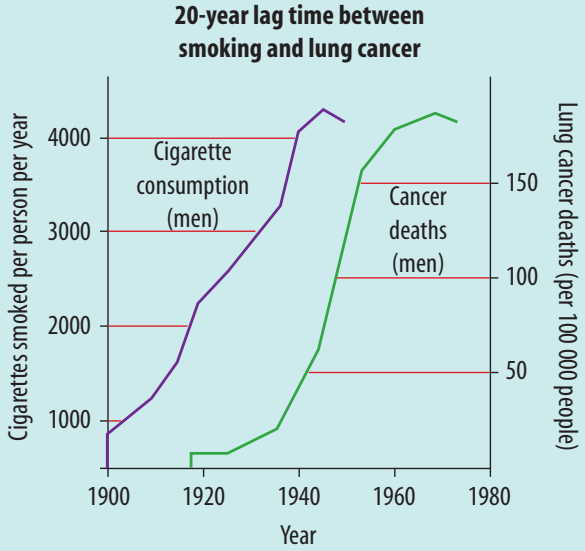
(ii) People who smoke 30 cigarettes a day are _____ times more likely to develop lung cancer than people who smoke 10 cigarettes a day.

(b) If a packet of cigarettes costs $15 and contains 20 cigarettes, calculate how much a person smoking 40 cigarettes a day spends on smoking:

(i) each day

(ii) each week

(iii) each year.



Graph 2: This graph shows that the number of deaths from lung cancer has risen as cigarette consumption has increased but there is a 20-year lag time because lung cancer takes many years to develop.

14 Study graph 2 above.

(a) Describe how the incidence of lung cancer changed between 1900 and 1980.

(b) Identify when the number of male smokers peaked.

(c) Identify when the number of deaths from lung cancer peaked.

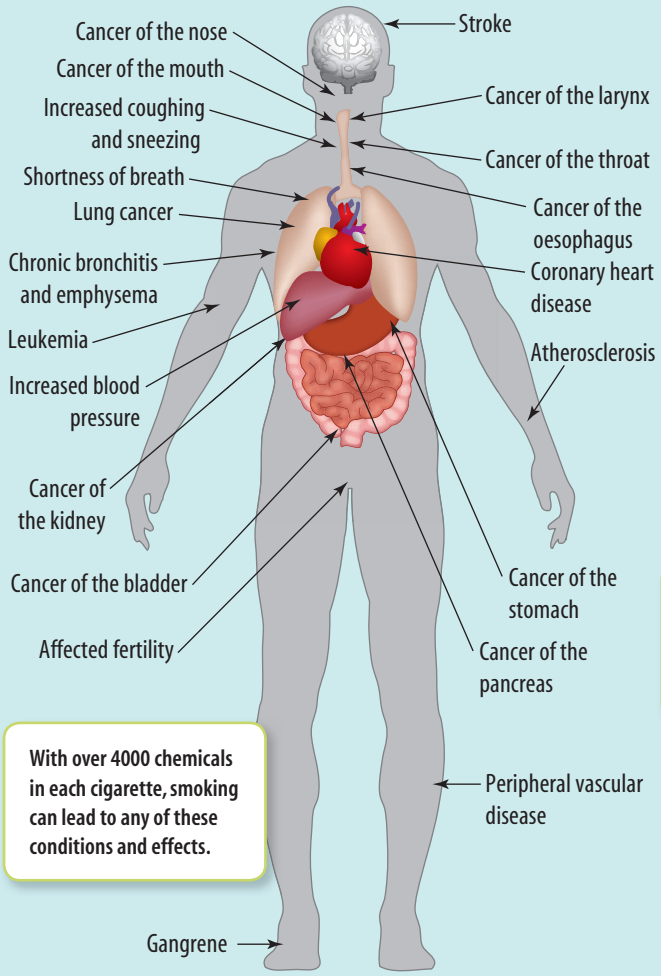
(d) Explain why there is a 20-year gap between the two numbers.

(e) The graph shows data for male smokers only. Predict when the number of cases of lung cancer in women peaked (use the graph you drew for question 1 to answer this).

THINK

15 Draw up a two-column table. The first column should be headed 'Reasons for smoking'; the second column should be headed 'Reasons for not smoking'. With at least one other person, complete the table. Then compare your table with others. You might be able to construct a large table for the whole class.

16 Smoking-related diseases cost taxpayers many millions of dollars because hospitals are mostly paid for by governments. Write down your opinion of each of the proposals below. Give reasons for your opinion.
- (a) The cost of hospital treatment for diseases caused by smoking should be paid for by the patient because it was their fault that they got sick.
- (b) Cigarettes should cost more. The extra money made from them could then be given to hospitals to help pay for treating smoking-related diseases.
- (c) Cigarette companies who make profits from smoking should be made to pay for hospital treatment of patients with diseases caused by smoking.



17 Although smoking is now banned in many places, including public transport vehicles, workplaces and restaurants, it is still legal. Why do you think smoking has not been made illegal when it causes so much damage?

INVESTIGATE AND CREATE

18 Draw a poster that sends one single important message about smoking.

19 Design an experiment that would investigate the effect of passive smoking on heart rates. Investigate whether any relevant research has been performed in this area. If so, share it with others in your class.

20 Research the evidence showing the effect of smoking on human lungs and construct a model to show its impact that could be used in a campaign against smoking.

21
- (a) In a small team, research media articles that relate to smoking.
- (b) In your team, discuss your findings.
- (c) As a team, evaluate the claims made. How accurate are they? How true are they? Are there any biases?

22
- (a) Is it ever illegal to smoke a cigarette in Australia? If so, what are the conditions?
- (b) Do you agree with Australia's laws and guidelines on smoking? Explain.

23
- (a) Research the structure and function of an alveolus.
- (b) Suggest how the structure of an alveolus is related to its function.
- (c) Suggest how smoking affects the ability of an alveolus to perform its function.

eBookplus

24 Use the **Quit Now** weblink in your eBookPLUS to learn about the National Tobacco Campaign. Create a poster that sends one single important message about smoking.

work sheet 4.4 Smoking and diseases

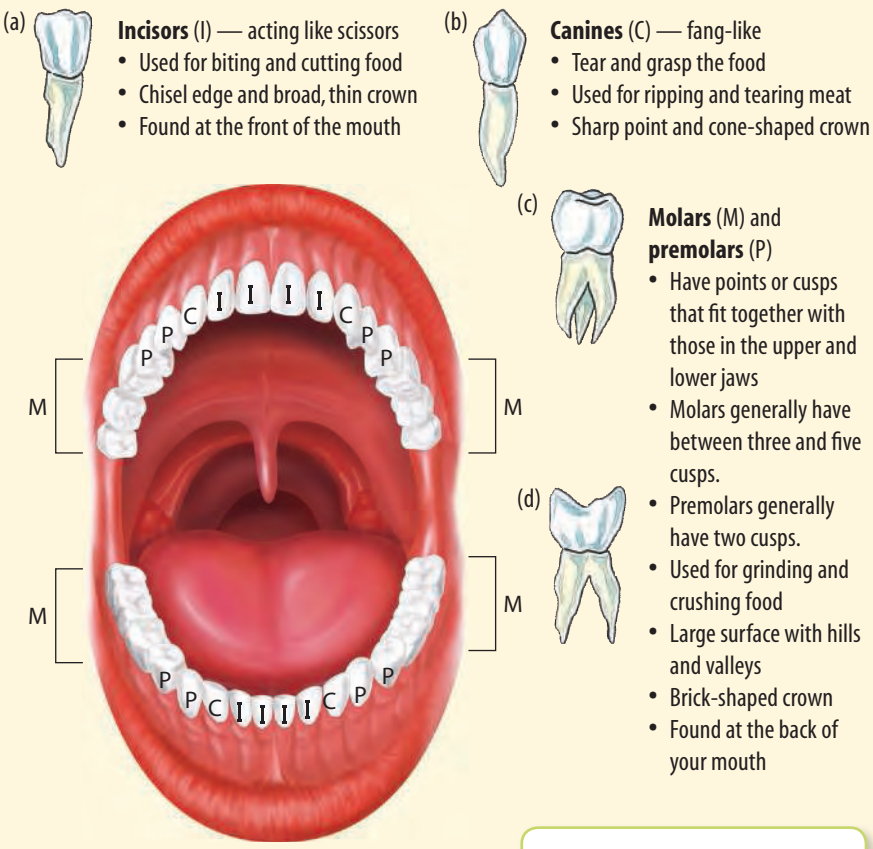
Break it down

Digestion is the breaking down of food so that the nutrients it contains can be absorbed into your blood and carried to each cell in your body.

Five processes are important in supplying nutrients to your cells:
- ingestion — taking food into your body
- mechanical digestion
- chemical digestion
- absorption of the broken-down food into your cells
- assimilation — converting the broken-down food into the chemicals in your cells.

Mechanical and chemical digestion

Mechanical digestion involves physically breaking down the food into smaller pieces. Most of this process takes place in your mouth when your teeth bite, tear, crush and grind food. **Chemical digestion** occurs when chemicals in your body react with the food. The chemical reactions that take place change the substances in food into simpler chemicals that are more easily absorbed into your blood.



Four different types of human teeth

Look at those teeth!

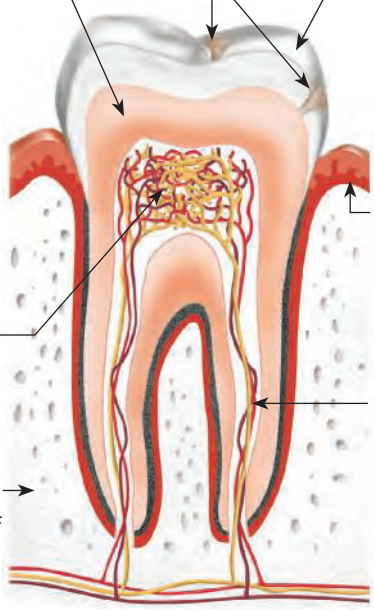
In many vertebrates, mechanical digestion begins with the teeth. There are four main types of teeth in humans, each type having a different function and position in your mouth as shown below. Your teeth are your very own set of cutlery.

Do you look after your teeth?

Your teeth can decay when bacteria in your mouth turn sugar from your food into acid. This acid can 'eat' a hole in your tooth enamel and dentine. Once this hole reaches a nerve, you get a toothache. The illustration opposite shows the structure of a tooth and where decay usually occurs — at the top of large back teeth and at the side where one tooth touches another.

If you don't clean your teeth regularly (at least once a day), they can become covered with a thin film of food, saliva and bacteria. This is called plaque. As this plaque rots, it causes your gums to swell and bleed. This is known as gum disease.

Our water supply and toothpaste often contain fluoride, which helps prevent tooth decay. Fluoride protects the enamel and helps repair or rebuild the enamel in your teeth. Damaged or missing teeth can make it difficult for you to chew your food properly and therefore may affect digestion of foods.



Structure of a tooth showing where tooth decay occurs

What do they eat?

HERBIVORE (WOMBAT)

Herbivores eat plants. They have large incisors for biting and cutting. Herbivores do not have canines. They have large premolars and molars because the fibrous plant material needs a lot of grinding.

CARNIVORE (TASMANIAN DEVIL)

Carnivores eat other animals. Because their prey is alive and moving, they have large canines for stabbing and holding on to it. Their incisors are used for tearing meat. The molars and premolars in carnivores have cutting edges.

OMNIVORE (HUMAN)

Omnivores eat both plants and animals. They have all of the different types of teeth needed to break down both meat and plants.

INQUIRY: INVESTIGATION 4.9

How well do you brush your teeth?

KEY INQUIRY SKILLS:

- planning and conducting
- processing and analysing data and information
- evaluating

Equipment:

3 nutrient agar plates	*toothbrush*
3 small labels	*toothpaste*
3 cotton buds	*incubator set at 35°C*
sticky tape	*dissection microscope*

Note: This is best done after lunch, prior to cleaning teeth.

- Label the three agar plates as 'Dirty', 'Toothpaste' and 'Brush only'.

- Wipe a cotton bud carefully over your uncleaned teeth and gums.
- Gently wipe in a zigzag pattern over the agar plate labelled 'Dirty'.
- Replace the lid and seal around the edges with sticky tape.

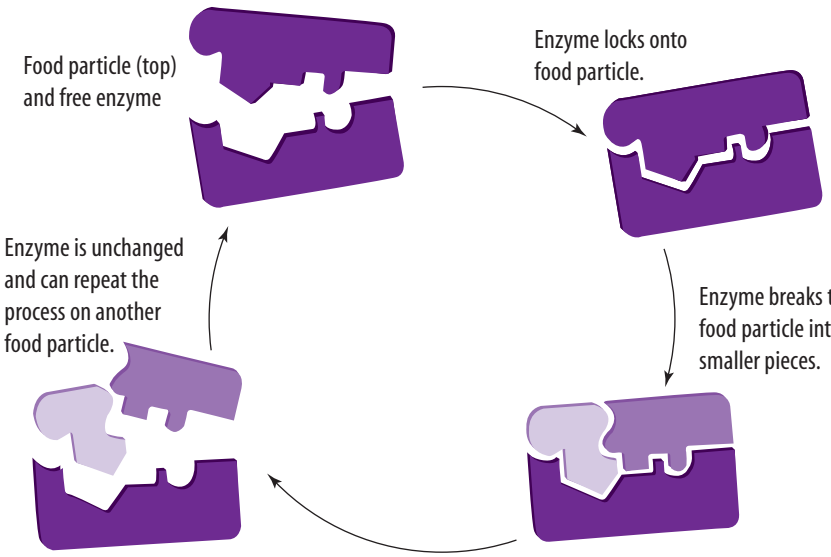
- Repeat the steps above after brushing with no toothpaste and then with toothpaste.
- Incubate the agar plates for 24–48 hours.
- Observe the agar plates using the dissection microscope. Do not open the agar plates. Observe through the plastic dish.

DISCUSS AND EXPLAIN

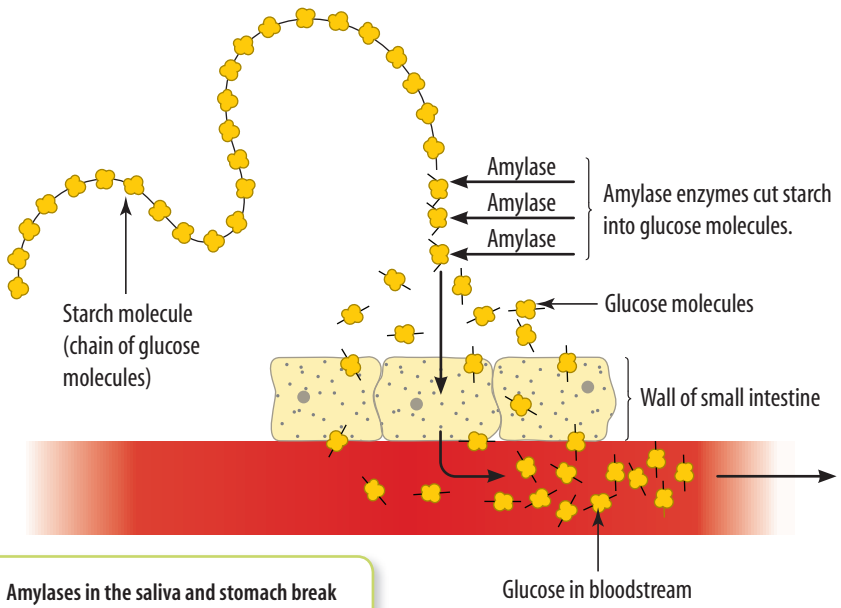
1. Sketch the three agar plates, showing any growth.
2. Which agar plate had the most growth? Why?
3. Suggest why we brush our teeth with toothpaste.
4. Suggest improvements to the design of this investigation.

INSECTIVORE (NUMBAT)

Insectivores are carnivores that eat only insects. Their teeth are small and pointed so that they can crush the exoskeleton of the insect. Insects are then swallowed whole.



Enzymes speed up chemical reactions in the body, but are not changed so can be reused again and again.



Amylases in the saliva and stomach break starch down into glucose molecules.

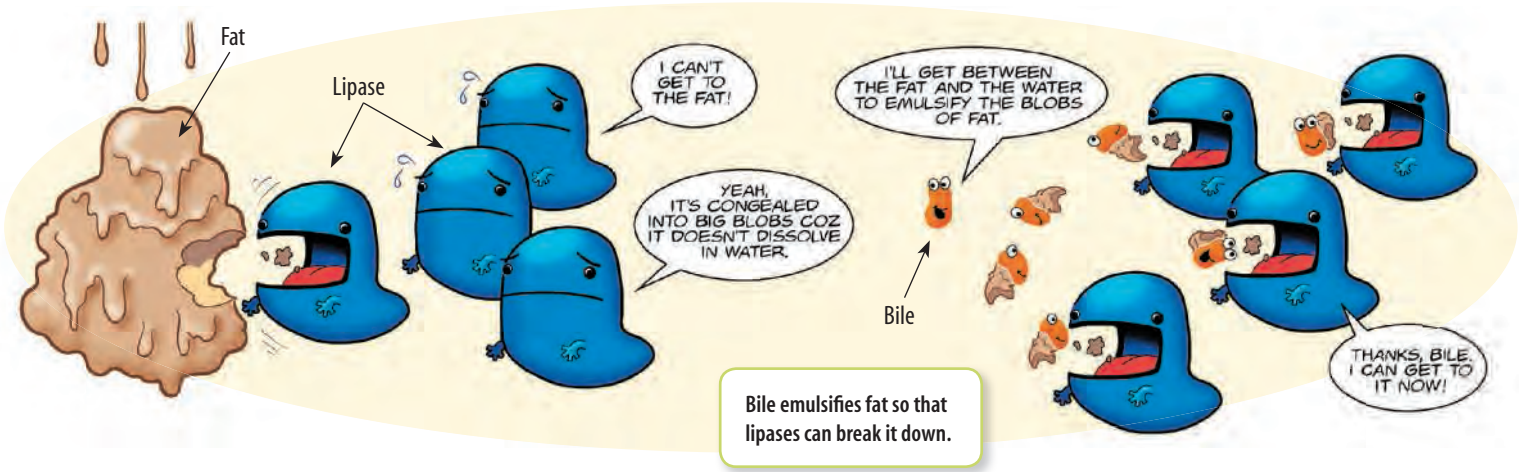
Enzymes — licensed to speed

Chemical digestion is usually assisted by compounds called **enzymes** that increase the rate of the chemical reactions. Without enzymes, a single meal could take many years to break down. Mechanical digestion increases the rate of chemical digestion because it increases the surface area of the food particles. This exposes more of the food surface to the digestive chemicals and enzymes.

Chemical digestion begins in your mouth where enzymes in saliva begin to break down some of the carbohydrates in the food that you eat.

NOT TOO HOT!

Enzymes are made of protein. That is why it is important that they are not overheated. If they are too hot, they can become **denatured**. It's a bit like cooking meat — once they are denatured, they can't go back to how they were before, so they can't work as enzymes anymore. Different enzymes operate best within specific temperature ranges.



Bile emulsifies fat so that lipases can break it down.

'ASE' ENDINGS

There are specific enzymes for specific tasks. The substance that they are breaking down is called a **substrate**. The name given to the resulting substance is referred to as the **product**. As the diagram above left shows, the enzyme remains unchanged at the end of the process.

Enzymes that break down carbohydrates, such as starch, into glucose are called **amylases** (this process is shown in the diagram at left). Those that break down fats and oils into fatty acids and glycerol are called **lipases**. **Proteases** are enzymes that break down proteins into amino acids.

Each enzyme has specific conditions in which it works. For example, amylases that break down carbohydrates in the stomach work in acidic environments, while those in the small intestine work best in alkaline conditions.

Fat stuff

Breaking down lipids, such as fats and oils, is hard work! Because lipids are insoluble in water, they tend to clump together into large blobs. A substance called **bile** helps solve this problem. Bile is produced by your liver and stored in your gall bladder. As half of the bile molecule is attracted to water and the other half attracted to lipids, it helps to **emulsify** or separate the lipids so the lipase enzymes can gain access to them and do their job. This is an example of mechanical digestion (bile) and chemical digestion (lipase) working together to get the job done!

INQUIRY: INVESTIGATION 4.10

Does temperature affect enzymes?

KEY INQUIRY SKILLS:

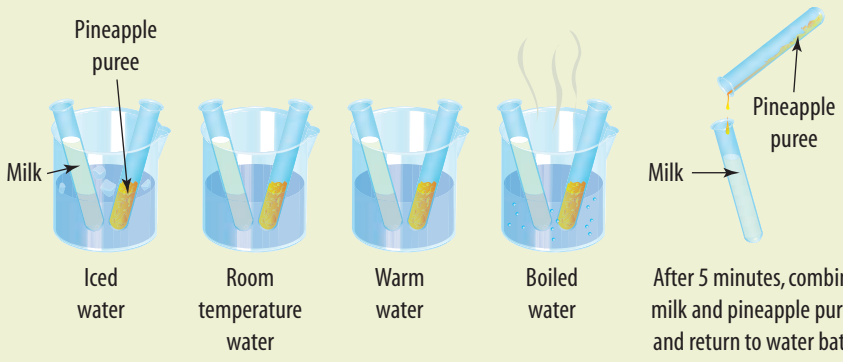
- planning and conducting
- processing and analysing data and information
- evaluating

Equipment:

4 beakers *8 test tubes* *milk* *4 thermometers*

fresh pineapple puree (Fresh pineapple can be pureed using a food processor. If fresh pineapple is not available, use junket powder or a junket tablet dissolved in 10 mL water.)

- Add water to the beakers so that they are two-thirds full. Use cold tap water and ice for beaker 1, cold tap water for beaker 2, hot tap water for beaker 3 and boiling water (from a kettle) for beaker 4. These are the 'water baths'.



- Half-fill four test tubes with milk and put one test tube in each water bath.
- Place one teaspoon of fresh pineapple puree (or 1 mL junket solution) in the other four test tubes. Put one of these test tubes in each water bath.
- Allow the test tubes to stand in the water baths for at least 5 minutes.
- For each water bath, pour the fresh pineapple puree into the milk and stir briefly. Quickly record the temperature of the milk and pineapple mixture and then allow it to stand undisturbed. The mixture will eventually set. Record the time taken to set. If the milk has not set after 15 minutes, record the time as 15+.
- Copy and complete the table of results below.

Water bath	Temperature of milk and pineapple mixture (°C)	Time taken to set (min)

DISCUSS AND EXPLAIN

1. Pineapple juice and junket contain an enzyme that causes a protein in milk (casein) to undergo a chemical reaction and change texture; that is why the milk sets. At what temperature did the enzyme work best? Explain your answer.
2. Did the enzyme work well at very high temperatures? Explain your answer.
3. Which variables were controlled in this experiment?
4. Do you think that the same results would be obtained if tinned pineapple puree was used instead of fresh pineapple? Explain your answer.

A future in teeth

Dentistry is only one example of many different 'tooth pathway' careers that you may be aware of. Examples of other dental specialities include oral and maxillofacial surgeons, dental–maxillofacial radiologists, endodontists, oral physicians, oral pathologists, orthodontists, paediatric dentists, periodontists, prosthodontists, public health dentists and special needs dentists.

Missing a tooth? A synthetic replacement for a tooth root is used in a tooth implant.

Oral surgeon is one of the many tooth-related careers you can choose from.

UNDERSTANDING AND INQUIRING

REMEMBER

1 Explain why it is important to break down the food that we eat.
2 Use a Venn diagram to compare mechanical digestion and chemical digestion.
3 Identify four types of teeth and describe their functions.
4 Construct a matrix table (see page 340) to compare the diets and teeth of herbivores, carnivores, omnivores and insectivores.
5 Outline what enzymes are and what they do.
6 State the name of the:
(a) type of digestion that enzymes are involved in
(b) enzymes that break down fats
(c) enzymes that break down proteins
(d) substance that enzymes act on.
7 Describe what happens to enzymes when they get too hot.
8 Describe how bile helps lipase enzymes get their work done.

THINK

9 Which teeth do you use to bite into a pear?
10 Which teeth are used to crush and grind nuts?
11 Why don't herbivores have canine teeth?
12 How do we know what different types of dinosaurs ate, even though they haven't existed for about 65 million years?

THINK AND DESIGN

13 Design an investigation to test the following hypotheses:
- Drinking fluoridated water reduces tooth decay.
- Mouthwash prevents the growth of bacteria that cause tooth decay.
- Drinking bottled water rather than tap water increases tooth decay.

14 Design an investigation to test the following hypotheses:
- Fresh pineapple results in a faster enzyme reaction than canned pineapple.
- The length of time that pineapple puree is kept in ice affects the rate of enzyme reaction.
- Different coloured junket tablets result in different rates of enzyme reaction.

INVESTIGATE AND CREATE

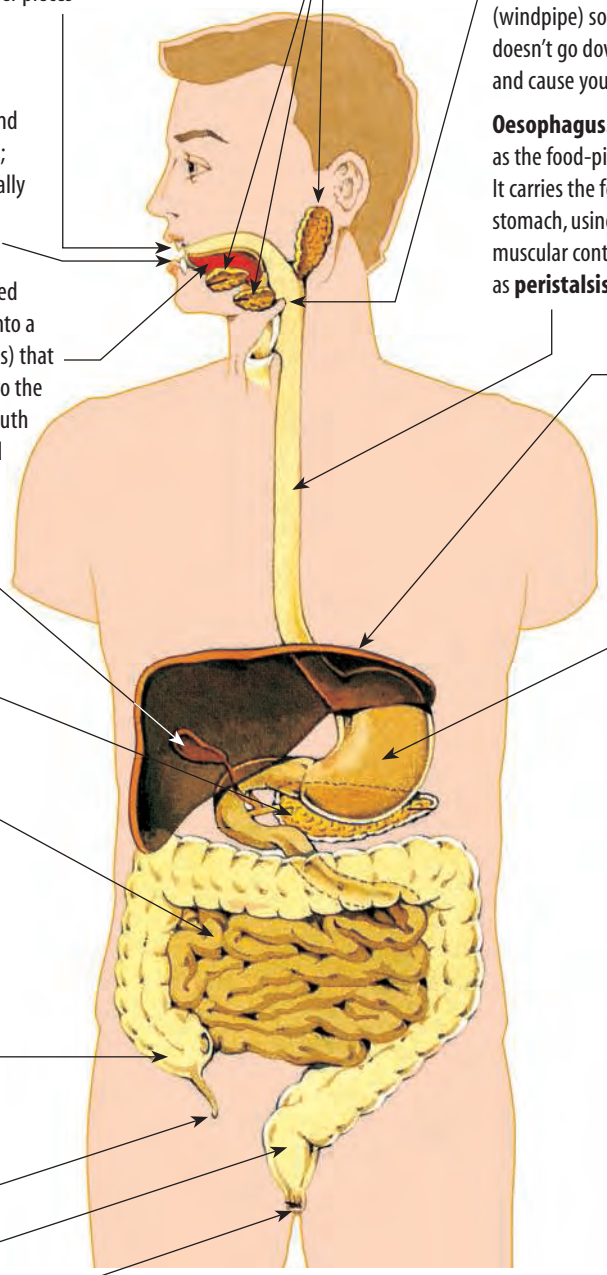
15 Research an enzyme of your digestive system. Find out what it does and where, then construct a poster or model to show how it works.
16 Select one of the 'tooth pathway' careers in the text 'A future in teeth'. Find out details of the training required and what a career in this pathway would entail. Present your findings in a brochure and include a section that describes what 'a day in the life of …' this career would be like.

work sheet 4.5 Mechanical and chemical digestion

Down we go...

The human digestive system

Your main digestive highway is called the **alimentary canal**. It consists of a long tube with coils, large caverns and thin passageways. Other organs that provide chemicals to break down the food or absorb nutrients are attached to the alimentary canal. The alimentary canal begins at the mouth and ends at the anus, where waste products are removed. Excretion of waste products produced by the body cells can also involve other organs, such as the skin, lungs and kidneys.



Salivary glands. Make about 1.5 litres of saliva each day. Saliva contains enzymes that begin breaking down starch in food.

Teeth. Used to bite and chew your food to break it down into smaller pieces

Epiglottis. A flap of tissue closes off your trachea (windpipe) so that food doesn't go down to your lungs and cause you to choke.

Mouth. Food and saliva are mixed; teeth mechanically break food into smaller pieces.

Oesophagus. Also known as the food-pipe or gullet. It carries the food to the stomach, using involuntary muscular contractions known as **peristalsis**.

Tongue. Involved in rolling food into a round ball (bolus) that is then pushed to the back of your mouth to be swallowed

Liver. The largest internal organ. It makes bile, which breaks down fats; controls blood sugar; destroys poisons; and stores vitamin A, vitamin D and iron.

Gall bladder. Bile made in the liver is stored here; bile breaks up fats into droplets small enough to be transported to the rest of the body.

Stomach. A temporary food storage area, which can expand to hold between 2 and 4 litres of food. Muscle movements in the stomach wall mix the food with gastric juice, which helps to break down proteins. The stomach also contains dilute hydrochloric acid, which kills germs and provides a suitable environment for protein digestion.

Pancreas. Makes pancreatic juice, which is alkaline (base dissolved in water) so it neutralises the stomach acid. Enzymes that break down proteins, fats and carbohydrates are also made here.

Small intestine. A tube about 6 m long. Food moves through it by peristalsis. The small intestine makes enzymes that complete the digestion process. The cells in the wall of the small intestine release over 5 litres of mucus and water each day. It is in the small intestine that nutrients from now almost totally digested food are absorbed into your bloodstream. The blood then carries the nutrients to all of the cells of your body.

Large intestine (colon). Undigested material passes into the large intestine. As the material is pushed through it by peristalsis, water, salts, vitamins and any remaining sugars are absorbed so that they can be reused by the body.

Appendix. Plays no part in digestion in humans. However, it is believed to play a role in fighting some diseases.

Rectum. The final part of the large intestine. This is where the faeces are stored.

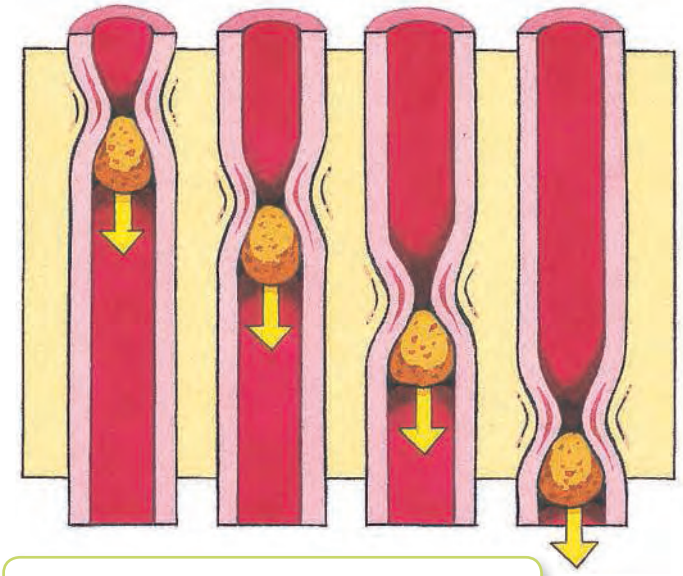
Anus. The faeces pass through here when you go to the toilet.

eBook plus eLesson

From dinner plate to sewerage system

Watch the amazing journey of food through the human body.

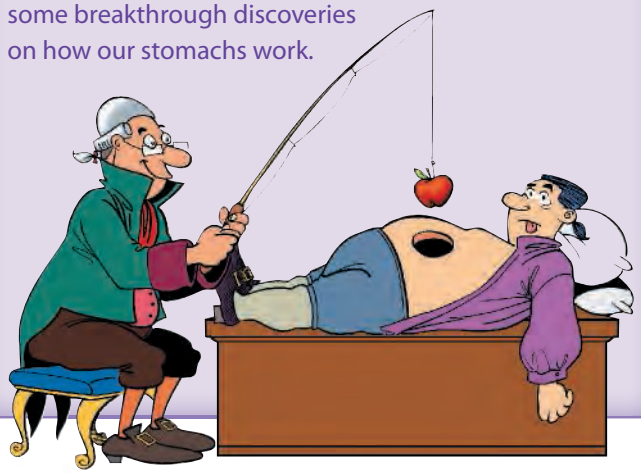
eles-0056



Partly digested food is forced along the oesophagus by peristalsis — a wave of involuntary muscular contractions.

HOW ABOUT THAT!

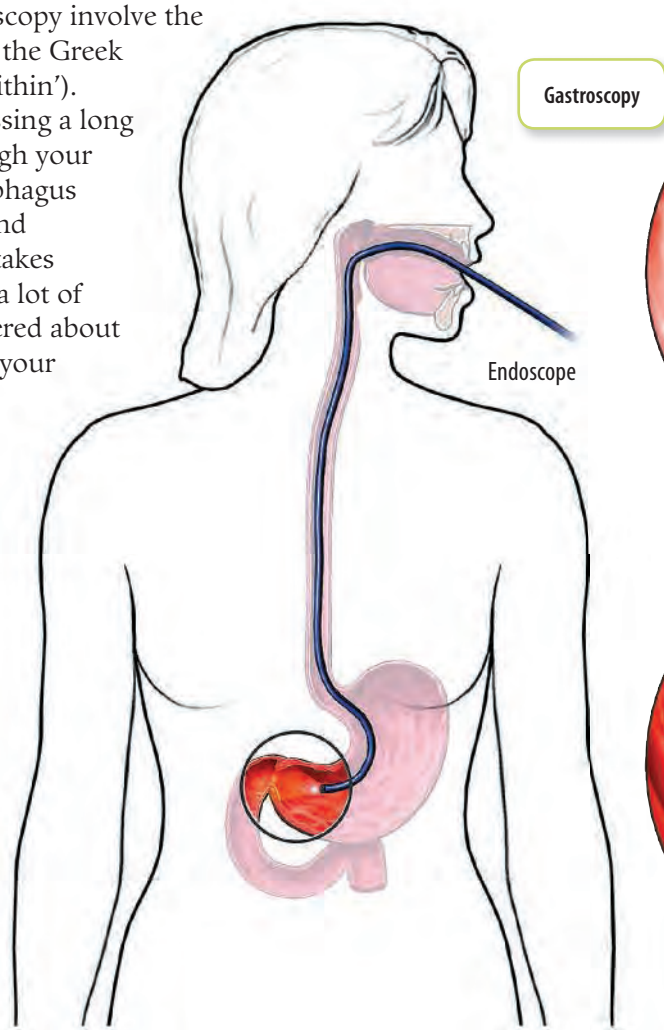
Just to whet your appetite . . . have you heard the story of Alexis St Martin? In 1822, he was shot in the stomach at close range. Wrong place at the wrong time! His wound healed, but left an open hole that showed the inside of his stomach! By dangling food suspended on a silk thread, his doctor William Beaumont made some breakthrough discoveries on how our stomachs work.

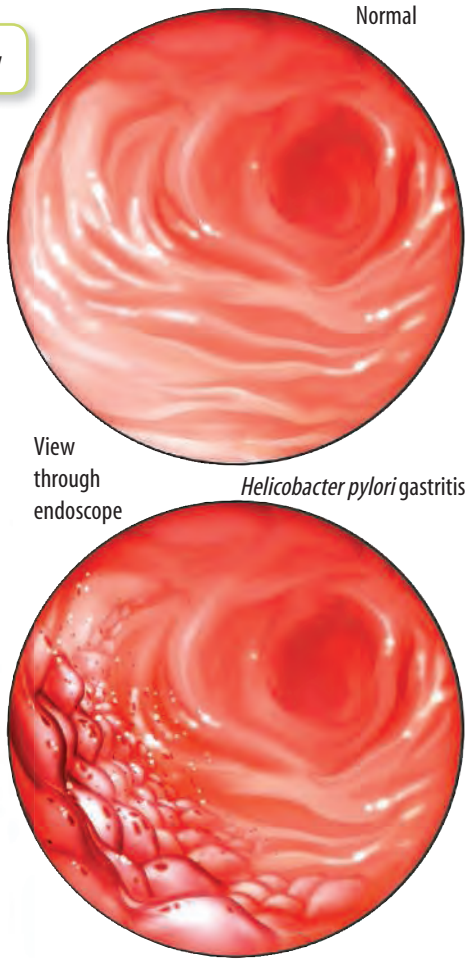


Going up or down?

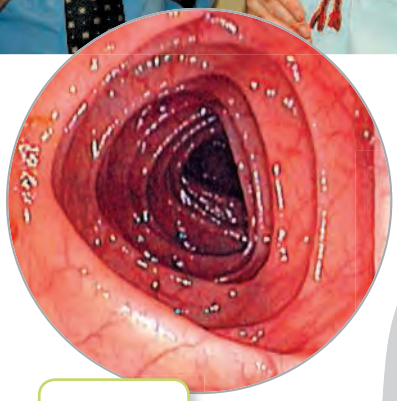
Gastroscopy and colonoscopy involve the use of endoscopes (from the Greek words meaning to 'see within'). Gastroscopy involves passing a long flexible endoscope through your mouth, down your oesophagus and into your stomach and duodenum. Although it takes only about five minutes, a lot of information can be gathered about the health of this part of your digestive system.

Colonoscopy enables the doctor to look directly at the lining of your colon or bowel. In this case the endoscope is inserted through your rectum. This procedure may take about thirty minutes. The results may be used to investigate abnormalities or detect the presence of colon polyps, which in some cases may turn into cancer.

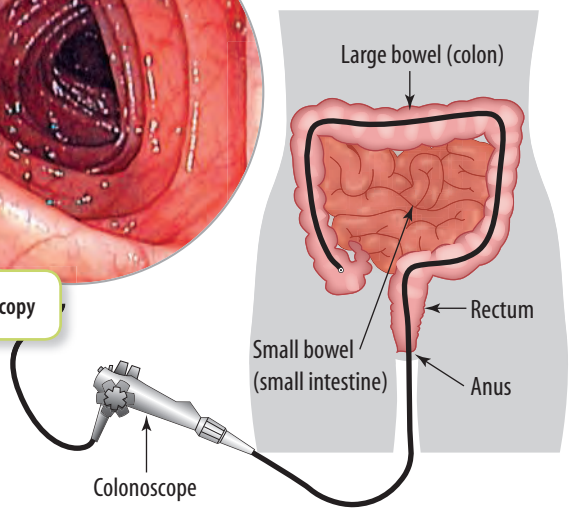




Australian scientists Dr Barry J. Marshall and Dr Robin Warren received the 2005 Nobel Prize in Medicine for their discovery that linked *Helicobacter pylori* bacteria to gastroduodenal disease. Their discovery dramatically improved the treatment of peptic ulcer disease.

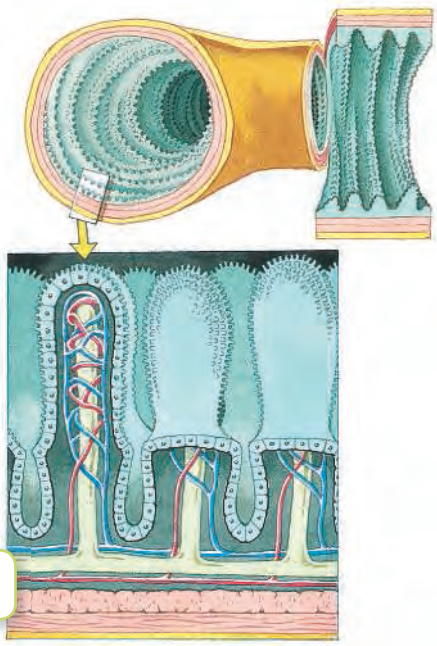


Colonoscopy



'Fingers' of absorption

Most nutrients are absorbed into your bloodstream when they get to the last section of your small intestine. The walls of this part of your intestine are lined with finger-like projections called villi.

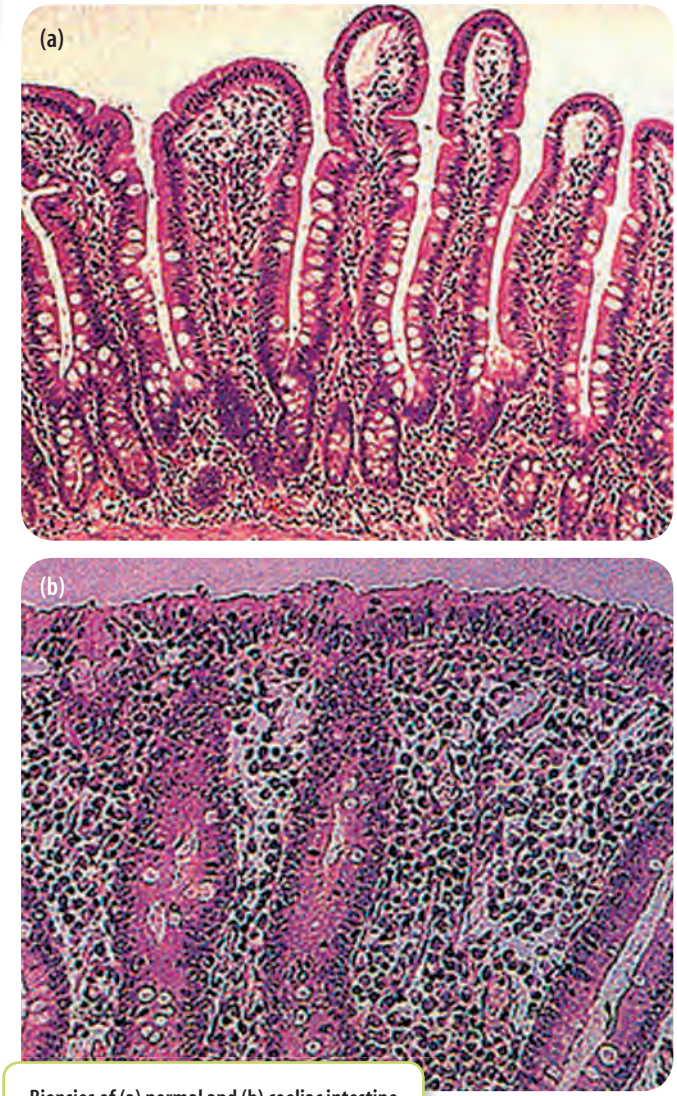


Villi

The shape of villi increases the surface area through which nutrients can diffuse through them into tiny blood vessels called capillaries. Nutrients are then transported to other parts of your body.

VILLI ALERT!

Coeliac disease is an auto-immune disease — that is, one in which your body produces antibodies to attack your own tissues. In this case, it is the villi of the small intestine that are damaged. This interferes with the absorption of nutrients.



Biopsies of (a) normal and (b) coeliac intestine

People with coeliac disease are intolerant to gluten. This protein is found in wheat, rye, barley and oats. Eating these foods triggers their immune system to damage the villi in their small intestine. The Coeliac Society of Australia refers to the condition as a 'hidden epidemic'. Coeliac disease affects approximately 1 in 100 people in Australia, with

many (about 75 per cent) not even knowing that they have it. If left undiagnosed, more severe consequences such as a variety of nutritional deficiencies, bowel cancer and osteoporosis may result.

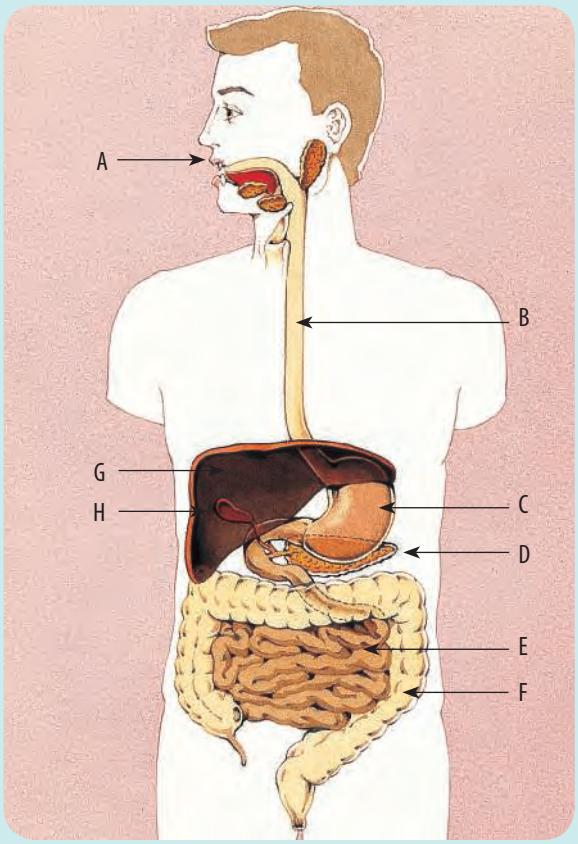
Australian researchers are attempting to develop a vaccine as a new treatment for coeliac disease. In 2009, Dr Robert Anderson and his team at the Walter and Eliza Hall Institute of Medical Research in Melbourne began the world's first trials of a coeliac vaccine. If this treatment is successful, it could mean the end of gluten-free diets for people with the condition.

Dr Robert Anderson and his team at the Walter and Eliza Hall Institute of Medical Research in Melbourne began the world's first trials of a coeliac vaccine.

UNDERSTANDING AND INQUIRING

REMEMBER

1 (a) Construct a table with the column headings 'Letter', 'Organ', 'Description', 'Function' and 'Interesting comment'.



(b) Write the letter label from the figure on the left in the 'Letter' column in your table.
(c) Complete the remaining columns.

2 Describe the process of peristalsis and suggest why it occurs.

3 Describe what happens to:
(a) digested food
(b) undigested food.

4 Outline the relationship between diet, coeliac disease and the digestive system.

5 Approximately how many people in Australia are affected by coeliac disease?

6 Use a Venn diagram to compare gastroscopy and colonoscopy.

7 Describe the discovery that led to two Australian scientists winning the 2005 Nobel Prize in Medicine.

THINK AND DISCUSS

8 Explain why it is necessary for your body to digest the food that you eat.

9 Suggest how you could still swallow food even if you were positioned upside down.

IMAGINE, INVESTIGATE AND CREATE

10 (a) Imagine that you are either a cheese and tomato sandwich or a hamburger.
(b) List the ingredients of the food you chose in part (a).
(c) Research what happens (and where) to each of these ingredients when eaten.

(d) Construct a flowchart to show the process of digestion in the human body including events and locations.
(e) Use this information to write a story in either a cartoon or picture book format.
(f) Convert your story into a play.
(g) Perform your play to the class using animations, team members or puppets.

11 Research one of the digestion-related diseases below and report on the cause, symptoms, treatment or cure, possible consequences and current research.

- Heartburn
- Inflammatory bowel disease
- Irritable bowel syndrome
- Appendicitis
- Constipation
- Crohn's disease
- Diverticulosis
- Gallstones
- Haemorrhoids
- Pancreatitis
- Peptic ulcer

12 Use information in this section and other resources to relate structural features to the functions of the following parts of the digestive system.

Part of system	Structural features	Function
Oesophagus		
Stomach		
Small intestine		
Villi		
Large intestine		

13 Imagine that you have invited two friends over for a sleepover. One of them has coeliac disease and the other is lactose intolerant.
(a) Find out the cause and/or symptoms associated with each of these conditions.
(b) Find out what sorts of foods you could offer your friends.
(c) Design a dinner and breakfast menu that includes foods that each of your friends would be able to eat.

14 Construct a 'working' model of the human digestive system.

15 Design and construct a 'trivial pursuit'–type board game about the digestive system. Include questions in three different categories. Create game pieces that are relevant to the digestive system.

16 Create a picture book or poster that would help explain to primary students how food is digested.

17 Construct a plasticine model (to scale) of the human digestive system, with each organ being a different colour.

INVESTIGATE AND REPORT

18 Find out more about one of the following digestion-related scientific careers and report your findings as a diary journal entry.

- Gastroenterologist
- Endoscopist
- Colorectal surgeon

19 Research the symptoms, diagnosis and treatment of coeliac disease and investigate Australian research into the condition. Report your findings in a PowerPoint presentation, a poster or an article. Cite your references and include the **Coeliac Society** and **Australian Gastronoenterology Institute** weblinks in your eBookPLUS in your research.

eBookplus

20 Find out more about gluten-free diets and coeliac disease using the weblinks in your eBookPLUS. Report your findings to your team or use them to support your response to question 12.
(a) Use the **Coeliac dietary advice** weblink to find out more about gluten-free diets.
(b) Use the **Bob Anderson** weblink to find out more about coeliac disease, diet and research.

21 Test your knowledge of the digestive system by completing the **Digestive jigsaw** interactivity in your eBookPLUS. **int-0216**

22 Recently, scientists have suggested a link between the presence of bacteria *Helicobacter pylori* and cancer protection. Find out more about this research and suggest possible implications that it may have.

work sheet 4.6 The digestive system

Keeping your balance

Blood vessels and the heart are not just involved in the delivery of needed nutrients and gases to the cells of the body. They also take away the body's wastes so that they may be disposed of. If the wastes were allowed to build up inside us, we would die.

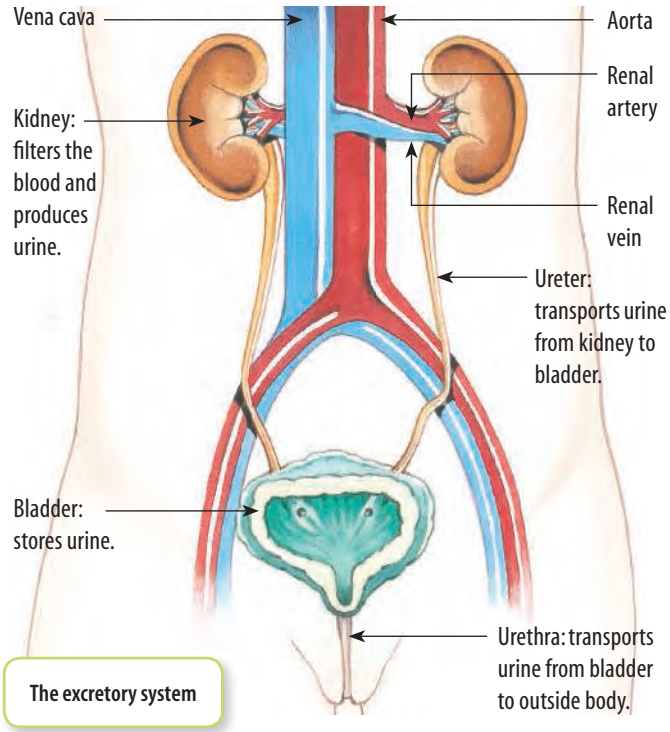
Getting rid of wastes

Excretion may be defined as any process that gets rid of unwanted products or wastes from the body. The main organs involved in human excretion are the **lungs**, the **liver** and the **kidneys**.

If you put your hands on your hips, your kidneys are close to where your thumbs are. You have two of these reddish-brown, bean-shaped organs. Without them you would survive only a few days.

Kidneys play an important role in filtering your blood. About a quarter of the blood that your heart pumps is sent to your kidneys. These small organs filter about 50 litres of blood each hour. Your kidneys remove waste products of chemical reactions, like **urea**, from your blood. Accumulation of these substances would otherwise be toxic or dangerous to your body. Other substances, like salts and water, which may be in excess, can also be removed. This keeps their concentration in the blood constant. If this did not occur, your cells would not work properly.

Urine is produced by your kidneys. This watery fluid contains unwanted substances. Tubes called **ureters** transport urine from your kidneys to your **bladder** to be stored temporarily. As it fills, your bladder expands like a balloon. It can hold about 400 mL of urine. **Urination** occurs when urine moves from your bladder through a tube called the **urethra** and out of your body.



The excretory system

HOW ABOUT THAT!

The human kidneys remove excess salt from the blood to help keep levels constant. Different types of animals have other ways of removing excess salt from their bodies. Turtles, for example, have salt-secreting glands behind their eyes. Hence you may see a turtle 'shedding tears'. On the other hand, penguins and some other seabirds, such as the Southern Giant Petrel, may appear to have runny noses because that is where their salt-secreting glands are located.

Blood, water and urine

Both blood and urine are mostly made up of water. Water is very important because it assists in the transport of nutrients within and between the cells of the body. It also helps the kidneys do their job because it dilutes toxic substances and absorbs waste products so that they may be transported out of the body.

A comparison of what is found in blood and in urine

Substance	Quantity (%)	
	In blood	In urine
Water	92	95
Proteins	7	0
Glucose	0.1	0
Chloride (salt)	0.37	0.6
Urea	0.03	2

Haemodialysis

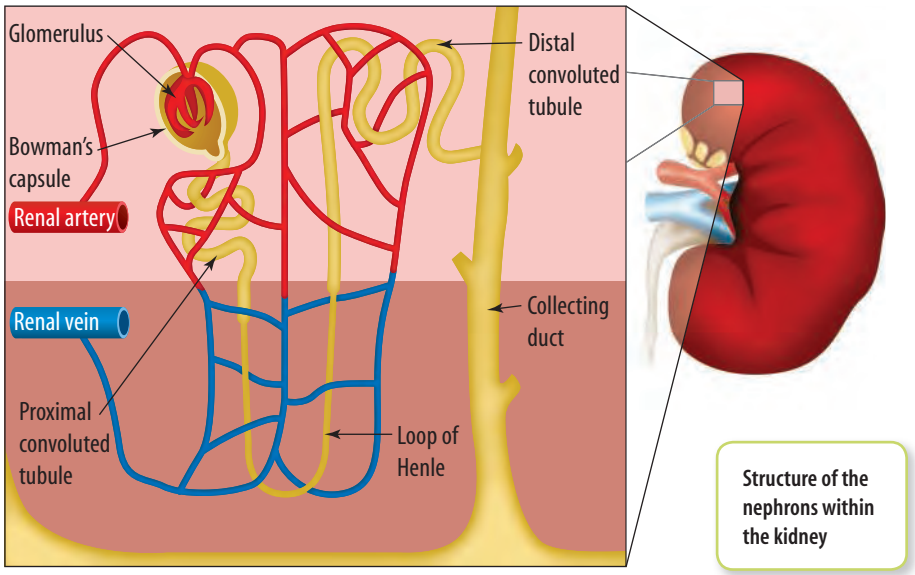
People with kidney disease may not be able to remove the waste materials from their blood effectively. They may be linked up to a machine that does this job for them. Their blood is passed along a tube that lets wastes, such as urea, pass out of it. However, useful substances, such as glucose, proteins and red blood cells, stay in the tube and are kept in the blood. This process is called **haemodialysis**.

Too much or too little

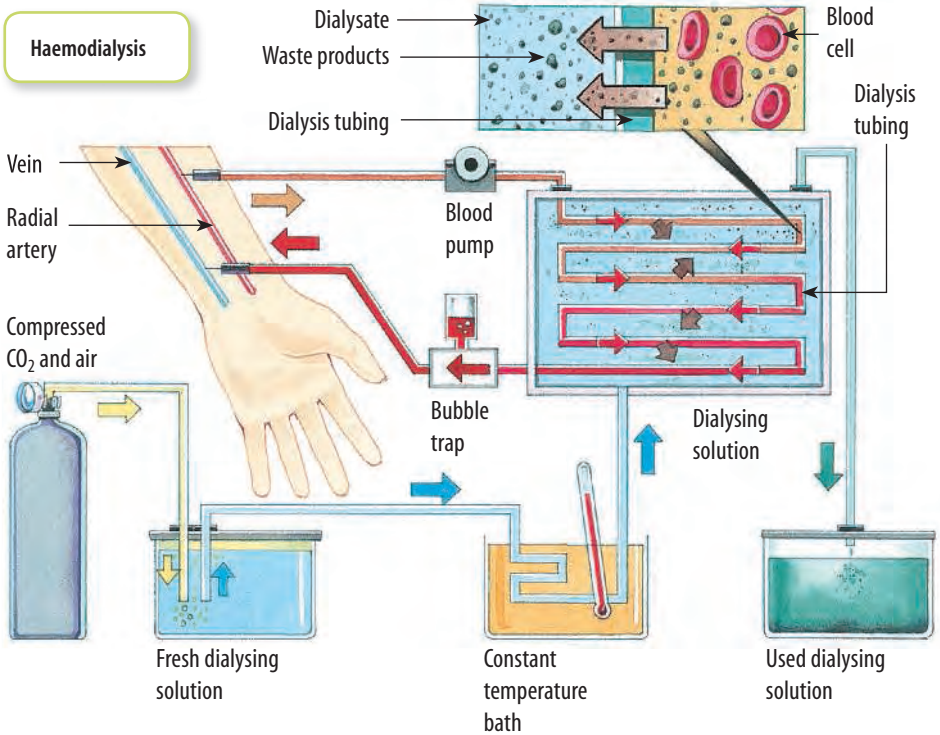
The concentration of substances in the blood is influenced by the amount of water in it. If you drink a lot of water, more will be absorbed from your large intestine and the kidneys will produce a greater volume of dilute urine. If you do not consume enough liquid, you will urinate less and produce more concentrated urine.

Millions of tiny filters?

Each of your kidneys is made up of about one million **nephrons**. Each nephron is made up of a long tubule (very fine tube) that forms a cuplike structure at one end. This structure is called the **Bowman's capsule** and it surrounds a cluster of capillaries called



Structure of the nephrons within the kidney



the **glomerulus** (from an ancient Greek word meaning 'filter').

'Dirty' blood containing wastes travels to the glomerulus within each nephron in your kidneys. It is here that the blood is filtered, with the wastes and excess water moving into the Bowman's capsule. From here, this waste fluid moves along the tubule and any useful substances are reabsorbed back into the circulatory system into capillaries that are 'twisted' around it. The remaining fluid becomes urine, which eventually travels in your ureters to your bladder.

UNDERSTANDING AND INQUIRING

REMEMBER

1 Define the term 'excretion'.

2 Draw and label a diagram of the kidneys showing the following attachments: renal arteries, renal veins, ureters, bladder.

3 State features that blood and urine have in common.

4 Outline what happens when you drink a lot of water.

5 Describe one way in which excess salt is removed from your body.

6 Explain how haemodialysis assists people with kidney disease.

ANALYSE AND EVALUATE

7 Use the table on the previous page and the other information in this section to answer the following questions.
   (a) Draw two bar graphs to show the quantity of water, proteins, glucose, salt and urea in blood and in urine.
   (b) Which substance is in the greatest quantity? Suggest a reason for this.
   (c) Which substances are found only in blood?
   (d) Which substances are found in urine in a greater quantity than in blood? Suggest a reason for this.
   (e) When would the amount of these substances in the urine become greater or less than in the blood?

THINK

8 Carefully observe the haemodialysis diagram on the previous page. Suggest reasons the following are included in the process:
   (a) blood pump
   (b) bubble trap
   (c) constant temperature bath.

9 Suggest what you would expect to find in used dialysing solution.

10 Suggest why red blood cells don't pass through the dialysis tubing.

11 Use a Venn diagram to compare haemodialysis with real kidneys.

INVESTIGATE AND CREATE

12 Research and report on one of these conditions: urinary incontinence, kidney stones, kidney transplants, cystitis, blood in urine, proteinuria, nephritis.

13 Find out and report on:
   (a) the differences between the urethra in human males and females
   (b) why pregnant women often need to urinate more frequently
   (c) how the prostate gland in males may affect urination in later life
   (d) which foods can change the colour or volume of urine
   (e) which tests use urine in the medical diagnosis of diseases.

14 Find out more about nephrons and how they work. Construct a model of a nephron that shows how it is linked to blood vessels and how urine gets to your bladder from it.

15 Research the nephrons of animals that live in different environments (for example, deserts, oceans and rivers). Comment on similarities and differences in their structure. Suggest reasons for the differences.

eBook*plus*

16 Use the **Kidney Health Australia** weblink in your eBookPLUS to watch the presentation by Professor Allan Collins. Use this website or other resources to find the answers to the following questions:
   (a) What can you do to reduce your risks of developing kidney disease?
   (b) How much water should you drink each day to keep your kidneys healthy?
   (c) Is there any evidence to support the claim that cranberry juice can reduce your risk of urinary tract infections? If so, what is it?
   (d) Some people donate a kidney to another person. How can they survive with only one kidney? Are there any side effects?
   (e) Can you buy or sell a kidney in Australia?
   (f) How can you reduce your risk of developing a urinary tract infection?
   (g) What are kidney stones? How do you get them? What are the symptoms? What is the treatment?

17 Design and create a 'picture book' story that will teach primary school children about how their urinary system works and how to look after it.

Same job, different path

Similar, but different?

Patterns, order and organisation

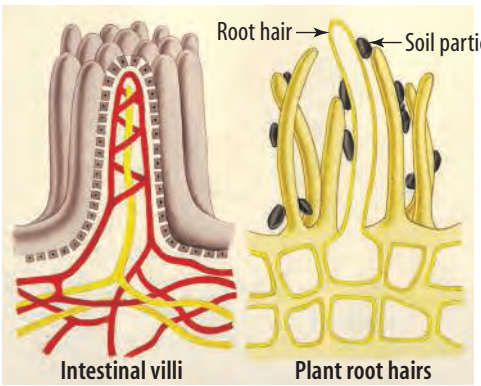
Organisms possess a variety of structures that help them to obtain the resources that they need to survive. While there are similarities and patterns in some of these structures, there are also differences. These differences provide examples of wonderful creative solutions to the continual challenge of staying alive.

See the human systems and tissues mind maps on page 79 for more examples of organisation in the body. Can you suggest other examples to include in the systems mind map? How would this map be different if it was for a different animal?

Organisms have different solutions to life's challenges within the same pattern of organisational framework.

Shaping clues

The structures of cells and tissues often provide clues to their function. For example, structures that are involved in absorption often have shapes that increase the surface area to volume ratio. Intestinal villi in humans and plant root hairs are examples of this. Can you see the similarities in the figures below? Can you think of other cells or tissues that also share this pattern?



Similar structures? What might their function be?

Respiratory routes

glucose + oxygen ⟶ carbon dioxide + water + energy

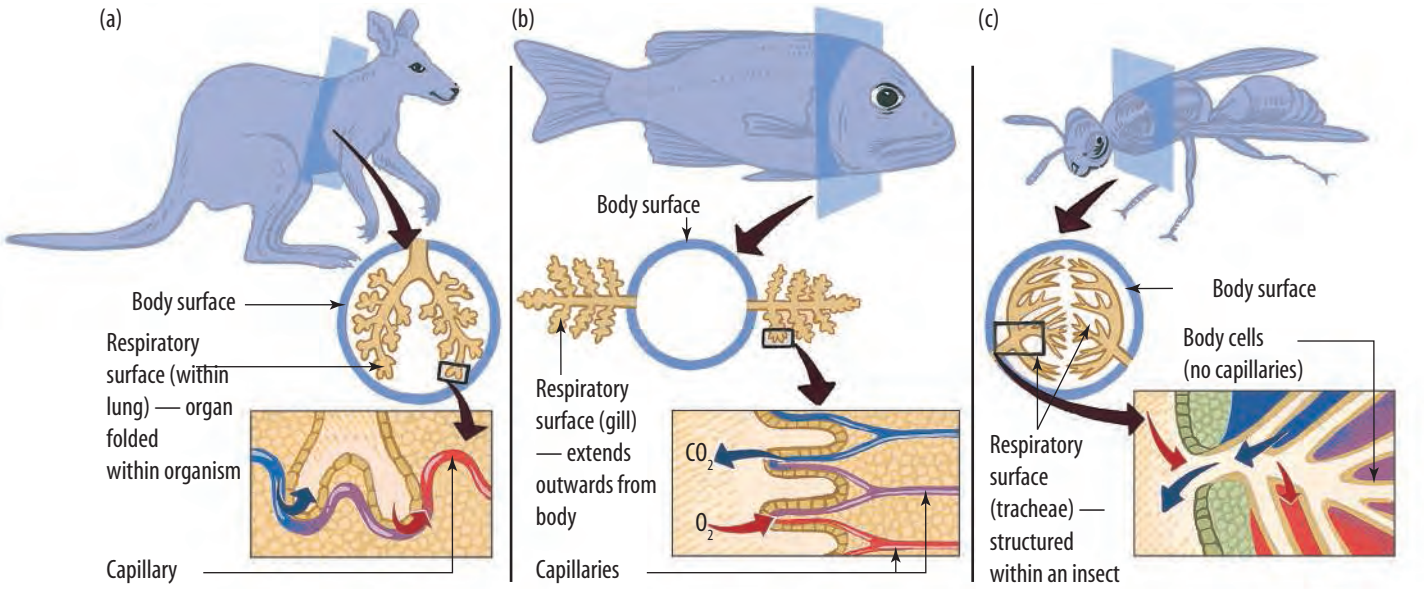
Cellular respiration is essential for life. Organisms require a supply of oxygen for this process, and they also need a way to remove the carbon dioxide that is produced as a waste product. Three main kinds of organs that are involved in this type of gas exchange are lungs in mammals and amphibians, gills in fish and tracheae in insects. Examine the figure below that shows the structures of these organs. How are they similar? How are they different?

Digestive differences

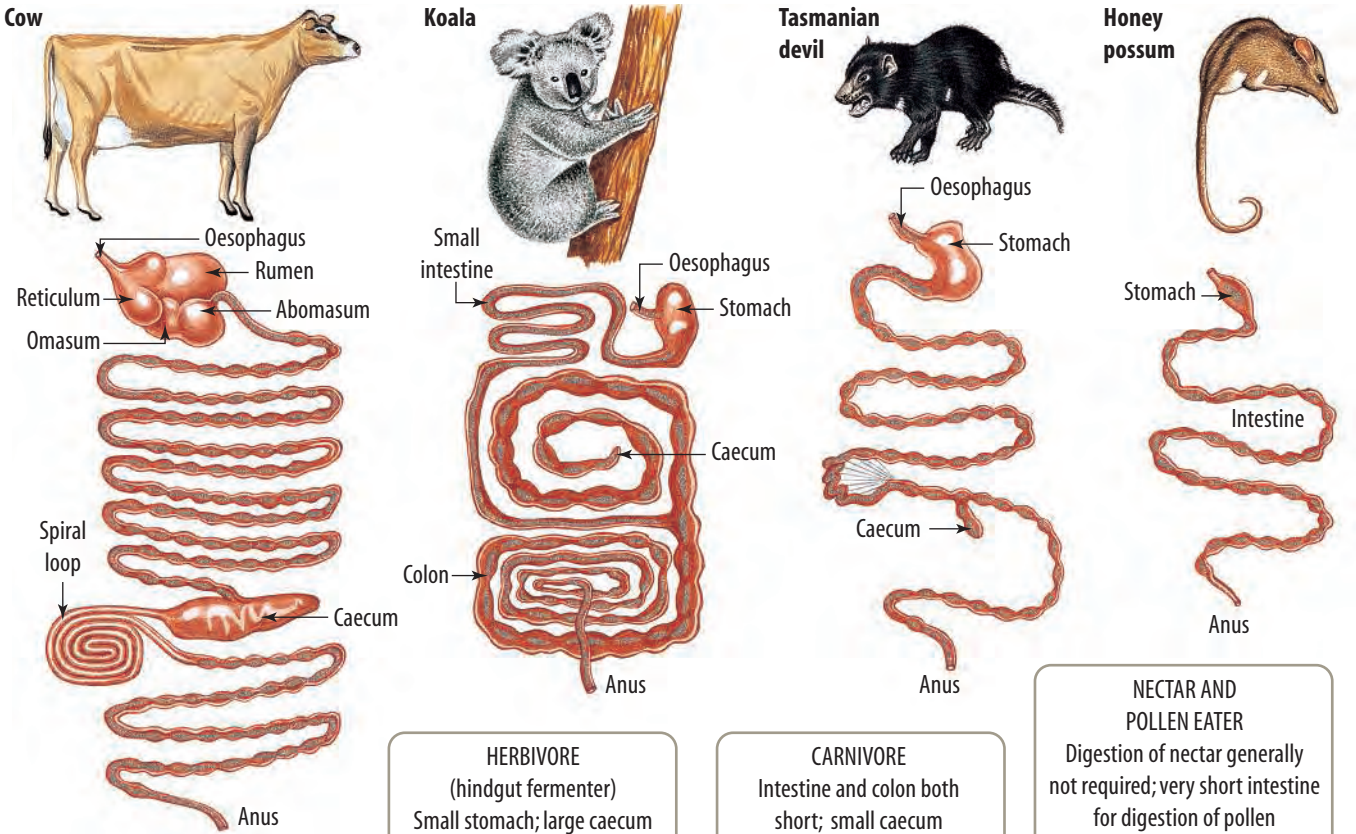
Although most vertebrates possess a digestive system that has a similar pattern, order and organisation, there may be differences that are related to nutritional needs and diet. Consider, for example, differences in the digestive systems of animals with diets that are high in plant material with lots of cellulose, such as diets of herbivores compared with those of carnivores with lots of animal flesh, high in protein. How would these compare with the digestive system of an organism that ate only nectar and pollen?

Heart count?

Two, three or four? Not all animals have a four-chambered heart like you. Fish have a heart with two chambers and blood passes through the heart only once each time around the body. The hearts of amphibians and most reptiles are three chambered and allow oxygenated and deoxygenated blood to mix. Birds and mammals are similar to amphibians and most reptiles in that blood flows through the heart twice in each circulatory trip, but they possess a heart with four chambers that does not allow the mixing of blood. What do they share? How are they different?



Notice any similarities or differences in these gas exchange surfaces?



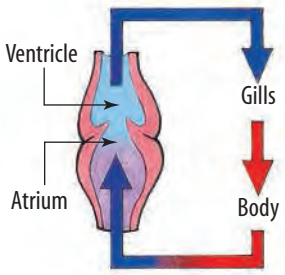
HERBIVORE (foregut fermenter)
Four-chambered stomach with large rumen; small and large intestines are long

HERBIVORE
(hindgut fermenter)
Small stomach; large caecum

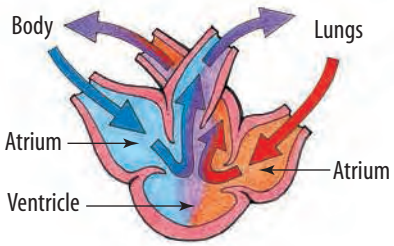
CARNIVORE
Intestine and colon both short; small caecum

NECTAR AND POLLEN EATER
Digestion of nectar generally not required; very short intestine for digestion of pollen

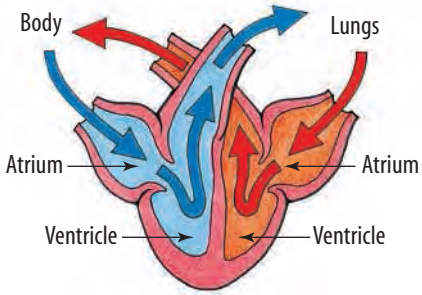
Notice any similarities or differences in these digestive systems?



(a) A fish heart has two chambers. Note that blood passes through the heart only once for every circulation within the body.



(b) Amphibians and most reptiles have a three-chambered heart. Oxygenated and non-oxygenated bloods mix in the single ventricle as blood flows through the heart twice for every circulation within the body.



(c) Birds and mammals have a four-chambered heart and blood flows through the heart twice for every circulation within the body.

Around we go … but which route do we take?

Congratulations!

The survival of a species depends on its members surviving the challenges related to both their internal and external environments. Over many generations, the selection of 'what works' and 'what doesn't work' has led to the diversity of organisms on our planet. Every organism is a celebration of the creative variety of solutions to challenges of survival.

INQUIRY: INVESTIGATION 4.11

Making a burp model

KEY INQUIRY SKILLS:

- planning and conducting
- evaluating

Equipment:

vinegar

baking soda

medium/large balloon

funnel

- Pour a small amount of vinegar into the bottom of the balloon 'stomach'.
- Add some baking soda to the balloon 'stomach' using a funnel.
- Using your fingers, pinch the balloon closed at its neck.
- Watch as your 'stomach' expands with gas.
- Unpinch the top of the balloon (or 'oesophagus/food tube') to release the gas (or burp).
- Try to make your model sound like the real thing!

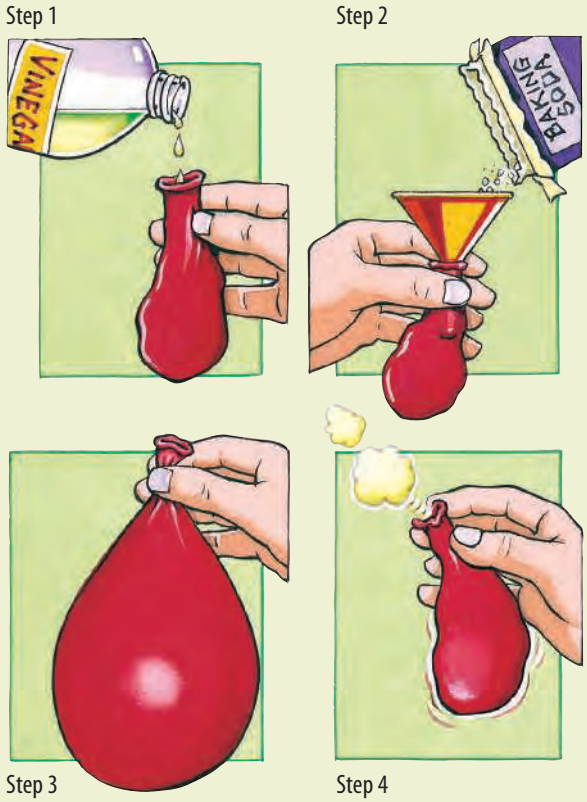
DISCUSS AND EXPLAIN

1 Select an organ belonging to an animal of your choice.

2 Find out more about the structure and function of your selected organ and how it does its job.

3 Design and make a simple model (such as the one used for this experiment) to show how your selected organ achieves its function, or what happens when something goes wrong.

4 Comment on the challenges you experienced during the design and construction of your model, and suggest ways that you could overcome these if you were to do it again.



UNDERSTANDING AND INQUIRING

REMEMBER

1 Place the following terms in order of simplest to most complex: cell, organ, system, multicellular organism, tissue.

2 Provide an example of how structure can give clues about function.

3 Write the word equation for cellular respiration.

4 Describe two key functions of gaseous exchange.

5 Suggest why there are differences between herbivores and carnivores in the structures of their digestive systems.

6 Construct a table to summarise similarities and differences between the hearts and circulation of fish, amphibians and mammals.

7 Name two organs belonging to each of the following systems.
(a) Respiratory system
(b) Circulatory system
(c) Excretory system

THINK

Use the mind maps on page 79 and information in this section to answer the following questions.

8 What kind of tissue:
(a) has the function of conducting and coordinating messages around the body?
(b) binds and connects tissues?
(c) supports the body?
(d) carries oxygen around the body?

9 Which body system has the function of:
(a) detecting stimuli?
(b) supporting and moving the body?
(c) taking in oxygen and getting rid of carbon dioxide?
(d) conducting messages from one part of the body to another?

INVESTIGATE, CREATE AND DESIGN

10 Research and prepare a poster on the hearts of different types of animals.

11 (a) Outline the key differences between the structures of the digestive systems of a cow, a koala, a Tasmanian devil and a honey possum.
(b) Suggest reasons for the differences.

12 Complete the following table.

Feature	Mammal	Fish
Number of chambers in the heart		
Times blood travels through heart in each circulatory trip		
Name of gaseous exchange respiratory organ		

13 Find out about the different tissues and systems that exist in plants. Present your information, using diagrams and lots of colour, on a poster or PowerPoint presentation. Be as creative as you can.

14 (a) Select one of the systems in the mind map on page 79 and find out more about what it does and how it works.
(b) Use your findings to write a brief play that other students in the class can act out.

15 In a small team, formulate scientific questions about how the structure of the heart, kidney or lungs is related to the function of that organ. Research these questions and present your findings to the class.

16 Select an organ (for example, heart, lungs or stomach) and find out the answers to the following questions:
(a) What is the function of the organ?
(b) Which system does it belong to?
(c) What other organs are in the same system?

17 Design and construct a model of one of the following systems: respiratory, excretory, reproductive, digestive.

INVESTIGATE AND DISCUSS

18 (a) In teams of three, use a 'lucky dip' system to match each member of the team with one of these organisms: earthworm, grasshopper, fish.
(b) Each member should investigate the respiratory system of their matched organism and report their findings to their team.
(c) As a team, construct a cluster map that summarises the key details and features of the respiratory systems of each of these organisms.
(d) As a team, construct a matrix table (see page 340) that summarises the similarities and differences between the respiratory systems of humans, earthworms, grasshoppers and fish.

19 Select an animal of your choice.
(a) Find out how it:
(i) detects stimuli
(ii) supports itself and moves
(iii) takes in oxygen and removes carbon dioxide
(iv) conducts messages from one part of its body to another.
(b) Construct a model or animation to demonstrate one of the functions above.

20 There are differences in the form in which groups of animals excrete nitrogenous wastes. Find out the differences between humans, freshwater fish, saltwater fish and insects. Communicate your findings using models or in a puppet play, 'documentary' or animation.

4.7 Organ systems

4.12 THINKING TOOLS

Flowcharts and cycle maps

1. Decide in which direction your flowchart will be read — from left to right, from the bottom up or from the top down.
2. Write the first action of the process you are describing inside a box.
3. Write the next event in another box and join this by an arrow to the first box.
4. Repeat until you have reached the final event.

how to ...?

Flowchart

- First event
- Next event
- Next event
- Next event
- Last event

why use?

Helps to show the order in which events happen

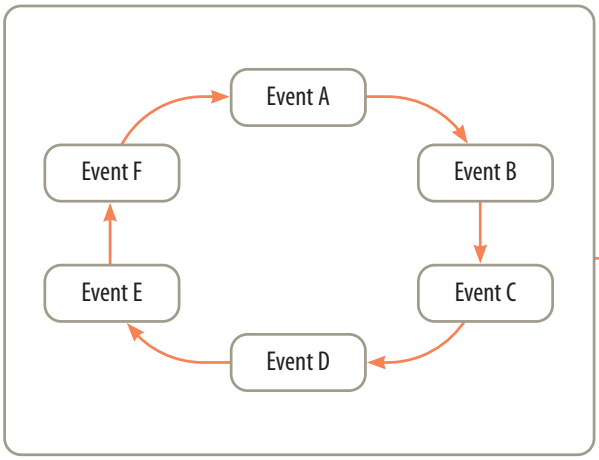
question

In which order did the stages occur?

also called

Flow map; sequence chart; chain of events

example



comparison

Similarity

Both show a sequence of events.

Difference

Cycle maps show the sequence of events repeated in the same order; flowcharts show only a linear sequence.

UNDERSTANDING AND INQUIRING

THINK AND CREATE

1 (a) Read through the information in section 4.4 to refresh your memory on the structure and function of your heart.
(b) Use a flowchart to show the movement of blood through your body using the following labels: left atrium, right atrium, right ventricle, left ventricle, pulmonary artery, pulmonary vein, lungs, aorta, vena cava, from body, to body.

2 (a) Use the cardiac cycle diagram below to answer the following questions.
(i) In which stage do the atria contract?
(ii) In which stage do both the atria and ventricles relax?
(b) Construct flowcharts to help you summarise the information in the diagram.

3 Systole is the contraction of your heart muscle and diastole is the relaxation of your heart muscle. What do you think the following mean?
(a) Atrial systole
(b) Ventricular systole
(c) Atrial diastole
(d) Ventricular diastole

4 (a) Construct a cycle map to outline the overall movement of blood through the heart.
(b) Use this information to design a working model of a human heart.
(c) Use a flowchart to plan the construction of your heart model.

5 (a) Find out more about Leonardo da Vinci's models of the human heart.
(b) Use a flowchart to map changes in scientific ideas about the heart throughout history.

One complete cardiac cycle can take about 0.8 seconds in an adult human with a pulse of about 75 beats per minute.

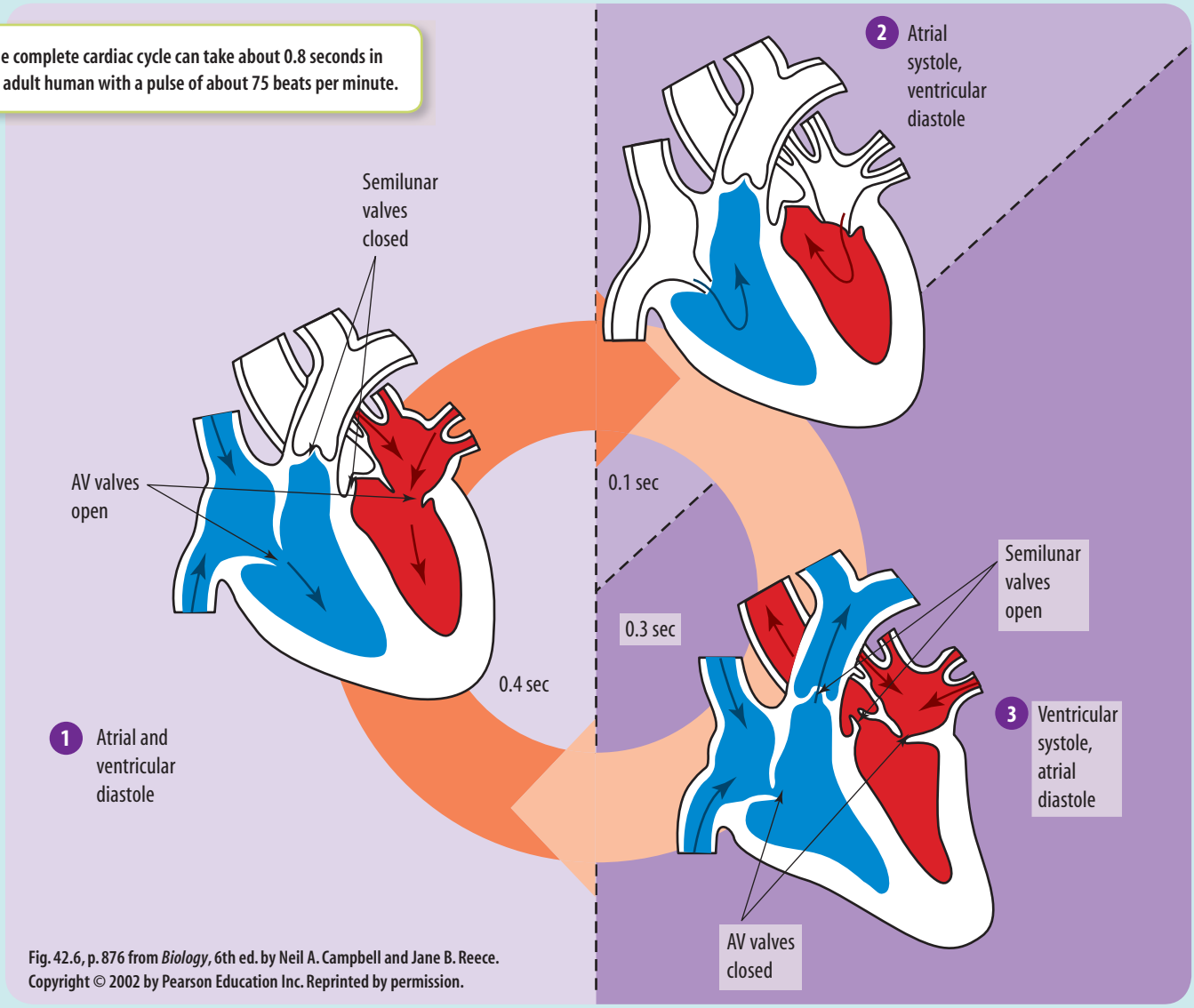


Fig. 42.6, p. 876 from *Biology*, 6th ed. by Neil A. Campbell and Jane B. Reece.
Copyright © 2002 by Pearson Education Inc. Reprinted by permission.

STUDY CHECKLIST

MULTICELLULAR ORGANISMS

- state the relationship between cells, tissues, organs, systems and multicellular organisms
- describe the key functions of the following body tissues: connective, epithelial, skeletal, blood, muscle, nerve
- identify the organs and overall function of a system of a multicellular organism
- describe the structure of each organ in a system and relate its function to the overall function of the system
- compare similar systems in different organisms

MUSCULOSKELETAL SYSTEM

- describe the structure of a human bone
- describe the relationship between bones, joints, ligaments and muscles

CIRCULATORY SYSTEM

- identify the components that make up blood
- compare red blood cells and white blood cells
- state the relationship between blood, heart, arteries, veins and capillaries
- give examples of technology related to the human heart

RESPIRATORY SYSTEM

- use a flowchart to describe the relationship between the trachea, alveoli, lungs, capillaries, oxygen and carbon dioxide

DIGESTIVE SYSTEM

- sequence the structures of the digestion system and state the function of each
- describe how the structure of the teeth, oesophagus and villi in the small intestine assist their function
- describe how the tongue, gall bladder, pancreas and liver are involved in the digestive process
- outline the importance of peristalsis
- distinguish between the processes of ingestion, mechanical digestion, chemical digestion, absorption of nutrients, assimilation and egestion

EXCRETORY SYSTEM

- explain how wastes are removed from the human body

INDIVIDUAL PATHWAYS

eBookplus

Activity 4.1 Systems **doc-6081**	**Activity 4.2** Investigating systems **doc-6082**	**Activity 4.3** Analysing systems **doc-6083**

ICT

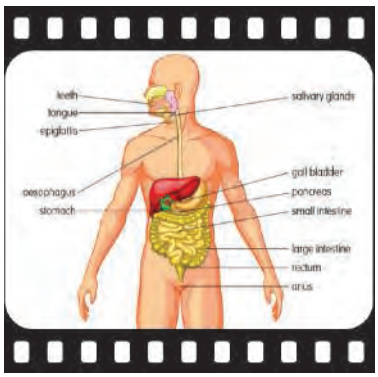
eBookplus

Summary

eLESSONS

From dinner plate to sewerage system

This video lesson explains the amazing journey of food through the human body, from dinner plate to sewerage system. Learn how our bodies release chemicals to break down food and absorb energy-giving nutrients, all without us even being aware of the process. A worksheet is attached to further your understanding.



Searchlight ID: eles-0056

Heart valve

This video lesson shows a revolutionary way of replacing heart valves.

Searchlight ID: eles-0858

INTERACTIVITY

Digestive jigsaw

This interactivity looks at the jigsaw puzzle that is the digestive system. Test your knowledge by re-creating the human digestive system. Instant feedback is provided.

Searchlight ID: int-0216

Beat it!

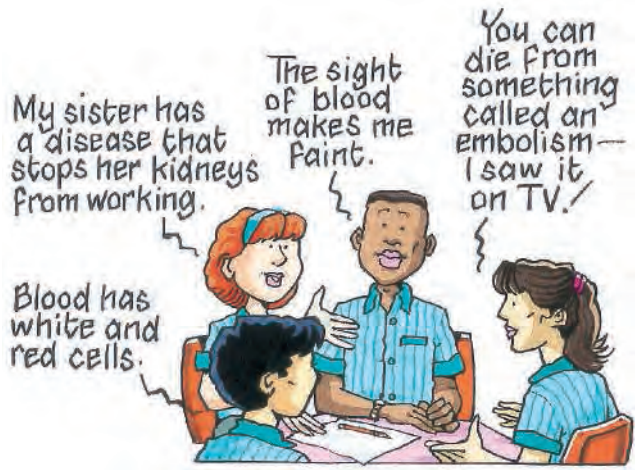
The heart is one of the most important organs in the human body. This interactivity tests your ability to label the parts of the heart. Instant feedback is provided.

Searchlight ID: int-0210

LOOKING BACK

1 In this chapter you have seen the importance of your transport system.
- (a) With your team, create mind or concept maps that summarise what you know about the following:
  - blood
  - blood vessels
  - heart
  - lungs
  - kidneys
  - liver.
- (b) For each of the six parts of your body listed above, brainstorm in your team as many questions as you can. Then select one question for each body part to do your own research. Report your findings back to the group.
- (c) Summarise your group findings in a creative way — this could be as a cartoon, song, poem, play, model or simulation.

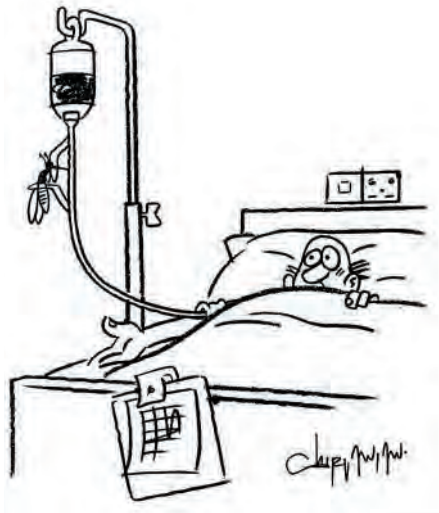
2 (a) In a pair, select one of the statements from the cartoon below and continue the conversation, taking turns asking relevant questions and answering them.
- (b) Find a new partner and repeat activity (a) with the same statement or another one.
- (c) In a team, create at least five other statements that could have been made on:
  - (i) animal transport systems
  - (ii) transport technology
  - (iii) transport system disorders or diseases.

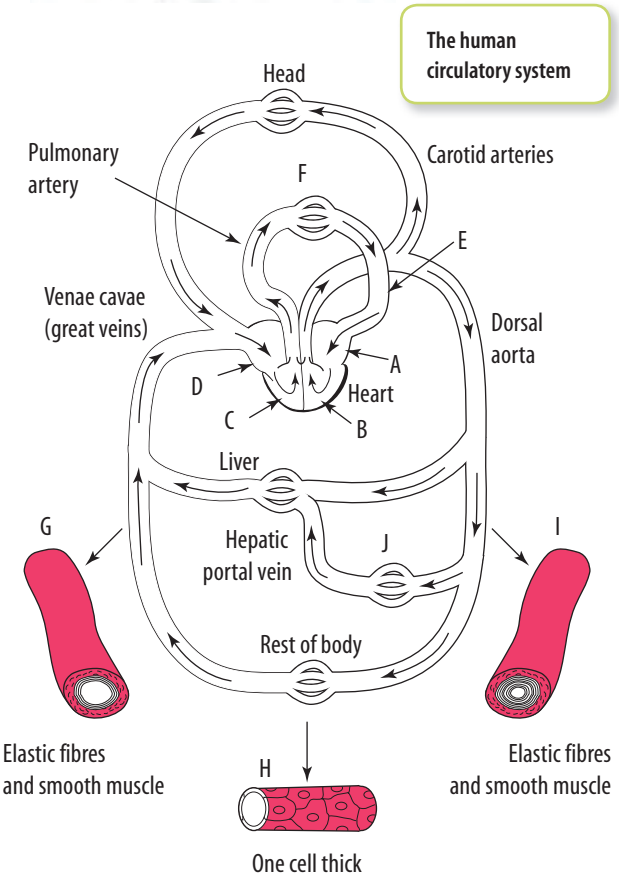


3 In a team, brainstorm ideas to come up with the following for transport systems.
- (a) Words for an alphabet key
- (b) Five 'what if' keys

4 Use your six thinking hats (see section 2.1) for three of the following issues or statements.
- (a) Drinking of any alcohol in Australia should be illegal.
- (b) Smoking in public should be punishable by a 10-year prison sentence.
- (c) Donating blood at least four times a year should be compulsory for all over the age of 16.
- (d) Only people under the age of 40 should be allowed to have a heart transplant.
- (e) Smokers should not be allowed to have surgery.
- (f) Blood transfusions should be illegal.
- (g) Everyone should have the right to a blood transfusion.
- (h) Organ donation should be compulsory.
- (i) Overweight people should not be allowed to have surgery on their circulatory system.

5 Using the cartoon below as a beginning, write a story that tells of the life of a red blood cell.





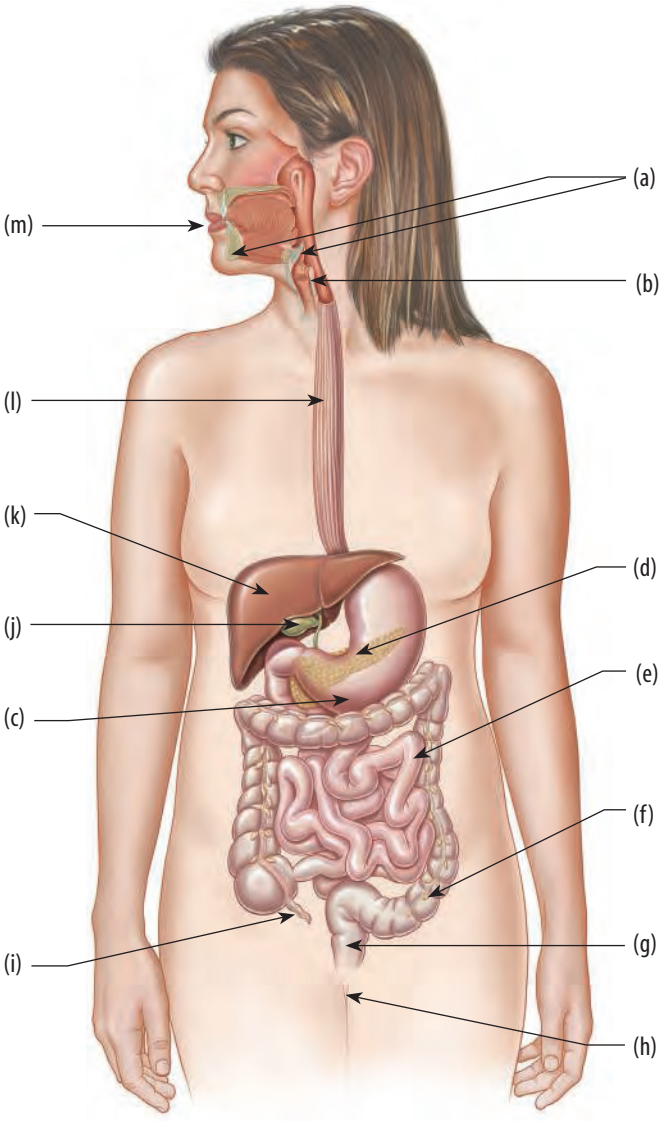
6 Make a copy of the diagram above for your workbook.
- (a) Label the lettered parts (A to J) of the human circulatory system and blood vessels on your diagram.

(b) Use a red pencil to colour in the blood vessels with oxygenated blood, and a blue pencil for those with deoxygenated blood.
(c) State whether the blood in the following blood vessels is deoxygenated or oxygenated:
  (i) aorta
  (ii) pulmonary artery
  (iii) pulmonary vein
  (iv) venae cavae
  (v) carotid arteries.
(d) Draw up a table that shows the differences in structure and function of the arteries, veins and capillaries.

7 Outline the purpose of digestion.

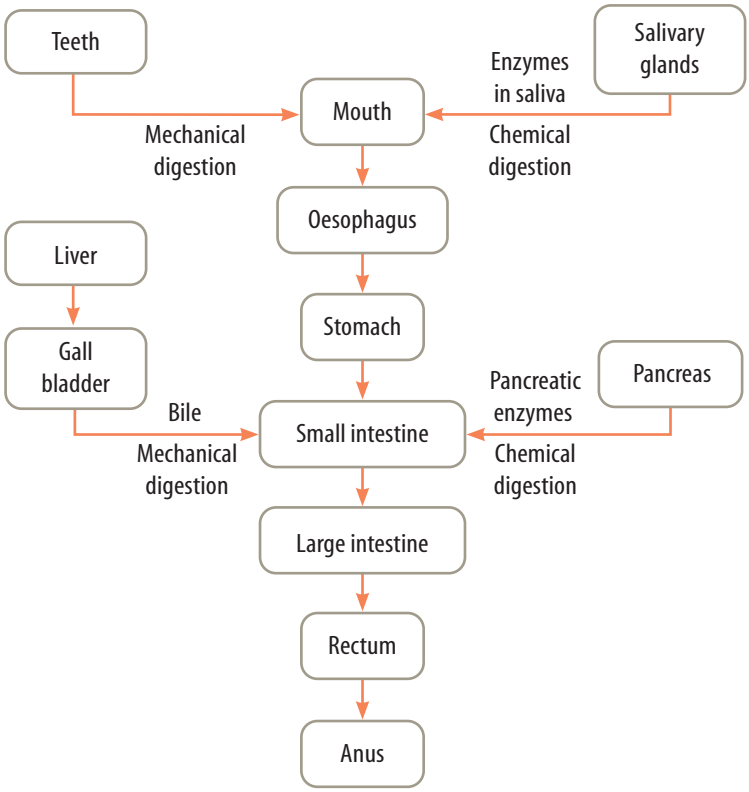
8 Construct a Venn diagram to show similarities and differences between mechanical and chemical digestion.

9 Label this diagram of the human digestive tract.

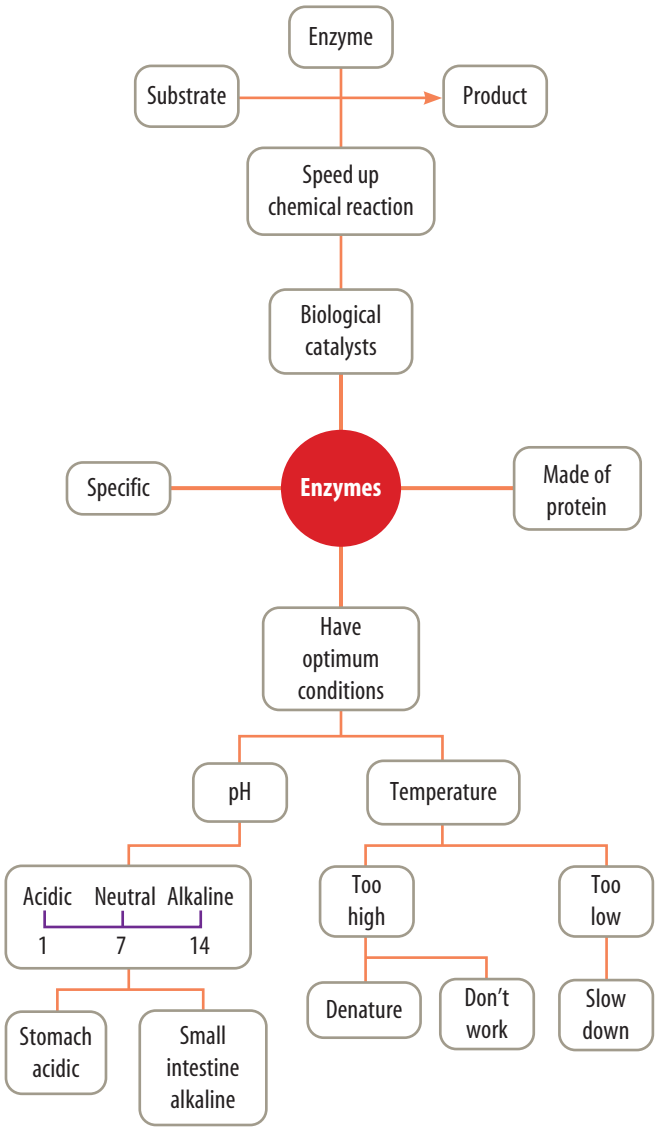


The human digestive tract

10 (a) Using the links and ideas in the two visual thinking maps on this and the next page (as well as your own), construct a board for the game 'Nutridigest', which you will create.
(b) In a team of four, brainstorm as many questions as you can that could be placed on each of the squares. Creatively write these onto your Nutridigest cards.
(c) As a team, discuss the rules for your Nutridigest game. Write down those you agree on.
(d) Make a brochure that explains how to play your game.
(e) Trial playing the game with your team.
(f) Make any alterations to your game that you think would improve it.
(g) Play the game that has been created by another team.
(h) In a team of eight, discuss the good things (strengths) and not-so-good things (limitations) of each game. Also, suggest ways that they could be improved in the next 'edition'.



What happens to food in each part of the digestive system?



What other points about enzymes could you add to this map? Can you suggest any more links between points already on the map?

11 (a) Have a lucky dip with names of parts of the digestive system and three different nutrients inside.
(b) As a class, make your selections from the lucky dip.
(c) Think about the function of your selection. Also, think about what sorts of actions or sounds it might have.
(d) As a class, act out your roles in digestion.

12 (a) Construct a mind or concept map to summarise what you have learned during your study of 'living connections'.
(b) Share your map with others in your team.
(c) Create another mind or concept map that incorporates the learning of all of your team members.

13 As a team, create a song, poem, cartoon or play about something that you have learned.

14 (a) Discuss in your team what you have learned or found interesting in this chapter.
(b) On sticky notes, write down other questions that you or your team may have about areas related to those in this chapter. Place the questions on a class question gallery board with those of other class members.
(c) Once all of the questions are on the board, organise them into groups or themes.
(d) In pairs, select one of the questions in the gallery.
(e) Research the question and report your findings to the class.

15 The process of replacing oxygen with carbon dioxide is called gas exchange. Some animals exchange gases through lungs or gills. Some other animals exchange gases though their skin, and yet others through rows of air holes along both sides of their bodies.
Investigate how animals and insects exchange gases to create a mind map and answer the following questions.
(a) Construct a mind map, poster or PowerPoint presentation that summarises your findings on how at least five different animals achieve their exchange of gases.
(b) Why do frogs need lungs when they can exchange gases through their skin?
(c) Why can't fish survive in the air?
(d) Why can't humans breathe under water without air tanks?
(e) How is gas exchange in insects similar to that in humans?
(f) What are two major differences between gas exchange in insects and humans?

16 (a) In teams, construct mind maps about each of the circulatory, respiratory and excretory systems.
(b) Use the information in these mind maps to construct matrix grids (see page 340) to summarise the key information for each system.

17 (a) Name the organs in each of the following systems.
(i) Circulatory system
(ii) Respiratory system
(iii) Excretory system
(b) Create a word-find with these terms and then swap it with another team member to see how many words each of you can find.

work sheet 4.8 Summing up

5 Systems — reproduction

Sex is fascinating. It has to be! It is the basic foundation for the continuation of life for most organisms on Earth. It can also be dangerous, desperate and competitive, as many insects and spiders would agree. The changes that occur at puberty in humans can be scary and exciting. But they all have the same purpose. They are the means by which you become an adult with the potential of passing on your genetic information to your offspring. It is all a part of the cycle of life. Around and around we go . . .

OVERARCHING IDEAS

- Patterns, order and organisation
- Form and function
- Stability and change
- Systems

SCIENCE UNDERSTANDING

Multicellular organisms contain systems of organs that carry out specialised functions that enable them to survive and reproduce.

Elaborations

Identifying the organs and overall function of a system of a multicellular organism in supporting the life processes

Describing the structure of each organ in a system and relating its function to the overall function of the system

Examining the specialised cells and tissues involved in structure and function of particular organs

Distinguishing between asexual and sexual reproduction

Comparing reproductive systems of organisms

THINK ABOUT REPRODUCTION

- How fast can a sperm swim?
- Why is a dinner date a bad idea for a male red-back spider?
- What do a clitoris and a penis have in common?
- Can sperm build up and burst your testicles?
- If females have about 400 000 eggs at birth, is there a possibility that they can have that many children?
- How big is an 8-week-old fetus?
- Which contraceptives work the best?
- What's a 'test-tube baby'?
- Which foods contain the swollen ovaries of plants?
- Which vertebrates were the first to have a penis?
- Do 'virgin births' really exist?
- What is a hermaphrodite?

Which sperm will fertilise the ovum?

YOUR QUEST

Perhaps you've noticed you're changing

During puberty, your body goes through some interesting, exciting and, sometimes, scary changes.

Investigate

1 In teams, find out the answers to the five questions asked by the students shown below.
2 Share and discuss your findings with those of other teams.

INQUIRY: INVESTIGATION 5.1

What's happening?

KEY INQUIRY SKILLS:

- questioning and predicting
- communicating

DISCUSS AND EXPLAIN

1 In teams of two or more, sketch a figure of a girl and a boy on separate sheets of paper.
2 Add labels to show the changes for each during puberty.
3 Compare and discuss your figures with those of other teams in the class.
4 Make any changes or additions you wish to your diagram.
5 (a) As a team, suggest changes and possible additions to your diagrams.
(b) As a class, collate examples of changes and possible additions to your diagrams.
6 (a) Suggest reasons why these changes occur.
(b) As a class, suggest reasons why these changes may occur.

INQUIRY: INVESTIGATION 5.2

The timeline of our lives

KEY INQUIRY SKILL:

- processing and analysing data and information

DISCUSS AND EXPLAIN

1 Create a timeline to show how you have changed over time: from the ages of 6 months, 2 years, 5 years, 8 years and 11 years to now. You may use photographs, cartoon sketches, or gingerbread or plasticine models.
2 As we are all individuals, our timelines vary. Some of this variation is due to our genes that are inherited; some is due to our environments and lifestyles. Variation is very important for the survival of the species. Compare your timeline with those of others in your class.
3 Record the similarities and differences.
4 Pose questions prompted by your observations and reflections.

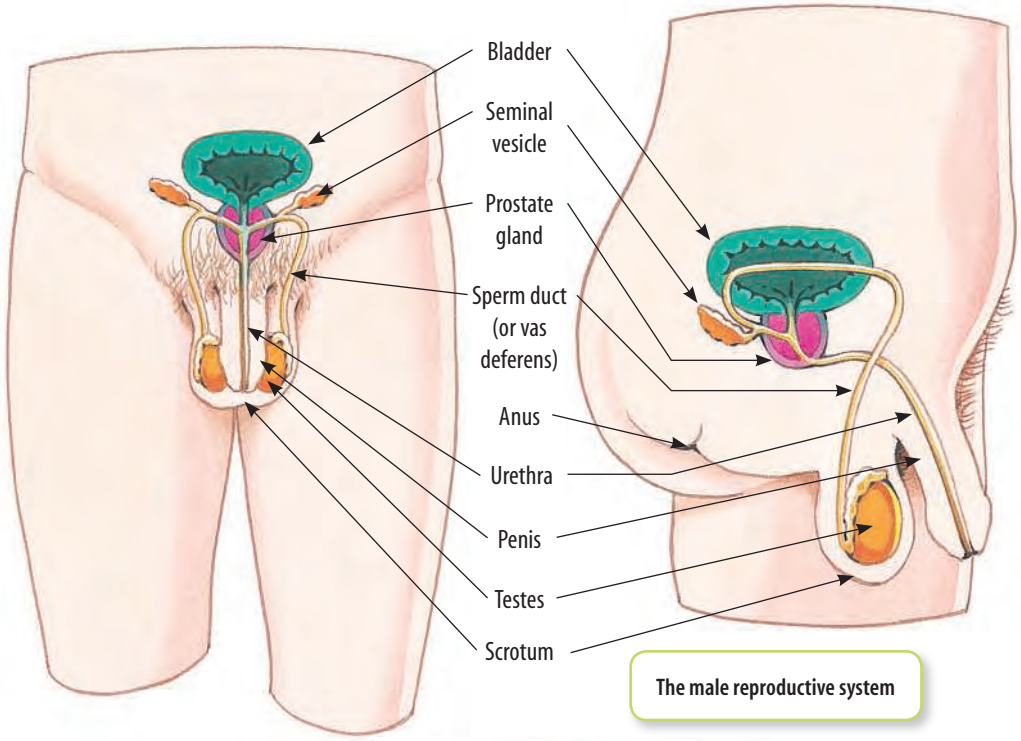
Private parts

We've all got private parts. Let's find out where they are and what they do.

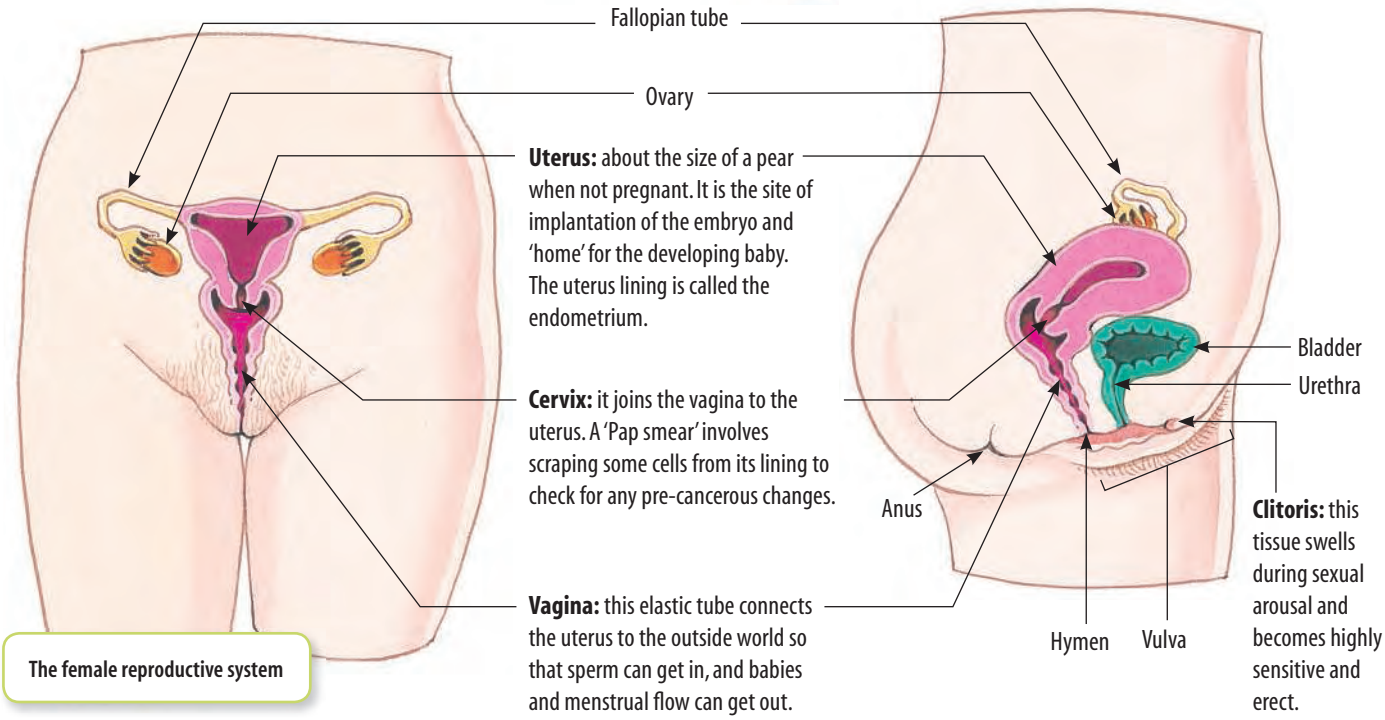
Gamete factories . . .

Although testes and ovaries may look different, these two organs have the same job. They both make gametes.

Sperm are made in the **testes** of a male when he is sexually mature. Testes hang from the body within the **scrotum** to maintain sperm at a temperature of about 3°C below that of the rest of the body. This temperature difference is essential for successful sperm production. Tight underwear or jeans can increase the temperature of the testes and so increase the number of damaged sperm produced.



The male reproductive system



The female reproductive system

Ova (the female's sex cells) are made in the ovaries. Females are born with about 400 000 eggs, or ova, in their ovaries. These eggs are in sacs called **follicles**. Usually only one ovum (singular = ova) is ripened and released into the **fallopian tube** (or oviduct) each month, once the female is sexually mature.

Fallopian tubes are about the diameter of a human hair. They form a tunnel in which the sperm

and ovum meet and are hence the site of **fertilisation**. Damage to these tubes can prevent the sperm and egg from meeting.

If the fertilised egg does not move down into the uterus, but remains in the tube, an ectopic pregnancy may result. If this is not detected and treated within about eight weeks, the tube may rupture and threaten the woman's life.

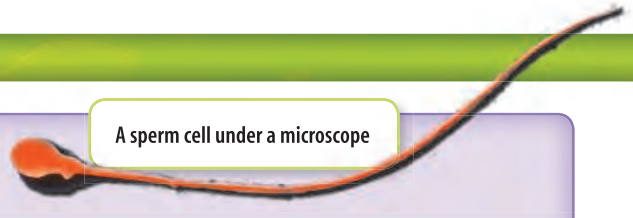
HOW ABOUT THAT!

Did you know that not all plants or animals have separate sexes? Some invertebrates are both male and female at once. These interesting combinations are called hermaphrodites. This enables an individual to achieve greater reproductive efficiency than if it was just the one sex.

Snails have been around for 600 million years and have developed intriguing methods of reproduction. Each snail has an organ called an ovotestis, which makes both sperm and eggs, and a single tube to carry both the sperm and the eggs.

After a complex courtship, in which hermaphrodite snails rear up, each pressing its muscular foot against its partner, and stroking each other with their tentacles, they simultaneously insert their sex organ into the other's body. In this manner, each snail gives sperm to the other and each has its eggs fertilised by the other.

HOW ABOUT THAT!



A sperm cell under a microscope

Amazing sperm

Sperm cells are less than half a millimetre long. Viewed through a microscope lens, spermatozoa (sperm, for short) remind you of tadpoles — a big head and a thin, whippy tail. They form in the testes, but only when the temperature is just right — a few degrees lower than body temperature. This is where the scrotum — a natural thermostat — does its job. It shrivels and scrunches up closer to the body when you are cold (keeping sperm warmer) and hangs away from your body when you are hot (cooling them down).

Parts of the male reproductive system

Part	Function
Testes	Produce sperm cells
Scrotum	Where the testes are located. Keeps the testes at a slightly lower temperature than body temperature.
Vas deferens	The tube through which sperm cells travel from the testes to the penis
Prostate gland	Secretes some of the liquid that is added to sperm cells to form semen. The fluid secreted by the prostate gland is alkaline and contains many chemicals including enzymes. It plays an important role in keeping sperm cells alive once they enter the female reproductive system.
Seminal vesicle	Also contributes some of the liquid that makes up semen. The fluid produced by the seminal vesicle contains proteins, enzymes, sugar, vitamin C and other substances. The sugar provides a source of energy for sperm cells.
Urethra	The tube inside the penis through which semen leaves the male's body
Penis	The penis swells during sexual arousal. Semen containing sperm cells is ejaculated (released) from the penis into the vagina.

Parts of the female reproductive system

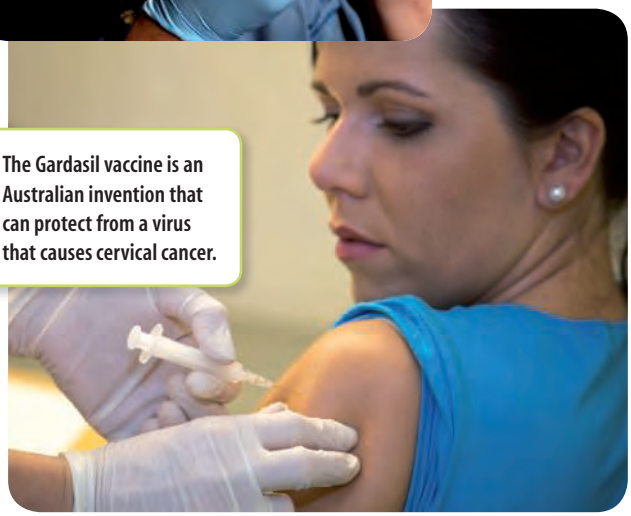
Part	Function
Ovaries	Produce ova. One egg is produced about once a month from one of the ovaries.
Fallopian tubes/ oviducts	Tubes through which ova must travel to reach the uterus. Fertilisation occurs in the fallopian tubes.
Uterus	Where the baby grows and develops
Cervix	The passageway between the vagina and the uterus. During childbirth the cervix needs to dilate (open up) to allow the baby to come out of the uterus.
Vagina	Semen enters the female's reproductive system via the vagina. When babies are born they are pushed out of the uterus and must pass through the vagina to enter the world.
Clitoris	Swells during sexual arousal and becomes highly sensitive when erect

HOW ABOUT THAT!

Sperm by the millions

The average amount of semen produced during an ejaculation (about a teaspoonful) contains about 200–500 million sperm cells! You might think it would take a long time for the testes to make 400 million sperm. Not so. Some 200 million sperm cells are manufactured each day by a fertile adult male. That's around 73 billion sperm cells in a year!

Australian immunologist Professor Ian Frazer developed the cervical cancer vaccine, Gardasil.



The Gardasil vaccine is an Australian invention that can protect from a virus that causes cervical cancer.

Cervical cancer breakthrough

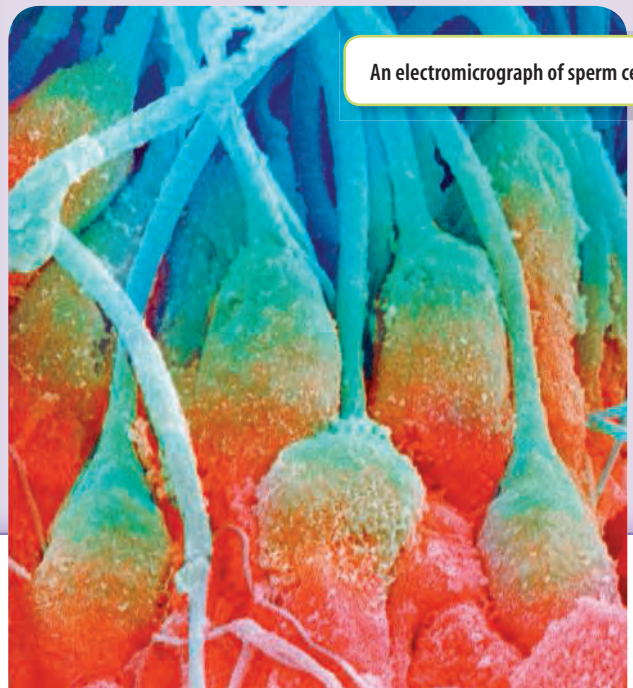
Cervical cancer occurs when the cells of the cervix start to divide abnormally. In its early stages cervical cancer has no symptoms. As the cancer spreads it may cause symptoms such as unusual vaginal bleeding or discharge and lower back or leg pain. When the cancer is discovered in its later stages it is sometimes necessary to remove some of the patient's reproductive organs so that she may no longer have children. The cancer may eventually spread to other parts of the body and cause death. Early detection of cervical cancer greatly increases the chance that a woman will make a full recovery.

A Pap smear is a medical procedure that can detect cervical cancer in its early stages. It involves scraping cells from the cervix and examining them to check for any abnormalities. It is recommended that women have a Pap test every two years.

Cancers can have a variety of causes but most cases of cervical cancer are caused by a virus called human papillomavirus, or HPV. It spreads by sexual contact. While men do not get cervical cancer, they can carry HPV and pass it on to their sexual partners. Professor Ian Frazer, an Australian scientist, recently succeeded in developing a vaccine against HPV. The vaccine, called Gardasil, has the potential to greatly cut down the rate of cervical cancer and save many lives. It is best to be vaccinated when you are young, before you have had a chance to become infected by HPV. For this reason, the Australian government is funding the cervical cancer vaccine for women aged 16 to 26. Different types of HPV can cause cervical cancer and the vaccine protects against about 80 per cent of these, so women who have been vaccinated should still have Pap tests.

HOW ABOUT THAT!

Why do men produce sperm and women produce eggs? Josephine Bowles at the Institute for Molecular Bioscience, University of Queensland, has been researching this question — and she has found an answer. A substance called retinoic acid, a relative of vitamin A, triggers a special type of cell division called meiosis. This results in the production of female gametes or ova. Cells in the testis of the developing male fetus produce a protein that degrades this substance, and so meiosis does not occur and sperm are produced rather than ova.



An electromicrograph of sperm cells

UNDERSTANDING AND INQUIRING

REMEMBER

1 Construct Venn diagrams to compare the following pairs:
   (a) ovaries and testes
   (b) vas deferens and fallopian tubes
   (c) penis and clitoris
   (d) seminal vesicles and prostate gland
   (e) uterus and vagina.

2 Why is tight underwear not recommended for males?

3 What is a 'Pap smear' and why do doctors do it?

4 What is an ectopic pregnancy?

5 Draw a table as shown below. Classify the following organs and list them in the correct column of your table.

fallopian tube, penis, urethra, testes, prostate gland, bladder, uterus, seminal vesicle, ovary, vas deferens, scrotum, cervix, vagina.

Found in males only	Found in females only	Found in both males and females

6 Match each organ with its function. There may be more than one organ with the same function.

Organ	Function
(a) Seminal vesicle	(i) Produces gametes
(b) Ovary	(ii) Where the baby grows and develops
(c) Scrotum	(iii) Where fertilisation occurs
(d) Testes	(iv) Keeps the testes slightly cooler than the rest of the body
(e) Prostate gland	(v) The passageway between the vagina and the uterus
(f) Uterus	(vi) Produces some of the liquid found in semen
(g) Cervix	
(h) Fallopian tubes	

INVESTIGATE, THINK AND DISCUSS

7 Suggest why women with blocked fallopian tubes are unable to have babies.

8 The Australian government currently funds HPV vaccine injections to young women so that it is supplied at no cost to them.
   (a) State what HPV stands for.
   (b) Find out more about the symptoms, diagnosis and consequences of cervical cancer.
   (c) Find out how Gardasil vaccine works.
   (d) Find out more about Professor Ian Frazer's research in Australia.
   (e) Research immunology as a career and then write a diary entry of 'a week in the life of an immunologist'.
   (f) Only young women are obtaining free access to the Gardasil vaccine. Why is this the case? Is this fair? Give reasons for your responses.

INVESTIGATE AND CREATE

9 (a) Construct a model of the reproductive system of:
   (i) a human
   (ii) another animal of your choice.
   (b) Describe how they are similar and how they are different.
   (c) Design and create a third model that has the best features of each.

10 Imagine that you are a sperm or an egg. Find out more about how you are produced, stored and used in sexual reproduction.
   (a) Write a dramatic story about your life.
   (b) Construct puppets, animated cartoons or 'fancy dress' actors to recreate your story as a play.

eBook*plus*

11 Test your knowledge of the reproductive organs by completing the **Know your privates** interactivity. **int-0671**

12 Find out more about sperm and ova and represent your findings in the following formats.
   (a) Venn diagram
   (b) PMI chart
   (c) Flowchart

13 Find out more about the structure and function of the human male and female reproductive systems, and design one of the following.
   - A Trivial Pursuit–style game with questions on each system, a board, dice and a rule book
   - An 'information wheel' made of two pieces of circular card connected with a paper fastener ('foldback pin') in the centre. (For an example of how to create an information wheel, see page 182.)

Why the changes?

Perhaps you have noticed that you are changing. Have you noticed any hairs in parts where they weren't before, changes in your body shape or height, changes in your interests . . .?

Changes can be very exciting

Changes that you may currently be aware of are indications that you are becoming an adult. These physical changes are called **puberty**. The term 'puberty' comes from the Latin word *pubertas*, which means adulthood.

The main purpose of the changes that occur in puberty is to enable you to start producing children. Your sex organs grow and develop. Males begin to produce sperm and females begin to develop the ova they were born with. When combined, these gametes can produce babies.

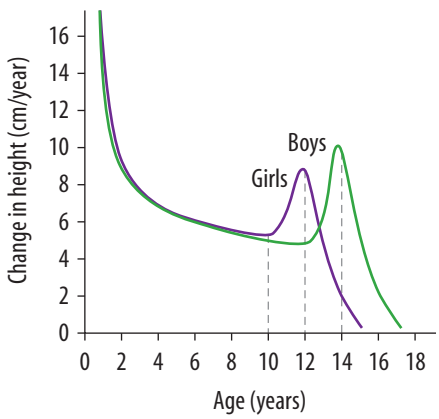
We are all different

It's okay for the changes to occur at different times and at different rates, because everyone is different. We are all individuals. People reach puberty at different ages. Girls reach puberty between the ages of 8 and 17 years and boys between 10 and 18 years. A message from your brain to your sex glands triggers all of these changes. When these glands get the message, they produce substances called **hormones**. These hormones travel around your bloodstream and trigger lots of changes.

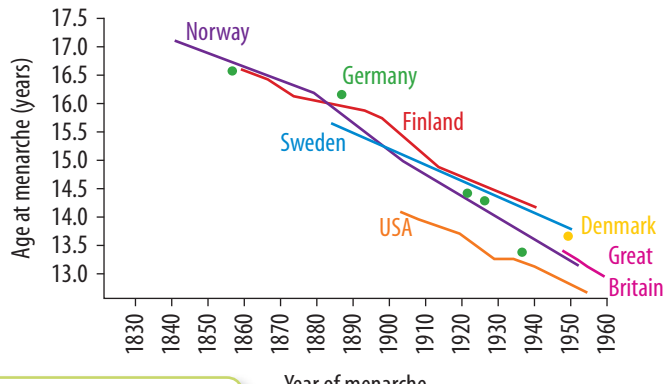
Some common questions

Q: *Why am I getting more pimples?*
A: The sex hormones (testosterone, for example) make the glands in the skin produce extra oils. This can cause the pores in the skin to become blocked and may result in pimples.

Growth rates of girls and boys



Both boys and girls experience a growth spurt around the time of puberty.

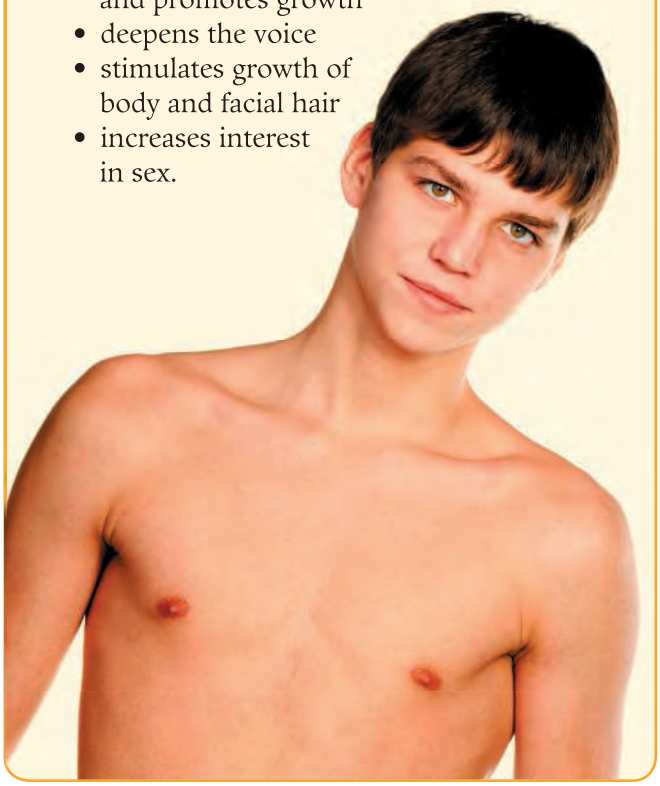


The average age of menarche (the first menstrual period) has decreased over the last two centuries.

Q: *What causes the changes in a boy's voice during puberty?*
A: The voice box, or larynx, gets bigger, making the voice deeper. Males develop a larger larynx so they develop deeper voices. The squeak often heard during this time of change is due to the muscles of the larynx getting momentarily out of control. Although female voices can also change, the final result is not usually so noticeable.

Q: *What role does the male hormone testosterone play in puberty?*
A: Testosterone:

- is needed for sperm production
- not only enlarges the penis, testes and scrotum, but also increases their sensitivity
- increases body muscle bulk and promotes growth
- deepens the voice
- stimulates growth of body and facial hair
- increases interest in sex.



Q: *In which new places am I likely to grow hair during puberty?*
A: If you are female, it's likely to be around your pubic region and under your arms. Males may notice an increase in these regions and also on their legs, arms, faces and chests.

Q: *What is masturbation?*
A: Masturbation involves touching or rubbing your genitals to give sexual pleasure. It can result in an orgasm — a throbbing feeling that brings intense pleasure. Masturbation is a harmless and natural way of exploring your sexuality on your own. Australian surveys show that by their late teens most males and females have masturbated. Some people don't feel the need to masturbate at all, and that's normal too.

Q: *What's the average penis size?*
A: The length of a fully grown penis usually ranges between 8 and 10 cm when soft, and between 12 and 18 cm when erect. As you can see, penis sizes vary greatly (vaginas vary in size too). The size of the penis has nothing to do with a person's masculinity or sexual performance.

Q: *I seem to have erections all the time! Why?*
A: Waking up with an erection can be caused by a full bladder or the effects of a dream. Raised levels of testosterone can also cause erections.

Q: *What are wet dreams?*
A: These dreams happen to a boy during sleep and result in ejaculation of semen. They are quite normal (although not all boys have them) and are an indication that you are becoming sexually mature.

Q: *Can sperm build up and burst your testes?*
A: No, unused sperm are stored for a while and then reabsorbed into your body. New sperm replace them.

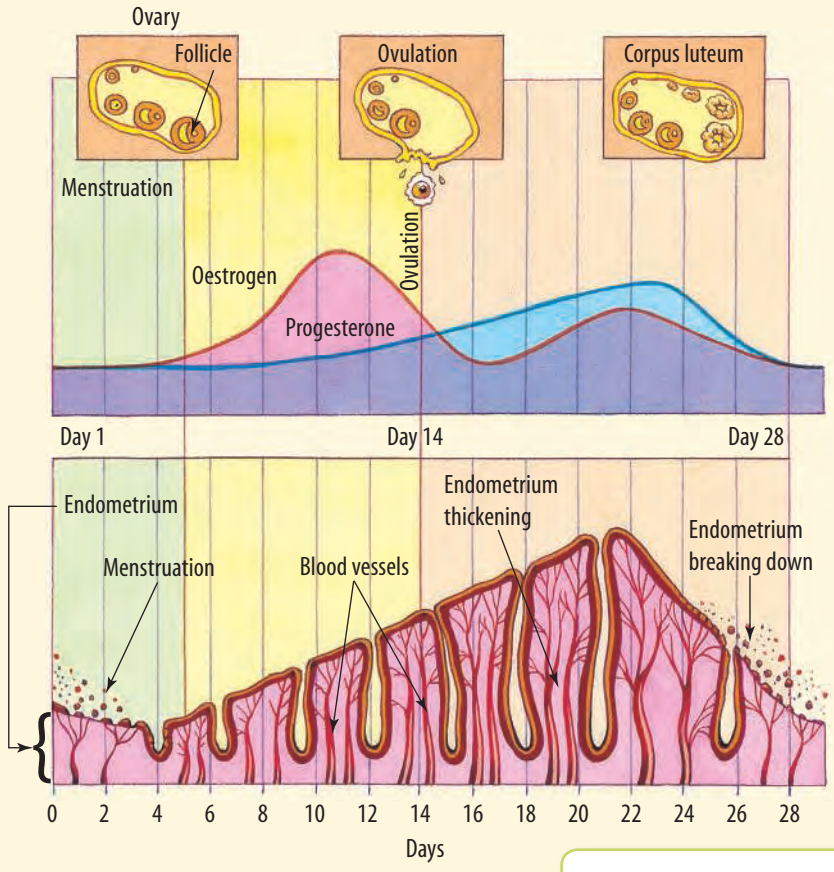
Q: *What is a breast bud?*
A: This is a little button of tissue just under the nipple from which the breast develops. Sometimes boys also get breast buds, but in their case they go away and do not develop.

Q: *At what age do you get periods and how long do they last?*

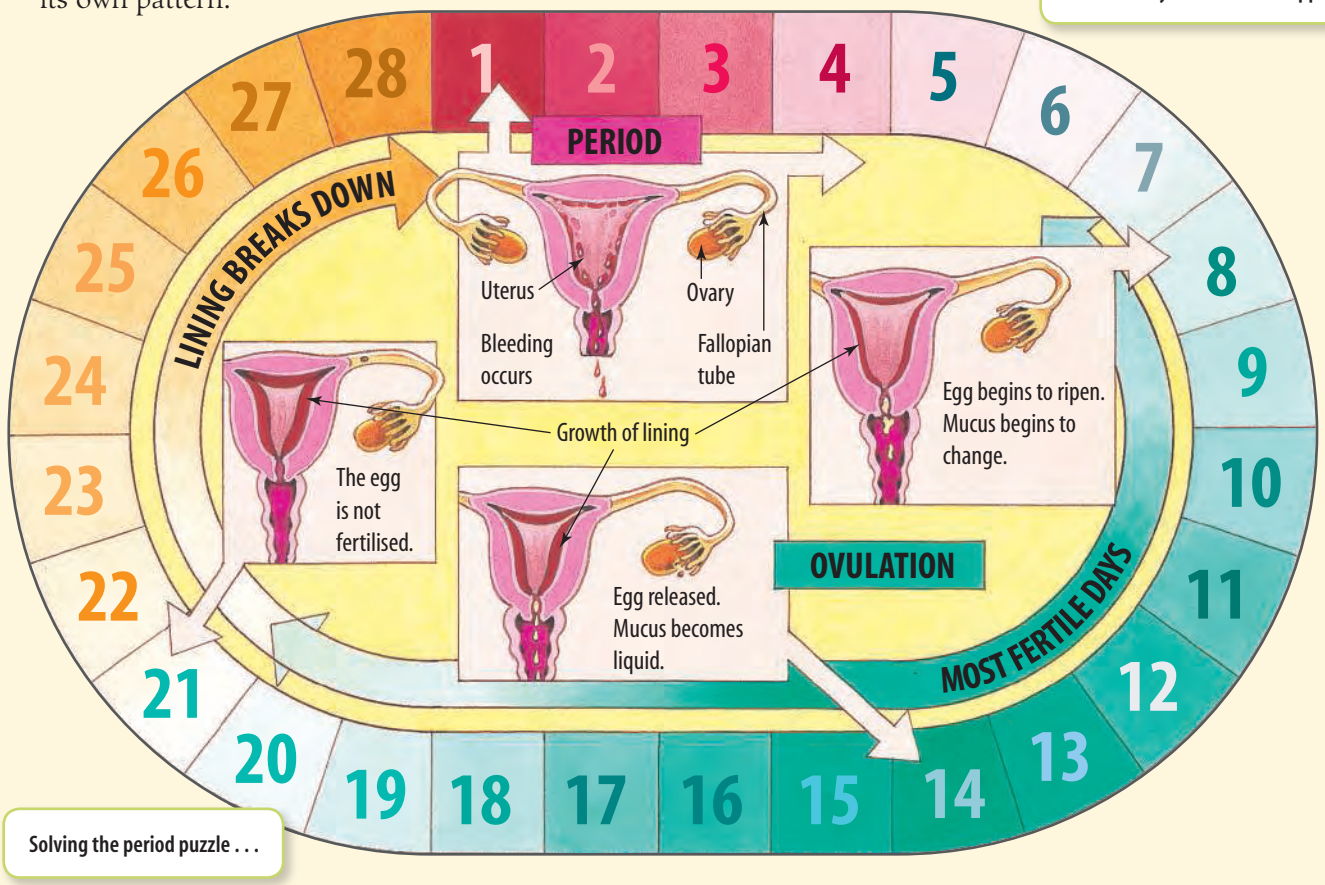
A: Most girls experience their first period between the ages of 11 and 14, although others may have it up to two years earlier or later. The first menstrual period is called **menarche**.

A period generally lasts about four to six days, with varying amounts of discharge over this time. The discharge contains cells from the lining of your uterus that was built up in preparation for a baby.

Your **menstrual cycle** is the time from the first day of one period to the first day of the next. It is usually about 28 days, but this varies in different women. It is quite common for your menstrual cycle to be irregular at first — until your body settles into its own pattern.



Menstrual cycle — what's happening?



Solving the period puzzle . . .

Q: *What is ovulation?*
A: Although all the eggs or ova are present at birth, it is only during puberty that they begin to ripen. Each month, one ovum matures and is released from the ovary into the fallopian tube. The release of the ovum is called ovulation. This continues about every 28 days until the woman enters menopause.

Q: *How many eggs are girls born with and about how many develop?*
A: Girls are born with about 400 000 eggs. Only about 400 of these will mature and be released during her menstrual cycles; the others will not mature.

Q: *What are periods?*
A: Each month, the lining of the uterus prepares itself for the fertilised egg. If the egg is not fertilised, the uterus lining is shed through the vagina. This monthly discharge, or shedding, is called **menstruation**, or a period. Sometimes there may be some discomfort or abdominal cramps during the first few days of your period. If you are having periods, it is a sign that you are also ovulating. This means that you are physically able to get pregnant and have a baby.

Q: *Is it true that you can't wash your hair, eat ice-cream or play sport during your periods?*
A: No. Having periods shouldn't affect your normal activities.

Q: *What is the difference between menstrual pads and tampons?*
A: Pads, or sanitary towels, are made of soft cotton materials with one waterproof side. They are worn outside the body and press on to fit inside your knickers. Tampons, however, are a tight roll of cotton wool with a string at one end. These are inserted into the vagina. Both of these sanitary products soak up menstrual blood.

Q: *FSH, LH, oestrogen and progesterone: what are they?*
A: FSH (follicle-stimulating hormone), LH (luteinising hormone), oestrogen and progesterone are all hormones. FSH and LH are hormones that make ova develop in girls' ovaries and begin sperm production in boys' testes. Oestrogen and progesterone are produced in the ovaries. They are the most important female hormones and are involved in changes in the lining of the uterus. The hormones control the menstrual cycle as shown in the diagram on the previous page.

HOW ABOUT THAT!

The feeling of 'being in love' is not a product of the heart. That happy, dreamy feeling experienced when you 'fall in love' is partly due to a chemical produced in your brain, called phenylethylamine. As chocolate also contains phenylethylamine, it is no wonder many people describe themselves as 'chocolate lovers'.

UNDERSTANDING AND INQUIRING

REMEMBER

1 Use a table to give definitions of the following terms: puberty, menstruation, ovulation, masturbation, menarche, menstrual cycle, hormones, testosterone, wet dreams, breast buds.

2 Which hormones are responsible for triggering sperm production in males and ovum development in females?

THINK AND REASON

3 Use the illustration of menstruation and other information in this section to answer the following questions.
   (a) What is ovulation?
   (b) At what day in a 28-day cycle is ovulation likely to occur?
   (c) At which time in the cycle is sperm most likely to meet (and fertilise) an egg?

4 Describe the changes in oestrogen and progesterone levels throughout the menstrual cycle.
   (a) Which hormone is found in the highest concentration just before ovulation?
   (b) Which hormone is found in the highest concentration when the uterine lining is thickest?
   (c) At which time would the lining provide the best 'home' for a fertilised egg?

5 Translate the graph on page 180 into a 28-day calendar.

THINK AND DISCUSS

6 Suggest why the hips become 'fleshier' and the pelvic bones widen in females during puberty.

7 The testes continue making sperm for the rest of a man's life. How is this different from gamete production in a woman? What are the consequences of this?

8 If a female has menstrual cycles, is she potentially able to have babies? Explain.

9 Why aren't all menstrual cycles, penises and breasts the same?

INVESTIGATE

10 Write down ten questions that you have about puberty or reproduction. Use a variety of texts and resources to find the answers to them. Report the findings to your friends.

CREATE

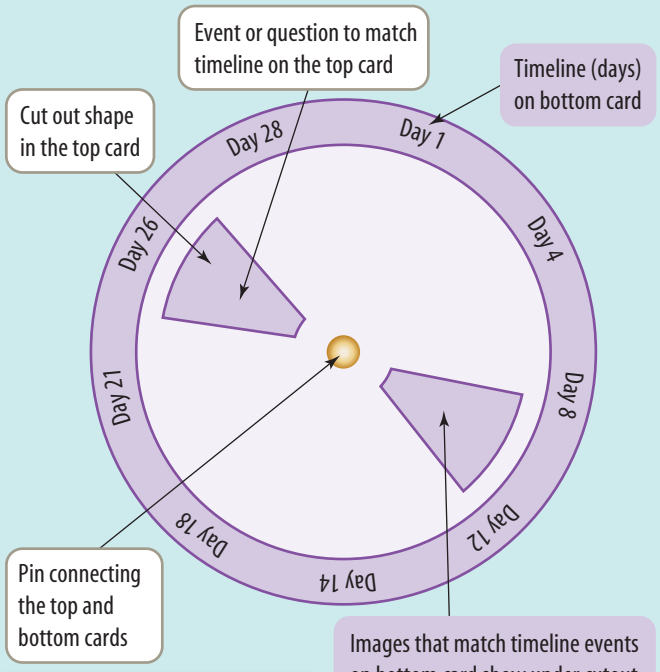
11 Construct a crossword with your own clues and answers, from information found in this section.

12 Write and act out a play to demonstrate the menstrual cycle. Include the following stages: egg ripening, ovulation, movement of the unfertilised egg into the fallopian tube and through the uterus, and both the egg and the uterine lining being shed through the vagina.

13 Get everyone in the class to write down two to five questions about the changes that occur during puberty and reproduction in a 'Dear Ethel' magazine format. After your teacher has collated these, select at least two to investigate. Report your findings by compiling a class 'Dear Ethel' magazine.

14 Copy and complete the table below. Use this table to help you construct an 'information wheel' about the human menstrual cycle.

Day in cycle	Key events	Hormonal changes	Possible images to include
1			
4			
8			
12			
14			
18			
21			
25			
28			



This is an example of how you might design your information wheel on the menstrual cycle, but there are many others. In this example, the bottom circle of the card is larger than the top piece. You may decide to reverse this or have both pieces the same size and add holes in the top piece to see the days of the timeline.

work sheets
- 5.1 Puberty and boys
- 5.2 Puberty and girls
- 5.3 Menstruation

5.3 SCIENCE UNDERSTANDING

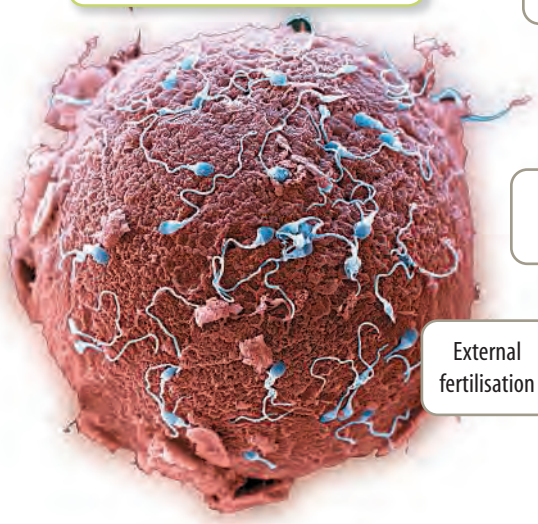
Getting together

Let's talk about sex. Sexual reproduction is not something to be embarrassed about, but something incredible and fascinating.

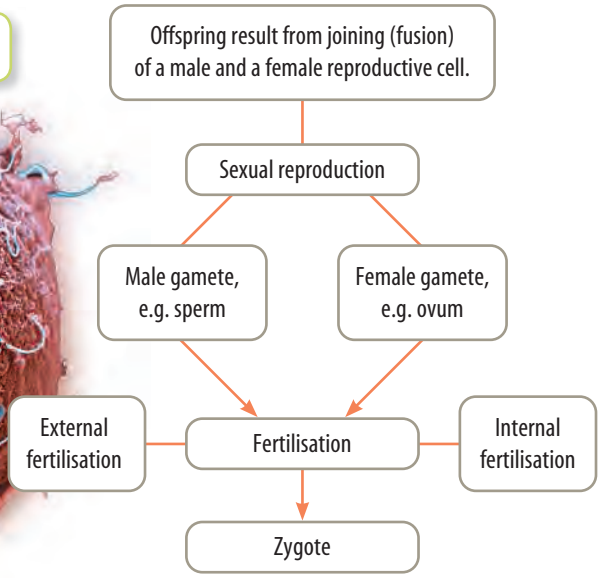
Snap!

Sexual reproduction occurs when offspring result from the joining (fusion) of a male reproductive cell and a female reproductive cell. These special reproductive cells are called **gametes** and are made in the reproductive organs of organisms. In animals, male gametes are called **sperm** and female gametes are called **ova** (singular = **ovum**) or egg cells.

Reproductive systems are designed to bring the male and female gametes together. The joining of sperm and egg cells is called fertilisation. This process mixes the genetic material from the nucleus of each parent together and results in the formation of a **zygote**.



Human sperm cells surround an ovum.

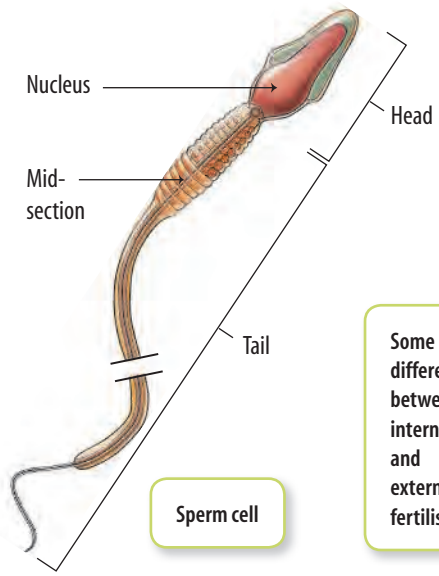


MEETING OUTSIDE . . .

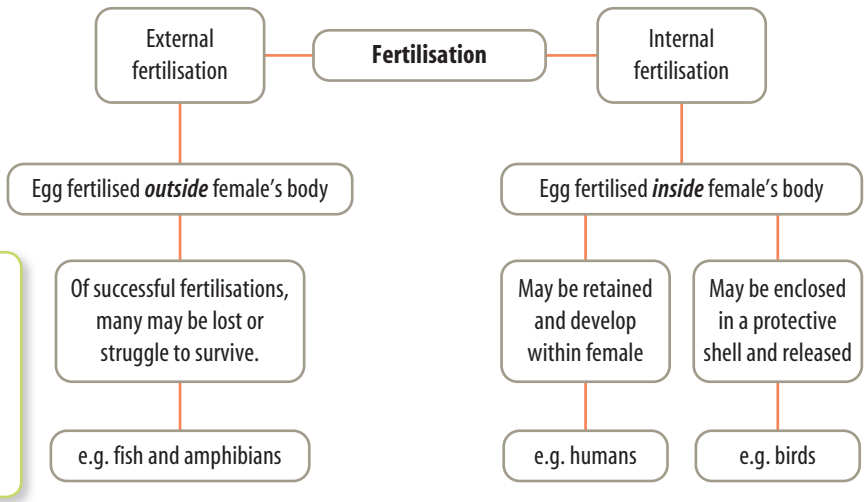
In some animals, especially those that live or breed in water such as fish and amphibians, fertilisation occurs *outside* the female's body. This is called **external fertilisation**. In this situation, the female releases her unfertilised eggs into the water to be fertilised by the male's sperm, which are also released into the water.

MEETING INSIDE . . .

In animals that live and breed on land, **internal fertilisation** occurs. This keeps the gametes inside the body so there is no chance of dehydration occurring. In this situation, sperm are introduced into the female by a process called **copulation** (or **sexual intercourse**).



Sexual reproduction involves fusion of gametes.



Some differences between internal and external fertilisation

Ova

Like sperm, ova are produced by a special type of cell division called **meiosis**. Unlike sperm, however, the ova that will be released throughout the female's reproductive years are already present at her birth. This brings differences in terms of **epigenetics** — an exciting new branch of science that involves studying the effect of our experiences on the expression of our genetic information.

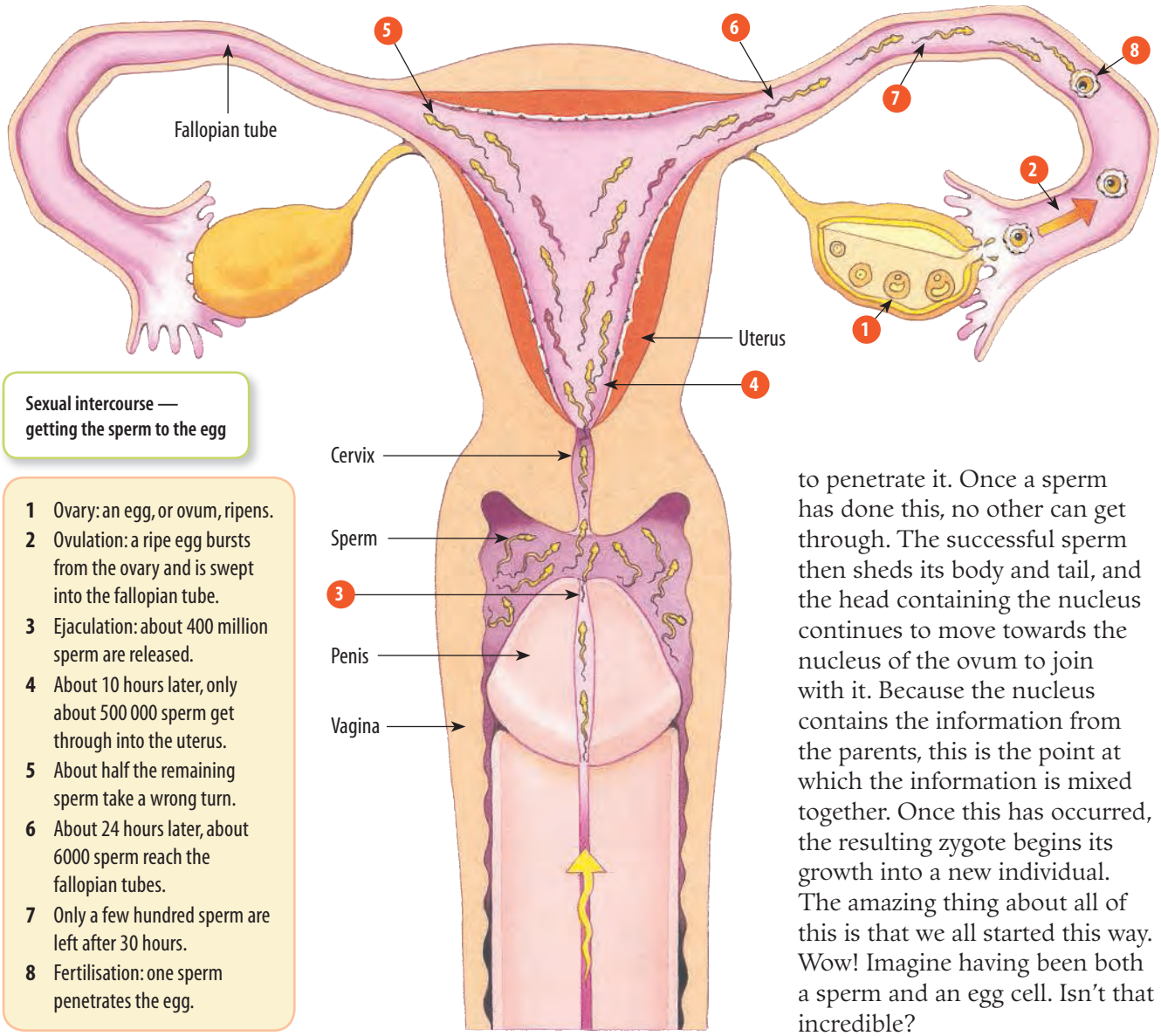
Although the resulting zygote will contain a mixture of the genetic material (**nuclear DNA**) from the nucleus of both the sperm and the ovum, it will contain the genetic material from mitochondria (**mitochondrial DNA** or **mtDNA**) only from the mother's ovum. MtDNA forms the basis of many new genetic and evolutionary studies.

Sperm

In humans, the sperm are mixed with a nutrient-rich fluid before their release, or **ejaculation**, from the male's penis. This combination of fluid and sperm is referred to as **semen**. The sperm make up less than one per cent of the semen. An ejaculation may release about 400 million sperm. Sperm are made up of three distinctive parts: the head, the middle section and the tail. Swishing its tail, and powered by the middle section, the sperm swims like a tadpole at about 4 mm per hour to reach its goal — the egg.

SUCCESS, AT LAST!

Upon reaching a ripe egg, hundreds of sperm surround it and try to break through. The sperm release an enzyme that dissolves the covering of the egg and enables one of them to penetrate it. Once a sperm has done this, no other can get through. The successful sperm then sheds its body and tail, and the head containing the nucleus continues to move towards the nucleus of the ovum to join with it. Because the nucleus contains the information from the parents, this is the point at which the information is mixed together. Once this has occurred, the resulting zygote begins its growth into a new individual. The amazing thing about all of this is that we all started this way. Wow! Imagine having been both a sperm and an egg cell. Isn't that incredible?

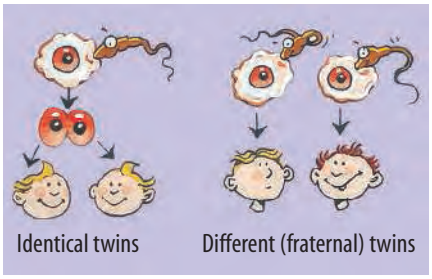


Sexual intercourse — getting the sperm to the egg

1. Ovary: an egg, or ovum, ripens.
2. Ovulation: a ripe egg bursts from the ovary and is swept into the fallopian tube.
3. Ejaculation: about 400 million sperm are released.
4. About 10 hours later, only about 500 000 sperm get through into the uterus.
5. About half the remaining sperm take a wrong turn.
6. About 24 hours later, about 6000 sperm reach the fallopian tubes.
7. Only a few hundred sperm are left after 30 hours.
8. Fertilisation: one sperm penetrates the egg.

Two or more?

Sometimes in the very early stages of division following fertilisation the embryo splits in two, so that two identical offspring are produced. This happens in the case of **identical twins**. They will always be the same gender as each has the same genetic make-up as the other.


Identical twins Different (fraternal) twins

Usually, only one ovum is released at a time. However, if several are released, twins or more can result from fertilisation by different sperm. In this case, the babies are not identical because they have different genetic combinations. These are called **fraternal twins**.

The use of fertility drugs and treatments has resulted in an increase in the number of multiple births. This is because they can affect the ovulation, so that more than one egg is released at a time. Some of these drugs can increase the chance of twins by 25 times and of triplets up to 350 times!

HOW ABOUT THAT!

Which animal has the longest sperm? Not an elephant, whale or human, but a fruit fly. Fruit flies of the species *Drosophila bifurca* have sperm about 5.8 cm long! That's about 20 times longer than their own body and around 1000 times longer than human sperm.

UNDERSTANDING AND INQUIRING

REMEMBER

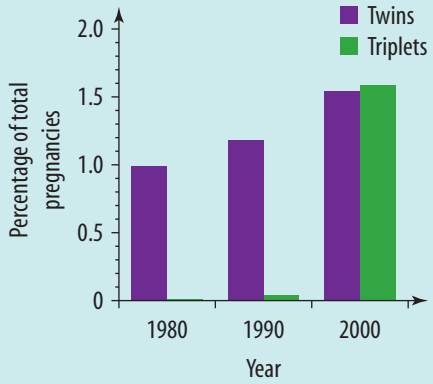
1 Define the term 'sexual reproduction'.
2 Put the following words into a sentence: gametes, sperm, ovum, zygote, fertilisation.
3 Describe the difference between external and internal fertilisation.
4 Draw a labelled diagram of a sperm.
5 Describe the events that take place when the sperm reach a ripe ovum.
6 Construct a Venn diagram to compare sperm and egg (ova) cells.
7 Explain the difference between identical and fraternal twins.
8 Construct a flowchart that includes ovulation, ejaculation and fertilisation.

THINK AND DISCUSS

9 With a partner or in a team, discuss the following questions.
(a) Why is internal fertilisation generally more efficient than external fertilisation?
(b) Why doesn't fertilisation occur each time a couple have sexual intercourse?
Send a team member to other teams to share your discussion points.

INVESTIGATE

10 Find out more about the gametes of at least four different animals. Display your findings either as models or as diagrams on a poster.
11 There is a theory that, by wearing tight jeans, human males may affect the development of their sperm. Find out if there is any scientific evidence to support this theory.



Incidence of multiple births in Australia since 1980 (as a percentage of total number of pregnancies)

Source: Based on ABS data

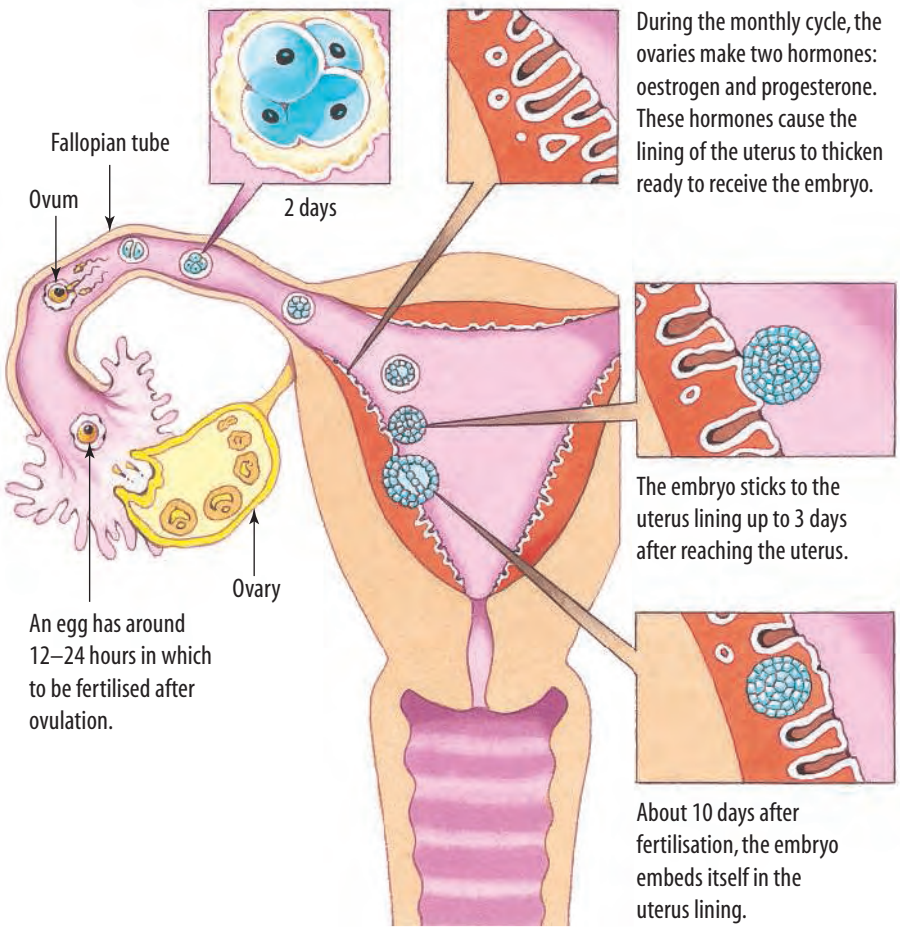
12 Carefully observe the graph above on the incidence of multiple births in Australia.
(a) Suggest reasons for any patterns in the graph.
(b) Suggest what the graph would look like if this year's data was added. Provide supporting information for your suggestion.
13 Research further into either epigenetics or mitochondrial DNA and share your findings with others.

INVESTIGATE AND CREATE

14 Write a story, play or poem about the successful sperm, from ejaculation to when it fertilises the ovum.
15 Design a board game that incorporates information about sperm and eggs.
16 Draw a descriptive timeline that includes: ovulation, ejaculation, sexual intercourse, the various stages of the sperm's travels through the female's reproductive tract, and fertilisation.

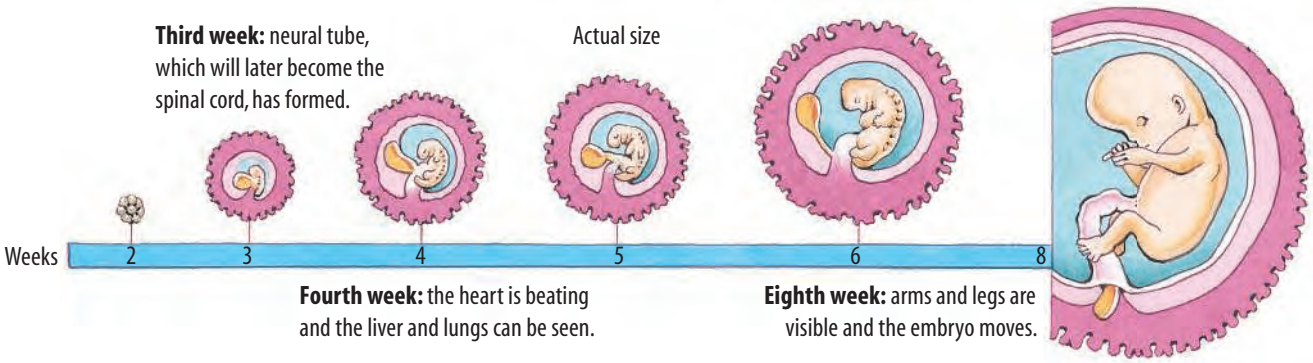
Making babies

Multiplication is the name of the reproductive game. Sometimes one and one can make three!



The first eight weeks

Conception occurs when the egg cell and sperm unite to form a zygote. When the zygote has divided into many more cells, it is known as an **embryo**. About ten days after fertilisation, the embryo completely embeds itself in the uterus lining (endometrium). This process is called **implantation**.



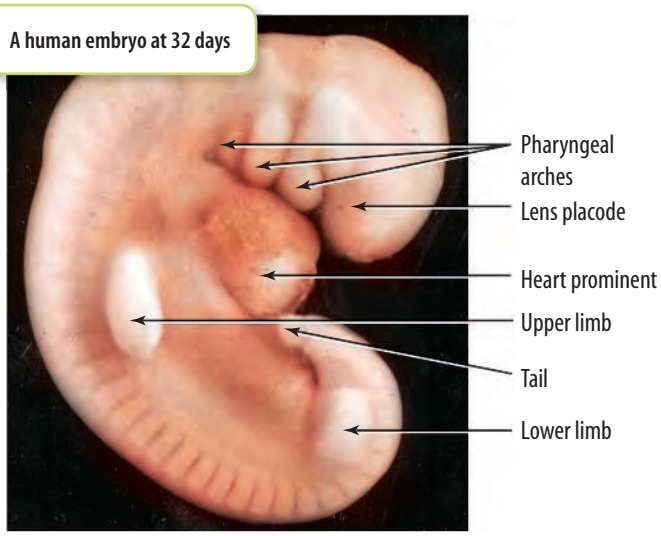
'After eight'

In humans, at about eight weeks, when the embryo has developed a distinct head, arms and legs, it is called a **fetus**. The fetus obtains its nutrients and oxygen through a special organ called the **placenta**. This organ is connected to the mother's blood vessels through the uterus. The placenta also absorbs fetal waste products and acts as a barrier against harmful substances. The unborn child continues to develop inside a sac that is filled with fluid (called amniotic fluid) for the rest of its time within the uterus. The total time spent in the uterus is often called the **gestation period**. In humans this is usually about 40 weeks. If a baby is born before 37 weeks, it is called **premature** and usually requires extra care and assistance.

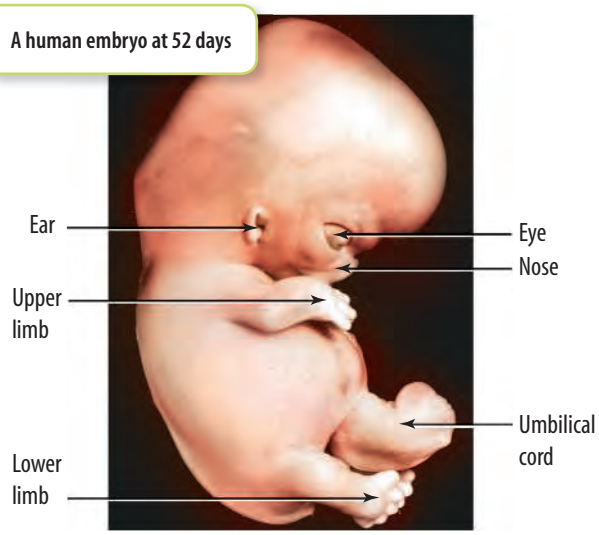
Approximate size of a fetus at different stages of development

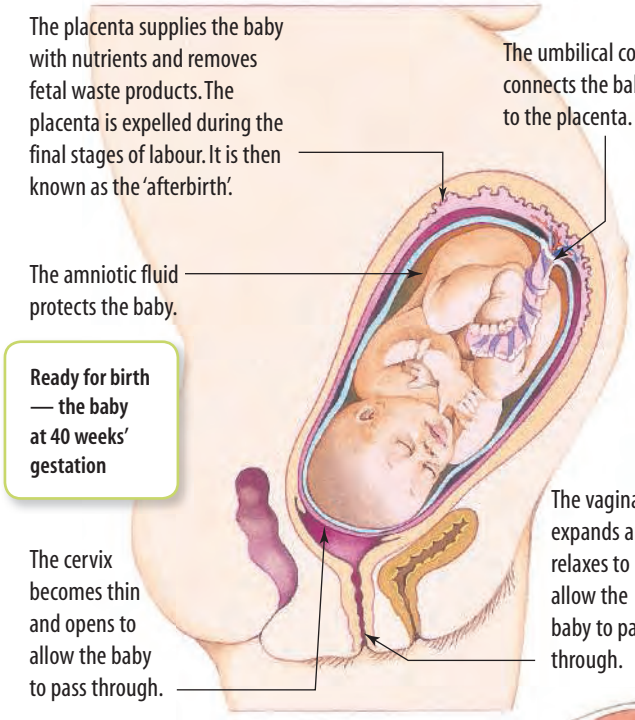
Development (weeks)	Length (cm)	Mass (g)
8	3	2
12	7.5	18
16	16	140
40	51	3400

A human embryo at 32 days



A human embryo at 52 days





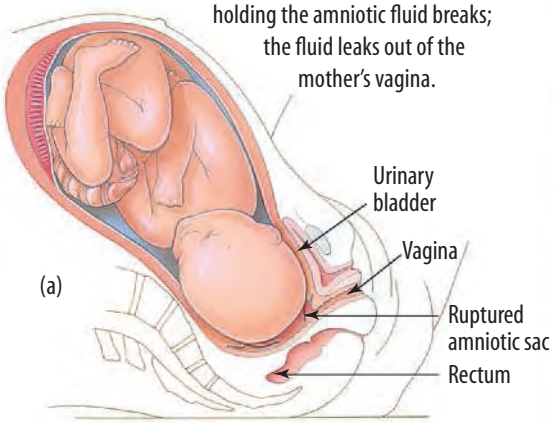
Giving birth

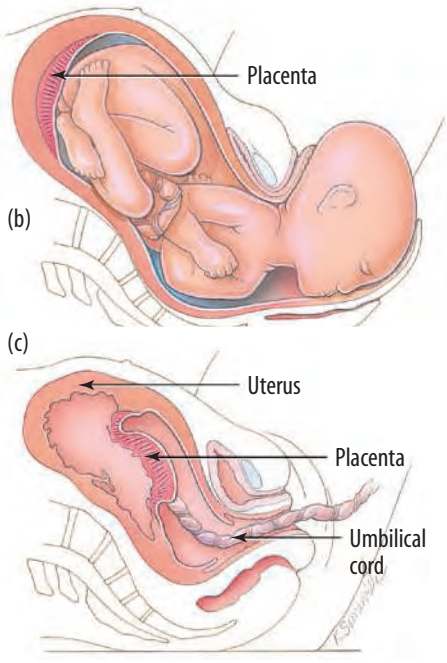
There are three stages involved in giving birth to a baby. Giving birth is usually referred to as **labour** because it can be a lot of hard work for the mother. During the first stage, the cervix gradually widens. In the second stage, the woman feels a strong urge to push with each contraction of the uterus. It is during this stage that the baby is born through the vagina, or birth canal. Usually the baby is born head first. Sometimes, however, the baby is born bottom or feet first; this is referred to as a **breech** birth and is often more difficult. The third stage lasts from the delivery of the baby until the placenta is delivered.

In some cases, the baby or the mother need extra assistance. A **caesarean** operation may be performed in which doctors surgically remove the baby by cutting through the mother's abdomen to her uterus.

(a) First stage

Uterus begins to contract at regular intervals that get closer and closer together. These contractions begin pushing down on the baby. At some point, the sac holding the amniotic fluid breaks; the fluid leaks out of the mother's vagina.





As contractions continue, the cervix stretches open more and more, until it is about 10 cm wide. This stage can last for many hours, especially for first-time mothers.

(b) Second stage

The mother gets a fierce urge to push (a bit like with a bowel motion) every time the uterus contracts. Bit by bit, this pushes the baby further down the vagina (birth canal).

(c) Third stage

The placenta is delivered after the baby is born. By this stage of the pregnancy it is a flattish, dinner-plate-shaped organ that looks a bit like a large piece of liver.

HOW ABOUT THAT!

Oxytocin — the 'trust' hormone

Hormones can have a variety of effects on our bodies. **Oxytocin** is an example of a hormone that not only has the potential to change how we feel, but also has important reproductive roles. This hormone causes the uterus to contract during childbirth and has a key role in breastfeeding. When a baby suckles on the mother's nipple, oxytocin is released in the mother, triggering the 'let down' response in which milk is released for the baby.

Oxytocin is also thought to be involved in the promotion of trust, love, empathy and social recognition. It has been described as the 'cuddle chemical', as it is released when mothers cuddle their babies. The release of oxytocin may also assist in the formation of bonds not only between mothers and their babies, but also between people in close relationships.

With trust comes power! Nasal sprays containing oxytocin have been marketed as 'trust sprays'. These are being advertised as having commercial value as they may contribute to feelings of trust in potential clients and customers. The development of oxytocin nasal sprays also provides an opportunity for researchers to investigate the potential use of this hormone in the development of treatments for specific autism spectral disorders (ASD) and in treatments to increase empathy and learn successful face recognition.

Cuddling and breastfeeding can result in the release of oxytocin, which can promote feelings of trust, love and bonding.

UNDERSTANDING AND INQUIRING

REMEMBER

1 State the differences between a zygote, an embryo and a fetus.
2 List the following in the correct order: birth, fertilisation, ovulation, growth, implantation.
3 In which part of the female reproductive system does the fetus develop?
4 Explain why sexual intercourse doesn't always result in fertilisation and pregnancy.
5 What is the difference between:
   (a) fertilisation and implantation?
   (b) fraternal and identical twins?

USING DATA

6 Construct a graph, using information in this section, to show the changes in length of the embryo from 2 to 8 weeks and the length and weight of the fetus from 8 to 40 weeks.

CREATE

7 Make scale models of the fetus at each age shown in the table on page 186.

THINK

8 Correctly match each key event with its day of occurrence in the menstrual cycle.

Day	Key event	Day	Key event
14	implantation	16	fertilisation
15	first cell division	23	ovulation

9 Suggest why menstruation must stop during pregnancy. What would happen if this was not the case?

INVESTIGATE

10 Research and report on one of the following: endometriosis, prolapse of the uterus, cervical cancer, hysterectomy, ectopic pregnancy.
11 Research and report on one of the following antenatal tests: ultrasound scanning, amniocentesis, chorionic villus sampling.
12 (a) Investigate the commercial availability and uses of the hormone oxytocin.
   (b) In your team, construct a PMI chart based on your findings.
   (c) Discuss the ethics and issues regarding the use of oxytocin in conditions not involving childbirth and breastfeeding.
   (d) If you were on an ethics committee or governing body, what regulations would you suggest be considered concerning the availability and use of synthetic versions of hormones (such as oxytocin)?
   (e) Organise your team's discussion and findings into a format that enables it to be shared with others.

To breed or not to breed

Preventing pregnancies

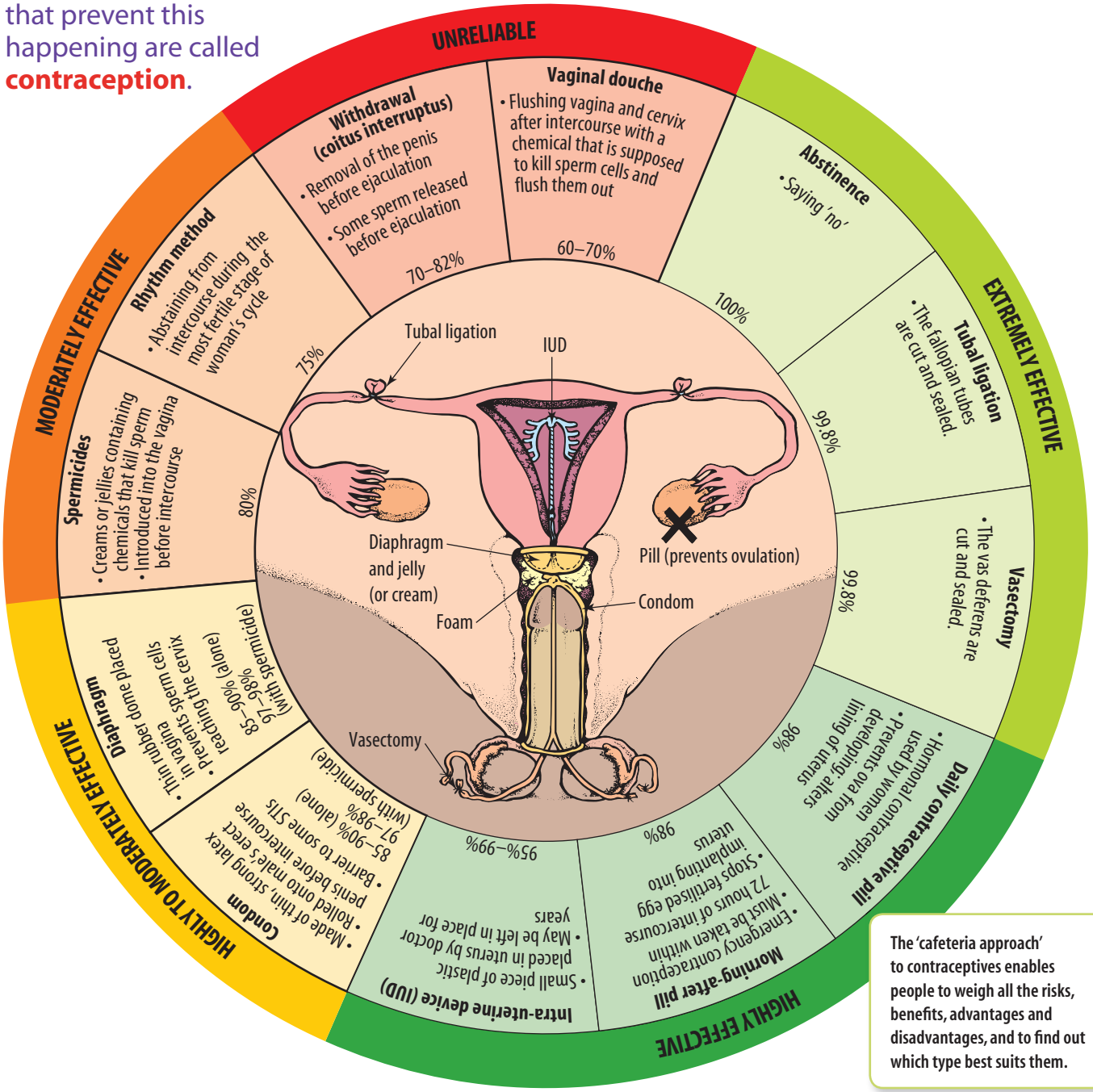
Conception involves the production of a zygote and its implantation into the wall of the uterus. Techniques that prevent this happening are called **contraception**.

Contraceptives are the devices or substances used to prevent unplanned pregnancies. There are two main types of contraceptives: those that prevent fertilisation taking place and those that prevent the fertilised ovum from implanting in the uterus.

eBookplus eLesson

Methods of contraception
This elesson will help you understand the many different forms of contraception and how effective they are in preventing conceptions.

eles-0127



The 'cafeteria approach' to contraceptives enables people to weigh all the risks, benefits, advantages and disadvantages, and to find out which type best suits them.

The World Health Organization estimates that, worldwide, half of all pregnancies are unplanned. The information that follows is given to you so that you are informed and know the facts about a variety of contraceptives. Quite often, television shows and magazines introduce sexual activity to young adults without giving them the full story. This can deliver a distorted message. But remember, the most effective method of contraception of all is to say 'no'!

New and improved products

Throughout history, people have tried to find methods that would enable them to have sex but not make babies — for example, people have tried swallowing tadpoles in spring, using lemons as a 'diaphragm' and using pig intestines as condoms. Because there is still no such thing as the perfect contraceptive, many new products are being invented and tested.

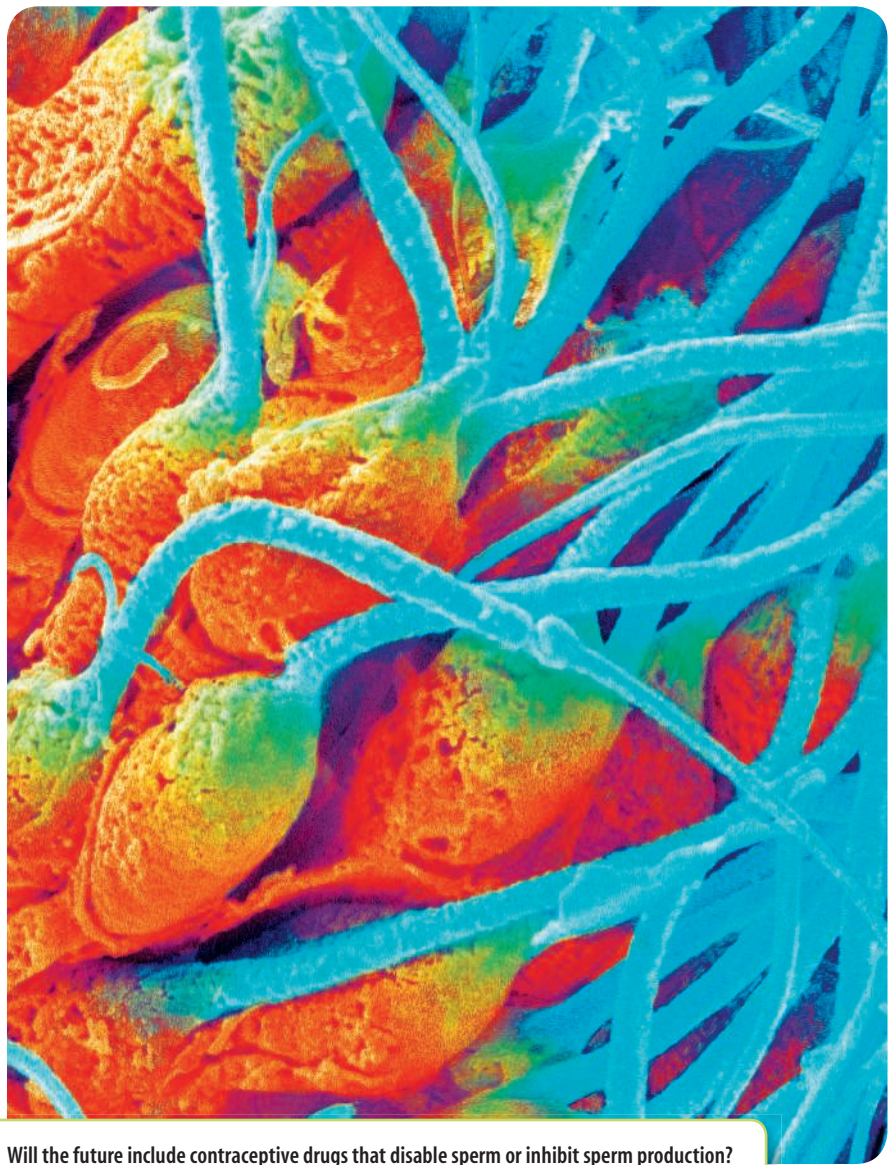
For women, some of these include **transdermal patches** that stick onto the skin and release hormones, **daily vaginal pills** that dissolve into spermicide when inserted into the vagina before intercourse, and the **Filshie clip**, a type of fallopian tube clamp. In some countries, even a female condom is available.

New products for men include **testosterone injections** or implants to reduce sperm levels, **male anti-fertility vaccines** that regulate sperm and testosterone production, **sperm duct plugs** that inject liquid plastic into the vas deferens, **chemical sterilisation**, and **gossypol**, a chemical that reduces sperm production.

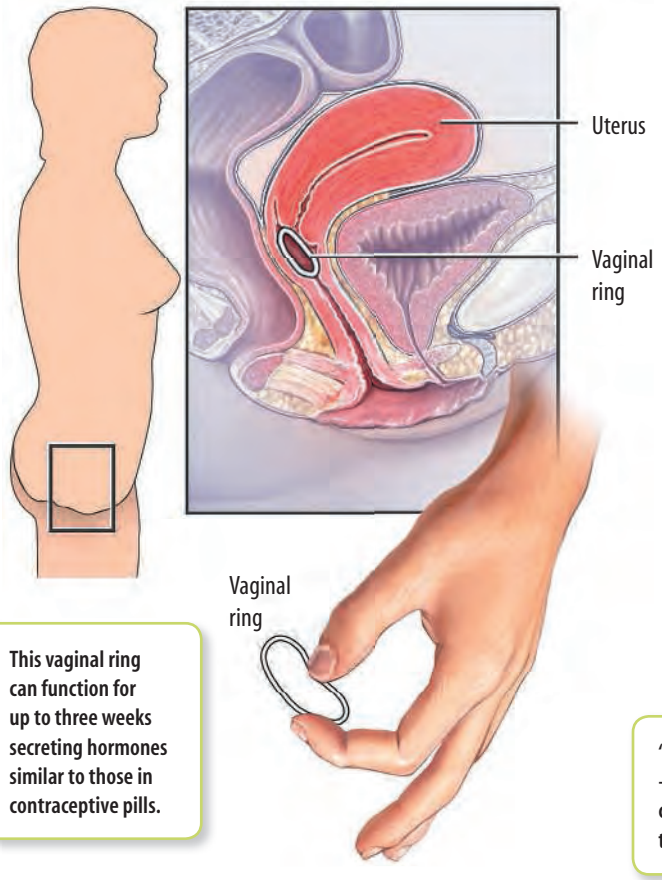
In for the long haul

There are also a variety of long-acting contraceptive methods available. Once 'introduced', these require no further action by the user for a long time. A disadvantage of most of them is that they require medical intervention for insertion and removal. Examples include:

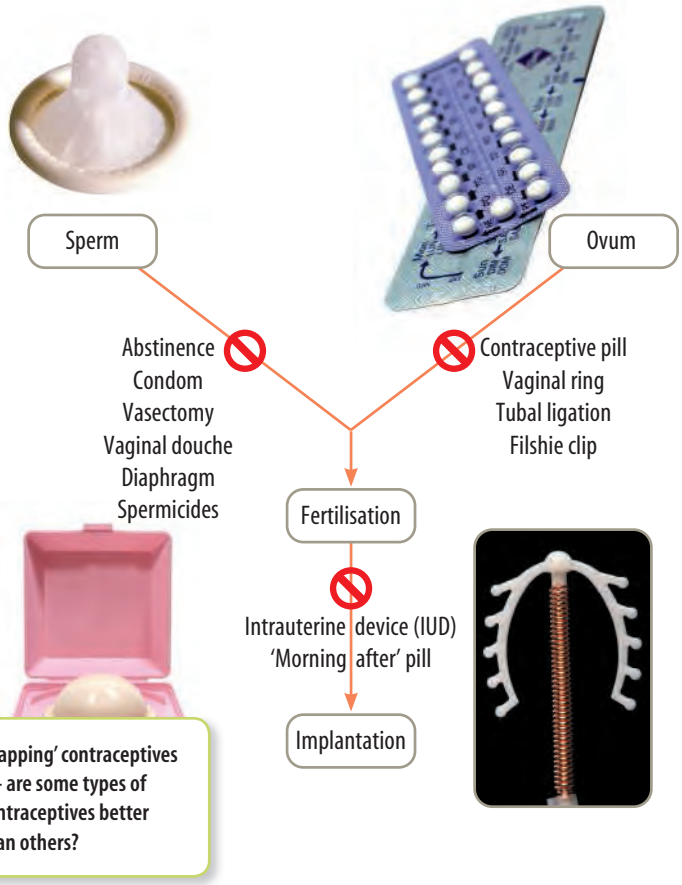
- *Depo injections:* Also known as Depo-Provera, this is a hormone injected into the user's buttock muscles that prevents ovulation for about three months.
- *implants:* A contraceptive implant (about the size of a matchstick) called Implanon is inserted under the skin of the inner, upper arm and prevents ovulation for about three years.
- *hormone releasing intra-uterine devices (IUD):* A Mirena is a T-shaped plastic device that releases hormones, acting directly on the lining of the uterus to make it thin and unreceptive to implantation of the fertilised egg. It also changes the fallopian tube lining, the mucus produced by the cervix and can stop ovulation in some women. It provides continuous contraceptive protection for about five years.



Will the future include contraceptive drugs that disable sperm or inhibit sperm production?



This vaginal ring can function for up to three weeks secreting hormones similar to those in contraceptive pills.



'Mapping' contraceptives — are some types of contraceptives better than others?

UNDERSTANDING AND INQUIRING

REMEMBER

1 Which methods of contraception:
   (a) prevent the sperm from reaching the egg?
   (b) prevent the release of the egg?

2 Construct a table that has six columns, with headings for: the type of contraceptive, a summary of how it works, who uses it (male or female), suggested advantages, suggested disadvantages, and a prediction of how many pregnancies may occur if 100 sexually active, fertile couples were to use it.

THINK

3 After a vasectomy:
   (a) does a male still produce sperm?
   (b) what does the ejaculate contain?

4 After a tubal ligation, does a woman:
   (a) still ovulate?
   (b) menstruate?

5 Suggest reasons why some contraceptives may be more effective than others.

INVESTIGATE AND DISCUSS

6 Research a method of contraception and present your findings to your team or class as a poster, PowerPoint presentation or concept map.

7 Find out about four of the following types of contraceptives and then present your findings with your team as a poster, PowerPoint presentation or concept map.

- Combined oral contraceptive pill
- Progesterone-only pill (or the mini pill)
- Depo-Provera (injectable contraceptive)
- Hormonal implants
- Morning-after pill (MAP)
- Today sponge
- Lng-Levonova IUD
- Vaginal ring
- Female condoms

8 Find out the advantages and disadvantages of each type of contraceptive list above. Present your findings as a matrix (see page 340). Compare and discuss your findings with others in the class.

CREATE

9 Design a 'future' contraceptive, using your imagination and knowledge of reproductive systems. Decide on how you would scientifically test the safety, effectiveness and popularity of your contraceptive. Produce a brochure that promotes your invention.

Reproductive technologies

Making babies is not always easy and simple. Not everyone can make their own babies.

The term **infertility** describes the inability to conceive or carry a pregnancy to a live birth. About 20 per cent of all couples are infertile. One of the commonest causes of infertility is the inability of either the male or the female to produce gametes. Such a person is **sterile**.

Some of the other reasons couples may not be able to have children are listed in the table below. Reproductive technologies have been developed to help people overcome some of these problems.

Some of the reasons couples may not be able to have children

Type of problem	Definition/reason
Gametes	Sperm or ova are not produced in sufficient quantity or quality.
Impotence	Some men cannot maintain an erection during sexual intercourse.
Blockage or damage	Some women may have blockages in their reproductive system (e.g. fallopian tubes), preventing fertilisation.
Miscarriage	The zygote or embryo is not maintained until the full term of the pregnancy.

Artificial insemination (AI)

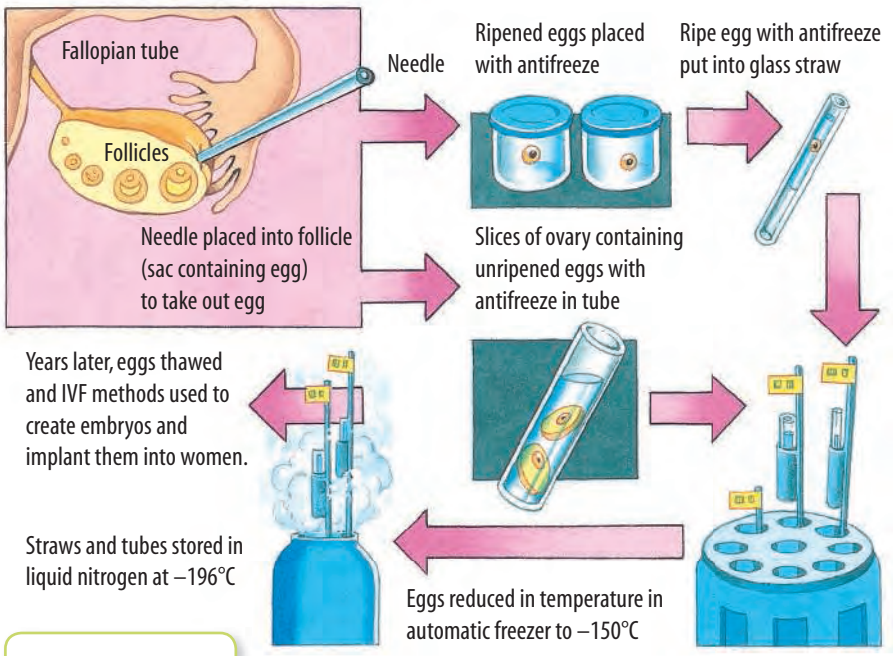
This technique involves injection of sperm into the woman's uterus close to the time of ovulation. The sperm may be collected from her partner, or from another male if her partner is sterile. Artificial insemination has also been used in agriculture in the production of prime farm animals, and in the breeding programs for endangered species.

In-vitro fertilisation (IVF)

In IVF, the sperm and the egg are fertilised outside the female's body. The fertilised egg is incubated until it develops into an embryo, which is then introduced into the female's uterus.

In this technology, eggs are surgically removed using a needle, laparoscope and forceps. A laparoscope has the lens of a microscope. To improve the chances of success, the woman is often treated with drugs that cause super-ovulation, resulting in several ova maturing at the same time (instead of one, as is usually the case). It is possible to freeze any fertilised eggs that are not used so that they may be available at a future time. This could enable 'twins' to be born years apart. Some women may use **donor eggs** (ova from other women).

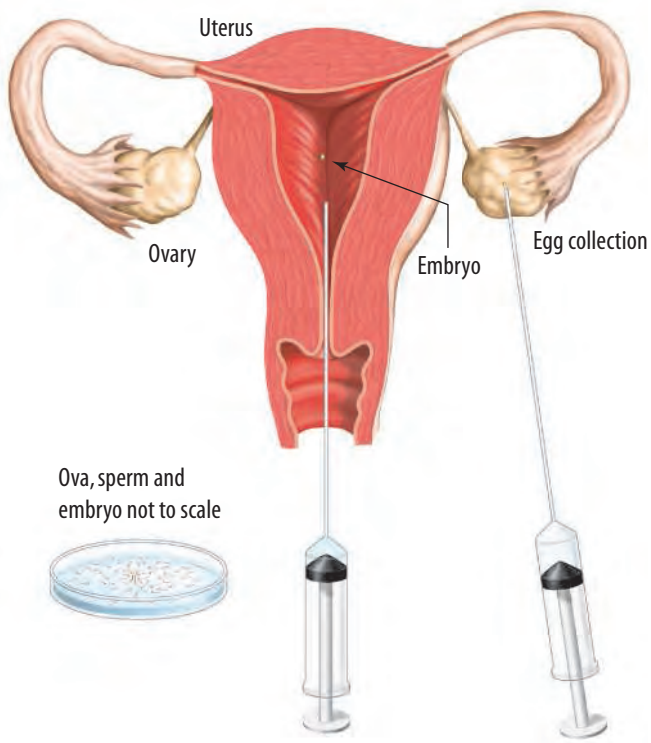
The removed egg is then fertilised outside the mother's body in a small glass tube or dish. The sperm used has been treated to remove its outer protein coat (an event that usually occurs in the female's reproductive tract). The fertilised egg is incubated in the laboratory until it is at



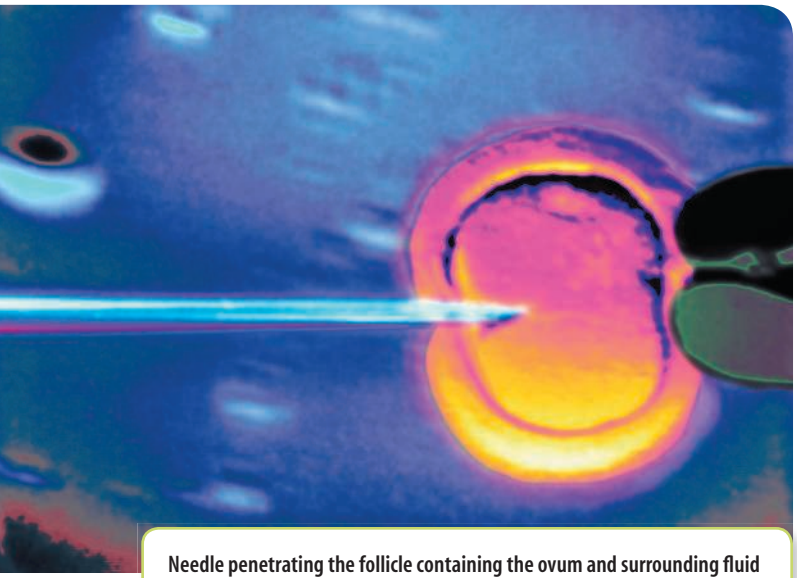
Frozen egg technology

the two- or four-cell stage. A four-cell embryo is obtained at about 35–46 hours after fertilisation.

The embryo is then placed into the woman's uterus for implantation. Babies born using this technique are often referred to as **test-tube babies**. In 1980, Australia's first IVF baby, a girl, was born in Melbourne; the first frozen embryo baby was also born in Melbourne, in 1984.



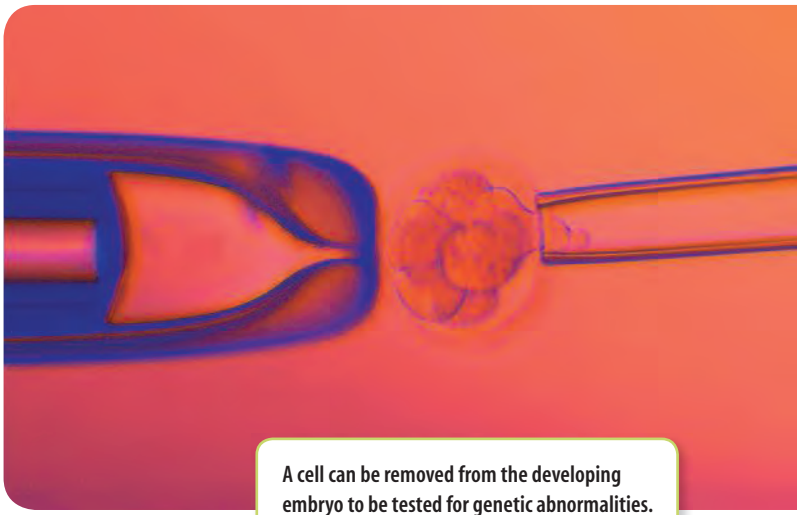
Hormone injections are used to increase ovulation. Collected eggs may be incubated and mixed with sperm to allow the possibility of fertilisation and formation of an embryo. The embryo may be inserted into the uterus. Surplus eggs or embryos may be frozen.



Needle penetrating the follicle containing the ovum and surrounding fluid

Although Louise Brown's conception using IVF in 1978 was considered a miracle of science at the time, she is now one of more than 80 000 Australians who started life in a dish. Louise is now in her thirties and is a mother herself, with her daughter being conceived naturally.

Some women are unable to maintain the growing embryo inside their uterus. A woman may, for example, be born without a uterus. The development of IVF technology has also opened up the field of **surrogacy**. In this situation, eggs are surgically removed from one woman and fertilised using IVF techniques. After this they are placed into another woman who undergoes the pregnancy.

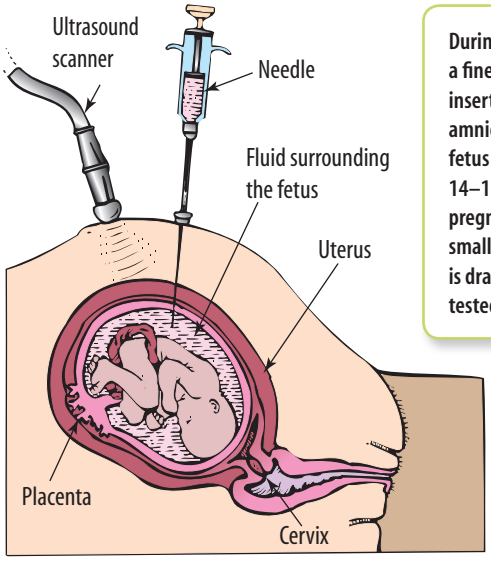


A cell can be removed from the developing embryo to be tested for genetic abnormalities.

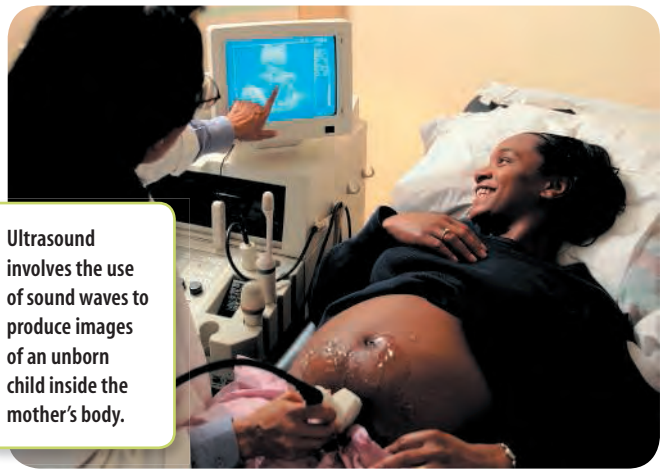
Testing the unborn child

There are a variety of technologies that can be used to test genetic composition and development. Some of these can be performed when the new life is in its very early stages. **Pre-implantation genetic diagnosis** (PGD) may be used to diagnose and exclude genetic abnormalities in embryos before potential implantation.

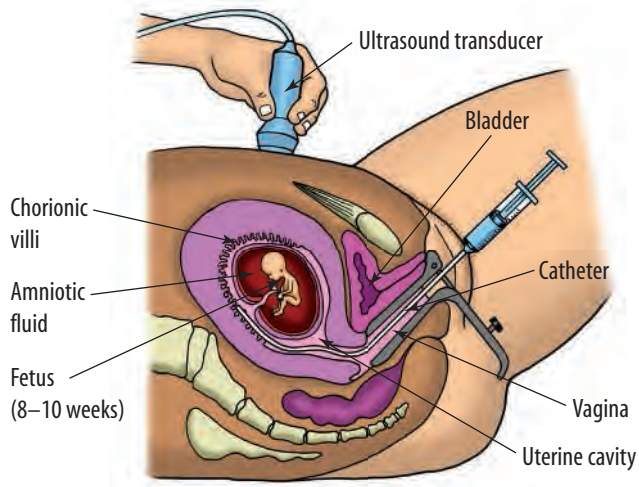
Other techniques, such as **ultrasound**, **amniocentesis** and **chorionic villus sampling** can be used at later stages of development. These techniques enable the gender and a variety of abnormalities to be identified.



During amniocentesis, a fine needle is inserted into the amniotic sac of the fetus at around 14–16 weeks of the pregnancy and a small amount of fluid is drawn out to be tested.



Ultrasound involves the use of sound waves to produce images of an unborn child inside the mother's body.



In chorionic villus sampling, cells from the developing placenta are removed for testing at around 10–12 weeks of pregnancy.

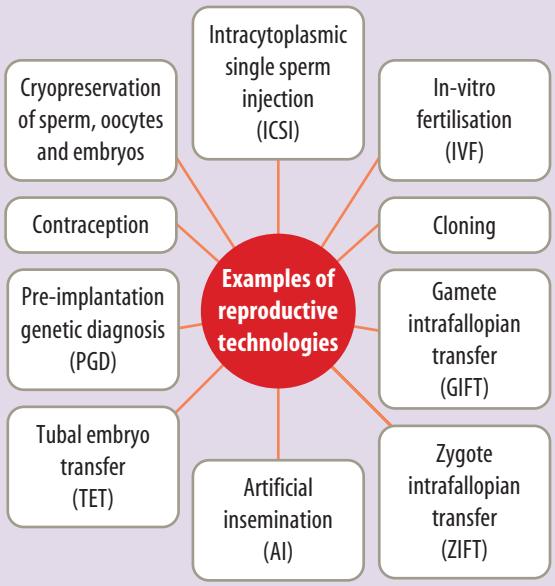
HOW ABOUT THAT!

Cadence Minge, a University of Adelaide researcher, was a winner of South Australia's Young Investigator of the Year award in 2007. Her research provided scientific evidence that high-fat diets could cause infertility in obese women. Her investigation involved using mouse eggs and showed that diets high in fat caused damage to eggs stored in the ovaries. She also found that a particular protein called PPAR-gamma could reverse the effects, but warned that it should not be considered a 'quick fix' for infertile women.

Cadence Minge (winner of South Australia's Young Investigator of the Year award in 2007)

HOW ABOUT THAT!

There are many different types of reproductive technologies. While assisted reproductive technologies (ART) can be used to treat infertility, other types (such as contraceptives) may be used to reduce fertility. Other types of reproductive technologies may be used to determine the likelihood of developing a particular genetic disease or to increase the number of offspring with particular features.



What's the difference between a GIFT and a ZIFT? Which of the reproductive technologies in this figure increase fertility and which decrease it? What's cryopreservation and why bother with it?

UNDERSTANDING AND INQUIRING

REMEMBER

1 Discuss the techniques that would help a couple reproduce if:
   (a) the male was infertile
   (b) the male was impotent
   (c) the female had blocked fallopian tubes
   (d) the female had a history of miscarriages.

2 Distinguish between:
   (a) artificial insemination and in-vitro fertilisation
   (b) ultrasound and amniocentesis.

3 Outline, in point form, the steps involved in IVF.

4 What are 'test-tube' babies? Is this an adequate name for them? Explain.

THINK, INVESTIGATE AND DISCUSS

5 (a) In groups of four or more, discuss each issue statement in the table below.
   (b) Write a list of people's 'gut reactions' or immediate responses to each statement.
   (c) Make a list of arguments for, and a list of arguments against, each statement.
   (d) Suggest what factors influenced your opinions on these issues.
   (e) Did the opinions differ between members of your group? Suggest reasons why.
   (f) Report your findings back to the class, or organise a debate.
   (g) Write a summary paragraph about the class's overall response to each statement.

6 Who should decide who is entitled to access to these technologies? Discuss this with your team and report back to the class.

7 What are the risks linked to reproductive technologies?

8 (a) Find out about the South Australian Young Investigator of the Year award and outline examples of scientific research that winners have been involved in.
   (b) Find out more and report on one of these research areas.

9 (a) Find out similarities and differences between the fields of obstetrics and gynaecology.
   (b) Select a topic that interests you in the area of reproductive technologies and investigate and report on research in that field.

10 Research and report on the scientific contributions of two leading reproductive technology pioneers:
   (a) Carl Wood (IVF pioneer, assisted reproductive technology)
   (b) Alan Trounson (IVF and stem cell pioneer).

eBookplus

11 Use the **IVF** weblink in your eBookPLUS to watch an animation of in-vitro fertilisation.

12 Developments in reproductive technology rely on scientific knowledge from different areas. Find examples that provide evidence for this claim.

work sheet 5.4 IVF — discussing the issues

Technique	Issue statement	Your opinion (Explain your response with arguments for and against the issue.)
AI	• Sperm should be used only from males with a high IQ, blue eyes and red hair. • All women should be artificially inseminated with sperm selected by their parents.	
IVF	• The IVF program is too expensive and should be abandoned. • IVF technology should be used to build a superior race.	
Donor gametes; surrogacy	• Donors and surrogates should be anonymous and have no rights over the offspring produced. • Sperm should be collected from all males at the age of 18 and only this is to be used for fathering children.	
Frozen embryos	• These embryos should be available to other couples if they are not used within six months. • These embryos should be developed so that they provide a supply of blood and organs for transplants.	
Ultrasound; amniocentesis	• These tests should be made compulsory for all women. Any abnormalities should result in immediate removal of the fetus. • These techniques should be used to select the gender of the child.	

Reproduction in the news

Research in the area of reproductive technology has brought along rapid change. There is a big difference between what is scientifically possible and what is socially acceptable.

The news snippets in this section provide examples of situations where the use of reproductive technology has raised ethical, social, legal or economic issues. Each snippet is a summary of a news story that was either published in a newspaper or was presented on the news.

UK woman pregnant by dead husband's sperm

Semen can be collected from a recently deceased man and frozen for later use, but is it ethical to do so?

Diane Blood confirmed that she was pregnant by her dead husband's sperm. Diane's husband died from meningitis in 1995. Some sperm samples were collected and frozen. Diane was told that she could not use the sperm samples for artificial insemination because her husband had not given his consent in writing before he died. She went to court and won the right to use the semen samples.

Source: BBC News, 28 June 1998

At 67, Romanian becomes the oldest mother

Adriana Iliescu gave birth at age 67.

Adriana Iliescu had IVF treatment using donor eggs. She started fertility treatment at the age of 58. After many years she finally succeeded in falling pregnant with twins. One of the twins died in the womb; the other baby was then delivered by caesarean section 6 weeks before the due date weighing 1.4 kg.

Source: The Age, 18 January 2005

Mother urges legalisation of surrogacy

Surrogacy is when a woman carries and delivers a baby for a childless couple. In some cases sperm from the father is used to artificially inseminate the surrogate mother. In other cases an embryo from the childless couple is produced by IVF and then transferred into the womb of the surrogate. In commercial surrogacy the surrogate is paid, sometimes large amounts of money, to carry the child. Altruistic surrogacy does not involve the exchange of money. A friend or sister might carry a child for a couple for example, and not expect payment in return. At the time the article went to press, altruistic surrogacy was legal in all states except Queensland where women could be fined or sent to jail for acting as a surrogate. A hearing was under way to decide whether it should become legal in Queensland.

Source: The Sydney Morning Herald, 7 July 2008

India's baby farms

Dr Kakoli Ghosh Dastidar works in a clinic in India where local women are paid to act as surrogates for other couples.

Indian clinics specialising in surrogacy are offering rich childless couples from around the world the opportunity to use a poorer Indian woman as a surrogate. The women are paid to act as surrogates. They live together while they are pregnant and they receive regular health checks and have all their meals prepared. For American couples (another country where commercial surrogacy is legal) the cost of an Indian surrogate is much less than an American surrogate.

Source: Sydney Morning Herald, 6 January 2008

Designer babies

Professor Julian Savulescu, an ethicist from Oxford University, argues that parents should be able to use genetic testing combined with IVF to choose the genetic characteristics of their children. Currently it is possible to test embryos created by IVF to find out which embryos carry certain disease-causing genes. Embryos found to carry the genes are not implanted. Professor Savulescu argues the technique should be further developed to allow parents to select genes for anything ranging from hair colour to intelligence or sporting ability.

Source: Channel 9 'Today' show, 7 August 2007

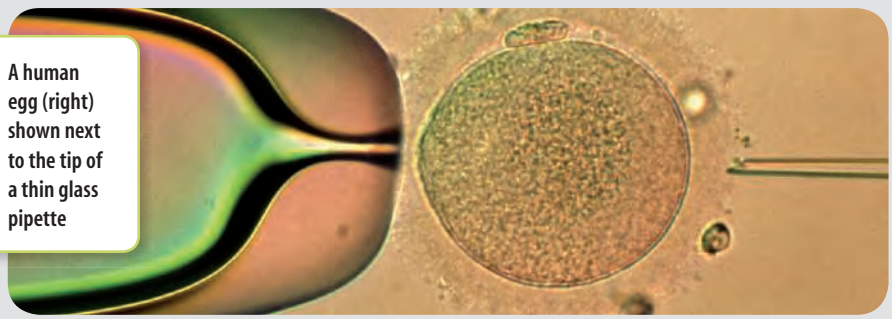
Should parents have the right to select certain characteristics in their children?

Freezing eggs is 'highly experimental'

The option of freezing some eggs is increasingly offered to women for a variety of reasons. Treatment for certain types of cancer can make women infertile or damage their eggs. Having some eggs collected and frozen before starting cancer treatment would give these women a chance to have children after they recover from the cancer. For women who have not met the right partner or do not yet feel ready to be mothers by the time they reach their late thirties, egg freezing might offer the possibility of extending a woman's reproductive years.

The chance of producing a baby from frozen eggs is not very high at this stage. In 2007 the American Society for Reproductive Medicine calculated that for every 100 frozen and thawed eggs only 2 to 4 pregnancies would result. Eggs are more difficult to freeze than semen or embryos because they contain more water and ice crystals can form inside the egg.

Source: Sydney Morning Herald, 26 October 2007



A human egg (right) shown next to the tip of a thin glass pipette

Parents sue over IVF son: Report

A Victorian couple are suing doctors at an IVF clinic. The couple decided to use IVF to conceive their child because they wanted to avoid giving birth to a child with haemophilia. The mother knew she was carrying a gene for haemophilia. Haemophilia is a disease where blood does not clot properly. A person who has severe haemophilia will usually require a transfusion of a special component of blood any time they have even a minor injury such as a cut or bruise. If a woman is a carrier for haemophilia, she does not have haemophilia herself. If she has a daughter and the father does not have haemophilia, the daughter will not have haemophilia either. If the same couple have a son, however, there is a 50 per cent chance that he will have haemophilia.

The couple in this news story used IVF because they wanted the doctors to test the embryos to find out whether they were boys or girls before transferring them to the mother's womb. The doctors made a mistake and transferred a male embryo. The couple gave birth to a son who has severe haemophilia.

Treatment for haemophilia usually includes regular transfusions of a product obtained from donated blood.

The couple are suing the doctors who carried out the IVF treatment. They are arguing that the unexpected arrival of a boy caused them shock and anxiety. They also want to be compensated for the cost of medical treatment for their son as well as the pay they have lost as a result of not being able to go to work when their son has needed treatment.

Source: ABC News, 23 March 2008

UNDERSTANDING AND INQUIRING

REMEMBER

1 Define the following terms:

surrogacy, surrogate mother, legalisation, altruistic, commercial.

2 Outline some situations where women may consider having their eggs frozen.

3 What is the chance of producing a baby from a frozen egg? Why is the success rate so low?

THINK

4 Discuss whether there should be an age limit for IVF treatment. Should this age limit apply to the mother only or to both parents? Justify your answer.

5 Create a PMI chart about the following statement:
A woman should be allowed to use her dead husband's sperm to conceive a child.

6 Using IVF and genetic testing it is currently possible for parents to choose certain characteristics in their children. The technology can be used to screen out certain genetic diseases and to select the sex of the child. In the future it may be possible to select a much greater number of characteristics.
   (a) Discuss whether this particular technology is harmful or beneficial.
   (b) Should parents be allowed to select any characteristics for which there is a test available or should there be restrictions on the characteristics that parents can select? Justify your answer.
   (c) IVF and genetic testing are expensive procedures, so they may not be available to poorer couples. Explain how this could have an impact on society.

INVESTIGATE

7 Use EBSCO or another database to locate other news stories about reproductive technology. Summarise the key points in each article.

Form and function: Comparing connections

Reproduction can be a risky business — but when the stakes are high it can be worth it! Some animals have some pretty tricky ways of reproducing . . .

Big families

Many organisms produce more eggs than can survive. Imagine what would happen if the 2000 eggs laid by a female house fly all survived! Environmental factors and predators may cause the death of many offspring before they get a chance to develop to the stage at which they themselves can reproduce. Sea urchins, for example, may discharge millions of gametes into the sea at one time. This coordinated timing of the release increases the chances of fertilisation occurring. However, most of the young sea urchins die. These deaths may be caused by many factors, such as competition for food and resources, and predation by other animals. If this reduction in the numbers of sea urchins did not occur, they would soon over-populate the oceans. A high juvenile death rate is quite common in many other organisms.

Dad's having a baby

Seahorses are very unusual fish, especially when it comes to making babies! It is the female that inserts part of her body (an ovipositor) into the male. She pumps eggs into a pouch at the front of his body and he then fertilises them with his sperm. Labour can sometimes take two days. Dad gives birth to 50–100 little seahorses, squeezing them out one at a time. No wonder he's called a big-bellied seahorse.

There are some amazing stories to tell about other types of seahorses. The male *Photocorynus* seahorse never grows larger than 10 cm and leads a parasitic life in which he is permanently attached to the female, hanging on by his mouth! This is useful to the female because it means that she doesn't have to search dark ocean depths to find a mate when her eggs are ready for fertilisation.

Male *Hippocampus abdominalis* seahorses try to get females to select them to carry eggs by inflating their pouches into a white balloon.

Guess who's coming to dinner?

In some fish species in which the male is in charge of protecting a clutch of eggs, it is not unusual for him to indulge in eating some of his own offspring. Honey, I ate the kids, indeed!

This trend also appears in some spider groups. The male Australian red-back spider, for example, is usually eaten by his sexual partner while mating with her. He is even considerate enough to position his body directly in front of her jaws after he has inserted his coil-shaped sexual organ into her. Male red-backs have a short lifespan; locating a female is extremely competitive and often the tip of their sexual organ breaks off during sex!

Recent studies have found that males that are consumed increase their chances of fertilising the female's eggs. By being eaten, they distract the female so that they may mate for longer. It was found that males that were eaten were able to mate for 25 minutes compared with 11 minutes for those that escaped. Hence, the eaten males had twice the chance of fertilising the eggs with their sperm. So, although being eaten for dinner seems like a high price to pay for sex, it does have some long-term rewards.

Some male damselflies have a penis with a special hook on the end. He uses it to remove other sperm left inside his mate by previous lovers before he makes his own deposit.

Did you know that . . .

- Some reptiles and rodents actually 'cement' up the female's genitalia by using some of the semen, which sets into a hard plug, not allowing other sperm to get in.
- Male starworms are 'live-in lovers' spending their entire lives within the female's vagina. Her eggs are fertilised by these parasitic males (which live off her vaginal fluids) as soon as they are released.
- Some butterflies have eyes on their genitals to help guide the hooks and claspers of the male to the appropriate nooks and crannies in the female during copulation.
- The Australian gastric brooding frog (now thought to be extinct) swallowed its externally fertilised eggs and then developed them in its stomach. A special chemical produced by the eggs stopped them from being digested. More than 25 little baby frogs would crawl out of the female's stomach and into her mouth.

Sending out signals

USING SMELL

Chemicals called **pheromones** can play an important role in communications between members of the same species. This type of communication makes it very easy for animals to locate a mate, even in sparsely populated areas.

When a female dog is about to ovulate, she comes 'on heat'. During this time she releases a pheromone into her urine to notify male dogs that she is ready for mating. Likewise, female moths use scented chemicals that sexually attract male moths from as far away as 8 kilometres.

USING LIGHT

Fireflies can make part of their body glow different colours. A chemical reaction produces a bright yellow, green or blue colour, which is used to help males and females find each other so that they can mate. Not all females, however, have reproduction on their minds. Females of a particular type of firefly have a different activity in mind. They flash their glowing abdomens on and off in a particular pattern, usually suggestive of a mating invitation. Sadly, instead of a romantic rendezvous, the males become a tasty meal.

USING SOUND

Whales may become separated by long distances, so in order to reproduce it is important that they can communicate. The male humpback whale sings a song during the mating season to advertise his sexual availability to females.

Birds also use their songs to attract potential mates. Frogs and crickets may not sound so melodic but they have their own way of making it known that they are available for sex. Male crickets make their chirping song by

rubbing their forewings together. Often they build their own version of a stereo amplifier by digging an underground nest with a twin-horned tunnel entrance. By sitting at the junction of the horns they can beam out their message loud and clear for all to hear.

Tammar trends

Researchers are studying the reproductive biology of the Tammar wallaby — this may help us to understand more about ourselves.

A baby Tammar wallaby is born about 26 days after conception. At birth, it weighs only 400 mg (about the size of the end of your little finger), and is blind and hairless. After leaving the birth canal, it crawls up into its mother's pouch and attaches itself to one of her teats. At this stage, its external sex organs have not yet developed; researchers already know that these develop in stages quite different from those in many other mammals.

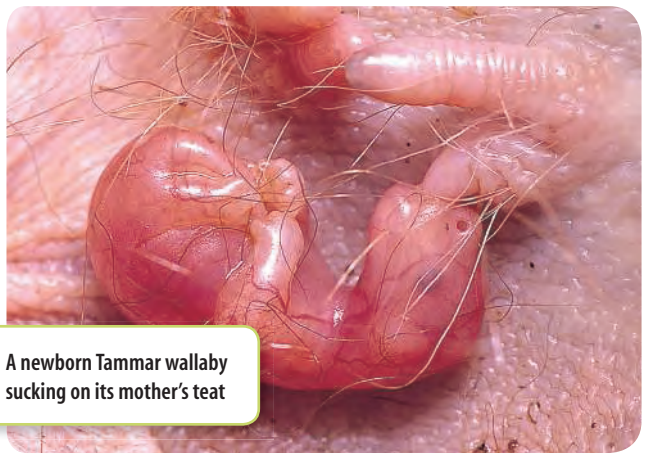
After suckling for about five months, it emerges from the pouch as a young joey. Although a joey can continue to suckle for up to a year, the mother can suckle another wallaby at a different stage of development at the same time. She does this by simultaneously producing two different types of milk. Research on how she does this could help us to improve milk production in farmed animals and our own human nutrition.

Will Tammar wallabies provide clues to our future reproductive technologies?

The mother Tammar wallaby can suspend the development of a fertilised egg until its older brother or sister has left the pouch, or until environmental conditions are more suitable. Finding out how she achieves this may help us develop new fertility and development technologies for other mammals, including humans.

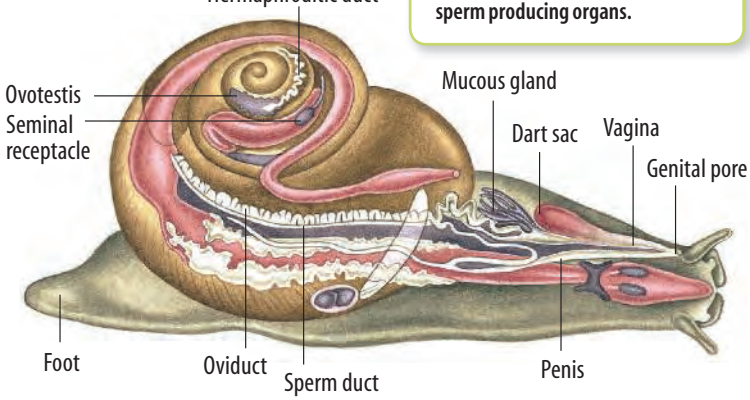
Get a look at that!

Did you know that ancestral reptiles were the first vertebrates to have a penis and that snails contain both male and female reproductive organs? While there is considerable diversity in the organisation of reproductive systems in organisms, there are also patterns and similarities. Although reproductive organs may appear structurally different, they often perform similar specialised functions that enable their species to survive and reproduce. In the figures of the snail and the turtle below, can you identify similarities to our human reproductive systems? If so, what are they?



A newborn Tammar wallaby sucking on its mother's teat

Snails are **hermaphrodites** because they have both egg-producing and sperm producing organs.



UNDERSTANDING AND INQUIRING

REMEMBER

1 Explain why a male red-back spider has a difficult life.
2 Describe why male seahorses are unusual fish.
3 Why didn't the babies in the Australian gastric brooding frog get digested in their mother's stomach?
4 Describe one way in which the following males may increase the chances of their sperm fertilising a female gamete:
   (a) some butterflies
   (b) starworms
   (c) damselflies
   (d) some reptiles and rodents.
5 What is the name of the group of chemicals that can play an important role in communications between members of the same species?
6 Suggest three ways in which smell is important to reproduction.
7 What does it mean when a dog is 'on heat'?
8 How do fireflies advertise their 'sexual availability'?
9 Which animals use sounds as a key invitation for a sexual interlude?
10 Describe what the Tammar wallaby looks like when it is born.
11 Outline some ways that Tammar wallaby research may assist studies in reproductive biology.

THINK AND DISCUSS

12 Why is reproduction worth the 'risks' that may be involved?
13 Why is internal fertilisation generally more efficient than external fertilisation?
14 Describe three ways in which animals may increase their chance of successful reproduction by having specialised reproductive structures or techniques.

INVESTIGATE

15 Find out more about the reproductive systems and mating behaviour of two different animals. Draw up a summary table to describe how are they similar and how are they different.
16 Research the reproductive system of an animal of your choice. Describe how it reproduces and draw a diagram of its reproductive parts. Present your information in a poster.
17 Research some other methods that plants and animals use to increase their chances of producing offspring. Report your findings to the class.
18 Find out more about Tammar wallaby research and present your findings in a PowerPoint presentation, mind map, newspaper article or poster.
19 What is a hermaphrodite? Suggest possible advantages and disadvantages of this condition.
20 Find out more about the evolution of reproductive structures in vertebrates, and report to others on current scientific research in this field.
21 Research the reproductive systems of at least three different animals and then suggest modifications to the design of reproductive systems that could improve their efficiency.
22 (a) Investigate the reproductive systems of the following animals so that you can complete the following table.

Features of:	Mammal	Fish	Turtle	Snail	Insect
Gametes					
Male reproductive structures					
Female reproductive structures					
Fertilisation					

   (b) Comment on the (i) similarities and (ii) differences between animals recorded in your table.
   (c) Which animals are most similar? Which animals are most different? Suggest possible reasons for this pattern.
   (d) Identify a reproductive research question that could be investigated for each animal.
   (e) Select one of the animals and build a model of its reproductive system.

CREATE

23 Write a story, play or poem about the life cycle (from gamete production to death) of an animal of your choice.
24 Make up a crossword with some amazing reproductive stories discovered from your own research.
25 Design a new breed of organisms. Make your 'organisms' out of plasticine or bread dough. Create a booklet that describes their lifestyle, how they find their mates and how they reproduce.

eBookplus

26 Use the **Animal video** weblink in your eBookPLUS to access the National Geographic video site. Type the word 'courtship' into the search bar to view examples of courtship displays. List some animals that carry out courtship displays.

5.9 SCIENCE UNDERSTANDING

The sex life of plants

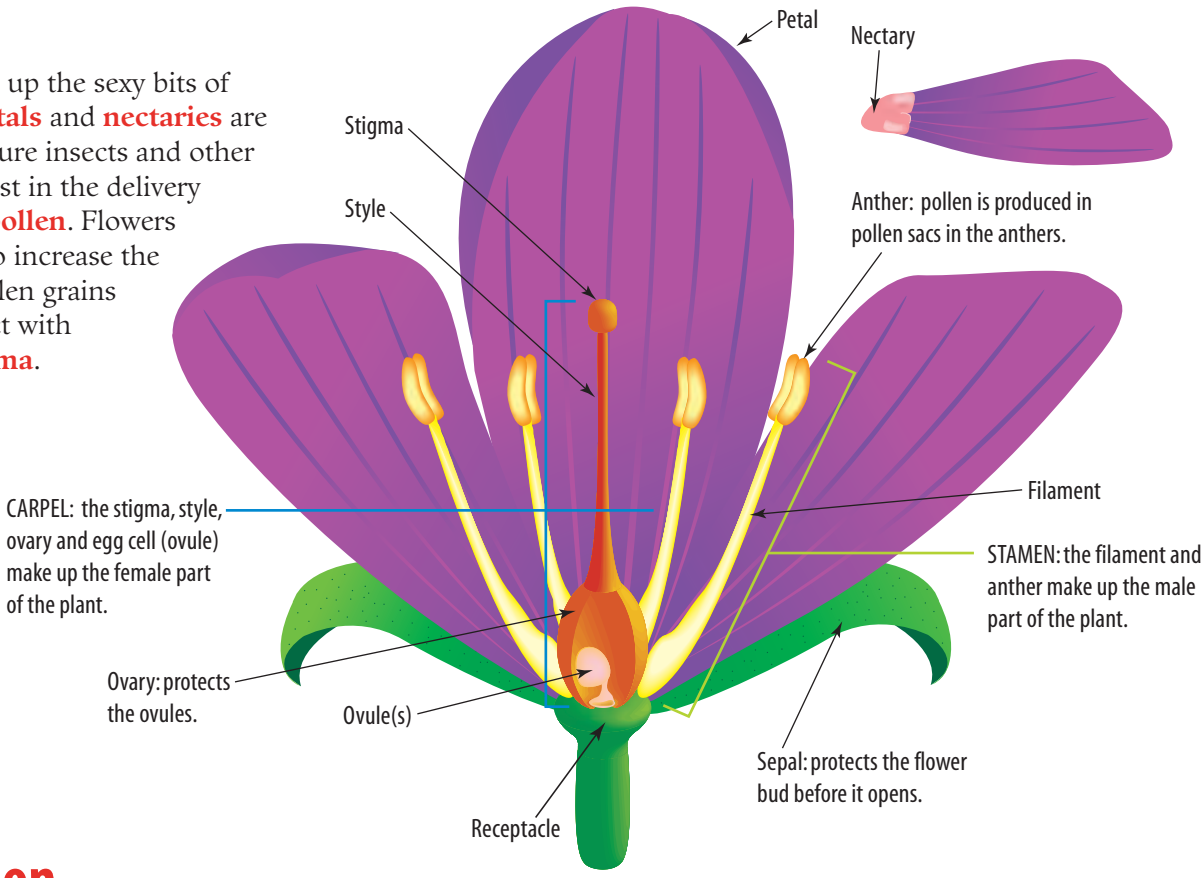
Like animals, many plants can reproduce sexually. Flowering plants (angiosperms) have their reproductive structures located in their flowers.

eBookplus eLesson

Growing plants in Australia
This video lesson is presented by a top Australian horticulturalist and provides you with tips for successfully growing plants in Australia. Watch this video as an introduction to your experiments with plants.
eles-0055

Flowers

Flowers make up the sexy bits of plants. The **petals** and **nectaries** are often used to lure insects and other animals to assist in the delivery of 'sperm' or **pollen**. Flowers are designed to increase the chances of pollen grains making contact with the sticky **stigma**.

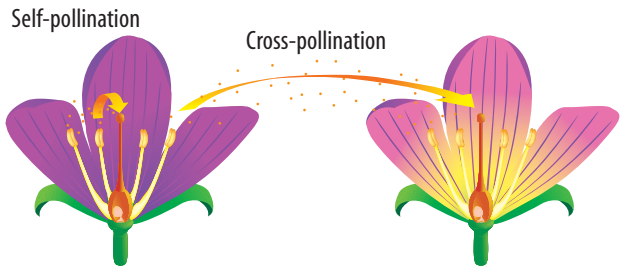


Pollination

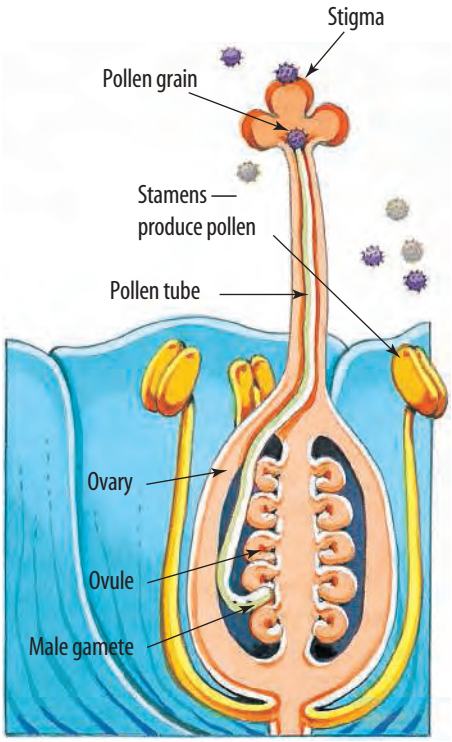
Pollination describes the way in which pollen grains reach the stigma. Plants may pollinate themselves (**self-pollination**). More often, however, they obtain the pollen from the flower of a different plant of the same species (**cross-pollination**). Cross-pollination increases the variation among the offspring and gives them a better chance of survival. The pollen grains may be transferred to other flowers by wind, insects and other animals.

Insect-pollinated flowers usually have attractive, brightly coloured petals and nectaries. The pollen grains themselves may be in a shape that makes them become easily attached to the insect.

Wind-pollinated flowers are usually less conspicuous and have no large scented petals or nectar. Their shape enables small, light pollen grains to be shaken from the plant and carried away with even the slightest gust of wind. The **anthers** hang outside the flower and the feathery stigmas spread out to catch airborne pollen grains.



The difference between self-pollination and cross-pollination



Pollination

As in animals, only a few of the pollen grains produced actually fertilise an egg cell. Pollen grains that reach the stigma of a flower become trapped by a sticky, sugary fluid, that provides them with nutrients. This fluid also triggers the pollen grain to burst open and develop a long hollow tube called a **pollen tube**. This tube grows down through the style until, finally, it reaches the **ovule**. It is here that the male gamete moves from the pollen grain and swims towards the ovum. If it fuses with the ovum, fertilisation has occurred and a zygote is formed.

Plant babies

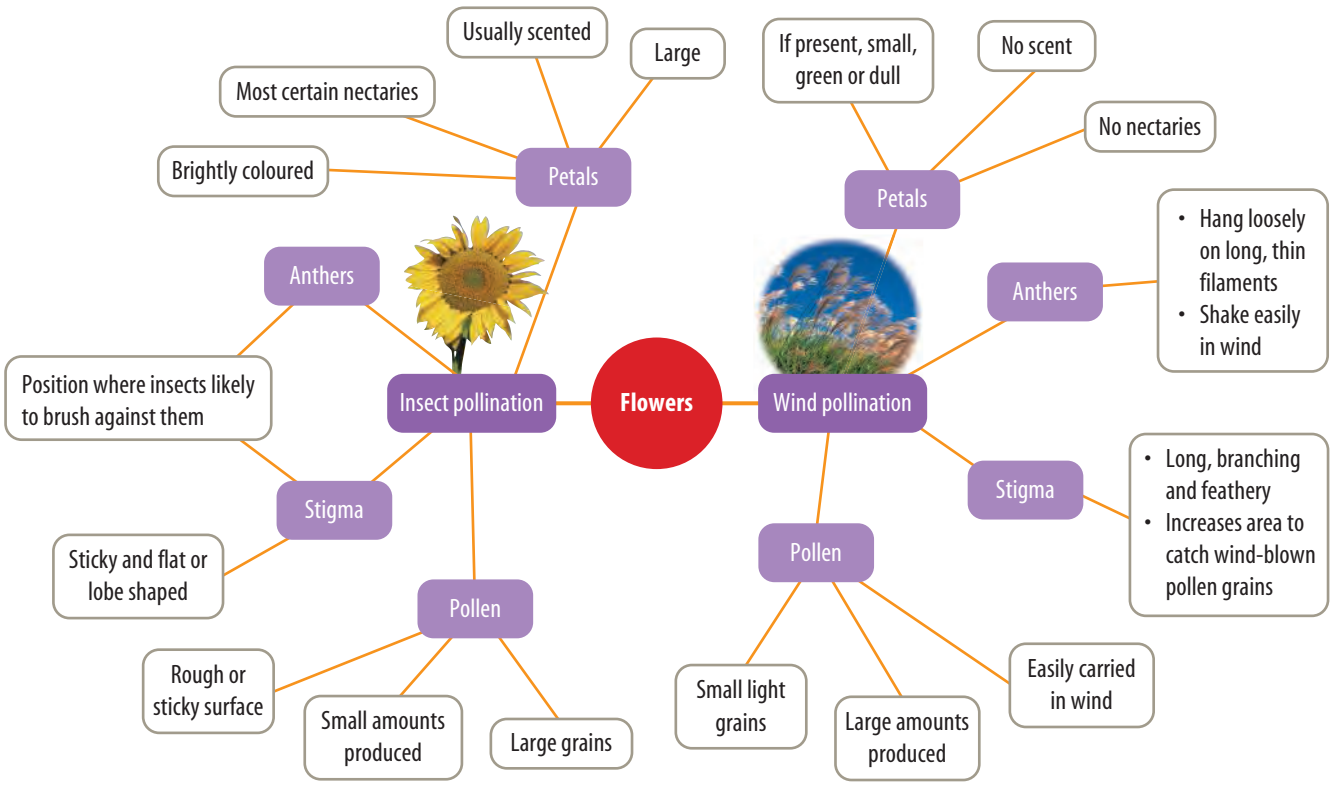
Inside the ovule, the fertilised egg, or zygote, divides into a little ball of cells that becomes an embryo. Special tissue called **endosperm** surrounds the embryo and supplies it with food. The ovule becomes the **seed**, and tissue forms around it to provide a protective seed coat. During the formation of the seed, the ovary expands and turns into a fruit.

Eggs, embryos, seeds and fruit

Are you aware that when you bite into an apple, cherry or orange you are actually eating the enlarged ovary of the plant? Did you know that these swollen ovaries contain the plant's 'babies' in their embryonic form? The plants are using you as a way of distributing their 'young' out into the world.

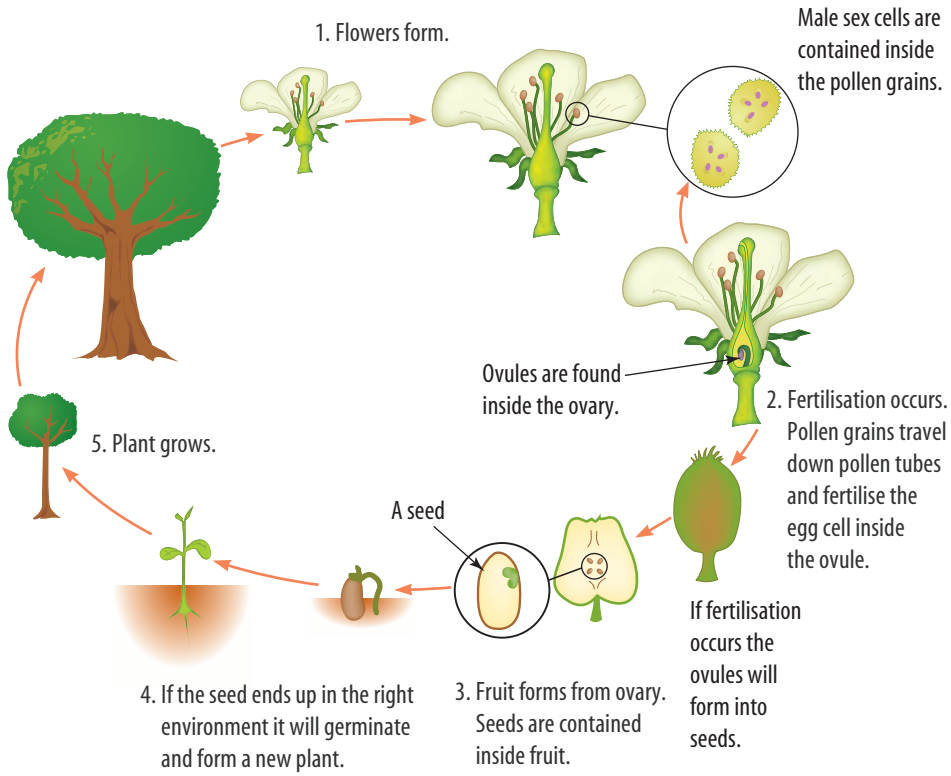
Once the flower has done its job and the egg cell has been fertilised by the pollen nucleus, another sequence of events takes place. The fertilised egg, in the middle of the ovule, divides into a little ball of cells that becomes an embryo. Special tissue called endosperm surrounds the embryo and supplies it with food.

The ovule becomes the seed and tissue forms around it to provide a protective **seed coat**. During the formation of the seed, the ovary expands and turns into a **fruit**.

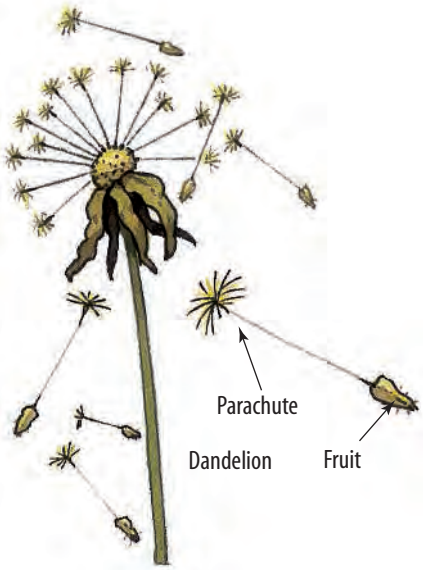


Seed dispersal

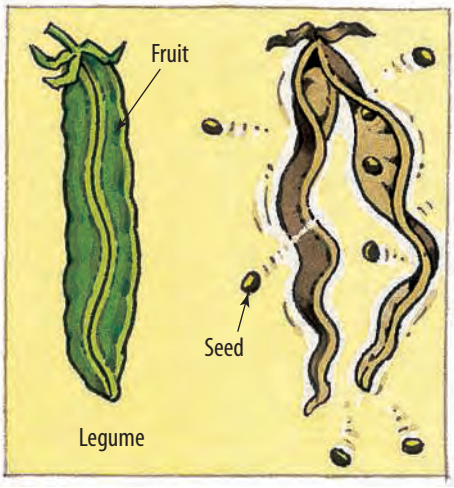
One of the main jobs that fruits do is to help **disperse** or spread the seeds. There is a variety of ways in which plants disperse their seeds: dispersal may involve animals, including birds (such as in tomatoes, grapes and apples); water (such as in coconuts); or wind (such as in grasses and dandelions). Some plants can disperse their seeds by themselves. For example, the fruits of some plants in the pea family (legumes) split open suddenly when they are ripe and dry, throwing the seeds out for long distances.



The life cycle of a flowering plant



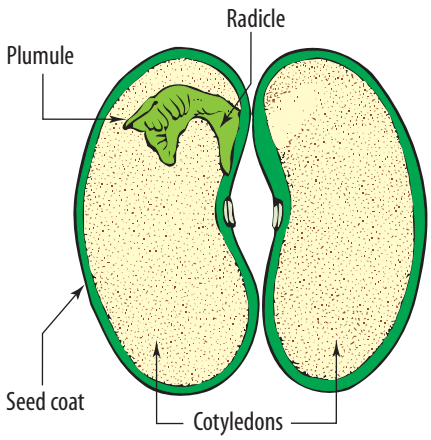
Wind dispersal (*top*) and self dispersal (*bottom*)



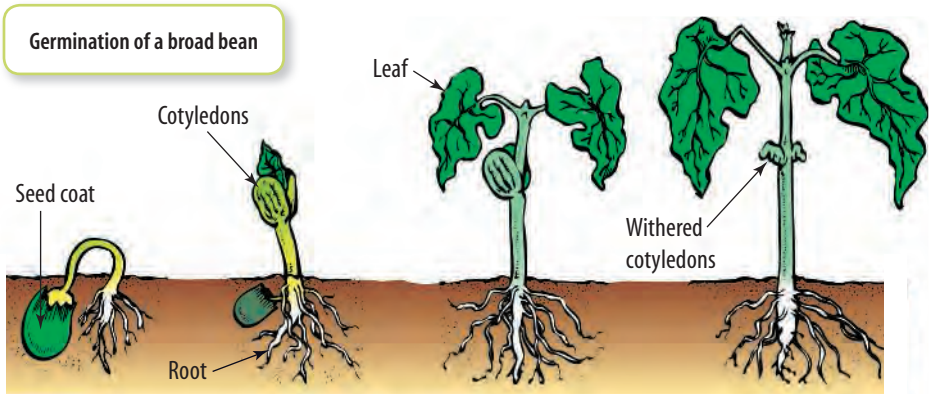
Seeds and germination

The embryo, inside the seed, is made up of three different parts: the baby shoot (**plumule**), the baby root (**radicle**) and one or two thick, wing-like **cotyledons**.

When the conditions are right, the seed bursts open and a new plant grows out. This process is called germination. When **germination** is complete, the embryo has become a young plant or **seedling**.



Germination of a broad bean



INQUIRY: INVESTIGATION 5.3

What's in a flower?

KEY INQUIRY SKILLS:

- planning and conducting
- processing and analysing data and information

Equipment:

flowers

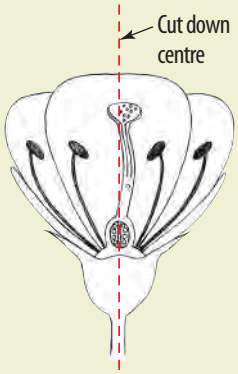
sharp knife or razor blade

cutting board

hand lens

tweezers

- Draw a picture of your flower. Locate, count and label the petals and sepals.



- Identify and label the male and female parts you can see.
- Place the flower on the cutting board and hold it with the tweezers.
- Carefully cut the flower in half down the middle (a vertical cross-section).
- Use the hand lens to look at the ovary and eggs.
- Draw the cross-section and label the female parts inside the flower.

DISCUSS AND EXPLAIN

1 Which parts of the flower become the seeds?

2 Which part of the flower do you think will grow into the fruit?

INQUIRY: INVESTIGATION 5.4

Angiosperm anatomy

KEY INQUIRY SKILLS:

- planning and conducting
- processing and analysing data and information
- evaluating

Equipment:

5 pieces of blank A4 paper

pencil

SOME WORDS OF WARNING:

1 Be responsible in your fieldwork and handle the plant parts very gently and carefully.

2 Do not pick, break, tread, trample or climb the plants.

3 Remember that you are dealing with living things.

- Find five plants, each with different types of flower.
- Using a separate page for each plant:
  - (a) at the top of the page
    - — record your name and the date
    - — record the plant's name, or, if unknown, record it as 'specimen A, B, C' etc.
    - — give a general description of the location in which the plant is found.
  - (b) divide the rest of your A4 sheet into three sections:
    - (i) half-page sketch of a flower
      - — Try to show the parts listed in the table on the opposite page and label them.
      - — Count or estimate how many stamens, stigma, petals and sepals are present.
    - (ii) quarter-page sketch of a leaf — include any veins that you see.
    - (iii) quarter-page sketch of the plant's overall appearance.
- Record the colour, scent (also give a mark out of 10 for its strength), and the texture and shape of the flowers, leaves and stems next to your diagrams.

INVESTIGATE, DISCUSS AND EXPLAIN

1 Identify key similarities and differences between the observed flowers. Suggest possible reasons for these.

2 Comment on your colour and scent data. Suggest a hypothesis to investigate a possible relationship between these.

3 Research your observed plants using databases and the internet. Construct a table, field guide, cluster map or multimedia format to summarise your findings on the following:
- (a) possible identification
- (b) labelled sketch or image of flower and fruit
- (c) type of pollination and type of seed dispersal
- (d) an interesting fact.

4 Suggest strengths, limitations and improvements for this investigation.

UNDERSTANDING AND INQUIRING

REMEMBER

1 Construct a mind or concept map to summarise the functions of different parts of a flower.

2 Draw a flowchart to show how flowering plants reproduce.

3 Which part of the plant is the fruit?

4 Match the words in the left-hand column (below) with those in the right-hand column.

sepal	sperm
petal	sugar
pollen	leaflet
nectary	colour
ovule	egg cell

5 Explain the difference between pollination and fertilisation.

6 Which part of the plant is the fruit?

THINK

7 Suggest reasons plants produce so much pollen.

8 Suggest why some orchid flowers closely resemble female wasps.

9 Use storyboards, cartoons or timelines to summarise how plants reproduce.

10 Use a bubble or mind map to show some foods that are seeds or products of seeds.

INVESTIGATE AND CREATE

11 Find and research examples of wind-pollinated and insect-pollinated plants. Construct flower hats that show what you have found out about their structures.

12 What does pollen have to do with hay fever? Make a model to show the relationship.

13 What are the conditions needed by most plants for germination?

14 Find four examples of different ways that the seeds of plants can be dispersed. Construct a story or play that includes these examples. Make puppets for the various roles in your story and present your play to the class.

15 Find out more about the life cycle of a plant of your choice. Report your findings as models, puppets or a poster.

16 Construct a Gantt chart or storyboard that includes seed dispersal, pollination, fertilisation, germination and development into a seedling.

17 Write a poem about the sex life of plants. Include as many of the bold-typed words in this section as possible.

work sheets
- 5.5 Science battles weeds
- 5.6 The sex life of plants
- 5.7 Those fabulous plants!

INVESTIGATE

18 Is there a relationship between the colour of a flower and the strength of its scent? Design and then carry out an investigation to determine whether the colour of the flower influences how strong the scent is.

19 Do some insects prefer some types of flower colour to others? Research or devise your own investigation. Share your conclusions with others in your class.

20 Complete the **Sex life of plants** interactivity in your eBookPLUS to answer questions about how plants reproduce. Success rewards you with a video of pollination. **int-0211**

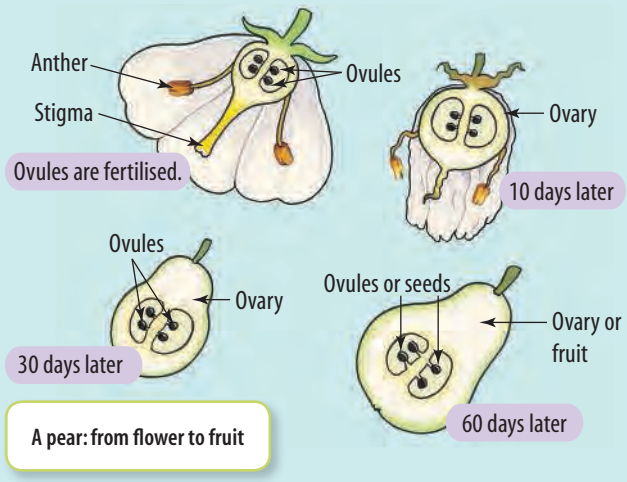
eBookplus

21 Find out more about the seed dispersal of five different types of plants and report your findings in a visual map.

22 Use information in this section and other resources to relate structural features of the following parts of the reproductive system of a flowering plant to their functions.

Part of system	Structural features	Function
Stigma		
Style		
Petal		
Ovary		
Pollen		
Pollen tube		
Seed		
Nectary		

23 The figure below shows stages in the formation of a pear (fruit) from the flower. Use the internet to find details of this process for three other fruits. Present your findings in labelled diagrams.



A pear: from flower to fruit

When one becomes two

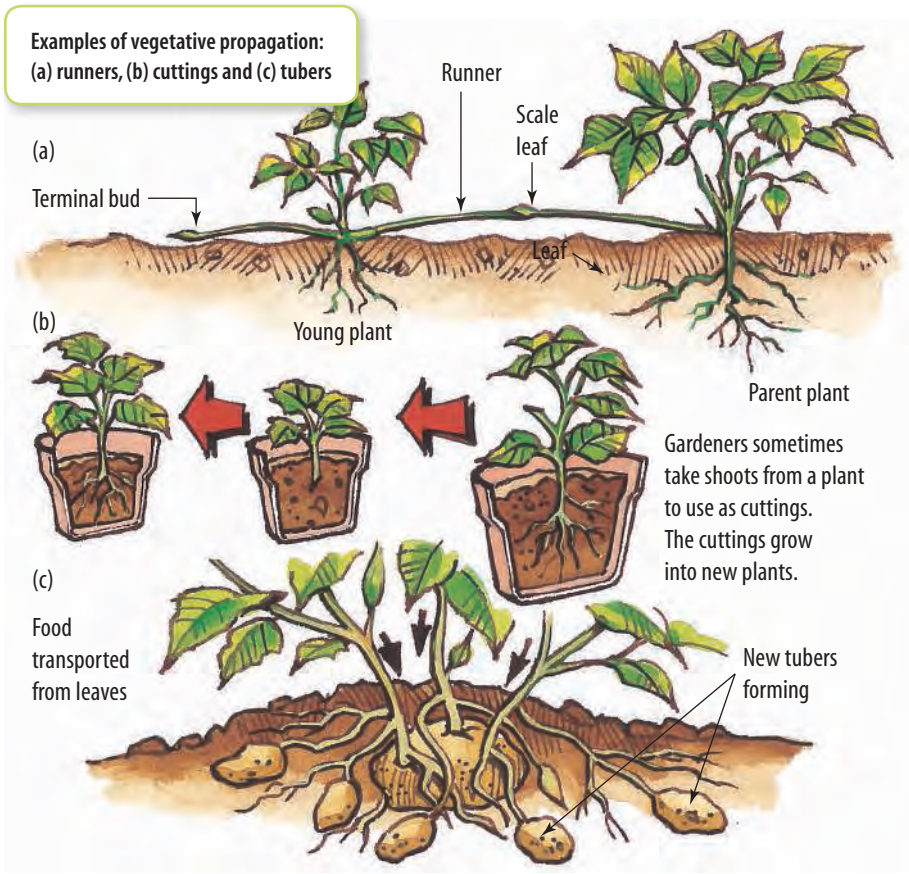
It might take two to tango, but there are times when it takes only one to multiply into many.

Not all animals reproduce themselves by sexual methods. Some can produce new individuals all by themselves. This is called **asexual reproduction**. It does not require the sex cells of another organism. Unlike the offspring resulting from sexual reproduction, all the offspring of asexual reproduction are identical to each other. These offspring are also identical to the original cell or organism from which they came. This difference between the two types of reproduction can have both advantages and disadvantages. Can you think of what these might be?

There are many different processes that involve asexual reproduction. Some examples include vegetative propagation, binary fission, budding, regeneration, parthenogenesis, and tissue and organ growth. New technologies, such as cloning, also involve the use of this type of reproduction.

Taking a short cut

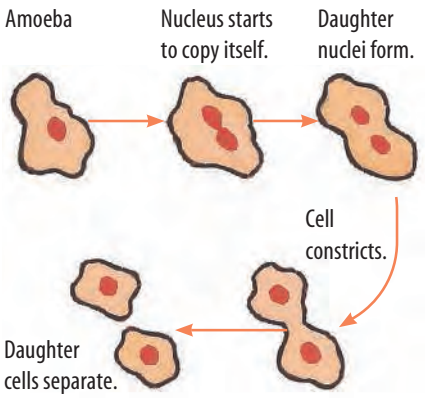
In **vegetative propagation**, the non-sexual parts of the plant are used to develop new individuals of the same type. Examples include bulbs (e.g. daffodils), stem tubers (e.g. potatoes), runners (e.g. native violets) and cuttings (e.g. roses).



Examples of vegetative propagation: (a) runners, (b) cuttings and (c) tubers

Let's split

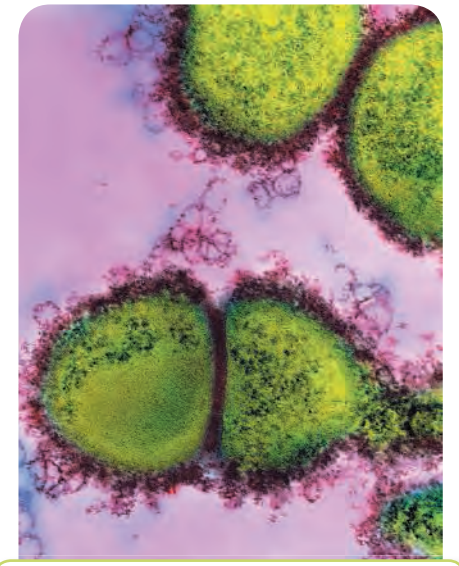
Some bacteria and other single-celled organisms such as amoebae reproduce by a type of asexual reproduction called **binary fission**. One parent divides into two new organisms, which each divides into two new organisms. Other bacteria may produce offspring by multiple fission, in which more than two daughter cells are produced with each division. This form of reproduction is very efficient, as it allows a rapid increase in numbers.



When one becomes two: binary fission in an amoeba

HOW ABOUT THAT!

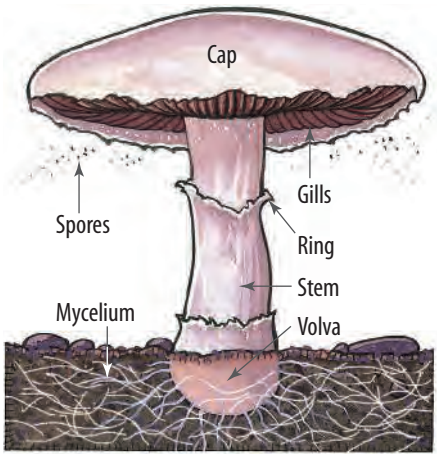
Flatworms and starfish have an unusual habit for animals in that they can reproduce asexually by regeneration. If the parent breaks into two or more pieces, each of these pieces may grow into a new organism that is a copy of the parent.



***Streptococcus pyogenes* bacteria, carried harmlessly by many people in their nose or throat. The bacterium at lower centre is dividing into two cells by binary fission.**

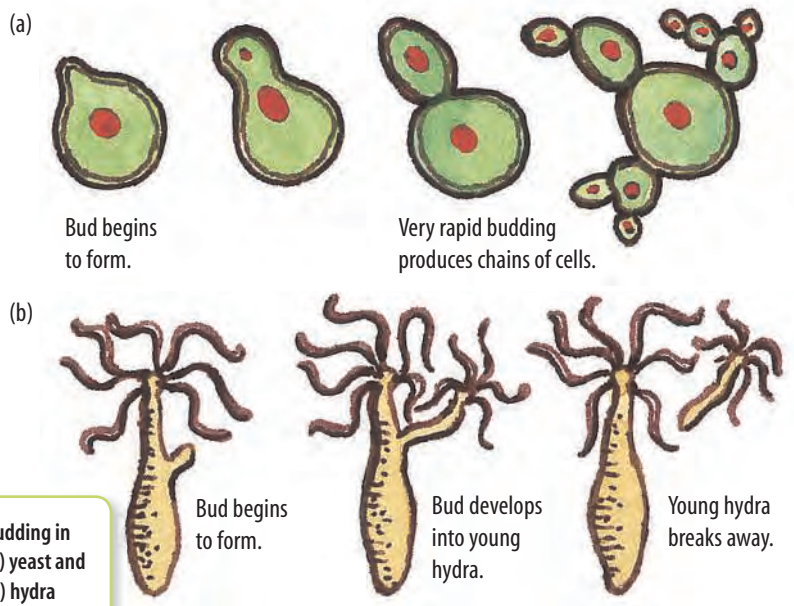
Spores on the wind

Some fungi (such as mushrooms, and bread and fruit mould) have spores that, when released, may develop into offspring identical to the parent fungi. These spores are merely a group of unspecialised body cells, combined with a source of nutrients and packaged in a resistant coat. They can provide an effective means of dispersing future generations and may also overcome adverse conditions by waiting until conditions are favourable before they begin to grow.



Budding offspring!

Imagine your offspring beginning as a simple swelling on your side and then developing its own mouth and features. When its development is complete it merely detaches itself and independently continues its own life. This is the sequence of events that happens in yeasts, and also in freshwater hydra. The initial swelling is called a bud and hence this process is often called **budding**.



Budding in (a) yeast and (b) hydra

Tiny new jellyfish (medusae) that have formed by budding

Girls only?

In some animals, the females produce eggs, but these develop into embryos without fertilisation taking place. The scientific name for the development of new individuals from an unfertilised egg is **parthenogenesis**. Worker bees, for example, develop from unfertilised eggs laid by the queen bee.

Some gecko lizard groups are parthenogenetic and form all-female families. An example is Bynoe's gecko (*Heteronotia binoei*), which is found only in Australia. A population of these geckos would contain only females. Births that result without any meeting between eggs and sperm are often referred to as **virgin births**.

INQUIRY: INVESTIGATION 5.5

Asexual reproduction

KEY INQUIRY SKILLS:

- planning and conducting
- processing and analysing data and information

Equipment:

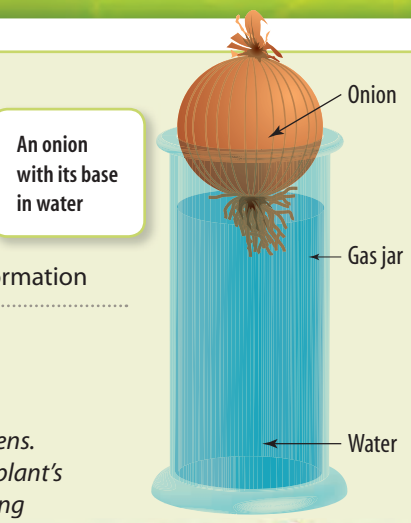
large onion *potato*

grass runner

leaf–stem cutting from geranium or impatiens. Note: a leaf–stem cutting is a piece of the plant's stem that is cut just below a joint or growing point and has a least three leaves.

leaf from an African violet, jade plant or snake plant

rooting medium (this can be purchased from a nursery)



An onion with its base in water

A leaf–stem cutting

- Fill a gas jar almost to the top with water and place the onion in the mouth of the gas jar so that its base is sitting in the water as shown in the diagram.
- Leave the potato in a dark cupboard.
- Remove the lower leaves from the leaf–stem cutting. Quarter fill a beaker or glass jar with water and place the cutting in the water.
- Place some rooting medium in a pot. Add water to the rooting medium until it feels moist. Cut a 3 cm section from the leaf of the African violet, jade or snake plant. Stand the piece of leaf upright in the rooting medium.
- Cut a piece of the grass runner. Ensure the section you have cut has at least one growing point. Press the piece of grass runner into the rooting medium (laying it flat on the surface).
- Leave all the plant parts undisturbed for two weeks. You may need to top up the water over that time.

DISCUSS AND EXPLAIN

1 Copy and complete the table below. You may need to dig the leaf–stem cutting and the runner from the rooting medium and wash them to see what has happened to them.

Plant part	Description after two weeks	Diagram
Onion		
Potato		
Leaf–stem cutting		
Leaf		
Runner		

2 Explain why each of the examples in the table above are forms of asexual reproduction.

3 What are the advantages of growing plants using one of the techniques described above rather than growing them from seeds?

4 Suggest improvements to the design of the investigation.

UNDERSTANDING AND INQUIRING

REMEMBER

1 What is asexual reproduction?

2 Describe the differences between the following types of asexual reproduction: budding, fission, cuttings, regeneration.

3 (a) What is parthenogenesis?

(b) Does this process involve sexual or asexual reproduction?

THINK AND DISCUSS

4 Suggest a reason why the parent involved in parthenogenesis is female.

5 Make a list of the advantages and disadvantages of parthenogenesis.

6 Suggest why many insects, which would usually reproduce sexually, use parthenogenesis to produce offspring in favourable conditions.

INVESTIGATE

7 (a) Place a carrot top on moist cotton wool until leaves appear, then transfer the plant to a plastic pot containing moist potting mix. Record what happens.

(b) Try this with a variety of other vegetables. Summarise your findings.

8 Find out more about parthenogenesis and virgin births.

9 Investigate and report on the impact of plant cloning techniques in agriculture in one of the following areas: horticulture, fruit production, vineyards.

10 Investigate the use of cloning technology in Australia. Organise a debate on any relevant issues.

Storyboards and Gantt charts

1. Decide how many scenes you need in your story. Often, 6–8 is a good number. Divide your page into this number of equal sections.
2. Consider which will be the three main events in your story and draw them roughly in the first, middle and last sections of your page.
3. Brainstorm the scenes that fit between these. Select the most appropriate and add them as intermediate scenes.
4. Mentally stand back and examine your story outline; make any desired changes to enhance its dramatic impact.

how to ...?

Helps you to use both your imagination and organisational skills to capture and share thoughts and ideas

why use?

What are the main scenes in a story or event?

question

Storyboard

A	B	C
Outline of scene 1	Outline of scene 2	Outline of scene 3
D	**E**	**F**
Outline of scene 4	Outline of scene 5	Outline of scene 6

also called

Comic strip

comparison

Similarity

Both show the sequence of events.

Difference

Storyboards use sketches or diagrams while Gantt charts use tables.

Gantt chart

example

Action	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
1							
2							
3							
4							
5							
6							
7							
8							

UNDERSTANDING AND INQUIRING

THINK AND CREATE

1 Construct storyboards for the following:
   (a) how you have changed between birth and ages two, four, six, eight and ten, and your current age
   (b) the 'life of a sperm' or the 'life of an egg'
   (c) how you could tell a Year 3 primary student 'the facts of life'.

2 (a) Read through the information in the 'Week by week' article below.
   (b) Mind map what you consider to be the key points.
   (c) Construct storyboards to show:
      (i) the changes experienced by the mother
      (ii) the baby's development.
   (d) Construct a Gantt chart to sequence your key points.

A baby's arms and legs start to form very early on. Soon, her limbs will lengthen, fingers and toes will develop and she'll be able to move them.

WEEK BY WEEK

Week 0–3
You … Seven days after implantation, chemical signals are sent to stop your period.

And your baby … Your fertilised egg moves down the fallopian tube. By day seven, it has implanted in the uterus wall.

Week 4
You … A pregnancy test will be able to detect that you are pregnant about 12 days after fertilisation.

And your baby … By the time she's reached 14 days after fertilisation, your baby has developed into an individual (twins can develop up until week four).

Week 5
You … may notice some signs that you are pregnant, such as changes to your sense of smell.

And your baby … is gently floating in the amniotic fluid attached to you only by the umbilical cord. She measures around 7 mm — about the length of a grain of rice.

Week 6
You … may feel the first symptoms of morning sickness, caused by the amount of pregnancy hormones in your body.

And your baby … Her heart forms a tube measuring half a centimetre — and already it can drive blood around her body.

Week 7
You … may feel completely exhausted because of all the hormonal changes in your body. You might go off tea and coffee altogether.

And your baby … Her head and tail have formed, as have tiny rudiments of eyes, kidneys and lungs. She's about the size of a five cent piece.

Week 8
You … Your uterus is now the size of an orange.

And your baby … The outline of your baby's face now appears — the two halves of her face will come together and meet in the middle. She's also been growing rapidly and now measures around 2.5 cm.

Week 9
You … probably won't look different to other people but you may notice your breasts are getting bigger and your waist is starting to thicken.

And your baby … has toes! Her liver is producing red blood cells which are pumped around her body by her tiny heart.

Week 10
You … are a quarter of the way through pregnancy and the time when you are most at risk of having a miscarriage is coming to a close.

And your baby … Her face is recognisably human and external sex organs have begun to form. She measures 6 cm.

Week 11
You … may have gained 900 g–1.3 kg (which is about 10 per cent of your total weight gain).

And your baby … is 7 cm and would fit into your palm. All her major body organs have formed and the most critical period in her development has passed.

Week 12
You … Your uterus is too large to remain in your pelvis and pushes above your pelvic bone. You'll feel hotter because extra blood is circulating in you.

And your baby … weighs about 14 g and is around 8 cm long. You can see her arm and leg movements on scans.

Source: Australian Pregnancy and Birth, September/October 2004

STUDY CHECKLIST

ASEXUAL REPRODUCTION

- contrast sexual and asexual reproduction
- outline the role of cell division in asexual reproduction
- describe some types of asexual reproduction

SEXUAL REPRODUCTION

- describe fertilisation
- contrast internal and external fertilisation
- outline the life cycle of flowering plants
- compare the reproductive systems of three different animals
- contrast mammalian and plant sexual reproduction

HUMAN REPRODUCTION

- recall the name and function of the organs of the male and female human reproductive system
- relate the structure of the organs of the male and female reproductive system to their function
- explain the role of the placenta
- outline the role of hormones in reproduction
- outline some technological advances that have been used in the fight against cervical cancer
- describe the process of fertilisation
- outline some changes that occur to the zygote between the time of fertilisation and implantation
- outline some changes that occur to a fetus as it develops inside the womb
- describe the birth process in humans
- describe some birth control techniques

LIFE CYCLE OF FLOWERING PLANTS

- recall the names and functions of the parts of flowering plants
- distinguish between pollination and fertilisation in flowering plants
- describe how fruit and seeds are formed from flowers
- outline ways in which seeds can be dispersed

REPRODUCTIVE TECHNOLOGIES

- evaluate the benefits and disadvantages of a number of reproductive technologies
- identify some causes of infertility
- assess the impact of reproductive technologies
- describe fertility treatments including artificial insemination and in-vitro fertilisation

INDIVIDUAL PATHWAYS

eBookplus

Activity 5.1
Growth and reproduction
doc-6087

Activity 5.2
Investigating growth and reproduction
doc-6088

Activity 5.3
Analysing growth and reproduction
doc-6089

ICT

eBookplus Summary

eLESSONS

Methods of contraception
This eLesson will help you understand the many different forms of contraception and how effective they are in preventing conception.

Searchlight ID: eles-0127

Growing plants in Australia
This video lesson is presented by a top Australian horticulturalist and provides you with tips for successfully growing plants in Australia. Watch this video as an introduction to your experiments with plants.

Searchlight ID: eles-0055

INTERACTIVITIES

Know your privates
In this interactivity, your knowledge of the male and female reproductive organs is tested by challenging you to identify private parts from the clues provided.

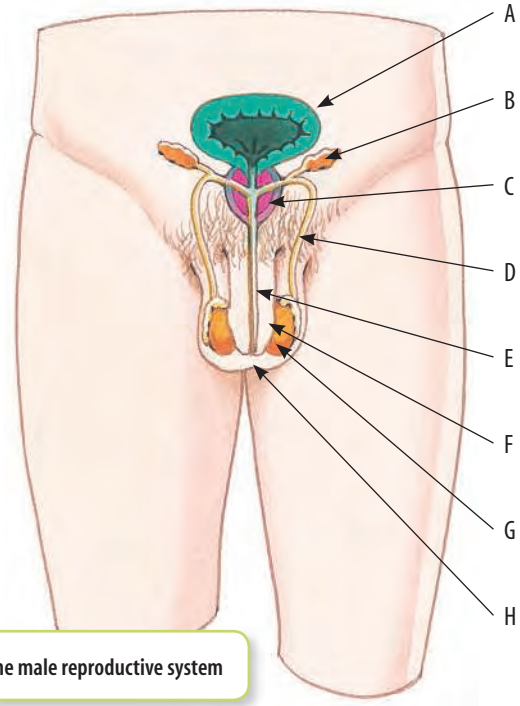
Searchlight ID: int-0671

Sex life of plants
This interactivity delves into the seedy world of the sex life of plants. Play the revelation game and answer questions about how plants reproduce. Success rewards you with an animation of the sex life of plants.

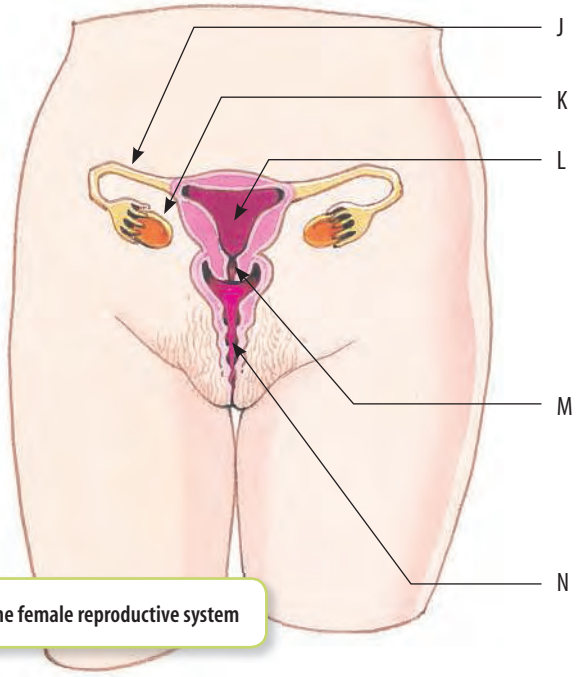
Searchlight ID: int-0211

LOOKING BACK

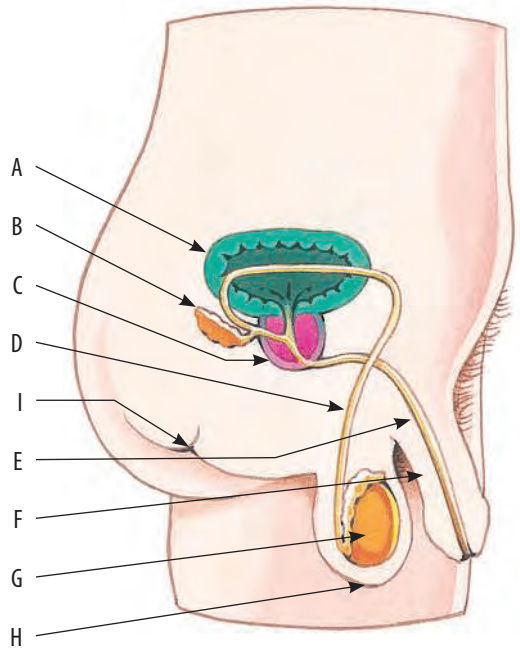
1 Identify the parts labelled A–T in the diagrams below. Write one function of at least two parts in each diagram.

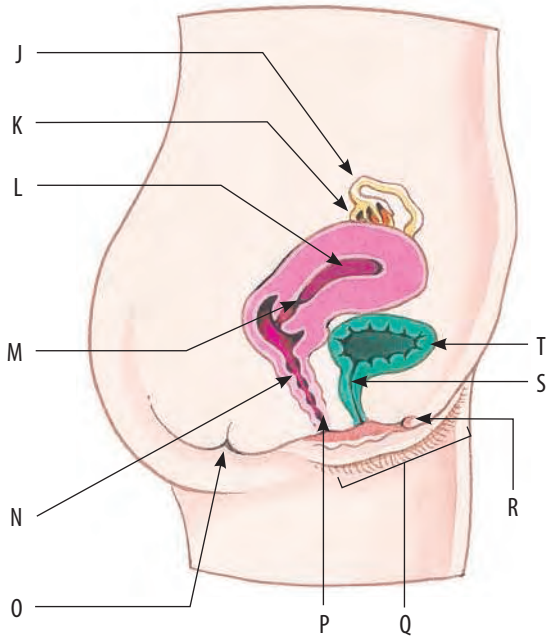


The male reproductive system



The female reproductive system





2 Design a calendar of the menstrual cycle and then outline the events that occur at each stage on your calendar.

3 Unscramble the following types of asexual reproduction.

taevvegeti gatponproai — gatienreoner
narybi sfionis — sheneipartognes

4 Summarise the disadvantages and advantages of sexual and asexual reproduction.

5 Invent, design and make your own creature. Describe its courting and mating behaviour, and give details about the way it reproduces.

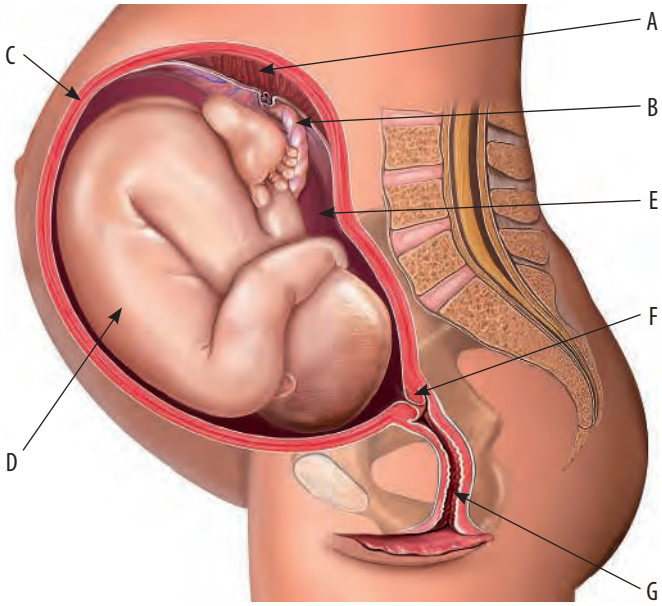
6 A paramecium is a single-celled organism that reproduces asexually.

(a) Make a list of the advantages and disadvantages of reproducing this way.
(b) Compare your list with that of your team. Discuss any differences.
(c) Find out more about paramecia and, as a team, write and perform a paramecium puppet play about their lives.

7 Match the contraceptives below with the way they prevent conception and their effectiveness.

Contraceptive	How it prevents conception	Effectiveness
Condom with spermicide	Prevents ova from developing	Extremely effective
Diaphragm without spermicide	The fallopian tubes or vas deferens are cut and sealed	Unreliable
Daily contraceptive pill	Keeps sperm and semen from entering the woman's vagina after ejaculation	Highly effective
Surgical: vasectomy and tubal ligation	Removal of male's penis from the vagina before ejaculation	Highly effective
Coitus interruptus	Prevents sperm cells from reaching the cervix	Moderately effective

8 Label the parts A–G in the diagram below:



9 (a) Using information from this chapter, make at least 20 'reproduction game cards' with a question on one side and the answer on the other side.
(b) Write a list of any questions that you still have about reproduction.
(c) Research at least three of these questions and summarise your findings on 'reproduction game cards' with the question on one side and the answer on the other.
(d) Design and construct a game board with at least one diagram idea from this chapter.
(e) Design a game that uses the cards and the board that you have made. Create a game rules book so that others will know how to play it. Create any other materials that you need for your game.
(f) Play each other's games.

10 Construct a working model that simulates some aspect of this chapter.

11 It has been said that we are currently in the midst of a biotechnological revolution with new technologies offering us many more reproductive options. Is this true for all parts of the world? Hold a discussion about the global impact of reproductive technologies.

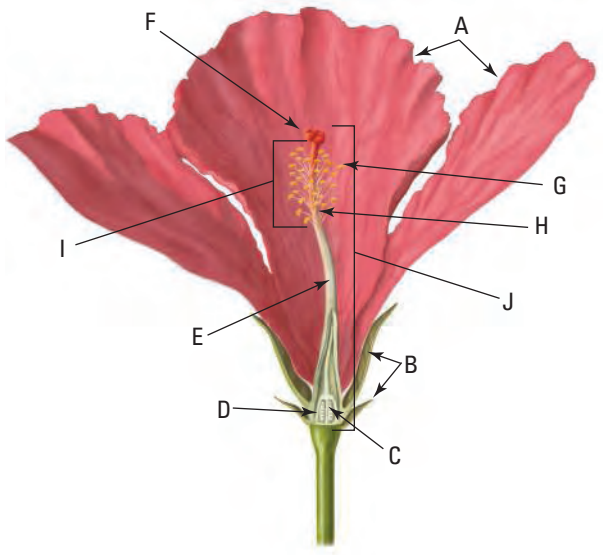
12 Construct a table naming the organs of the human male and female reproductive systems. For each organ, describe its structure and function.

13 Suggest how scientific knowledge about the life cycles of plants and animals can be used to develop regulations about importation of foodstuffs into Australia. Suggest reasons for these regulations.

14 Suggest how knowledge of the life cycle of a particular plant or animal may influence the practices of an agriculturalist.

15 On the basis of what you have learned in this section of your studies, suggest responses to the following questions:
(a) How can there be weeds in the garden if I didn't plant them there?
(b) Why don't twins always look the same?
(c) Why doesn't my caged bird lay eggs that can hatch into baby birds?

16 Write down in your workbook which letter in the following diagram corresponds to each of these terms.
ovules, sepals, filament, style, stigma, ovary, anther, petals, stamen, carpel



Heights (cm) of seedlings

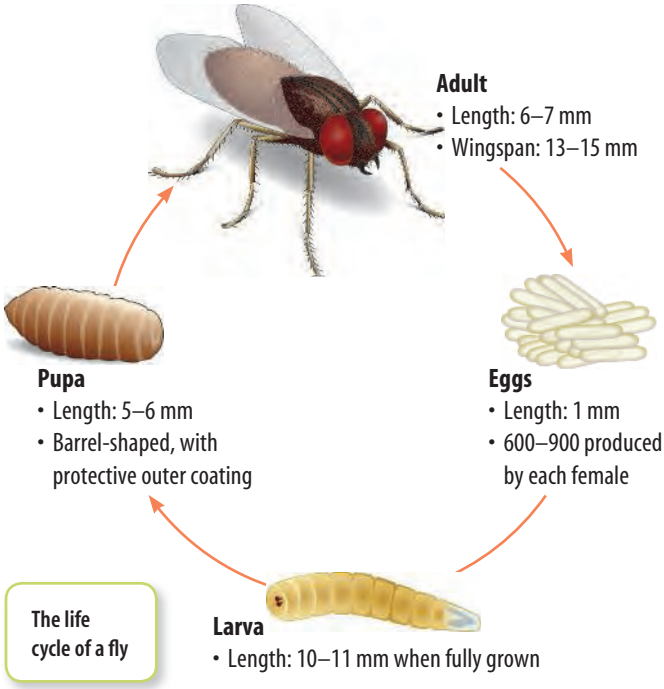
Position	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	Day 8	Day 9	Day 10
Fridge	5.0	5.5	6.0	6.2	6.6	7.0	7.3	7.5	7.7	8.0
Garage	5.0	5.6	6.2	6.6	7.0	7.3	7.6	7.9	8.4	8.8
Windowsill	5.0	6.0	6.7	7.5	8.0	8.5	9.0	9.6	10.2	10.6
Desk	5.0	5.8	6.3	7.0	7.5	8.0	8.5	9.1	9.6	10.0

17 Charlotte wanted to find out if temperature affects the growth of plants. She bought four seedlings. She put one seedling in the fridge and one in her garage (which has no windows so is dark and cooler than her house). She put the third seedling on the windowsill (in full sun) and the fourth seedling on her desk (out of the sun but in daylight). Charlotte measured the height of each seedling every day for 10 days. Her results are shown in the table above.
(a) Write an aim for Charlotte's experiment.
(b) Suggest three improvements to Charlotte's experiment.
(c) Graph Charlotte's results.
(d) Write a conclusion for this experiment.

Complete the following activities to produce a learning and thinking journal for this chapter.

18 Draw a diagram of an insect-pollinated flower and use descriptive labels to show what each part does.

19 (a) Use a table to show differences between the sizes, shapes and structures of a fly during each stage of its life cycle.
(b) Construct a graph to show the differences in length during the adult, egg, larval and pupal stages of the life cycle.
(c) Suggest possible survival advantages for the differences throughout the life cycle.

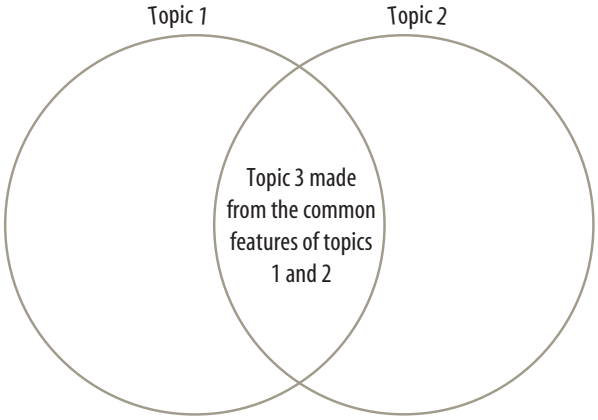


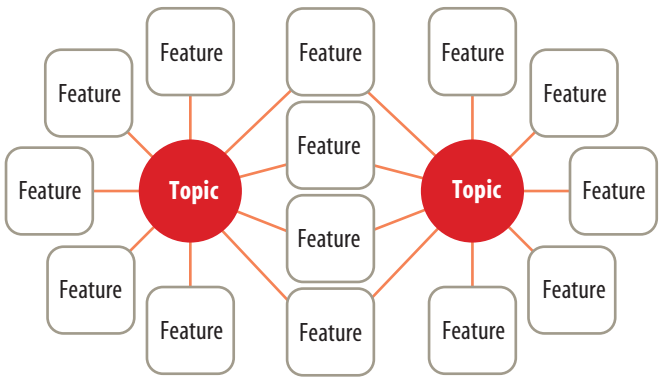
The life cycle of a fly

20 Use a matrix to compare flowers that are wind pollinated with those that are insect pollinated.

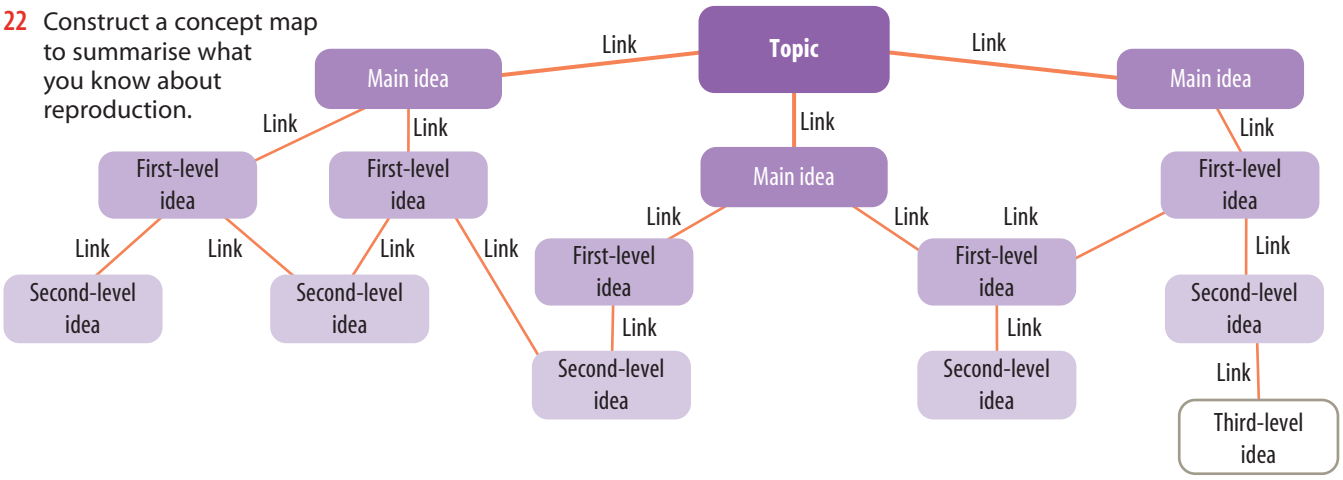
Topic	Feature A	Feature B	Feature C	Feature D	Feature E
1	✓		✓	✓	✓
2		✓			✓
3		✓		✓	✓
4			✓	✓	✓

21 Construct a Venn diagram or double bubble map to show the similarities and differences between:
(a) structures involved in plant and animal reproduction
(b) fertilisation in plant and animal reproduction
(c) fusion of gametes in plants and animal reproduction
(d) embryo development in plants and animals
(e) 'birth' in plants and animals.



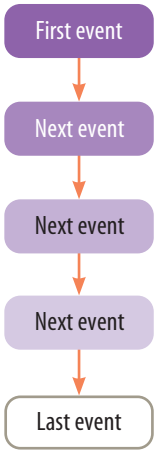


22 Construct a concept map to summarise what you know about reproduction.

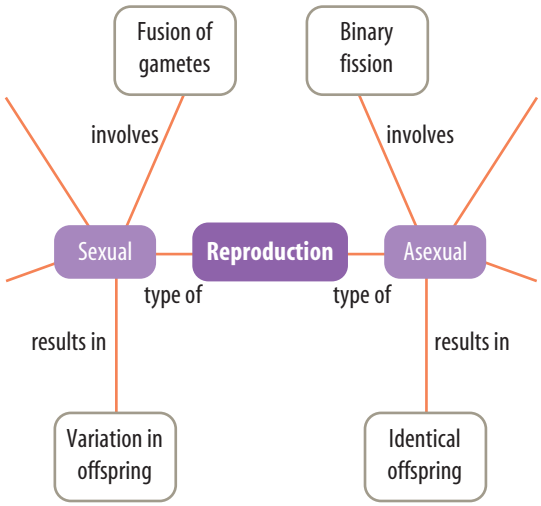


23 Increased knowledge and understanding of reproductive processes have led to the development of new reproductive technologies. Construct a PMI for issues associated with one of these technologies.

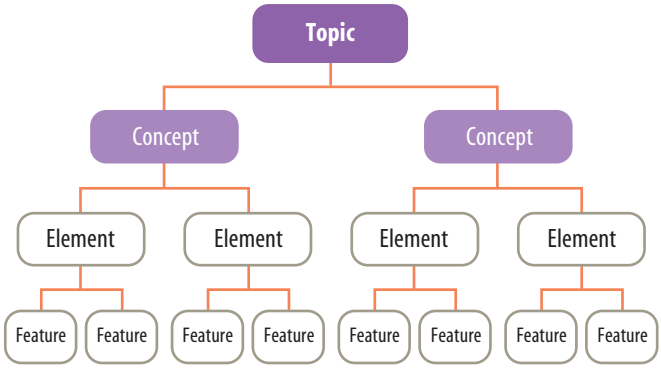
24 Use a flowchart to show an example of a life cycle of a flowering plant. Include pollination, fertilisation, development, seed dispersal and germination.



25 Use the figure below to help you construct a summary of the differences between sexual and asexual reproduction. What other features can you add?

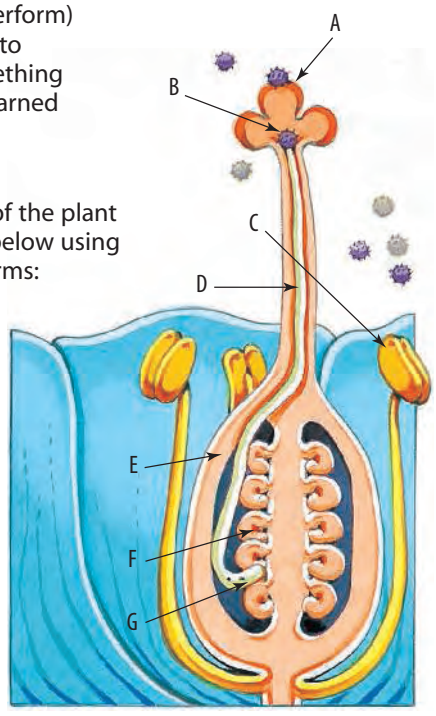


26 Use a tree map to show two sides of a discussion about plant reproduction and animal reproduction.



27 Make up (and perform) a song or poem to summarise something that you have learned in this chapter.

28 Label the parts of the plant in the diagram below using the following terms:
stigma,
male gamete,
pollen grain,
pollen tube,
stamen,
ovary,
ovule.



work sheets
5.8 Science literacy 1
5.9 Science literacy 2
5.10 Reproduction: Crossword

6 States of matter

All substances on Earth can be grouped as solids, liquids or gases. By comparing the properties of solids, liquids and gases, you can begin to answer questions such as 'what are substances made of?'. This question has fascinated people for thousands of years, and scientists are still looking for more answers to that same question.

Water exists as a liquid in oceans, lakes and rivers, as solid icebergs in the oceans, and as water vapour in the air. Without it plants and animals could not exist. Each of the forms of water has its own different properties and uses.

THINK ABOUT PARTICLES

- Why does ice melt?
- What is dry ice and why doesn't it melt?
- Why do car windows fog up in winter?
- What are clouds made of?
- What is the difference between hail and snow?
- Why are there small gaps in railway lines?

OVERARCHING IDEAS

- Patterns, order and organisation
- Form and function
- Stability and change
- Matter and energy
- Systems

SCIENCE UNDERSTANDING

The properties of the different states of matter can be explained in terms of the motion and arrangement of particles.

Elaborations

Modelling the arrangement of particles in solids, liquids and gases

Linking the energy of particles to temperature changes

YOUR QUEST

Bathroom science

1 Why does the mirror fog up in the bathroom after someone has had a hot shower?

2 On really hot days, you may have a cold shower to cool down. Does the bathroom mirror fog up when you do this?

3 Some showers have shower curtains rather than glass shower screens. When people have warm showers, the curtain tends to move in towards the person in the shower and stick to them. Give possible explanations for why this happens.

4 When you have a hot shower, the bathroom fills with water vapour. Is this water vapour a gas or a liquid or both? Explain your reasoning.

What is water vapour — a gas, a liquid, or both?

5 How hot does water have to be before it can burn you?

6 Does water vapour always rise?

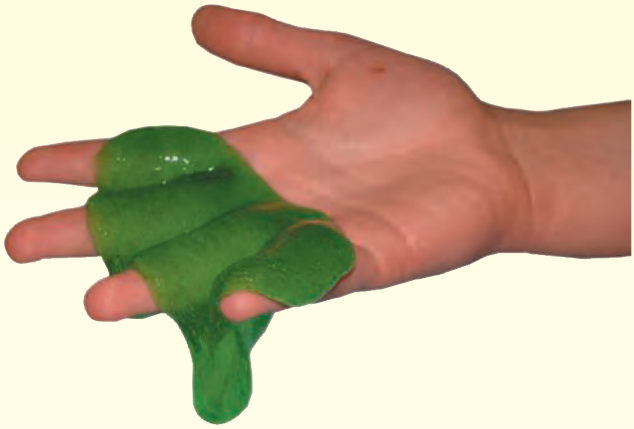
7 Are water vapour and steam the same thing?

8 Can you see water vapour or steam?

Ranking substances

1 Working in small groups, rank the following substances in order from most solid-like to most liquid-like to most gas-like.

- a brick
- jelly
- sugar
- Vegemite®
- orange cordial
- steam
- plasticine
- tomato sauce
- air
- green slime



Green slime — is it solid or liquid? How do you know?

2 Compare your rankings with those of other groups. Comment on any differences between the rankings.

3 Which substances were most difficult to classify as solid, liquid or gas? Explain why they were difficult to classify.

4 Draw a three-column table, like the one below, and separate the substances into three categories — solid, liquid or gas.

Solid	Liquid	Gas

States of matter

Every substance in the universe is made up of matter that can exist in a number of different forms called states. Almost all matter on Earth exists in three different states: **solid**, **liquid** and **gas**. These states of matter have very different **properties**. That is, they are different in the way they behave and appear.

Iodine diffusing in a fume cupboard

INQUIRY: INVESTIGATION 6.1

Comparing solids, liquids and gases

KEY INQUIRY SKILLS:

- questioning and predicting
- processing and analysing data and information

Equipment:

ice cube *plastic syringe* *spatula*
balloon *beaker of water*

- Pick up an ice cube and place it on the bench. Using a spatula, try to squash it or compress it to make it smaller.
- Take the beaker of water and draw a small amount up into the syringe. Place your finger over the opening at the end of the syringe and press down on the plunger.
- Partially inflate a balloon with air and hold the opening tightly closed. Try to squeeze the balloon.
- Release your hold on the opening of the balloon.

DISCUSS AND EXPLAIN

1 Copy the table below and use your observations to complete it.

Properties of solids, liquids and gases

Substance	State of substance	Can the shape be changed easily?	Does it take up space?	Can it be compressed?
Ice	Solid			
Water	Liquid			
Air	Gas			

2 Where did the air in the balloon go when you released the opening?

Solids

Solids such as ice have a very definite shape that cannot easily be changed. They take up a fixed amount of space and are generally not able to be compressed.

Most solids cannot be poured, but there are some, such as salt, sand and sugar, that can be poured.

Liquids

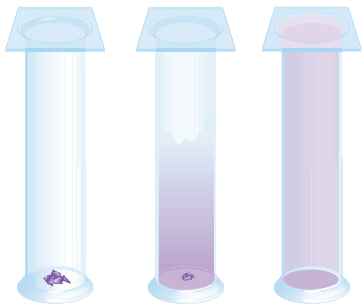
Water is a liquid and its shape changes to that of the container in which it is kept. Like solids, liquids take up a fixed amount of space.

If a liquid is poured into a glass, it will take up the shape of the glass. If you continue to pour, it will eventually overflow onto the bench or floor.

Gases

Gases spread out and will not stay in a container unless it has a lid. Gases move around, taking up all of the available space. This movement is called **diffusion**. In the illustration below, iodine gas is being formed and is spreading, or diffusing, throughout the gas jar.

Gases, unlike solids and liquids, can be compressed, making them take up less space. An inflated balloon can be compressed by squeezing it.



The purple iodine gas diffuses, taking up all of the available space. What will happen to the gas if the lid is removed?

WHAT DOES IT MEAN?

The word *diffusion* comes from the Latin word ***diffusio***, meaning 'spreading out'.

MEASURING MATTER

The amount of matter in a substance, whether solid, liquid or gas, is called **mass**. The most commonly used unit of mass is the kilogram (kg), which is equal to 100 grams (g). Mass is measured with an electronic scale or beam balance.

The amount of space taken up by a substance is called its **volume**. The volume of solids is usually measured in cubic metres (m^3) or cubic centimetres (cm^3). The volume of fluids is measured in millilitres (mL). One millilitre occupies the same volume as 1 cm^3. A **fluid** is a substance that can flow. All liquids and gases are fluids.

INQUIRY: INVESTIGATION 6.2

Measuring the volume of an irregular-shaped solid

KEY INQUIRY SKILLS:

- processing and analysing data and information
- evaluating

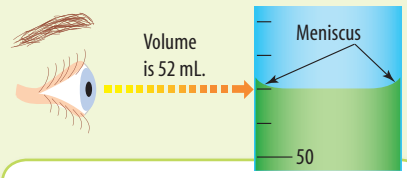
Equipment:

100 mL beaker

100 mL measuring cylinder

stone or pebble that will fit into the measuring cylinder

- Half-fill (approximately) a 100 mL beaker with water.
- Carefully pour the water into the measuring cylinder.
- Read and record the volume of water in the measuring cylinder using the technique shown in the diagram above right.
- Carefully place the pebble into the measuring cylinder. Take care not to spill any water.
- Read and record the new volume.



Reading the volume of a liquid in a measuring cylinder. The curved upper surface is called the meniscus. Your eye should be level with the flat part in the centre of the meniscus.

DISCUSS AND EXPLAIN

1 What was the volume of the solid in mL?

2 What was the volume of the solid in cm^3?

3 Suggest another way of measuring the volume of the solid object.

UNDERSTANDING AND INQUIRING

REMEMBER

1 List as many as you can remember of the solids, liquids and gases you came into contact with before leaving for school today. Organise them into a table under three headings, 'Solids', 'Liquids' and 'Gases', or into a cluster, mind or concept map.

2 (a) Write down three properties that most solids have in common.
(b) Would liquids have the same three properties? If not, what differences might be expected?

3 Which properties of gases are different from those of liquids?

4 What is the unit used for measuring small volumes such as that of liquid medicines? How could you measure such a volume?

THINK

5 Both steel and chalk are solids. What properties of steel make it more useful than chalk for building bridges?

6 Are plasticine and playdough solids or liquids? Explain why.

7 What is diffusion? Give two examples of this occurring around your house.

8 Is it possible for a solid to behave like a fluid? Explain your answer.

9 At the petrol station, the safety sign asks for the car engine to be switched off before you fill the petrol tank. Why is this necessary?

IMAGINE

10 You are designing a new type of armchair. It needs to be comfortable and capable of fitting in different positions or spaces around the room.
(a) What properties would you want in the chair?
(b) Would you need to develop a new material to match these properties? If so, would it be a solid or a liquid, or perhaps a combination of states?

INVESTIGATE

11 Different liquids pour or flow in different ways. Test this by pouring honey, shampoo, cooking oil and water from one container to another. Time how long they take to pour. Make sure it is a fair test. Record the results in a table and write a conclusion based on your observations and results.

Changing states

Water is the only substance on Earth that exists naturally in three different states at normal temperatures. It is in the oceans, in the polar ice and in the air as water vapour. Water is constantly moving and changing states. You can also observe water changing states in the kitchen. To change the state of any substance, including water, it must be heated or cooled.

Unfortunately, the ice sculpture in the photograph won't last for very long. Even as the sculptor works, it is melting as heat moves into it from the warmer air around it.

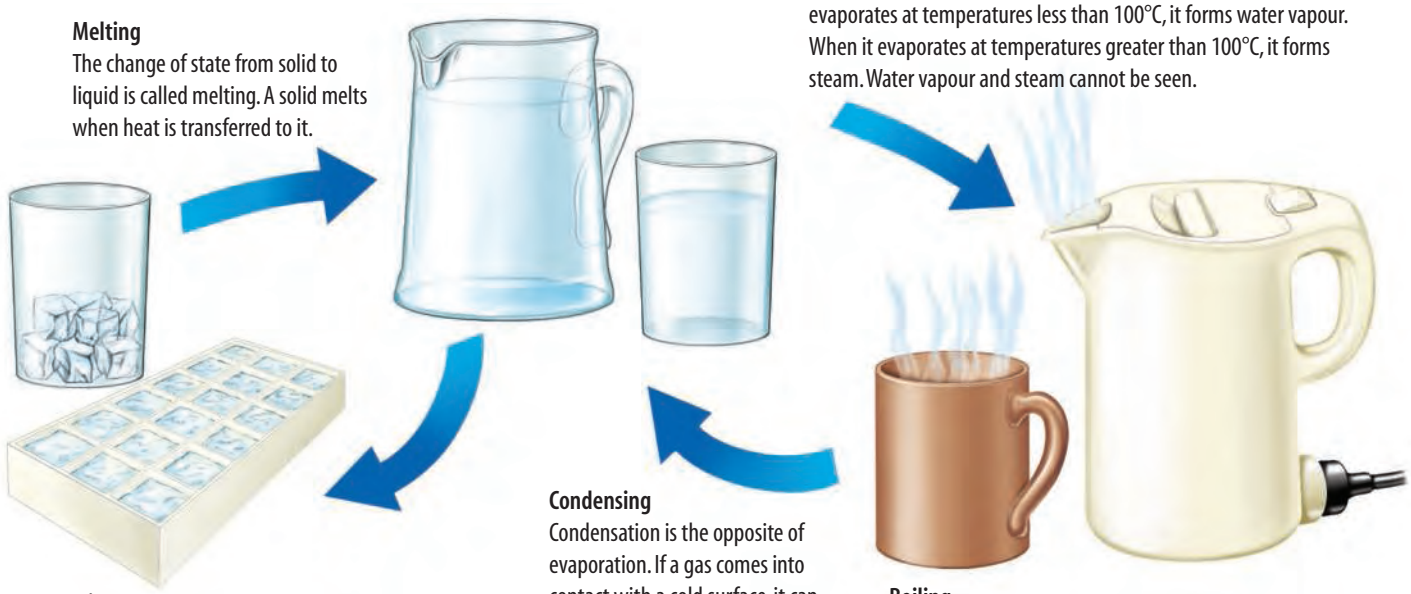
Melting point and boiling point

The state of matter of any substance depends on its temperature. The temperature at which a substance changes from a solid into a liquid (melts) is called its **melting point**. A liquid changes into a solid (freezes) at the same temperature. Water has a melting point of 0°C, so to melt ice it has to be heated to a temperature of 0°C. To freeze water it has to be cooled to a temperature of 0°C.

The **boiling point** is the temperature at which a substance boils. At this temperature, the substance changes from liquid into gas (evaporates) quickly. At the same temperature, a gas changes into a liquid (condenses). The boiling point of water is 100°C. The melting and boiling points of some common substances are shown in the table on the next page.

Melting
The change of state from solid to liquid is called melting. A solid melts when heat is transferred to it.

Evaporating
Evaporation occurs when a liquid changes to a gas. When water evaporates at temperatures less than 100°C, it forms water vapour. When it evaporates at temperatures greater than 100°C, it forms steam. Water vapour and steam cannot be seen.



Freezing
The change of state from a liquid to a solid is called freezing. A liquid turns into a solid when heat is transferred away from it.

Condensing
Condensation is the opposite of evaporation. If a gas comes into contact with a cold surface, it can turn into a liquid.

Boiling
During boiling, the change from liquid to gas (evaporation) happens quickly. The change is so fast that bubbles form in the liquid as the gas rises through it and escapes. During boiling, the entire substance is heated. A liquid remains at its boiling point until it has all turned into a gas.

Melting and boiling points of some common substances at sea level

Substance	Melting point (°C)	Boiling point (°C)
Water	0	100
Table salt	804	1413
Iron	1535	2750
Aluminium	660	1800
Oxygen	–218	–183
Nitrogen	–210	–196

HOW ABOUT THAT!

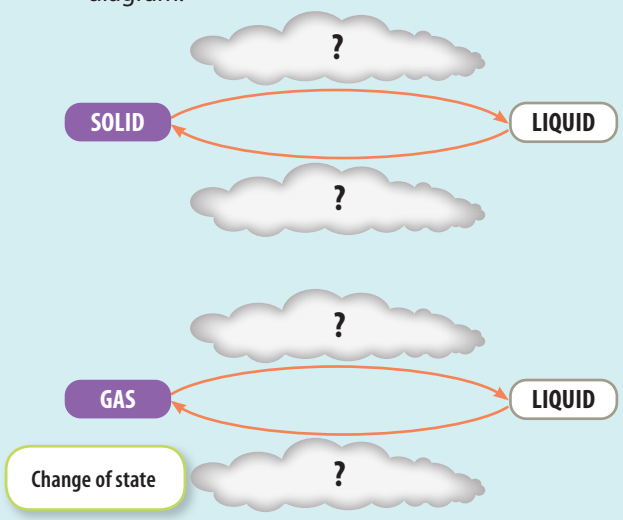
Melting and boiling points change with the height above sea level. This is because the air gets thinner as you move away from the Earth's surface. If you were climbing Mount Everest and made a cup of coffee near its peak, you would find that the water boiled at about 70°C instead of 100°C.

At a concert or special event, you may have seen a thick 'smoke' used for effect. This smoke is produced when solid carbon dioxide, called 'dry ice', changes state from a solid directly to a gas. This very unusual change of state is called **sublimation**. The 'smoke' is actually tiny droplets of water that condense from the air as the cold dry ice sublimes. Dry ice sublimes at a temperature of –78.5°C. Iodine also sublimes. Diamonds sublime at a temperature of 3550°C.

UNDERSTANDING AND INQUIRING

REMEMBER

1 (a) Copy and complete the diagram below, labelling the changes of state.
(b) Use a labelled arrow to add 'sublimation' to your diagram.



2 What is the name given to the change of state from liquid water to steam? What happens to make this occur?

3 What happens to liquid water when it is cooled below 0°C? Has heat moved into or out of the liquid?

4 When water evaporates it can change state from liquid to a gas in the form of either steam or water vapour. Explain the difference between steam and water vapour.

ANALYSE

Use the table above to answer these questions.

5 At what temperature would you expect table salt to melt? At what temperature would it freeze?

6 Would you expect aluminium to be found as a solid, liquid or gas at:
(a) 200°C? (b) 680°C? (c) 1900°C?

7 Which substance — oxygen or nitrogen — would freeze first if the temperature were gradually lowered?

THINK

8 Explain the difference between evaporation and boiling.

9 Why is dry ice useful to produce a 'smoke' effect? What other uses are there for dry ice?

INVESTIGATE

10 Dry the outside of a very cold can of soft drink or carton of milk with a towel and allow it to stand on a table or bench for about ten minutes. (Don't forget to put it back in the fridge afterwards.)
(a) What change occurred on the outside of the can?
(b) Where did the water come from?
(c) What change of state has occurred?

work sheet 6.1 Changing the boiling point of water

The state of the weather

Two-thirds of the Earth's surface is covered with water. But not all of it is in liquid form. About 2 per cent of the Earth's water exists as ice, mostly in Antarctica. A much smaller amount exists as water vapour in the atmosphere. Energy from the sun melts ice and causes liquid water to evaporate. Low temperatures in the atmosphere and on the surface freeze liquid water and cause water vapour in the air to condense.

Water and the weather

Rain, hail, snow and sleet are all types of **precipitation**. Precipitation is falling water, whether in solid or liquid form. The type of precipitation depends mostly on the temperature in the clouds and the air around them.

RAIN

Rain forms when water vapour condenses in cold air, forming tiny droplets of water. These droplets are so small that they are kept up by moving air, forming clouds.

As the droplets join together they become too heavy to remain in the air. They fall to the ground as rain. When air currents are low, very tiny drops of rain may fall as a fine mist known as drizzle.

Clouds are formed by tiny droplets of water, kept up by air currents.

HAIL

If drops of rain freeze, they may form hailstones. Air currents within clouds move raindrops from the bottom of the cloud upwards to the top of the cloud. The top of the cloud is much colder than the bottom of the cloud. The rising raindrops freeze very quickly. The frozen raindrops fall back towards the bottom of the cloud. If the air currents are strong enough, the rain drops rise again, adding a new layer of ice around the frozen raindrop. It falls again, then rises again to form another layer of ice. This can happen over and over again, each time adding a new layer of ice. When the ice has built up many layers, it gets heavy enough to fall to the ground as a hailstone. Hailstones can be extremely large and cause extensive damage.

In summer, warm rising air helps to keep the hailstones in the clouds for longer, forming even more layers than usual. These hailstones can reach masses of over one kilogram before they fall.

SNOW

Snow consists of crystals of ice that have frozen slowly in clouds. Many different shapes and patterns can be found in snowflakes. The shape and size depend on how cold the cloud is, its height and the amount of water vapour in the cloud. Crystals of ice form when clouds have temperatures of less than –20°C. The crystals join together and fall. As they fall, they become wet with moisture but then refreeze as snowflakes.

If the air between the cloud and the ground is colder than 0°C, the snowflakes fall as very powdery, dry snow. If the air is warmer, the ice crystals melt and fall as rain or sleet.

Snowflakes form many different shapes and patterns but always have six 'sides'.

SLEET

Sleet is snow that is melting or raindrops that are not completely frozen. Sleet forms when the air between the clouds and the ground is warm enough to melt ice.

eBookplus eLesson

Understanding a weather forecast

Interpret weather maps for yourself and find out how isobars indicate air pressure.

eles-0161

PREDICTING THE WEATHER

The scientists who predict, or forecast, the weather are **meteorologists**. Meteorology is the study of the atmosphere and includes the observation, explanation and prediction of weather and climate. Numerous observations of temperature, precipitation, wind speed, air pressure, humidity and more need to be made in order to make weather forecasts. Humidity is a measure of the amount of water vapour in the air.

Before the first weather balloon was launched in 1882, observations with instruments such as thermometers, barometers and rain gauges could be made only on land or ships. Not long after the invention of the first 'flying machine' in 1903, weather instruments were attached to the wings of planes, allowing them to be taken higher in the atmosphere.

As new technology becomes available, the number and quality of observations improve. Improved weather balloons, together with radar, satellite images and computer modelling allow meteorologists to make predictions further ahead and more accurately than ever before.

A meteorologist releases a weather balloon in Antarctica.

UNDERSTANDING AND INQUIRING

REMEMBER

1 What are clouds made of?

2 Explain using words or a labelled diagram how hailstones are formed.

3 How can hailstones get as large as those in the photograph on the previous page?

4 Explain the difference between snow and sleet.

5 What is meteorology concerned with?

6 What is humidity a measure of?

THINK

7 Suggest why extra-large hailstones are more common in summer than winter.

8 Ski resort operators have suffered a shortage of snow in recent years. What conditions would they be looking for to predict coming snowfalls?

INVESTIGATE

9 Make a list of leisure activities that rely on predictions about the weather.

10 In which occupations does each of the following types of weather prevent activity?
   (a) Extreme heat
   (b) Heavy rain
   (c) Thunderstorms

11 Record the predictions of the maximum temperature of your nearest capital city made in a 7-day forecast. For each day of the 7-day period, also record the maximum temperature predicted on the day before.

These forecasts can be found online on the Bureau of Meteorology website (www.bom.gov.au), on the TV news or in daily newspapers.

Then record the actual maximum temperature for each day as reported on the evening news or the following day's newspaper. Use a table like the one below to record your data.

Daily maximum temperatures (°C)

	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
Prediction in 7-day forecast							
Forecast the day before							
Actual maximum temperature							

   (a) How does the accuracy of the 7-day forecast compare with the accuracy of the previous day's forecast?
   (b) State your opinion about the accuracy of the forecast made on the day before.
   (c) Apart from temperature, what other aspects of the weather forecast are reported in newspapers and the TV news?

12 Find out what relative humidity is and with which instrument it is measured.

13 Research and report on what a hydrologist does.

14 Find out the difference between weather and climate.

6.4 OVERARCHING IDEA

Matter and energy: The particle model

How do you explain why ice has properties that are different from those of water or steam? Scientists use a model to explain the different properties of solids, liquids and gases. This model is called the **particle model**.

WHAT DOES IT MEAN?

The word *particle* comes from the Latin word ***particula***, meaning 'part'.

According to the particle model:

- all substances are made up of tiny particles
- the particles are attracted towards other surrounding particles
- the particles are always moving
- the hotter the substance is, the faster the particles move.

Particles in a solid

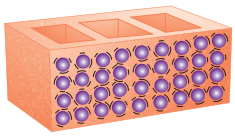
In solids the particles are very close together, so they cannot be compressed. The attraction between neighbouring particles in a solid is usually strong. Because there are such strong bonds between the particles, solids usually have a fixed shape and a constant volume. The particles in solids cannot move freely. They vibrate in a fixed position.

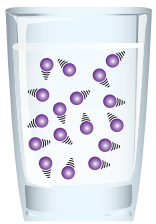
Particles in a liquid

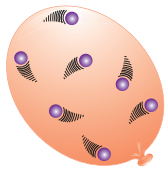
In liquids the particles are held together by attraction, but the bonds between them are not as strong as those in solids. The weak particle attraction allows the particles to roll over each other. But they can't 'escape' from each other. For this reason, liquids have a fixed volume but the rolling motion of the particles allows liquids to take up the shape of their container. As in solids, the particles in liquids are still very close together; liquids cannot be compressed into smaller spaces.

Particles in a gas

The particles in a gas have much more energy than those in solids and liquids. They are in constant motion. The attraction between the particles in a gas is so weak that they are able to move freely in all directions. They spread out to take up any space that is available. This means that gases have no fixed shape or volume. Because of the large spaces between particles, gases can be compressed.

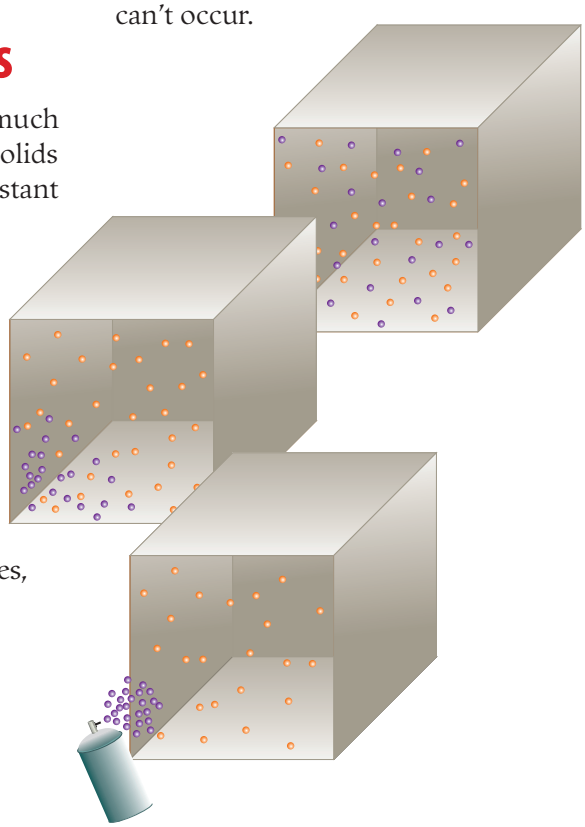






Spreading out

The spreading of one substance through another is called diffusion. This can happen only when the particles of one substance can spread through the particles of another substance. Diffusion is possible in liquids and gases because the particles are moving around. You would expect diffusion to happen faster in gases than in liquids because the particles are moving faster. Particles in a solid are vibrating in a fixed position, so diffusion can't occur.



The particles of two gases spread through each other until they are evenly mixed.

The particle model and balloons

The particle model can be used to explain what happens to a balloon when you inflate it. Particles of air inside the balloon

are constantly moving in all directions. They are colliding with each other and with the inside wall of the balloon. But the wall is not rigid. It can stretch as more particles are added. The balloon **expands** until it can't stretch any more. When you let some of the air out of the balloon, fewer particles are colliding with the inside wall of the balloon. It gets smaller or **contracts**.

INQUIRY: INVESTIGATION 6.3

Investigating diffusion

KEY INQUIRY SKILL:

- processing and analysing data and information

Equipment:

250 mL beaker
water
food colouring
eye-dropper
fragrant spray

- Place a drop of food colouring into a beaker of water and record your observations for several minutes, making sure the beaker is not moved.
- Release some of the fragrant spray in one corner of the classroom. Move away and observe by smell.

DISCUSS AND EXPLAIN

1 Draw a diagram to show the movement of the food colouring through the water.
2 Explain how the fragrant spray moved through the air.
3 This investigation shows diffusion in a liquid (water) and in a gas (air).
   (a) In which state does diffusion occur faster — liquid or gas?
   (b) Why do you think this is so?

Heating solids, liquids and gases

When a substance is heated, the particles gain energy, move faster, become further apart and take up more space. The substance will expand.

The tyres on a moving car get quite hot. This makes the air inside expand. This may even cause a blowout in extreme circumstances. Gases usually expand much more than solids or liquids. Gases expand easily because the particles are spread out and are not attracted to each other strongly. Solids, liquids and gases contract when they are cooled again because the particles lose energy, slow down, need less space to move in and become more strongly attracted to each other.

Hot air balloons rise when the air inside them expands. The particles in the heated air move faster and take up more space. This makes each cubic centimetre of air inside the balloon lighter than each cubic centimetre of air outside the balloon, so the air inside the balloon rises, taking the balloon with it.

These hot air balloons rise when the air inside them expands. How do they get back down to the ground?

Architects and engineers allow for expansion and contraction of materials when designing bridges and buildings. Bridges have gaps at each end of large sections so that in hot weather, when the metal and concrete expand, they will not buckle. Railway lines also have gaps to allow for expansion in hot weather. Electrical wires are hung from poles loosely so that, when the weather cools, they will not become too tight and break as they contract. The amount by which each structure will expand or contract depends on the material it is made from; so when choosing a material for a special purpose, it is important to find out how much that material will expand or contract. The table on page 230 shows how much some commonly used materials expand when the temperature increases by 10°C.

Thermometers

Liquids expand more than solids. This property makes them useful in thermometers. Most thermometers consist of thin tubes and a bulb that contains a liquid. As the temperature rises, the liquid expands, moving up the tube. In a thermometer, the tube is sealed at the top.

The two most commonly used liquids for thermometers are mercury and alcohol. Mercury has a low freezing point (–39°C) and a high boiling point (357°C). Alcohol, however, is much more useful in very cold conditions because it does not freeze until the temperature drops to –117°C. On the other hand, alcohol boils at 79°C so it cannot be used for measuring higher temperatures.

The temperature of the human body ranges between 34°C and 42°C; it is normally about 37°C. A clinical thermometer is especially designed to measure human body temperature.

INQUIRY: INVESTIGATION 6.4

Explaining gases

KEY INQUIRY SKILLS:

- planning and conducting
- processing and analysing data and information

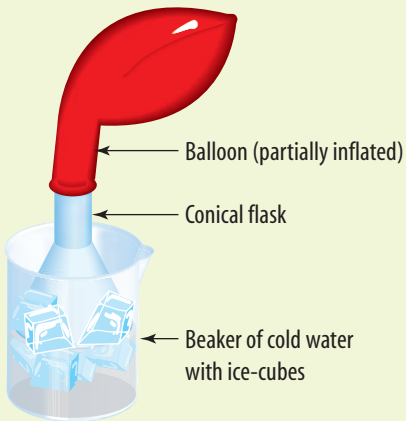
Equipment:

balloon *piece of string* *ruler* *small conical flask*
2 large beakers *ice cubes* *hot and cold water*

- Copy the table below into your workbook.
- Inflate the balloon to its maximum size. Then deflate it. This makes it easier to stretch.
- Inflate the balloon again, to a size slightly larger than an orange. Fit the neck of the balloon over the conical flask to seal it.
- Wrap the string once around the widest part of the balloon to find its circumference. With a ruler, measure the length of the string that encircled the balloon.
- Record the measurement in your table.
- Half-fill one of the beakers with ice cubes and a small amount of cold water.
- Place the conical flask in the ice-water beaker and observe the balloon. After a few minutes, use the string to measure the circumference of the balloon again. Record your measurement in your table.
- Put some hot water into the second beaker. Take the conical flask from the ice-water and place it into the hot water. Leave for a few minutes, then measure and record the balloon's circumference.

DISCUSS AND EXPLAIN

1 Was any air added to or removed from the balloon after it was placed over the conical flask?
2 After being in ice-water and hot water, were there any changes in the size of the balloon?
3 Using the particle model, try to explain what might have made the balloon contract and expand.
4 What quantity was varied or changed in this experiment? What things were kept the same?

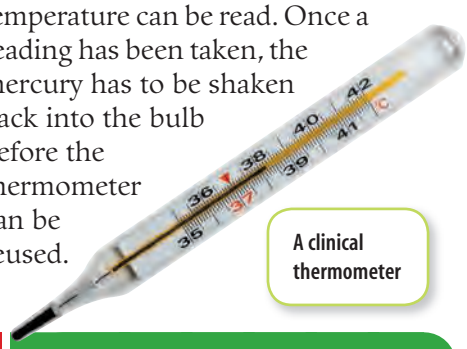


What happens to the air in the balloon when it gets cold?

Effect of temperature on air

Temperature of surroundings	Circumference of balloon (cm)
Room temperature	
Cold (ice-water)	
Hot (hot water)	

Look at the photograph below. The tube narrows near the bulb. Once the mercury has expanded, this narrowing prevents the mercury contracting and moving back into the bulb before the temperature can be read. Once a reading has been taken, the mercury has to be shaken back into the bulb before the thermometer can be reused.



A clinical thermometer

INQUIRY: INVESTIGATION 6.5

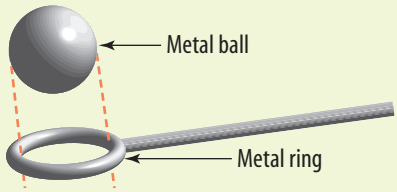
Expansion of solids

KEY INQUIRY SKILL:

- processing and analysing data and information

Equipment:

metal ball and ring set
Bunsen burner and heatproof mat
tongs



A ball and ring set

- Try to put the ball through the ring.
- Use the Bunsen burner to heat the ring and use tongs to try to put the ball through it. Take care not to touch the hot metal.
- Let the ring cool and try to put the ball through the ring again.

DISCUSS AND EXPLAIN

1 What has happened to change the size of the ring?
2 Use the particle model to explain the change that took place in the ring.

Gases under pressure

The fire extinguishers used to put out electrical fires are filled with carbon dioxide gas.

Carbon dioxide can be used in this way only because huge amounts of it can be compressed, or squeezed, into a container. Gases can be compressed because there is a lot of space between the particles. Gases compressed into cylinders are used for barbecues, scuba diving, natural gas in cars and aerosol cans.

1. Gases, including carbon dioxide, have lots of space between their particles.
2. The carbon dioxide is compressed into the cylinder. The particles are squashed closer together.
3. The carbon dioxide particles are now under increased pressure. This means that the particles in the gas collide frequently with the walls of the cylinder. The particles push outwards on the walls of the cylinder. The particles are trying to escape, but are held in by the container.
4. When the nozzle is opened, the pressure forces the carbon dioxide gas out very quickly through the opening.
5. The particles of gas quickly spread out over the fire. The gas smothers the fire, stopping oxygen from the air getting to it. Fires cannot burn without oxygen, so the fire goes out.

eBookplus eLesson

Under pressure

Learn about the factors that affect the pressure of a gas and how compressed gases are used to make fire extinguishers and aerosol cans

eles-0058

INQUIRY: INVESTIGATION 6.6

Expansion of liquids

KEY INQUIRY SKILLS:

- planning and conducting
- processing and analysing data and information

Equipment:

500 mL conical flask *narrow glass tube*
rubber stopper with one hole to fit the tube
Bunsen burner, heatproof mat and matches
tripod and gauze mat *food colouring*
eye-dropper *marking pen*

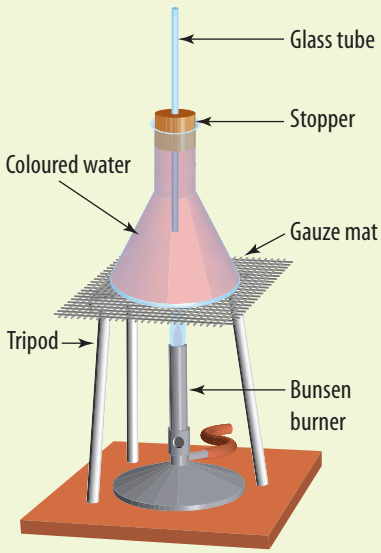
- Use an eye-dropper to place two or three drops of food colouring in the flask and fill it with water right to the top.
- Place the stopper in the flask with the tube fitted. Some coloured water should rise into the glass tube. Mark the level of the liquid in the tube with the marking pen.
- Place the flask on the tripod and gauze mat, light the Bunsen burner and gently heat the liquid.
- After about five minutes of heating, turn off the Bunsen burner and watch what happens to the level of the liquid in the glass tube.

DISCUSS AND EXPLAIN

1 What happens to the level of the liquid while it is being heated?

2 What happens to the level of the liquid while it is cooling down?

3 Use the particle model to explain why liquids expand.



Investigating the expansion of liquids

UNDERSTANDING AND INQUIRING

REMEMBER

1 List the four main ideas of the particle model.
2 What is diffusion?
3 Explain why solids generally expand when they are heated.
4 The following statements are incorrect. Rewrite them correctly.
   (a) Heating a liquid might make its particles stick closer together.
   (b) Solids have a definite shape because their particles are free to move around.
   (c) You can compress a gas because its particles are close together.
5 When a substance is heated, its temperature increases. What other change might be observed?
6 (a) Describe what change you expect to see when hot metal objects are cooling.
   (b) Why does this happen? Explain, using the particle model.
7 List two examples of structures that contain gaps to prevent them from buckling in hot weather.
8 Give one reason why overhead electric power lines are not hung tightly.
9 Use a double bubble map to show the similarities and differences between alcohol and mercury thermometers.
10 What happens to the particles in carbon dioxide gas when they are compressed into a fire extinguisher?

ANALYSE

Expansion of materials

Substance	Expansion (mm) of 100 m length when temperature increases by 10°C
Steel	11
Iron	12
Platinum	9
Brass	19
Concrete	11
Glass — soda	9
Glass — Pyrex	3
Lead	29
Tin	21
Aluminium	23
Bronze	18

Use the table above to answer the following questions.

11 If a steel rod of 10 metres in length were heated so that its temperature rose by 10°C, how long would the rod become?
12 Why is Pyrex, rather than soda glass, used in cooking glassware such as casserole dishes and saucepans?
13 Concrete is often reinforced with steel bars or mesh to make it stronger. Why is steel a better choice than another metal, such as aluminium or lead?

THINK

14 Describe an everyday example of diffusion occurring.
15 Use the particle model to explain why:
   (a) perfume can be smelled from a few metres away
   (b) steam can be compressed while ice cannot
   (c) water vapour takes up more space than the same amount of liquid
   (d) solids do not mix well, but gases and liquids mix easily in most cases.
16 Use the particle model to predict what happens to the length and width of a solid substance when it is heated (without melting).
17 A jar with the lid jammed on tightly can be hard to open. If hot water is run over the lid, it becomes easier to open. Why?
18 Hot air balloons have a gas heater connected to them.
   (a) What happens to the particles inside the balloon when the heater is turned on?
   (b) Explain why the balloon rises.
19 Under what conditions might you use an alcohol thermometer rather than a mercury thermometer?
20 Use the particle model to explain what keeps car or bicycle tyres in the right shape when they are pumped up to a high air pressure.

INVESTIGATE

21 The mercury thermometer was invented by a German named Gabriel Fahrenheit (1686–1736). A different set of markings is used to scale Fahrenheit thermometers. At what temperatures does water boil and freeze on this scale?
22 Why do icebergs float in the Arctic and Antarctic waters? Do you think there is much of the iceberg under the water, or is it mostly above? How could you test out your hypothesis? Design a suitable experiment.
23 Use the internet to investigate the safe storage of gas cylinders. Make a list of requirements and state the reason for each of them.
24 eBookplus Use the **States of matter in containers** weblink in your eBookPLUS to watch how solids, liquids and gases behave differently within a container.
25 All materials expand when heated and contract when cooled, right? Use the **Mystery expansion** weblink in your eBookPLUS to learn about a substance that breaks all the rules.

work sheets
6.2 Fire! Fire!
6.3 Expansion of liquids
6.4 Particles in our lives

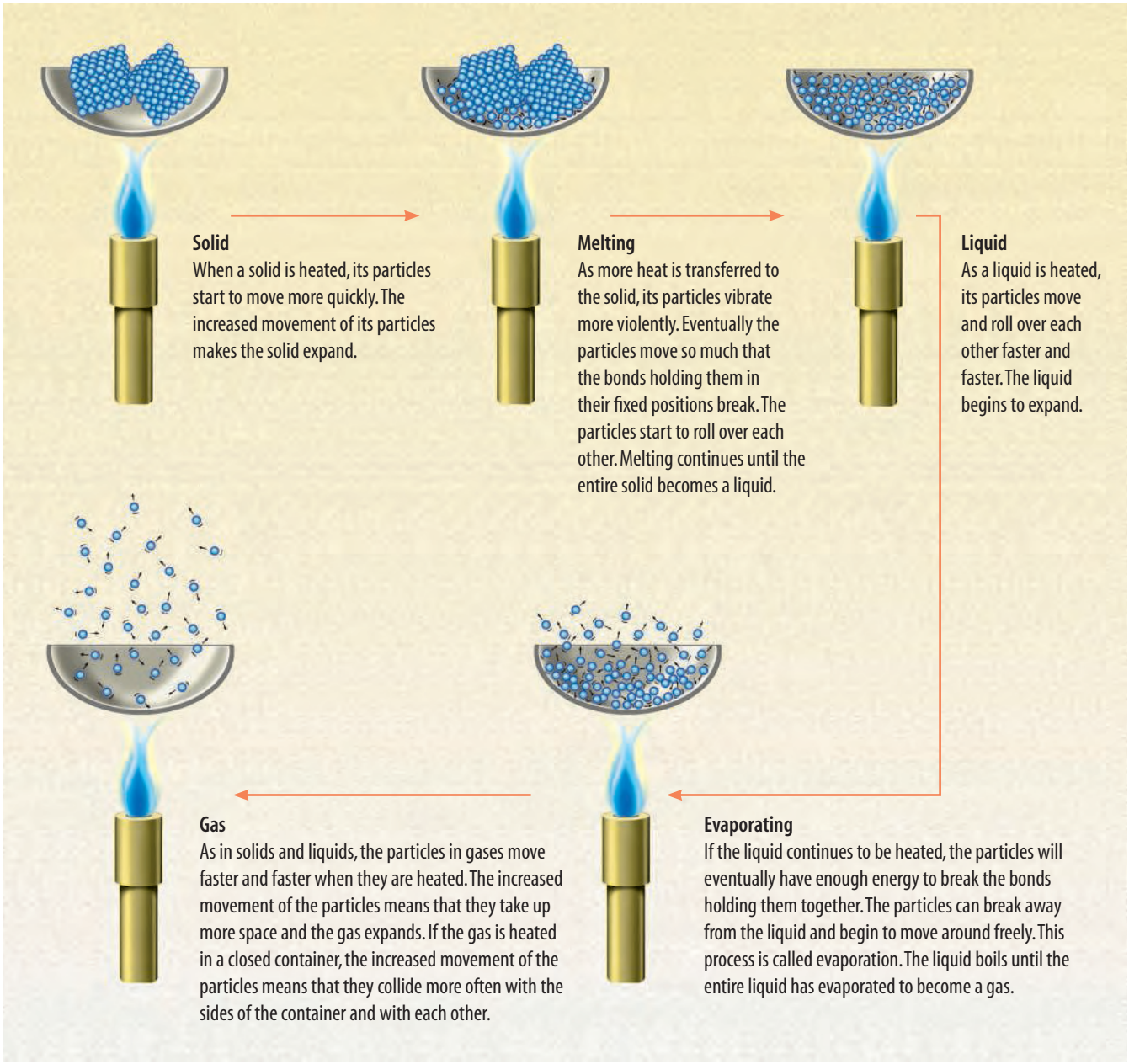
6.5 SCIENCE UNDERSTANDING

Energy in, energy out

A change of state involves the heating or cooling of matter. As a substance is heated, energy is transferred to it. When a substance cools, energy moves away from it to another substance or to the environment. The change in energy causes the particles in the substance to move at different speeds.

The flowchart below shows what happens to the particles that make up a substance when it changes from a solid state into the gas state.

When a gas is cooled, the direction of the flowchart can be reversed as the substance changes from the gas state into a solid state.



Solid
When a solid is heated, its particles start to move more quickly. The increased movement of its particles makes the solid expand.

Melting
As more heat is transferred to the solid, its particles vibrate more violently. Eventually the particles move so much that the bonds holding them in their fixed positions break. The particles start to roll over each other. Melting continues until the entire solid becomes a liquid.

Liquid
As a liquid is heated, its particles move and roll over each other faster and faster. The liquid begins to expand.

Evaporating
If the liquid continues to be heated, the particles will eventually have enough energy to break the bonds holding them together. The particles can break away from the liquid and begin to move around freely. This process is called evaporation. The liquid boils until the entire liquid has evaporated to become a gas.

Gas
As in solids and liquids, the particles in gases move faster and faster when they are heated. The increased movement of the particles means that they take up more space and the gas expands. If the gas is heated in a closed container, the increased movement of the particles means that they collide more often with the sides of the container and with each other.

Foggy mirrors

Have you noticed how the mirror in the bathroom 'fogs up' after a hot shower? The 'fog' is actually formed when invisible water vapour in the air cools down when it contacts the cold glass. It condenses to become water.

Invisible gas
Water vapour forms when particles in the hot water gain enough energy to escape from each other and become a gas. You can't see water vapour. The particles in the water vapour move around freely. They have more energy than the particles in the liquid water.

Fog in the air
Some of the energy of the particles in the water vapour is transferred away from the vapour to the air. The transfer of energy leaves the water vapour with less energy — so much less energy that its particles slow down. The transfer of energy away from the water vapour means it cools down and turns into tiny droplets of water. These tiny droplets form clouds. This process is called condensation.

Fog on the mirror
The energy from some of the particles in the water vapour is transferred to the cold mirror. This causes the water vapour to condense on the mirror.

UNDERSTANDING AND INQUIRING

REMEMBER

1 Describe the motion of the particles in a liquid.

2 When a substance changes state from a solid into a liquid:
   (a) what happens to the bonds between the particles?
   (b) how does the motion of the particles change?

3 Describe what happens to the movement of particles as a substance changes its state from a gas to a liquid.

THINK

4 In movies, you sometimes see a mirror being held up to the mouth and nose of someone who is unconscious to check whether they are breathing. Explain why this would work.

5 Construct a flowchart like the one on the previous page to show how a gas changes state to become a liquid and then a solid. Include the names and descriptions of the two changes of state that take place.

6 For each of the following changes of state of a substance, identify whether it involves adding energy to the particles or transferring energy away from the particles.
   (a) Melting
   (b) Condensation
   (c) Boiling
   (d) Freezing
   (e) Sublimation
   (f) Evaporation

7 Use the particle model to explain why an ice cube changes shape when it melts.

eBook*plus*

8 Simulate heating matter over a Bunsen burner by using the **Changes of state** interactivity in your eBookPLUS. **int-0222**

work sheet 6.5 Changes of state

Concept maps and mind maps

1. On small pieces of paper, write down all the ideas you can think of about a particular topic.
2. Select the most important ideas and arrange them under your topic. Link these main ideas to your topic and write the relationship along the link.
3. Choose ideas related to your main ideas and arrange them in order of importance under your main ideas, adding links and relationships.
4. When you have placed all of your ideas, try to find links between the branches and write in the relationships.

how to ...?

question

How can I explain this topic to someone else?
What do I understand about this particular topic?

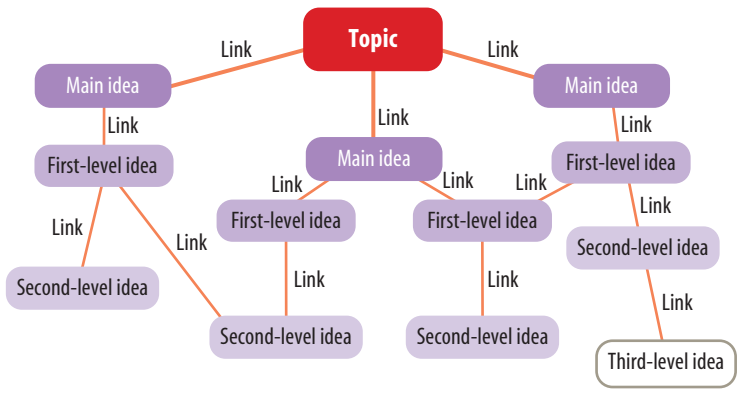
To show what you understand about a particular topic

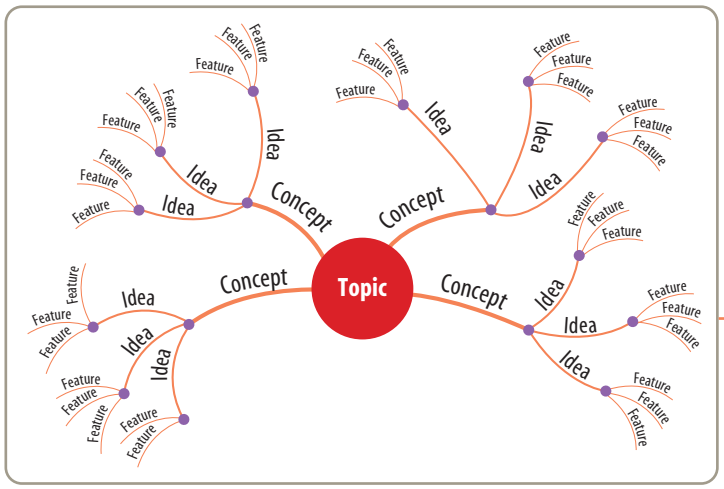
why use?

also called

Knowledge map; concept web

Concept map





comparison

Mind map

example

Similarity

Both show graphically the structure of a topic in a hierarchical way.

Difference

Concept maps explain the relationship between the parts or elements with statements on the links.

UNDERSTANDING AND INQUIRING

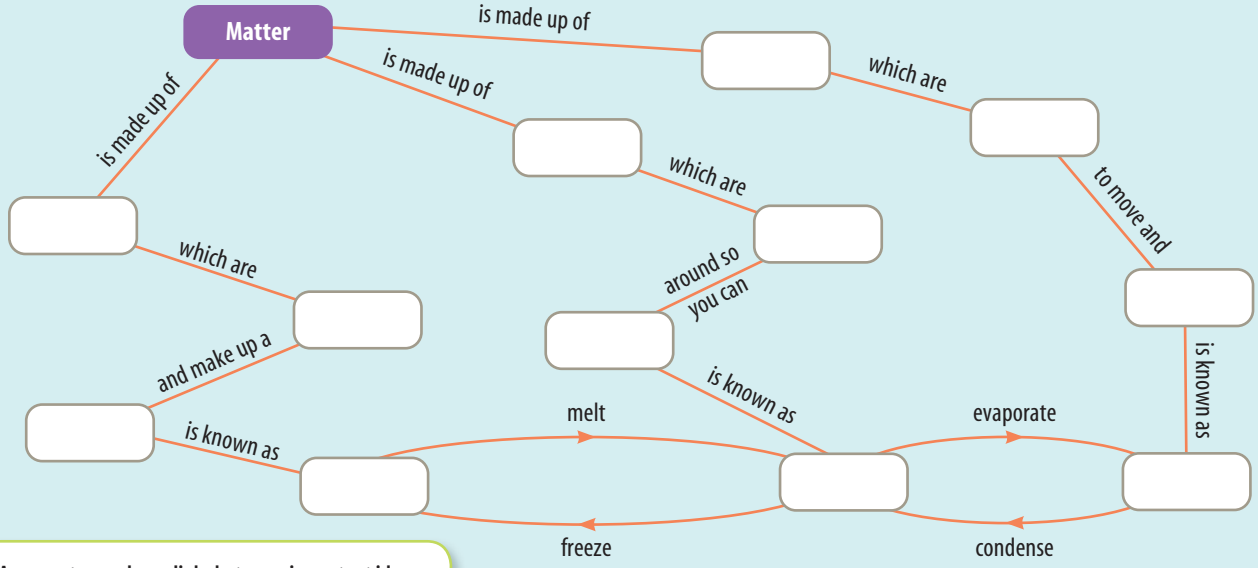
THINK AND CREATE

1 The concept map below represents some of our knowledge about the states of matter. This concept map is just one way of representing ideas about matter and how they are linked. However, all but one of the keywords in the boxes are missing.

Copy the concept map and complete it by writing in suitable keywords in the boxes.

Select the keywords from the list below. One keyword is used three times.

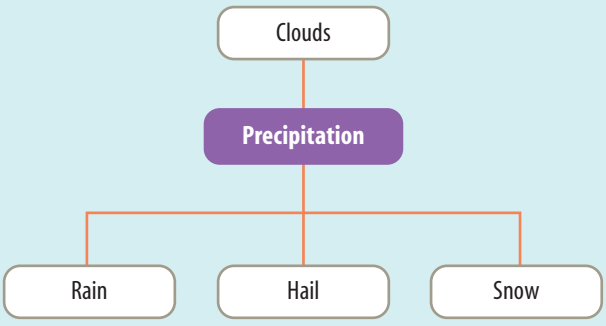
fill space	liquid	sliding
fixed shape	particles	solid
free	pour	vibrating
gas		



A concept map shows links between important ideas.

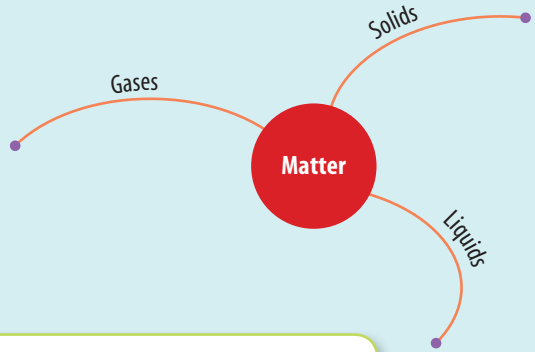
2 Using sticky notes, create a concept map similar to the one above to represent your knowledge about the three states of water. Start with the word 'water' at the top.

3 Create a concept map of your own to represent your knowledge of how water in the atmosphere affects precipitation. The diagram below shows one way of starting your concept map. You'll need to write in suitable link words yourself. (*Hint:* Start by writing down all of the important words or terms related to precipitation that you can think of and use as many of them as you can in your concept map.)



A concept map of precipitation could begin like this.

4 A mind map is similar to a concept map, but the topic is placed in the centre instead of at the top. There are no other boxes — just branches on which the keywords and terms are listed. Complete the mind map below to represent your knowledge of the states of matter.



A mind map begins with the topic in the centre.

5 The states of matter can be represented by a concept map or a mind map. Which map did you find easier to construct? Explain why.

STUDY CHECKLIST

STATES OF MATTER

- describe the properties and physical behaviour of solids, liquids and gases
- define fluid as a substance that flows
- measure the volume of solids and liquids
- explain how mass is measured

CHANGES OF STATE

- describe the changes in the physical properties of substance during melting, freezing, evaporation, boiling, condensation and sublimation
- relate changes of state to heating and cooling
- define melting point and boiling point

THE PARTICLE MODEL OF MATTER

- list the four major assumptions of the particle model of matter
- describe the arrangement and movement of particles in solids, liquids and gases
- describe the diffusion of gases and why it occurs in terms of the particle model
- use the particle model to explain why solids, liquids and gases expand when they are heated
- describe the behaviour of gases under pressure in terms of the particle model
- link the energy of particles to heating and cooling
- link changes in state to the flow of energy into or out of a substance and the subsequent changes in the behaviour of the particles of the substance

SCIENCE AS A HUMAN ENDEAVOUR

- describe the role of meteorologists in observing, explaining and predicting the weather
- relate weather events such as rain, hail and snow to changes of state
- outline the implications of expansion and contraction of materials for engineers and architects

INDIVIDUAL PATHWAYS

eBook*plus*

Activity 6.1
Investigating particles
doc-6066

Activity 6.2
Analysing particles
doc-6067

Activity 6.3
Investigating particles further
doc-6068

ICT

eBook*plus* Summary

eLESSON

Under pressure
In this video lesson, you will see animations that reflect the behaviour of gas particles and learn about the factors that affect the pressure of a gas. You will also learn how compressed gases are used to make fire extinguishers and aerosol cans. A worksheet is attached to further your understanding.

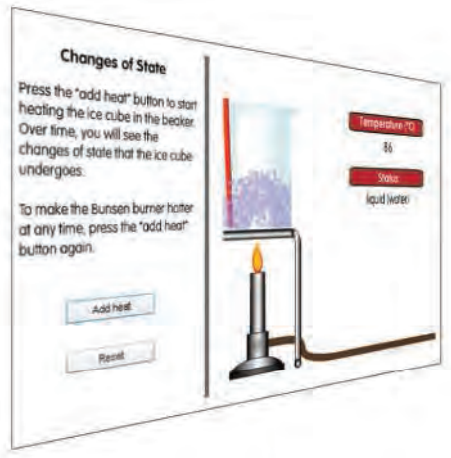
Searchlight ID: eles-0058

Understanding a weather forecast
You might watch the weather report every day on the evening news, but do you really understand what all the information means? Just what is a high or low pressure system, and what are all those swirly lines and colours all over Australia? This eLesson will help you interpret weather maps for yourself and provide you with an insight into how the experts predict future weather conditions.

Searchlight ID: eles-0161

INTERACTIVITY

Changes of state
This interactivity allows you to simulate heating an ice cube over a Bunsen burner. As you add more heat, you will see the effect on the particles as the ice changes state to become boiling water. A worksheet is attached to further your understanding.



Searchlight ID: int-0222

LOOKING BACK

1 Use the particle model to explain why steam takes up more space than liquid water.

2 In which state are the forces of attraction between the particles likely to be greatest?

3 List all of the changes of state that take place in the water cycle.

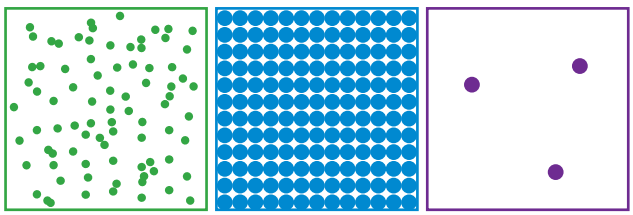
4 In which state — solid, liquid or gas — do the particles have:
(a) the most energy?
(b) the least energy?

5 Explain why perfume or aftershave lotion evaporates more quickly than water.

6 Copy and complete the table below to summarise the properties of solids, liquids and gases. Use a tick to indicate which properties each state *usually* has.

Property	Solid	Liquid	Gas
Has a definite shape that is difficult to change			
Takes up a fixed amount of space			
Can be poured			
Takes up all of the space available			
Can be compressed			
Is made of particles that are strongly attracted to each other and can't move past each other			
Is made of particles that are not held together by attraction			

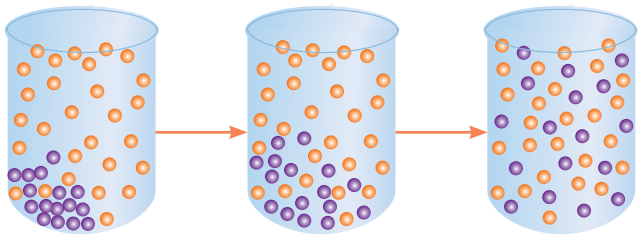
7 Copy and label the three diagrams below to show which represent solids, liquids and gases. Make an improvement to each of the diagrams so that they describe the particle model more fully.



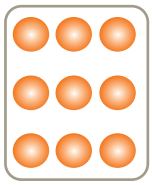
Which states are represented by these diagrams?

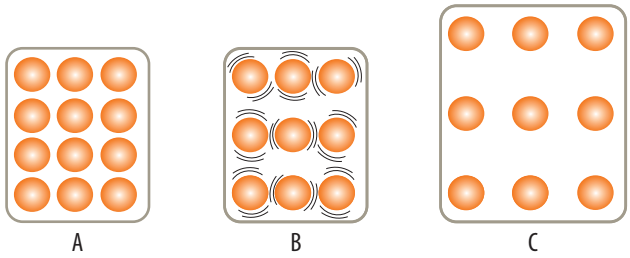
8 Snow and hail are water in a solid state. Describe the difference between snow and hail and explain how each of them is formed.

9 Name the process that is taking place in the following diagrams and explain why it occurs only in liquids and gases.

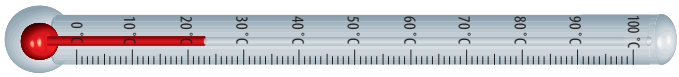


10 Which of the diagrams below (A, B or C) best represents the particles of a solid after expanding?

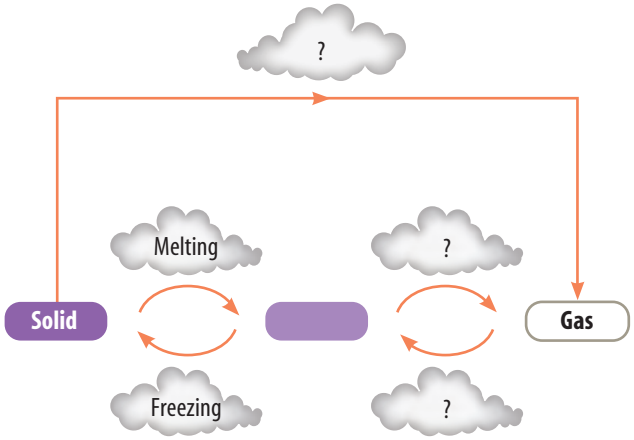




11 Explain how mercury and alcohol thermometers are able to provide a measure of temperature.



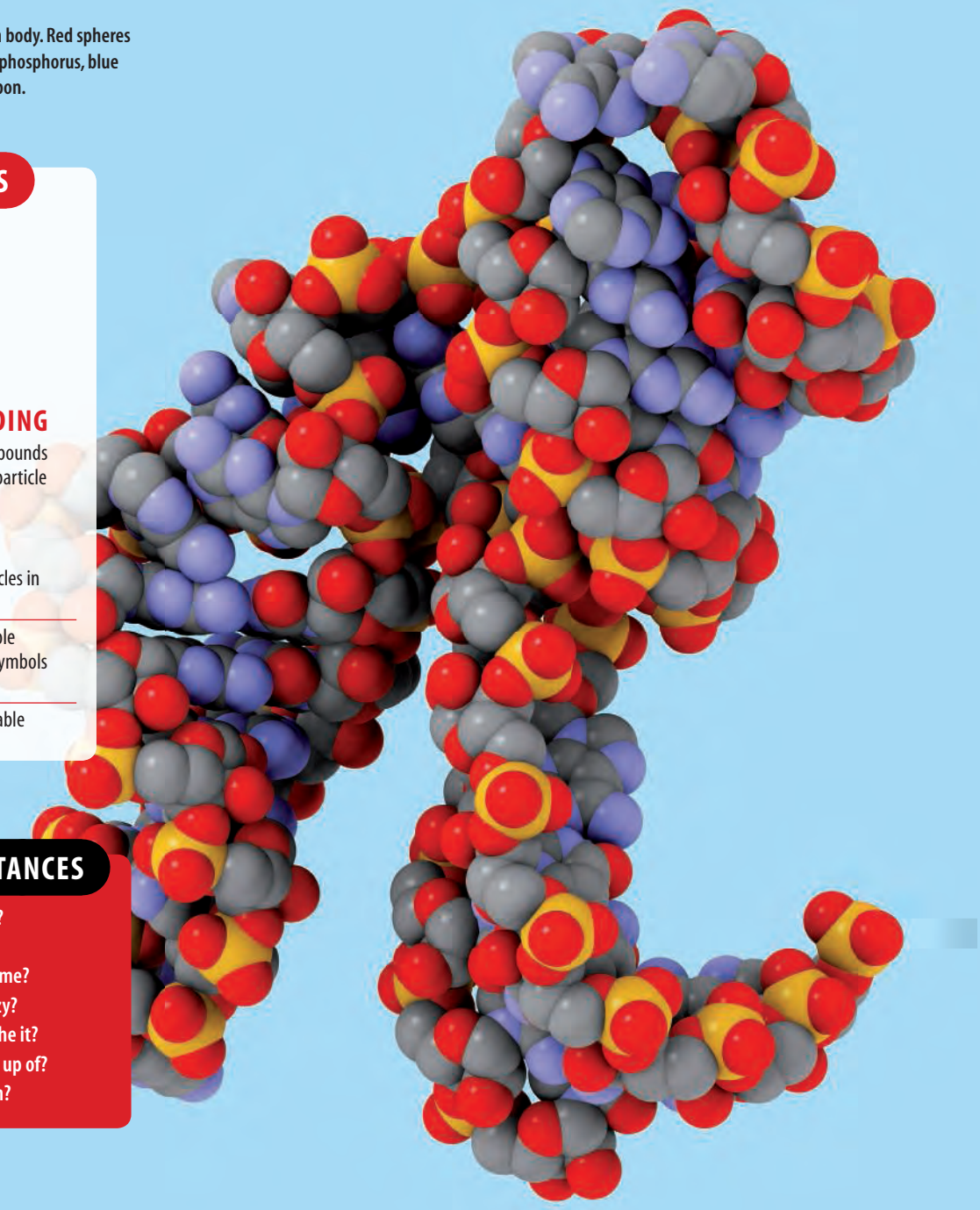
12 Copy and complete the diagram below, labelling the missing state and changes of state.



work sheet 6.6 Summing up

7 Elements, compounds and mixtures

There are millions of different substances in the world. Some, like water, occur naturally. Others, like paper and plastic, are made in factories. Some substances, like sugar and blood, are made by living things. All substances have one important thing in common; they are all made of the tiny building blocks of matter that we call atoms.



Model of a large molecule in the human body. Red spheres indicate oxygen atoms, yellow indicate phosphorus, blue indicate nitrogen and grey indicate carbon.

OVERARCHING IDEAS

- Patterns, order and organisation
- Form and function
- Stability and change
- Scale and measurement
- Matter and energy
- Systems

SCIENCE UNDERSTANDING

Differences between elements, compounds and mixtures can be described at a particle level.

Elaborations

Modelling the arrangement of particles in elements and compounds

Recognising that elements and simple compounds can be represented by symbols and formulae

Locating elements on the periodic table

THINK ABOUT SUBSTANCES

- Can lead be changed into gold?
- Can water be split?
- How did plumbers get their name?
- Which metal can drive you crazy?
- Nitrogen gas — can you breathe it?
- What is most of an atom made up of?
- Just what is 'plastic' made from?

YOUR QUEST

How small are the bits that matter?

INQUIRY: INVESTIGATION 7.1

KEY INQUIRY SKILL:

- questioning and predicting

Equipment:

a strip of paper cut from A4 paper (about 30 cm long)

pair of scissors

ruler

- Construct a table like the one on the right and record the length of the strip of paper.
- Cut the strip of paper in half. Put one half aside. Measure the length of the other half.
- Cut the measured half in half again. Again, put one half aside and measure and record the length of the other half.
- Before you go any further, predict how many times you will be able to cut the strip in half.
- Continue this process until you can no longer cut the strip in half.

DISCUSS AND EXPLAIN

1 How many cuts were you able to make? Was it more or less than your prediction?

2 Estimate the number of cuts you would need to make before the strip would be too small to see.

How small are the bits?

Number of cuts	Length of strip (approximate)
0	30 cm
1	15 cm
2	7.5 cm (easy?)
3	
4	
5	
6	
7	
8	1 mm (you're doing well to get this far!)
9	
10	
12	
14	
18	1 micron (1 millionth of a metre, one thousandth of a millimetre)
22	
26	
31	The size of a single atom

What's inside?

How do you know what's inside a substance when you can't actually see inside it and when it is so small that you can't see it even with the most powerful microscope? It seems impossible — but it can be done!

INQUIRY: INVESTIGATION 7.2

KEY INQUIRY SKILL

- questioning and predicting

Equipment:

sealed box 1, containing a mystery object

sealed box 2, which is empty

- Together with a partner, and without opening the boxes, work out what is inside box 1.
- Ask your teacher for any equipment that you think might be helpful.

But remember that you are not allowed to open the boxes!

DISCUSS AND EXPLAIN

1 Write down any information you can find out about the object.

2 What do you think is in box 1?

3 What are the reasons for your decision about what is in the box?

4 What was the reason for having box 2 in this activity?

It's elementary

About 1000 years ago, when kings and queens lived in castles and were defended by knights in shining armour, there lived the **alchemists**.

They chanted secret spells while they mixed magic potions in their flasks and melted metals in their furnaces. They tried to change ordinary metals into gold. They also tried to find a potion that would make humans live forever. They studied the movements of the stars and claimed to be able to see into the future. The kings and queens took the advice of the alchemists very seriously.

The alchemists never found the secrets they were looking for, but they did discover many things about substances around us. There were other people of these ancient times whose work has also helped us to understand the substances around us. Blacksmiths worked with metals to make stronger and lighter swords and armour, fabric dyers learned how to colour cloth, and potters decorated their work with glazes from the Earth. Without the knowledge passed down by these people, the world as we know it would be very different! They discovered twelve important substances: gold, iron, silver, sulfur, carbon, lead, mercury, tin, arsenic, bismuth, antimony and copper. Five of these were discovered by the alchemists.

Real science

In about the seventeenth century, people stopped thinking about magic and instead carried out **investigations** based on careful **observations**. These new seekers of knowledge were called **scientists**. They discovered that the twelve substances could not be broken down into other substances. Scientists investigated many common everyday substances as well, including salt, air, rocks, water and even urine! They discovered that nearly everything around us could be broken down into other substances. They gave the name 'element' to any substance that could not be broken down into other substances. Between 1557 and 1925, another seventy-six **elements** were discovered. We now know that ninety-two elements exist naturally. In recent years scientists working in laboratories have been able to make at least another 24 artificial elements.

Warning! Danger!

Many elements are safe to handle. However, there are also many that are not. For example, the elements sodium, potassium and

HOW ABOUT THAT!

In days gone by, substances containing the element mercury were used to make hats. In those days it was not known that mercury is a very poisonous element. Poisoning by mercury can affect your nervous system and your mind. This sometimes happened to people who made hats and were exposed to mercury for a long time; hence the expression 'mad as a hatter'!

Lewis Carroll's Mad Hatter character in *Alice's Adventures in Wonderland* was mad because mercury was used in the making of hats.

mercury need special care and handling. Sodium and potassium are soft metals that can be cut with a knife. They both get very hot if they come into contact with water. They are stored under oil so that water in the atmosphere cannot reach them.

Elements are rare

Most of the substances around you are made up of two or more elements. You will not be able to find many of the ninety-two naturally occurring elements in their pure form. It is possible, however, to examine many of the elements in the school laboratory.

INQUIRY: INVESTIGATION 7.3

Checking out appearances

KEY INQUIRY SKILL:

- processing and analysing data and information

Equipment:

samples of chemical elements (e.g. carbon, sulfur, copper, iron, aluminium, silicon)

- Copy the table below into your workbook, or obtain a copy from your teacher.
- Carefully examine each of the elements in the set (look for colour, appearance, hardness).
- Complete the table by filling in the description and discussing with other students the substances that might include the element. One example is completed for you.

Elements	State	Description	In which substances might the element be present?
Hydrogen	Gas	Clear, colourless, explosive	Acids, water

UNDERSTANDING AND INQUIRING

REMEMBER

1 What problems were the alchemists of ancient times trying to solve?
2 What is an element?
3 Which types of substances did blacksmiths help us to understand?
4 How did the scientists differ from the alchemists?
5 Why do sodium and potassium need to be stored under oil?
6 State one harmful effect of mercury on humans.

THINK

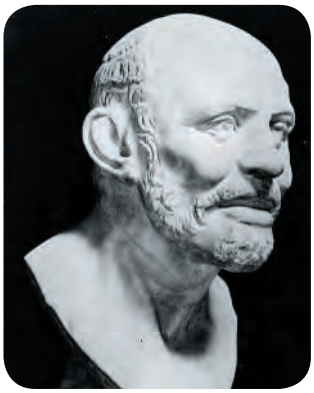
7 Is water an element? Give a reason for your answer.
8 Give one reason for displaying chemical safety symbols at the entrances of many buildings.

INVESTIGATE

9 Many years ago, balloons were filled with hydrogen so that they could float high in the sky. However, hydrogen is no longer used in balloons because it explodes too easily. At fairs, carnivals and in florists' shops, you can often buy colourful gas-filled balloons that fly high into the sky if you let them go. These balloons are filled with another element called helium. Find out who discovered the gas helium, where it was discovered and when.
10 The element mercury was known to ancient people and was very important to the alchemists. Find out all you can about this liquid metal. What does its name mean? Where is it found? What has it been used for in the past? What is it used for now? What is the safety procedure if mercury is spilt?

work sheet 7.1 How big is an atom?

Elements: the inside story



About 2500 years ago a teacher named Democritus lived in ancient Greece. He walked around the gardens with his students, talking about all sorts of ideas.

Democritus suggested that everything in the world was made up of tiny particles so small that they couldn't be seen. He called these particles *atomos*, which means 'unable to be divided'. Other thinkers at the time disagreed with Democritus. It took about 2400 years for evidence of the existence of these **atoms** (as we now call them) to be found.

Atoms and elements

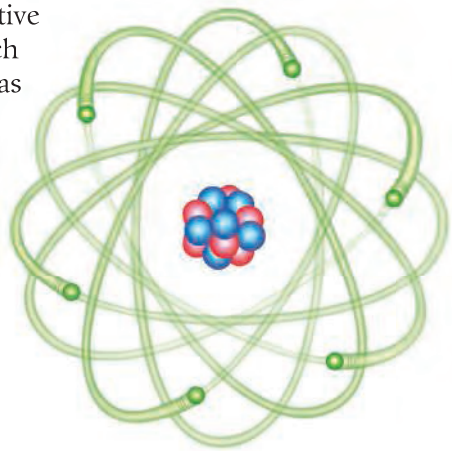
We now know that each element is made of its own particular kind of atom. Gold contains only gold atoms, oxygen contains only oxygen atoms, carbon contains only carbon atoms and so on. But what is it that makes atoms different from one another? To answer this question we need to know a little bit more about the atom.

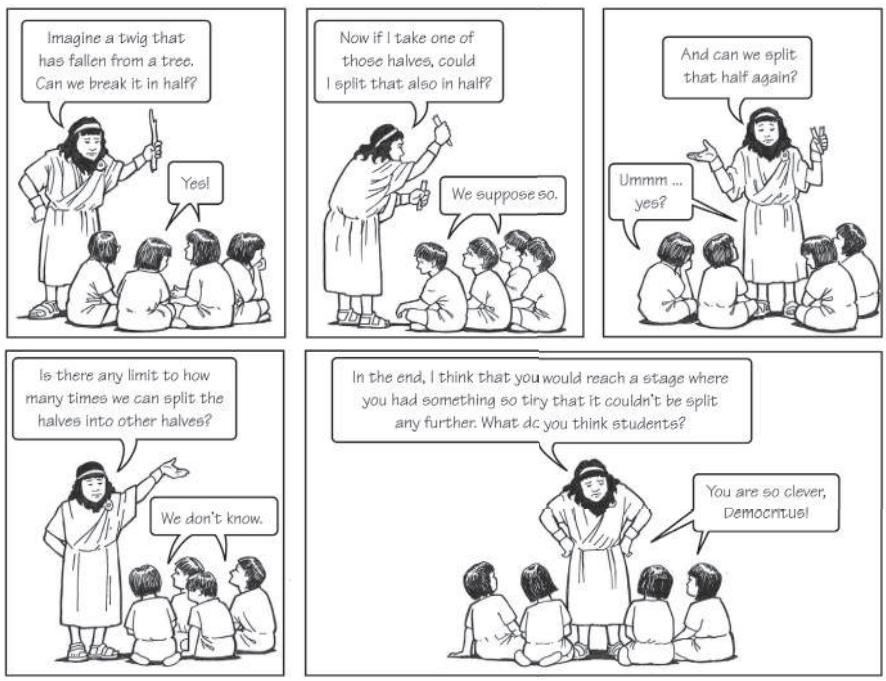
For scientists, the atom was like the mystery box on page 238. Even though the atom couldn't be seen, scientists did experiments over many years and they thought carefully about the information they gathered.

Until 1911, scientists thought the atom was a solid sphere. Then Sir Ernest Rutherford proved them wrong. He fired extremely tiny particles at a very thin sheet of gold. Most of the particles went straight through. Only sometimes did they bounce off as if they had hit something solid. He concluded that the tiny particles could be getting through only if most of each atom consisted of empty space.

Inside the atom

It is now understood that all atoms are made up of even smaller particles. The particles in the centre of the atom are called **protons** and **neutrons**. Together they form the **nucleus**. Each proton carries a positive electric charge. Neutrons have no electric charge. Moving very rapidly around the nucleus are **electrons**. Electrons are much smaller in size and weight than both protons and neutrons. Each electron carries a negative electric charge. The amount of negative charge carried by each electron is the same as the amount of positive charge carried by each proton. In an atom, the number of protons is equal to the number of electrons, so there is no overall electric charge.





The ancient Greeks and their students didn't test their ideas with experiments like Investigation 7.1 on page 238. They thought deeply about them and discussed them.

INQUIRY: INVESTIGATION 7.4

Modelling an atom

KEY INQUIRY SKILL:

- processing and analysing data and information

Equipment:

1 hula hoop
rice grains
table tennis ball
broom and dustpan
1 straw
cotton thread
sticky tape

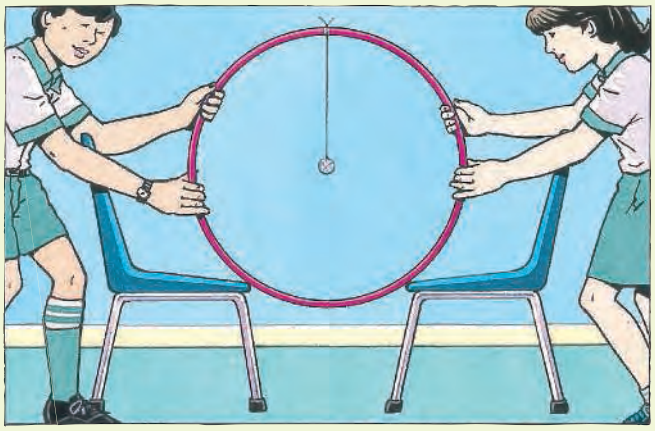
- Set up the equipment as shown in the diagram.
- From across the room, but within target distance, use the straw as a peashooter to fire rice grains at the table-tennis ball.

CAUTION: Ensure that the rice grains are not fired towards any person.

- Count how many grains go right through and how many hit the table tennis ball. (*Note:* Hits to the cotton thread do not count!)
- Use the broom and dustpan to clean up the mess you've left on the floor.

DISCUSS AND EXPLAIN

1 Construct a bar graph to display your results.
2 Construct a bar graph to show the class results.
3 Which part of the atom does the table-tennis ball represent in this model?
4 What does the hula hoop represent?



Atomic numbers

The number of protons in an atom is called its **atomic number**. Each element has a different atomic number. This blimp is filled with helium, which has an atomic number of 2. Helium atoms are lighter than all others except hydrogen atoms. All carbon atoms have six protons inside the nucleus, so the atomic number of carbon is 6. For each proton in the carbon atom it also has one electron, meaning a carbon atom has six electrons. Carbon atoms can have 6, 7 or 8 neutrons in their nuclei. The lightest element is hydrogen, with one proton in each atom and an atomic number of 1. The heaviest natural element is uranium with 92 protons in each atom.

Helium atoms are lighter than all others except hydrogen atoms. This blimp is filled with helium.

What's in a name?

As the early scientists discovered more and more elements, it became more important that they all agreed on names for the elements. Each element was given a name and a **chemical symbol**.

The chemical symbols of most elements are very easy to understand. The symbol sometimes starts with the capital letter that is the first letter of the element's name. For some elements that is the complete symbol. For example:

O = oxygen, C = carbon,
N = nitrogen, H = hydrogen.

When there is more than one element starting with the same capital letter, a small letter is also used. For example:

Cl = chlorine, Ca = calcium,
Cr = chromium, Cu = copper.

If an element has a symbol that doesn't match its modern name, that's because the symbol is taken from the original Greek or Latin name.

For example:

Na = sodium (*natrium*)
Pb = lead (*plumbum*)
Hg = mercury (*hydro argyros*)
Ag = silver (*argentum*)
K = potassium (*kalium*)
Fe = iron (*ferrum*).

The names and symbols of some of the elements have some interesting origins.

- Einsteinium (Es) is named after the famous scientist Albert Einstein.
- Polonium (Po) was discovered by another famous scientist, Marie Curie. She named polonium after Poland, the country of her birth.
- Helium (He) was first discovered in the sun. It is named after Helios, the Greek god of the sun.
- Sodium (Na) was first called by the Latin name *natrium*.
- Lead (Pb) also used to have a Latin name, *plumbum*. That's where the word 'plumber' comes from. The ancient Romans, who spoke Latin, used lead metal to make their water pipes.

UNDERSTANDING AND INQUIRING

REMEMBER

1 What idea did Democritus have about 2500 years ago about what substances were made up of?
2 How many types of natural atoms are there?
3 Name the three parts of an atom and state the location of each part.
4 What does the atomic number of an element tell you?
5 Explain why electrons don't fly off their atoms.
6 In what way is an atom of carbon different from an atom of uranium?
7 What makes up most of every atom?
8 List the symbols of each of the following elements: hydrogen, carbon, oxygen, nitrogen, iron, calcium, copper, lead, mercury.

THINK

9 What type of electric charge does the nucleus of every atom have?
10 What is the atomic number of uranium?
11 To which element does the atom illustrated on page 241 belong?
12 Why do carbon atoms have six electrons?
13 Draw a diagram of an atom that has three protons and one neutron in its nucleus. How many electrons will it have?
14 Why is it important for scientists around the world to agree on the names and chemical symbols of the elements?

CREATE

15 Construct a model of an atom with three protons, three neutrons and three electrons. Use any appropriate materials. For example, you could use plastic balls, papier-mache and wire, or perhaps a bowl of jelly with lollies in it.

INVESTIGATE

16 Find out the names, atomic numbers and uses of the elements with the following symbols. Construct a table in which to display your findings.

Sn Au Cu N Ne Sr Ca

17 Research and report on what nanotechnology is and what connection it has with atoms.

eBookplus

18 Play the **It's elementary! revelation game** in your eBookPLUS and test your ability to identify the symbols of common elements. **int-0229**

Grouping elements

It is often convenient to group objects that have features in common. Shops provide a good example of this. In a department store, the goods are grouped so that you know where to buy them. You go to the clothing section for a new pair of jeans, to the jewellery section for a new watch and to the food section for a packet of potato chips.

Scientists also organise objects into groups. Biologists organise living things into groups. Animals with backbones are divided into mammals, birds, reptiles, amphibians and fish. Geologists organise rocks into groups. The elements that make up all substances can also be organised into groups.

Metals and non-metals

Scientists have divided the elements into two main groups: the **metals** and the **non-metals**.

METALS

The metals have several features in common:

- They are solid at room temperature, except for mercury, which is a liquid.
- They can be polished to produce a high shine or lustre.
- They are good conductors of electricity and heat.
- They can all be beaten or bent into a variety of shapes. We say they are malleable.
- They can be made into a wire. We say they are ductile.
- They usually melt at high temperatures. Mercury, which melts at –40°C, is one exception.

NON-METALS

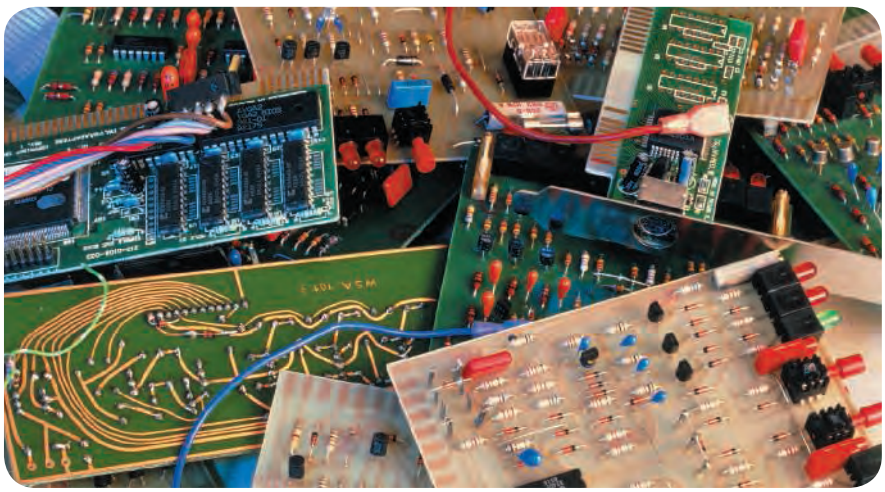
Only twenty-two of the elements are non-metals. At room temperature eleven of them are gases, ten are solid and one is liquid. The solid non-metals have most of the following features in common:

- They cannot be polished to give a shine like metals; they are usually dull or glassy.
- They are brittle, which means they shatter when they are hit.
- They cannot be bent into shape.
- They are usually poor conductors of electricity and heat.
- They usually melt at low temperatures.
- Many of the non-metals are gases at room temperature.

Common examples of non-metals are sulfur, carbon and oxygen.

METALLOIDS

Some of the elements in the non-metal group look like metals. One example is silicon. While it can be polished like a metal, silicon is a poor conductor of heat and electricity and cannot be bent or made into wire. Those elements that have features of both metals and non-metals are called **metalloids**. There are eight metalloids altogether: boron, silicon, arsenic, germanium, antimony, polonium, astatine and tellurium.



Metalloids are important materials often used in electronic components of computer circuits.

INQUIRY: INVESTIGATION 7.5

Looking for similarities

KEY INQUIRY SKILL:

- processing and analysing data and information

Equipment:

safety glasses
samples of sulfur, zinc, tin, carbon, silicon, copper
steel wool or very fine sandpaper
battery or power pack
wires with alligator clips
light globe

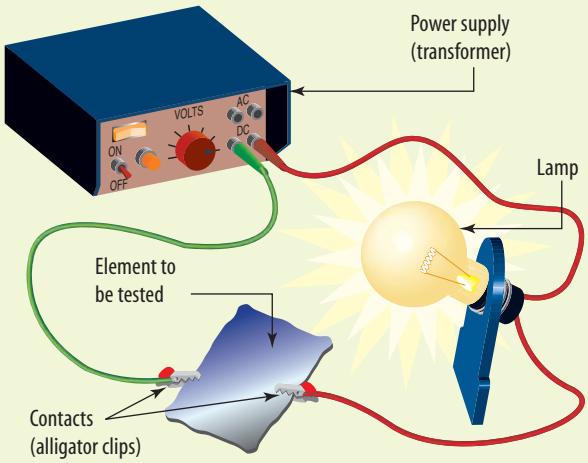
- Make a copy of the table on the right and use it to record your observations.
- Rub each of the elements with the fine sandpaper and observe whether they are shiny or dull.
- Try to bend the metal.
- Connect the circuit shown in the diagram, to determine whether electricity passes through each of the elements.

DISCUSS AND EXPLAIN

1 Which of the six elements have a shiny surface when polished?
2 Which of the six elements do not have a shiny surface when polished?
3 Which of the six elements can be bent?
4 Which of the six elements cannot be bent?
5 Which of the six elements allow electricity to pass through?
6 Which of the six elements do not conduct electricity?
7 Attempt to divide the elements into two groups on the basis of your observations. Suggest names for these two groups.
8 Which of the six elements tested does not seem to fit into either of these two groups?

Characteristics of some elements

Element	Shiny or dull?	Does it bend?	Does it conduct electricity?
Sulfur			
Zinc			
Tin			
Carbon			
Silicon			
Copper			



Connect your element sample into this circuit.

UNDERSTANDING AND INQUIRING

REMEMBER

1 Outline four features that metals have in common.
2 Outline four features that non-metals have in common.
3 What is a metalloid? List some examples.
4 Which metal is liquid at room temperature?
5 What does 'metallic lustre' mean?

THINK

6 While all metals have similar characteristics, there are also differences between them. List three ways in which metals can differ from each other.
7 Silicon is used in the 'chips' of computer circuits, but it is never used in the connecting wires of electric circuits. Why not?

IMAGINE

8 Imagine that you are a scientist who has discovered what appears to be a new element. It is golden in colour and very shiny. What experiments would you do to test if it was a metal or non-metal? What results would you expect to get if it was a metal?

INVESTIGATE

9 Polonium is a metal discovered by Marie Curie. She also discovered another metal. Find out its name and the important role it played in medicine.

work sheet 7.2 Metals and non-metals

Patterns, order and organisation: Chemical name tags

As more and more elements were being discovered, the early scientists began to find that some of them had things in common.

Because of this they decided to organise the elements into groups. It took a long time and a lot of experimenting to work out the groups. The person who finally worked out a system for sorting the elements into groups was a Russian scientist called Dmitri Mendeleev. His system is called the **periodic table** and a modern version is used by scientists today.

	Group I	Group II							
Period 2	3 Lithium Li	4 Beryllium Be							
Period 3	11 Sodium Na	12 Magnesium Mg							
Period 4	19 Potassium K	20 Calcium Ca	21 Scandium Sc	22 Titanium Ti	23 Vanadium V	24 Chromium Cr	25 Manganese Mn	26 Iron Fe	27 Cobalt Co
Period 5	37 Rubidium Rb	38 Strontium Sr	39 Yttrium Y	40 Zirconium Zr	41 Niobium Nb	42 Molybdenum Mo	43 Technetium Tc	44 Ruthenium Ru	45 Rhodium Rh
Period 6	55 Caesium Cs	56 Barium Ba	57–71 Lanthanides	72 Hafnium Hf	73 Tantalum Ta	74 Tungsten W	75 Rhenium Re	76 Osmium Os	77 Iridium Ir
Period 7	87 Francium Fr	88 Radium Ra	89–103 Actinides						

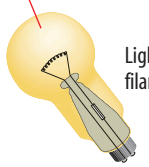
The periodic table. Elements 1–92 all occur naturally. Those from element 93 onwards have been made in laboratories and are all radioactive. Those for element 104 onwards are not shown in this table.

Period 1: 1 Hydrogen H; 2 Helium He

Key: Atomic number, Name, Symbol

Transition metals

X-rays



Light filament

Nail

The group number corresponds to the number of electrons in the outer shell.

The period number refers to the number of the outermost shell containing electrons.

New radioactive elements are still being produced — the most recent one at the time of publication was element 118.

Lanthanides

57 Lanthanum La	58 Cerium Ce	59 Praseo-dymium Pr	60 Neodymium Nd	61 Promethium Pm	62 Samarium Sm

Actinides

89 Actinium Ac	90 Thorium Th	91 Protactinium Pa	92 Uranium U	93 Neptunium Np	94 Plutonium Pu

Looking for similarities

A vertical column on the periodic table is called a **group**. Elements in the same group on the periodic table always have some features in common. Sometimes these common features are easy to observe, but some of the similarities are not so obvious. For example, neon and argon are gases that do not change when mixed with other elements except under extreme circumstances. They are said to be **inert**. These two gases are found in the last group of the periodic table along with three other inert gases. The group containing the inert gases is called the **noble gas** group.

UNDERSTANDING AND INQUIRING

REMEMBER

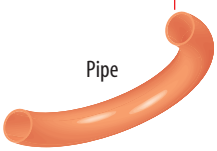
1. Obtain a copy of the periodic table from your teacher. Colour in the elements that you have already seen in the laboratory.
2. Write down the symbols for the following elements: hydrogen, carbon, oxygen, nitrogen, iron, tin, calcium, sulfur, copper and krypton.
3. What is similar about all of the gases in the noble gas group of the periodic table?

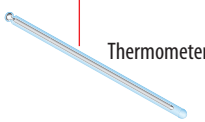
CREATE

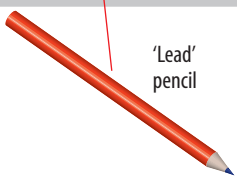
4. Make up a 'Guess the element' card game, finding out and using information about at least twenty of the elements.

	Group III	Group IV	Group V	Group VI	Group VII	Group VIII
Metals	5 Boron B	6 Carbon C	7 Nitrogen N	8 Oxygen O	9 Fluorine F	10 Neon Ne
Metalloids	13 Aluminium Al	14 Silicon Si	15 Phosphorus P	16 Sulfur S	17 Chlorine Cl	18 Argon Ar
Non-metals	31 Gallium Ga	32 Germanium Ge	33 Arsenic As	34 Selenium Se	35 Bromine Br	36 Krypton Kr
	49 Indium In	50 Tin Sn	51 Antimony Sb	52 Tellurium Te	53 Iodine I	54 Xenon Xe
	81 Thallium Ti	82 Lead Pb	83 Bismuth Bi	84 Polonium Po	85 Astatin At	86 Radon Rn

28 Nickel Ni	29 Copper Cu	30 Zinc Zn
46 Palladium Pd	47 Silver Ag	48 Cadmium Cd
78 Platinum Pt	79 Gold Au	80 Mercury Hg


Pipe


Thermometer


'Lead' pencil

Neon sign

Lanthanides (cont.)

63 Europium Eu	64 Gadolinium Gd	65 Terbium Tb	66 Dysprosium Dy	67 Holmium Ho	68 Erbium Er	69 Thulium Tm	70 Ytterbium Yb	71 Lutetium Lu

Actinides (cont.)

95 Americium Am	96 Curium Cm	97 Berkelium Bk	98 Californium Cf	99 Einsteinium Es	100 Fermium Fm	101 Mendelevium Md	102 Nobelium No	103 Lawrencium Lr

Compounding the situation

There are millions and millions of different substances in the world. They include the paper of this book, the ink in the print, the air in the room, the glass in the windows, the wool of your jumper, the cotton and polyester in your shirt or dress, the wood of your desk, the paint on the walls, the plastic of your pen, the hair on your head, the water in the taps and the metal of the chair legs. The list could go on and on.

All substances can be placed into one of three groups: elements, compounds or mixtures.

- Elements are substances that contain only one type of atom. Very few substances exist as elements. Most substances around us are either compounds or mixtures.
- **Compounds** are usually very different from the elements of which they are made. In compounds, the atoms of one element are **bonded** very tightly to the atoms of another element or elements. The elements that make up a compound are completely different substances from the compound. For example, common table salt (sodium chloride) is a compound made up of the elements sodium (a silvery metal) and chlorine (a green, poisonous gas).
- **Mixtures** can be made up of two or more elements, two or more compounds or a combination of elements and compounds. The substances that make up mixtures can usually be easily separated from each other. When the parts of a mixture are separated, no new substances are formed. Fizzy soft drink is a good example of a mixture. It contains water, gas, sugar and flavours. If you shake the soft drink, the gas bubbles separate from the water and go into the air. You still have the water in the bottle and the gas in the air; they are just not mixed together any more. The parts of the mixture can be separated quite easily. The gas escapes when the lid of the container is opened, and the water can be separated by evaporation, leaving behind sugar and some other substances.

When the atoms of different elements bond together, a compound is formed. When heated together, the elements iron and sulfur form a new compound called iron sulfide. Iron sulfide has the formula FeS. Every compound has a formula comprising the symbols of the elements that make it up. Unlike mixtures, the elements within a compound cannot easily be separated from each other.

A compound is completely different from the elements of which it is made. Table salt consists of the elements sodium and chlorine.

Some common substances

Substance	Type	Composed of:	Scientific name
Gold	Element	Gold	Gold
Diamond	Element	Carbon	Carbon
Water	Compound	Hydrogen and oxygen	Dihydrogen oxide
Table salt	Compound	Sodium and chlorine	Sodium chloride
Brass	Mixture	Copper and zinc	Brass
Soft drink	Mixture	Water, sugar, carbon dioxide and other compounds	
Sea water	Mixture	Water, sodium chloride and other compounds	

WHAT DOES IT MEAN?

The word *compound* comes from the Latin word ***componere***, meaning 'to put together'.

HOW ABOUT THAT!

Carbon dioxide is the gas that is added to soft drinks to give them their fizz. Solid carbon dioxide, commonly known as 'dry ice', is used to keep things cold at outdoor events.

There are several ways in which elements can be separated from compounds. These include:

- passing electricity through a compound
- burning the compound
- mixing the compound with other chemicals.

Each of these methods involves a chemical reaction in which completely different substances are formed.

Splitting water

We are surrounded by water. It is in our taps, in our bodies, in the rivers, in the sea, in the air, and it comes down as rain. We wash in it, cook in it and drink it. We cannot live without water. Water is not an element — it can be broken down into simpler substances. The illustration on the next page shows a piece of apparatus called a Hofmann voltameter. Water is placed in the voltameter, which is connected to a battery. The electricity splits the water into the elements of which it is made: **hydrogen** and **oxygen**.

Hydrogen and oxygen are both elements. They are both gases, and they look the same; they have no colour and no smell. Oxygen is necessary for substances to burn — even hydrogen does not burn without it. Hydrogen is a much less dense gas than oxygen. This means that a balloon filled with hydrogen will float up very high, but one filled with oxygen will not.

The element hydrogen is present in almost all acids. By placing a piece of metal in an acid, the hydrogen is forced out. The hydrogen can be collected and tested with a flame.

The element oxygen is present in water, air, rocks and even hair bleach. Oxygen is the gas that all living things need to stay alive. It is also necessary

INQUIRY: INVESTIGATION 7.6

Making a compound from its elements

KEY INQUIRY SKILL:

- processing and analysing data and information

Equipment:

4–5 cm strip of clean, shiny magnesium ribbon. It can be coiled to fit in the crucible.

crucible with lid

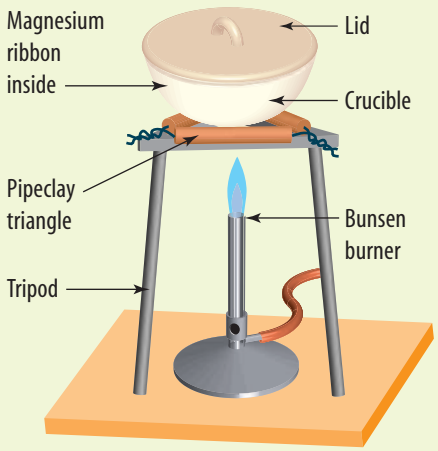
pipeclay triangle, tongs and safety glasses

Bunsen burner, heatproof mat and matches

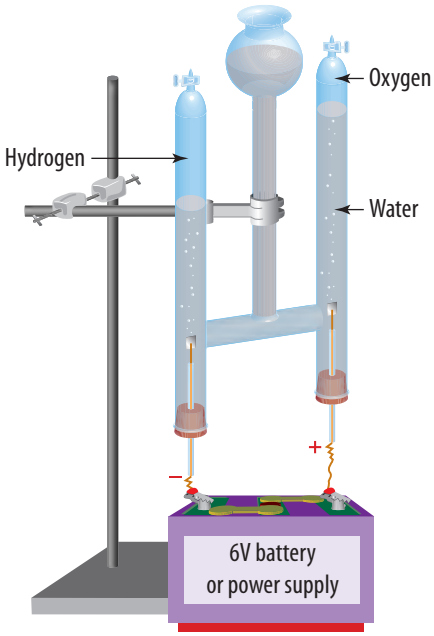
- Examine the piece of magnesium and note its appearance before placing it in the crucible and covering it with the lid.
- Put the crucible on the pipeclay triangle as shown in the diagram.
- Heat the crucible with a strong blue flame, monitoring the reaction by occasionally lifting the lid a little with tongs.
- When all the magnesium ribbon has been changed, turn off the flame and leave the crucible on the tripod to cool.

DISCUSS AND EXPLAIN

1. Describe the substance in the crucible.
2. Is magnesium an element or a compound? Give a reason for your decision.
3. Magnesium is one of the reactants in this experiment. What is the other reactant?
4. Is the substance remaining in the crucible an element or a compound? What is its name?
5. What is the evidence that a new substance has been made?
6. Apart from observing whether the reaction is complete, give another reason for lifting the lid of the crucible a little with tongs during the burning.



for all substances to burn. When hydrogen gas is burned, it combines with the oxygen in the air to form water. This releases a lot of energy. If large amounts of hydrogen and oxygen are used, enough energy can be released to lift a space rocket.



Water is split in a Hofmann voltameter. The clear gas in the left tube is hydrogen. The gas in the right tube is oxygen. What do you notice about the amounts of hydrogen and oxygen that are produced?

INQUIRY: INVESTIGATION 7.7

Let's collect an element

KEY INQUIRY SKILL:

- processing and analysing data and information

Equipment:

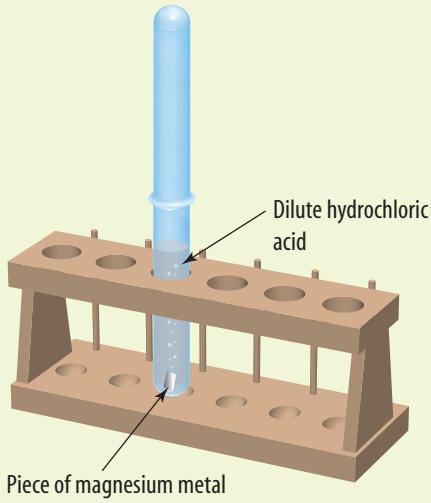
safety glasses	*2 test tubes and test-tube rack*
matches	*dilute hydrochloric acid*
measuring cylinder	*magnesium metal*

- Measure 10 mL of hydrochloric acid and pour it into the test tube.
- Add a piece of magnesium and place the second test tube on top of the first as shown in the diagram. Carefully observe what happens.
- After one minute take the second test tube off the first. While it is still inverted, immediately light the gas in the second test tube with a match.

DISCUSS AND EXPLAIN

1 Describe what happened in the test tube containing the metal and the acid.
2 What does hydrogen gas look like?
3 What happened when you lit the gas?
4 Look closely at the second test tube. Describe what you see inside it.

Collect the hydrogen gas by placing the second test tube over the first.



UNDERSTANDING AND INQUIRING

REMEMBER

1 How do compounds differ from elements?
2 What are the important differences between a mixture and a compound?
3 Fizzy soft drink is a mixture of several compounds. List three of the compounds and suggest how each of them could be separated from the mixture.
4 What happens when atoms are bonded together?
5 Which elements are bonded together to form table salt?
6 List three ways in which elements can be separated from their compounds.

THINK

7 How do you know that water is not simply a mixture of hydrogen and oxygen?
8 Magnesium oxide is a compound of magnesium and oxygen. How do you know that it is a completely different substance from each of the two elements it is made up of?

INVESTIGATE

9 Joseph Priestley was one of the first scientists to discover the element oxygen. He also discovered many compounds that are gases. Research and report on the life of Joseph Priestley.

work sheet 7.3 Pure substances and mixtures

7.6 SCIENCE UNDERSTANDING

Making molecules

The naturally occurring elements are the building blocks of everything in our world. The atoms of various elements can be joined in a wide variety of ways to produce many compounds. Elements and compounds can be combined in many ways to make countless mixtures.

Atoms can join, or bond, in many different ways. In some substances, atoms are joined in groups called **molecules**. For example, in oxygen gas, oxygen atoms are joined in groups of two. In the compound carbon dioxide there are one carbon and two oxygen atoms joined in every molecule. Atoms can join to form small or large molecules of many different shapes.

INQUIRY: INVESTIGATION 7.8

Mix 'n' match

KEY INQUIRY SKILLS:

- questioning and predicting
- communicating

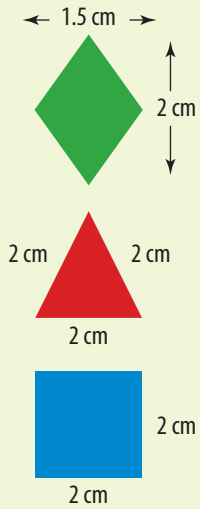
Equipment:

green, red and blue sheets of paper

scissors *pencil* *ruler*

1 large sheet of cartridge paper

- Cut out 25 diamonds, each 2 cm long and 1.5 cm wide, from the green sheet of paper.
- Cut out 30 equilateral triangles, with each side 2 cm, from the red sheet of paper.
- Cut out 15 squares, with each side 2 cm, from the blue sheet of paper.



Cut these shapes from coloured paper.

- Imagine that different types of atoms are represented by particular shapes:

  a blue square = carbon

  a green diamond = oxygen

  a red triangle = hydrogen

  and that, by placing them side by side on the sheet of paper, you are joining them.
- Place two green diamonds next to each other on the sheet. This represents the oxygen molecule, as shown in the diagram below.

A green diamond represents an atom of oxygen. Together, two diamonds represent a molecule of oxygen.

- Place one blue square on the sheet between two green diamonds. This represents the compound carbon dioxide. Label it with its name and symbol.
- Represent and label the following substances:
  (a) water, which contains 1 oxygen and 2 hydrogen atoms
  (b) methane (natural gas), which contains 1 carbon and 4 hydrogen atoms
  (c) benzene (in petrol), which contains 6 carbon and 6 hydrogen atoms
  (d) glucose (sugar), which contains 6 carbon, 12 hydrogen and 6 oxygen atoms
  (e) hydrogen peroxide (found in hair bleach), which contains 2 oxygen atoms and 2 hydrogen atoms.

DISCUSS AND EXPLAIN

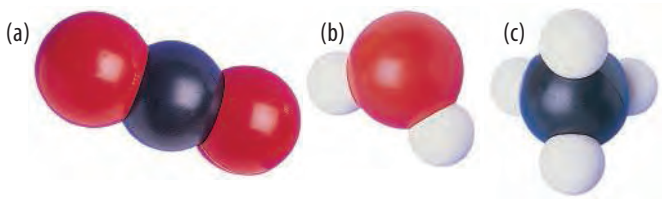
1 Which of these compounds contain only hydrogen and carbon atoms?

2 In what ways are these two substances different from each other?

3 Which of the compounds contain only oxygen and hydrogen? Do these compounds have the same characteristics?

4 Think about the appearance of the compound sugar. How does it differ in appearance from the elements from which it is made?

Some compounds are not made up of molecules. Instead the atoms bond by lining up one after the other. Sodium bonds to chlorine, which bonds to sodium and so on. Common table salt is an example of a substance that is bonded in this way



Models representing the molecules of the compounds (a) carbon dioxide, (b) water and (c) methane. The black balls represent carbon, the red, oxygen, and the white, hydrogen.

Compounds of today and tomorrow

A **polymer** is the name given to a compound made of molecules that are long chains of atoms. Most polymers are made up of chains containing carbon atoms. **Plastics** are synthetic polymers. Cotton and rubber are examples of natural polymers. Although scientists first developed polymers in laboratories in the 1800s, it was not until after World War II that most of the modern polymers were invented. Modern polymers are used in food wrapping, paint, plastic 'glass', polystyrene foam for packaging and cups, note money, cases for electronic appliances such as computers and televisions, clothing, glues, shopping bags, sports equipment and even tea bags!

WHAT DOES IT MEAN?

The word *polymer* comes from the Greek word ***polymeres***, meaning 'of many ports'.

HOW ABOUT THAT!

- Nitrogen is an element. It is a clear, colourless gas made up of molecules. Each molecule is made up of a pair of atoms. Nitrogen makes up 80 per cent of the atmosphere. That means that four-fifths of each breath you take in is nitrogen. Our bodies cannot use this nitrogen so we breathe it straight out again! The gases oxygen, hydrogen and chlorine also exist as molecules made up of pairs of atoms.
- Gold is the only metal element to be found in large amounts in its pure form, rather than bonded in compounds with other elements.

UNDERSTANDING AND INQUIRING

REMEMBER

1 What is a molecule? Name two compounds that are made up of molecules.
2 Are all compounds made up of molecules? Explain.
3 Name four elements that are made up of molecules.
4 What are polymers?

THINK

5 What is the difference between an atom and a molecule?
6 Copy and complete the table below. Use the formula of each compound to work out how many elements are present and which ones they are. (The formula of a compound not only tells you which elements are present, but also indicates the ratio of atoms of the different elements. For example, in the compound NH_3 there are three hydrogen atoms for each nitrogen atom.)

Compound	Formula	Number of elements	Names of elements
Copper sulfate	$CuSO_4$	3	Copper, sulfur, oxygen
Zinc sulfide	ZnS		
Ammonia	NH_3		
Sulfuric acid	H_2SO_4		
Hydrochloric acid	HCl		
Table salt	NaCl		

INVESTIGATE

7 Australia has led the way in the production of polymer money. Find out all you can about how these notes are made.

eBook*plus*

8 Complete the **Making molecules** interactivity in your eBookPLUS by creating the correct model of the molecule as each chemical formula appears.
int-0228

work sheet 7.4 The periodic table — atomic structure

Carbon: it's everywhere

That's carbon?

Carbon is a most amazing element. It is found naturally in three different forms. One form is diamond, another is graphite (the 'lead' in lead pencils), and the third is called amorphous carbon (coal, charcoal and soot). Diamond is the hardest substance known and is used to make drill tips and cutting tools. The three forms are different from each other because the carbon atoms are joined in different ways.

Carbon is found combined with other elements in a huge range of compounds. No other element forms as many different types of compounds as carbon. Carbon is found in everything from the skin of an elephant to paint on the walls!

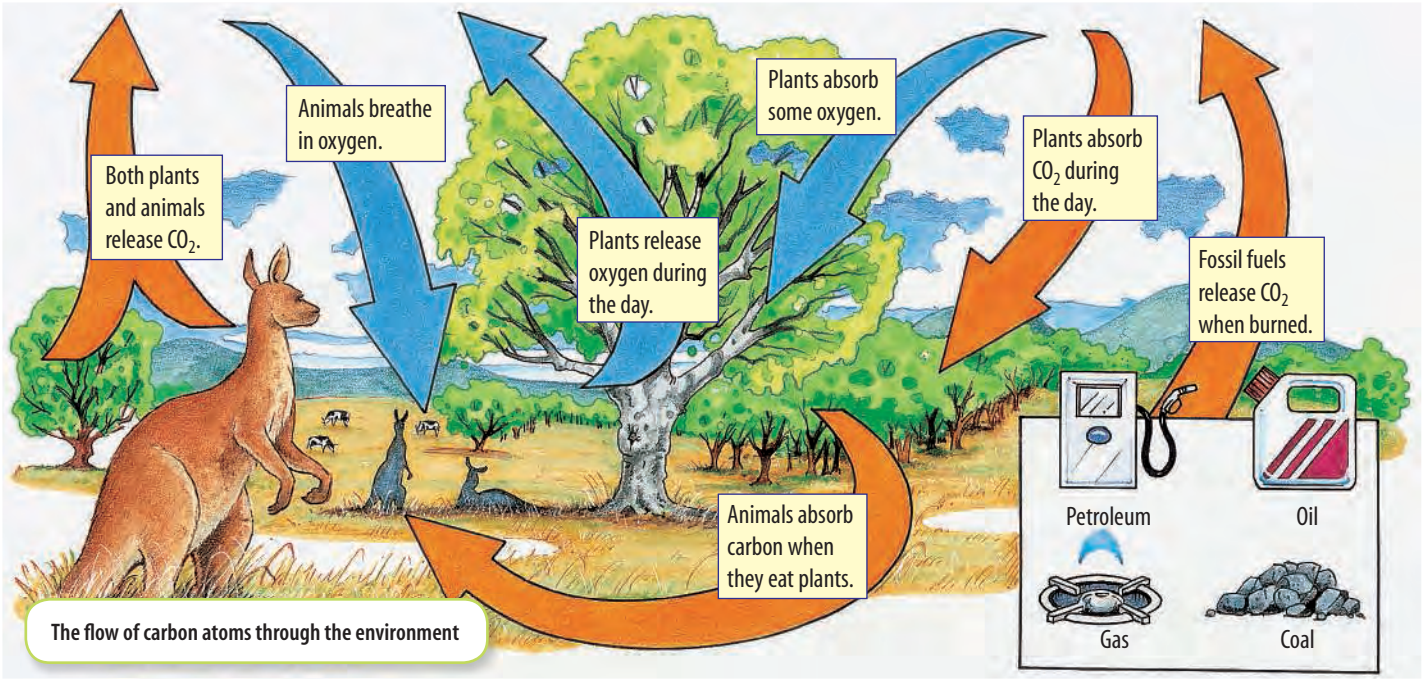
The chemistry of life

All living things are made up of compounds including proteins, fats and carbohydrates. The main element in these compounds is carbon. Carbon is not found only in living things. It is also found in the air in carbon dioxide and under the sea in limestone. The carbon atoms in carbon dioxide were once carbon atoms in living things. The carbon atoms in living things will eventually become carbon atoms in the air or carbon atoms in limestone under the sea. The illustration below shows how nature constantly recycles carbon atoms.

Plants take in carbon dioxide through their leaves and, in a process known as **photosynthesis**, use the carbon dioxide and water to make sugar. Sugar is a compound made up of carbon, hydrogen and oxygen atoms. Plants use the sugar to make other substances and for energy to grow. Animals eat plants or plant-eating animals. The carbon atoms then become part of the animals' bodies.

Carbon atoms in the bodies of living things return to the air in several ways. These ways are **respiration**, **decomposition** and **burning**.

- Respiration is a process that occurs in the cells of every living thing, from a microscopic water plant to a humpback whale. Respiration releases energy and produces carbon dioxide. The carbon dioxide released by the cells in your body is taken by your blood to your lungs. The carbon dioxide that you breathe out contains carbon atoms that were once part of your body.
- Decomposition is what happens when plant or animal material breaks down, such as in a compost heap or after something is buried. Microscopic living creatures called decomposers absorb some of the substances in the dead material and release carbon dioxide to the air by respiration.



The flow of carbon atoms through the environment

- When substances containing carbon are burned, carbon dioxide is released. Coal, natural gas and oil are all **fuels** formed from living things, and contain carbon atoms. Fuels are **combustible**; that is, they are easily ignited. When these fuels are burned in homes, cars, factories and power stations, carbon dioxide is released into the air. Bushfires also release carbon dioxide back to the air.

INQUIRY: INVESTIGATION 7.9

Looking for carbon

KEY INQUIRY SKILL:

- processing and analysing data and information

Equipment:

safety glasses
Bunsen burner, heatproof mat and matches
metal tongs
small samples of substances to investigate: woollen cloth, cotton wool, sugar cube, wood, bread, peanut, steel wool, glass, paper, aluminium foil

Your task in this investigation is to find out whether the element carbon is present in some common substances. If carbon is in a substance, the substance turns black when it is burnt. Your teacher may allow you to burn some plastics, including artificial fabrics such as nylon and rayon, in the fumehood.

CAUTION: Burning plastics produce poisonous fumes. A fume cupboard must be used.

- Hold a small piece of the substance you are going to test in the metal tongs.
- Put the substance in the blue flame of the Bunsen burner.
- When it catches alight, take it out of the flame and, keeping it above the heatproof mat, allow it to burn slowly. Does it turn black?
- Draw up a table like the one below and record your observations.

DISCUSS AND EXPLAIN

1 In which of the substances tested is carbon present?
2 Can you be sure that, if the substance went black, carbon was present? Give a reason for your answer.
3 Can you be sure, if a substance didn't go black, that it didn't contain carbon? Give a reason for your answer.

Substance	Observations	Is carbon present?
Wood		
Cotton wool		

UNDERSTANDING AND INQUIRING

REMEMBER

1 List and describe the three different forms of the element carbon.
2 How do plants get the carbon that they need to make sugar?
3 Describe three ways in which carbon can return to the atmosphere.
4 Where does respiration take place?
5 Some fabrics are more combustible than others. What does this mean?

THINK

6 How do animals obtain carbon?
7 Where does the carbon come from to form limestone at the bottom of the sea?
8 The amount of carbon dioxide in the Earth's atmosphere is increasing. Why is this happening?
9 Many different materials are used to provide heating. The table below shows how much carbon is in some of them. The last column indicates how much heat (in therms) 50 kg of each material provides.

Material	Carbon content (%)	Heat production (therms)
Wood	11	8.5
Brown coal	73	14
Black coal	80	17
Natural graphite	90	18.5

(a) Draw a bar graph showing the percentage carbon content of each material.
(b) Which is the best material for heating?
(c) Does the table indicate any relationship between the amount of carbon in a material and the amount of heat that it provides? Explain your answer clearly.

INVESTIGATE

10 Fuels may be solids, liquids and gases. Search the internet or use a library to find as many examples as possible of solid, liquid and gas fuels, and complete a table like the one below.

	State		
	Solid	Liquid	Gas
Examples			
Uses			
Advantages			
Disadvantages			

Affinity diagrams and cluster maps

1. Think about a topic and write any ideas you have onto small pieces of paper.
2. Examine your pieces of paper and put similar ideas into groups. Feel free to rearrange your groups until you are happy with them.
3. Think of names for your groups.
4. Now you are ready to draw an affinity diagram like the one below.

how to ...?

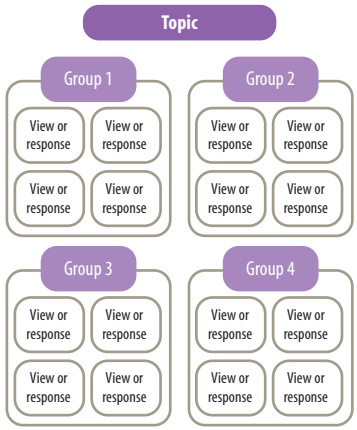
What do I/we feel about this issue?

question

Allows you to become aware of both your and others' feelings and thoughts about issues

why use?

Affinity diagram



also called

'JK method', named after its developer Jiro Kawakita

comparison

Similarity

Both organise ideas or features into groups.

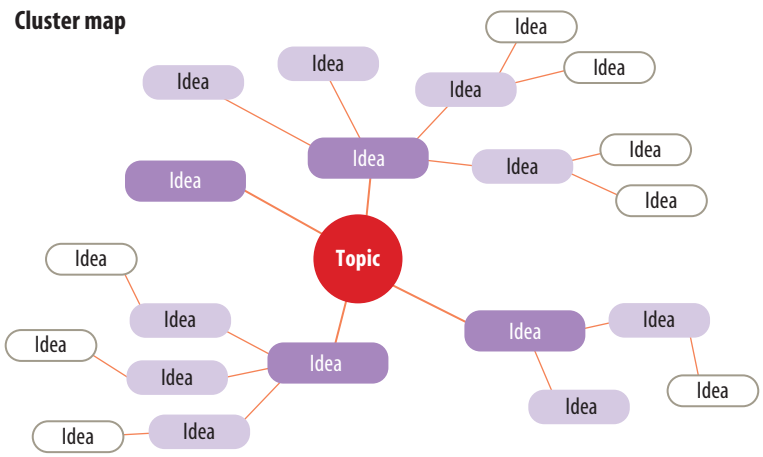
Difference

Related features radiate out in cluster maps but are boxed in affinity diagrams.

Cluster map

example

Cluster map



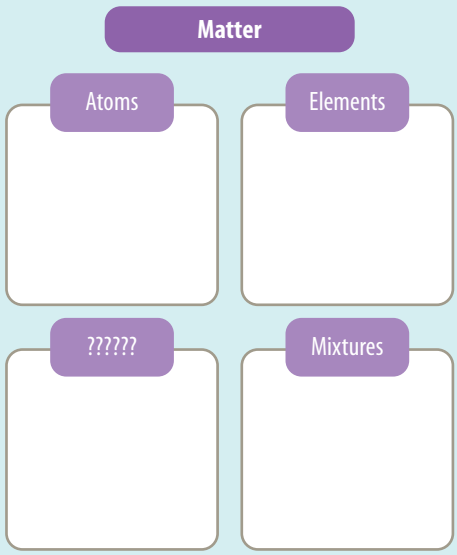
UNDERSTANDING AND INQUIRING

THINK AND CREATE

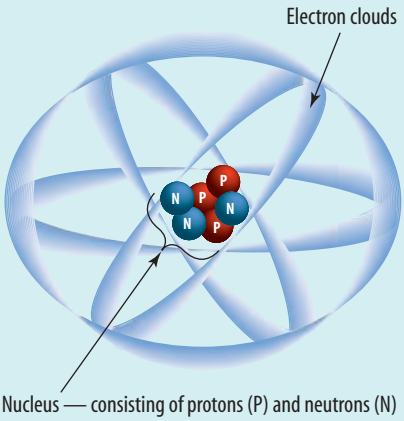
1 (a) Write each of the ideas, objects or substances listed below on a small card or sticky note.

air hydrogen protons oxygen water periodic table METALS bond sea water nucleus easily separated electrons salt carbon CHOCOLATE THICK SHAKE neutrons H_2O orbit

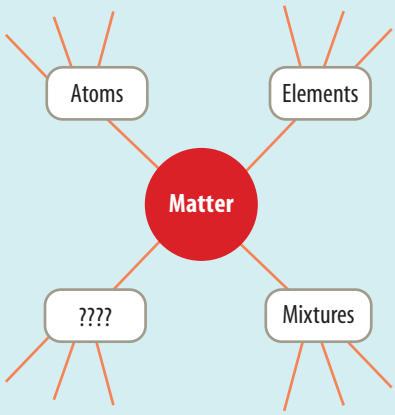
(b) Arrange the ideas, objects or substances on the cards or sticky notes into four categories in an affinity diagram like the one below. You will need to work out the name of the missing category.



2 Which element is shown below?



3 Use the ideas, objects and substances from question 1 to create a cluster map using the four categories as the main associations. Add as many associations as you can to the diagram. Don't forget that you can sometimes make links between the different arms of your cluster map.

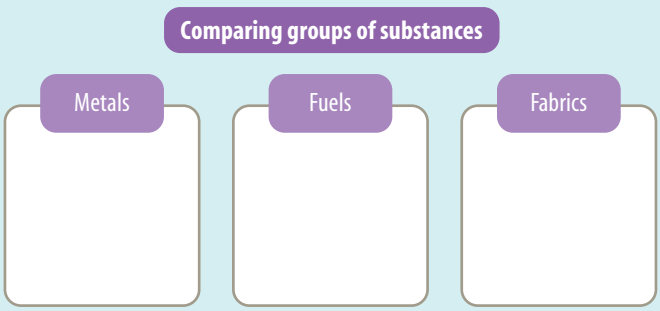


4 About 2500 years ago, when the Greek teacher Democritus suggested that all matter was made of atoms, other Greek thinkers proposed that there were four elements.

These elements were earth, air, fire and water. All other substances were combinations of these four elements.

Work in a small group to create a cluster map called 'elements' using 'earth', 'air', 'fire' and 'water' as the main associations. Add as many common substances as you can to your map.

5 Create an affinity diagram like the one below that illustrates the properties and uses of metals, fuels and fabrics. Include as many statements as you can about each group of substances.



STUDY CHECKLIST

ELEMENTS AND ATOMS

- describe some common chemical elements
- recall the chemical symbols of some common elements
- identify some of the dangers associated with some chemical elements
- model the structure of the atom and describe the characteristics of the three main particles
- recall that each chemical element is identified with a unique atomic number, which is equal to the number of protons in its nucleus
- distinguish between metals, non-metals and metalloids
- identify similar properties of groups of elements in the periodic table

COMPOUNDS AND MIXTURES

- distinguish between elements, compounds and mixtures
- recall that the atoms in compounds are bonded very tightly together
- recall that elements can be separated from compounds only as a result of a chemical reaction
- recognise that the properties of compounds are different from the elements that make them up
- use the formulae of simple compounds to identify the elements that make them up
- model the arrangement of atoms in the molecules of some compounds
- identify and describe some common compounds and their uses

SCIENCE AS A HUMAN ENDEAVOUR

- describe how an understanding of the particles that make up all matter has changed over time
- explain how the ideas about elements and the atom have changed over time
- describe the contributions of some of the scientists who have added to our knowledge of the atom and the elements

INDIVIDUAL PATHWAYS

eBookplus

Activity 7.1
Investigating substances
doc-6069

Activity 7.2
Analysing substances
doc-6070

Activity 7.3
Investigating substances further
doc-6071

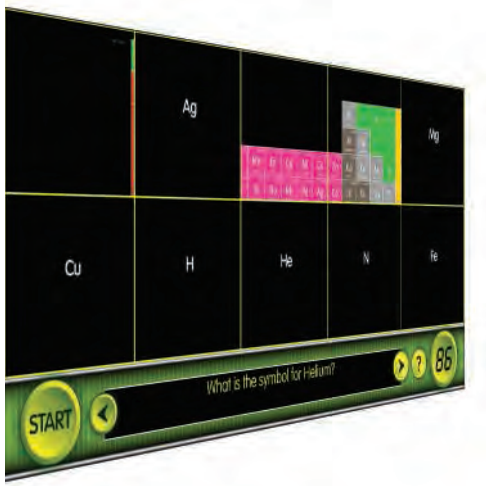
ICT

eBookplus **Summary**

INTERACTIVITIES

It's elementary! revelation game

In this revelation game, you must identify common elements from their symbols to reveal the full periodic table. You must answer quickly to complete the game in time.



Searchlight ID: int-0229

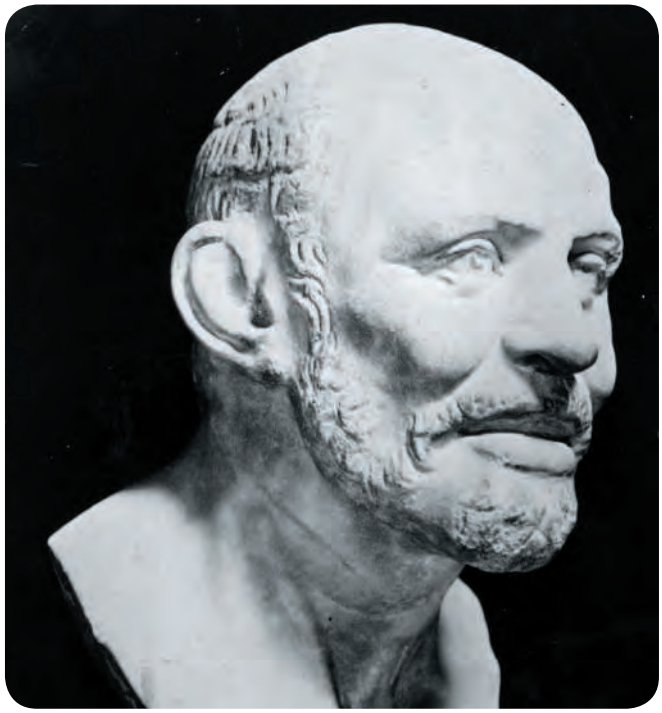
Making molecules

In this interactivity, you will use carbon, chlorine, hydrogen, nitrogen and oxygen atoms to create the correct models of a series of chemical formulae. Instant feedback is provided.

Searchlight ID: int-0228

LOOKING BACK

1 About 2500 years ago Democritus suggested what all substances were made up of.
(a) In what way was Democritus' idea about substances the same as the model that scientists currently use to describe substances?
(b) Suggest why most thinkers of the time disagreed with Democritus.



2 Copy and complete the following table, which describes the structure of atoms.

Part of atom	Location	Size and weight (relative)	Electric charge
		Large	Positive
Neutron			
	Outside the nucleus		

3 If a neutral atom has 12 protons, how many electrons does it have?

4 What takes up most of the space in an atom?

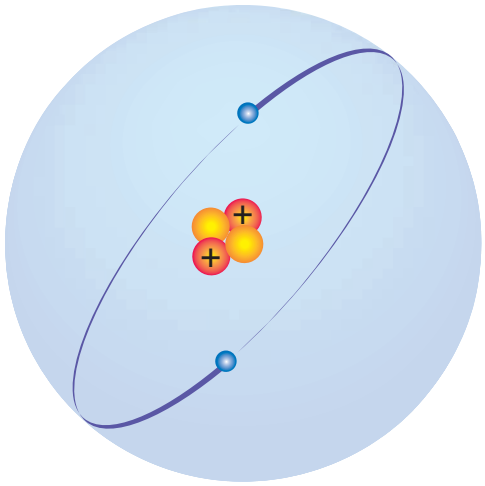
5 Identify the one feature that every single atom of the element sodium has in common.

6 What is the atomic number of each of the following elements?
(a) Hydrogen
(b) Carbon
(c) Uranium

7 How many protons does each of the elements listed in question 6 have in their nuclei?

8 How many electrons does each of the elements listed in question 6 have in their nuclei?

9 Make a copy of the diagram of the atom below and label an electron and the nucleus. Answer the following questions.
(a) How many protons does this atom have?
(b) How many neutrons does this atom have?
(c) How many electrons does this atom have?
(d) What is the atomic number of this atom?
(e) Describe one use of the element that is made up of these atoms.



10 Complete the following table to summarise what you know about metals and non-metals.

	Metals	Non-metals
Conduct electricity well		
Conduct heat well		
Surface features		
State at room temperature		
Malleable		
Ductile		
Brittle		

11 Which of the elements iron, lead, hydrogen, oxygen, silicon, uranium and sodium are:
(a) metals?
(b) metalloids?
(c) non-metals?

12 (a) Which element is used inside illuminated signs like the one below?

(b) To which group in the periodic table does this element belong?

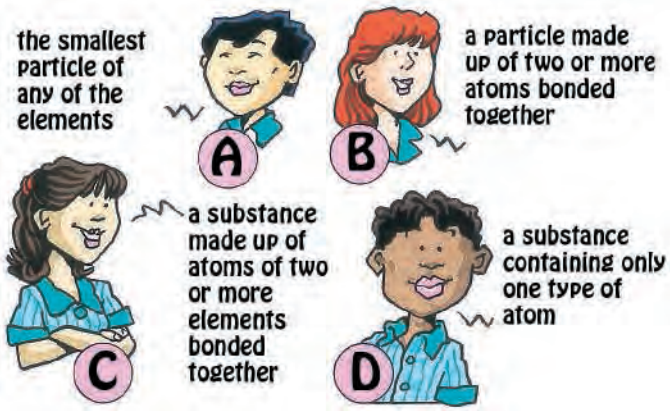
13 What event must take place in order to separate a compound into separate elements?

14 How are the molecules in polymers different from the molecules of other compounds?

15 Complete the table below to indicate whether the substances listed are elements, compounds or mixtures. Also indicate why you made that decision.

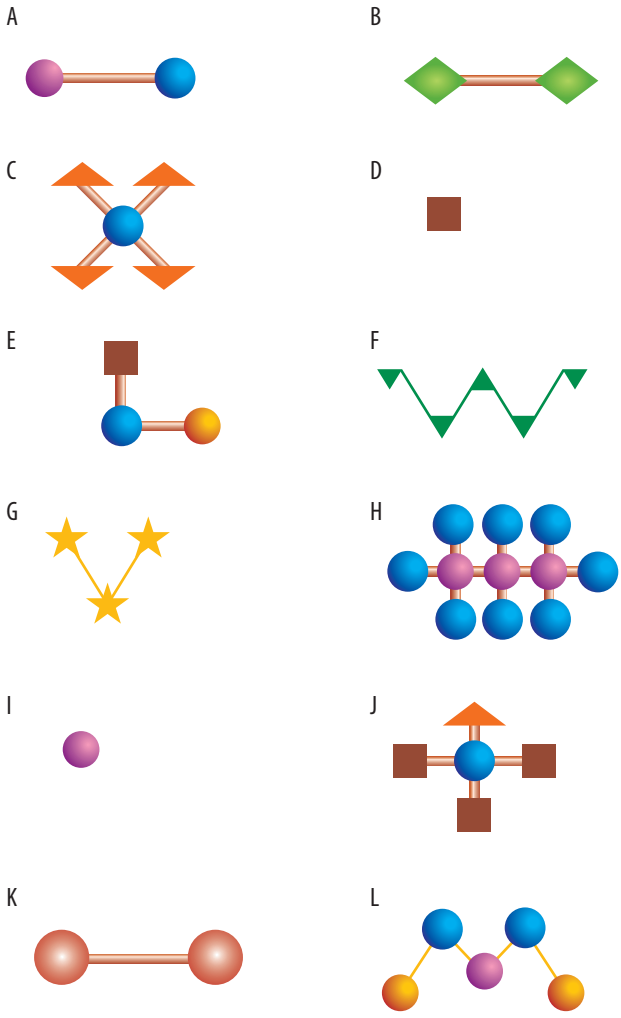
Substance	Element, compound or mixture?	Why do you think so?
Gold		
Diamond		
Carbon dioxide		
Air		
Sea water		
Pure water		
Iron		
Ammonia		
Table salt (NaCl)		

16 Which of 'the bits that matter' is represented by each of the cartoons below?



17 What do diamonds, the 'lead' in pencils and coal have in common?

18 Each of the diagrams below represents one of 'the bits that matter' that make up substances.



Which of the diagrams represents:
(a) an atom of an element?
(b) a molecule of an element?
(c) a molecule of a compound?

19 Most of the substances around you are compounds and mixtures.
(a) What differences could be observed between a mixture of hydrogen and oxygen and a compound of hydrogen and oxygen?
(b) Explain the difference between a compound and a mixture in your own words.

20 Respiration is a chemical reaction in which carbon dioxide is produced.
(a) Where in your body does respiration take place?
(b) What is released during respiration apart from carbon dioxide?
(c) Suggest how the carbon atoms in carbon dioxide enter your body.

21 Why doesn't water appear in the periodic table?

work sheet 7.5 Summing up

projectsplus

Science TV

SEARCHLIGHT ID: PRO-0090

Scenario

In the media world, programs that combine entertainment and education are known as 'edutainment'. With the success of edutainment programs such as *Mythbusters* (SBS), *Scope* (Network 10) and *The FundaMentals* (ABC), it seems that science is attracting a bigger share of the television market than many network executives would have expected. Now, your local TV network — Channel 55 — has decided to jump on the 'science as edutainment' bandwagon and has announced that it will be developing a program called *Science TV* next year.

To make *Science TV* more appealing to a younger audience, the developing executives of the program want it to be presented by a team of school students, who will do all of the introductions, explanations and experiments for each of the segments. It is important that the right team of students is found or the program will be canned after only a few episodes, so Channel 55 has announced that it is accepting online audition files from groups of students who think they have what it takes to be the *Science TV* stars.

Your task

Your group is going to put together a video submission that you could send to the Channel 55 developers to showcase how suitable you would be as the stars of *Science TV*.

The guidelines for the video submission from the Channel 55 website are as follows.

- The video must be between four and five minutes in length.
- The target audience of *Science TV* is between 8 and 14 years old.
- At least two people must be shown on camera.
- The video must be in the form of a chemistry segment that explains one of the following:
  (a) What is the difference between a physical change and a chemical change?
  (b) What are elements, compounds and mixtures?
  (c) How would we separate a mixture of iron filings, sand, copper sulfate and chalk dust?
- At least one experiment must be shown being performed in the segment — the experiment must be relevant to the segment and safe to perform (i.e. no explosions and no dangerous fumes produced).

The segment should be engaging and informative. It should have an introduction (either a scenario played out or a discussion between the presenters), an experiment to either test or demonstrate an idea, an explanation of the main concepts involved and a resolution that ties back into the original scenario or discussion. Remember: the main idea is to show that science is FUN!

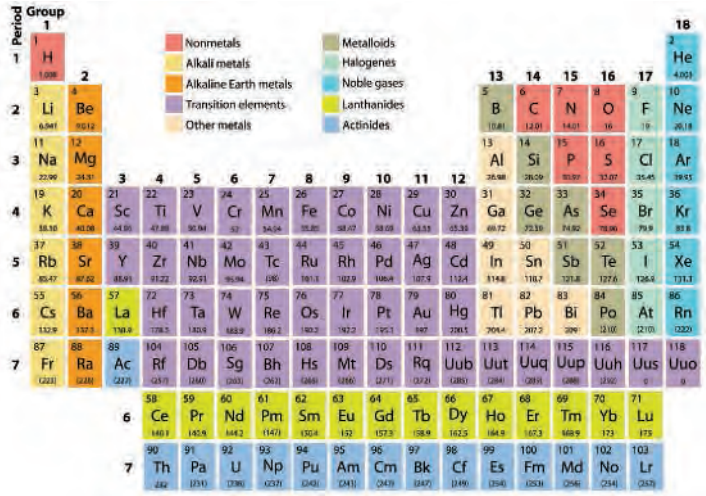
Process

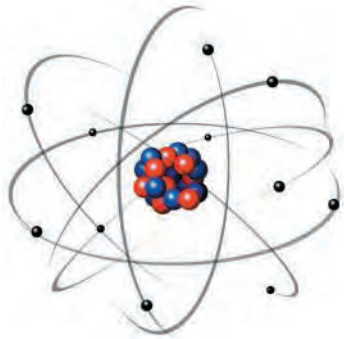
- Open the ProjectsPLUS application for this chapter located in your eBookPLUS. Watch the introductory video lesson and then click the 'Start Project' button to set up your project group. Save your settings and the project will be launched.
- Navigate to your Research Forum. Here you will find a number of pre-loaded research topics that will help you organise your presentation ideas. You can also add other research topics that you think may help you in your task.
- Research. Make notes of ideas that you can use in your segment, including interesting facts, good experiments and ways in which you could make your topic easier to understand — remember that the audience of *Science TV* is between 8 and 14 years old. Enter your findings as articles under your topics in the Research Forum. You should each find at least two sources (other than the textbook and at least one offline such as a book or encyclopedia) to help you discover extra information. You can view and comment on other group members' articles and rate the information they have entered. When your research is complete, print out your Research Report to hand in to your teacher.
- Visit your Media Centre and download the storyboard template to help you plan your audition segment. Your Media Centre also includes images and video clips that you may find useful to add to your audition segment where appropriate.
- Film the scenes for your segment using a webcam, digital camera or camcorder.
- Use video editing software such as Movie Maker to put the segment together for submission.

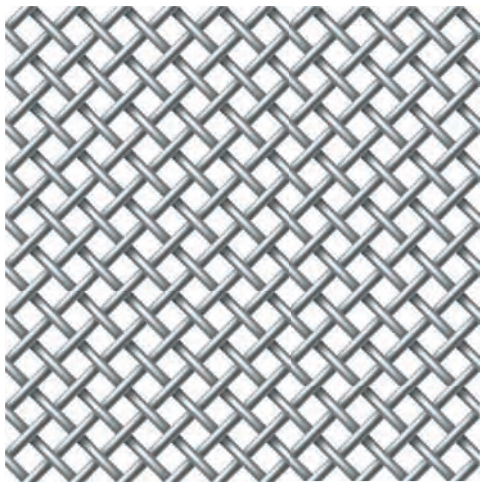
MEDIA CENTRE

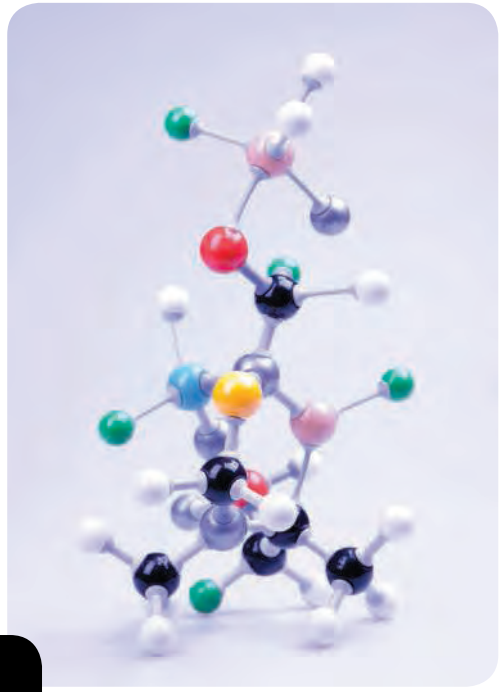
Your Media Centre contains:

- a storyboard template
- a selection of images
- a selection of video clips
- an assessment rubric.









SUGGESTED SOFTWARE

- ProjectsPLUS
- Word processing software (e.g. Word)
- Movie Maker (PC) or iMovie (Mac) or other video editing software
- Internet access

Your ProjectsPLUS application is available in this chapter's Student Resources tab inside your eBookPLUS. Visit www.jacplus.com.au to locate your digital resources.

Weblinks

eBookplus

- Use the **Scope** weblink in your eBookPLUS to look at episodes of the Channel 10 children's science program *Scope*.
- Use the **ExperiMentals** weblink in your eBookPLUS to see episodes of the ABC Kids' science program *The ExperiMentals*.
- Use the **Slime** weblink in your eBookPLUS to see the transcript of the 'Slime' segment from the *Scope* episode 'Solids, Liquids, Gases'.
- Use the **Storyboard** weblink in your eBookPLUS to find out how to develop your storyboard.
- Use the **Movie Maker** weblink in your eBookPLUS if you are a PC user, to get help with using Movie Maker.
- Use the **iMovie** weblink in your eBookPLUS if you are a Mac user, to get help with using iMovie.

8 Chemical change

Chemical reactions are happening everywhere. Chemical reactions in your body digest food, decay your teeth and much more. Chemical reactions occur in batteries to provide electricity, in the oven when you bake a cake, in your hair when it is bleached or coloured and in your car when it burns fuel. The list goes on — and on — and on.

Chemical reactions can be explosive and colourful.

OVERARCHING IDEAS

- Patterns, order and organisation
- Form and function
- Stability and change
- Scale and measurement
- Matter and energy

SCIENCE UNDERSTANDING

Chemical change involves substances reacting to form new substances.

Elaborations

Identifying the differences between chemical and physical changes

Identifying evidence that a chemical change has taken place

Investigating simple reactions such as combining elements to make a compound

Recognising that the chemical properties of a substance, for example its flammability and ability to corrode, will affect its use

THINK ABOUT REACTIONS

- Why does a half-eaten apple go brown?
- How is an explosion different from other chemical reactions?
- Why does a spoonful of sugar dissolve more quickly than a sugar cube?
- What makes a nail rust?
- Why is the Sydney Harbour Bridge continually being painted without a break?
- What is a backdraught and what causes it?
- What is the difference between soaps and detergents?
- What makes Lycra® so special?

Why is recycling so important?

YOUR QUEST

What is a chemical reaction?

You've probably already heard a lot about chemical reactions — at school, on television, at the movies or in books. But what is a chemical reaction — and how do you know whether a chemical reaction has taken place?

THINK

Check out the images on this page and answer the questions based on what you already know about chemical reactions.

The boiling liquid in the pot below is changing colour. It began as a mixture of reds, greens and blues and, after stirring, is changing into a dangerous-looking yellow soup.

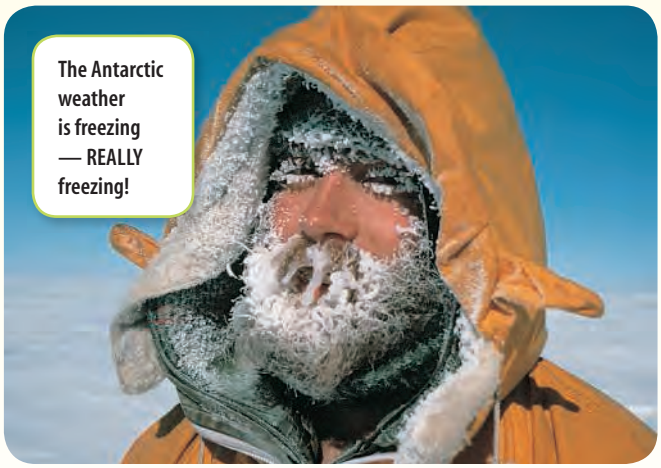
1 Write down your opinion about whether or not a chemical reaction is taking place.

2 Explain how you know whether a chemical reaction has taken place.

3 Is there a chemical reaction taking place underneath the pot? Explain your answer.

4 Clouds are forming above the pot. Is this evidence of another chemical reaction? Explain your answer.

5 Is the frozen substance in this man's beard and the inside edge of his hood the result of a chemical reaction? Explain your answer.

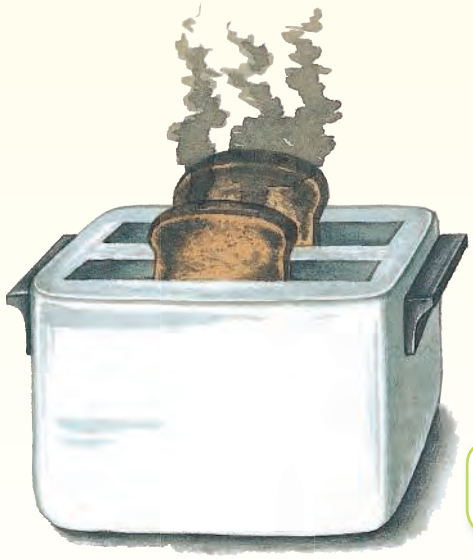


6 Runners in long-distance races sweat heavily. The water lost due to sweating evaporates from the skin. Is this evaporation an example of a chemical reaction? Explain your answer.

It was a long, tough and hot cross-country race. Just as well this runner drank lots of water along the way.

7 Does a chemical reaction take place when you burn toast? What observations support your answer?

8 Does a chemical reaction take place when you toast bread without burning it? Explain your answer.



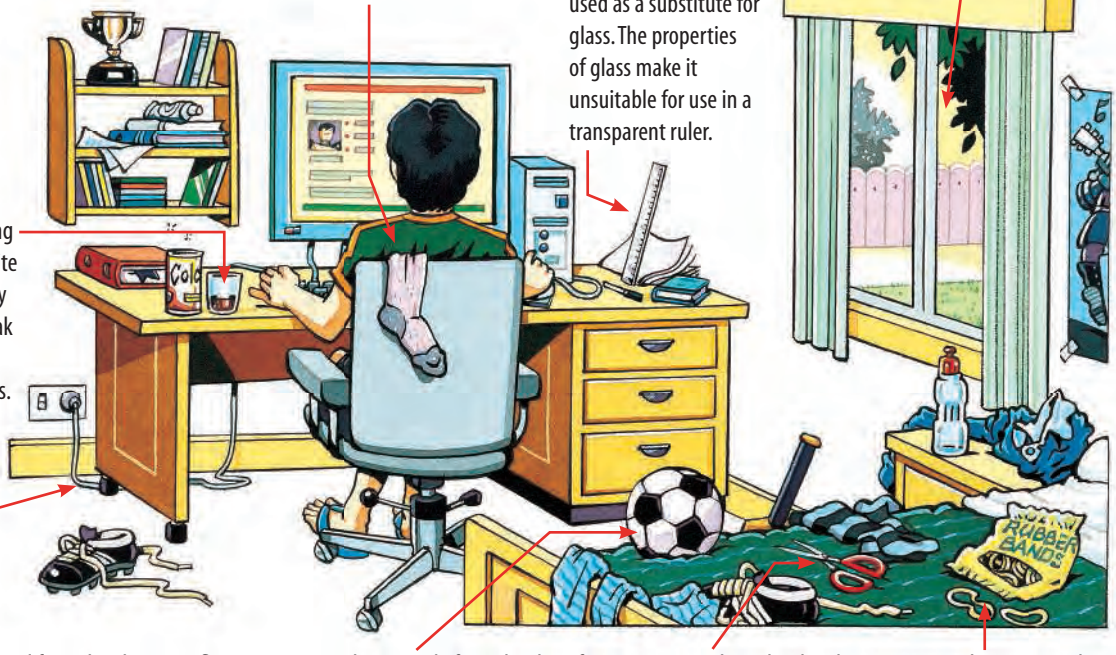
Patterns, order and organisation: Physical and chemical properties

Thousands and thousands of different substances are used in the objects that surround you. Each substance shown in the scene below has physical and chemical properties that make it useful for a particular purpose.

Physical or chemical?

The properties of most substances fall into two categories — physical or chemical. **Physical properties** are those that you can either observe using your five senses — seeing, hearing, touching, smelling and tasting — or measure directly. Examples include colour, size, shape, texture, temperature, malleability and ductility, but there are many, many more. **Chemical properties** are those that describe how a substance combines with other substances to form new chemicals or how a substance breaks up into two or more different substances. Examples of chemical properties include **flammability**, **reactivity** and **toxicity**.

- Flammability is an indicator of how easily a substance catches fire. When a substance burns, it creates new substances.
- Reactivity is a measure of how easily a substance combines with other substances to produce new substances.
- Toxicity refers to the danger to your health caused when poisonous substances combine with chemicals in your body to produce new substances and damaging effects.



You are protected from the electricity flowing through the metal wires in the centre of this cable by its outer coat of plastic. Most metals are **ductile**. That means that they can be drawn into wires. Electricity flows easily through metal but not through most plastics.

Leather is made from the skin of animals. It is treated with chemicals to preserve it, remove the hair and give it colour. Its properties of toughness and **elasticity** make it ideal for use in soccer balls.

Metals are hard and **malleable**. This means that they can be beaten or bent into shape — and flattened.

The property that makes the rubber in a rubber band useful is its elasticity. This property also makes it useful for car and bicycle tyres.

INQUIRY: INVESTIGATION 8.1

Checking out properties

KEY INQUIRY SKILLS:

- planning and conducting
- communicating

Equipment:

a range of small items that might include a tennis ball, a table-tennis ball, a table-tennis paddle, a dishwashing sponge, assorted fabrics (for example, wool from a jumper, nylon socks and stockings, polyester, cotton), a magnifying glass or lens, a roll of sticky tape, a candle, paper clips, small springs, polystyrene cups, foam rubber, aluminium foil, a clear plastic bottle of dishwashing detergent, a bottle of perfume

- Work in groups of three or four so that you can discuss the properties. Work on one item at a time.
- For each item, list all the physical properties that you can think of. Some items will consist of more than one substance. In those instances, list the physical properties of each substance.
- For each physical property of each substance in the item, explain how that property makes the substance useful for its purpose.

DISCUSS AND EXPLAIN

List some tests that you could perform to discover some of the chemical properties of the substances in some of the items.

UNDERSTANDING AND INQUIRING

REMEMBER

1 Most metals can be described as ductile and malleable. How does each of these properties make metals useful?

2 Which of the two categories of properties can be directly observed with your five senses?

3 Some substances have the chemical property of toxicity. What does this mean?

THINK

4 What property of leather makes a soccer ball easy to grip?

5 List the properties of the packaging of potato chips.

6 List at least two physical properties that you can describe using your sense of hearing.

7 Three chemical properties are listed on the previous page — can you think of another one? Explain that chemical property using an example.

8 Look at the illustration on the previous page. Choose three items and explain how the properties of the materials make them suitable for their purpose.

IMAGINE

9 Imagine that you are designing a spacecraft that will take astronauts to the moon and back. List the properties that the outer surface of the spacecraft would need to have. Include at least two chemical properties.

Why are potato chip bags pumped full of air? Why is foil often used for the packaging? Could a different material be used?

INVESTIGATE

10 Road bike frames for serious and competitive cyclists are made from aluminium, carbon fibre or a combination of both substances. Find out:

(a) what properties both aluminium and carbon fibre have that make them suitable for the frames of road racing bikes

(b) which properties make aluminium bikes more suitable than carbon fibre for some purposes. (Note that cost is not a property!)

work sheet 8.1 Properties of materials

Time for some changes

Chemical changes

When you hard-boil an egg, a **chemical change** takes place. At about 100°C the eggwhite and yolk undergo chemical changes that alter their chemical make-up. Bonds between atoms or molecules are broken or new bonds between these particles are formed. Unlike cooling melted chocolate, which brings about another physical change, cooling the egg will not change it back to its raw state. In fact, most chemical changes are difficult to reverse.

When paper is burnt, it combines with oxygen to form ash and smoke. This is a chemical reaction, because new substances are formed. Burning gas in a Bunsen burner is also a chemical change. The methane gas burns with oxygen in the air to form two new substances: carbon dioxide and water vapour. During this chemical reaction heat is also produced.

How does a candle burn?

When you try to light a piece of solid wax it melts, but it does not burn. If solid wax doesn't burn, how does a candle burn? Is it the string wick that is in the middle of the candle that burns? String will burn, but it doesn't burn like a candle does. How then does a candle burn?

When you light the wick of a candle, the wax at the top of the candle melts. The molten wax is drawn up the wick just as water soaks into a paper towel. As the liquid wax flows up the wick and gets closer to the heat of the flame it **evaporates**. The wax vapour mixes with oxygen in the air and burns.

How does a candle burn?

If you enjoy eating chocolate you'll know that it's not so easy to eat on a hot summer's day. Energy transferred from the hot air surrounding the chocolate causes it to melt. The chocolate changes **state** from solid to liquid. The change in state of the chocolate is reversible. The melted chocolate can be cooled and solid chocolate will form again.

INQUIRY: INVESTIGATION 8.2

A burning candle

KEY INQUIRY SKILLS:

- processing and analysing data and information
- communicating

Equipment:

safety glasses *candle*
jar lid *matches*
heatproof mat

CAUTION: Do not smell the vapour directly. Fan the odour to your nose with your hand.

- Place a jar lid on a heatproof mat.
- Light a candle and allow a drop of wax to drip onto the jar lid. Place the candle on the drop of wax and fix it to the jar lid.
- Observe the candle and write down as many observations of the burning candle as you can.
- Discuss your observations with others in your group.
- Blow out your candle and you will see the white vapour rising from the top of the wick.
- To confirm that the white vapour is not smoke, carry out the following test: Relight the candle. Once it is burning properly, blow it out. Quickly light the top of the vapour trail. The flame should run down the vapour to the wick and relight the candle.

DISCUSS AND EXPLAIN

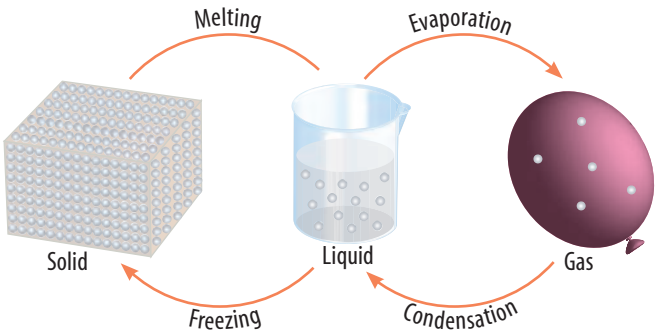
1. How far is the flame from the solid wax?
2. The solid wax has formed a little pool of liquid wax around the wick. Why has this happened?
3. Describe the odour of the vapour that is present after the candle is blown out.
4. Draw a diagram of a candle and its flame. Label this diagram to explain how a candle burns.
5. Explain why lighting the wax vapour causes the candle to relight.

Physical changes

The changes to the chocolate described on the previous page are **physical changes**. Melting, evaporation, condensation and freezing are all physical changes. Changes of state are reversible physical changes.

Changes in the shape or size of a substance are also physical changes. These are not always reversible. For example, if you drop an egg, its shape is changed forever. But when you stretch an elastic band, it can quickly return to its original shape when you let it go.

A physical change does not break any bonds between the atoms of a substance, nor does it create any new bonds. No new substances are formed.



Changes of state are reversible physical changes.

Describing change

In a burning candle, there are both physical and chemical changes. The melting of the solid wax to form liquid wax and the evaporation of liquid wax to form the wax vapour are physical changes. The burning of the wax vapour is a chemical change. The wax vapour reacts with oxygen in the air to form new substances including carbon dioxide and ash.

Physical and chemical changes can be described using word equations.

Melting chocolate can be described by the equation:

solid chocolate ⟶ liquid chocolate

The burning of paper can be described by the equation:

paper + oxygen ⟶ smoke + ash

UNDERSTANDING AND INQUIRING

REMEMBER

1 What is the difference between a physical and a chemical change?

2 Describe two examples of a physical change.

3 Describe two examples of a chemical change.

4 Match the word with the definition.

Change from solid to liquid	Freezing
Change from gas to liquid	Melting
Change from liquid to solid	Condensation
Change from liquid to gas	Evaporation

5 Which type of physical change can always be reversed by heating or cooling?

THINK

6 Copy and complete the table below.

7 Write two word equations to describe the changes of state that take place when a candle burns.

8 Write a word equation to describe the chemical change that takes place when a candle burns.

9 When you hard-boil an egg, the inside of the egg gets hard. Why is this a chemical change and not a physical change?

CREATE

work sheet 8.2 Changing states

10 Candles are a good example of both physical change and chemical change. Write a poem about a candle burning.

Observation	Physical or chemical change
Water freezing to form snow	
A cake cooking	
Lighting the gas on the stove	
Petrol evaporating at the petrol pump	
Lighting a match	
Steam condensing on the bathroom mirror	
Melting gold to cast gold bars	
Dynamite exploding	
Bleaching a stain	
Dissolving eggshell in acetic acid	

Chemical reactions

A **chemical reaction** is a chemical change in which a completely new substance is produced.

Almost all the products you use or wear each day are made by chemical reactions; from cosmetics to concrete, plastics to paper, glass to graphite, stainless steel to shampoo, fibres to food additives, margarine to medicines and many, many more.

Reactants and products

The substances that you begin with in a chemical reaction are called the **reactants**; the substances that are produced are called the **products**. When you wash the dishes, a chemical reaction occurs between the detergent and the mess on the dishes. When you shampoo your hair, some of the chemicals in the shampoo react with the greasy substances on your scalp that contain dust, dirt and tiny organisms such as bacteria that can make your hair unhealthy.

WHAT DOES IT MEAN?

The word *product* comes from the Latin word ***productum***, meaning 'thing produced'.

WHERE'S THE EVIDENCE?

You can usually tell whether a chemical reaction has taken place by identifying one or more of these clues:

- a **precipitate** (cloudiness caused by a solid substance) appears in a liquid or gas
- an odour is detected
- bubbles appear
- there is an increase or decrease in temperature
- light is emitted or a flame appears
- there is a change in colour.

However, the only way to be certain that a chemical reaction has taken place is to identify one or more new chemical products.

The bright yellow precipitate in this flask is a product of the chemical reaction that takes place when colourless solutions of lead nitrate and potassium iodide are mixed together.

Chemical reaction experiments

Before you start each of the four investigations on the next page, design a suitable table for recording your observations.

As you perform the experiments:

1. Make a note of the appearance of each of the reactants you are starting with.
2. Carry out the experiment and observe carefully to detect any changes that occur.
3. Describe the changes that take place and products of the reaction.

Safely glasses should always be worn during experiments involving chemical reactions.

INQUIRY: INVESTIGATION 8.3

Magnesium metal in hydrochloric acid

KEY INQUIRY SKILLS:

- planning and conducting
- processing and analysing data and information

Equipment:

heatproof mat

safety glasses

test tube and test-tube rack

1 cm piece of magnesium ribbon

dropping bottle of 0.5M hydrochloric acid

- Put the magnesium in the test tube.
- Add 20 drops of hydrochloric acid to the test tube.

CAUTION: The test tube may become quite hot.

- Record your observations.

INQUIRY: INVESTIGATION 8.4

Heating copper carbonate

KEY INQUIRY SKILLS:

- planning and conducting
- processing and analysing data and information

Equipment:

Bunsen burner, heatproof mat and matches

safety glasses

test tube and test-tube rack

test-tube holder

spatula

copper carbonate powder

- Pour 2 spatulas of copper carbonate into the test tube.
- Using the test-tube holder, heat the test tube in the Bunsen burner flame. Remember to move the test tube in and out of the flame and point the test tube away from people.
- Stop heating when the copper carbonate has changed colour.
- Record your observations.

INQUIRY: INVESTIGATION 8.5

Sodium sulfate and barium chloride

KEY INQUIRY SKILLS:

- planning and conducting
- processing and analysing data and information

Equipment:

heatproof mat

safety glasses

test tube and test-tube rack

test-tube holder

dropping bottle of 0.1M sodium sulfate solution

dropping bottle of 0.1M barium chloride solution

- Put 20 drops of sodium sulfate solution in the test tube.
- Add 20 drops of barium chloride solution to the test tube.
- Record your observations.
- What observation provides evidence that a chemical reaction has taken place?

INQUIRY: INVESTIGATION 8.6

Steel wool in copper sulfate solution

KEY INQUIRY SKILLS:

- planning and conducting
- processing and analysing data and information

Equipment:

heatproof mat

safety glasses

test tube and test-tube rack

glass stirring rod

1 cm ball of steel wool

dropping bottle of 0.5M copper sulfate solution

- Put the steel wool in the test tube, using the glass stirring rod to push it gently to the bottom of the test tube.
- Add copper sulfate solution to the test tube to a depth of 2 cm.
- Record your observations.
- What observation provides evidence that a chemical reaction has taken place?

FEELING HUNGRY?

This hamburger is an incredible mixture of chemicals. Every part of it has been produced by chemical reactions. The most important chemical reaction in growing the lettuce is photosynthesis, in which the reactants are carbon dioxide and water. The products are glucose (a type of sugar) and oxygen. That chemical reaction cannot take place without light and a chemical called chlorophyll, which gives plants their green colour. In fact, none of the other components of the hamburger could be grown or produced without photosynthesis.

The substance used to make cheese is the product of a chemical reaction in which a protein, casein, in cow's milk reacts with acetic acid when heated. Acetic acid is found in orange and lemon juice and is more commonly known as vinegar.

Writing word equations

Each of the chemical reactions in Investigations 8.3–8.6 can be described by a chemical word equation. In each case the reactants are on the left side of the equation and the products are on the right side of the equation.

1. When magnesium metal reacts with hydrochloric acid, hydrogen gas and magnesium chloride are formed:

   magnesium + hydrochloric acid $\longrightarrow$ hydrogen + magnesium chloride

2. Heating copper carbonate forms copper oxide and carbon dioxide:

   copper carbonate $\xrightarrow{\text{heat}}$ copper oxide + carbon dioxide

   Although heat is required for this chemical reaction to take place, it is not a substance and therefore is not a reactant. It is written above the arrow for this reason.

3. Sodium sulfate and barium chloride in solution react to form solid barium sulfate and sodium chloride, which remains dissolved in the solution:

   sodium sulfate solution + barium chloride $\longrightarrow$ solid barium sulfate + sodium chloride solution

4. Steel wool (which is made of iron) dissolves in copper sulfate solution to form iron sulfate solution and copper metal:

   iron + copper sulfate solution $\longrightarrow$ iron sulfate solution + copper

UNDERSTANDING AND INQUIRING

REMEMBER

1 Write down four observations that could provide evidence that a chemical reaction has taken place.

2 When magnesium metal reacts with hydrochloric acid, hydrogen gas and magnesium chloride are formed.
   (a) What are the products?
   (b) What are the reactants?

3 What is the only real proof that a chemical reaction has taken place?

THINK

4 Write word equations for the following chemical reactions.
   (a) Octane gas is burnt with oxygen in a car engine to produce carbon dioxide and water.
   (b) Sodium metal reacts with chlorine gas to form sodium chloride.
   (c) Hydrogen gas and oxygen gas combine to form water.
   (d) Zinc metal dissolves in hydrochloric acid to form hydrogen gas and zinc chloride.

5 Explain why the reaction that takes place when copper carbonate is heated is called a decomposition reaction.

6 Explain why the tomato, cheese, bread and meat in a hamburger cannot be grown or produced without photosynthesis.

7 Describe the evidence that one or more chemical reactions takes place when meat is grilled.

CREATE

8 Performing some chemical reactions can be dangerous. Design a safety poster for one of the experiments you have done.

9 Choose one of the materials below and find out how it is manufactured. Write a report about the chemical reactions used in its production.
   - Glass
   - Soap
   - Margarine
   - Paper
   - Nylon
   - Polyethylene

work sheets 8.3 Physical and chemical changes
8.4 Describing chemical changes

8.4 SCIENCE UNDERSTANDING

Fast and slow reactions

Explosions

Explosions are chemical reactions that take place very quickly. Explosions also release a lot of heat, light and noise. In less than 10 milliseconds, a dynamite blast in a large mine can produce 5 billion litres of gas and release 20 billion joules of energy — enough energy to tear any rock apart.

WHAT DOES IT MEAN?

The word *explosion* comes from the Latin word ***explosio***, meaning 'driven off by clapping or hooting'.

Explosions are fast chemical reactions.

In contrast, the chemical reactions that cause concrete to set are very slow. It can take several days for concrete to set hard. Rusting is another example of a slow chemical reaction.

The **rate of a reaction** is a measure of how quickly a chemical reaction occurs. How can the rate of a reaction be changed to make a slow reaction happen quickly or make a fast reaction slow down?

Speeding up a reaction with heat

Heating a substance adds energy to its particles. They move more rapidly and collide more frequently. When they collide, bonds between the particles are broken and new ones are more easily formed with the particles of other substances. Heating substances, therefore, usually causes the rate of a chemical reaction to increase.

Stay cool

Food 'goes off' because micro-organisms cause chemical reactions in the food that make it rot. These chemical reactions can be slowed by lowering the temperature of the food. Imagine what life would be like without a refrigerator or freezer.

Catalysts

A **catalyst** is a chemical that can speed up a chemical reaction but is still present once the reaction has finished. Catalysts are not reactants because they are not changed by the reaction.

Catalytic converters in car exhausts use a precious metal, such as platinum, as a catalyst. This enables nitrogen oxide to react with toxic gases, such as carbon monoxide, to form the less harmful carbon dioxide and nitrogen gases; this reaction would not occur in the absence of the catalyst. This reaction can be shown as:

$$\text{carbon monoxide} + \text{nitrogen oxide} \xrightarrow{\text{platinum}} \text{carbon dioxide} + \text{nitrogen}$$

The catalysts in living things are called **enzymes**. Enzymes in the human body help to digest the food you eat more quickly.

Apples and other fruits go brown because chemicals in them, called phenolics, react with oxygen in the air. The brown chemical products are called quinones. Enzymes in the fruit speed up the reaction. The chemical word equation for this reaction is:

$$\text{phenolics} + \text{oxygen} \xrightarrow{\text{enzymes}} \text{quinones}$$

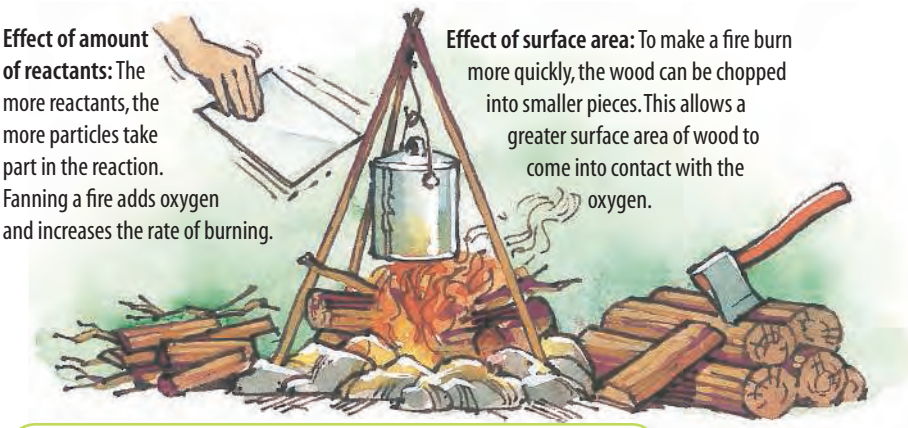
Apples go brown when phenolics react with oxygen in the air. Enzymes speed up the reaction.

Altering the reactants

No doubt you have been in situations where you have wanted to increase the rate of a chemical reaction. Perhaps you wanted a faster burning camp fire, or a tablet to dissolve or a stain to be removed more quickly. What would you do in each case to make the reaction faster?

One solution is to add more reactant. In the case of the camp fire you can add more wood, or more oxygen by fanning the fire. To make the stain disappear more quickly, you could add more bleach.

Another solution is to increase the surface area of the reactants so that they can mix more easily. In the case of the camp fire, you can chop the wood or use smaller pieces of twigs and leaves. To help the tablet dissolve, you could crush it.



The effects of amount of reactants and surface area on the rate of reaction

INQUIRY: INVESTIGATION 8.7

The effect of temperature on a reaction

KEY INQUIRY SKILLS:

- processing and analysing data and information
- evaluating

Equipment:

safety glasses
test-tube holder
heatproof mat
Bunsen burner
matches
marble chips
dropping bottle of 1M hydrochloric acid
test tube
test-tube rack

- Carefully slide one or two marble chips to the bottom of the test tube.
- Add 1M hydrochloric acid to half-fill the test tube.
- Observe the reaction.
- Now gently heat the test tube and observe the reaction.

DISCUSS AND EXPLAIN

1 Has a chemical reaction occurred? Describe the evidence that you have observed.

2 What effect did heating the test tube have on the rate of this reaction?

3 In this chemical reaction, the calcium carbonate that makes up the marble chips reacts with the hydrochloric acid to produce calcium chloride, water and carbon dioxide gas.

(a) List the reactants.
(b) List the products.
(c) Write a word equation for this chemical reaction.

4 Suggest a different method of increasing the rate of reaction.

INQUIRY: INVESTIGATION 8.8

Changing the rate of reaction

KEY INQUIRY SKILLS:

- questioning and predicting
- planning and conducting
- communicating

Equipment:

safety glasses
test tubes and test-tube rack
mortar and pestle
0.5M hydrochloric acid
measuring cylinder
heatproof mat
white chalk
spatula
1M hydrochloric acid

Hydrochloric acid reacts with chalk to produce carbon dioxide gas, water and calcium chloride.

- Place a small amount of chalk in a test tube and add enough hydrochloric acid to cover it. Observe the chemical reaction.
- Discuss with your partner how you could use this reaction to demonstrate one of the hypotheses below.
  (a) That increasing the concentration or amount of reactants will speed up a chemical reaction
  (b) That increasing the surface area of reactants will speed up a chemical reaction
  (c) That decreasing the concentration or amount of reactants will slow a chemical reaction
- Design your experiment and write down the method that you have chosen.
- Predict the results you would expect to obtain that would support the hypothesis you have chosen.
- Perform the experiment.
- Prepare a report of your findings.

HOW ABOUT THAT!

Have you ever had a composite resin filling in your tooth? The dentist uses blue light or ultraviolet (UV) radiation to set this type of filling. The visible or UV light speeds up the reactions that cause the materials in the filling to harden. Without the UV light, you would be waiting for hours for this type of filling to set.

UV light can speed up the setting of a composite resin filling.

UNDERSTANDING AND INQUIRING

REMEMBER

1 What is the rate of a reaction?
2 State five different methods of changing the rate of a reaction.
3 How does heating increase the rate of a reaction?
4 What is a catalyst?
5 How do you know that a catalyst is not a reactant?
6 What is an enzyme?

THINK

7 Does a refrigerator stop food from rotting or does it just slow the rotting? Explain your answer.
8 What is the point of adding catalysts to washing powders?
9 Which will dissolve more quickly — a sugar cube or the same amount of sugar in a teaspoon? Explain your answer.

CREATE

10 Write and act out a short dramatic performance to show how a piece of chalk takes longer to react with an acid than the same amount of crushed chalk.

INVESTIGATE

11 Change the temperature, concentration and surface area in the **Reaction rates** interactivity in your eBookPLUS to see how they affect the rate of a reaction; then decide how the rates of particular reactions could be changed. **int-0230**

12 Test your ability to identify different types of reactions by completing the **Time Out: 'Reactions'** interactivity in your eBookPLUS. **int-0759**

work sheet 8.5 Speeding up reactions

Rusting out

Rusting is an example of **corrosion**. Corrosion is a chemical reaction that occurs when substances in the air or water around a metal 'eat away' the metal and cause it to deteriorate.

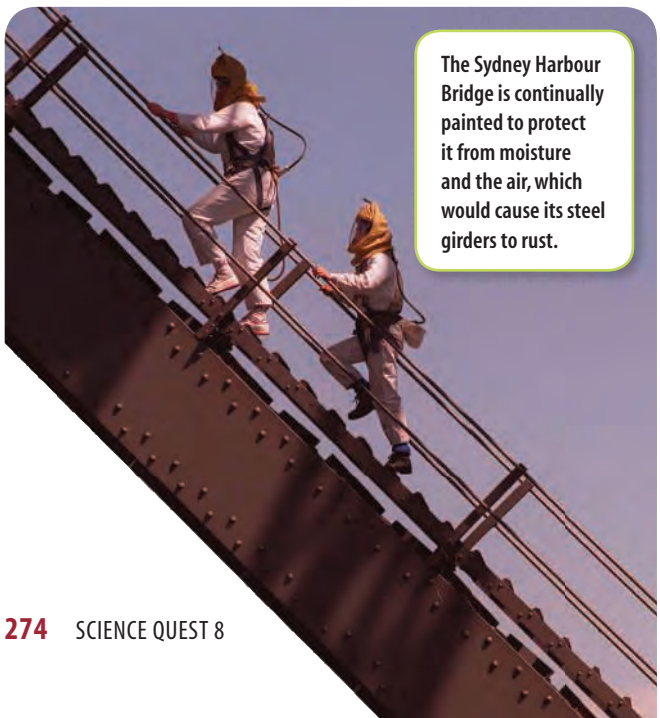
There are many examples of corrosion: silver tarnish; the green film that forms on copper or brass objects; and, the most common one, the rusting of iron. Corrosion causes enormous damage to buildings, bridges, ships, railway tracks and cars.

Rust

Rust is the flaky substance that forms when iron corrodes. Iron reacts with water and oxygen in the air to form iron oxide and other iron compounds that make up the familiar red-brown substance known as rust. Rusting is a slow chemical reaction that can be represented by the following word equation:

iron + water + oxygen ⟶ rust

Even strong buildings and bridges that are made from steel, an alloy of iron, are weakened by rusting. The Sydney Harbour Bridge, for example, is continually painted to protect it from moisture and the air, which would cause its steel girders to rust. Ships and cars are also constructed largely of steel. Despite the strength of steel, it needs to be protected from the corrosive effects of the environment.



The Sydney Harbour Bridge is continually painted to protect it from moisture and the air, which would cause its steel girders to rust.

INQUIRY: INVESTIGATION 8.9

Observing rusting

Steel wool is made from iron. You can observe rusting of the iron in steel wool by performing the following experiment.

KEY INQUIRY SKILL:

- processing and analysing data and information

Equipment:

glass Petri dish

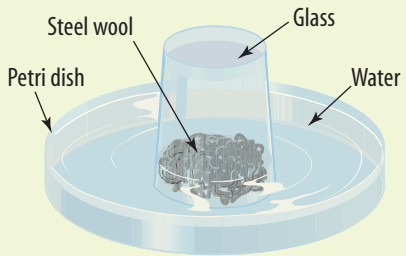
water

steel wool (without any soap)

small glass

permanent marker

- Pour some water into the Petri dish.
- Place the steel wool in the middle of the Petri dish.
- Cover the steel wool by placing the glass over it upside-down.
- Mark the level of the water on the outside of the glass with a permanent marker.
- Leave for several days, adding water as required to keep the level at the mark on the glass.



Observing the rusting of iron

- Construct a table in which you can record your observations over several days.

DISCUSS AND EXPLAIN

1 What did you observe about the level of water inside the glass? Can you explain why this happened?

2 Write down a word equation for the chemical reaction that occurred inside the glass.

Speeding up rusting

Some substances in the environment make rusting happen much more quickly. One of the most effective of these is salt. Steel dinghies that are used in the ocean rust much faster than those that are used only in fresh water. This is because the salt in the sea water speeds up the reaction between oxygen in the air and the iron in the steel.

Some chemicals released from factories also increase the rate of rusting. A CSIRO study conducted in Melbourne found that rusting rates were high near airports and sewage treatment plants.

Rusting is much slower in dry environments like deserts, where the rainfall is nearly zero and there is very little water vapour in the air.

INQUIRY: INVESTIGATION 8.10

Investigating the corrosion of different metals

KEY INQUIRY SKILLS:

- planning and conducting
- processing and analysing data and information
- evaluating
- communicating

Equipment:

small strips of a range of metals such as copper, aluminium, zinc and magnesium

sandpaper

other equipment approved by your teacher

- Design and carry out an investigation to study the resistance of a selection of different metals to corrosion. Ensure that appropriate variables are controlled. Before commencing, clean the metal strips with sandpaper to ensure that any coatings already caused by corrosion are removed.
- Write a report on your investigation that includes your aim, method, results (including a table), discussion and a clear conclusion listing the metals in order of resistance to corrosion, from most resistant to least resistant. Include the answers to the questions below in your discussion.

DISCUSS AND EXPLAIN

1 Identify the independent and dependent variables in your investigation.
2 Name the variables that you controlled.
3 Suggest how you might be able to improve or speed up the investigation.

HOW ABOUT THAT!

In the Mojave Desert of Southern California, hundreds of unused aircraft are stored out in the open air. Due to the dry air, rusting occurs extremely slowly. As a result, some of the aircraft are still structurally sound after being exposed in the open air for about twenty years.

Rust protection

The layer of rust that forms on an iron object flakes off the metal, allowing air and moisture to get through to the iron below. This causes more rusting to occur and eventually the iron becomes a heap of rust. It is important to protect iron and steel from corrosion, especially if it is part of a bridge or the hull of a ship.

There are several ways to protect iron and steel from rusting. One way is to prevent oxygen or moisture from contacting the metal. This is called **surface protection**. The metal can be protected by coating it with paint, plastic or oil. If the surface protection becomes scratched or worn off, the metal below can be attacked by moisture and oxygen and rusting will occur. Examine the painted surface of an old car. Wherever the paint has chipped off you will find that corrosion has occurred and rust can be seen.

Another way to protect iron from rusting is to coat it with a layer of zinc. This is called **galvanising**. Zinc is a more reactive metal than iron, and in the presence of moisture and oxygen the zinc layer corrodes, leaving the iron unaffected. Many roofing materials and garden sheds are made from galvanised iron. You can also buy galvanised nails.

INQUIRY: INVESTIGATION 8.11

Rusting and salt water

KEY INQUIRY SKILLS:

- planning and conducting
- processing and analysing data and information
- evaluating

Equipment:

test tubes and test-tube rack
iron nails
salt (sodium chloride)
measuring cylinder
water

- Design an experiment to test the effect of the saltiness of water on the time taken for an iron nail to rust.
- Propose a hypothesis.
- Discuss your experimental design with a partner. You will need to consider which conditions must be kept the same and which condition will be varied.
- You will need to set up a control test tube. Find out what the purpose of a control is.
- Write down your method. It should be clear enough for someone else to follow without any help.
- Construct a table in which to record your observations over the next few days.

DISCUSS AND EXPLAIN

1 What effect did salt have on the time taken for the iron nail to rust?
2 Was your hypothesis supported?
3 How do your results compare with those of others in your class?
4 Write a report of your findings.

This wrecked car has rusted quickly because of its proximity to the sea.

Rusting can be useful

Not all rusting is bad. You can buy, from a pharmacy, hand warmers, which are commonly used by skiers and campers. These packages will produce heat when you shake them. The contents of the packet include powdered iron, water, salt and sawdust. When the packet is shaken vigorously, the iron rusts quickly, which produces heat.

HOW ABOUT THAT!

City councils face problems caused by the action of dogs on metal lampposts. The corrosive properties of the dogs' urine rusts the steel of the lampposts a few centimetres above the ground.

UNDERSTANDING AND INQUIRING

REMEMBER

1 What is corrosion?
2 What is rusting?
3 What is surface protection?
4 What is galvanised iron and what advantage does it have over iron?

THINK

5 Why does rusting occur faster in coastal regions than in areas further away from the sea?
6 How can galvanising protect iron from rusting when the zinc coating corrodes more quickly than the iron?
7 Suggest why the powdered iron in hand warmers used by skiers and campers rusts much more quickly than an iron nail.

INVESTIGATE

8 Corrosion is found in many places. Survey your school for rust spots. Write a report about your findings.
9 If you have access to an old car, survey it carefully and record all the rust spots on the car. Why are some parts of the car more likely to rust?
10 Aluminium corrodes quite quickly, yet it is used to make soft drink cans. Find out why aluminium cans are not corroded by the drinks they store.

eBookplus

11 Use the **How rust works** weblink in your eBookPLUS to watch a video about rust and the corrosion of iron.

work sheet 8.6 Rusting

Burning is a chemical reaction

Burning is a chemical reaction. It involves the combination of oxygen with a fuel and always produces heat and gases. Reactions that involve combination with oxygen are examples of **oxidation** reactions.

There are many other oxidation reactions that can occur. The rusting of iron to form iron oxide is an oxidation reaction. Rusting could correctly be described as a very slow type of burning reaction.

HOW ABOUT THAT!

A backdraught occurs when a fire in a closed room dies down because it has been starved of oxygen, but flammable gases continue to stream out of the hot materials in the room. When a door to the room is opened, air is quickly drawn inside, restoring the supply of oxygen and allowing the fire to reignite. The resulting fire consumes all the flammable gases in a few seconds and produces sufficient heat to ignite any remaining materials in the room. This is very dangerous to firefighters.

Burning fossil fuels

When a **fossil fuel** reacts with oxygen, heat is produced, along with carbon dioxide and water vapour. Fossil fuels are those fuels that are formed from the remains of living things. Petrol, natural gas, coal, wood and even paper are fossil fuels.

THE OXYACETYLENE TORCH

To obtain temperatures as high as 3000°C — hot enough to melt iron and weld metals — acetylene fuel is mixed with pure oxygen in an oxyacetylene torch.

acetylene + oxygen ⟶ carbon dioxide + water

An oxyacetylene torch is used in construction work.

THE CAR ENGINE

Burning is also known as **combustion**. Car engines work by the combustion of petrol or gas in the cylinders. A mixture of air and fuel is drawn into each cylinder and ignited by a spark from the spark plug. The fuel reacts rapidly with oxygen in the air. The resulting explosion pushes the piston, which turns the drive shaft. The products of the reaction, carbon dioxide and water vapour, leave the car engine through the exhaust pipe.

ROCKET FUELS

Liquid and solid fuels are used in the NASA space shuttle program. When these fuels are burnt, they provide sufficient thrust to place a space shuttle in orbit hundreds of kilometres from Earth. Liquid hydrogen and liquid oxygen react to power the shuttle's main engines.

hydrogen + oxygen ⟶ water

Most of the thrust required to place the shuttle in its desired orbit comes from chemical reactions in the solid fuel, which is located in the solid rocket boosters. In space, liquid fuel such as hydrazine is oxidised to produce an enormous volume of gas. As the gas is released, the rocket is thrust forward. By controlling the direction of the thrust, it is possible to steer the rocket.

Oxidation reactions provide the thrust to launch the space shuttle.

INQUIRY: INVESTIGATION 8.12

Burning magnesium

KEY INQUIRY SKILLS:

- planning and conducting
- processing and analysing data and information

Equipment:

safety glasses

Bunsen burner, heatproof mat and matches

tongs

2 cm piece of magnesium ribbon

sandpaper

- If the magnesium ribbon is dull, use the sandpaper to remove the dull layer.
- Hold the magnesium ribbon with tongs in the Bunsen burner flame.

CAUTION: Do not look directly at the flame — eye damage may occur.

- After burning the magnesium metal, observe the product of burning that remains.

DISCUSS AND EXPLAIN

1 Describe the magnesium metal before burning.
2 During burning, the magnesium reacted with the oxygen in the air by combining with it to form magnesium oxide. Describe the magnesium oxide.
3 How do you know that a chemical reaction has taken place?
4 Write a word equation for the chemical reaction.

INQUIRY: INVESTIGATION 8.13

Burning paper

KEY INQUIRY SKILLS:

- planning and conducting
- processing and analysing data and information

Equipment:

safety glasses

Bunsen burner, heatproof mat and matches

tongs

gas jar

limewater

paper

deflagrating spoon

- Place 10 mL of limewater in the bottom of the gas jar.
- Put a ball of scrunched-up paper into the deflagrating spoon.
- Light the paper and lower it into the gas jar.
- When burning has stopped, remove the deflagrating spoon and cover the jar.
- Shake the gas jar and observe the colour of the limewater.

DISCUSS AND EXPLAIN

1 What happened to the limewater?
2 What gas was given off by the burning paper?

UNDERSTANDING AND INQUIRING

REMEMBER

1 Define the term 'burning'.
2 What evidence is there that burning is a chemical reaction?
3 What is a fossil fuel? List three examples of fossil fuels.
4 List and describe three examples of useful oxidation reactions.
5 Write a word equation for each of the three examples listed in question 4.
6 Is rusting an example of burning? Explain.

THINK

7 Complete this word equation:
fuel + _____ ⟶ _____ + water vapour
8 Name at least one fuel that is not a fossil fuel.
9 What are the three different one-word names given to the chemical reactions in which fuels react with oxygen?

CREATE

10 Choose one fuel from the list below and prepare a poster on the use of this particular fuel. Include in your poster details of where the fuel comes from, what it is used for, and a word equation for its oxidation reaction.
- methane • ethanol • butane • propane
- kerosene • lignite • diesel • acetylene

INVESTIGATE

11 Fire has always been present in Australia and is a major cause of change in the environment. Find out how Australian Aboriginals have traditionally used fire to benefit themselves and the environment.
12 Research and report on the chemical reaction that takes place in a soda–acid fire extinguisher when it is used to fight fires. Also, explain how the reaction is started.

8.7 Combustion

8.7 SCIENCE UNDERSTANDING

A new breed of materials

The scientists and engineers who develop new plastics for spacesuits that allow astronauts to walk in space need a knowledge of chemistry to create materials that are strong, light and heat resistant. Developing new materials for a particular purpose requires an assessment of the required properties and an understanding of chemical reactions.

The spacesuits worn by astronauts when they are walking in space contain many layers of materials developed by scientists and engineers.

WHAT DOES IT MEAN?

The word *plastic* comes from the Greek word ***plastikos***, meaning 'able to be moulded'.

Fantastic plastic

Metals, paper and ceramics have been used for thousands of years. But **plastics** have been around for less than 100 years. Plastics are synthetic (manufactured) materials that can be easily moulded into shape. Some plastics are flexible and soften when they are heated. They can be easily moulded into products such as milk and fruit juice containers, rubbish bins, spectacle lenses, electrical insulation and laundry baskets. Others are quite hard and rigid. These plastics are used to make items such as toilet seats, electrical switches, bench tops and outdoor furniture. Most plastics are the products of chemical reactions with crude oil, from which petrol and bitumen are also produced, as the main reactant.

HOW ABOUT THAT!

Australia was the first country in the world to use only plastic notes for currency. The notes are more difficult to forge and last much longer than the old paper notes.

The clothes you wear

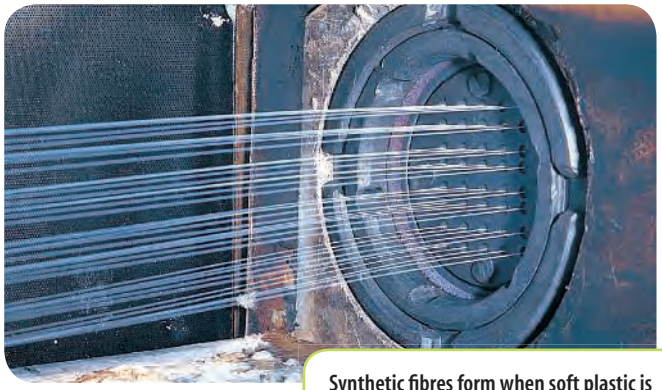
Until the development of nylon in 1938, just in time to make parachutes for World War II, the world relied almost completely on fabrics made from **natural fibres** such as wool, cotton, linen and silk.

Animal-based products include wool from sheep and silk from silkworms. Cotton is derived from cotton bushes and linen comes from flax plants. Today, it would be impossible to provide clothing and bedding for the world's population with purely

natural fibres because of the amount of land and water that would be needed for crops and sheep.

Synthetic fibres such as those used in compression sports gear have many desirable qualities that natural fibres lack, including easy care, colour-fastness and light weight.

Of the many synthetic fibres, the most widely used are **nylon** and **polyester**. Synthetic fibres are made by pushing softened plastic materials through tiny holes in a nozzle called a **spinneret**, which looks a little like a shower head.



Synthetic fibres form when soft plastic is forced through the holes of a spinneret.

Would you use this sporting clothing if it were made from wool or cotton? The properties of those fibres do not suit this type of clothing. Materials have been specially developed with their use in mind.

eBookplus eLesson

The future of clothing
Find out abut the clothing of the future. eles-0859

HOW ABOUT THAT!

The spinneret gets its name from the organ used by spiders to spin their webs. Liquid silk flows through the spider's spinneret. It hardens into a fibre as it passes through. Most spiders have six spinnerets.

The spider pictured here has used the fibre to bind and preserve its prey.

Each fibre, whether natural or synthetic, has advantages and disadvantages. Some of these are outlined in the table below.

Fibre	Advantages	Disadvantages
Wool	Warm in cold weather Crease resistant Burns slowly Retains its shape well	Shrinks when washed Turns yellow in sunlight
Cotton	Absorbs moisture Soft Cool in hot weather	Creases easily Burns quickly
Nylon	Dries quickly Light Strong Elastic	Builds up static electricity Melts rather than burns Turns yellow in sunlight
Polyester	Dries quickly Crease resistant Resistant to many chemicals	Builds up static electricity Melts rather than burns

The best of both worlds

Many of today's fabrics are made from blends of natural and synthetic fibres to make the best of the properties of each of the fabrics in the blend. A blend of polyester and cotton is commonly used for shirts and dresses. The cotton helps keep the wearer cool, while the polyester reduces creasing.

Rayon is shiny, easy to dry and cool in summer. On the 'down' side it has low **durability** and is not **elastic**. Elastic describes the ability of a material to return to its original size after being stretched. Rayon is neither a natural nor a synthetic fibre. To make it, cellulose fibres from spruce and eucalyptus trees are mixed with chemicals that soften them. The mixture is then passed through a spinneret.

A new breed of fibres

When you watch the feats of Olympic athletes, cyclists, skiers and skaters, it's almost certain that they are wearing Lycra. Lycra is not a fabric. Lycra is the registered trademark of a synthetic fibre called spandex. Spandex was invented in 1958. Spandex is lightweight, durable, retains its shape and fits snugly. It even pulls moisture away from the wearer's skin. Spandex is very elastic. It can be stretched to up to

INQUIRY: INVESTIGATION 8.14

Putting fibres to the test

KEY INQUIRY SKILLS:

- planning and conducting
- processing and analysing data and information
- evaluating

Equipment:

a range of threads of different fibres, e.g. cotton, polyester, wool, nylon, rayon

uniformly sized fabric samples made from different fibres or blends of fibres

equipment decided upon by the group

- Work in groups of three or four to complete this investigation.
- Start by listing the properties of either the fabric samples or the fibres that can be tested by experiments. Some examples to help get you started include durability, elasticity and the ability to absorb water.
- Devise an experiment that will allow you to compare one property of threads of either different fibres or different fabric samples.
- Make a list of the equipment you will need.
- Have your experiment plan and equipment checked by your teacher.

CAUTION: Obtain your teacher's approval before carrying out any tests. Synthetic fibres or blends should be burned only in a fume cupboard.

- Carry out your experiment and keep a record of your measurements and observations.
- Write a report about your experiment.

DISCUSS AND EXPLAIN

Include answers to the following questions in the 'Discussion' section of your report.

1. List the variables that you were able to control in your experiment.
2. Identify any variables that you didn't feel the need to control.
3. Identify the most useful properties of each of the fibres or fabrics that you tested.
4. Suggest at least one improvement that you could have made to your experiment.

seven times its normal length and spring back to its initial length when released. Spandex is always blended with other fibres. As little as 2 per cent of this material in a blend makes a difference to the properties of the fabric. Lycra suits usually consist of between 3 per cent and 10 per cent spandex.

Cleaning up

Whether you're washing dishes, clothes or yourself, it's obvious that water alone is not good enough. That's because dirt, grease and oil don't dissolve in water. You need to use soaps or detergents which, when mixed with water, loosen or remove soil and greasy and oily substances from surfaces. They can then be rinsed away by the action of running water or a wash cloth.

COMPARING SOAPS AND DETERGENTS

Soaps and detergents share many properties. Both dissolve grease and oil (a chemical property), both can be found as solids and liquids (a physical property) and both are very, very slippery (another physical property). Most soaps and detergents on the market are biodegradable (a chemical property), which means that they will break down naturally after their journey down the plughole.

The biggest difference between soaps and detergents is the way they are made. Soaps are made from natural fats and oils. No one is certain about when the first soap was made, but a recipe for a substance that would have resembled soap has been found in the Middle East, dating back to the Babylonian civilisation about 4800 years ago. The ingredients included ashes from fires, oils from trees and sesame seed oil. Detergents, like most plastics, are made from chemicals obtained from crude oil. They are heavily scented with fragrances to disguise the odour, and preservatives are added so that they don't spoil.

Cleansing agents like hair shampoo release soil, grease and oil from the scalp. Almost all hair shampoos are detergents.

Lycra is used extensively in sport because of its light weight, tight fit and ability to pull moisture away from the skin.

The perils of packaging

Just about everything you buy at the supermarket comes in a package. Even if it doesn't come in a package, you usually put it in a bag to take it home. The type of packaging needed depends on the properties of the product inside. For example, you can't package tomato sauce in a paper bag. The most commonly used materials in packaging are paper (or cardboard), plastic, metal and glass. For a consumer, it's not just the properties of the packaging that are important. At least two questions should be asked when you make a choice about buying a product:

- Is the packaging recyclable?
- Is the packaging biodegradable?

If the packaging is glass, aluminium or steel it is probably recyclable, saving energy and water. If it is a plastic bottle, it is also likely to be recyclable. If the packaging is not recyclable, think about whether it is **biodegradable**; that is, it can be broken down by natural chemical reactions in the bodies of worms or other small **organisms** that live in the soil. Plastics, metals and glass are not biodegradable. If they are thrown out with other household rubbish such as food waste, they end up in rubbish tips and will not break down. This creates the need for more rubbish tips. Of course, there is a limit to how much land can be used for rubbish tips in or near major towns and cities.

Paper is mostly biodegradable. Paper packaging that has been contaminated by food or oils cannot be recycled. But at least when it gets to the rubbish tip it can be broken down in the soil. If you have a choice, choose items with packaging that is either recyclable or biodegradable.

UNDERSTANDING AND INQUIRING

REMEMBER

1 Identify the single property that all plastics have.
2 From which substance found beneath the ground are most plastics made?
3 Where do all natural fibres come from?
4 Why is woollen clothing popular in winter whereas cotton clothing is popular in summer?
5 Why are cotton and polyester blends so commonly used for shirts and dresses?
6 From which group of materials are synthetic fibres made?
7 What is a spinneret used for?
8 Which fibre do Lycra suits always contain?
9 How do soap and detergents remove grease, oil and dirt from surfaces?
10 What is the major difference between soap and detergents?
11 Identify two properties that soap and detergents have in common.
12 Plastics, metals and glass are not biodegradable.
   (a) What does this mean?
   (b) If non-biodegradable rubbish cannot be recycled, what happens to it?

THINK

13 Which properties make plastic more suitable for use in outdoor furniture than:
   (a) wood? (b) metal?
14 Which properties of plastic currency notes make them more suitable than the old paper ones?
15 Which properties of nylon made it suitable for making parachutes during World War II?
16 Explain why rayon is neither a natural fibre nor a synthetic fibre.
17 Explain why spandex is not a fabric.
18 How would blending with spandex change the properties of a pure cotton fabric?
19 Which properties make Lycra suitable for:
   (a) the clothes of speed skaters?
   (b) toddlers' clothing?
   (c) underwear?
20 Why is it important to rinse your hair thoroughly after shampooing?
21 State the properties of the packaging that make it suitable for each of the following supermarket items.
   (a) Tomato sauce in a glass bottle
   (b) Tomato sauce in a red plastic bottle
   (c) Lemonade in an aluminium can
   (d) Lemonade in a clear plastic bottle

IMAGINE

22 Make a list of at least ten items in your house made from plastic. Imagine that they could no longer be made from plastic. For each item, write down:
   (a) the most important properties that the items must have
   (b) which other material could be used to make them.

INVESTIGATE

23 Find out why the group of chemicals known as phosphates is no longer used in most detergents.
24 Research and report on how each of the following materials was discovered or developed, by whom and what they are used for.
   (a) Polyester (c) PSZ (partially stabilised zirconia)
   (b) Kevlar®

Use it again

The material that you throw out as household rubbish is buried in landfill tips like the one pictured below.

The food scraps that make up almost half of your household rubbish are biodegradable. They will be broken down by microbes and other decomposers such as worms in the soil. Chemical changes take place when these organisms digest the scraps, returning nutrients to the soil. You can use compost bins or compost heaps to allow this to happen in your own backyard. Even paper and cardboard break down fairly quickly in the soil.

However, materials such as plastic, glass and metals take hundreds or even thousands of years to break down. The properties of these materials allow them to be **recycled**.

Looking after your PET

There are two very good reasons for recycling plastics:

- Plastics are non-biodegradable. That is, they are not broken down naturally by micro-organisms. Plastics add thousands of tonnes of new rubbish to the environment every year.
- Plastics are made from oil — a resource that is expensive and dwindling. The continued production of new plastic is not **sustainable**.

Household waste contains many different types of plastic, which need to be separated. The plastics industry has introduced a code system to help consumers identify recyclable plastics. The symbols shown on the right make the sorting of plastics before recycling easier and cheaper. Some plastics are more easily recycled than others because of differences in the structure of the chains of molecules of which they are made.

Dumping rubbish in a landfill tip is not just unsightly. Poisonous chemicals in the rubbish can leak into the soil and end up in rivers and streams, endangering living things.

PETE

PET or PETE (polyethylene terephthalate) is used to make plastic soft-drink bottles. Most commonly known as PET, this plastic is recycled to make carpet fibre and flower tubes. Empty PET bottles completely and remove the lids before placing them in a home recycling bin. The lids are recyclable, but their small size makes the sorting process awkward.

HDPE

HDPE (high-density polyethylene) is used to make plastic milk and fruit juice bottles. This plastic is recycled to make bottles, crates, pipes, 'wheelie' bins and playground equipment. Empty HDPE bottles completely and remove the lids before placing them in a home recycling bin.

V

Polyvinyl chloride, more commonly known as PVC, is used to make pipes, fencing and bottles containing substances other than food. It is not easily recycled and can be dangerous to your health and the environment. PVC bottles can be placed in your home recycling bin as long as the lids are removed. They are separated from the more easily recyclable plastics and sent to a separate plant for processing.

LDPE

LPDE (low-density polyethylene) is used to make wash bottles and other containers. These are recyclable and can be placed in your recycling bin. LPDE is also used to make supermarket plastic bags, which should not be placed in your home recycling bin. These bags interfere with the automatic sorting machines in recycling plants. It is best to avoid using them by using reusable bags for shopping.

PP

PP (polypropylene) is used in synthetic fibres to make clothing, industrial fibres, car batteries, bumper bars and other car parts. Although these products are not appropriate for your recycling bin, polypropylene can be recycled for use in carpet, furniture, white goods and even polymer bank notes. Most car workshops, scrap metal dealers and service stations will accept used car batteries for recycling.

PS

Polystyrene is a very light plastic used in solid form to make plastic cutlery, toys and cases for CDs and DVDs. In its softer foam form, it is used for disposable drinking cups and packing materials. Polystyrene products should not be placed in home recycling bins. The lightness of polystyrene foam makes it difficult to sort and recycle. Although all polystyrene can be recycled, it is a very expensive process.

OTHER

Other plastics, including nylon, fibreglass and polycarbonate, are not generally recycled and should not be placed in your home recycling bin.

See-through recycling

About 45 per cent of the glass packaging used in Australia is recycled. Used glass bottles, known as **cullet**, are collected and melted down in a furnace to produce new products. The overall energy saving is only 8 per cent of that used in making new glass. This is because of the high cost of collecting and melting down the bottles. In some countries, milk is sold in bottles that can be sterilised and reused up to 50 times before they need melting down, which saves a large amount of energy.

INQUIRY: INVESTIGATION 8.15

Design your own waste disposal system

Your group is responsible for preparing a report on ways to improve the household waste management and disposal for the shire of Green Valley.

The shire currently collects rubbish from its 134 500 ratepayers using large green bins that are emptied by compactor trucks. The rubbish is taken to the local tip and used as landfill, at a cost to the council of $60 per tonne. The tip is nearing capacity and will be closed within twelve months. Waste paper is collected separately by a private recycling company.

Your report could be produced in written form or as an audio tape or videotape. It should address the following issues:

1. How will the shire encourage each household to produce less waste?
2. Is recycling too costly?
3. If recycling occurs, will recyclable wastes such as plastics, glass and metals be separated at a disposal station after collection, or collected in separate containers from households?
4. What measures will be used to encourage households to use compost bins?
5. How will the shire dispose of rubbish when the landfill site closes?

Saving trees

Over a million tonnes of paper, about a third of our annual consumption, is recycled in Australia. Paper is made out of fibres of the chemical cellulose and is relatively easy to recycle. Waste paper is first mixed with water to separate the fibres. Additives such as ink and adhesives are then removed, producing low-quality fibres that can be used to make cardboard and other products. Steam rollers are used to improve the quality of the finished paper. Recycling paper reduces the amount of new paper needed, saving millions of trees.

This compost bin, made from recycled polypropylene (PP), is 220 L in capacity and is sold to various councils across Australia. The compost decreases in volume as it breaks down. Almost 50 per cent of domestic waste in Australia is material suitable for composting.

And metal too!

Metals such as steel and aluminium are easily recycled as long as they can be cheaply separated from other rubbish. Steel cans, aerosol containers, jar lids and bottletops can be recycled. The recycling of aluminium cans saves huge amounts of energy. Twenty aluminium cans can be recycled with the same amount of energy needed to produce just one new can.

Recycling aluminium cans. About 70 per cent of the aluminium cans used in Australia each year are recycled.

Sorting it out

The separation of the items in your recycling bin relies on differences in their physical properties including size, weight, magnetic properties and even colour. For example, items of different weights can be separated using blasts of air or a centrifuge that works like the spin dryer of a washing machine. Steel can be separated from other metals by a large magnet.

Special recycling programs

There are separate recycling programs for some products that cannot be placed in home recycling bins. These recycling programs are generally used to collect products containing substances that would endanger the environment or the community if they were dumped in landfill tips. For example, printer cartridges can be placed in recycling boxes at many Australia Post outlets and retail stores that sell computers and printers. Mobile phones can be left at most mobile phone outlets for recycling. Use the **Recycling** weblink in your eBookPLUS to find out where computers and other electronic equipment, white goods such as fridges and washing machines, corks, light globes and many other items are collected for recycling. This site also provides information about how to dispose of chemical wastes from home, school or industry. Oil, paints and unused medicines should not be placed in rubbish bins or flushed down the sink.

You can make a difference

The three bin collection system used by many city and shire councils throughout Australia makes it very easy for you to make a difference to the environment by recycling.

During the twelve months to June 2009, Hornsby Shire in NSW recycled 18 000 tonnes of paper. That's the equivalent of:

- saving over 299 000 gigajoules of energy. That's enough to power 13 870 homes for a year!
- preventing 9032 tonnes of greenhouse gas from entering the atmosphere. That's like taking 2169 vehicles off our roads — permanently!
- saving over 298 million litres of water. That's enough to fill 119 Olympic-sized swimming pools!

Imagine the combined effect of all cities and shires Australia wide!

UNDERSTANDING AND INQUIRING

REMEMBER

1 How are biodegradable substances different from those that are not biodegradable?
2 Where do the chemical changes that break down biodegradable waste take place?
3 What are the main benefits of recycling plastics?
4 Why are plastics such as PET now identified by a code?
5 What is cullet?

THINK

6 The first part of the process of recycling paper involves mixing it with water. What is the major purpose of mixing the paper with water?
7 List three problems associated with the disposal of waste in landfill sites.
8 Why is only a small amount of energy saved when glass is recycled?
9 List two benefits of recycling aluminium.
10 What factors influence the decision as to whether it is worth the trouble of recycling a resource?
11 Ink is removed from paper when it is recycled. Is the combination of ink and paper a compound or a mixture? Explain your answer.
12 If a plastic bag manufacturer claimed that the bags it produced were biodegradable, what evidence would you need to be satisfied that the claim was correct?

CREATE

13 Draw a poster that could be used to encourage people to do one of the following.
   (a) Recycle all plastic products.
   (b) Reduce their use of LDPE plastic bags.
   (c) Separate household rubbish into different bins for more efficient recycling.
   (d) Recycle paper used at school.
   (e) Recycle newspaper.
   (f) Recycle aluminium cans.

INVESTIGATE

14 Use the **Recycling** weblink in your eBookPLUS to find out how you can dispose of each of the following substances at a nearby location.
   (a) Unwanted paint
   (b) Used motor oil
   (c) Unwanted and out-of-date medicines

8.9 THINKING TOOLS

Target maps and single bubble maps

1. Draw three concentric circles on a sheet of paper.
2. Write the topic in the centre circle.
3. In the next circle, write words and phrases that are relevant to the topic.
4. In the outer circle, write words and phrases that are not relevant to the topic.

how to ...?

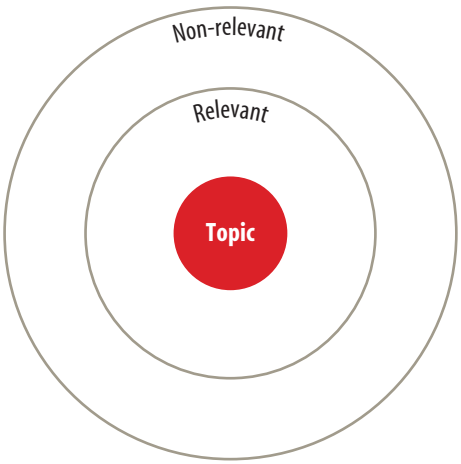
To identify (target) what is part of (relevant to) the topic and what is not

why use?

How can we find out what is relevant?

question

Target map



Circle map

also called

comparison

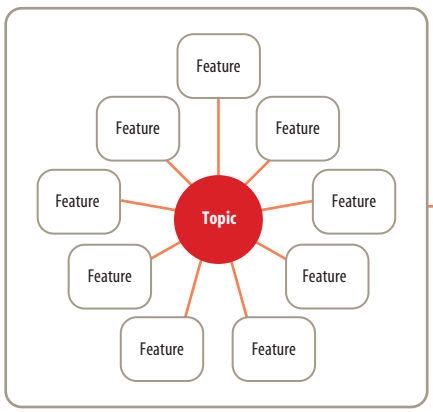
Similarity

Both identify and describe the range of the content.

Difference

Single bubble maps do not identify the non-relevant material.

Single bubble map



example

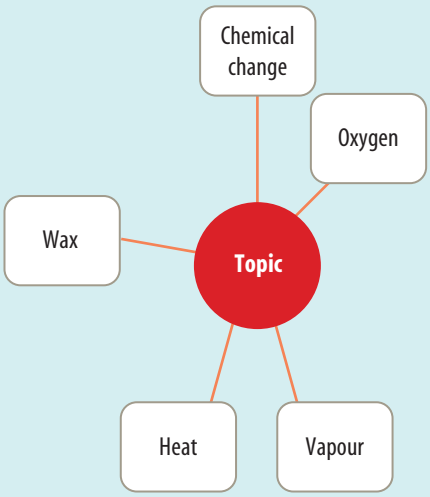
UNDERSTANDING AND INQUIRING

THINK AND CREATE

1 Use the words listed below to construct a target map about physical change.

melting
evaporation
rusting
reactant
burning
condensation
stretching
precipitate
freezing
explosion
catalyst

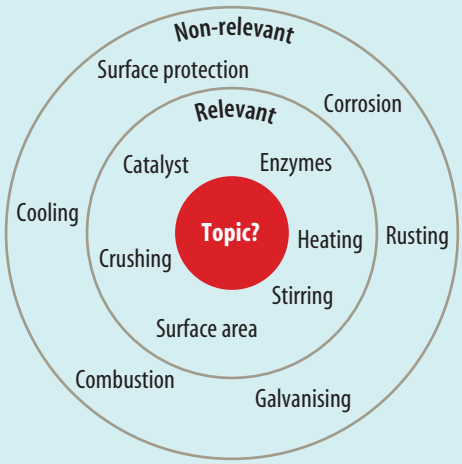
2 The single bubble map below identifies some of the ideas associated with a burning candle.

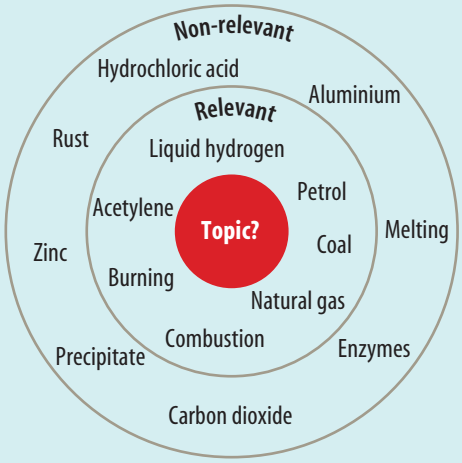


(a) Draw your own single bubble map about the topic 'a burning candle', adding as many additional bubbles as you can.
(b) Construct a single bubble map that identifies clues that provide evidence that a chemical reaction has taken place.

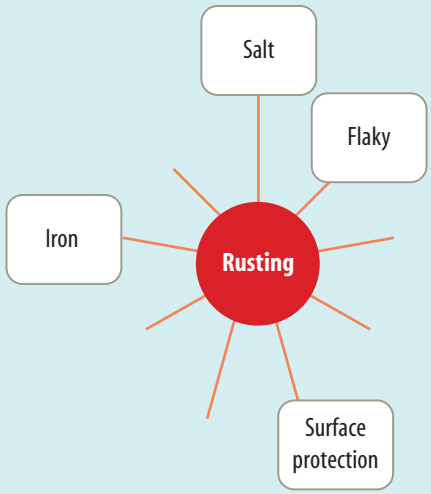
3 (a) In teams, brainstorm as many single words as you can that are associated with chemical reactions.
(b) Use the brainstorm list to create a team single bubble map on chemical reactions.
(c) Compare your list with those of other teams and then work together to construct a class single bubble map on chemical reactions. Suggest other types of thinking tools that could be useful.

4 Suggest a topic for each of the target maps below.





5 The single bubble map below identifies some of the ideas associated with rusting.
(a) Complete the bubble map by adding as many ideas as you can.
(b) Use the ideas in your single bubble map to create a target map on rusting.



STUDY CHECKLIST

PHYSICAL AND CHEMICAL PROPERTIES

- distinguish between the physical and chemical properties of substances
- outline some examples of physical and chemical properties
- recognise that the chemical properties of a substance affect its use
- outline the benefits and disadvantages of plastics
- compare the properties and method of manufacturing soaps and detergents
- explain the difference between natural and synthetic fibres, and discuss their advantages and disadvantages

PHYSICAL AND CHEMICAL CHANGE

- define chemical change as a change in which the bonds between atoms or molecules are broken or new bonds between these particles are formed
- distinguish between physical changes and chemical changes

CHEMICAL REACTIONS

- define a chemical reaction as a chemical change in which a new substance is produced
- identify evidence that a chemical reaction has taken place
- distinguish between the reactants and products of a chemical reaction
- describe simple chemical reactions using word equations
- describe a variety of methods of speeding up or slowing down chemical reactions
- recognise that corrosion and burning (combustion) are chemical reactions
- describe rusting as an example of corrosion
- identify burning in oxygen as an oxidation reaction
- outline examples of the use of fuels in combustion reactions
- recall that most plastics are made from crude oil
- recall that biodegradability depends on chemical reactions within living organisms

SCIENCE AS A HUMAN ENDEAVOUR

- investigate the use of fire by traditional Aboriginal people
- investigate the development by scientists of a variety of new materials
- distinguish between biodegradable and non-biodegradable materials
- evaluate the suitability of different types of packaging for recycling
- outline the requirements and development of systems for the collection and recycling of household waste

INDIVIDUAL PATHWAYS

eBookplus

Activity 8.1
Investigating reactions
doc-6072

Activity 8.2
Analysing reactions
doc-6073

Activity 8.3
Investigating reactions further
doc-6074

ICT

eBookplus Summary

eLESSON

The future of clothing
Find out how developments in the properties of materials and fibres could affect what you'll be wearing in the future.

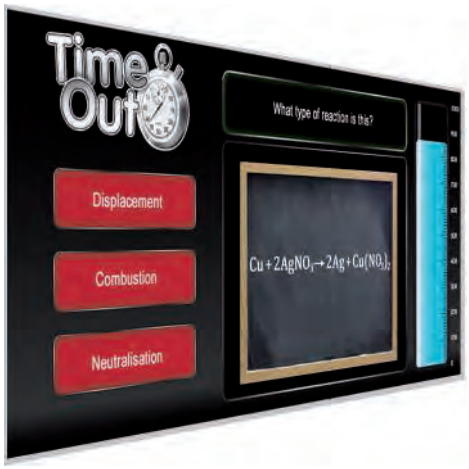
Searchlight ID: eles-0859

INTERACTIVITIES

Reaction rates
This interactivity allows you to change the temperature, concentration and surface area of reagents to see how they affect the rate of a reaction, and then decide how the rates of a number of reactions could be changed.

Searchlight ID: int-0230

Time Out: 'Reactions'
Use this exciting interactivity to test your ability to classify different types of reactions before time runs out.



Searchlight ID: int-0759

LOOKING BACK

1 List two useful properties of:
(a) glass
(b) metals
(c) plastics
(d) paper.

2 Explain, using examples, the difference between physical and chemical properties.

3 In your own words, express the meaning of each of these terms that describe the properties of substances.
(a) Elastic
(b) Ductile
(c) Reactive
(d) Malleable
(e) Lustrous
(f) Toxic
(g) Transparent
(h) Flammable
(i) Melting point

4 Identify the following as either chemical or physical changes.
(a) The wax on a burning candle melts.
(b) The wax vapour at the top of the candle wick burns with oxygen to produce carbon dioxide, water vapour and heat.
(c) Calcium carbonate is dissolved by hydrochloric acid to form calcium chloride, water and carbon dioxide gas.
(d) Hydrogen gas explodes with oxygen gas to form water.

5 Write word equations for each of the changes in question 4.

6 How do you know that:
(a) toasting bread is not a physical change?
(b) rusting of a nail is not a physical change?

7 Some chemical reactions can be useful. Write down three examples of useful chemical reactions.

8 Catalysts are sometimes added to the reactants taking part in a chemical reaction.
(a) What is a catalyst?
(b) When a word equation is written to describe a chemical reaction, catalysts are not included either as reactants or products. Why?

9 When a lead nitrate solution is added to a potassium iodide solution, a chemical reaction takes place. A bright yellow solid appears. It is the compound lead iodide. Another compound, potassium nitrate, remains in the solution and is not visible.
(a) Name the reactants in this chemical reaction.
(b) What are the products of the reaction?
(c) The yellow lead iodide will eventually settle to the bottom of the flask. What 11-letter word beginning with 'p' is given to a substance that behaves like the lead iodide?
(d) Write a chemical word equation for the reaction.

10 Rusting is an example of a slow chemical reaction.
(a) What are the three reactants of rusting?
(b) What is the product of the rusting reaction?

11 For each of the reactions below, suggest ways that the reaction could be made to happen more quickly.
(a) Burning a pile of dry leaves
(b) Cooking potatoes
(c) Dissolving marble chips in acid
(d) Removing a stain using bleach
(e) Making an iron nail go rusty
(f) Milk going sour

12 Some chemical reactions can be destructive. Write down three examples of harmful chemical reactions.

13 Children's steel swing sets in beachside towns and suburbs rust much faster than those further from the coast.
(a) Explain why this happens.
(b) Suggest two methods of slowing down or preventing the rusting of steel swing sets.

14 The oxyacetylene torch shown below is used to melt metals to allow them to be joined together.

(a) What type of chemical reaction takes place in the oxyacetylene torch?
(b) What evidence is there in the photo that a chemical reaction has taken place?

15 Just as chemicals can be grouped or classified, so can chemical reactions. What name is given to the following chemical reactions?
(a) The corrosion of iron
(b) The reaction of substances with oxygen
(c) Burning

16 The illustration below shows a camper boiling water in a billy over a camp fire.

(a) What physical change in the wood has taken place during the preparation of the camp fire?
(b) What physical change is shown taking place?
(c) What chemical change is shown taking place?
(d) List two ways in which the chemical reaction taking place has been sped up.

17 Which properties of the plastic used to make light switches and power points make it right for the job?

18 What is the difference (other than their properties) between natural and synthetic fibres?

19 Identify each of the fibres listed below as natural, synthetic, or neither natural nor synthetic.
(a) Nylon
(b) Cotton
(c) Rayon
(d) Lycra
(e) Wool
(f) Polyester

20 How are synthetic fibres such as nylon made?

21 Why are many garments made of a blend of two or more different fibres?

22 Soap and detergents can be used for the same purpose — including washing your own body.
(a) What is the main difference between soap and detergents?
(b) What properties do soap and detergents have in common?

23 Which properties are essential for the packaging of the following products?
(a) Pool chemicals
(b) Eggs
(c) Soft drink
(d) Peanuts

24 Some plastic containers are marked with this symbol.

(a) What substance would you expect to find in bottles made from this type of plastic?
(b) What two things should you do before placing bottles made from this type of plastic in a recycling bin?
(c) State two uses for this type of plastic after it has been recycled.

25 Describe the 'three bin system' used by many cities and shires, and explain how it helps the environment.

9 Sedimentary, igneous and metamorphic rocks

The Earth's surface is constantly changing. Volcanoes and earthquakes can cause quick changes, but most of the changes to the Earth's surface happen slowly. Rocks on and below the surface of the Earth are slowly and constantly being changed by natural events. Rocks also provide a valuable record of past events.

OVERARCHING IDEAS

- Patterns, order and organisation
- Form and function
- Stability and change
- Scale and measurement
- Matter and energy
- Systems

SCIENCE UNDERSTANDING

Sedimentary, igneous and metamorphic rocks contain minerals and are formed by processes that occur within Earth over a variety of time scales.

Elaborations

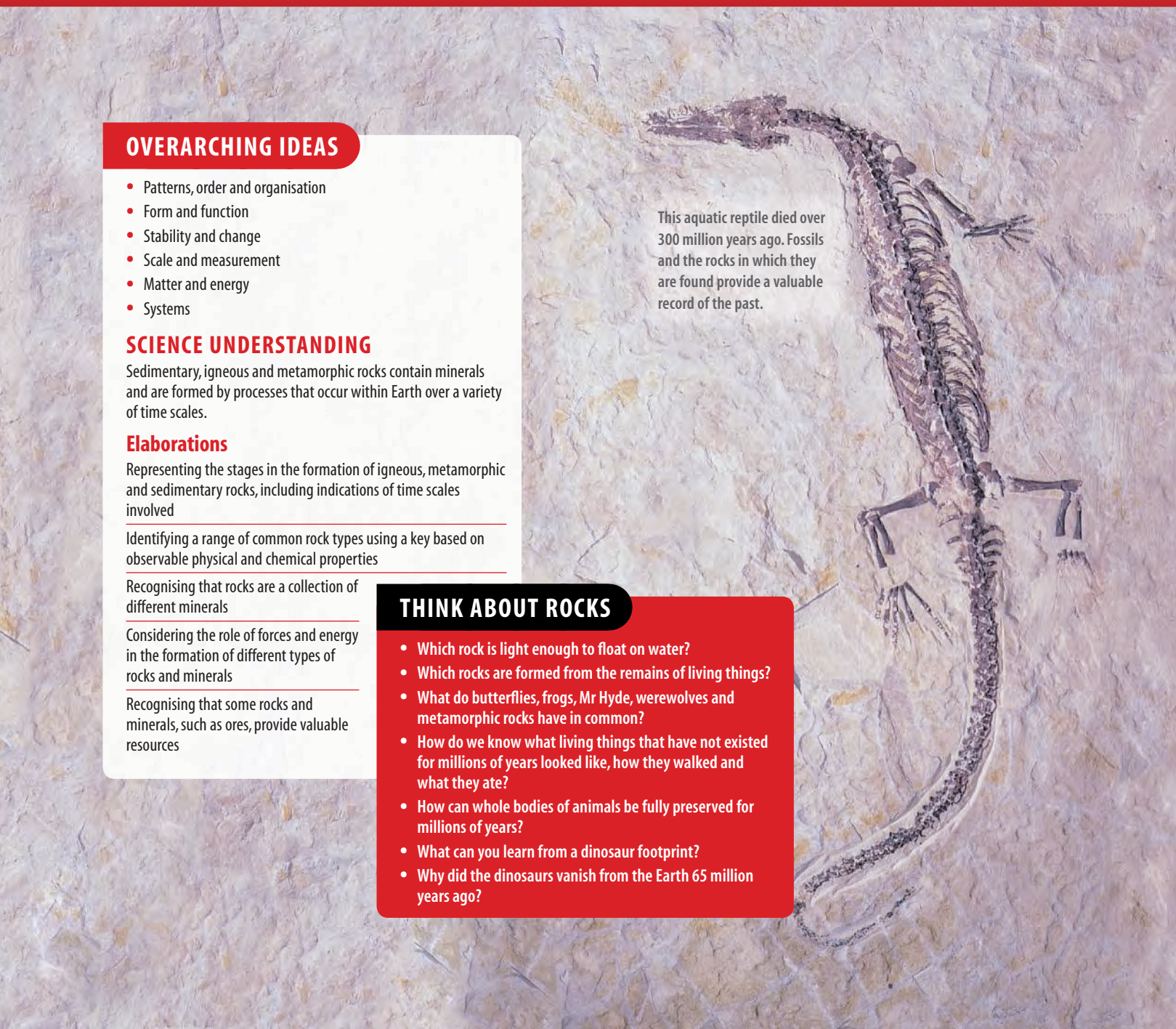
Representing the stages in the formation of igneous, metamorphic and sedimentary rocks, including indications of time scales involved

Identifying a range of common rock types using a key based on observable physical and chemical properties

Recognising that rocks are a collection of different minerals

Considering the role of forces and energy in the formation of different types of rocks and minerals

Recognising that some rocks and minerals, such as ores, provide valuable resources



This aquatic reptile died over 300 million years ago. Fossils and the rocks in which they are found provide a valuable record of the past.

THINK ABOUT ROCKS

- Which rock is light enough to float on water?
- Which rocks are formed from the remains of living things?
- What do butterflies, frogs, Mr Hyde, werewolves and metamorphic rocks have in common?
- How do we know what living things that have not existed for millions of years looked like, how they walked and what they ate?
- How can whole bodies of animals be fully preserved for millions of years?
- What can you learn from a dinosaur footprint?
- Why did the dinosaurs vanish from the Earth 65 million years ago?

YOUR QUEST

Bathroom rocks

Taps, tiles, mirrors and even soap. Where do the materials needed to produce these come from?

When you last used the bathroom, it's not likely that you were thinking about rocks. After all what does a bathroom have to do with rocks?

But where did the materials to make the shower recess come from? What about the taps and pipes that delivered the water? Where do the materials to make tiles come from? And what about the soap, talcum powder and toothpaste — where do they come from?

The answers to all of these questions lead back to rocks. For example, metals are extracted from rocks. Talcum powder is made from a mineral called talc, which comes from a rock. Some of the ingredients of toothpaste come from rocks.

Even food and clothes can be traced back to rocks. Plants grow in soil, which is made up mostly of weathered rock. Glass is made using sand, which is weathered rock.

THINK

Work in small groups to answer the following questions.

1 What materials are mirrors made from?
2 Where does the metal used to make bathroom taps come from?
3 What are bathroom tiles made from?
4 List some building materials that are:
   (a) made directly from rocks
   (b) not made directly from rocks but can be traced back to rocks.
5 The terms 'igneous', 'sedimentary' and 'metamorphic' are used to describe the three main groups of rocks. Just from looking at the words, suggest how each of these groups of rocks is formed.
6 Make a list of the names of rocks that you know. Attempt to classify them as igneous, sedimentary or metamorphic.

'Hot' rocks

Lava is released from erupting volcanoes at temperatures of 1000°C or more. At that temperature, it could take weeks to cool down to become solid rock. But lava surging violently into the air from explosive volcanoes cools much faster. The lava from volcanoes erupting under water on the ocean floor also cools quickly to form solid rock.

Rocks that form from the cooling of lava or magma below the Earth's crust are called **igneous rocks**. The igneous rocks that form from red-hot lava above the Earth's surface are called **extrusive rocks**. Igneous rocks that form from the lava spilling from underwater volcanoes are also classified as extrusive rocks.

The appearance of all extrusive igneous rocks depends on two major factors:

- how quickly the lava cooled
- what substances it contains.

Frothy rocks

Some violent volcanic eruptions shoot out lava filled with gases. The lava cools quickly, while it is still in the air, and traps the gases inside. Rocks that form this way are full of holes. Two examples of these rocks are **pumice** and **scoria**.

PUMICE

Pumice is a pale-coloured rock. It is very light because it is full of holes. It floats on water and sometimes washes up on beaches. Powdered pumice is used in some **abrasive** cleaning products.

eBookplus eLesson

Volcanoes
Learn how volcanoes are formed, and what happens whey they blow their tops.

eles-0130

SCORIA

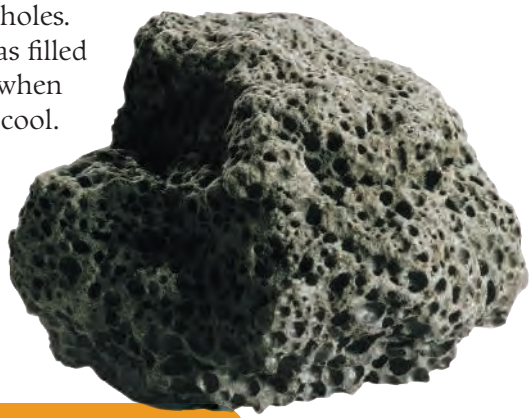
Scoria is heavier than pumice, and darker because it has more iron. It is usually found closer to the volcano's crater than pumice. Scoria is a reddish-brown or grey rock that can be used in garden paths or around drainage pipes.

BASALT

Basalt is an extrusive rock that can take on many appearances. One big difference between samples of basalt is the size of the crystals that make up the rock.

Basalt with bubbles

When viewed under a microscope, it is apparent that the crystals in this basalt are large. This is because they formed from lava on the ground. The crystals had time to grow before the rock became solid. Notice the holes. The lava was filled with gases when it began to cool. The gases have since escaped.



WHAT DOES IT MEAN?

The word *igneous* comes from the Latin word ***ignis***, meaning 'fire'. The words *ignite* and *ignition* also come from the same Latin word.

Pillow basalt

This rock formation came from a volcano that was once under water. Rocks formed from underwater volcanoes are smooth and round. The crystals in this basalt are so small that they are difficult to see.

OBSIDIAN

Obsidian is a smooth, black rock that looks like glass. It is formed when lava cools almost instantly. This rock is different from basalt because it cooled so quickly that no crystals formed. Sometimes very fine air bubbles are trapped in the rock, which give it a coloured sheen.

INQUIRY: INVESTIGATION 9.1

Does fast cooling make a difference?

KEY INQUIRY SKILLS:

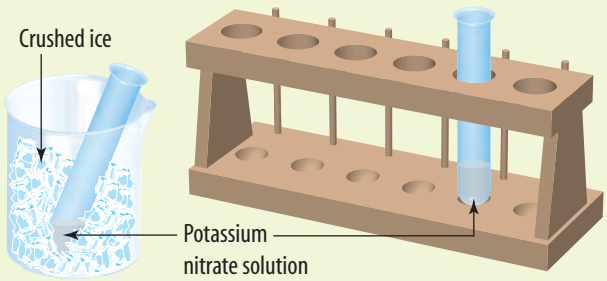
- planning and conducting
- processing and analysing data and information
- evaluating

Equipment:

freshly made saturated solution of potassium nitrate
potassium nitrate
spatula
250 mL beaker
3 test tubes and test-tube rack
test-tube holder
Bunsen burner, heatproof mat and matches
crushed ice
safety glasses
hand lens

CAUTION: Safety glasses must be worn during this experiment.

- Half-fill a beaker with crushed ice.
- Quarter-fill a clean test tube with saturated potassium nitrate solution. Add a spatula of potassium nitrate to the solution.
- Gently heat the solution over a Bunsen burner flame until the added potassium nitrate has dissolved or until the solution starts to boil.
- Pour half the warm solution into each of two clean test tubes.
- Place one test tube in the beaker of crushed ice and the other test tube in the rack to cool.



Cool one solution quickly and the other one slowly.

- When crystals have formed in each test tube, examine them with a hand lens.

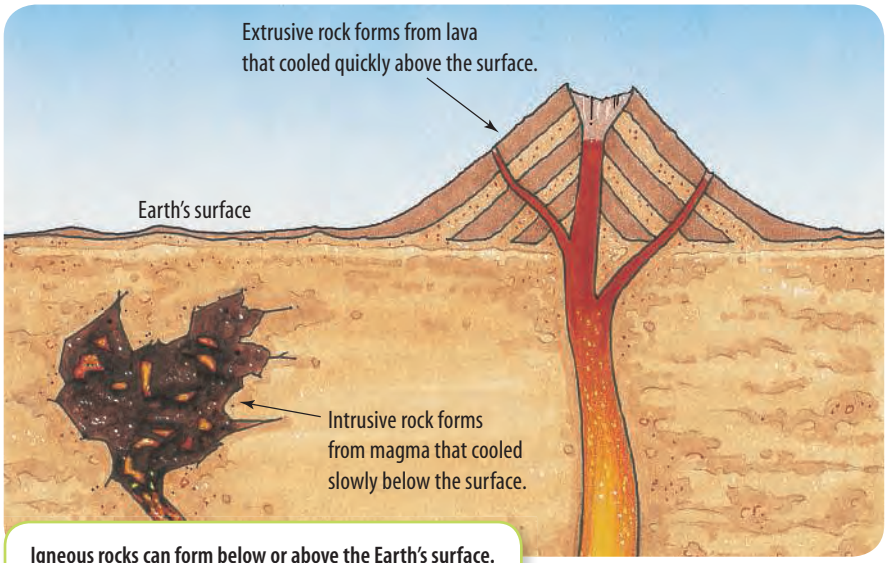
DISCUSS AND EXPLAIN

1 Draw a labelled diagram of some crystals in each test tube, concentrating on their shape and size.
2 Which test tube contained the larger crystals — the one that cooled quickly or the one that cooled slowly?
3 Which type of extrusive rock would you expect to have the larger crystals — those that cool slowly on the surface or those that cool quickly underwater?
4 Why do safety glasses need to be worn during this experiment?

Cooling underground

Igneous rocks can form below the surface of the Earth. Those that have formed from magma that cooled below the surface are called **intrusive rocks**. They cool very slowly and become visible only when the rocks and soil above them erode. Large bodies of intrusive rock are called **batholiths**. They can stretch over distances of up to 1000 kilometres.

Intrusive rocks (sometimes called plutonic rocks) have larger crystals than extrusive rocks because the crystals have had more time to grow.



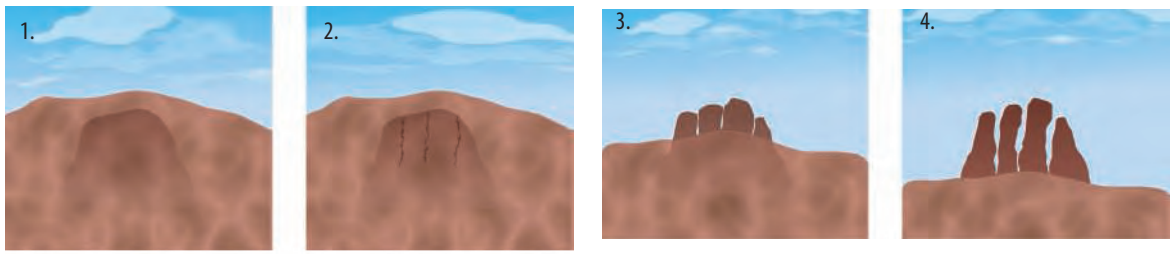
Igneous rocks can form below or above the Earth's surface.

GRANITE

Granite is a common intrusive rock. The crystals in granite form over long periods of time and grow large enough to make them easy to see with the naked eye. Granite is very hard and can be used for building. Headstones and other monuments can be made from granite that has been polished to give it a glossy finish.

The crystals found in granite are a mixture of white, pink, grey, black and clear minerals. These are quartz (clear to grey), feldspar (white and pink) and mica (black). Feldspar is made of aluminium silicate, and black mica is aluminium silicate combined with potassium, magnesium and iron.

If a batholith is exposed to the environment, it will start to wear away along the cracks. Over time, the batholith may break down completely. The breakdown of rocks is called weathering.



UNDERSTANDING AND INQUIRING

REMEMBER

1 What causes the frothy appearance of pumice and scoria?
2 Why are the crystals in pillow basalt smaller than those in many other examples of basalt?
3 Batholiths form well below the ground. Explain how they become visible on the Earth's surface.
4 Distinguish between the ways in which extrusive and intrusive igneous rocks are formed.
5 Describe two major differences between the appearance of granite and basalt.
6 List the three minerals found in granite.

THINK

7 Explain how you would decide whether an igneous rock formed as a result of a volcano.
8 Rhyolite is an extrusive rock that contains the same minerals as granite. In what ways would you expect it to be different from granite?

INVESTIGATE

9 Locate a building, statue or memorial in your area that is made from granite. Describe the granite in the structure, and suggest why it was the chosen material.
10 Use the **Who am I?** weblink in your eBookPLUS to play the Rock Game and identify rocks from a series of clues.

eBookplus

work sheets 9.1 Igneous rocks

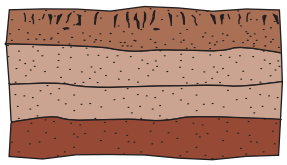
Sedimentary rocks

Rocks that are formed from the particles of sediments are called **sedimentary rocks**.

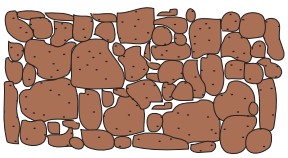
Most sedimentary rocks are formed from weathered rock that has been deposited due to erosion. Grains of sediment are cemented together to form a solid rock. The process is shown in the diagrams below.

WHAT DOES IT MEAN?

The word *conglomerate* comes from the Latin word ***conglomerare***, meaning to 'roll together'.

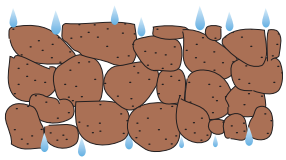
1 

Sediments are laid down by ice, wind or water, in horizontal layers called beds.

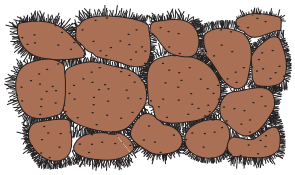
2 

Within each bed, the sediment grains are squashed together so that they are in close contact.

Many sedimentary rocks form in this way.

3 

Water seeps in between the grains, bringing with it many dissolved chemicals.

4 

When the water evaporates, these chemicals are left behind as crystals around the edges of the grains. These crystals cement the grains of sediment together to form rock.

HOW ABOUT THAT!

Chalk is a sedimentary rock. It is similar to limestone, but not as hard. Chalk is formed from very fine grains of calcium carbonate that have separated from sea water and settled to become a white, muddy sediment on the sea floor. The sediment hardens over time to form chalk. This process takes millions of years. The remains of shellfish and other sea animals are also found in the sediment that forms chalk, but most of these remains are microscopic.

The white cliffs of Dover that overlook the English Channel are composed of chalk.

Sandstone is formed from grains of sand that have been cemented together over a period of time. **Mudstone** and **shale** are formed from finer grains of sediment deposited by calm water in the form of mud. **Siltstone** has grains slightly larger than those of mudstone. **Conglomerate** contains grains of different sizes that have been cemented together.

Conglomerate is formed from sediments that might be deposited by a fast-flowing or flooded river.

Rocks from living things

Limestone is a sedimentary rock that is formed from deposits of the remains of sea organisms such as shellfish and corals. The hard parts of these dead animals contain calcium carbonate. These deposits are cemented together over a period of time in very much the same way as sedimentary rocks form from weathered rock.

This limestone, rich in corals and shells, is many metres above sea level. How did it get there?

Coal is formed from the remains of dead plants that are buried by other sediments. In dense forests, layers of dead trees and other plants build up on the forest floor. If these layers are covered with water before rotting is completed, they can become covered with other sediments. The weight of the sediments above compacts the partially decayed plant material. Over millions of years the compacting increases the temperature of the sediment and squeezes out the water, forming coal.

Rocks from chemicals

Some sedimentary rocks form when water evaporates from a substance, leaving a layer that is compressed after being buried by other sediments. **Rock salt** is an example of a rock formed in this way. It forms from residues of salt that remain after the evaporation of water from salt lakes or dried-up seabeds.

Rocks in layers

Layers of sedimentary rock are often clearly visible in road cuttings and the faces of cliffs. The limestone in the photograph at bottom left was formed on the ocean floor. Layers of sediments and sedimentary rocks can be pushed upwards by the same forces below the Earth's surface that cause mountains to be formed. Those forces can also bend and tilt the layers.

Layers of sedimentary rock can be pushed upwards, bent and tilted by forces beneath the Earth's surface.

Using sedimentary rocks

Sandstone and limestone are often used as external walls of buildings. These sedimentary rocks are well suited to carving into bricks of any shape. Shale can be broken up and crushed to make bricks.

Limestone is broken up to produce a chemical called lime. Lime is used to make mortar, cement and plaster, in the treatment of sewage and on gardens to neutralise acid in the soil.

Coal is used as a fuel. It is burned in electric power stations to boil water. The steam is then used to drive the turbines that produce electricity. In some countries, coal is burned in home heaters.

INQUIRY: INVESTIGATION 9.2

Identifying sedimentary rocks

KEY INQUIRY SKILLS:

- processing and analysing data and information
- evaluating

Equipment:

several examples of unlabelled sedimentary rocks, including limestone

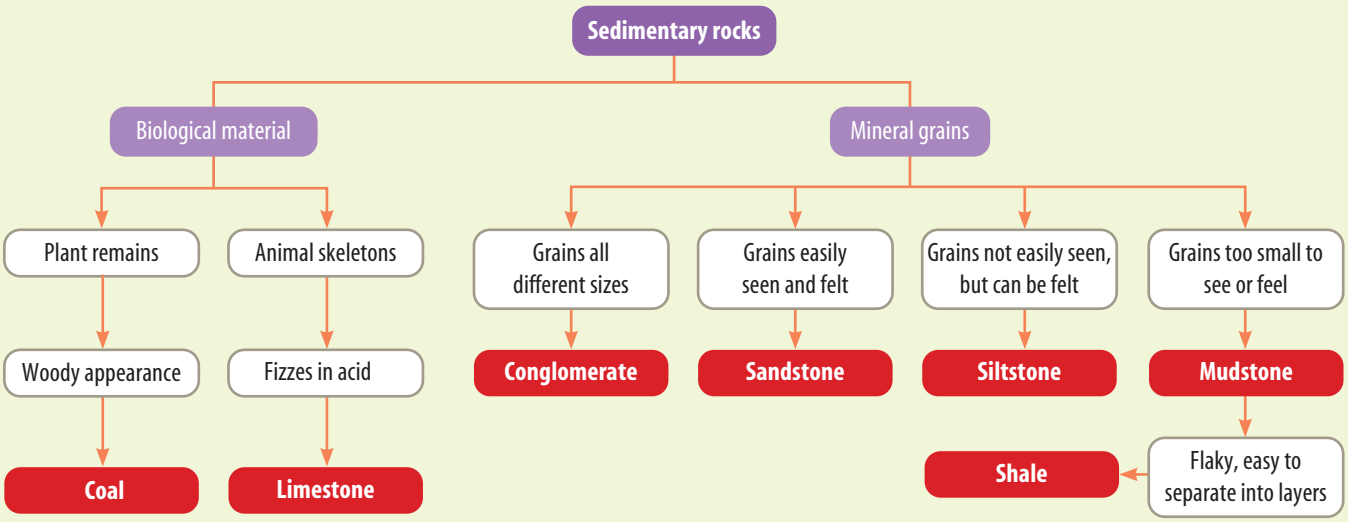
dropping bottle of dilute hydrochloric acid

- Use the key below to identify the samples of sedimentary rocks that you are provided with.
- To do the acid test, just add one drop of dilute hydrochloric acid onto the sample.

DISCUSS AND EXPLAIN

1 How many of the unlabelled rocks did you correctly identify?

2 Which of the rock samples were the most difficult to identify?



UNDERSTANDING AND INQUIRING

REMEMBER

1 From what are all sedimentary rocks formed?

2 Explain, with the aid of a diagram, how grains of sand can become part of a sedimentary rock.

3 Sedimentary rocks formed from the erosion of weathered rock cannot be identified by their colour. What feature allows you to identify them?

4 Explain how a rock can be formed from the remains of animals.

5 How is coal formed?

6 Explain why sedimentary rocks are found in layers.

THINK

7 A road cutting reveals a layer of sandstone beneath a layer of mudstone. Between them is a much thinner layer of conglomerate.

(a) Which layer would have formed from sediments beneath the sea?

(b) Which layer would have formed while the area was flooded by a swollen, fast-flowing river?

(c) Which layer would have formed while the land was the floor of a still lake?

(d) Which layer was formed most recently?

8 Explain why limestone and coal are sometimes referred to as biological rocks.

9 Limestone is mostly formed on the ocean floor. Explain why hundreds of limestone caves are found around the Nullarbor Plain.

10 In which type of sedimentary rock would you be most likely to find embedded seashells?

INVESTIGATE

11 What do peat, brown coal and black coal have in common? How are they different from each other?

work sheet 9.2 Sedimentary rocks

Stability and change: Metamorphic rocks

The thing that never changes about the Earth is that it never stops changing. Igneous rocks and sedimentary rocks are constantly undergoing change. If they are on the surface, they are weathered, cracked or even bent.

Igneous and sedimentary rocks deep below the Earth's surface are buried under the huge weight of the rocks, sediments and soil above them. They are also subjected to high temperatures. The temperature increases by about 25°C for every kilometre below the surface. This heat and pressure can change the composition and appearance of the minerals in rocks.

The process of change in the rocks is called **metamorphism** and the rocks that are formed by these changes are called **metamorphic rocks**.

Shale (pictured below) is a common type of sedimentary rock. It has fine grains and crumbles easily along its layers. When shale is exposed to moderate heat and pressure, it forms **slate**.

Marble (pictured below) forms from limestone under heat and pressure. It contains the same minerals as limestone.

The changes that take place during the formation of metamorphic rocks depend on:

- the type of original rock
- the amount of heat to which the original rock is exposed
- the amount of pressure caused by the weight of the rocks above
- how quickly the changes take place.

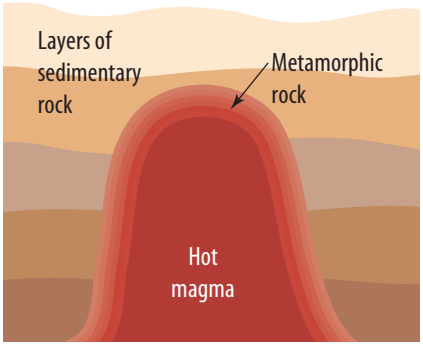
Metamorphic rocks that are mainly the result of great pressure can often be identified by bands or flat, leaf-like layers. These bands are evident in the sample of **gneiss** (pronounced 'nice') pictured below.

Gneiss is formed mainly as a result of great pressure on granite.

HOW ABOUT THAT!

Have you ever tried to lift one end of a pool table and noticed how incredibly heavy it is? It's really heavy because the flat surface under the felt is not wood as you may have thought — it's actually made of slate. Because of its natural hardness and its flat face, it makes an ideal even surface!

The diagram below shows how rocks can be changed by the high temperatures that result from contact with hot magma.



The formation of metamorphic rock by contact with hot magma

Other common examples of the formation of metamorphic rocks are:

Shale (sedimentary)	mainly pressure ⟹	Slate
Sandstone (sedimentary)	mainly heat ⟹	Quartzite
Limestone (sedimentary)	mainly heat ⟹	Marble

WHAT DOES IT MEAN?

The word *metamorphic* comes from the Greek words ***meta***, meaning 'change', and ***morph***, meaning 'form'.

Clues from metamorphic rocks

The nature of metamorphic rocks above and below the ground can provide clues about the history of an area. Think about why the presence of quartzite or marble high in a mountain range would suggest that the area was once below the sea.

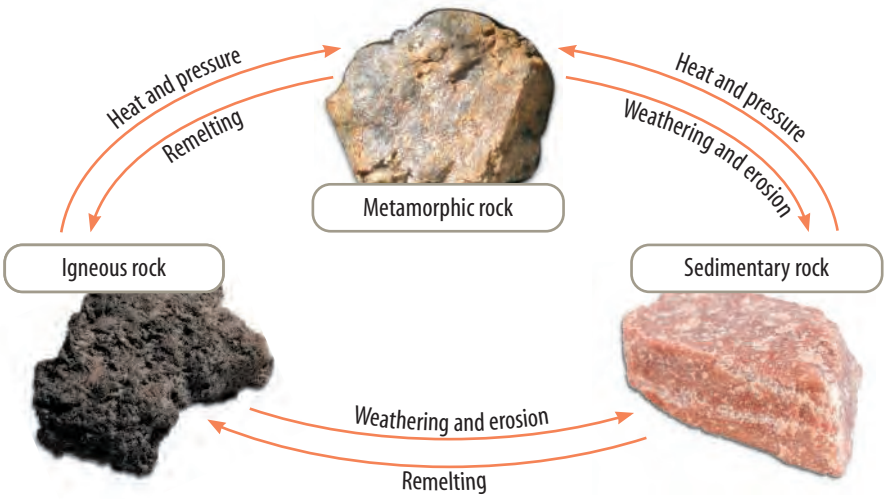
The presence of slate might suggest that the area was once the floor of a still lake or river mouth. The sediments were probably buried under many other sediments, and cemented together to form shale. The shale was transformed, or metamorphosed, into slate as a result of new rock formed above it.

Uses of metamorphic rock

The strength, resistance to weathering and appearance of marble make it suitable for use in statues and the walls and floors of buildings (inside and outside). It is usually highly polished. The hardness, flat structure and strength of slate make it ideal for use in buildings, especially in roofing and floor tiles. The sedimentary rocks from which marble and slate are formed could not be used for these purposes.

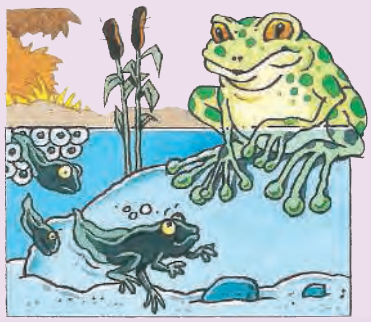
The rock cycle

The rock cycle below describes how rocks can change from one type to another. Weathering, erosion, heat, pressure and remelting are processes that help change rocks. The rock cycle is different from other cycles because there is no particular order in which the changes happen. Some rocks have been unchanged on Earth for millions of years and may not change for millions more. Some rocks change very quickly, especially near the edges of the plates that make up the Earth's crust.



HOW ABOUT THAT!

A tadpole grows into a frog, female frogs lay eggs, and eventually more tadpoles emerge from the eggs. That's a life cycle. Some of the changes in rocks can be described as cycles too. Weathered rock is moved by erosion and the particles form sediments, which can be cemented together to form sedimentary rocks, which in turn may eventually change into metamorphic rocks. Once those rocks are exposed at the surface the weathering starts all over again. A complete cycle normally takes millions of years, but sometimes never takes place at all. Why?



There are many cycles in nature. Some happen faster than others.

INQUIRY: INVESTIGATION 9.3

Rocks — the new generation

KEY INQUIRY SKILLS:

- questioning and predicting
- processing and analysing data and information

Equipment:

labelled samples of granite, gneiss, limestone, marble, sandstone, quartzite, shale and slate

hand lens

- Try to sort the rocks into pairs of 'parent' rock and corresponding metamorphic rock. Use the descriptions and examples on the previous two pages if you have trouble pairing the rocks.
- Examine each pair of rocks with a hand lens. Take particular note of grain or crystal size and banding.
- If necessary, re-sort the rocks into different pairs.
- Copy and complete the table below by noting the similarities and differences between the 'parent' and metamorphic rock of each pair.

DISCUSS AND EXPLAIN

1 Why is the term 'parent' rock used to describe the rock before metamorphosis?

2 Use the last column of your table to suggest whether the main cause of metamorphism was heat or pressure.

Comparing 'parent' and metamorphic rocks

'Parent' rock	Metamorphic rock	Similarities	Differences	Main cause of metamorphism
Shale				
	Gneiss			
Sandstone				
	Marble			

UNDERSTANDING AND INQUIRING

REMEMBER

1 Rocks are classified into three groups. Metamorphic rocks make up one of these. Identify two groups of rock from which metamorphic rocks are formed.

2 What can cause rocks to change form and become metamorphic rocks?

3 Describe the differences between gneiss and granite.

4 What causes granite to be transformed into gneiss?

5 Slate is commonly used in floor tiles. Why?

THINK

6 If an igneous or sedimentary rock gets so hot that it melts completely, it does not become a metamorphic rock. Explain why.

7 Why is limestone referred to as the 'parent' rock of marble?

8 Metamorphic rocks are generally formed deep below the surface of the Earth. However, they are often found above the ground — even high in mountain ranges. How can this be so?

INVESTIGATE

9 Find out more about the uses of marble and slate. Where are they obtained? What are they used for? Why are they expensive?

CREATE

10 Devise a 'buildings trail' in your city or town to locate buildings made of different kinds of rock. Draw a map to show the location of the buildings and the type of rock used in constructing them.

eBook*plus*

11 Apply mainly heat or pressure to a series of rocks and watch them change with the **Metamorphic rocks** interactivity in your eBookPLUS. **int-0234**

12 Use the **Rock cycle** weblink in your eBookPLUS to watch an animation of how rocks undergo change.

work sheets
9.3 Metamorphic rocks
9.4 The rock cycle

What's in a rock?

Rocks are made up of substances called **minerals**. Any naturally occurring solid substance with a definite chemical composition is called a mineral.

Elements found naturally in their uncombined form are also minerals. These elements, called **native elements**, include diamonds (pure carbon) and gold.

Native elements, such as the pure carbon that forms this diamond, are rare and valuable.

Most minerals in rocks are compounds of the elements oxygen and silicon, together with one or more of the metal elements. The colours, shapes and textures of the minerals in rocks tell us what they are made of and how they were formed, and also provide clues about the past.

Cool shapes

The atoms that join together to form minerals make up regular geometric shapes in particles called **crystals**. The way in which crystals grow when a mineral is formed depends on the speed of the cooling process of the molten material from which they form and how much space is available. The quartz crystals shown below have cooled slowly and have had a lot of time and space to grow. Quartz, one of the most common minerals, consists of hexagonal-shaped crystals of silicon dioxide (SiO_2).

Quartz is one of the most common minerals in the Earth's crust.

Fast or slow?

The effect of cooling rate on crystal size can be seen by comparing the crystals of two common igneous rocks. Basalt forms when volcanic lava cools quickly on the ground or under the ocean. There is not enough time for large crystals to form. Granite is formed when molten rock cools more gradually under the ground. The large crystals of quartz, mica and feldspar of which granite is made up can be easily identified.

Identifying minerals

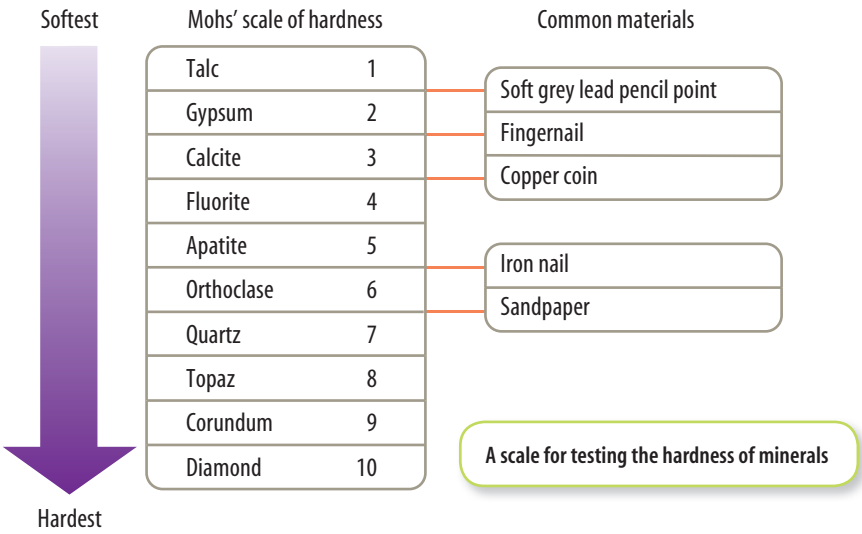
Although colour might seem to be the quickest way to identify a mineral, it is not reliable. Many different minerals have similar colours. Some minerals, even though they have the same chemical composition, can have different colours. Quartz, for example, can be colourless like glass, or may be pink, violet, brown, black, yellow, white or green. Other properties can be used to identify minerals.

The **lustre** of a mineral describes the way that light is reflected from it. Minerals could be described, for example, as dull, pearly, waxy, silky, metallic, glassy or brilliant.

The **streak** is the powdery mark left by a mineral when it is scraped across a hard surface like an unglazed white ceramic tile.

The **hardness** of a mineral can be determined by trying to scratch one mineral with another. The harder mineral leaves a scratch on the softer mineral. Friedrich Mohs' scale of hardness is a numbered list of ten minerals ranked in order of hardness. Higher numbers correspond to harder minerals. The hardness of a mineral is determined by comparing it with the minerals in Mohs' scale. For example, a mineral that can be scratched by quartz but not by orthoclase has a hardness between 6 and 7.

The diagram at the top of the next page shows that, if the minerals in Mohs' scale are not available, some more common materials can be used to determine the hardness of a mineral.



A scale for testing the hardness of minerals

INQUIRY: INVESTIGATION 9.4

Which mineral is it?

KEY INQUIRY SKILLS:

- processing and analysing data and information
- communicating

Equipment:

mineral kit

common materials to substitute for unavailable Mohs' scale minerals

hand lens

white ceramic tile

- Construct a table like the one shown below and use it to record your observations as you work through the following steps for each mineral.
- Describe the colour and lustre of the mineral.
- Use the magnifying glass to look closely at the mineral and describe the shape and size of its crystals.
- Scrape the mineral across the unglazed side of a white ceramic tile. Record the colour of the streak.
- Use the Mohs scale minerals or the common materials to estimate the hardness of the mineral by trying to scratch it. An approximate range, such as 5–6, is sufficiently accurate.

DISCUSS AND EXPLAIN

1 Other than those already described, what additional properties of minerals could be used to identify them?
2 If two unlabelled mineral samples have the same colour and lustre, can you be sure that they are the same mineral? Explain how you would find out.

Mineral	Colour	Lustre	Crystal shape and size	Streak	Hardness

UNDERSTANDING AND INQUIRING

REMEMBER

1 What is a mineral?
2 What is a native element? List two examples.
3 Which minerals are present in granite?
4 List at least five properties that you could observe to help you identify an unknown mineral.
5 What is the approximate hardness on Mohs' scale (to the nearest whole number) of a mineral that can be scratched by sandpaper but not by an iron nail?

THINK

6 Explain the difference between a rock and a mineral.
7 What does the size of the crystals in a rock tell you about the way the rock was formed?
8 You have a sample of each of two minerals but no other equipment to test them for hardness. How could you tell which mineral was harder?
9 A mineral can be scratched by a copper coin but not by a fingernail. You know that the mineral is one of quartz, fluorite or calcite. Which mineral is it?
10 Is table salt a mineral? Think carefully about your answer and suggest reasons for and against classifying it as a mineral.

IMAGINE

11 If only rocks could talk! Imagine you are a rock lying by the side of a newly made roadside cutting. Imagine that you were magically given the ability to walk and talk. Write a speech that you could deliver to a class of Year 8 students about your life — from your formation to the present time.

CREATE

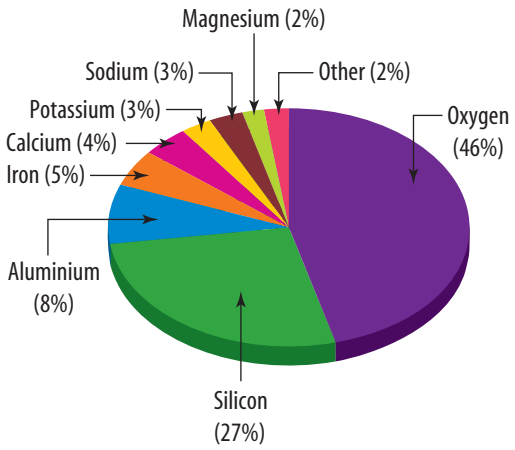
12 Find out how crystals can be artificially grown and then grow a crystal garden.

work sheet 9.5 Identifying and classifying minerals

Mining for metals

Metals play an important part in our lives every day. You probably slept on a mattress with metal inner springs. The alarm clock that woke you up has metal parts. You use metal cutlery to eat food. The bus, car or bike that you may have used to get to school is made from metal.

The metal elements used to make these things are all found in minerals in the Earth's crust. The pie chart below shows that almost three-quarters of the Earth's crust (by weight) is made up of the non-metals oxygen and silicon. Most of the metal elements are combined with other elements in compounds, usually with oxygen, silicon or other non-metals. It is these compounds that are the minerals.



The elements in the Earth's crust. The metal elements are relatively rare compared with oxygen and silicon.

Minerals containing metals of value are called **mineral ores**. It takes a lot of time, effort and money to get the rocks that contain the mineral ores out of the ground, separate the mineral ores from the rock and extract the metal element from the mineral ore. The **mining** of a mineral ore can take place only if enough of it is found at a single location.

The mining industry makes a major contribution to Australia's economy. Apart from the profits that go to shareholders in mining companies and to the government in taxes, the mining industry employs many thousands of Australians. Scientists and engineers are involved at every stage of the mining process.

Mineral exploration

Geologists make use of satellites equipped with cameras, radar and other sensors to search for geological features that are likely to contain high concentrations of minerals. Minerals in the crust dissolve in rain and running water and get washed into creeks and rivers. A chemical analysis of the sediments and surface water of lakes and streams, therefore, provides evidence of the presence of minerals in the area.

INQUIRY: INVESTIGATION 9.5

Searching without disturbing

KEY INQUIRY SKILLS:

- questioning and predicting
- processing and analysing data and information

Equipment:

a tray of sand	*10 paperclips*
blindfold (optional)	*compass*
paper and clipboard	*ruler*

- Find a partner. Begin by each of you drawing identical maps of the sand tray. Use the ruler to construct a grid on each map. The grids need to be labelled across the top and down the side (e.g. A–J across the top, 1–15 down the side) and should consist of at least 100 equal-sized rectangles or squares.
- Without showing your partner, hide the paperclips in the tray of sand and mark the location of the ten clips on your map.
- Your partner's task is to locate the ten paperclips and mark them on the map without disturbing the sand. You might wish to set a time limit.
- Swap roles and repeat the steps above.

DISCUSS AND EXPLAIN

1. What property of the paperclips was being used to allow them to be located?
2. How could your predictions of the location be checked with a pencil?
3. After checking, can the sand be restored to its initial condition?

The magnetic properties of large bodies of rocks containing some minerals can be detected from aircraft or by geological surveyors on the ground. Samples of soil and rocks are taken using portable equipment. On average, only one in 1000 sites that are sampled is eventually mined.

If there is sufficient evidence of useful mineral deposits that might be worth mining, a licence must be obtained before any clearing is done and heavy drilling equipment is brought in. Helicopters are sometimes used to bring in heavy equipment to protect sensitive ecosystems. The drilling allows mining companies to have a detailed look at what lies beneath the surface. Mining companies are required by law to clean up exploration drill sites and ensure they are left as they were found.

You can't start until . . .

In the past, mining was often carried out without considering its long-term effect on the environment and the people who live and work in the area. Today, however, an **environmental impact statement (EIS)** must be prepared before a mining operation can commence. This statement outlines how the mining company intends to manage all environmental aspects of the proposed mine. It also outlines how the land will be **rehabilitated** or reconstructed, so that it can be used again after the completion of the mining.

The environmental impact statement, along with any other relevant information, is studied by the government before permission to proceed is granted.

The EIS will report on:

- the existing flora, fauna and soils
- existing towns and roads in the area
- proposed new towns, roads and other developments
- how the new development might affect the local community and environment
- alternative plans to complete the development that might have less impact on the environment
- the measures that will be put in place to monitor and control air, water and noise pollution during the project and while rehabilitation is undertaken
- the rehabilitation proposals for the area.

Taking out the mineral ore

In order to obtain mineral ore from the ground, it is necessary to remove large amounts of rocks and soil. The way this is done depends on how close the mineral ore is to the surface. If it is close to the surface the vegetation and topsoil are removed. The waste rock removed from beneath the topsoil is called the **overburden**. The removed topsoil and overburden are used to fill areas that have already been mined, or they are left in a pile to restore the newly mined area when mining is completed. This method of mining is called **open-cut mining**.

If the mineral ores are deep below the surface, **underground mining** is undertaken. This method of mining is more dangerous and expensive than open-cut mining. Shafts and tunnels are dug up to four kilometres into the ground to reach the rocks containing the mineral ore. The development of open-cut and underground mining is overseen by mining engineers.

Geological surveyors use equipment like pick hammers to collect samples of rock.

Getting the metal

Obtaining the metal element takes place in two stages:

- Mineral extraction is the separation of the mineral ore from the rock taken from the ground. This involves crushing, grinding and washing the rock to separate the minerals from the unwanted rock.
- Metal extraction is the separation of the metal element that is obtained from the mineral. This always involves chemical reactions. The nature of these reactions depends on a number of factors, including the chemical composition of the mineral ore. Chemical engineers are involved in the design of this process.

Rehabilitation

Before mining of a new site begins, seeds of the natural vegetation of the area are collected so that seedlings may be cultivated at a later stage. The seedlings are grown in special nurseries until they are mature enough to return to the site of the mine.

During open-cut mining, the overburden (the material removed from the site to expose the mineral ore or coal) is used to fill holes left from earlier stages of the mining operation. Fresh topsoil is used to cover the overburden to ensure that new vegetation will grow. The soil surface is shaped to fit in with the surroundings, fertilised and sown with seeds or planted with seedlings. Care is taken to shape the new surface to prevent the newly sown soil from being eroded or washed away by wind or rain.

Prior to rehabilitation of the CSR Building Materials PGH mining site in Cooroy, Queensland (July 1994)

Six months into the rehabilitation process. The ongoing rehabilitation of this site earned CSR an environmental excellence award.

UNDERSTANDING AND INQUIRING

REMEMBER

1 Where are all mineral ores found?
2 In the Earth's crust, where are the metal elements found?
3 Describe the method of open-cut mining for the removal of mineral ores from ground.
4 Outline the two stages involved in obtaining a metal element from rock.
5 What is an EIS?
6 Outline the information that is included in an EIS.
7 How do mining companies rehabilitate the land used for mining?

THINK

8 The most common single element in the Earth's crust is oxygen. This element is a gas except at extremely low temperatures. In what form is the element found in the Earth's crust?
9 Explain why it is important that we recycle metals as much as possible.
10 In a table like the one below, make a list of the benefits and disadvantages of mining.

Benefits	Disadvantages

11 Summarise the reasons for and against allowing mining to take place in Australia's national parks.

BRAINSTORM

12 In a small group, discuss and list:
(a) the factors that should be considered by a mining company when it decides whether to start a mining project
(b) the different tasks that scientists and engineers might perform from the beginning of mining exploration until mining rehabilitation is complete.
Compare the lists of your group with those of others in your class.

Stone technology

Stone technology began about two million years ago when early humans started using rocks to make simple chopping tools. This was the beginning of the period known as the **Stone Age**. For the great civilisations of Asia, Europe and North Africa, the Stone Age ended around 3000 BC with the discovery of bronze, an **alloy** of copper and tin.

Sharp edges

The most commonly used resource in the Stone Age was a fine-grained sedimentary rock called **flint**. When flint breaks, it leaves a razor-sharp edge, so it was ideal for making sharp tools like knives, axes and spearheads.

Flint arrowheads were attached to wooden shafts with twine or animal sinews.

Small tools were made by striking tool stones like flint or the glass-like igneous rock obsidian with harder stones, such as quartzite, a metamorphic rock. To remove large flakes from the tool stone, a sharp blow was delivered by the harder rock. If the tool stone was struck correctly, a flake sheared from it. This process is called **percussion flaking**. The toolmaker continued to remove flakes from the stone until the desired shape was obtained. The flakes were then used to make tools such as knife blades, scrapers and engravers.

Larger items such as axeheads and spearheads were made with a combination of techniques, such as percussion flaking, grinding stones against each other and chiselling against the edge of a stone with tools made of bone or wood.

Indigenous ingenuity

Aboriginals and Torres Strait islanders were still using Stone Age tools when Europeans began to settle in Australia in 1788. They were highly skilled at working with stone. In fact, indigenous Australians were the first people to use ground edges on cutting tools and to grind seed.

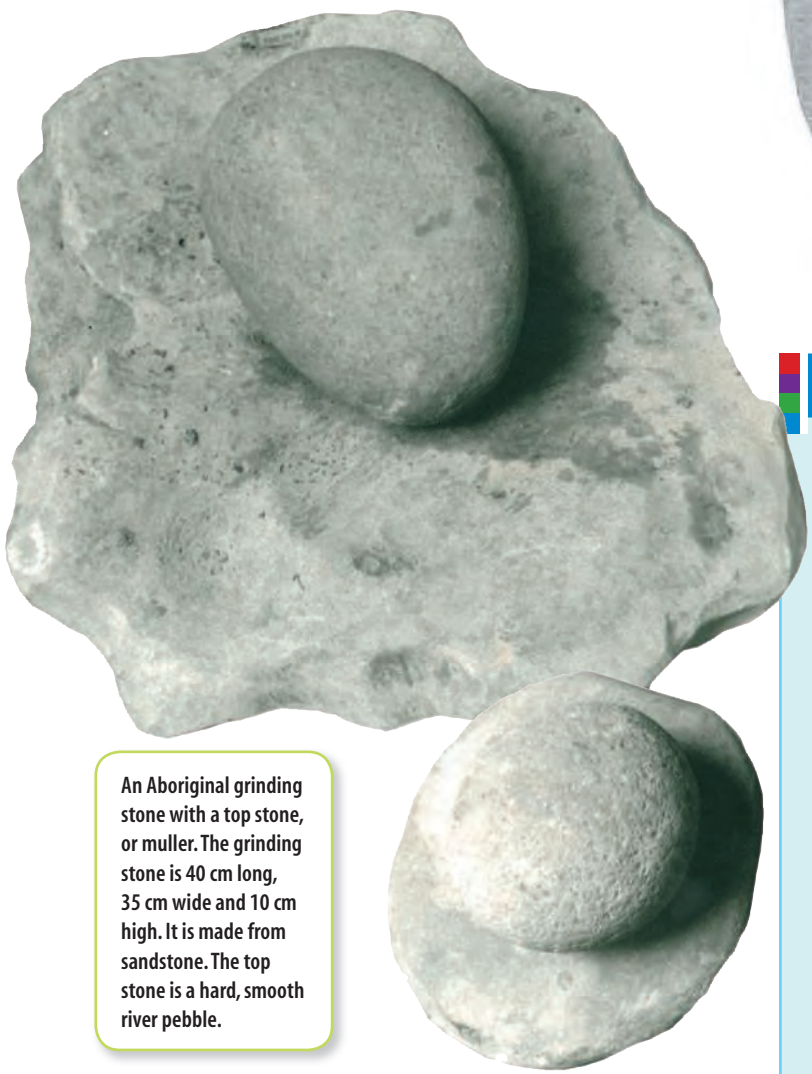
Their stone axes and other sharp tools were used to cut wood, shape canoes, chop plants for food, skin animals and make other tools out of stone or wood. The sharpened stones were often attached to wooden handles with twine from trees or with animal sinews.

Hand axes made and used by the Ngadojonji people of the tropical rainforests of northern Queensland

Grinding stones are slabs of stone used with a smaller, harder top stone to grind seeds such as corn and wheat, berries, roots, insects and many other things to prepare food for cooking. Leaves and bark were sometimes ground to make medicines. Grinding stones were also used to grind various types of soil and rock to make the powders used to paint shields and other wooden implements with traditional patterns.

Grinding stones were rough and usually made from sandstone, basalt with large crystals or quartzite. The smaller top stone was usually a hard, smooth river pebble.

A food grater made from stone by the Ngadjonji people of northern Queensland



An Aboriginal grinding stone with a top stone, or muller. The grinding stone is 40 cm long, 35 cm wide and 10 cm high. It is made from sandstone. The top stone is a hard, smooth river pebble.

The tools and the type of stone used to make them varied from group to group, depending on the location. Aboriginal people were skilled at making good use of the available resources. Apart from grinding stones, axes and other cutting tools, they made items such as bowls, cups and food graters out of stone.

UNDERSTANDING AND INQUIRING

REMEMBER

1 List one example of each of the following types of rock that were used in the Stone Age to make tools.
   (a) Igneous
   (b) Sedimentary
   (c) Metamorphic

2 Which alloy replaced stone as the substance chosen to make tools when the Stone Age ended?

3 What role did animal sinews play in toolmaking by indigenous Australians?

4 List three different uses of grinding stones.

THINK

5 What properties of flint made it so useful during the Stone Age?

6 List some properties that you would look for when selecting a suitable top stone for a grinding stone.

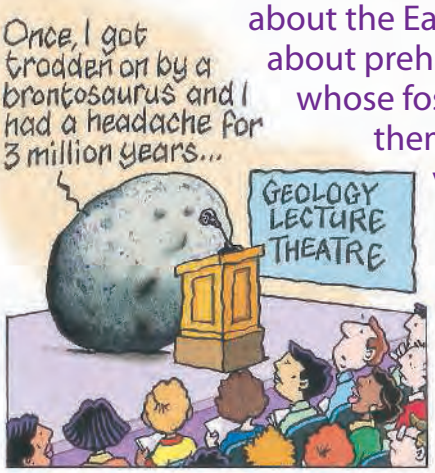
7 Suggest how the process of percussion flaking got its name.

INVESTIGATE

8 Research and report on a range of tools, weapons and other devices made from rocks and other natural materials used in the daily lives of Aboriginal and Torres Strait Islander peoples.

Every rock tells a story

If only rocks could talk! They would have so much to say. They would tell us about the Earth's history — about prehistoric creatures whose fossils lie within them, about explosive volcanoes, earthquakes, about flooded rivers that washed them away and about what it is like inside the Earth.



Although rock can't talk, geologists are able to use them to answer questions such as:

- What did the first humans on Earth eat?
- How has the Earth's climate changed over millions of years?
- What caused the extinction of the dinosaurs?

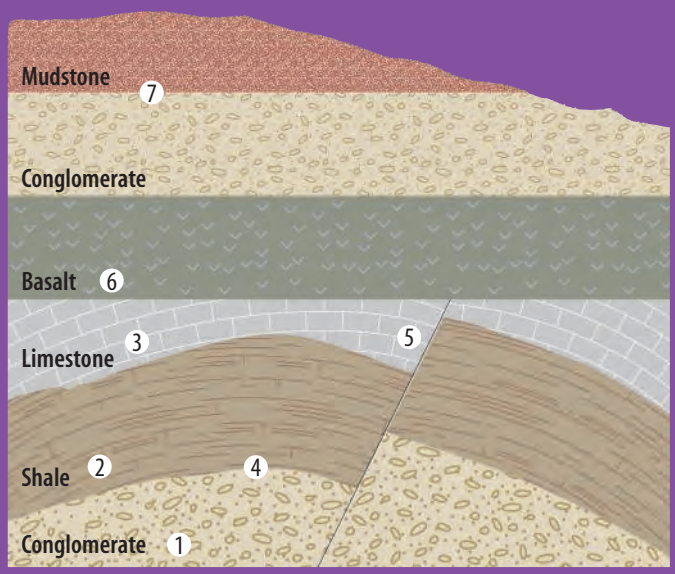
The clues lie in the appearance of the rocks, how they feel and the different layers of rock beneath the Earth's surface. There are also clues in fossils. A **fossil** is evidence of living things preserved in rocks.

Fossils provide clues about life in the past. This is a fossil of an ancient reptile.

Layers of clues

Over very long periods of time, rivers change their course; mountains form where seas once existed; the climate changes. As these changes take place, different layers of sediment can be deposited at the same location. Some layers will be thicker than others. Sedimentary rocks, formed by the different layers of sediments, provide many clues about the order in which events took place. Sudden events, like erupting volcanoes or earthquakes, are also recorded in the layers.

7. These layers were deposited last. They have started to weather and erode.
6. A long period of weathering and erosion left the layer of limestone with a flat surface. When a volcano then erupted nearby, lava from the volcano cooled to form basalt on the flat surface.
5. A sudden event, like an earthquake, has occurred to break the layers of rocks like this. This event took place after the lower layers were folded. A break like this is called a fault.
4. A slow event has caused the lower levels to buckle. This is called folding. Folding can occur when rock layers are under pressure from both sides.
3. The third event to occur was the deposition of limestone. It tells us that there were probably marine organisms present in the area during this time.
2. This is the second layer deposited. Shale is a fine-grained rock that is deposited in a quiet environment such as a swamp, lake or the slow-flowing part of a river.
1. Conglomerate was deposited first in this rock sample. This layer was deposited by a glacier or an active environment — like a very fast flowing river.



Slow movements caused by the forces beneath the surface have tilted, curved and pushed up the layers. Weathering and erosion have also exposed some layers that were below the ground millions of years ago.

It's all relative

Fossils provide a way of finding out how living things have changed over time. Evidence of the very oldest living things is buried within the deepest and oldest layers of rock. Scientists who study fossils are called **palaeontologists**.

By comparing fossils found in rocks in different areas, including different continents, it is possible to compare the **relative age** of rocks throughout the world. The relative age of a rock simply indicates whether it was formed before or after another rock. In any particular location it is almost certain that a layer of sedimentary rock is older than the rocks above it and younger than those below it. The relative ages of some igneous rocks and metamorphic rocks can be determined in the same way. It can also be assumed that the fossils in lower layers are older than those in the layers above.

How fossils form

The remains of most animals and plants decay or are eaten by other organisms, leaving no trace behind. However, if the remains are buried in sediments before they disappear, they can be preserved, or fossilised. Fossils can take several forms.

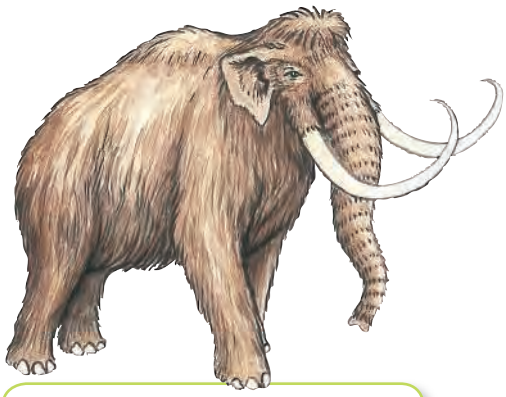
HARD PARTS

The hard parts of plants and animals are more likely to be preserved than the softer parts. Wood, shells, bones and teeth can be replaced or chemically changed by minerals dissolved in the water that seeps into them. Fossils formed in this way are the same shape as the original remains but are made of different chemicals; petrified wood is an example. Animal bones and shells can be preserved in sediments or rock for many years without changing. The types of bones, shells and other remains found in the layers of sedimentary rock provide clues about the environment, behaviour and diets of ancient animals.

This insect was trapped in the sap of a tree millions of years ago.

WHOLE BODIES

Sometimes, fossils of whole organisms, including the soft parts, are preserved. Such fossils are rare and valuable. Insects that became trapped in the sap (called amber) of ancient trees have sometimes been wholly preserved. Similarly, if the remains of animals or plants are frozen and buried in



An ancient woolly mammoth. Whole bodies of these ancient animals have been discovered in the ice of Siberia and Alaska.

INQUIRY: INVESTIGATION 9.6

Making a fossil

KEY INQUIRY SKILLS:

- questioning and predicting
- communicating

Equipment:

small seashell	*small box (shoebox or milk carton)*
fine sand	*plaster of Paris*

CAUTION: Do not put plaster of Paris down the sink.

- Half-fill the box with fine, damp sand.
- Make a clear imprint of a small seashell in the sand.
- Mix some plaster of Paris and pour it carefully into the imprint.
- Once the plaster has set, remove the plaster cast carefully from the sand.

You have two records of the seashell — the mould or imprint in the sand and the plaster cast.

DISCUSS AND EXPLAIN

1 Which parts of animals are most likely to be preserved as casts?
2 Is the fossil of a fern leaf more likely to be found as a cast or a mould? Why?
3 Dinosaur fossils are found in casts and moulds. What evidence of dinosaurs is likely to be found as a mould?

ice, they can be fully preserved. Whole bodies of ancient woolly mammoths (including skin, hair and internal organs) have been found trapped in the ice of Siberia and Alaska. These remains provide clues to the way in which living things have changed since ancient times. Whole bodies and preserved skulls of animals can even reveal evidence of their last meal before death.

MAKING AN IMPRESSION

The remains of animals or plants sometimes leave an impression, or imprint, in hardened sediments or newly formed rock. It is also possible for remains trapped in rock to be broken down by minerals in water, leaving a **mould** in the shape of the organism.

JUST A TRACE

Some fossils, called **trace fossils**, provide only signs of the presence of animals or plants. For example, footprints preserved in rock can provide clues about ancient animals, including dinosaurs, and how they lived. By studying the shape, size and depth of footprints, hypotheses can be made about the size and weight of **extinct** animals as well as how they walked or ran. Plant, leaf and root imprints, and feather impressions are other examples of trace fossils.

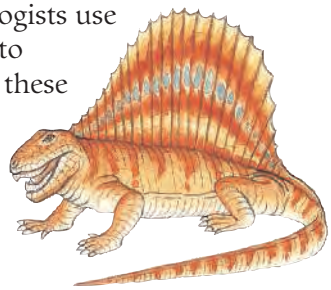
The imprint of the leaf of an ancient fern left in mudstone is a trace fossil.

Delving into dinosaurs

It is about 65 million years since the last non-flying dinosaurs existed on the Earth.

- What did they look like?
- What colour were they?
- How fast could they move?
- How did they behave?
- What did they eat?

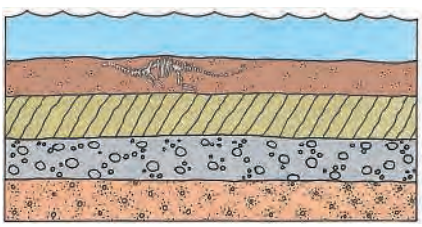
Palaeontologists use fossils to try to answer all of these questions and more!



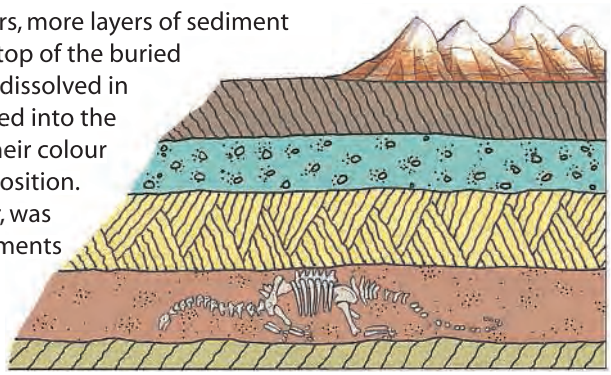
DINOSAURS PRESERVED IN ROCK

1 After the death of a dinosaur, its body would usually be eaten by meat-eating animals (**carnivores** or **scavengers**). Its bones would be crushed or weathered, leaving no remains. If, however, the remains of a dinosaur were buried in sediment, the bones could be preserved.

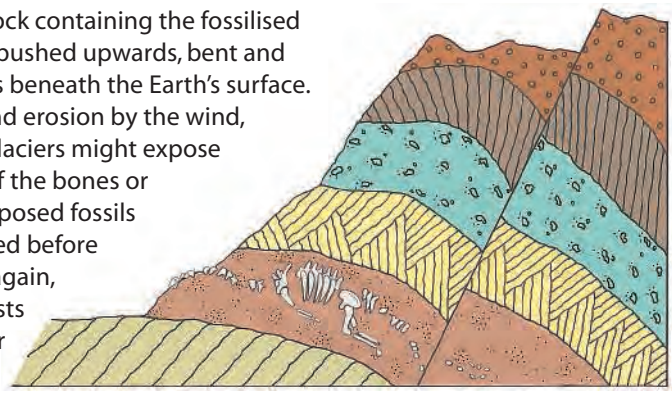
2 If a dinosaur died near a muddy swamp, shallow lake or riverbed, its remains sank in the mud or were washed into a river in a flood. The bones were quickly buried in sediment.



3 Over millions of years, more layers of sediment were deposited on top of the buried remains. Chemicals dissolved in the water that seeped into the remains changed their colour and chemical composition. The shape, however, was preserved. The sediments were gradually transformed into sedimentary rock.



4 The layers of rock containing the fossilised remains were pushed upwards, bent and tilted by forces beneath the Earth's surface. Weathering and erosion by the wind, sea, rivers or glaciers might expose one or more of the bones or teeth. If the exposed fossils were discovered before being buried again, palaeontologists might discover the remains.



NOT JUST A PILE OF BONES

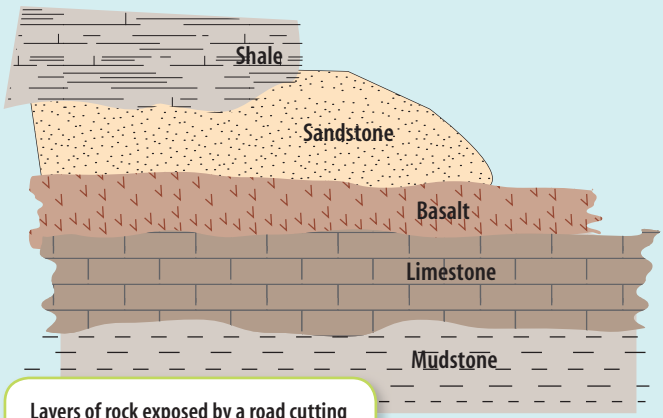
Dinosaur fossils are not all bones. They may include the following.

- Fossilised teeth: The shape of the teeth and the way they are arranged provide vital clues about the diets of dinosaurs. Flat-surfaced grinding teeth would have belonged to a dinosaur with a plant diet. When fossilised teeth like these are examined under a microscope, scratches caused by the grinding of the teeth are sometimes visible. Sharp-pointed teeth suited to tearing flesh would have belonged to a meat-eating dinosaur.
- Footprints: Dinosaur footprints are often preserved in rock. Footprints from a single dinosaur provide clues about the size and weight of the dinosaur. They also indicate whether the dinosaur walked on two legs or four, and how its weight was spread. The distance between footprints enables palaeontologists to estimate how fast the dinosaur moved. Footprints also provide clues about the behaviour of dinosaurs and whether they lived in herds or alone.
- Impressions of skin may be left in mud that has been hardened.

UNDERSTANDING AND INQUIRING

REMEMBER

1 A road cutting reveals the layers of rock shown in the diagram. Which of the rocks in the cutting is:
   (a) the oldest rock?
   (b) the youngest rock?
   (c) evidence of volcanic activity?



Layers of rock exposed by a road cutting

2 Explain why some layers of sedimentary rock are tilted, even though the sediments that formed them were laid in horizontal beds.
3 What does a palaeontologist study?
4 What clues about life in the past do fossils provide?
5 Under what circumstances can whole ancient living things be preserved as fossils?
6 Describe trace fossils and how are they useful.
7 What is the difference between a cast and a mould?
8 List the information about dinosaurs that can be obtained from fossils.
9 Fossils of dinosaurs form when their remains are buried under many layers of rock. Explain why fossils are often discovered in rocks and soil on the surface.

work sheet 9.6 Tracking changes in the rock

THINK

10 In which rocks shown in the diagram on the left would you be most likely to find the fossil of:
   (a) a seashell?
   (b) the leaf of a fern usually found in swamps?
11 Why are some layers in the diagram on the left thicker than others?
12 Explain why the hard parts of plants and animals are more likely to be preserved than the softer parts.
13 Explain how it is possible to use preserved dinosaur footprints to form hypotheses about:
   (a) whether dinosaurs lived alone or in herds
   (b) the way that dinosaurs walked
   (c) the weight of different kinds of dinosaurs
   (d) the walking or running speed of dinosaurs.

CREATE AND EXPLORE

14 Use plasticine to construct a sample of sedimentary rocks. Apply a gentle force to the sides of the layers. Describe how the layers fold under gentle pressure.

INVESTIGATE

15 Find out how the actual age of a rock in years is determined. This actual age is known as the absolute age.
16 Even an animal's droppings can become fossilised. Use the internet or books to research and report on the following.
   (a) Which animal was responsible for a huge fossilised dropping found in Canada in 1998?
   (b) How long was the dropping?
   (c) What can palaeontologists find out from it?
17 Use the **Australian Museum** weblink in your eBookPLUS or other resources to report on a major Australian fossil site where palaeontologists are working or have worked to increase our knowledge of ancient life.

eBookplus

18 Rate the rock formations in order from the oldest to the most recent with the **Relative age of rocks** interactivity in your eBookPLUS. **int-0233**

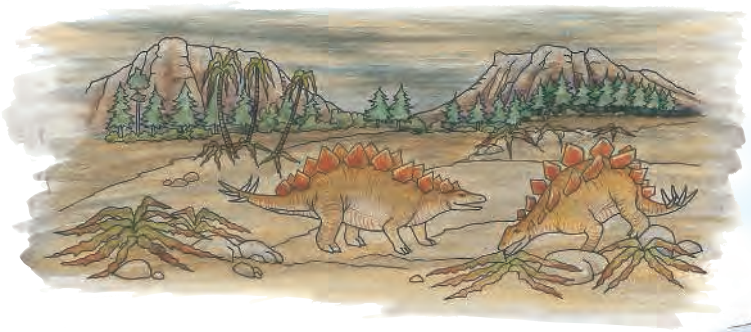
Questioning and predicting

Two of the inquiry skills that geologists and other scientists use are questioning and predicting. The question of how the dinosaurs died out has intrigued scientists for many years. In answering this question, scientists are using scientific knowledge to make 'predictions' about what happened many millions of years ago.

Solving the dinosaur riddle

Between about 250 million and 65 million years ago, dinosaurs were the most successful animals on Earth. In fact, those years are known as 'the age of the dinosaurs'. Dinosaurs thrived and dominated the land while mammals lived in their shadow. Fossil evidence indicates that the last of the dinosaurs died about 63 million years ago. There are several theories about the extinction of the dinosaurs. Scientists and others argue about whether the end of the dinosaurs was sudden or gradual. Scientists do generally agree that the riddle of the dinosaur remains unsolved. Palaeontologists and other scientists continue to look for clues that might provide the final solution.

THE ASTEROID THEORY



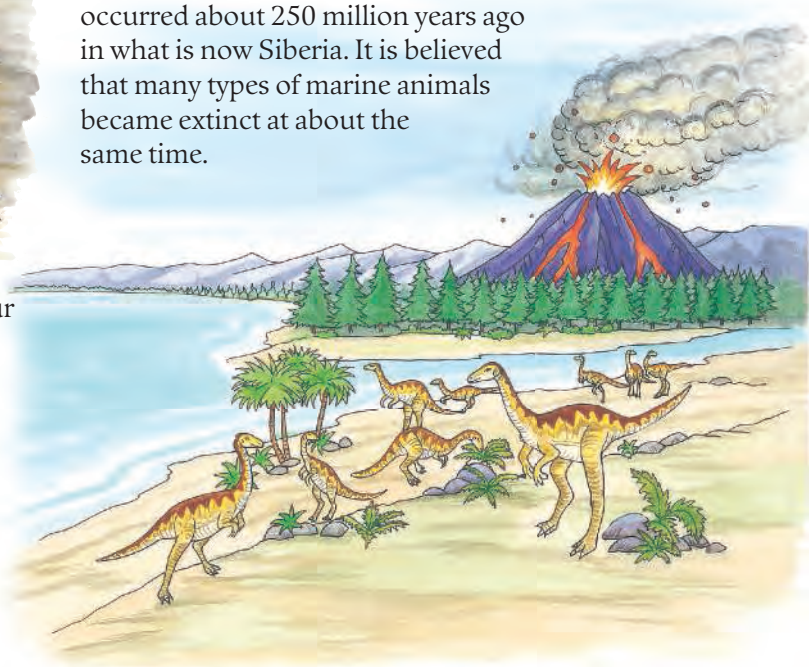
The most widely accepted solution to the dinosaur riddle is that an asteroid collided with the Earth around 65 million years ago. The impact of the asteroid threw billions of tonnes of dust into the air, blocking out sunlight and plunging the Earth into darkness for two or three years. Plants stopped growing but their seeds remained intact. The temperature dropped. The large plant-eating dinosaurs would have died quickly of starvation. The meat-eating dinosaurs would probably have died next, having lost their main food supply but surviving for a while by eating smaller animals. Many smaller animals would have survived by eating seeds, nuts and rotting plants.

As the debris began to settle and sunlight filtered through the thinning dust clouds, many of the plants began to grow again. The surviving animals continued to live as they did before the impact. The surviving mammals were no longer competing with dinosaurs for food. It was the beginning of the age of mammals.

THE VOLCANO THEORY

The eruption of Mount Pinatubo in the Philippines in June 1991 showed that ash and gases from volcanoes could reduce average temperatures all over the world. The average global temperatures during the years 1992 and 1993 were almost 0.2°C less than expected. While this is not a large drop in temperature, the size of the eruption of Mount Pinatubo was very much smaller than those of many ancient volcanoes.

The ash from a large volcano could have the same effect on sunlight and the Earth's temperature as an asteroid impact. If there was an unusually large amount of volcanic activity about 65 million years ago, the extinction of the dinosaurs could be explained. The largest known volcanic eruption occurred about 250 million years ago in what is now Siberia. It is believed that many types of marine animals became extinct at about the same time.

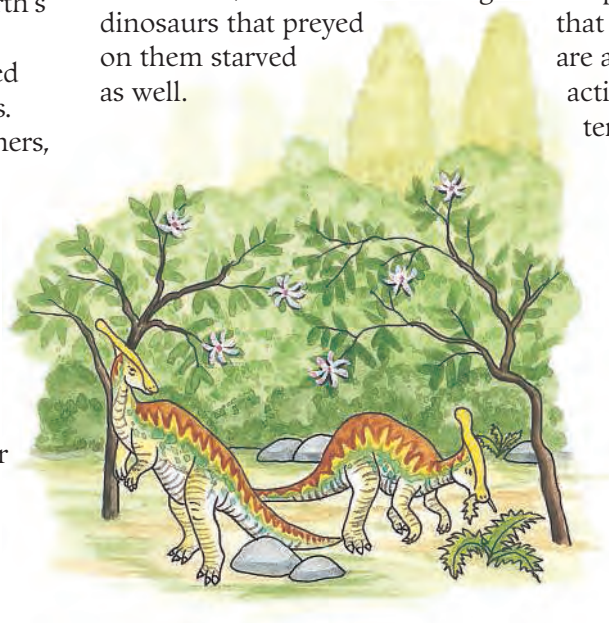


THE COOLING CLIMATE THEORY

The gradual cooling of the Earth's climate due to changes in the sun's activity could have caused the extinction of the dinosaurs. Dinosaurs, with no fur or feathers, had less protection from cold weather than mammals and birds. The larger dinosaurs would have found it very difficult to shelter from the cold conditions. Many smaller animals could burrow below the ground or shelter in the hollow trunks of trees or in caves. Many mammals and birds would have been able to migrate to warmer regions closer to the equator.

THE EMERGING PLANTS THEORY

During the Cretaceous period (140 million to 65 million years ago), new types of plants began to appear. Flowering plants evolved, competing with the more primitive plants such as ferns for nutrients, water and sunlight. The plant-eating dinosaurs did not eat flowering plants. As their traditional food supply became more scarce, the plant-eating dinosaurs could not survive, and the meat-eating dinosaurs that preyed on them starved as well.



COLD-BLOODED OR WARM-BLOODED?

Until recently, it was believed that dinosaurs were **ectothermic**. Ectothermic animals have body temperatures that depend on the temperature of their surroundings. As the surrounding temperature decreases, their body temperature decreases and they become less active.

Mammals are **endothermic**. Endothermic animals are able to maintain a constant body temperature that is usually above that of their surroundings. They are able to remain warm and active in lower surrounding temperatures.

If dinosaurs were in fact ectothermic, a cooler climate would have made it more difficult for them to compete with other animals for food. However, many scientists now believe that dinosaurs may have been endothermic.

The question of whether dinosaurs were cold-blooded or warm-blooded needs to be answered before the riddle of the dinosaurs can be solved.

UNDERSTANDING AND INQUIRING

REMEMBER

1 What is the most widely accepted theory about the extinction of the dinosaurs?
2 Why would smaller animals be more likely to survive the effects of an asteroid impact or large volcanic eruption than larger animals?
3 What is the difference between an ectothermic animal and an endothermic animal?

THINK

4 In what ways were the dinosaurs different from mammals?
5 How could volcanic eruptions affect life throughout the whole world?
6 How could meat-eating dinosaurs be endangered by the evolution of new types of plants?
7 Which group of animals benefited the most as a result of the extinction of the dinosaurs?
8 List as many weaknesses as you can in each of the four theories presented.
9 Which theory of the extinction of the dinosaurs do you think is most likely to be correct? Explain your answer.

IMAGINE

10 Imagine what it would have been like 65 million years ago if an asteroid plunged into the Earth. Write a story about the first 24 hours after the impact.
11 Which animals and plants do you think would be most likely to survive if an asteroid struck central Australia now? Explain your answer.

Fishbone diagrams and tree maps

1. Think of an event that you do not know the causes of.
2. Brainstorm as many possible causes as you can for this event.
3. In pairs or a team of four, use an affinity diagram to organise your list of causes into groups.
4. Write the event that you are analysing as the 'fish's head' of a fishbone diagram. Your groups of causes then become the main 'bones' of the diagram, one bone for each group.
5. Write the title for each of your groups of causes on its relevant 'fishbone'.
6. Write the causes on the smaller 'fishbones' that are joined to the sides of the main bones. (You can attach causes to more than one bone or group of causes.)

how to ...?

why use?

To show visually the reasons for what is happening/has happened

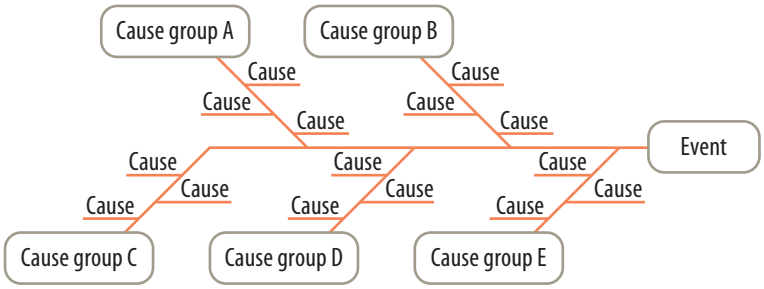
also called

Cause-and-effect analysis; Ishikawa diagram

question

What could have caused this to happen?

Fishbone diagram



comparison

Similarity

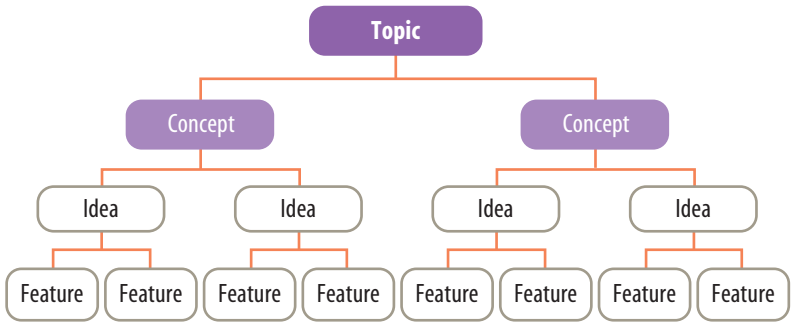
Both divide ideas into groups or categories.

Difference

Fishbone diagrams always categorise causes but tree maps do not.

Tree map

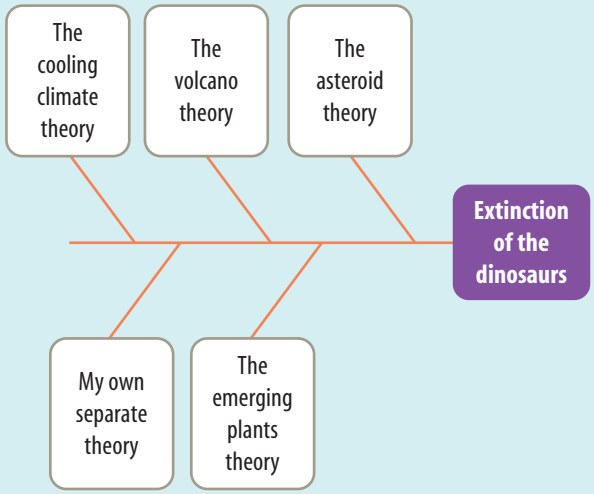
example



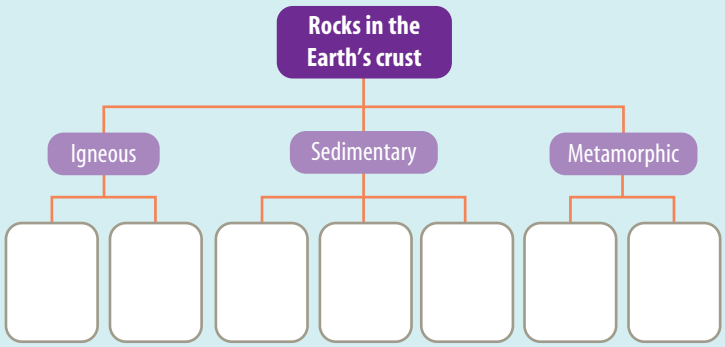
UNDERSTANDING AND INQUIRING

THINK AND CREATE

1 Create a fishbone diagram that shows the possible causes of the extinction of the dinosaurs. Use the one on the right as a template, adding the smaller 'bones' and causes yourself. The four theories about the cause discussed on pages 314–15 are already on the template. Include a separate theory of your own on the diagram. It could be a combination of the four other theories, or something completely different.



2 Working with a partner, copy the tree map below onto A3 paper and complete it to represent the three main types of rock found in the crust of the Earth. Add further branches below the existing ones if you can.

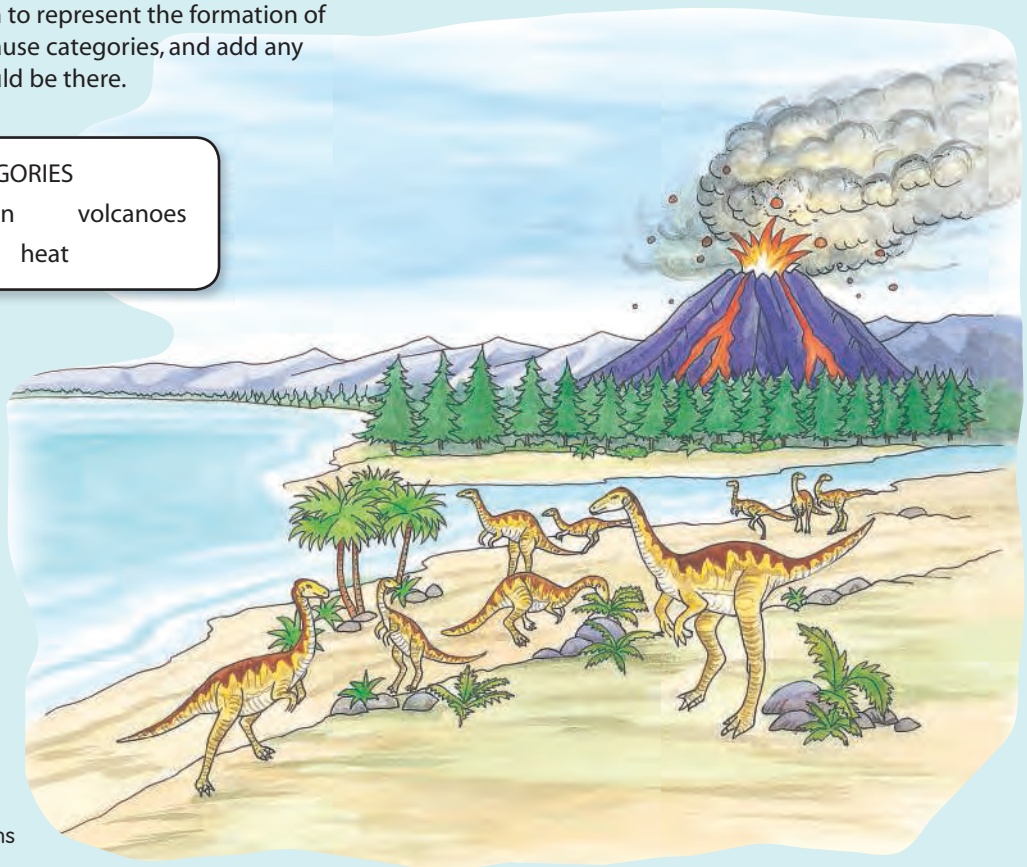


3 Create a fishbone diagram to represent the formation of rocks. Use the following cause categories, and add any others that you think should be there.

CAUSE CATEGORIES

weathering erosion volcanoes

pressure heat



work sheet 9.7 Thinking tools: Fishbone diagrams and tree maps

STUDY CHECKLIST

CLASSIFYING ROCKS

- describe the formation of igneous rocks
- distinguish between extrusive and intrusive igneous rocks
- explain how cooling rate affects crystal size
- describe the formation of sedimentary rocks
- explain the role of water in the formation of many sedimentary rocks
- identify a range of sedimentary rocks using a key based on the observation of physical and chemical properties
- describe the roles of heat and pressure in the formation of metamorphic rocks
- identify the cyclic nature of the formation of igneous, sedimentary and metamorphic rocks
- outline the uses of igneous, sedimentary and metamorphic rocks, including as building materials

MINERALS

- recall that all rocks are made up of substances called minerals
- describe the physical properties of a variety of minerals
- recognise that the minerals in some rocks provide valuable resources
- describe the processes involved in the mining of mineral ores

READING THE EARTH'S HISTORY IN ROCKS

- explain how layers of sedimentary and other rocks, together with fossils, reveal information about past environments and life on Earth
- use geological cross-sections to interpret simple geological histories
- outline the use of fossil evidence for investigating dinosaurs and their behaviour

SCIENCE INQUIRY SKILLS

- examine evidence in order to evaluate the theories about the extinction of the dinosaurs

SCIENCE AS A HUMAN ENDEAVOUR

- explain how the expertise of scientists and engineers is used in mineral exploration, the extraction of mineral ores and metals and the rehabilitation of mining sites
- describe some examples of the use of traditional stone technology in the daily lives of Aboriginal and Torres Strait Islander peoples
- discuss the environmental and community issues associated with the mining of mineral resources

INDIVIDUAL PATHWAYS

eBookplus

Activity 9.1
Investigating rocks and minerals
doc-6057

Activity 9.2
Analysing rocks and minerals
doc-6058

Activity 9.3
Investigating rocks and minerals further
doc-6059

ICT

eBookplus **Summary**

eLesson

Volcanoes

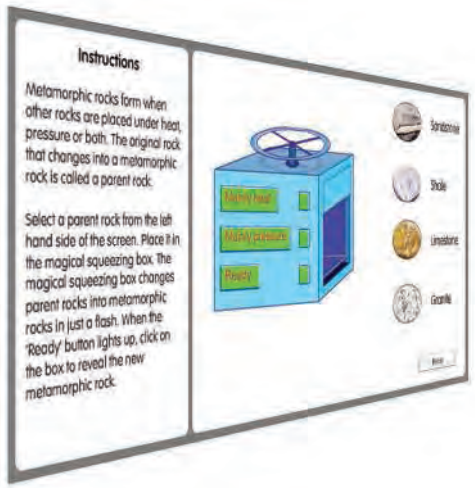
In this eLesson you will learn how volcanoes are formed, what happens when they blow their tops and how volcanic eruptions change the face of the Earth.

Searchlight ID: eles-0130

Interactivities

Metamorphic rocks

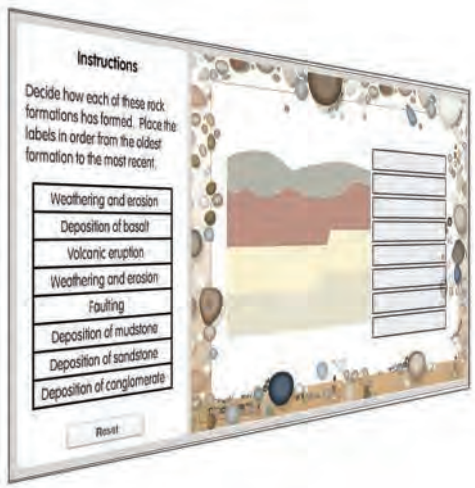
Metamorphic rocks form when other rocks are placed under heat, pressure or both. The original rock that changes into metamorphic rock is called a parent rock. This interactivity enables you to apply heat or pressure to a series of rocks and watch them change.



Searchlight ID: int-0234

Relative age of rocks

This interactivity tests your knowledge of how rocks are formed. Arrange a series of rock formations in order from the oldest to the most recent. Instant feedback is provided.



Searchlight ID: int-0233

LOOKING BACK

1 How are all igneous rocks formed?

2 Explain the difference between the ways in which extrusive igneous rocks and intrusive rocks are formed.

3 List three examples of extrusive igneous rocks.

4 What clues does the size of the crystals in an igneous rock provide about how the rock was formed?

5 What is the difference between:
(a) erosion and deposition?
(b) weathering and erosion?

6 What are sediments?

7 Describe the three different ways in which sedimentary rocks can be formed.

8 Suggest a way of checking that a rock sample that appears to have seashells embedded in it is limestone.

9 Explain why some layers of sedimentary rocks are titled or bent.

10 While studying sedimentary rocks in a railway cutting, a geologist discovers a bed of rock with ripple marks in its surface. How could the ripple marks have been made in the rock?

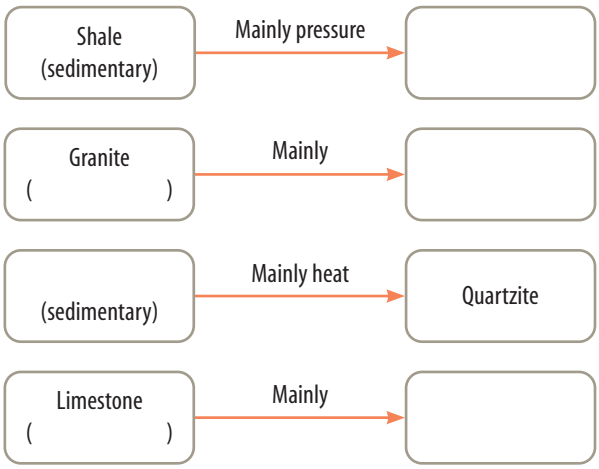
11 Describe two ways in which igneous and sedimentary rocks can be transformed into metamorphic rocks.

12 What is a parent rock?

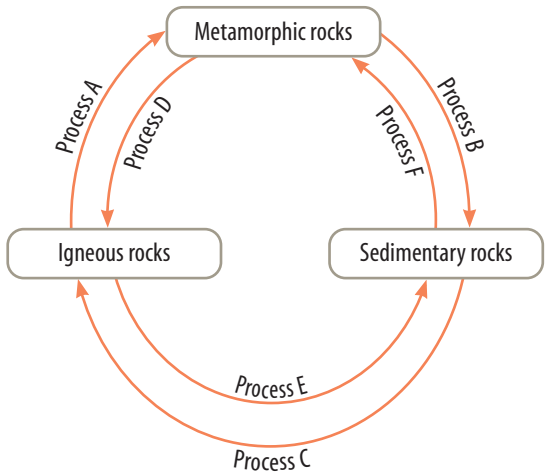
13 Copy and complete the table below to summarise what you know about igneous, sedimentary and metamorphic rocks.

Type of rock	How it is formed	Special features	Example	Uses
Igneous				
Sedimentary				
Metamorphic				

14 Copy and complete the diagram below to show how some common metamorphic rocks are formed.



15 The changes that lead to the formation of the three main groups of rocks can be drawn as a cycle, as shown below.



Which of processes A–F involve:
(a) weathering and erosion?
(b) heat and pressure?
(c) remelting?

16 Explain why the crystals in granite are larger than those in basalt.

17 What characteristic of minerals do the following terms describe?
(a) Lustre
(b) Streak
(c) Hardness

18 What is a native element? List two examples of native elements.

19 What property of minerals does Mohs' scale provide an approximate measure of?

20 If you were given a sample of each of two different minerals, how could you tell which one had the greater hardness?

21 Which sedimentary rock was the most important material for toolmaking in the Stone Age? What properties made it so useful?

22 In the Stone Age, tools were made by a process called percussion flaking.
(a) Describe the process of percussion flaking.
(b) Which property of the rock used to make the tool must be different from those of the stone from which the tool is formed?

23 What is the most common element in the Earth's crust and where is it found?

24 One factor that determines the way in which mineral ores are mined is their depth. Compare the mining processes used for mineral ores located near the Earth's surface with those used for mineral ores located deeper in the Earth's crust.

25 According to many geologists, parts of Antarctica are rich in mineral resources, similar to those found in Australia. Use a two-column table to list reasons why these mineral resources should be mined and why they should not be mined.

26 The mining industry provides employment for many Australians. Make a list of occupations that are involved in the mining industry. (*Hint:* Think about what happens before, during and after mining is undertaken.)

27 Imagine that the set of fossilised dinosaur footprints shown in the illustration below were found in a layer of sedimentary rock.
(a) Use the footprints to write a description of what might have happened millions of years ago.
(b) Compare your interpretation of the footprints with others.
(c) Does each person interpret the evidence in the same way?
(d) If there are differences of opinion about what happened, is there any way of knowing who is right?
(e) List as many differences as you can between the two types of dinosaurs making these footprints.

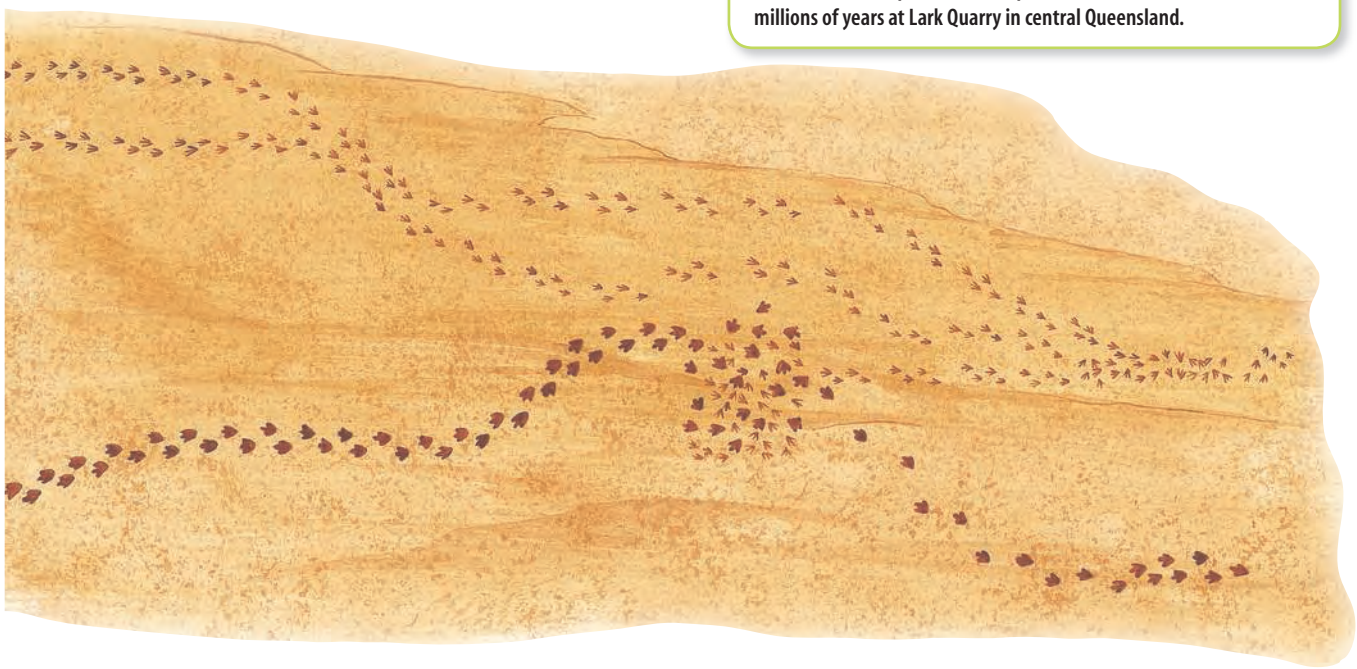
28 Not all fossils are the actual remains of living things. Name and describe two types of fossils that are not preserved remains.

29 Normally, old layers of rock are found below younger layers. Sometimes, however, younger layers are found beneath older layers. Explain how this could happen.

30 The photograph below is of a giant dinosaur footprint that has been preserved in rock at Lark Quarry in central Queensland.
(a) What type of fossil is it?
(b) Why is it classified as a fossil even though it could be described as a dent in a rock?
(c) Have all dinosaur footprints been preserved? Why has this one been preserved for hundreds of millions of years?
(d) What can be learned about the features of the dinosaur that left this footprint?
(e) What forms of evidence, apart from preserved footprints, can be used to gather knowledge about dinosaurs?

This dinosaur footprint has been preserved in rock for hundreds of millions of years at Lark Quarry in central Queensland.

work sheet 9.8 Summing up



10 Transferring and transforming energy

A fireworks display is one of the most spectacular energy transformations; you can not only see it but also hear, feel and smell it. When fireworks are ignited, the energy stored in the substances inside them is quickly transformed into movement (kinetic energy), light energy, sound energy and thermal energy (more commonly called heat). Energy that is stored is known as potential energy.

OVERARCHING IDEAS

- Patterns, order and organisation
- Form and function
- Stability and change
- Matter and energy
- Systems

SCIENCE UNDERSTANDING

Energy appears in different forms including movement (kinetic energy), heat and potential energy, and causes changes within systems.

Elaborations

Recognising that kinetic energy is the energy possessed by moving bodies

Recognising that potential energy is stored energy, such as gravitational, chemical and elastic energy

Investigating different forms of energy in terms of the effects they cause, such as gravitational potential energy causing objects to fall and heat energy transferred between materials that have different temperatures

Recognising that heat energy is often produced as a by-product of energy transfer such as brakes on a car and light globes

Using flow diagrams to illustrate changes between different forms of energy

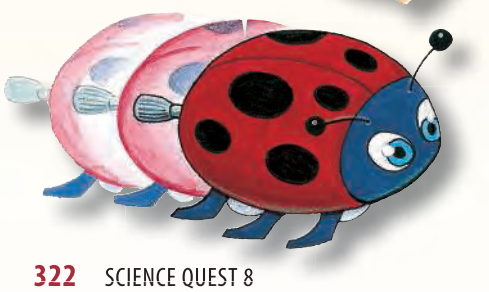
THINK ABOUT ENERGY

- Which type of energy do you find in chocolate?
- When you drop a tennis ball to the ground, why doesn't it return to its initial height?
- How much electrical energy is wasted as heat by a traditional light globe?
- How does ceiling insulation keep your house warmer in winter?
- From where does a firefly get the energy to make it light up?
- How do glow-in-the-dark stickers work?
- How does a didgeridoo player create its unique sound?

YOUR QUEST

Potential energy and kinetic energy

All substances and objects possess **potential energy**. But you can't tell unless something happens to transform the potential energy into a different type of energy. In the case of fireworks it's obvious when they explode. When a diver dives from a platform or diving board, the **kinetic energy** they gain on the way down is transformed from the energy stored in them because of their height above the ground. And the energy stored in the stretched string of the bow is transformed into the kinetic energy of the arrow when it is released.



Think

1 Copy and complete the table below. One example has been completed for you.

Object	What to do to release the stored energy	Potential energy is usefully transformed into . . .
Torch battery	Switch it on	electrical energy and light energy
Chocolate		
Petrol		
Dynamite		
Olympic diver on platform		
Match		
Stretched elastic band		

2 Answer the following questions about the wind-up toy shown on the left.

(a) Where is the energy stored when it is wound up?

(b) What do you have to do to allow the stored energy to be transformed into different forms?

(c) Name two forms of energy into which the potential energy is transformed.

(d) From where does the energy come that allows the user to wind up the toy?

Matter and energy: Making things happen

What is energy?

Energy is a word that you sometimes use to describe how active you feel. Sometimes you don't seem to have any energy. At other times you feel like you have enough energy to do just about anything. Energy is defined in most dictionaries as 'the ability to do work'. That is, it is the ability to make something observable happen.

We know that:

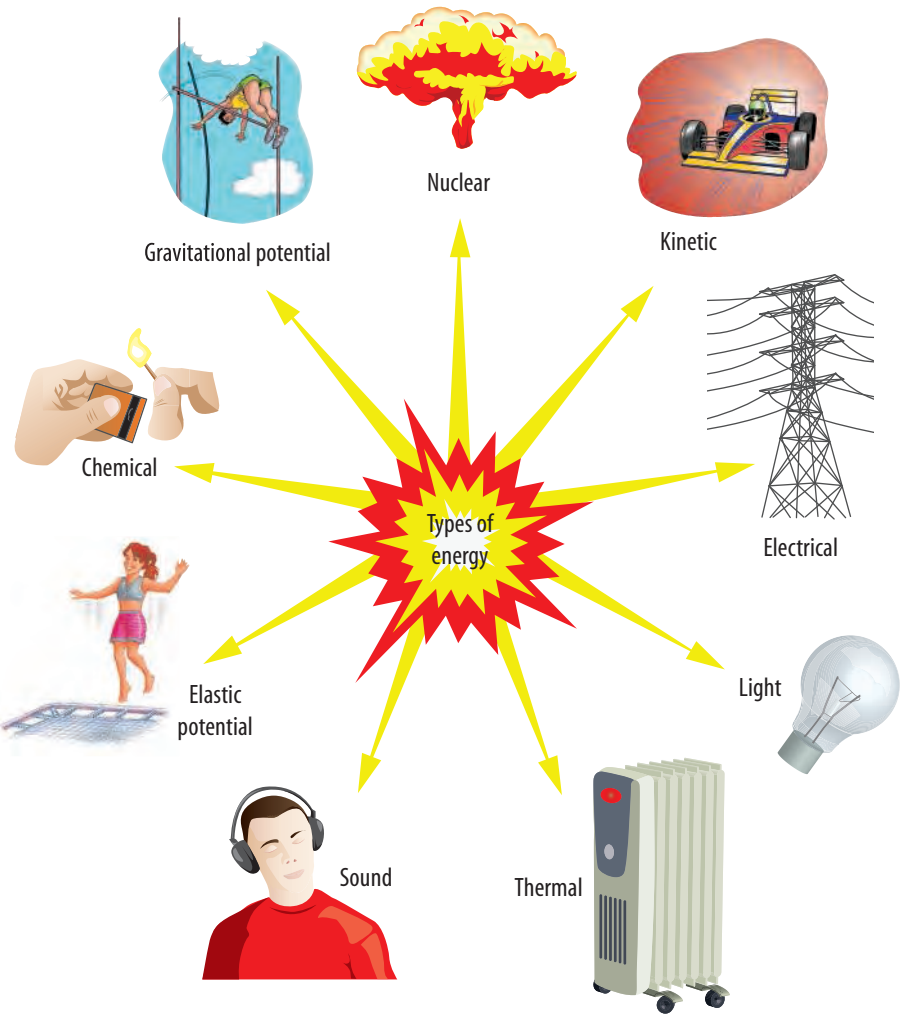
- all things possess energy even if they are not moving
- energy cannot be created or destroyed. This statement is known as the **Law of Conservation of Energy**. It means that the amount of energy in the universe is always the same.
- energy can be transferred to another object (for example, from a cricket bat to a ball) or transformed into a different form (for example, from electrical into sound)
- energy can be stored.

Types of energy

Light energy, sound energy, thermal energy and kinetic energy are all very easily observed. All objects that move have kinetic energy. **Electrical energy** can be seen if there is a spark or a lightning strike, but you can't see it when it's moving in wires. You become aware of it when it is changed into other forms, for example into light in a fluorescent tube or into sound in an iPod.

Stored energy is known as potential energy because it has the 'potential' to make something happen. There are several different forms of potential energy:

- A ball held above your head has **gravitational potential energy**. This form of energy becomes noticeable when you drop the ball and its stored energy is transformed into kinetic energy.
- A battery contains **chemical energy** but this is not noticeable until the battery is connected in an electric circuit. When that happens the chemical energy is transformed into electrical energy, which in turn is transformed into other types of energy — to make things glow, get hot, produce sounds or move. It is the chemical energy stored in food and drinks that gives you the energy to live and be active.



The chemical energy in fuels is transformed to operate cars and other vehicles, keep you warm and generate electricity.

- The **elastic potential energy** stored in a stretched elastic band is released when you let go of one end. The stored energy is transformed into kinetic energy.

1 At the very top of a jump, the bouncer is momentarily stopped — he has no kinetic energy. But he does have gravitational potential energy due to his height above the trampoline. As the force of gravity pulls the bouncer down, some of his potential energy is transformed into kinetic energy.

2 As the bouncer strikes the trampoline, his kinetic and gravitational potential energy are transferred to the trampoline's surface and springs. The energy is momentarily stored in the springs. It is called elastic potential energy.

3 At this point, the bouncer pushes off the trampoline. The elastic potential energy is transformed back into kinetic energy and some gravitational potential energy.

4 As the bouncer rises again, his kinetic energy is transformed into gravitational potential energy again. At the top of the jump, the bouncer has no kinetic energy, just gravitational potential energy.

Types of energy changes involved in bouncing on a trampoline

- **Nuclear energy** is the energy stored at the centre of atoms, the tiny particles that make up all substances. The energy we receive on Earth from the sun has been transformed from nuclear energy. Under the right conditions nuclear energy can be transformed into electrical energy in a nuclear power station. Unfortunately it has also been transformed into thermal energy in nuclear weapons.
- Electrical energy can also be stored. For example, if you rub a plastic ruler with a cloth it can become charged. You can't see the stored electrical energy but you can tell it's there when it bends a slow stream of water from a tap.

An unavoidable loss

Every electrical appliance you use, whether powered by batteries or plugged into a power point, converts electrical energy into other forms of energy. Most of that energy is usually converted into useful energy — but some is converted into forms of energy that are wasted or not so useful. But all of the electrical energy is converted — that's the Law of Conservation of Energy in action. The table below shows some examples of energy conversion by electrical appliances. None of the wasted energy is actually lost.

Energy conversion by appliances

Appliance	Electrical energy usefully converted to . . .	Electrical energy wasted . . .
Microwave oven	thermal energy of food	heating air in the oven, plates and cups etc.
Television	light and sound	heating the television and the surrounding air
Hair dryer	thermal energy and kinetic energy of air	as sound
Electric cooktop	thermal energy of food	as light and heating the surrounding air

The loss of useful energy is also evident when you apply the brakes in a car or on a bicycle, hit a cricket or tennis ball and jump on a trampoline. On a larger scale it is seen in power stations, where the fuel, falling water, solar energy or any other energy source is used to produce electricity; some of the energy of the source is transformed to heat, warming the power equipment, the surrounding air and the water used as coolant. The 'loss' of useful energy is unavoidable.

Some types of lighting waste more energy than others. Old-fashioned incandescent light bulbs convert more energy to wasted heat than to light. They emit light only when the filament inside gets white hot. Fluorescent lights and LEDs (light-emitting diodes) waste very little energy. Almost all of the electrical energy is converted to light, so you use much less energy to produce the same amount of light than you would using an incandescent bulb.

Efficiency

The **efficiency** of a car, light bulb, gas heater, car, power station, solar cell or any other energy converter is a measure of its ability to provide useful energy. Efficiency is usually expressed as a percentage. The efficiency of the traditional light globe pictured on the next page is 10% because 10% of the total electrical energy input is usefully transformed into light. The efficiency of the compact fluorescent light is 70%.

The efficiency of every device that uses fossil fuels is very important for the environment and life on the Earth. Scientists and automotive engineers are constantly working on methods of reducing fuel consumption by:

- increasing the efficiency of burning petrol and other fossil fuels such as diesel by reducing the amount of energy wasted as heat
- changing the external design of cars to reduce the amount of energy needed to overcome air resistance
- searching for alternative fuels such as ethanol that can be produced from sugar cane and grain crops.

eBookplus eLesson

Energy in disguise

Did you know that all energy is constantly being transformed and transferred from one object to another? There's more going on in your world than meets the eye.

eles-0063

eBookplus eLesson

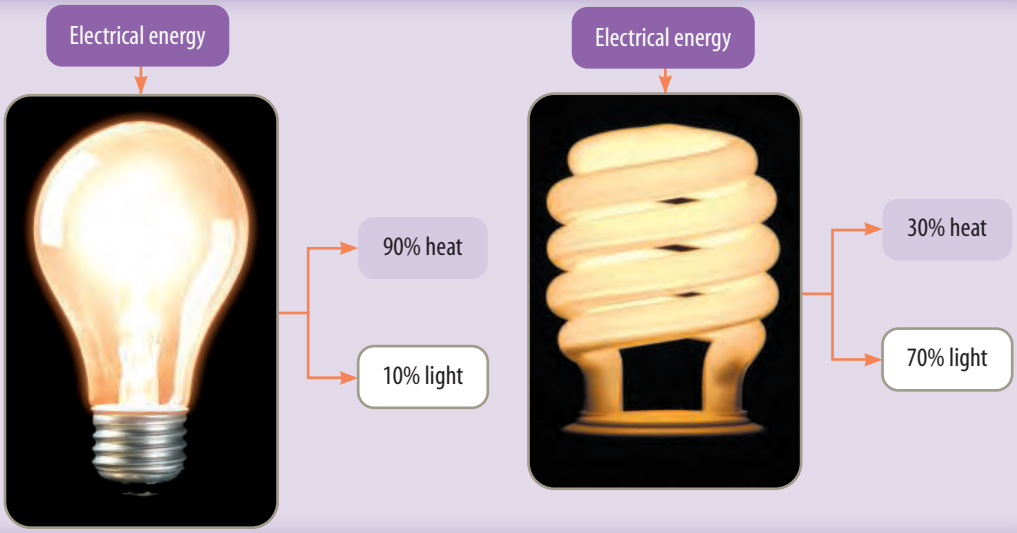
Australian International Model Solar Challenge

Learn about the exciting annual event where Australian high school students compete by building and racing model cars and boats.

eles-0068

HOW ABOUT THAT!

In traditional light bulbs, electricity passes through a thin filament in the bulb, causing it to glow white hot. The light is a useful form of energy but the heat is a wasted form of energy. Compact fluorescent lights (CFL) offer a more energy-efficient form of lighting as they generate less wasted heat.



UNDERSTANDING AND INQUIRING

REMEMBER

1 State the Law of Conservation of Energy.
2 Which form of energy is observed when:
(a) an athlete runs?
(b) a spring is squashed?
3 List five types of stored energy.
4 Describe the difference between gravitational potential energy and elastic potential energy.
5 Define the efficiency of an energy converter.
6 Outline three of the methods being used by scientists and automotive engineers to reduce the fuel consumption of cars.

THINK

7 Identify four types of energy that are present during a lightning strike.
8 From which type of energy does the sound of a cymbal come?
9 From which two types of energy does the sound of a bass drum come?
10 How can you tell that a high diver has gravitational potential energy?
11 Construct a table similar to the one below and use it to list the useful energy and the wasted energy converted by the following devices.
(a) A torch
(b) A wind-up toy
(c) A pop-up toaster
(d) A gas cooktop
(e) A car engine

Device	Source of energy	Energy usefully converted to . . .	Forms of energy wasted

12 When a tennis ball is bounced on the ground, it never returns to its original height. Does this break the Law of Conservation of Energy? Explain your answer.
13 Outline at least three reasons why efficiency is important for devices that use fossil fuels.
14 Suggest some methods that drivers could use to increase the fuel efficiency of their vehicles.

CREATE

15 Illustrate the energy transfers and transformations of the person on the trampoline shown on page 324 in the form of a flowchart.
16 Create a poster-sized flowchart to show the energy transformations that take place to produce lightning and thunder. (Think first about how the clouds become electrically charged during an electrical storm.)

INVESTIGATE

17 Research and report on the methods used by motor companies to increase the fuel efficiency of their cars since 1975.
18 Are solar-powered cars a realistic alternative to cars that run on fossil fuels or biofuels like ethanol? Find out what scientists, engineers and members of the public have contributed to the design of solar-powered vehicles.
19 Find out the purpose of the Australian International Solar Challenge and how you could become involved in it.
20 Use the **Coaster** interactivity in your eBookPLUS to identify the positions on a roller coaster rider where the car has more kinetic energy and where it has more gravitational potential energy. **int-0226**

eBookplus

work sheets
10.1 Skateboard flick cards
10.2 Types of energy

Hot moves

If you accidentally touch a hotplate you'll find out quickly — and painfully — that heat travels from warm objects to cooler objects.

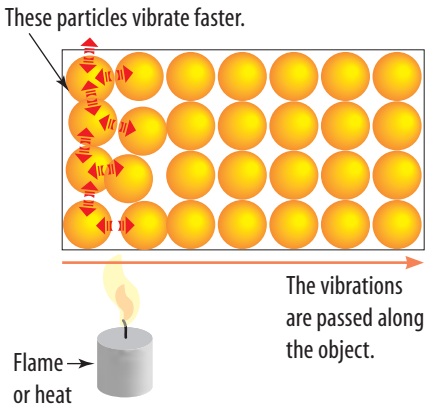
It is the rapid transfer of energy into your hand that causes the pain. Sports people sometimes use ice baths to assist with injury. The body heat is transferred quickly into the cold ice. If you touch something that has the same temperature as your hand you won't feel any sensation of heat transfer into or away from your hand.

Heat always travels from areas of high temperatures to areas of lower temperatures.

Heat is energy in transit from an object or substance to another object or substance with a lower temperature. There are three different ways in which heat can move from one place to another — by conduction, convection or radiation.

Conduction

If you've ever picked up a metal spoon that has been left in a hot saucepan of soup you will know that heat moves along the spoon as it reaches the handle. This is an example of **conduction** of heat. Metals are very good conductors of heat. Like all substances, metals are made up of tiny particles. The particles in all solid substances are vibrating. Of course you can't see the vibrations because the particles are far too small to see — even with a microscope. When you heat a solid object its temperature increases. The particles vibrate faster and bump into each other. The vibrations are passed from particle to particle along the object until the whole object is hot.

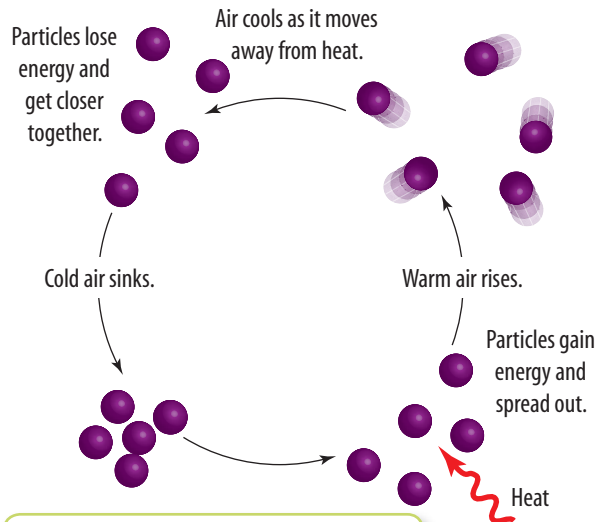


Conduction of heat occurs as a result of vibrating particles.

Not all solids conduct heat at the same rate. Metals, for example, are much better conductors than most other solids. Some solid substances are very poor conductors of heat. Glass, wood, rubber and plastic are all poor conductors of heat, and are called **insulators**. Many metal saucepans have a plastic or wooden handle. Suggest a reason for this.

Convection

The particles that make up solids are close to each other and held together tightly. They can vibrate faster only when heated. However, in liquids and gases the particles are further apart and can move around. So when liquids and gases are heated, rather

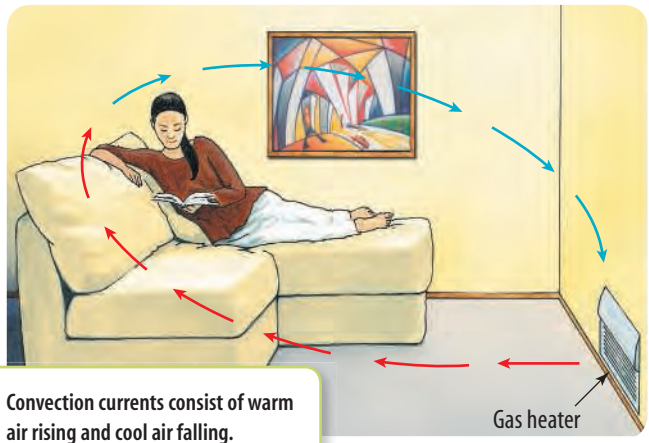


Modelling heat transfer in air by convection

than the vibration passing between particles, the particles themselves can move. Heat can travel through liquids by a process called **convection**.

The figure at the bottom of the previous page shows how convection takes place. Heat causes the particles of air to gain energy, move faster and spread out. This warmer air is less dense than the air around it so it rises. As it rises it begins to cool. The particles lose some of the energy gained, slow down and move closer together. This cooler air is denser than the air around it so it falls. The whole process then starts again, creating a pattern of circulation called a convection current.

Gas wall heaters create convection currents with the aid of a fan that pushes warm air out near floor level so that it heats the entire room as the air rises.



Convection currents consist of warm air rising and cool air falling.

Radiation

Heat from the sun cannot reach the Earth by either conduction or convection because there are not enough particles in space to transfer heat by moving around or passing on vibrations. Heat from the sun reaches the Earth by **radiation**. Heat transferred in this way is called **radiant heat**. Heat transfer by radiation is much faster than heat transfer by conduction or convection.

Place your hand near the base of the globe of a lamp. Turn on the lamp. You feel the heat from the globe almost instantly. There is not enough time for conduction or convection. The heat reaches your hand by radiation.

INQUIRY: INVESTIGATION 10.1

Moving particles

KEY INQUIRY SKILLS:

- processing and analysing data and information
- evaluating

Equipment:

250 mL beaker
single small crystal of potassium permanganate
Bunsen burner and heatproof mat
tripod and gauze mat
tweezers
drinking straw

- Fill the beaker with water and place it on a gauze mat and tripod.
- Use the tweezers to drop a crystal of potassium permanganate down the drinking straw into the water at the bottom of the beaker.

Drop the coloured crystal down the drinking straw.

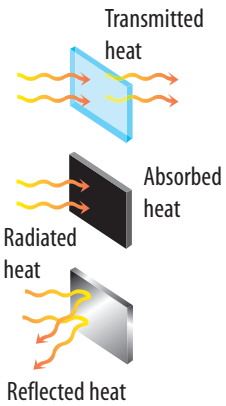
- Slowly remove the straw, taking care not to disturb the water.
- Light the Bunsen burner and turn it to a blue flame. Be sure not to disturb the beaker.

DISCUSS AND EXPLAIN

1 Draw a diagram to show the movement of colour through the beaker. This will show the currents within the beaker.
2 Explain why the colour moved in the way it did.
3 Is this experiment successful at modelling convection? Explain why or why not.

TRANSMISSION, ABSORPTION AND REFLECTION

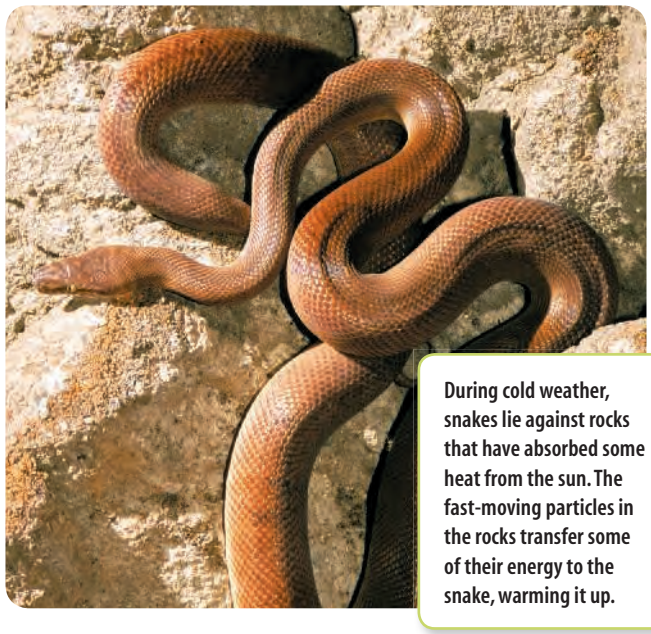
When radiant heat strikes a surface, it can be **reflected**, **transmitted** or **absorbed**. Most surfaces do all three; some surfaces are better reflectors, others are better absorbers and some transmit more heat.



Transmitted radiant heat
Clear objects, like glass, allow light and radiant heat to pass through them. The temperature of these objects does not increase quickly when heat reaches them by radiation.

Absorbed radiant heat
Dark-coloured objects tend to absorb light and radiant heat. Their temperatures increase quickly when heat reaches them by radiation.

Reflected radiant heat
Shiny or light-coloured surfaces tend to reflect light and radiant heat away. The temperature of these objects does not change quickly when heat reaches them by radiation.



During cold weather, snakes lie against rocks that have absorbed some heat from the sun. The fast-moving particles in the rocks transfer some of their energy to the snake, warming it up.

UNDERSTANDING AND INQUIRING

REMEMBER

1 Copy and complete the table below.

Type of heat transfer	Describe briefly how heat moves	Substances in which heat moves in this way
Conduction		
Convection		
Radiation		

2 What is an insulator? Name three different materials that can act as insulators.

3 Heat can travel through empty space (for example, between the sun and the Earth). How does the heat move?

4 What three things can happen to radiated heat when it arrives at any surface?

THINK

5 Conduction occurs in solid materials like metals but is not an effective way of transferring heat in liquids and gases. Explain why this is so.

6 Draw a diagram similar to the one on the previous page (top left) to show how air-conditioners push cool air out near the ceiling to create convection currents that cool rooms in hot weather.

7 When you hold a mug of coffee or hot soup, your hands feel warm. How is the heat transferred to your hands? Use a storyboard, cartoon or flowchart to illustrate your response.

8 Would it be hotter to sit in a black or a white car during summer? Why?

INVESTIGATE

9 Compare the advantages and disadvantages of evaporative and refrigerated air-conditioners.

10 How quickly do things cool? The rate at which substances cool is determined by many factors. A cup of hot chocolate will cool more rapidly than the same cup filled with thick vegetable soup. The material in the cup is one variable that affects how quickly cooling takes place. The size of the container, the temperature around the outside of the container, and the type of container are other variables that affect the rate of cooling.

Choose one variable to investigate. All other variables must remain the same so that the test is fair. If, for example, you decide to investigate the effect of the shape of the cup, you must make sure that nothing but the shape changes. The two or three shapes of cup you choose to investigate would need to contain the same amount of liquid, start at the same temperature, be made from the same materials, and be in the same surroundings.

- Write down the aim of your investigation and state your hypothesis.
- List the set of steps that you will follow.
- Decide what equipment is needed and make a list of it.
- Decide how your results will be recorded and draw up any necessary tables.
- Check with your teacher before beginning.
- Use your results to write a conclusion. State whether your hypothesis was supported.

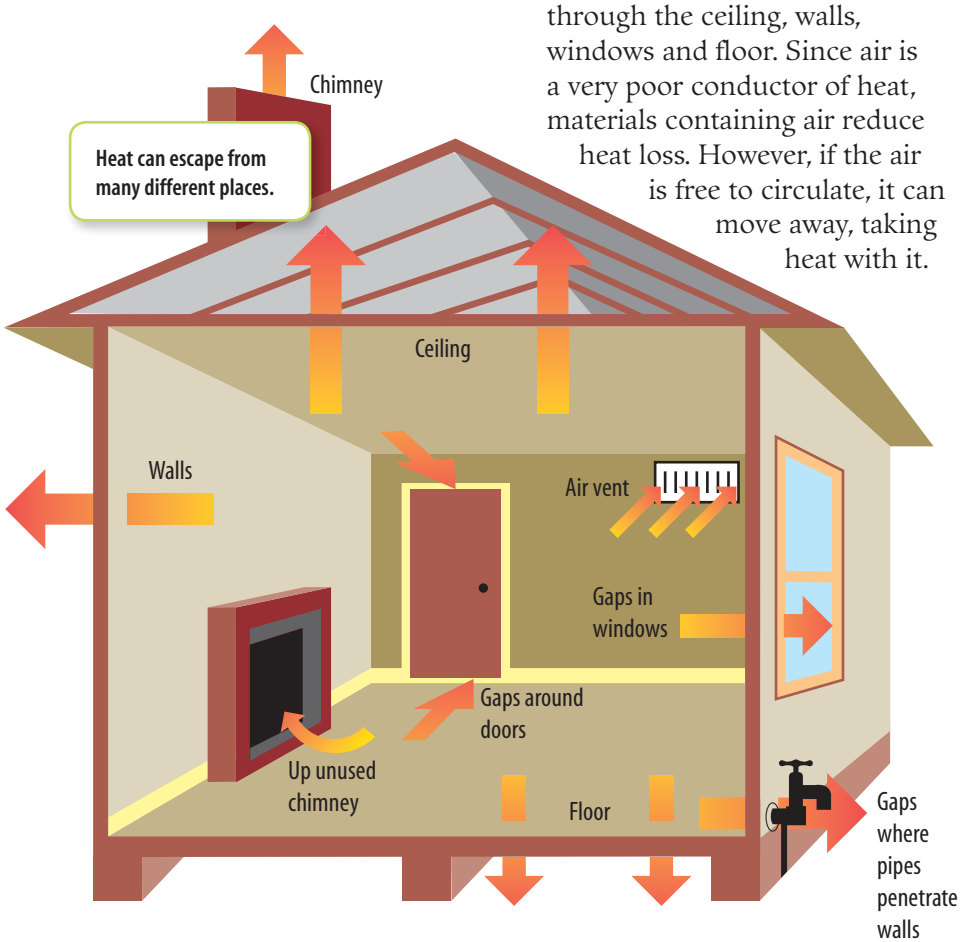
A costly escape

Knowledge of how heat moves from a warm place to a cooler place can help you to save on the energy that is used to heat and cool your home.

Using less energy for heating and cooling also conserves valuable resources such as coal (used to generate electricity) and natural gas.

Staying warm

In winter, heat leaves the inside of a warm, cosy home by conduction, convection and radiation. New homes are designed to reduce heat losses by all three methods.



However, there are also measures that occupants can take to reduce heat losses (and the bills that go with them).

USING THE SUN

The direction that a house faces, positioning of windows and skylights, and the types of trees planted around the house all affect the amount of sunlight and radiated heat that enter the home. **Deciduous** trees planted near north-facing windows allow radiated heat from the sun through in winter but block it out in summer.

INSULATION

Heat loss by conduction occurs through the ceiling, walls, windows and floor. Since air is a very poor conductor of heat, materials containing air reduce heat loss. However, if the air is free to circulate, it can move away, taking heat with it.

The best insulators, therefore, are those that contain air that is restricted from moving. Woollen clothes, birds' feathers and animal fur are all good insulators because they restrict heat loss by both conduction and convection.

Some ways in which insulation is used in the home include:

- ceiling insulation such as fibreglass batts, and loose rockwool that can be blown in. These materials contain pockets of air that provide insulation, and reduce the loss of warm air from the roof by convection.
- cavity wall insulation, a foam that can be sprayed in between the inside and outside walls
- heavy curtains, which trap a still layer of air between them and windows
- double glazing — the use of two sheets of glass in windows with a narrow gap of air between them
- cavity bricks, which have holes in them. The still air in the holes reduces heat loss by conduction and convection.

Do you feel a draught?

Preventing draughts is the easiest way to reduce heat loss in winter. There are many products available from hardware stores designed to seal small cracks and gaps to stop draughts. Draughts from chimneys and exhaust fans are difficult to control, but some exhaust fans have automatic shutters that close when the fan is not in use. Chimneys may have a metal plate to seal off air when there is no fire alight.

INQUIRY: INVESTIGATION 10.2

Investigating insulators

KEY INQUIRY SKILLS:

- processing and analysing data and information
- evaluating

Equipment:

6 empty soft drink cans
6 thermometers
newspaper
woollen cloth
cottonwool
polystyrene foam and sticky tape, or foam drink can holder
foam rubber
hot water
measuring cylinder
sticky tape (to tape on the materials)

(a) Plain (b) Newspaper (a few layers) (c) Woollen cloth (d) Cotton-wool (e) Foam drink can holder (f) Foam rubber

Investigating insulators

- Surround each can except one with a different material.
- Copy the table below into your workbook and use it to record your measurements.
- Measure out and pour 100 mL of hot water into each of the cans.
- Measure the temperature of the water in each can. Repeat the measurement of temperature every 5 minutes for 20 minutes.

DISCUSS AND EXPLAIN

1 Draw a bar graph that will allow you to compare the drop in temperature of the water in the cans after 20 minutes.
2 Which covering appears to be the most effective insulator?
3 Which one or more of the three methods of heat transfer does the most effective insulator reduce?
4 Use your data to suggest a good container for a mug of hot chocolate.
5 Why was one can left without a covering?
6 Are your conclusions reliable? Discuss the difficulties encountered in making sure that the comparison of insulators was fair.

Temperature of water in cans (°C)

Can covering	Time (minutes)				
	0	5	10	15	20
None					
Newspaper					
Woollen cloth					
Cottonwool					
Foam can holder					
Foam rubber					

Radiation

A warm house radiates heat in all directions. Heat loss by radiation can be reduced with shiny foil that reflects radiated heat. Foil can be added to insulation in the ceiling and is also used in external walls.

UNDERSTANDING AND INQUIRING

REMEMBER

1 What property makes a material a good insulator?
2 Installing insulation in the ceiling reduces which method (or methods) of heat transfer?
3 What is the cheapest way of reducing heat losses from your home in cold weather?

THINK

4 Foil placed in ceilings and walls is often referred to as 'insulation'. Is this term appropriate? Explain your answer.
5 What are convection currents? Draw a diagram to show how they move heat around a room.
6 Homes with central heating that are built on concrete slabs have heating ducts in the ceiling because they cannot be installed in the floor.
   (a) What is the disadvantage in having ducts in the ceiling?
   (b) Suggest a way of overcoming this disadvantage.
7 Loose clothing is recommended on hot days as it allows body heat to escape. Explain why loose clothing is better than close-fitting clothing for this purpose.

INVESTIGATE

8 What features of a thermos flask reduce heat loss by:
   (a) conduction?
   (b) convection?
   (c) radiation?

work sheet 10.3 A costly escape

Light energy

Like all stars, the sun changes some of the energy stored inside it into light energy. A burning candle converts some of the chemical energy stored in wax into light energy. Some living things are also able to change chemical energy stored in their bodies into light energy.

The firefly's light energy comes from a chemical reaction that takes place in its abdomen.

Without light from the sun, the world would be in darkness. Plants would not grow and no other life on Earth would exist. However, light makes up only a very small fraction of the energy that comes to us from the sun.

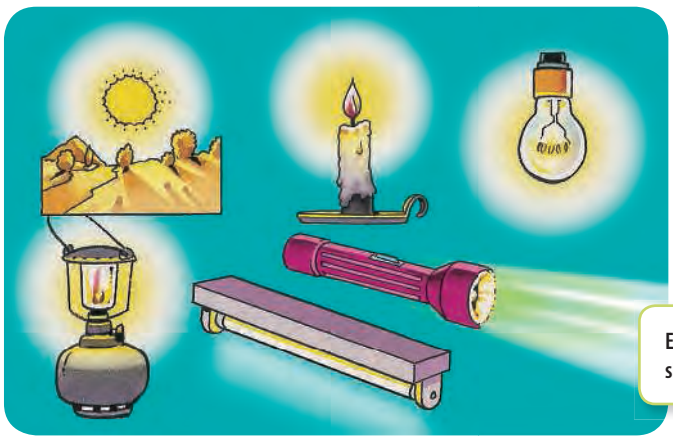
Light travels through space at about 300 000 kilometres per second, taking almost 10 minutes to get here.

Sources of light

The sun is not the only source of light. Any objects or substances that give off their own light are said to be **luminous**. Examples of some other sources of light are shown on the right.

Most of the light sources shown are **incandescent**. They emit light because they are hot. The sun and all other stars, light bulbs and flames are incandescent. Other sources, such as fluorescent tubes, the paint on the hands and numerals of clocks and watches, fireflies, glow-worms and some deep-sea fish, emit light without getting hot — they are not incandescent. Living things that emit light without heat are referred to as **bioluminescent**.

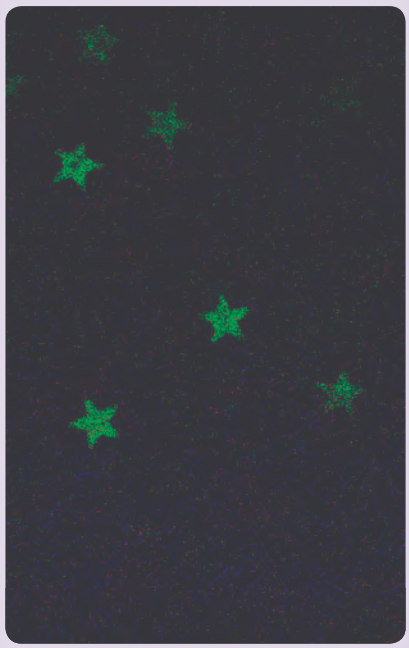
Most things that you see are not luminous. We see **non-luminous** objects because light from luminous objects is reflected from them. They do not emit their own light. Light from luminous objects, such as the sun, light globes or fluorescent tubes, strikes them and is reflected into your eyes. The moon is not a luminous object. Its surface reflects light from the sun.



Each of the light sources shown here is luminous.

HOW ABOUT THAT!

Glow-in-the-dark stickers are made with a chemical called phosphor, which absorbs light energy. It then slowly releases this extra energy as a single colour — usually green. Because the light energy is released more slowly than it is absorbed, the sticker releases green light for quite a long time. This process is called phosphorescence.



The angler fish, living in darkness about 4000 metres below the ocean surface, uses a luminous lure to attract its prey.

The deep black sea

Light from the surface does not reach deep below the ocean. From a depth of about 1000 metres downwards, the ocean is in complete darkness. Imagine the problems the fish that live here have in finding food. Some deep-sea fish swim closer to the surface to get their food, but others spend all of their time in the dark. The angler fish wiggles a luminous lure to attract its prey. The viperfish uses bioluminescent lights in its open mouth to entice prey directly into its stomach. The black dragonfish produces red light from a spot just beneath its eye. This allows the dragonfish to see its prey without being seen itself, as most of its prey can't see red light.

Seeing the light

Light is not normally visible between the source and any surface that it strikes. You can see a beam of light only if there are small particles in its path. The light is then **scattered** in many directions by the particles, some of it reaching your eye.

INQUIRY: INVESTIGATION 10.3

Observing a radiometer

A radiometer consists of four vanes, each of which is black on one side and silver on the other. The vanes are balanced on a vertical support so that they can turn with very little friction. The mechanism is encased inside a glass bulb from which air has been pumped out, making it almost a vacuum.

KEY INQUIRY SKILL:

- processing and analysing data and information

Equipment:

radiometer

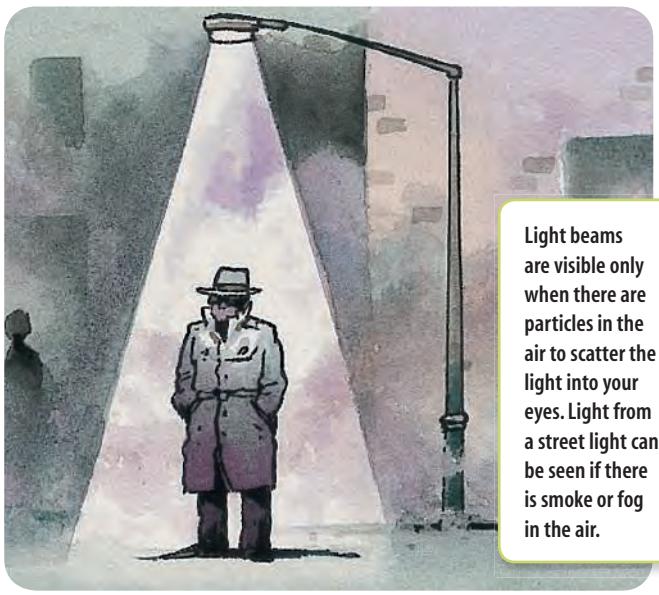
- Put the radiometer in direct sunlight. Record your observations.
- Put it in the shade. Record your observations again.

DISCUSS AND EXPLAIN

1 What effect does sunlight have on a radiometer?

2 How does this experiment demonstrate that sunlight is a form of energy?

3 Research a scientific theory to explain the effect of sunlight on the radiometer.



Light beams are visible only when there are particles in the air to scatter the light into your eyes. Light from a street light can be seen if there is smoke or fog in the air.

INQUIRY: INVESTIGATION 10.4

Seeing the light

KEY INQUIRY SKILL:

- developing explanations

Equipment:

moderately dark room

torch or projector

matches or a well-used chalk duster

- Shine the torch or projector on a nearby wall.
- Now hit the chalk duster with your hand, or light and blow out a match, so that chalk dust or smoke falls between the light source and the wall.

DISCUSS AND EXPLAIN

1 Can you see the light beam between the light source and the wall without the chalk dust or smoke?

2 What changes when the chalk dust or smoke is present?

3 Explain what happens to the light from the source to make it visible.

WHAT DOES IT MEAN?

The word *absorb* comes from the Latin word ***sorbere***, meaning 'to suck in'.

Meeting new substances

When light energy travels from one substance to another, three things can happen to it.

1 It can be transmitted; that is, the light energy can travel through the substance. For example, light is transmitted through clear glass.

2 It can be absorbed; that is, the light energy can be transferred to particles inside the substance. For example, the tinted glass in many cars contains a substance that absorbs some of the light energy passing through it.

3 It can be reflected from the surface of the substance or reflected (scattered) by small particles inside the substance. For example, light is reflected from opaque objects like a piece of wood and scattered by particles of water in fog. This is how you are able to see them.

You can't see the people in this car because most of the light energy coming from inside the car is absorbed by the tinted glass.

UNDERSTANDING AND INQUIRING

REMEMBER

1 What is light and how fast does it travel through space?

2 (a) What does 'incandescent' mean?
(b) List two examples of light sources that are incandescent.
(c) List two examples of light sources that are not incandescent.

3 Why do you see the beam of light from a torch if it is foggy?

4 Describe what can happen to light energy travelling through the air when it meets a new substance.

THINK

5 Which of the following objects are luminous?
(a) The sun
(b) The moon
(c) The stars
(d) A burning candle
(e) This page

6 Apart from light, what other form of energy comes to the Earth from the sun?

7 From what form of energy is the light produced by fireflies converted?

8 Explain how it is that you can see this page even though it does not emit light of its own.

9 How long does it take light to travel from the sun to the distant dwarf planet Pluto when it is 6000 million kilometres from the sun?

10 When light energy meets the surface of your sunglasses, what is the evidence that some of it has been:
(a) transmitted?
(b) reflected?
(c) absorbed?

INVESTIGATE

11 Find out how the energy that reaches the Earth is produced by the sun.

12 Research and report on an animal that produces light by bioluminescence.

work sheet 10.4 Light energy

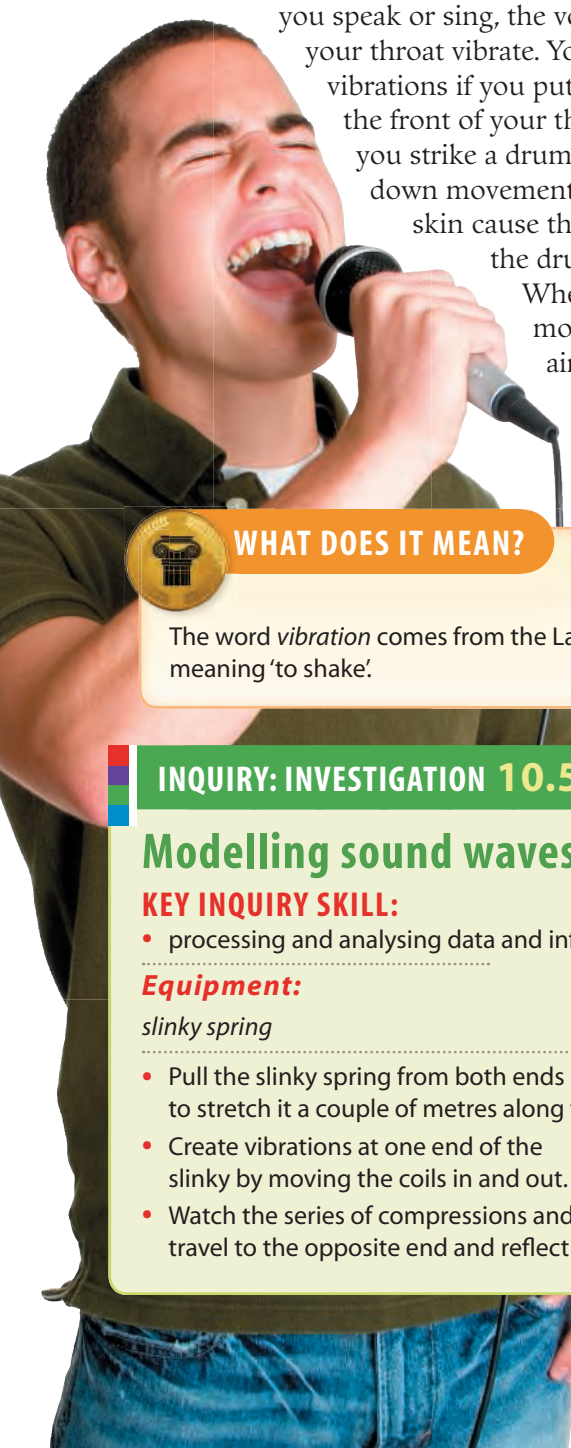
Sound energy

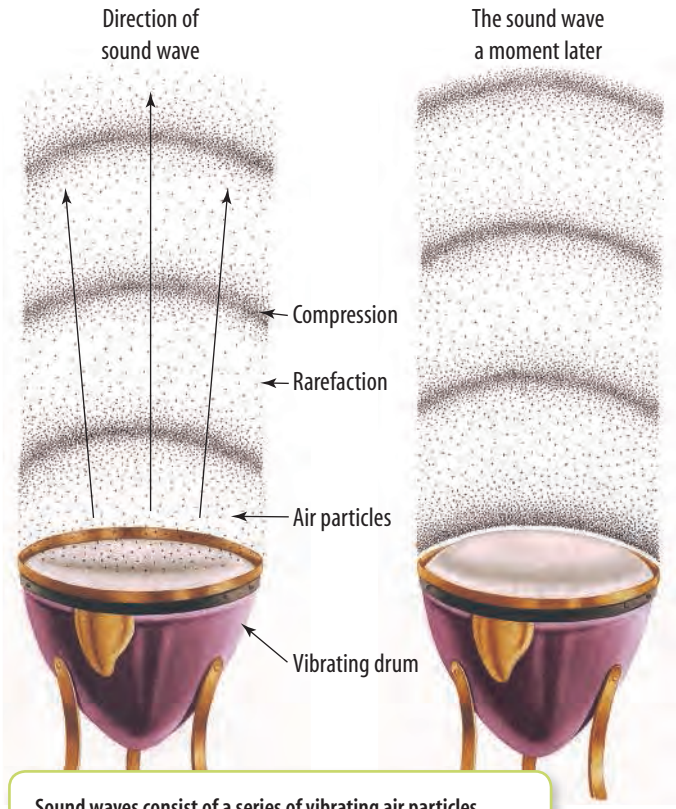
Humans and other animals rely heavily on sound energy to communicate with each other. You can use your voice, whistle or tap something to make a sound. How else can you make a sound?

Good vibrations

All sounds are caused by **vibrations**. When you speak or sing, the vocal cords in your throat vibrate. You can feel the vibrations if you put your hand over the front of your throat. When you strike a drum, the up and down movements of the drum skin cause the air around the drum to vibrate. When the drum skin moves down, the air particles near it are pulled back, spreading them out. A fraction of a second later the drum skin moves back up, squeezing the air particles together.

The energy of the air particles is transferred to nearby air particles, causing them to vibrate as well. This creates a moving series of **compressions** (air particles closer together than usual) and





Sound waves consist of a series of vibrating air particles.

WHAT DOES IT MEAN?

The word *vibration* comes from the Latin word ***vibrare***, meaning 'to shake'.

INQUIRY: INVESTIGATION 10.5

Modelling sound waves

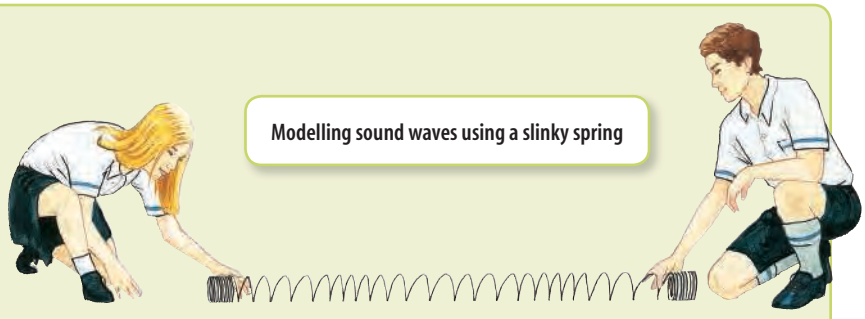
KEY INQUIRY SKILL:

- processing and analysing data and information

Equipment:

slinky spring

- Pull the slinky spring from both ends to stretch it a couple of metres along the floor.
- Create vibrations at one end of the slinky by moving the coils in and out.
- Watch the series of compressions and rarefactions travel to the opposite end and reflect back.



Modelling sound waves using a slinky spring

DISCUSS AND EXPLAIN

1. Describe how your model is similar to real sound waves.
2. Describe how your model is different from real sound waves.

rarefactions (air particles further apart than usual) that move away from the source of the sound. These moving compressions and rarefactions are what we know as **sound waves**. If enough energy is transferred to the vibrating air, the sound waves reach your eardrum and you hear sound.

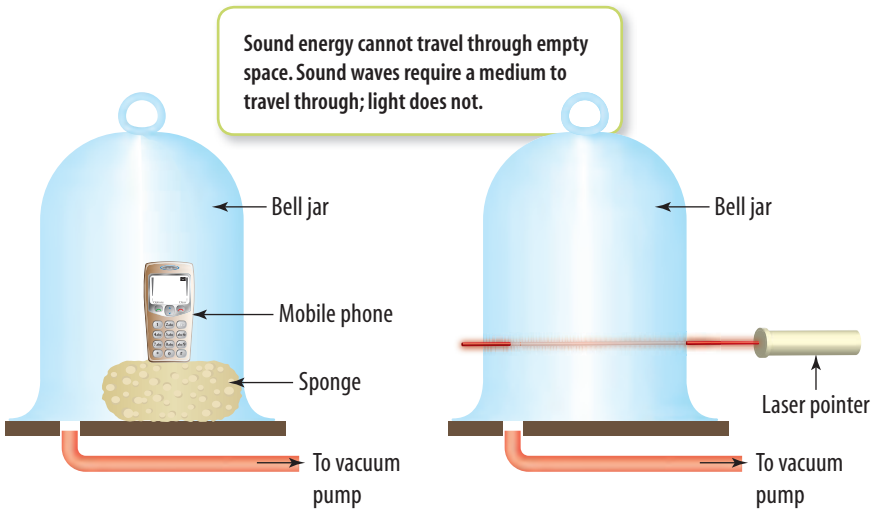
The highness or lowness of a sound is called its **pitch**. The faster an object vibrates, the higher the pitch of the sound it makes. A short string vibrates faster than a long one so has a higher pitch. When you blow across the top of a straw, the air inside it vibrates. If the straw is shorter, the air inside vibrates faster, producing a higher pitched sound.

The need for air

When a mobile phone rings in a bell jar, the sound can be heard clearly. But if the air inside is sucked out by a vacuum pump, the sound can't be heard. Sound energy cannot travel through empty space — it can travel only by making particles vibrate. In empty space there are no particles to vibrate. Light energy, unlike sound, can travel through empty space. It doesn't need particles. So you can still see the ringing phone in the bell jar, even if you can't hear it.

HOW ABOUT THAT!

During a thunderstorm, the flash of lightning and the crash of thunder occur only a tiny fraction of a second apart. So why do you always hear thunder one or more seconds after you see the lightning? The answer lies in one of the differences between sound and light. Sound energy travels through the air at a speed of about 340 m/s. Light energy travels through air at a speed of 300 000 km/s. The delay between seeing lightning and hearing thunder is about 3 seconds for each kilometre that you are away from the lightning.



Making it louder

If you pluck a stretched guitar string while it is not attached to a guitar, it vibrates but makes very little sound. If you strike a stretched drum skin while it is not attached to the drum, it makes very little noise. Even your own vocal cords make very little noise while they are vibrating. In each of these cases, a vibration is needed to create the sound but an enclosed region of air is needed to make the sound louder.

The air inside the body of an acoustic guitar is set vibrating by the strings. The air inside a drum vibrates when the drum skin is struck. The vibrating air inside your throat and mouth makes the sound created by your vocal cords loud enough to be heard.

The sounds of music

How do musical instruments produce sound? The energy comes from the person playing the instrument — but what does the instrument do to convert that energy into sound?

INQUIRY: INVESTIGATION 10.6

Vibrations and pitch

KEY INQUIRY SKILL:

- processing and analysing data and information

Equipment:

ruler
2 straws
scissors
spatula
small beaker
large beaker

- Hold a ruler over the edge of a table so that one end is firmly pushed down. Flick the overhanging end of the ruler.
- Move the ruler so that more of it is over the edge of the table and flick it again.
- Cut one straw into two so that one part is twice as long as the other part. Place the top of the uncut straw lightly against your bottom lip and blow gently across the opening. Listen to the sound made.
- Blow across the two shorter (cut) pieces of straw in the same way and listen to the sounds.
- Tap the side of a small beaker gently with a spatula and listen to the sound. Do the same with a larger beaker.

DISCUSS AND EXPLAIN

1. How does the sound change as the vibrating part of the ruler is made longer?
2. How does the sound change as the straws get shorter?
3. How does the sound of the large beaker compare with the sound of the smaller one?
4. How would you change each of the following to make a higher pitched sound?
   (a) The length of a vibrating strip of wood
   (b) The length of a tube of air
   (c) The size of a cymbal

INQUIRY: INVESTIGATION 10.7

Making it louder

KEY INQUIRY SKILL:

- processing and analysing data and information

Equipment:

guitar
guitar string
tuning fork

- Pluck a stretched guitar string. Listen to the sound it makes.
- Pluck a similar string attached to a guitar.
- Strike a tuning fork on the sole of your shoe and listen to the sound it makes. While it is still vibrating, place the base of the fork on a solid table surface.

DISCUSS AND EXPLAIN

1. How does the sound of a plucked string change when it is attached to a guitar?
2. How does the sound change when the tuning fork is placed on the table?
3. Explain why the sound changes.

With an acoustic guitar, the vibrations are made by plucking the strings. The air around the sound hole vibrates, causing the air inside the body of the guitar to vibrate. In an electric guitar, a microphone or pick-up detects the vibrating air and an amplifier is used to make the sound louder. The pitch of the sound made by a guitar is increased by shortening the strings using your fingers, tightening the strings or using lighter strings.

A saxophone's vibrations are first made by a thin wooden reed. The air inside the saxophone then vibrates, making a loud sound. The pitch can be changed by using keys to open or close holes. When all the holes are closed, the saxophone contains a long column of air, producing a low-pitched sound. As holes are opened, the length of the air column becomes shorter, and the pitch increases.

The didjeridu is a wind instrument that has no holes to change the length of the column of vibrating air. The player blows into the instrument using loosely vibrating lips to control how quickly the air inside vibrates.

Sounding great

Like light, sound energy can be transmitted, reflected or absorbed when it meets a new substance.

- All materials transmit some sound, some better than others. That's why you can sometimes hear conversations from the other side of a wall.
- Sound is reflected from hard substances like the tiles in your bathroom. Each note that

INQUIRY: INVESTIGATION 10.8

Making music

KEY INQUIRY SKILL:

- processing and analysing data and information

Equipment:

a small selection of musical instruments

- If musical instruments are available in your classroom, have someone demonstrate how they work.
- Look at the musical instruments illustrated below.

DISCUSS AND EXPLAIN

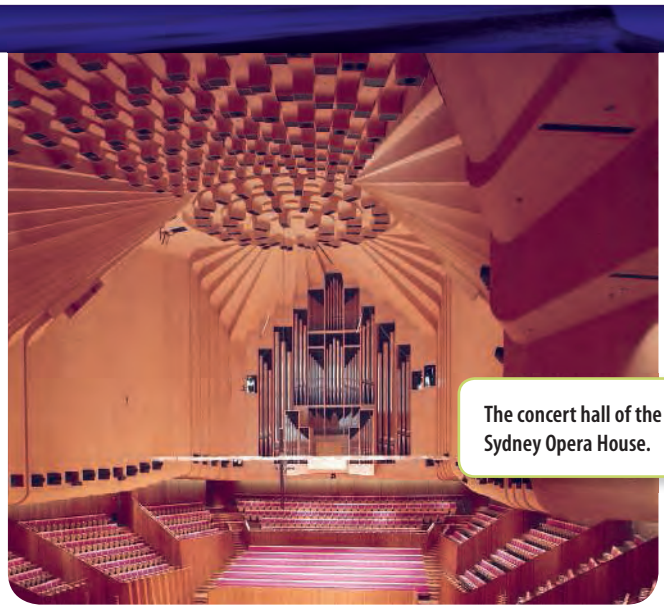
1 For each instrument, write down:
(a) what the player does to make the instrument work
(b) what vibrates to make musical sounds.

2 What do all of the musical instruments have in common about the way they make sound?

you sing in the shower lasts longer because it is reflected over and over again from hard surfaces. This effect is called **reverberation**.

- Soft materials, like curtains and carpet, absorb much more sound than walls of plaster or tiles.

Concert halls are designed to control the transmission, reflection and absorption of sound. For example, the timber in the panelling on the ceiling and walls of the concert hall in the Sydney Opera House was selected because it minimises reflection and reverberation. In the Melbourne Concert Hall, heavy curtains behind the audience can be closed to increase the amount of sound absorbed.



The concert hall of the Sydney Opera House.

UNDERSTANDING AND INQUIRING

REMEMBER

1 What is the cause of all sounds?

2 When compared with air particles in a silent room, how are the particles in compressions and rarefactions different?

3 Explain why sound can't travel through empty space.

4 If you blow across the top of a straw, a sound is made. How could you increase the pitch of the sound? How do we know, without taking any measurements, that light travels through air faster than sound?

5 Which vibrates more quickly — a long string or a short string made of the same material?

6 Describe what can happen to sound energy travelling through the air when it meets a new substance.

7 Which types of surfaces cause reverberation?

THINK

8 Is sound energy a form of kinetic energy? Explain your answer.

9 Explain why the mobile phone in the bell jar needed to be sitting on a sponge when the air was removed.

10 Imagine that light energy couldn't travel through empty space. What would you observe if the air was removed from a bell jar containing a ringing mobile phone?

11 How would you expect a carpeted classroom to sound compared with one with a hard vinyl floor? Give reasons for the differences.

12 How are different notes played on:
(a) a single string of a guitar?
(b) a recorder?
(c) a xylophone?

13 Complete the gaps in the following table.

Musical instrument	What vibrates first?	What makes the sound louder?
Guitar	Plucked string	Air inside guitar
Trumpet	Player's lips	
Drum		Air inside drum
Saxophone		Air inside saxophone
	String hit by hammers	Air inside instrument

CREATE AND EXPLAIN

14 Make a string telephone. You will need about five metres of string and two open and empty cans. Punch a small hole in the bottom of each can. Thread the string through each hole and tie a knot to keep the string in place. Hold the cans far enough apart so that the string is tight. Talk into the can at one end while your partner listens at the other end.
(a) How does the sound travel from one can to the other?
(b) Does the sound change if you make the string tighter or looser?
(b) Would a string telephone work without the cans? Why are the cans used?

INVESTIGATE

15 Find out more about the following careers that involve using and understanding sound energy.
(a) Audiologist
(b) Acoustic engineer
(c) Audio engineer

10.6 THINKING TOOLS

Matrixes and Venn diagrams

To show similarities and differences between topics

why use?

1. Write the topics in the left-hand column of the matrix.
2. Write the characteristics to be compared along the top row of the matrix.
3. If a characteristic applies to a topic, put a tick in the appropriate cell of the matrix.
4. The matrix now shows how the various topics are related.

how to ...?

In what ways are these topics similar and different?

question

Matrix

Topic	Feature A	Feature B	Feature C	Feature D	Feature E
1	✓		✓	✓	✓
2		✓			✓
3		✓		✓	✓
4			✓	✓	✓

also called

Table; grid; decision chart

Similarity

Both identify common points between two separate topics.

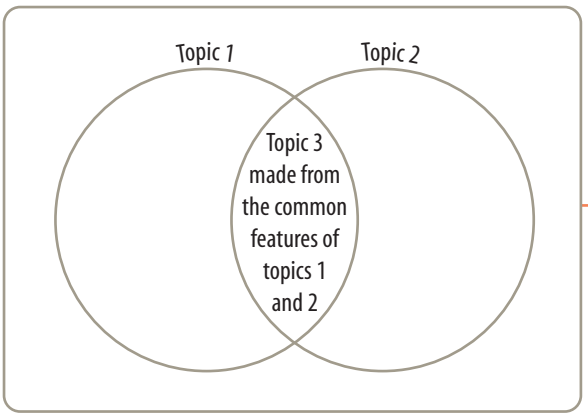
Difference

They use different graphic formats to show the common features.

comparison

Venn diagram

example



UNDERSTANDING AND INQUIRING

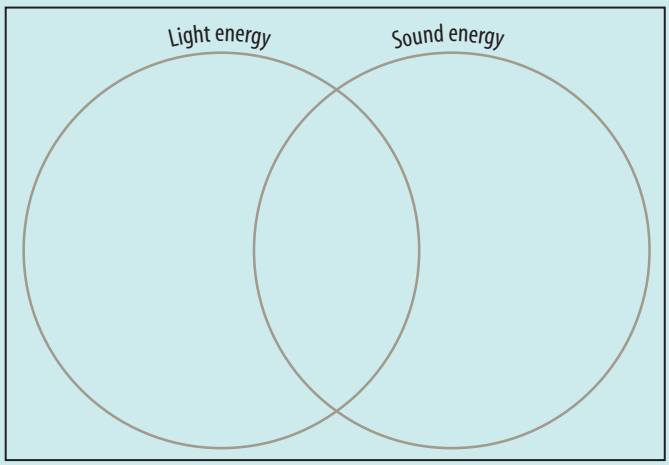
THINK AND CREATE

1 Copy and complete the matrix below. Use ticks to indicate the forms of energy that electrical energy is transformed into by each of the electrical devices listed.

Device	Electrical energy is converted into . . .				
	Light energy	Sound energy	Thermal energy	Kinetic energy	Potential energy
Hairdryer					
Television					
Desk lamp					
Vacuum cleaner					
Home computer					
Incandescent light bulb					
Air-conditioner					
Elevator going up					

2 (a) Copy and complete the matrix at right. Use ticks to show which statements refer to light and which refer to sound. Some of the statements refer to both light and sound.

(b) The information in the matrix at right can be represented in a Venn diagram. Convert the information in the matrix into a large Venn diagram based on the one below.



The overlapping section of the Venn diagram contains the properties of both light and sound.

Statement	Light energy	Sound energy
Travels through air at 300 000 kilometres per second		
Travels faster than a speeding bicycle		
Can be reflected		
Can be absorbed		
Is always caused by vibrating objects or substances		
Is observed in an electrical storm		
Can travel through empty space		
Can be produced from another form of energy		
Is used by physiotherapists to treat some muscle problems		

STUDY CHECKLIST

ENERGY TRANSFERS AND TRANSFORMATIONS

- define the term 'energy'
- identify bodies that possess kinetic energy because of their motion
- define potential energy as stored energy
- differentiate between gravitational, chemical and elastic potential energy
- outline examples of transformations from potential and kinetic energy into other forms of energy
- recognise that heat energy is always produced as a by-product of energy transfers
- use flow diagrams to illustrate changes between different forms of energy

HEAT, LIGHT AND SOUND ENERGY

- define heat as energy in transit from one object or substance to another object or substance with a lower temperature
- describe and compare the transfer of heat by conduction, convection and radiation
- differentiate between luminous and non-luminous objects
- relate the ability to see non-luminous objects to the scattering and reflection of light
- describe sound as a series of vibrating air particles
- recognise that heat, light and sound energy can be transmitted, reflected or absorbed

SCIENCE AS A HUMAN ENDEAVOUR

- explain how the unwanted transfer of heat can be decreased to reduce the amount of energy needed for home heating and cooling
- describe the energy transformations involved in playing a variety of musical instruments
- investigate how energy efficiency can reduce energy consumption
- investigate the development of more energy-efficient motor vehicles

INDIVIDUAL PATHWAYS

eBookplus

Activity 10.1
Transferring and transforming energy
doc-6093

Activity 10.2
Investigating energy
doc-6094

Activity 10.3
Analysing energy
doc-6095

ICT

eBookplus Summary

eLESSON

Energy in disguise
Did you know that all energy is constantly being transformed and transferred from one object to another? This eLesson helps you to discover that there's more going on in your world than meets the eye as you learn about the different types of energy and the laws that govern it.
A worksheet is attached to further your understanding.

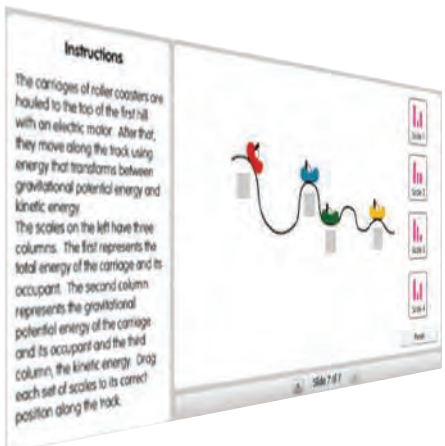
Searchlight ID: eles-0063

Australian International Model Solar Challenge
Learn about the exciting annual event where Australian high school students compete by building and racing model cars and boats.

Searchlight IN: eles-0068

INTERACTIVITY

Coaster
This interactivity helps you apply your knowledge of energy to an amusement ride. Identify the positions in a roller-coaster ride where the car would have more kinetic energy and where it would have more gravitational energy. Instant feedback is provided.



Searchlight ID: int-0226

LOOKING BACK

1 Replace each of the following descriptions with a single word.
   (a) Energy associated with all moving objects
   (b) Energy that is stored
   (c) The form of energy that causes an object to have a high temperature
   (d) The form of energy stored in a battery that is not connected to anything
   (e) The source of most of the Earth's light

2 Explain why the amount of energy in the universe never changes.

3 Describe an example of an object that has:
   (a) elastic potential energy
   (b) gravitational potential energy.

4 Draw a flowchart to illustrate the energy transformations that take place:
   (a) after you switch on a torch
   (b) when a firecracker is lit
   (c) when a ball rolls down a hill and then up another hill.

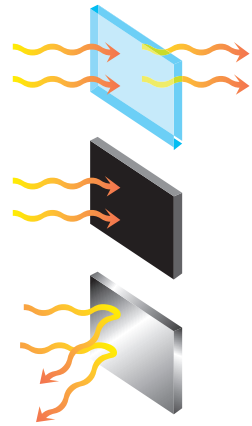
5 When a kettle of water is boiled on a gas cook-top, not all of the energy stored in the gas is used to heat the water. Where does the rest of the energy go?

6 Explain why it is not possible for an energy converted like a battery or car to have an efficiency of 100%.

7 Explain how fibreglass batts are able to reduce the loss of heat through the ceiling by both conduction and convection.

8 Heat moves from regions of high temperature to regions of low temperature by conduction, convection or radiation. In which of these three ways is heat most likely to be transferred:
   (a) from a frying pan to an egg being fried?
   (b) from the sun to the planets of the solar system?
   (c) through water in a saucepan on a hotplate or gas burner?
   (d) through a metal spoon being used to stir hot soup?
   (e) from a very hot and bright light globe near the ceiling to your body directly beneath?

9 Each of the diagrams on the right shows radiated heat falling on a solid object. Which diagram shows heat being:
   (a) absorbed?
   (b) reflected?
   (c) transmitted?



10 Explain how your body keeps its core temperature at 37°C even when the air temperature is greater than this.

11 Explain how you are able to see an object like a tree even though it doesn't produce its own light energy.

12 Make a list of as many luminous objects as you can.

13 Why is it incorrect to describe fluorescent lights as incandescent?

14 You can't normally see the beam of light coming from a car headlight. However you can see the beams if there is fog or smoke in the air. How does the fog or smoke make a difference?

15 When a sound is made, what happens to the particles in the regions of the air nearby that are called:
   (a) compressions?
   (b) rarefactions?

16 When an object making sound vibrates faster, what happens to the pitch of the sound?

17 When you sing in the shower the sound of your voice reverberates.
   (a) What happens to sound energy to cause reverberation?
   (b) Why don't you observe reverberation when you sing in a room with carpet and soft curtains?
   (c) In some outdoor places, if you speak loudly you can hear an echo. For example, you might say 'hello' and a second or two later you hear the word 'hello' again. Explain how an echo is different from a reverberation.

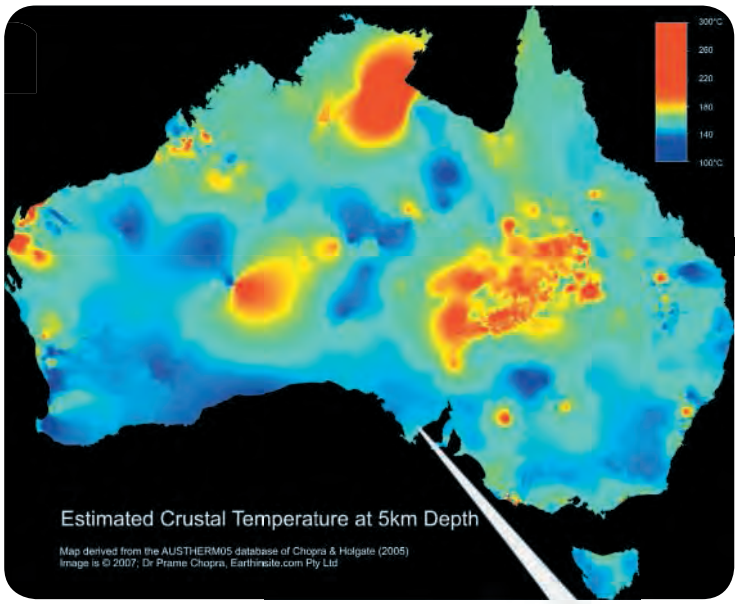
18 Explain what is wrong with this cartoon.

work sheet 10.5 Energy: Puzzles

projectsplus

Going green

SEARCHLIGHT ID: PRO-0093



Scenario

As the supply of fossil fuels dwindles, cities become more crowded and anthropogenic global warming becomes an unavoidable reality, an increasing number of people are opting in to a more self-sufficient lifestyle. To meet this need, there is an increasing number of architecture and building firms that specialise in the design and construction of houses that are energy efficient and that are able to exist off the electricity grid indefinitely because they use electricity generation systems that meet all of the household's needs using renewable energy sources.

You and your team at Sustainable Housing Solutions have been approached by a potential client who wants to build a series of sustainable eco-tourist cottages in remote locations across the country. To see whether your company should be awarded the lucrative contract to oversee the work on the whole chain of cottages, the client has asked you to make a presentation detailing how you would make one of these cottages as energy efficient and self-sustaining as possible. You can place this trial cottage anywhere in the country for your presentation purposes, provided that it is at least 100 km away from any town with a population greater than 5000 people. Other criteria must also be met as follows:

- All of the cottages will have the same layout and will be constructed of mud bricks and have tiled roofs (you will be given a copy of the plan). While you can change the orientation and location of the cottage, you cannot change the design or these construction materials.
- Each cottage must have the following appliances: refrigerator, washing machine, stove, microwave, TV set, DVD player and stereo system. Smaller appliances such as toasters, shavers, hairdryers and computers may occasionally be used by guests as well.
- The cottages must be cool in summer and warm in winter; the client is not opposed to the idea of a reverse-cycle air-conditioner or fans.
- There must be sufficient lighting to be able to read in every room.

- The cottages will not be attached to the national electricity grid — all of the electricity needs of each cottage must be met using a renewable energy source in its area. (Water will be provided from rainwater tanks and septic tanks will take care of the sewage.)

Your task

Your team will prepare and deliver a report for the client that provides the following information:

- The best location to place the trial cottage (keeping in mind that it can be placed somewhere close to a source of renewable energy)
- Suggestions as to how the cottage can be made as energy efficient as possible
- A detailed estimate of how much electricity will need to be generated to power the cottage and run appliances
- A justified recommendation as to which renewable energy system should be used to generate that amount of electricity and how it would be supplied to the trial cottage
- An estimate of how much the energy system will cost, using costs for similar systems available on the internet as a guideline.

The report will take the form of an oral presentation with visuals (which may include PowerPoint slides, overhead transparencies and models). The presentation should be between six and eight minutes long.

Process

- Open the ProjectsPLUS application for this chapter located in your eBookPLUS. Watch the introductory video lesson and then click the 'Start Project' button to set up your project group. You can complete this project individually or invite other members of your class to form a group. Save your settings and the project will be launched.
- Navigate to your Research Forum. Here you will find a number of different headings under which you will organise your research. You may delete those topics that you will not be considering or add your own topics if you find your research going in a different direction.
- Start your research. Make notes about information you think will be relevant to your project, entering your findings as articles under your topics in the Research Forum. You should each find at least three sources (other than the textbook, and at least one offline such as a book or encyclopaedia) to help discover extra information about energy efficiency and self-sufficiency. You can view and comment on other group members' articles and rate the information they have entered. When your research is complete, print out your Research Report to hand in to your teacher.
- Visit your Media Centre and download the client's house plan from the Documents section. Your Media Centre also includes images and weblinks that you may find helpful for your project.
- Start putting your presentation together.

SUGGESTED SOFTWARE

- ProjectsPLUS
- Word or other word-processing software
- PowerPoint
- Internet access

MEDIA CENTRE

Your Media Centre contains:

- a plan of the cottage in Word
- a selection of images
- a selection of useful weblinks
- an assessment rubric.

Your ProjectsPLUS application is available in this chapter's Student Resources tab inside your eBookPLUS. Visit www.jacplus.com.au to locate your digital resources.